



# Model Test Results Report

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 23-March-2026

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# 1 Executive Summary

This report documents the results of automated model tests run on **23 March 2026 at 12:58:50**.

Table 1: Overall Test Results

|                |         |
|----------------|---------|
| Total Tests    | 6419    |
| Passed         | 6405    |
| Failed         | 0       |
| Skipped        | 14      |
| Pass Rate      | 99.8%   |
| Total Duration | 277.33s |

## 1.1 Test Environment

|            |                                     |
|------------|-------------------------------------|
| Platform   | macOS-26.3.1-arm64-arm-64bit-Mach-O |
| Python     | 3.13.1                              |
| pytest     | 8.3.4                               |
| Coverage   | 98.8%                               |
| Git Commit | 73a9acc7b7dd                        |

## 1.2 Results by Model

Table 2: Test Results by Model

| Model ID   | Model Name                   | Total | Pass | Fail | Skip | Time (s) |
|------------|------------------------------|-------|------|------|------|----------|
| MKM-DD-001 | Flood Risk (Depth-Damage)    | 29    | 29   | 0    | 0    | 0.00     |
| MKM-GH-001 | GEV Hazard Model             | 16    | 16   | 0    | 0    | 0.27     |
| MKM-MP-001 | Mortgage Pricer              | 67    | 67   | 0    | 0    | 0.05     |
| MKM-PR-001 | PRS Pricing Model            | 61    | 61   | 0    | 0    | 6.47     |
| MKM-PV-001 | Property Valuation           | 55    | 55   | 0    | 0    | 19.07    |
| MKM-SG-001 | Storm-Gauge Response Model   | 81    | 81   | 0    | 0    | 0.04     |
| MKM-SI-001 | Storm Intensity Distribution | 105   | 105  | 0    | 0    | 0.24     |
| MKM-SS-001 | Storm Sequence Generator     | 52    | 52   | 0    | 0    | 0.21     |
| MKM-TD-001 | Trading Desk                 | 278   | 278  | 0    | 0    | 2.33     |
| CDM-ALL    | CDM Schema Validation        | 76    | 76   | 0    | 0    | 0.01     |
| PLATFORM   | Platform Infrastructure      | 5599  | 5585 | 0    | 14   | 169.91   |

## 2 Test Traceability Matrix

The following table maps every executed test to its assigned model component, providing evidence of coverage required under SR 11-7 §4.2 and SS1/23.

| Model      | Module                                    | Test Name                                   | Status |
|------------|-------------------------------------------|---------------------------------------------|--------|
| MKM-DD-001 | velocity.py::TestAttenuation              | test_at_length_scale                        | Pass   |
|            | velocity.py::TestAttenuation              | test_far_distance_near_zero                 | Pass   |
|            | velocity.py::TestAttenuation              | test_negative_distance                      | Pass   |
|            | velocity.py::TestAttenuation              | test_zero_distance_full_strength            | Pass   |
|            | velocity.py::TestAttenuation              | test_zero_length                            | Pass   |
|            | velocity.py::TestBuildPropertyHydrograph  | test_1d_hours_present                       | Pass   |
|            | velocity.py::TestBuildPropertyHydrograph  | test_depth_never_negative                   | Pass   |
|            | velocity.py::TestBuildPropertyHydrograph  | test_each_entry_has_required_keys           | Pass   |
|            | velocity.py::TestBuildPropertyHydrograph  | test_empty_readings                         | Pass   |
|            | velocity.py::TestBuildPropertyHydrograph  | test_high_floor_prevents_flooding           | Pass   |
|            | velocity.py::TestBuildPropertyHydrograph  | test_no_flooding_before_arrival             | Pass   |
|            | velocity.py::TestBuildPropertyHydrograph  | test_output_length_matches_input            | Pass   |
|            | velocity.py::TestBuildPropertyHydrograph  | test_peak_produces_flooding                 | Pass   |
|            | velocity.py::TestComputeSlope             | test_basic_slope                            | Pass   |
|            | velocity.py::TestComputeSlope             | test_min_slope_clamp                        | Pass   |
|            | velocity.py::TestComputeSlope             | test_slope_is_absolute                      | Pass   |
|            | velocity.py::TestComputeSlope             | test_zero_distance                          | Pass   |
|            | velocity.py::TestManningVelocity          | test_deeper_water_faster                    | Pass   |
|            | velocity.py::TestManningVelocity          | test_negative_depth_returns_zero            | Pass   |
|            | velocity.py::TestManningVelocity          | test_negative_slope_uses_abs                | Pass   |
|            | velocity.py::TestManningVelocity          | test_positive_depth_and_slope               | Pass   |
|            | velocity.py::TestManningVelocity          | test_slope_clamped_to_min                   | Pass   |
|            | velocity.py::TestManningVelocity          | test_steeper_slope_faster                   | Pass   |
|            | velocity.py::TestManningVelocity          | test_zero_depth_returns_zero                | Pass   |
|            | velocity.py::TestManningVelocity          | test_zero_roughness_returns_zero            | Pass   |
|            | velocity.py::TestTravelTime               | test_basic_travel_time                      | Pass   |
|            | velocity.py::TestTravelTime               | test_farther_takes_longer                   | Pass   |
|            | velocity.py::TestTravelTime               | test_zero_depth_infinite                    | Pass   |
|            | velocity.py::TestTravelTime               | test_zero_distance                          | Pass   |
| MKM-GH-001 | hazard_builder.py::TestGEVFit             | test_exceedance_probability_in_range        | Pass   |
|            | hazard_builder.py::TestGEVFit             | test_fit_returns_three_floats               | Pass   |
|            | hazard_builder.py::TestGEVFit             | test_force_gumbel_sets_shape_zero           | Pass   |
|            | hazard_builder.py::TestGEVFit             | test_higher_threshold_lower_probability     | Pass   |
|            | hazard_builder.py::TestGEVFit             | test_return_level_increases_with_period     | Pass   |
|            | hazard_builder.py::TestGaugeResponseModel | test_all_responses_returns_dict             | Pass   |
|            | hazard_builder.py::TestGaugeResponseModel | test_single_response_returns_gauge_response | Pass   |
|            | hazard_builder.py::TestGaugeResponseModel | test_response_threshold_flags               | Pass   |

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| Model      | Module                                    | Test Name                                    | Status |
|------------|-------------------------------------------|----------------------------------------------|--------|
|            | hazard_builder.py::TestHazardCurveBuilder | test_builder_flood_probabilities_in_range    | Pass   |
|            | hazard_builder.py::TestHazardCurveBuilder | test_builder_returns_dict_of_curves          | Pass   |
|            | hazard_builder.py::TestHazardCurveBuilder | test_builder_curve_is_gauge_hazard_curve     | Pass   |
|            | hazard_builder.py::TestTermStructure      | test_cumulative_prob_increasing              | Pass   |
|            | hazard_builder.py::TestTermStructure      | test_probabilities_sum_to_one                | Pass   |
|            | hazard_builder.py::TestTermStructure      | test_returns_correct_length                  | Pass   |
|            | hazard_builder.py::TestTermStructure      | test_returns_term_structure_points           | Pass   |
|            | hazard_builder.py::TestTermStructure      | test_survival_prob_decreasing                | Pass   |
| MKM-MP-001 | credit.py::TestCreditSpread               | test_custom_tax_rate                         | Pass   |
|            | credit.py::TestCreditSpread               | test_floor_at_10bps                          | Pass   |
|            | credit.py::TestCreditSpread               | test_higher_affordability_higher_spread      | Pass   |
|            | credit.py::TestCreditSpread               | test_negative_income_fallback                | Pass   |
|            | credit.py::TestCreditSpread               | test_spread_bounded_by_schedule              | Pass   |
|            | credit.py::TestCreditSpread               | test_zero_income_fallback                    | Pass   |
|            | credit.py::TestFloodRiskFactor            | test_batch_passes_flood_risk                 | Pass   |
|            | credit.py::TestFloodRiskFactor            | test_case_insensitive                        | Pass   |
|            | credit.py::TestFloodRiskFactor            | test_flood_factor_in_result                  | Pass   |
|            | credit.py::TestFloodRiskFactor            | test_flood_risk_increases_spread             | Pass   |
|            | credit.py::TestFloodRiskFactor            | test_flood_risk_multipliers [High-1.4]       | Pass   |
|            | credit.py::TestFloodRiskFactor            | test_flood_risk_multipliers [Low-1.05]       | Pass   |
|            | credit.py::TestFloodRiskFactor            | test_flood_risk_multipliers [Medium-1.2]     | Pass   |
|            | credit.py::TestFloodRiskFactor            | test_flood_risk_multipliers [Very High-1.75] | Pass   |
|            | credit.py::TestFloodRiskFactor            | test_flood_risk_multipliers [Very Low-1.0]   | Pass   |
|            | credit.py::TestFloodRiskFactor            | test_no_flood_risk_defaults_to_unity         | Pass   |
|            | credit.py::TestFloodRiskFactor            | test_none_returns_unity                      | Pass   |
|            | credit.py::TestFloodRiskFactor            | test_unknown_category_returns_unity          | Pass   |
|            | credit.py::TestLTVRiskFactor              | test_ltv_thresholds [0.5-1.0]                | Pass   |
|            | credit.py::TestLTVRiskFactor              | test_ltv_thresholds [0.8-1.0]                | Pass   |
|            | credit.py::TestLTVRiskFactor              | test_ltv_thresholds [0.85-1.1]               | Pass   |
|            | credit.py::TestLTVRiskFactor              | test_ltv_thresholds [0.9-1.1]                | Pass   |
|            | credit.py::TestLTVRiskFactor              | test_ltv_thresholds [0.91-1.3]               | Pass   |
|            | credit.py::TestLTVRiskFactor              | test_ltv_thresholds [0.95-1.3]               | Pass   |
|            | credit.py::TestLTVRiskFactor              | test_ltv_thresholds [0.96-1.5]               | Pass   |
|            | credit.py::TestLTVRiskFactor              | test_ltv_thresholds [1.0-1.5]                | Pass   |
|            | credit.py::TestLTVRiskFactor              | test_zero_property_value                     | Pass   |
|            | payment.py::TestMonthlyPayment            | test_higher_rate_higher_payment              | Pass   |
|            | payment.py::TestMonthlyPayment            | test_shorter_term_higher_payment             | Pass   |
|            | payment.py::TestMonthlyPayment            | test_standard_mortgage                       | Pass   |
|            | payment.py::TestMonthlyPayment            | test_zero_rate                               | Pass   |
|            | payment.py::TestTotalCost                 | test_positive_rate_exceeds_principal         | Pass   |
|            | payment.py::TestTotalCost                 | test_zero_rate_equals_principal              | Pass   |

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| Model      | Module                                                              | Test Name                                         | Status |
|------------|---------------------------------------------------------------------|---------------------------------------------------|--------|
|            | pricing.py::TestBatchPricing                                        | test.default_ids_assigned                         | Pass   |
|            | pricing.py::TestBatchPricing                                        | test.error_isolation                              | Pass   |
|            | pricing.py::TestBatchPricing                                        | test.prices_multiple_mortgages                    | Pass   |
|            | pricing.py::TestConstructor                                         | test.custom_tax_rate                              | Pass   |
|            | pricing.py::TestConstructor                                         | test.default_tax_rate                             | Pass   |
|            | pricing.py::TestDebugAndEdgeCases                                   | test.debug_also_in_credit_spread                  | Pass   |
|            | pricing.py::TestDebugAndEdgeCases                                   | test.debug_true_does_not_crash                    | Pass   |
|            | pricing.py::TestDebugAndEdgeCases                                   | test.debug_with_zero_income                       | Pass   |
|            | pricing.py::TestDebugAndEdgeCases                                   | test.flood_risk_none_with_debug                   | Pass   |
|            | pricing.py::TestDebugAndEdgeCases                                   | test.zero_interest_rate                           | Pass   |
|            | pricing.py::TestPortfolioMetrics                                    | test.all_errors_returns_error                     | Pass   |
|            | pricing.py::TestPortfolioMetrics                                    | test.high_risk_count                              | Pass   |
|            | pricing.py::TestPortfolioMetrics                                    | test.mixed_results_exclude_errors                 | Pass   |
|            | pricing.py::TestPortfolioMetrics                                    | test.valid_results                                | Pass   |
|            | pricing.py::TestPriceMortgage                                       | test.balance_converges_to_zero                    | Pass   |
|            | pricing.py::TestPriceMortgage                                       | test.expected_losses_non_negative                 | Pass   |
|            | pricing.py::TestPriceMortgage                                       | test.fair_value_less_than_par                     | Pass   |
|            | pricing.py::TestPriceMortgage                                       | test.hazard_rates_positive                        | Pass   |
|            | pricing.py::TestPriceMortgage                                       | test.high_ltv_increases_spread                    | Pass   |
|            | pricing.py::TestPriceMortgage                                       | test.higher_haircut_more_loss                     | Pass   |
|            | pricing.py::TestPriceMortgage                                       | test.input_validation_clamps                      | Pass   |
|            | pricing.py::TestPriceMortgage                                       | test.ltv_factor_in_result                         | Pass   |
|            | pricing.py::TestPriceMortgage                                       | test.raises_on_invalid_types                      | Pass   |
|            | pricing.py::TestPriceMortgage                                       | test.recovery_haircut_zero_low_ltv                | Pass   |
|            | pricing.py::TestPriceMortgage                                       | test.survival_first_period_less_than_one          | Pass   |
|            | pricing.py::TestPriceMortgage                                       | test.survival_monotonically_decreasing            | Pass   |
|            | mortgage_generator.py::TestMortgageGeneratorHasIdProperty           | test.mortgage_generator_has_id_property           | Pass   |
|            | mortgage_generator.py::TestMortgageGeneratorHasQuality              | test.mortgage_generator_has_quality               | Pass   |
|            | mortgage_generator.py::TestMortgageGeneratorHasOutputFile           | test.mortgage_generator_has_output_file           | Pass   |
|            | mortgage_generator.py::TestMortgageGeneratorOutputFileExists        | test.mortgage_generator_output_file_exists        | Pass   |
|            | mortgage_generator.py::TestMortgageGeneratorOutputFileIsValidJson   | test.mortgage_generator_output_file_is_valid_json | Pass   |
|            | mortgage_generator.py::TestMortgageGeneratorReturnsExpectedKeys     | test.mortgage_generator_returns_expected_keys     | Pass   |
|            | mortgage_generator.py::TestMortgageGeneratorOutputMatchesProperties | test.mortgage_generator_output_matches_properties | Pass   |
|            | mortgage_generator.py::TestMortgageGeneratorOutputFormat            | test.mortgage_generator_output_format             | Pass   |
|            | mortgage_generator.py::TestMortgageGeneratorOutputIsUnique          | test.mortgage_generator_output_is_unique          | Pass   |
| MKM-PR-001 | maturity.py::TestComputeMaturityDateFromFeb2026                     | test.compute_maturity_date_from_feb_2026          | Pass   |
|            | maturity.py::TestComputeMaturityDateFromFeb2026                     | test.compute_maturity_date_from_feb_2026          | Pass   |
|            | maturity.py::TestComputeMaturityDateFromJan2027                     | test.compute_maturity_date_from_jan_2027          | Pass   |
|            | maturity.py::TestComputeMaturityDateFromSep2026                     | test.compute_maturity_date_from_sep_2026          | Pass   |
|            | maturity.py::TestComputeMaturityDateFromFeb2026                     | test.compute_maturity_date_from_feb_2026          | Pass   |
|            | maturity.py::TestComputeMaturityDateFromSep2026                     | test.compute_maturity_date_from_sep_2026          | Pass   |
|            | maturity.py::TestComputeMaturityDateMaturityIsFriday                | test.compute_maturity_date_maturity_is_friday     | Pass   |
|            | maturity.py::TestComputeMaturityDateRollJumpAtAugust                | test.compute_maturity_date_roll_jump_at_august    | Pass   |
|            | maturity.py::TestCurrentRollDateAugPeriod                           | test.current_roll_date_aug_period                 | Pass   |
|            | maturity.py::TestCurrentRollDateDecUsesAug                          | test.current_roll_date_dec_uses_aug               | Pass   |

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| Model | Module                                            | Test Name                                 | Status |
|-------|---------------------------------------------------|-------------------------------------------|--------|
|       | maturity.py::TestCurrentRollDate                  | test_feb_period                           | Pass   |
|       | maturity.py::TestCurrentRollDate                  | test_jan_uses_prev_aug                    | Pass   |
|       | maturity.py::TestCurrentRollDate                  | test_jul_still_feb_roll                   | Pass   |
|       | maturity.py::TestCurrentRollDate                  | test_mar_still_feb_roll                   | Pass   |
|       | maturity.py::TestDefaultDates                     | test_compute_maturity_default             | Pass   |
|       | maturity.py::TestDefaultDates                     | test_current_roll_date_default            | Pass   |
|       | maturity.py::TestDefaultDates                     | test_maturity_table_default               | Pass   |
|       | maturity.py::TestLastFridayOfMonth                | test_aug_2026                             | Pass   |
|       | maturity.py::TestLastFridayOfMonth                | test_feb_2026                             | Pass   |
|       | maturity.py::TestLastFridayOfMonth                | test_may_2029                             | Pass   |
|       | maturity.py::TestLastFridayOfMonth                | test_nov_2029                             | Pass   |
|       | maturity.py::TestLastFridayOfMonth                | test_result_is_friday                     | Pass   |
|       | maturity.py::TestMaturityTable                    | test_actual_years_increasing              | Pass   |
|       | maturity.py::TestMaturityTable                    | test_all_may_for_feb_roll                 | Pass   |
|       | maturity.py::TestMaturityTable                    | test_all_nov_for_aug_roll                 | Pass   |
|       | maturity.py::TestMaturityTable                    | test_maturity_str_format                  | Pass   |
|       | maturity.py::TestMaturityTable                    | test_table_has_5_entries                  | Pass   |
|       | maturity.py::TestMaturityTable                    | test_tenors_sequential                    | Pass   |
|       | maturity.py::TestTenorFromMaturity                | test_all_tenors_round_trip                | Pass   |
|       | maturity.py::TestTenorFromMaturity                | test_closest_match_fallback               | Pass   |
|       | maturity.py::TestTenorFromMaturity                | test_exact_match                          | Pass   |
|       | basis.py::TestBasisWaterfall                      | test_basis_waterfall_in_generated_output  | Pass   |
|       | basis.py::TestBasisWaterfall                      | test_composition_flat_penalty             | Pass   |
|       | basis.py::TestBasisWaterfall                      | test_composition_pre2000_penalty          | Pass   |
|       | basis.py::TestBasisWaterfall                      | test_distance_basis_capped                | Pass   |
|       | basis.py::TestBasisWaterfall                      | test_distance_basis_scales_linearly       | Pass   |
|       | basis.py::TestBasisWaterfall                      | test_elevation_above_gauge_reduces_spread | Pass   |
|       | basis.py::TestBasisWaterfall                      | test_elevation_below_gauge_no_benefit     | Pass   |
|       | basis.py::TestBasisWaterfall                      | test_elevation_capped_at_3bp              | Pass   |
|       | basis.py::TestBasisWaterfall                      | test_model_uncertainty_is_fixed           | Pass   |
|       | basis.py::TestBasisWaterfall                      | test_terrain_zone1_low_risk               | Pass   |
|       | basis.py::TestBasisWaterfall                      | test_terrain_zone3_high_risk              | Pass   |
|       | basis.py::TestBasisWaterfall                      | test_total_basis_sums_components          | Pass   |
|       | basis.py::TestEdgeCases                           | test_empty_propertyts_dir                 | Pass   |
|       | basis.py::TestEdgeCases                           | test_no_gauge_hazard_curves               | Pass   |
|       | basis.py::TestEdgeCases                           | test_no_propertyts_dir                    | Pass   |
|       | generator.py::TestPropertyHazardCurvesCalculation | test_basis_calculation                    | Pass   |
|       | generator.py::TestPropertyHazardCurvesCalculation | test_depth_thresholds                     | Pass   |
|       | generator.py::TestPropertyHazardCurvesCalculation | test_generator_produces_output            | Pass   |
|       | generator.py::TestPropertyHazardCurvesCalculation | test_generator_structure                  | Pass   |
|       | generator.py::TestPropertyHazardCurvesCalculation | test_generator_mission_rate               | Pass   |
|       | generator.py::TestPropertyHazardCurvesCalculation | test_generator_structure                  | Pass   |
|       | generator.py::TestPropertyHazardCurvesCalculation | test_properties_with_gev                  | Pass   |
|       | generator.py::TestPropertyHazardCurvesCalculation | test_survival_decreases_with_tenor        | Pass   |

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| Model      | Module                                  | Test Name                                      | Status |
|------------|-----------------------------------------|------------------------------------------------|--------|
|            | generator.py::TestPropertyHazard        | test_curve_structure                           | Pass   |
|            | pricing.py::TestPRSSpreadComp           | test_positive_spread                           | Pass   |
|            | pricing.py::TestPRSSpreadComp           | test_spread_increases_with_hazard              | Pass   |
|            | pricing.py::TestPRSSpreadComp           | test_spread_values_at_different_tenors         | Pass   |
|            | pricing.py::TestPRSSpreadComp           | test_zero_hazard_rate                          | Pass   |
|            | pricing.py::TestRecoveryRates           | test_recovery_rates_by_trigger                 | Pass   |
|            | pricing.py::TestRecoveryRates           | test_recovery_reduces_spread                   | Pass   |
| MKM-PV-001 | property_generator.py::TestBuild        | test_non_dict_schema_returns_empty             | Pass   |
|            | property_generator.py::TestBuild        | test_skips_type_field                          | Pass   |
|            | property_generator.py::TestDate         | test_date_to_isoformat                         | Pass   |
|            | property_generator.py::TestDate         | test_float_to_isoformat                        | Pass   |
|            | property_generator.py::TestDate         | test_float_to_integer                          | Pass   |
|            | property_generator.py::TestDate         | test_float_to_array                            | Pass   |
|            | property_generator.py::TestDate         | test_unknown_type_raises                       | Pass   |
|            | property_generator.py::TestGenerate     | test_generate_has_required_keys                | Pass   |
|            | property_generator.py::TestGenerate     | test_generate_returns_correct_count            | Pass   |
|            | property_generator.py::TestGenerate     | test_generate_target_elevation                 | Pass   |
|            | property_generator.py::TestGenerate     | test_generate_properties_basic                 | Pass   |
|            | property_generator.py::TestGenerate     | test_generate_properties_output_file_created   | Pass   |
|            | property_generator.py::TestProperties   | test_generate_properties_range                 | Pass   |
|            | property_generator.py::TestProperties   | test_generate_properties_quality               | Pass   |
|            | property_generator.py::TestProperties   | test_generate_properties_exists                | Pass   |
|            | property_generator.py::TestProperties   | test_generate_properties_is_valid_json         | Pass   |
|            | property_generator.py::TestProperties   | test_generate_properties_count                 | Pass   |
|            | property_generator.py::TestProperties   | test_generate_properties_returns_expected_keys | Pass   |
|            | property_generator.py::TestProperties   | test_generate_properties_output_format         | Pass   |
|            | property_generator.py::TestProperties   | test_generate_properties_unique                | Pass   |
|            | property_generator.py::TestProperties   | test_generate_grid_reference_format            | Pass   |
|            | property_generator.py::TestProperties   | test_generate_owner_name_two_words             | Pass   |
|            | property_generator.py::TestProperties   | test_generate_past_date_format                 | Pass   |
|            | property_generator.py::TestProperties   | test_generate_past_date_range                  | Pass   |
|            | rand_property_mortgage.py::TestGenerate | test_generate_premium_flood_high               | Pass   |
|            | rand_property_mortgage.py::TestGenerate | test_generate_premium_no_area                  | Pass   |
|            | rand_property_mortgage.py::TestGenerate | test_generate_premium_no_value                 | Pass   |
|            | rand_property_mortgage.py::TestGenerate | test_generate_no_area                          | Pass   |
|            | rand_property_mortgage.py::TestGenerate | test_generate_no_value                         | Pass   |
|            | rand_property_mortgage.py::TestGenerate | test_generate_positive                         | Pass   |
|            | rand_property_mortgage.py::TestGenerate | test_generate_area_all_types                   | Pass   |
|            | rand_property_mortgage.py::TestGenerate | test_generate_invalid_date                     | Pass   |
|            | rand_property_mortgage.py::TestGenerate | test_generate_no_current_value                 | Pass   |
|            | rand_property_mortgage.py::TestGenerate | test_generate_positive                         | Pass   |
|            | rand_property_mortgage.py::TestGenerate | test_generate_with_sale_date_1_to_2_years      | Pass   |
|            | rand_property_mortgage.py::TestGenerate | test_generate_with_sale_date_over_2_years      | Pass   |

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| Model      | Module                                             | Test Name                                  | Status |
|------------|----------------------------------------------------|--------------------------------------------|--------|
|            | rand_property_mortgage.py::TestGeneralCalculations | Test calculate mortgage with sale date     | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test calculate mortgage financials return  | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test generate financial data keys          | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test generate financial range              | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test generate financial less than value    | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test mortgage financial format             | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test general financial consistency check   | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test calculate insurance premium positive  | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test calculate property area positive      | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test calculate property value in range     | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test generate bedrooms detached            | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test generate bedrooms flat                | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test generate bedrooms code format         | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test generate field value decimal          | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test generate field value text             | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test generate construction year in range   | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test generate property id format           | Pass   |
|            | rand_property_mortgage.py::TestGeneralCalculations | Test generate dict with required keys      | Pass   |
| MKM-SG-001 | forward_model.py::TestComputerAHR                  | Test response ids match                    | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test response gauge higher response        | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test response one response per gauge       | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test all kernels decrease with distance    | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test all kernels in 0.1 range              | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test exponential at half                   | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test exponential at one lambda             | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test gaussian at one sigma                 | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test gaussian at two sigma                 | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test linear beyond cutoff is zero          | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test linear midpoint                       | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test zero distance returns one all kernels | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test sift Arc uses last point              | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test sift Arc time is nonzero              | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test sift Arc start uses first point       | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test sift Arc track returns zero           | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test sift Arc track is lower               | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test accumulation nonnegative              | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test duration nonnegative                  | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test file coded flag correct               | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test high intensity storm floods           | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test peak level at least base              | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test peak level is max of series           | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test response ids match                    | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test response series length                | Pass   |
|            | forward_model.py::TestComputerAHR                  | Test roundtrip                             | Pass   |

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| Model | Module                                  | Test Name                                          | Status |
|-------|-----------------------------------------|----------------------------------------------------|--------|
|       | forward_model.py::TestComputeResponse   | test_predict_without_timeseries                    | Pass   |
|       | forward_model.py::TestFindNearbyTracks  | test_track_point_is_positive                       | Pass   |
|       | forward_model.py::TestFindNearbyTracks  | test_track_point_nearest_of_three                  | Pass   |
|       | forward_model.py::TestFindNearbyTracks  | test_track_point_outtrack_point                    | Pass   |
|       | forward_model.py::TestHaversine         | test_equatorial_degree_approx_111km                | Pass   |
|       | forward_model.py::TestHaversine         | test_london_to_paris_approx_340km                  | Pass   |
|       | forward_model.py::TestHaversine         | test_same_point_is_zero                            | Pass   |
|       | forward_model.py::TestHaversine         | test_symmetric                                     | Pass   |
|       | forward_model.py::TestIntensityLevel    | test_level_notone_increasing                       | Pass   |
|       | forward_model.py::TestIntensityLevel    | test_level_negative_intensity_returns_zero         | Pass   |
|       | forward_model.py::TestIntensityLevel    | test_level_positive_intensity_positive_level       | Pass   |
|       | forward_model.py::TestIntensityLevel    | test_level_sensitivity_scales_output               | Pass   |
|       | forward_model.py::TestIntensityLevel    | test_level_historical_high_when_set                | Pass   |
|       | forward_model.py::TestIntensityLevel    | test_level_zero_intensity_returns_zero             | Pass   |
|       | gauge_generator.py::TestDateTimeEncoder | test_encoder_datetime                              | Pass   |
|       | gauge_generator.py::TestDateTimeEncoder | test_encoder_numpy_float32                         | Pass   |
|       | gauge_generator.py::TestDateTimeEncoder | test_encoder_numpy_floating                        | Pass   |
|       | gauge_generator.py::TestDateTimeEncoder | test_encoder_numpy_integer                         | Pass   |
|       | gauge_generator.py::TestDateTimeEncoder | test_encoder_numpy_ndarray                         | Pass   |
|       | gauge_generator.py::TestDateTimeEncoder | test_encoder_fallback_raises_for_unknown           | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_add_data_ordered                              | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_add_data_unordered                            | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_save_data_caption_propagates                  | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_save_data_generate_hits_end_time_branch       | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_build_section_non_dict_schema_returns_empty   | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_build_section_skips_metadata_keys             | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_configure_section_available_points_capped     | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_configure_section_raises_value_error          | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_generate_gauges_convenience_function          | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_generate_gauges_single_gauge_failure_count    | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_log_method_works                              | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_set_edge_case_values_creates_missing_sections | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_set_edge_case_sets_warning_level              | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_output_file_exists                            | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_output_file_is_valid_json                     | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_output_file_json_has_correct_count            | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_main_block                                    | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_gauge_output_format                           | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_gauge_ids_unique                              | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_generate_output_correct_count                 | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_generate_output_returns_expected_keys         | Pass   |
|       | gauge_generator.py::TestGaugeGenerator  | test_locations_have_coordinates                    | Pass   |
|       | rand_gauge.py::TestFloodRiskAssessment  | test_choose_to_thames_high_risk                    | Pass   |
|       | rand_gauge.py::TestFloodRiskAssessment  | test_choose_from_thames_lower_risk                 | Pass   |

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| Model      | Module                                    | Test Name                                   | Status |
|------------|-------------------------------------------|---------------------------------------------|--------|
|            | rand_gauge.py::TestGenerateField          | test_date_field_returns_date_string         | Pass   |
|            | rand_gauge.py::TestGenerateField          | test_decimal_field_returns_float            | Pass   |
|            | rand_gauge.py::TestGenerateField          | test_gauge_name_includes_area               | Pass   |
|            | rand_gauge.py::TestGenerateField          | test_menu_field_returns_valid_option        | Pass   |
|            | rand_gauge.py::TestGenerateField          | test_text_field_returns_string              | Pass   |
|            | rand_gauge.py::TestGenerateGauge          | test_attachment_name_stored                 | Pass   |
|            | rand_gauge.py::TestGenerateGauge          | test_flow_data_stage_ordering               | Pass   |
|            | rand_gauge.py::TestGenerateGauge          | test_gaugeaid_format                        | Pass   |
|            | rand_gauge.py::TestGenerateGauge          | test_invalid_historical_high_level_range    | Pass   |
|            | rand_gauge.py::TestGenerateGauge          | test_invalid_date_format                    | Pass   |
|            | rand_gauge.py::TestGenerateGauge          | test_invalid_data_dict_with_required_keys   | Pass   |
| MKM-SI-001 | distribution.py::TestConstruction         | test_custom_parameters                      | Pass   |
|            | distribution.py::TestConstruction         | test_default_parameters                     | Pass   |
|            | distribution.py::TestConstruction         | test_from_params_object                     | Pass   |
|            | distribution.py::TestConstruction         | test_from_scenario_baseline                 | Pass   |
|            | distribution.py::TestConstruction         | test_from_scenario_unknown_raises           | Pass   |
|            | distribution.py::TestConstruction         | test_seed_gives_deterministic_output        | Pass   |
|            | distribution.py::TestConstruction         | test_tail_prob_between_0_and_1              | Pass   |
|            | distribution.py::TestExceedance           | test_above_max_is_zero                      | Pass   |
|            | distribution.py::TestExceedance           | test_above_threshold_uses_pareto            | Pass   |
|            | distribution.py::TestExceedance           | test_at_max_intensity_is_zero               | Pass   |
|            | distribution.py::TestExceedance           | test_at_min_intensity_is_one                | Pass   |
|            | distribution.py::TestExceedance           | test_at_threshold_continuous                | Pass   |
|            | distribution.py::TestExceedance           | test_below_min_is_one                       | Pass   |
|            | distribution.py::TestExceedance           | test_below_threshold_uses_normal            | Pass   |
|            | distribution.py::TestExceedance           | test_monotonically_decreasing               | Pass   |
|            | distribution.py::TestInverseExceedance    | test_cone_prob_returns_min                  | Pass   |
|            | distribution.py::TestInverseExceedance    | test_result_in_bounds                       | Pass   |
|            | distribution.py::TestInverseExceedance    | test_roundtrip_above_threshold              | Pass   |
|            | distribution.py::TestInverseExceedance    | test_roundtrip_below_threshold              | Pass   |
|            | distribution.py::TestInverseExceedance    | test_zero_prob_returns_max                  | Pass   |
|            | distribution.py::TestParetoInterpretation | test_fat_tail_higher_exceedance             | Pass   |
|            | distribution.py::TestParetoInterpretation | test_pareto_above_threshold_between_0_and_1 | Pass   |
|            | distribution.py::TestParetoInterpretation | test_pareto_at_threshold_is_one             | Pass   |
|            | distribution.py::TestParetoInterpretation | test_pareto_below_threshold_is_one          | Pass   |
|            | distribution.py::TestReturnPeriod         | test_100yr_greater_than_10yr                | Pass   |
|            | distribution.py::TestReturnPeriod         | test_ordering_10_50_100                     | Pass   |
|            | distribution.py::TestReturnPeriod         | test_result_in_bounds                       | Pass   |
|            | distribution.py::TestReturnPeriod         | test_storms_per_year_affects_result         | Pass   |
|            | distribution.py::TestSampling             | test_different_seeds_different_output       | Pass   |
|            | distribution.py::TestSampling             | test_sample_n_returns_correct_length        | Pass   |
|            | distribution.py::TestSampling             | test_sample_single                          | Pass   |
|            | distribution.py::TestSampling             | test_samples_within_bounds                  | Pass   |
|            | distribution.py::TestSampling             | test_seeded_reproducibility                 | Pass   |

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| Model | Module                                   | Test Name                                | Status |
|-------|------------------------------------------|------------------------------------------|--------|
|       | distribution.py::TestSerialisation       | test.describe_returns_string             | Pass   |
|       | distribution.py::TestSerialisation       | test.params_roundtrip                    | Pass   |
|       | distribution.py::TestSerialisation       | test.tail_probability_matches_internal   | Pass   |
|       | distribution.py::TestSerialisation       | test.to_dict_keys                        | Pass   |
|       | distribution.py::TestSummaryStatistics   | test.exceedance_probs_decreasing         | Pass   |
|       | distribution.py::TestSummaryStatistics   | test.fat_tail_higher_p99_than_thin       | Pass   |
|       | distribution.py::TestSummaryStatistics   | test.mean_in_reasonable_range            | Pass   |
|       | distribution.py::TestSummaryStatistics   | test.percentiles_ordered                 | Pass   |
|       | distribution.py::TestSummaryStatistics   | test.required_keys_present               | Pass   |
|       | storm_duration.py::TestDurationEstimator | test.all_categories_defined              | Pass   |
|       | storm_duration.py::TestDurationEstimator | test.catastrophic_extended               | Pass   |
|       | storm_duration.py::TestDurationEstimator | test.each_tier_doubles_approx            | Pass   |
|       | storm_duration.py::TestDurationEstimator | test.extreme_extended                    | Pass   |
|       | storm_duration.py::TestDurationEstimator | test.moderate_extended                   | Pass   |
|       | storm_duration.py::TestDurationEstimator | test.severe_extended                     | Pass   |
|       | storm_duration.py::TestHelpers           | test.get_duration_range                  | Pass   |
|       | storm_duration.py::TestHelpers           | test.get_max_duration                    | Pass   |
|       | storm_duration.py::TestSampleDuration    | test.invalid_category                    | Pass   |
|       | storm_duration.py::TestSampleDuration    | test.mode_is_most_common                 | Pass   |
|       | storm_duration.py::TestSampleDuration    | test.no_rng_uses_default                 | Pass   |
|       | storm_duration.py::TestSampleDuration    | test.reproducibility                     | Pass   |
|       | storm_duration.py::TestSampleDuration    | test.within_bounds[baseline]             | Pass   |
|       | storm_duration.py::TestSampleDuration    | test.within_bounds[catastrophic]         | Pass   |
|       | storm_duration.py::TestSampleDuration    | test.within_bounds[extreme]              | Pass   |
|       | storm_duration.py::TestSampleDuration    | test.within_bounds[minimal]              | Pass   |
|       | storm_duration.py::TestSampleDuration    | test.within_bounds[moderate]             | Pass   |
|       | storm_duration.py::TestSampleDuration    | test.within_bounds[severe]               | Pass   |
|       | storm_gap.py::TestFitsInWindow           | test.constants                           | Pass   |
|       | storm_gap.py::TestFitsInWindow           | test.exact_boundary                      | Pass   |
|       | storm_gap.py::TestFitsInWindow           | test.exceeds_window                      | Pass   |
|       | storm_gap.py::TestFitsInWindow           | test.simple_doublet_fits                 | Pass   |
|       | storm_gap.py::TestFitsInWindow           | test.single_storm                        | Pass   |
|       | storm_gap.py::TestGapParams              | test.long_gaps                           | Pass   |
|       | storm_gap.py::TestGapParams              | test.medium_gaps                         | Pass   |
|       | storm_gap.py::TestGapParams              | test.short_gaps                          | Pass   |
|       | storm_gap.py::TestGetGapRange            | test.doublet_medium                      | Pass   |
|       | storm_gap.py::TestGetGapRange            | test.isolated_zero                       | Pass   |
|       | storm_gap.py::TestSampleGap              | test.doublet_uses_medium                 | Pass   |
|       | storm_gap.py::TestSampleGap              | test.isolated_returns_zero               | Pass   |
|       | storm_gap.py::TestSampleGap              | test.no_rng_uses_default                 | Pass   |
|       | storm_gap.py::TestSampleGap              | test.persistent_uses_short               | Pass   |
|       | storm_gap.py::TestSampleGap              | test.reproducibility                     | Pass   |
|       | storm_gap.py::TestSampleGap              | test.within_bounds[SequenceType.CLUSTER] | Pass   |
|       | storm_gap.py::TestSampleGap              | test.within_bounds[SequenceType.DOUBLET] | Pass   |

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| Model      | Module                                         | Test Name                                                    | Status |
|------------|------------------------------------------------|--------------------------------------------------------------|--------|
|            | storm_gap.py::TestSampleGap                    | test_within_bounds[SequenceType.PERSISTENT]                  | PASS   |
|            | storm_gap.py::TestValidation                   | test_duration_mismatch                                       | Pass   |
|            | storm_gap.py::TestValidation                   | test_exceeds_precip_window                                   | Pass   |
|            | storm_gap.py::TestValidation                   | test_invalid_intensity_category                              | Pass   |
|            | storm_gap.py::TestValidation                   | test_invalid_sequence_type                                   | Pass   |
|            | storm_gap.py::TestValidation                   | test_overlapping_storms                                      | Pass   |
|            | storm_gap.py::TestValidation                   | test_storm_count_mismatch                                    | Pass   |
|            | storm_gap.py::TestValidation                   | test_too_many_storms                                         | Pass   |
|            | storm_gap.py::TestValidation                   | test_valid_doublet                                           | Pass   |
|            | storm_sequence.py::TestEnums                   | test_gap_types                                               | Pass   |
|            | storm_sequence.py::TestEnums                   | test_sequence_gap_mapping                                    | Pass   |
|            | storm_sequence.py::TestEnums                   | test_sequence_types                                          | Pass   |
|            | storm_sequence.py::TestEnums                   | test_storm_count_ranges                                      | Pass   |
|            | storm_sequence.py::TestIDGenerators            | test_ids_unique                                              | Pass   |
|            | storm_sequence.py::TestIDGenerators            | test_id_sequence_id_format                                   | Pass   |
|            | storm_sequence.py::TestIDGenerators            | test_id_sequence_ids_unique                                  | Pass   |
|            | storm_sequence.py::TestIDGenerators            | test_id_storm_id_format                                      | Pass   |
|            | storm_sequence.py::TestSequenceSet             | test_create_basic                                            | Pass   |
|            | storm_sequence.py::TestSequenceSet             | test_from_dict_round_trip                                    | Pass   |
|            | storm_sequence.py::TestSequenceSet             | test_precipitation_rounding                                  | Pass   |
|            | storm_sequence.py::TestSequenceSet             | test_no_dict                                                 | Pass   |
|            | storm_sequence.py::TestStormSequenceSet        | test_create_defaults                                         | Pass   |
|            | storm_sequence.py::TestStormSequenceSet        | test_create_doublet                                          | Pass   |
|            | storm_sequence.py::TestStormSequenceSet        | test_from_dict_round_trip                                    | Pass   |
|            | storm_sequence.py::TestStormSequenceSet        | test_isolated_no_gaps                                        | Pass   |
|            | storm_sequence.py::TestStormSequenceSet        | test_spatial_correlation_defaults                            | Pass   |
|            | storm_sequence.py::TestStormSequenceSet        | test_spatial_correlation_enabled                             | Pass   |
|            | storm_sequence.py::TestStormSequenceSet        | test_to_dict                                                 | Pass   |
| MKM-SS-001 | batch_generator.py::TestGenerateStormSequences | test_generate_storm_sequences                                | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_generate_storm_sequences_propagated                     | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_generate_storm_sequences_basic_count                    | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_generate_storm_sequences_seeds_differ                   | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_generate_storm_sequences_sequence_type_doublet          | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_generate_storm_sequences_sequence_type_isolated         | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_generate_storm_sequences_sensible_with_same_seed        | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_generate_storm_sequences_id_dist                        | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_generate_storm_sequences_highlights_change_distribution | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_generate_storm_sequences_categories_present             | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_generate_storm_sequences_distribution                   | Pass   |
|            | batch_generator.py::TestSaveLoadStormSequences | test_save_load_storm_sequences_parent_dirs                   | Pass   |
|            | batch_generator.py::TestSaveLoadStormSequences | test_save_load_storm_sequences_missing_file_raises           | Pass   |
|            | batch_generator.py::TestSaveLoadStormSequences | test_save_load_storm_sequences_returns_correct_count         | Pass   |
|            | batch_generator.py::TestSaveLoadStormSequences | test_save_load_storm_sequences_wrong_schema_raises           | Pass   |
|            | batch_generator.py::TestSaveLoadStormSequences | test_save_load_storm_sequences_sequences_still_valid         | Pass   |

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| Model                                  | Module                                 | Test Name                                | Status |
|----------------------------------------|----------------------------------------|------------------------------------------|--------|
| MKM-TD-001                             | batch_generator.py::TestSaveLoadH5     | test_round_trip_num_storms               | Pass   |
|                                        | batch_generator.py::TestSaveLoadH5     | test_round_trip_precipitation            | Pass   |
|                                        | batch_generator.py::TestSaveLoadH5     | test_round_trip_sequence_type            | Pass   |
|                                        | batch_generator.py::TestSaveLoadH5     | test_round_trip_storm_fields             | Pass   |
|                                        | batch_generator.py::TestSaveLoadH5     | test_round_trip_total_duration           | Pass   |
|                                        | batch_generator.py::TestSaveLoadH5     | test_save_creates_file                   | Pass   |
|                                        | batch_generator.py::TestSaveLoadH5     | test_schema_version_in_file              | Pass   |
|                                        | batch_generator.py::TestSaveSummary    | test_creates_file                        | Pass   |
|                                        | batch_generator.py::TestSaveSummary    | test_duration_stats_present              | Pass   |
|                                        | batch_generator.py::TestSaveSummary    | test_num_sequences_matches               | Pass   |
|                                        | batch_generator.py::TestSaveSummary    | test_precipitation_stats_present         | Pass   |
|                                        | batch_generator.py::TestSaveSummary    | test_schema_version                      | Pass   |
|                                        | batch_generator.py::TestSaveSummary    | test_type_counts_sum_to_total            | Pass   |
|                                        | batch_generator.py::TestSaveSummary    | test_type_fractions_sum_to_one           | Pass   |
|                                        | batch_generator.py::TestValidation     | test_all_sequences_valid                 | Pass   |
|                                        | batch_generator.py::TestValidation     | test_medium_batch_all_valid              | Pass   |
|                                        | batch_generator.py::TestValidation     | test_no_negative_durations               | Pass   |
|                                        | batch_generator.py::TestValidation     | test_no_negative_precipitation           | Pass   |
|                                        | batch_generator.py::TestValidation     | test_timing_constraint                   | Pass   |
|                                        | sequence_gen.py::TestBuildIsolated     | test_isolated_duration_clamped           | Pass   |
|                                        | sequence_gen.py::TestBuildIsolated     | test_isolated_has_one_storm              | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_agg_generates_stats                 | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_catchment_id_in_storms              | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_build_storm_sequence                | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_dont_buffer_force                   | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_isolated_force                      | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_in_generate_category Often isolated | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_num_generate_calls                  | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_persistent_sequence                 | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_returns_storm_sequence              | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_sequence_fits_in_window             | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_storms_have_correct_fields          | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_custom_catchment                    | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_default_catchment                   | Pass   |
|                                        | sequence_gen.py::TestSequenceGenerator | test_seeded_reproducibility              | Pass   |
| sequence_gen.py::TestSequenceGenerator | test_unknown_catchment_defaults_to_35  | Pass                                     |        |
| MKM-TD-001                             | leg_pvs.py::TestComputeLegPV           | test_higher_spread_means_higher_premium  | Pass   |
|                                        | leg_pvs.py::TestComputeLegPV           | test_positive_hazard_rate                | Pass   |
|                                        | leg_pvs.py::TestComputeLegPV           | test_zero_hazard_rate                    | Pass   |
|                                        | market_making.py::TestGenerateBook     | test_balanced_book                       | Pass   |
|                                        | market_making.py::TestGenerateBook     | test_balanced_notional                   | Pass   |
|                                        | market_making.py::TestGenerateBook     | test_counterparties_distributed          | Pass   |
|                                        | market_making.py::TestGenerateBook     | test_files_saved                         | Pass   |

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| Model | Module                                   | Test Name                                       | Status |
|-------|------------------------------------------|-------------------------------------------------|--------|
|       | market_making.py::TestGenerator          | test_payer_trades_below_fair                    | Pass   |
|       | market_making.py::TestGenerator          | test_receiver_trades_above_fair                 | Pass   |
|       | market_making.py::TestGenerator          | test_reproducible_with_seed                     | Pass   |
|       | market_making.py::TestGenerator          | test_small_mtm                                  | Pass   |
|       | market_making.py::TestGenerator          | test_valid_cdm_structure                        | Pass   |
|       | market_making.py::TestPrintBooks         | test_summary_runs                               | Pass   |
|       | market_making.py::TestTradeSpread        | test_bookmaking_spreads_are_integers            | Pass   |
|       | market_making.py::TestTradeSpread        | test_thames_central_spreads_are_integer         | Pass   |
|       | thames_central.py::TestThamesCentralBook | test_book_saved                                 | Pass   |
|       | thames_central.py::TestThamesCentralBook | test_book_calendar_spreads                      | Pass   |
|       | thames_central.py::TestThamesCentralBook | test_book_reproducible_with_seed                | Pass   |
|       | thames_central.py::TestThamesCentralBook | test_book_count                                 | Pass   |
|       | thames_central.py::TestThamesCentralBook | test_book_central_gauges                        | Pass   |
|       | thames_central.py::TestThamesCentralBook | test_book_valid_cdm_structure                   | Pass   |
|       | thames_central.py::TestThamesCentralBook | test_book_inster_is_net_short                   | Pass   |
|       | engine.py::TestDeltaEngine               | test_build_risk_grid                            | Pass   |
|       | engine.py::TestDeltaEngine               | test_enrich_receiver_trade_negative_dv01        | Pass   |
|       | engine.py::TestDeltaEngine               | test_enrich_trade                               | Pass   |
|       | engine.py::TestDeltaEngine               | test_payer_receiver_dv01_opposite               | Pass   |
|       | engine.py::TestHazardCurveReversion      | test_curve_change_moves_fair_spread             | Pass   |
|       | engine.py::TestHazardCurveReversion      | test_curve_change_moves_mtm                     | Pass   |
|       | engine.py::TestHazardCurveReversion      | test_curve_change_pnl_consistent_with_fair      | Pass   |
|       | engine.py::TestHazardCurveReversion      | test_curve_change_produces_daily_pnl            | Pass   |
|       | formulas.py::TestBasisDelta              | test_basis_delta_positive                       | Pass   |
|       | formulas.py::TestBasisDelta              | test_basis_delta_similar_to_gauge_delta         | Pass   |
|       | formulas.py::TestGaugeDelta              | test_delta_spread_approximately_one             | Pass   |
|       | formulas.py::TestGaugeDelta              | test_higher_notional_means_higher_dv01          | Pass   |
|       | formulas.py::TestGaugeDelta              | test_positive_dv01_for_long_position            | Pass   |
|       | formulas.py::TestGaugeDelta              | test_zero_hazard_rate_gives_zero_delta          | Pass   |
|       | formulas.py::TestMarkToMarket            | test_mtm_calculation_correct                    | Pass   |
|       | formulas.py::TestMarkToMarket            | test_mtm_negative_when_spread_widens_for_seller | Pass   |
|       | formulas.py::TestMarkToMarket            | test_mtm_positive_when_spread_widens_for_buyer  | Pass   |
|       | formulas.py::TestMarkToMarket            | test_mtm_zero_when_spreads_equal                | Pass   |
|       | formulas.py::TestRiskyAnnuity            | test_longer_tenor_means_higher_annuity          | Pass   |
|       | formulas.py::TestRiskyAnnuity            | test_positive_hazard_rate_reduces_annuity       | Pass   |
|       | formulas.py::TestRiskyAnnuity            | test_reasonable_range                           | Pass   |
|       | formulas.py::TestRiskyAnnuity            | test_zero_hazard_rate                           | Pass   |
|       | market_state.py::TestEffectiveRates      | test_effective_rates_include_all_gauges         | Pass   |
|       | market_state.py::TestEffectiveRates      | test_effective_rates_mark_adjusted              | Pass   |
|       | market_state.py::TestEffectiveRates      | test_effective_rates_merge_correctly            | Pass   |
|       | market_state.py::TestMarketState         | test_base_rates_match_gaugehc                   | Pass   |
|       | market_state.py::TestMarketState         | test_initial_load_creates_state                 | Pass   |
|       | market_state.py::TestMarketState         | test_load_persists                              | Pass   |
|       | market_state.py::TestMarketState         | test_reset_all                                  | Pass   |

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| Model | Module                                         | Test Name                                      | Status  |
|-------|------------------------------------------------|------------------------------------------------|---------|
|       | market_state.py::TestMarketStateBase           | test_reset_single_gauge                        | Pass    |
|       | market_state.py::TestMarketStateBase           | test_get_effective_rate_with_adjustment        | Pass    |
|       | market_state.py::TestMarketStateBase           | test_update_adjusted_gauge_returns_base        | Pass    |
|       | market_state.py::TestMarketStateBase           | test_update_gauge_rate                         | Pass    |
|       | market_state.py::TestMissingCoverage           | test_commit_yield_curve                        | Pass    |
|       | market_state.py::TestMissingCoverage           | test_get_yield_rate_above_max_tenor            | Pass    |
|       | market_state.py::TestMissingCoverage           | test_get_yield_rate_below_min_tenor            | Pass    |
|       | market_state.py::TestMissingCoverage           | test_get_yield_rate_exact_tenor                | Pass    |
|       | market_state.py::TestMissingCoverage           | test_get_yield_rate_interpolated               | Pass    |
|       | market_state.py::TestMissingCoverage           | test_load_base_curves_gauge_with_no_id skipped | Skipped |
|       | market_state.py::TestMissingCoverage           | test_load_base_curves_gauges_key               | Pass    |
|       | market_state.py::TestMissingCoverage           | test_load_base_curves_no_file                  | Pass    |
|       | market_state.py::TestMissingCoverage           | test_load_base_curves_unknown_format           | Pass    |
|       | market_state.py::TestMissingCoverage           | test_reset_yield_curve                         | Pass    |
|       | market_state.py::TestMissingCoverage           | test_update_gauge_rate_with_notes              | Pass    |
|       | daily_pnl.py::TestDailyPnL                     | test_closed_trades_excluded                    | Pass    |
|       | daily_pnl.py::TestDailyPnL                     | test_daily_pnl_without_previous_eod            | Pass    |
|       | daily_pnl.py::TestDailyPnL                     | test_pnl_decomposition_sums                    | Pass    |
|       | daily_pnl.py::TestDailyPnL                     | test_running_pnl_calculation                   | Pass    |
|       | daily_pnl.py::TestPhantomPnLFt                 | test_market_pnl_nonzero_with_eod_mark          | Pass    |
|       | daily_pnl.py::TestPhantomPnLFt                 | test_market_pnl_zero_for_missing_eod_trades    | Pass    |
|       | daily_pnl.py::TestPhantomPnLFt                 | test_new_trade_has_inception_pnl               | Pass    |
|       | daily_pnl.py::TestPhantomPnLFt                 | test_pnl_decomposition_with_prev_eod           | Pass    |
|       | eod.py::TestEodSnapshot                        | test_eod_history                               | Pass    |
|       | eod.py::TestEodSnapshot                        | test_generate_snapshot                         | Pass    |
|       | eod.py::TestEodSnapshot                        | test_idempotent_snapshot                       | Pass    |
|       | eod.py::TestEodSnapshot                        | test_pnl_series                                | Pass    |
|       | eod.py::TestEodSnapshot                        | test_snapshot_persists                         | Pass    |
|       | marks.py::TestTradeMarks                       | test_close_trade                               | Pass    |
|       | marks.py::TestTradeMarks                       | test_load_empty_marks                          | Pass    |
|       | marks.py::TestTradeMarks                       | test_update_and_load_mark                      | Pass    |
|       | stress_hydrograph.py::TestStressKestckOnLogic  | test_cash_stays_at_notional_after_roll         | Pass    |
|       | stress_hydrograph.py::TestStressKestckOffLogic | test_ko_event_triggers_ko_independently        | Pass    |
|       | stress_hydrograph.py::TestStressKestckOnLogic  | test_ko_level_below_trigger                    | Pass    |
|       | stress_hydrograph.py::TestStressKestckOnLogic  | test_ko_cash_is_negative                       | Pass    |
|       | stress_hydrograph.py::TestStressKestckOnLogic  | test_ko_clocks_out_at_trigger                  | Pass    |
|       | stress_hydrograph.py::TestSynthesisHydrograph  | test_base                                      | Pass    |
|       | stress_hydrograph.py::TestSynthesisHydrograph  | test_histogram_hours                           | Pass    |
|       | stress_hydrograph.py::TestSynthesisHydrograph  | test_histogram_rise                            | Pass    |
|       | stress_hydrograph.py::TestSynthesisHydrograph  | test_peak_level                                | Pass    |
|       | stress_hydrograph.py::TestSynthesisHydrograph  | test_histogram_after_peak                      | Pass    |
|       | stress_hydrograph.py::TestSynthesisHydrograph  | test_histogram_168_hours                       | Pass    |
|       | stress_hydrograph.py::TestSynthesisHydrograph  | test_histogram                                 | Pass    |
|       | stress_hydrograph.py::TestSynthesisHydrograph  | test_histogram_base_level                      | Pass    |

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| Model | Module                                            | Test Name                                        | Status |
|-------|---------------------------------------------------|--------------------------------------------------|--------|
|       | stress.hydrograph.py::TestSyntheticHydrograph     | test_hydrograph_creation                         | Pass   |
|       | blotter_routes.py::TestActiveGauges               | test_all_gauges_with_trades_are_returned         | Pass   |
|       | blotter_routes.py::TestActiveGauges               | test_empty_blotter_returns_empty_list            | Pass   |
|       | blotter_routes.py::TestActiveGauges               | test_gauge_without_trade_is_not_returned         | Pass   |
|       | blotter_routes.py::TestActiveGauges               | test_lambeth_appears_when_it_has_an_open_trade   | Pass   |
|       | blotter_routes.py::TestActiveGauges               | test_lambeth_excluded_after_its_trade_is_closed  | Pass   |
|       | blotter_routes.py::TestActiveGauges               | test_other_gauges_unaffected_by_lambeth_closeout | Pass   |
|       | blotter_routes.py::TestActiveGauges               | test_response_is_sorted                          | Pass   |
|       | blotter_routes.py::TestBlotterCloseout            | test_filters_closed_trades_by_default            | Pass   |
|       | blotter_routes.py::TestBlotterCloseout            | test_include_closed                              | Pass   |
|       | blotter_routes.py::TestBlotterErrorHandling       | test_active_gauges_corrupt_file_skipped          | Pass   |
|       | blotter_routes.py::TestBlotterErrorHandling       | test_blotter_engine_error_returns_500            | Pass   |
|       | blotter_routes.py::TestBlotterErrorHandling       | test_trade_engine_error_returns_500              | Pass   |
|       | blotter_routes.py::TestBlotterErrorHandling       | test_trade_pdf_regen_exception_caught            | Pass   |
|       | blotter_routes.py::TestBlotterGaugesNameInjection | test_gauge_name_injected_when_missing            | Pass   |
|       | blotter_routes.py::TestCloseOut                   | test_close_trade_has_final_pnl                   | Pass   |
|       | blotter_routes.py::TestCloseOut                   | test_close_trade_marks_closed                    | Pass   |
|       | blotter_routes.py::TestCloseOut                   | test_close_trade_not_found                       | Pass   |
|       | blotter_routes.py::TestCloseOut                   | test_close_trade_requires_closeout_spread        | Pass   |
|       | blotter_routes.py::TestCloseOut                   | test_close_trade_settlement_uses_closeout_spread | Pass   |
|       | blotter_routes.py::TestCloseOut                   | test_close_trade_success                         | Pass   |
|       | blotter_routes.py::TestGetBlotter                 | test_blotter_empty_when_no_trades                | Pass   |
|       | blotter_routes.py::TestGetBlotter                 | test_blotter_fs01_sign_payer_positive            | Pass   |
|       | blotter_routes.py::TestGetBlotter                 | test_blotter_fs01_sign_receiver_negative         | Pass   |
|       | blotter_routes.py::TestGetBlotter                 | test_blotter_gauge_name_injected                 | Pass   |
|       | blotter_routes.py::TestGetBlotter                 | test_blotter_has_pnl_fields                      | Pass   |
|       | blotter_routes.py::TestGetBlotter                 | test_blotter_has_prev_fair_spread                | Pass   |
|       | blotter_routes.py::TestGetBlotter                 | test_blotter_has_required_fields                 | Pass   |
|       | blotter_routes.py::TestGetBlotter                 | test_blotter_summary_fields                      | Pass   |
|       | blotter_routes.py::TestGetBlotter                 | test_get_blotter_returns_trades                  | Pass   |
|       | curves_routes.py::TestCurveHistory                | test_curve_history_empty_when_no_eods            | Pass   |
|       | curves_routes.py::TestCurveHistory                | test_curve_history_requires_gauge_id             | Pass   |
|       | curves_routes.py::TestCurveHistory                | test_curve_history_with_eod                      | Pass   |
|       | curves_routes.py::TestCurvesErrorHandling         | test_commit_hazard_ts_error_returns_500          | Pass   |
|       | curves_routes.py::TestCurvesErrorHandling         | test_curve_history_error_returns_500             | Pass   |
|       | curves_routes.py::TestCurvesErrorHandling         | test_hazard_ts_error_returns_500                 | Pass   |
|       | curves_routes.py::TestCurvesErrorHandling         | test_yield_curve_error_returns_500               | Pass   |
|       | curves_routes.py::TestCurvesErrorHandling         | test_physics_series_error_returns_500            | Pass   |
|       | curves_routes.py::TestCurvesErrorHandling         | test_commit_hazard_ts_error_returns_500          | Pass   |
|       | curves_routes.py::TestCurvesErrorHandling         | test_yield_curve_error_returns_500               | Pass   |
|       | curves_routes.py::TestCurvesErrorHandling         | test_update_yield_curve_error_returns_500        | Pass   |
|       | curves_routes.py::TestCurvesNullBody              | test_commit_hazard_ts_null_body                  | Pass   |
|       | curves_routes.py::TestCurvesNullBody              | test_commit_yield_curve_null_body                | Pass   |
|       | curves_routes.py::TestCurvesNullBody              | test_update_hazard_ts_null_body                  | Pass   |

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| Model | Module                                       | Test Name                                           | Status |
|-------|----------------------------------------------|-----------------------------------------------------|--------|
|       | curves_routes.py::TestCurvesNullBody         | test_update_yield_curve_null_body                   | Pass   |
|       | curves_routes.py::TestHazardTSConstruction   | test_commit_hazard_ts                               | Pass   |
|       | curves_routes.py::TestHazardTSConstruction   | test_commit_hazard_ts_missing_gauge                 | Pass   |
|       | curves_routes.py::TestHazardTSConstruction   | test_commit_hazard_ts_missing_rates                 | Pass   |
|       | curves_routes.py::TestHazardTSConstruction   | test_commit_hazard_ts_only_affects_target_gauge     | Pass   |
|       | curves_routes.py::TestHazardTSReset          | test_reset_hazard_ts_all                            | Pass   |
|       | curves_routes.py::TestHazardTSReset          | test_reset_hazard_ts_single_gauge                   | Pass   |
|       | curves_routes.py::TestHazardTermStructure    | test_get_hazard_term_structure                      | Pass   |
|       | curves_routes.py::TestHazardTermStructure    | test_update_hazard_ts_invalid_tenor                 | Pass   |
|       | curves_routes.py::TestHazardTermStructure    | test_update_hazard_ts_missing_gauge                 | Pass   |
|       | curves_routes.py::TestHazardTermStructure    | test_update_hazard_ts_single_point                  | Pass   |
|       | curves_routes.py::TestPnLSeries              | test_pnl_series_after_eod                           | Pass   |
|       | curves_routes.py::TestPnLSeries              | test_pnl_series_empty                               | Pass   |
|       | curves_routes.py::TestYieldCurve             | test_get_yield_curve                                | Pass   |
|       | curves_routes.py::TestYieldCurve             | test_get_yield_curve_default_shape                  | Pass   |
|       | curves_routes.py::TestYieldCurve             | test_update_yield_curve_invalid_tenor               | Pass   |
|       | curves_routes.py::TestYieldCurve             | test_update_yield_curve_no_body                     | Pass   |
|       | curves_routes.py::TestYieldCurve             | test_update_yield_curve_single_tenor                | Pass   |
|       | curves_routes.py::TestYieldCurveConstruction | test_commit_yield_curve                             | Pass   |
|       | curves_routes.py::TestYieldCurveConstruction | test_commit_yield_curve_affected_count              | Pass   |
|       | curves_routes.py::TestYieldCurveConstruction | test_commit_yield_curve_empty_rates                 | Pass   |
|       | curves_routes.py::TestYieldCurveConstruction | test_commit_yield_curve_no_body                     | Pass   |
|       | curves_routes.py::TestYieldCurveReset        | test_reset_yield_curve                              | Pass   |
|       | eod_routes.py::TestEODErrorHandling          | test_eod_history_engine_error_returns_500           | Pass   |
|       | eod_routes.py::TestEODErrorHandling          | test_get_eod_pdf_error_returns_500                  | Pass   |
|       | eod_routes.py::TestEODErrorHandling          | test_submit_eod_engine_error_returns_500            | Pass   |
|       | eod_routes.py::TestEODHistory                | test_eod_history_after_snap                         | Pass   |
|       | eod_routes.py::TestEODHistory                | test_eod_history_empty                              | Pass   |
|       | eod_routes.py::TestEODHistory                | test_multiple_eod_snaps                             | Pass   |
|       | eod_routes.py::TestEODNoDateBody             | test_submit_eod_no_json_uses_today                  | Pass   |
|       | eod_routes.py::TestEODPdf                    | test_eod_pdf_found_serves_file                      | Pass   |
|       | eod_routes.py::TestEODPdf                    | test_eod_pdf_not_found                              | Pass   |
|       | eod_routes.py::TestEODPdfGeneration          | test_submit_eod_import_error_still_succeeds         | Pass   |
|       | eod_routes.py::TestEODPdfGeneration          | test_submit_eod_pdf_exception_still_succeeds        | Pass   |
|       | eod_routes.py::TestPnLResetAfterEOD          | test_pnl_uses_eod_as_baseline                       | Pass   |
|       | eod_routes.py::TestSubmitEOD                 | test_eod_idempotent_same_day                        | Pass   |
|       | eod_routes.py::TestSubmitEOD                 | test_submit_eod                                     | Pass   |
|       | eod_routes.py::TestSubmitEOD                 | test_submit_eod_returns_summary                     | Pass   |
|       | eod_routes.py::TestSubmitEOD                 | test_submit_eod_saves_snapshot_file                 | Pass   |
|       | eod_routes.py::TestSubmitEOD                 | test_submit_eod_snapshot_has_market_state           | Pass   |
|       | eod_routes.py::TestSubmitEOD                 | test_submit_eod_snapshot_has_positions              | Pass   |
|       | market_state_routes.py::TestGetMarketState   | test_get_market_state                               | Pass   |
|       | market_state_routes.py::TestGetMarketState   | test_get_market_state_has_effective_rates           | Pass   |
|       | market_state_routes.py::TestGetMarketState   | test_get_market_state_initial_load_creates_defaults | Pass   |

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| Model | Module                                              | Test Name                                                 | Status |
|-------|-----------------------------------------------------|-----------------------------------------------------------|--------|
|       | market_state_routes.py::TestMarketStateErrorHandler | test_market_state_engine_error_returns_500                | Pass   |
|       | market_state_routes.py::TestMarketStateErrorHandler | test_market_state_engine_error_returns_500                | Pass   |
|       | market_state_routes.py::TestMarketStateErrorHandler | test_market_state_null_body_returns_400                   | Pass   |
|       | market_state_routes.py::TestResetMarketState        | test_market_state_all                                     | Pass   |
|       | market_state_routes.py::TestResetMarketState        | test_market_state_single_gauge                            | Pass   |
|       | market_state_routes.py::TestUpdateMarketState       | test_market_state_persists                                | Pass   |
|       | market_state_routes.py::TestUpdateMarketState       | test_market_state                                         | Pass   |
|       | market_state_routes.py::TestUpdateMarketState       | test_market_state_missing_gauge                           | Pass   |
|       | market_state_routes.py::TestUpdateMarketState       | test_market_state_no_body                                 | Pass   |
|       | market_state_routes.py::TestUpdateMarketState       | test_market_state_returns_pnl_impact                      | Pass   |
|       | risk_routes.py::TestRiskErrorHandler                | test_risk_grid_engine_error_returns_500                   | Pass   |
|       | risk_routes.py::TestRiskErrorHandler                | test_trade_map_engine_error_returns_500                   | Pass   |
|       | risk_routes.py::TestRiskGrid                        | test_risk_grid_has_gauges                                 | Pass   |
|       | risk_routes.py::TestRiskGrid                        | test_risk_grid_returns_grid                               | Pass   |
|       | risk_routes.py::TestRiskGrid                        | test_risk_grid_with_no_trades                             | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map_fs01_by_tenor                              | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map_gauge_has_coordinates                      | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map_includes_fs01                              | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map_includes_notional                          | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map_includes_pnl                               | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map_num_trades                                 | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map_returns_positions                          | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map_with_no_trades                             | Pass   |
|       | risk_routes.py::TestTradeMapWithProperties          | test_trade_map_gauge_not_in_gaugehc_loads_from_gauge_json | Pass   |
|       | risk_routes.py::TestTradeMapWithProperties          | test_trade_map_loads_property_locations                   | Pass   |
|       | risk_routes.py::TestTradeMapWithProperties          | test_trade_map_skips_closed_trades                        | Pass   |
|       | blotter_detail.py::TestBlotterEndpoint              | test_blotter_fetches_correct_endpoint                     | Pass   |
|       | blotter_detail.py::TestBlotterEndpoint              | test_blotter_close_endpoint                               | Pass   |
|       | blotter_detail.py::TestFilterBar                    | test_clear_filters                                        | Pass   |
|       | blotter_detail.py::TestFilterBar                    | test_external_filter_api                                  | Pass   |
|       | blotter_detail.py::TestFilterBar                    | test_filter_function_exists                               | Pass   |
|       | blotter_detail.py::TestFilterBar                    | test_get_filtered_trades                                  | Pass   |
|       | blotter_detail.py::TestFilterBar                    | test_remove_single_filter                                 | Pass   |
|       | blotter_detail.py::TestHazardRateArrow              | test_down_arrow                                           | Pass   |
|       | blotter_detail.py::TestHazardRateArrow              | test_prev_fair_spread_comparison                          | Pass   |
|       | blotter_detail.py::TestHazardRateArrow              | test_up_arrow                                             | Pass   |
|       | blotter_detail.py::TestMarketPnLColumn              | test_market_pnl_column                                    | Pass   |
|       | blotter_detail.py::TestMarketPnLColumn              | test_market_pnl_label                                     | Pass   |
|       | blotter_detail.py::TestPnLBar                       | test_pnl_bar_daily                                        | Pass   |
|       | blotter_detail.py::TestPnLBar                       | test_pnl_bar_from_market                                  | Pass   |
|       | blotter_detail.py::TestPnLBar                       | test_pnl_bar_net_notional                                 | Pass   |
|       | blotter_detail.py::TestPnLBar                       | test_pnl_bar_running                                      | Pass   |
|       | blotter_detail.py::TestSignedNotional               | test_is_payer_referenced                                  | Pass   |
|       | blotter_detail.py::TestSignedNotional               | test_assigned_notional_direction                          | Pass   |

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| Model | Module                                   | Test Name                                | Status |
|-------|------------------------------------------|------------------------------------------|--------|
|       | blotter_detail.py::TestSortAndNavigation | test_sort_toggle                         | Pass   |
|       | blotter_detail.py::TestSortAndNavigation | test_sort_view_click                     | Pass   |
|       | blotter_detail.py::TestTableStructure    | test_gauge_fs01_column                   | Pass   |
|       | blotter_detail.py::TestTableStructure    | test_maturity_formatting                 | Pass   |
|       | eod_detail.py::TestEODEndpoint           | test_eod_blotter_fetch                   | Pass   |
|       | eod_detail.py::TestEODEndpoint           | test_eod_history_fetch                   | Pass   |
|       | eod_detail.py::TestEODEndpoint           | test_eod_pnl_series_fetch                | Pass   |
|       | eod_detail.py::TestEODEndpoint           | test_eod_submit_endpoint                 | Pass   |
|       | eod_detail.py::TestEODFunction           | test_attribution_section                 | Pass   |
|       | eod_detail.py::TestEODFunction           | test_chart_rendering                     | Pass   |
|       | eod_detail.py::TestEODFunction           | test_cleanup_function                    | Pass   |
|       | eod_detail.py::TestEODFunction           | test_download_pdf_function               | Pass   |
|       | eod_detail.py::TestEODFunction           | test_submit_function                     | Pass   |
|       | eod_detail.py::TestEODUIStructure        | test_attribution_wrap                    | Pass   |
|       | eod_detail.py::TestEODUIStructure        | test_cards_container                     | Pass   |
|       | eod_detail.py::TestEODUIStructure        | test_date_input                          | Pass   |
|       | eod_detail.py::TestEODUIStructure        | test_history_wrap                        | Pass   |
|       | eod_detail.py::TestEODUIStructure        | test_submit_button                       | Pass   |
|       | market_detail.py::TestActionButtonFont   | test_button_font_matches_tabs            | Pass   |
|       | market_detail.py::TestActionButtonFont   | test_button_padding_matches_tabs         | Pass   |
|       | market_detail.py::TestActionButtonFont   | test_buttons_in_shared_container         | Pass   |
|       | market_detail.py::TestActionButtonFormat | test_commit_button_blue                  | Pass   |
|       | market_detail.py::TestActionButtonFormat | test_history_button_grey_not_blue        | Pass   |
|       | market_detail.py::TestActionButtonFormat | test_individual_border_radius_on_buttons | Pass   |
|       | market_detail.py::TestActionButtonFormat | test_phhist_button_grey_not_blue         | Pass   |
|       | market_detail.py::TestActionButtonFormat | test_reset_button_grey                   | Pass   |
|       | market_detail.py::TestCommitButton       | test_commit_button_calls_function        | Pass   |
|       | market_detail.py::TestCommitButton       | test_commit_button_id_present            | Pass   |
|       | market_detail.py::TestCommitButton       | test_commit_button_not_hidden            | Pass   |
|       | market_detail.py::TestCommitButton       | test_commit_function_defined             | Pass   |
|       | market_detail.py::TestCommitEndpoint     | test_commit_posts_hazard_ts              | Pass   |
|       | market_detail.py::TestCommitEndpoint     | test_commit_posts_yield_curve            | Pass   |
|       | market_detail.py::TestCommitEndpoint     | test_commit_shows_pnl_impact             | Pass   |
|       | market_detail.py::TestCommitEndpoint     | test_commit_switches_to_blotter          | Pass   |
|       | market_detail.py::TestDirtyState         | test_hazard_dirty_keys                   | Pass   |
|       | market_detail.py::TestDirtyState         | test_kingcommitted_label                 | Pass   |
|       | market_detail.py::TestDirtyState         | test_kingld_dirty_flag                   | Pass   |
|       | market_detail.py::TestMarketUIState      | test_cleanup_function                    | Pass   |
|       | market_detail.py::TestMarketUIState      | test_gauge_list_function                 | Pass   |
|       | market_detail.py::TestMarketUIState      | test_market_canvas                       | Pass   |
|       | market_detail.py::TestMarketUIState      | test_mode_selector                       | Pass   |
|       | market_detail.py::TestNewTradeButton     | test_new_trade_button_grey               | Pass   |
|       | market_detail.py::TestNewTradeButton     | test_new_trade_button_label              | Pass   |
|       | market_detail.py::TestNewTradeButton     | test_new_trade_button_present            | Pass   |

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| Model   | Module                               | Test Name                                           | Status |
|---------|--------------------------------------|-----------------------------------------------------|--------|
|         | market_detail.py::TestNewTradeButton | test_new_trade_calls_function                       | Pass   |
|         | market_detail.py::TestNewTradeButton | test_new_trade_function_opens_prs                   | Pass   |
|         | market_detail.py::TestNewTradeButton | test_new_trade_left_of_history                      | Pass   |
|         | market_detail.py::TestNewTradeButton | test_no_border_radius_on_new_trade_btn              | Pass   |
|         | market_detail.py::TestResetEndpoint  | test_reset_function_exists                          | Pass   |
|         | market_detail.py::TestResetEndpoint  | test_reset_hazard_ts_endpoint                       | Pass   |
|         | market_detail.py::TestResetEndpoint  | test_reset_yield_curve_endpoint                     | Pass   |
| CDM-ALL | gauge_mapping.py                     | test_all_cdm_fields_present                         | Pass   |
|         | gauge_mapping.py                     | test_all_types_valid                                | Pass   |
|         | gauge_mapping.py                     | test_all_values_valid                               | Pass   |
|         | mortgage_mapping.py                  | test_all_cdm_fields_present                         | Pass   |
|         | mortgage_mapping.py                  | test_all_types_valid                                | Pass   |
|         | mortgage_mapping.py                  | test_all_values_valid                               | Pass   |
|         | property_mapping.py                  | test_all_cdm_fields_present                         | Pass   |
|         | property_mapping.py                  | test_all_types_valid                                | Pass   |
|         | property_mapping.py                  | test_all_values_valid                               | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_required_fields_returns_list[gauge_cdm]    | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_required_fields_returns_list[mortgage_cdm] | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_required_fields_returns_list[property_cdm] | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_required_fields_returns_list[prscdm]       | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_required_fields_returns_list[storm_cdm]    | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_required_fields_returns_list[stormts_cdm]  | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_all_fields_returns_paths[gauge_cdm]        | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_all_fields_returns_paths[mortgage_cdm]     | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_all_fields_returns_paths[property_cdm]     | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_all_fields_returns_paths[prscdm]           | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_all_fields_returns_paths[storm_cdm]        | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_all_fields_returns_paths[stormts_cdm]      | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_repr_returns_string[gauge_cdm]             | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_repr_returns_string[mortgage_cdm]          | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_repr_returns_string[property_cdm]          | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_repr_returns_string[prscdm]                | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_repr_returns_string[storm_cdm]             | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_get_repr_returns_string[stormts_cdm]           | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_schema_returns_non_empty_dict[gauge_cdm]       | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_schema_returns_non_empty_dict[mortgage_cdm]    | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_schema_returns_non_empty_dict[property_cdm]    | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_schema_returns_non_empty_dict[prscdm]          | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_schema_returns_non_empty_dict[storm_cdm]       | Pass   |
|         | schema_basics.py::TestCDMSchema      | test_schema_returns_non_empty_dict[stormts_cdm]     | Pass   |
|         | schemas.py::TestFloodGaugeCDM        | test_create_mapping_returns_flat_dict               | Pass   |
|         | schemas.py::TestFloodGaugeCDM        | test_create_mapping_with_empty_input                | Pass   |
|         | schemas.py::TestFloodGaugeCDM        | test_get_field_info_for_gauge_id                    | Pass   |
|         | schemas.py::TestFloodGaugeCDM        | test_required_fields_include_gauge_id               | Pass   |

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| Model    | Module                              | Test Name                                             | Status |
|----------|-------------------------------------|-------------------------------------------------------|--------|
|          | schemas.py::TestFloodGaugeCDM       | test_schema_has_flood_gauge_root                      | Pass   |
|          | schemas.py::TestFloodGaugeCDM       | test_schema_has_header_section                        | Pass   |
|          | schemas.py::TestFloodGaugeCDM       | test_validate_empty_data_returns_errors               | Pass   |
|          | schemas.py::TestFloodGaugeCDM       | test_validate_valid_gauge_data                        | Pass   |
|          | schemas.py::TestMortgageCDM         | test_create_mapping_returns_flat_dict                 | Pass   |
|          | schemas.py::TestMortgageCDM         | test_required_fields_include_mortgage_id              | Pass   |
|          | schemas.py::TestMortgageCDM         | test_schema_has_mortgage_root                         | Pass   |
|          | schemas.py::TestMortgageCDM         | test_validate_valid_mortgage                          | Pass   |
|          | schemas.py::TestPhysicalRiskSwapCDM | test_create_mapping_returns_flat_dict                 | Pass   |
|          | schemas.py::TestPhysicalRiskSwapCDM | test_all_fields_has_gauge_set                         | Pass   |
|          | schemas.py::TestPhysicalRiskSwapCDM | test_required_fields_include_swap_id                  | Pass   |
|          | schemas.py::TestPhysicalRiskSwapCDM | test_schema_has_root                                  | Pass   |
|          | schemas.py::TestPropertyCDM         | test_create_mapping_returns_flat_dict                 | Pass   |
|          | schemas.py::TestPropertyCDM         | test_required_fields_not_empty                        | Pass   |
|          | schemas.py::TestPropertyCDM         | test_schema_has_property_root                         | Pass   |
|          | schemas.py::TestPropertyCDM         | test_validate_valid_property                          | Pass   |
|          | schemas.py::TestStormEventCDM       | test_create_mapping_exception_raises                  | Pass   |
|          | schemas.py::TestStormEventCDM       | test_create_mapping_full_data                         | Pass   |
|          | schemas.py::TestStormEventCDM       | test_create_mapping_returns_flat_dict                 | Pass   |
|          | schemas.py::TestStormEventCDM       | test_get_required_fields_returns_list                 | Pass   |
|          | schemas.py::TestStormEventCDM       | test_required_fields_include_storm_id                 | Pass   |
|          | schemas.py::TestStormEventCDM       | test_schema_has_storm_root                            | Pass   |
|          | schemas.py::TestStormEventCDM       | test_validate_empty_data                              | Pass   |
|          | schemas.py::TestStormEventCDM       | test_validate_exception_returns_error_dict            | Pass   |
|          | schemas.py::TestStormEventCDM       | test_validate_missing_catchment_id                    | Pass   |
|          | schemas.py::TestStormEventCDM       | test_validate_missing_storm_id                        | Pass   |
|          | schemas.py::TestStormEventCDM       | test_validate_valid_storm                             | Pass   |
|          | schemas.py::TestStormTimeSeriesCDM  | test_create_mapping_exception_raises_validation_error | Pass   |
|          | schemas.py::TestStormTimeSeriesCDM  | test_create_mapping_full_data                         | Pass   |
|          | schemas.py::TestStormTimeSeriesCDM  | test_create_mapping_returns_flat_dict                 | Pass   |
|          | schemas.py::TestStormTimeSeriesCDM  | test_create_reading_mapping                           | Pass   |
|          | schemas.py::TestStormTimeSeriesCDM  | test_get_required_fields_returns_list                 | Pass   |
|          | schemas.py::TestStormTimeSeriesCDM  | test_required_fields_include_ts_id                    | Pass   |
|          | schemas.py::TestStormTimeSeriesCDM  | test_schema_has_root                                  | Pass   |
|          | schemas.py::TestStormTimeSeriesCDM  | test_validate_empty_data                              | Pass   |
|          | schemas.py::TestStormTimeSeriesCDM  | test_validate_exception_returns_error_dict            | Pass   |
|          | schemas.py::TestStormTimeSeriesCDM  | test_validate_missing_catchment_id                    | Pass   |
|          | schemas.py::TestStormTimeSeriesCDM  | test_validate_missing_ts_id                           | Pass   |
|          | schemas.py::TestStormTimeSeriesCDM  | test_validate_valid_data                              | Pass   |
| PLATFORM | catchment_integration.py            | test_config                                           | Pass   |
|          | catchment_integration.py            | test_config_endpoints_include_health_check            | Pass   |
|          | catchment_integration.py            | test_html_file                                        | Pass   |
|          | catchment_integration.py            | test_routes                                           | Pass   |
|          | test_data_lineage_report_collect.py | test_test_collector_result_structure                  | Pass   |

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| Model | Module                                | Test Name                                     | Status |
|-------|---------------------------------------|-----------------------------------------------|--------|
|       | test_data_lineage_report_collect.py   | testEggsGoldenAdhes_step_details              | Pass   |
|       | test_data_lineage_report_collect.py   | testSumStepsAllpositive                       | Pass   |
|       | test_data_lineage_report_collect.py   | testCullusAllct                               | Pass   |
|       | test_data_lineage_report_collect.py   | testCpldetails_list                           | Pass   |
|       | test_data_lineage_report_collect.py   | testCreatedPDRReport                          | Pass   |
|       | test_data_lineage_report_collect.py   | testPdfheaderPdrReport                        | Pass   |
|       | test_data_lineage_report_collect.py   | testPdffromRptBy                              | Pass   |
|       | test_data_lineage_report_collect.py   | testPdfwithPdrEmptystep_details               | Pass   |
|       | test_data_lineage_report_collect.py   | testPdfwithPdrSuccess                         | Pass   |
|       | test_data_lineage_report_collect.py   | testPdfwithPdrLineage_results                 | Pass   |
|       | test_data_lineage_report_collect.py   | testReturnPdrRpt                              | Pass   |
|       | test_data_lineage_report_sectionstest | testTestConstantWhenhall_clear                | Pass   |
|       | test_data_lineage_report_sectionstest | testTestGadqautoHealth_steps                  | Pass   |
|       | test_data_lineage_report_sectionstest | testTestGadqautoHealth_failures               | Pass   |
|       | test_data_lineage_report_sectionstest | testTestChompoldCrapping                      | Pass   |
|       | test_data_lineage_report_sectionstest | testTestConsistentHealth_broken_chain         | Pass   |
|       | test_data_lineage_report_sectionstest | testTestChainantWhenall_pass                  | Pass   |
|       | test_data_lineage_report_sectionstest | testTestOnlylineage_results                   | Pass   |
|       | test_data_lineage_report_sectionstest | testTestComplianceVndir_broken_chain          | Pass   |
|       | test_data_lineage_report_sectionstest | testTestComplianceVndir_majority_failures     | Pass   |
|       | test_data_lineage_report_sectionstest | testTestComplianceVndir_missing_steps         | Pass   |
|       | test_data_lineage_report_sectionstest | testTestPartialtyCompilcant_on_test_failure   | Pass   |
|       | test_data_lineage_report_sectionstest | testTestPartialtyCompilcant_when_no_tests_run | Fail   |
|       | test_data_lineage_report_sectionstest | testTestGistBadgoldenlth                      | Pass   |
|       | test_data_lineage_report_sectionstest | testTestGradeBadgeHondifailures               | Pass   |
|       | test_data_lineage_report_sectionstest | testTestConsistentGradeHondibroken_chain      | Pass   |
|       | test_data_lineage_report_sectionstest | testTestFlicStruVerdict                       | Pass   |
|       | test_data_lineage_report_sectionstest | testTestExomSimplaryVerdict                   | Pass   |
|       | test_data_lineage_report_sectionstest | testTestExisSummaryVerdict                    | Pass   |
|       | test_data_lineage_report_sectionstest | testTestFailedflickBrenders_table             | Pass   |
|       | test_data_lineage_report_sectionstest | testTestKnownflickBrenders_bullet             | Pass   |
|       | test_data_lineage_report_sectionstest | testTestHelpcolour_fresh                      | Pass   |
|       | test_data_lineage_report_sectionstest | testTestHelpcolour_missing                    | Pass   |
|       | test_data_lineage_report_sectionstest | testTestHelpcolour_stale                      | Pass   |
|       | test_data_lineage_report_sectionstest | testTestHelpcolour_unknown                    | Pass   |
|       | test_data_lineage_report_sectionstest | testTestHelpdabel_unknown_uppercased          | Pass   |
|       | test_data_lineage_report_sectionstest | testTestHelpdabel_values                      | Pass   |
|       | test_data_lineage_report_sectionstest | testTestBackistingResults                     | Pass   |
|       | test_data_lineage_report_sectionstest | testTesttrandictwhenmissing                   | Pass   |
|       | test_data_lineage_report_sectionstest | testTestAndesManifest_json                    | Pass   |
|       | test_data_lineage_report_sectionstest | testTestAndesManifest_file                    | Pass   |
|       | test_data_lineage_report_sectionstest | testTestTrandManifest                         | Pass   |
|       | test_data_lineage_report_sectionstest | testTestMaineates.audit_dir                   | Pass   |
|       | test_data_lineage_report_sectionstest | testTestMainints.summary                      | Pass   |

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| Model | Module                                     | Test Name                                | Status |
|-------|--------------------------------------------|------------------------------------------|--------|
|       | test_data_lineage_report_sectionstest      | TestMainroduces_pdf                      | Pass   |
|       | test_data_lineage_report_sectionstest      | TestFailRecursioninPrinciples_key_tests  | Pass   |
|       | test_data_lineage_report_sectionstest      | TestDocTableifiesPrinciple_column        | Pass   |
|       | test_data_lineage_report_sectionstest      | TestCompanitarionVersitable              | Pass   |
|       | test_data_lineage_report_sectionstest      | TestReseridationVendible                 | Pass   |
|       | test_data_lineage_report_sectionstest      | TestRediationinNustistale_steps          | Pass   |
|       | test_data_lineage_report_sectionstest      | TestRediationinMistale_failure_met       | Pass   |
|       | test_data_lineage_report_sectionstest      | TestShowstrenghtsandweaknesses           | Pass   |
|       | test_data_lineage_report_sectionstest      | TestRetentioninPrinciplesImprugucibility | Pass   |
|       | test_data_lineage_report_sectionstest      | TestSectionBriders                       | Pass   |
|       | test_data_lineage_report_sectionstest      | TestSectionBesiders                      | Pass   |
|       | test_data_lineage_report_sectionstest      | TestTextSectionBridhealy                 | Pass   |
|       | test_data_lineage_report_sectionstest      | TestTextSectionBridissues                | Pass   |
|       | test_data_lineage_report_sectionstest      | TestTextSectionBridkshealthy             | Pass   |
|       | test_data_lineage_report_sectionstest      | TestTextSectionBridkswith_stale_details  | Pass   |
|       | test_data_lineage_report_sectionstest      | TestSectionBuilderno_results             | Pass   |
|       | test_data_lineage_report_sectionstest      | TestSectionBuilderwith_failures          | Pass   |
|       | test_data_lineage_report_sectionstest      | TestSectionBuilderwith_results           | Pass   |
|       | test_data_lineage_report_sectionstest      | TestSectionBuildercompliant              | Pass   |
|       | test_data_lineage_report_sectionstest      | TestSectionBuilderith_issues             | Pass   |
|       | test_data_lineage_report_sectionstest      | TestSectionPolicy                        | Pass   |
|       | test_data_lineage_report_sectionstest      | TestSectionBuilderatation                | Pass   |
|       | test_data_lineage_report_sectionstest      | TestSectionBuilderatation_empty_params   | Pass   |
|       | test_data_lineage_report_sectionstest      | TestTopologyBuilders                     | Pass   |
|       | test_data_lineage_report_sectionstest      | TestTopologyOovershipall_graph_steps     | Pass   |
|       | test_data_lineage_report_sectionstest      | TestTopologyrendersuper                  | Pass   |
|       | test_duplication_report.py::TestAnlyse     | empty_report                             | Pass   |
|       | test_duplication_report.py::TestAnlyse     | fmt_stats_has_python                     | Pass   |
|       | test_duplication_report.py::TestAnlyse     | largest_line_numbers_are_integers        | Pass   |
|       | test_duplication_report.py::TestAnlyse     | largest_line_numbers_not_question        | Mark   |
|       | test_duplication_report.py::TestAnlyse     | largest_sorted_descending                | Pass   |
|       | test_duplication_report.py::TestAnlyse     | sum_clones_correct                       | Pass   |
|       | test_duplication_report.py::TestAnlyse     | returns_dict                             | Pass   |
|       | test_duplication_report.py::TestAnlyse     | total_present                            | Pass   |
|       | test_duplication_report.py::TestAnlyse     | worst_populated                          | Pass   |
|       | test_duplication_report.py::TestMain       | main_handles_jscpd_error                 | Pass   |
|       | test_duplication_report.py::TestMain       | main_prints_stats                        | Pass   |
|       | test_duplication_report.py::TestMain       | main_produces_pdf                        | Pass   |
|       | test_duplication_report.py::TestMakePdf    | of_rating_for_low_pct                    | Pass   |
|       | test_duplication_report.py::TestMakePdf    | High_rating_report                       | Pass   |
|       | test_duplication_report.py::TestMakePdf    | header                                   | Pass   |
|       | test_duplication_report.py::TestMakePdf    | nonempty                                 | Pass   |
|       | test_duplication_report.py::TestMakePdf    | returns_bytes                            | Pass   |
|       | test_model_risk_report_cover.py::TestBuild | Coverasic                                | Pass   |

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| Model | Module                               | Test Name                                    | Status |
|-------|--------------------------------------|----------------------------------------------|--------|
|       | test_model_risk_report_cover.py::    | testBuildCoverbcbs                           | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildCovertests                          | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildCoverwith.bcbs                      | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildCoverwith.tests                     | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildSummaryall_pass                     | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildSummaryfailures                     | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildSummaryno_coverage                  | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildSummaryno_tests                     | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildSummaryrenders                      | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildInventoryempty                      | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildInventoryno_risk_ratings            | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildInventoryrag_colours                | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildInventoryrenders                    | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildInventoryrisk_ratings               | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildValidationall_reviewers_assigned    | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildValidationhigh_assumptions          | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildValidationno_assumptions            | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildValidationno_questions              | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildValidationpeer_reviewer_gaps        | Pass   |
|       | test_model_risk_report_cover.py::    | testBuildValidationrenders                   | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildAuditDate_range                     | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildAuditEmpty                          | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildAuditTimestamps                     | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildAuditTenders                        | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildDsempty                             | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildDscribenlist                        | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildDsrenders                           | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildRetentionsadditionclean             | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildRetentionsadditions_risk            | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildRetentionsoverrides_remediation     | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildRetentionsprevious_reviewer_gap     | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildRetentionsvalidation_backup_gaps    | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildRetentionsrenders                   | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildRetentionsvalidation_failures       | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildRetentionsvalidation_gap            | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildToolEvidence                        | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildToolSensitivity_generators          | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildToolFmodel_coverage                 | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildToolFvidence                        | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildToolFvidence_coverage               | Pass   |
|       | test_model_risk_report_evidence.py:: | testBuildToolFvidence_sensitivity_generators | Pass   |
|       | test_model_risk_report_mrc.py::      | testBuildBCBS239_compliant                   | Pass   |
|       | test_model_risk_report_mrc.py::      | testBuildBCBS239_risk                        | Pass   |
|       | test_model_risk_report_mrc.py::      | testBuildBCBS239_principles                  | Pass   |

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| Model                                             | Module                                            | Test Name                                       | Status |
|---------------------------------------------------|---------------------------------------------------|-------------------------------------------------|--------|
| test_model_risk_report_mrc.py::TestBuildMRC       | test_model_risk_report_mrc.py::TestBuildMRC       | TestBuildMRCHeaders                             | Pass   |
| test_model_risk_report_mrc.py::TestBuildMRC       | test_model_risk_report_mrc.py::TestBuildMRC       | TestBuildMRCCompleted_actions_excluded          | Pass   |
| test_model_risk_report_mrc.py::TestBuildMRC       | test_model_risk_report_mrc.py::TestBuildMRC       | TestBuildMRCDecisions                           | Pass   |
| test_model_risk_report_mrc.py::TestBuildMRC       | test_model_risk_report_mrc.py::TestBuildMRC       | TestBuildMRCMeetings                            | Pass   |
| test_model_risk_report_mrc.py::TestBuildMRC       | test_model_risk_report_mrc.py::TestBuildMRC       | TestBuildMRCOpen_actions                        | Pass   |
| test_model_risk_report_mrc.py::TestBuildMRC       | test_model_risk_report_mrc.py::TestBuildMRC       | TestBuildMRCHeaders                             | Pass   |
| test_model_risk_report_mrc.py::TestBuildMRC       | test_model_risk_report_mrc.py::TestBuildMRC       | TestBuildMRCDecisions                           | Pass   |
| test_model_risk_report_mrc.py::TestBuildRAG       | test_model_risk_report_mrc.py::TestBuildRAG       | TestBuildRAGEscalation_triggers_dict            | Pass   |
| test_model_risk_report_mrc.py::TestBuildRAG       | test_model_risk_report_mrc.py::TestBuildRAG       | TestBuildRAGEscalation_triggers_string          | Pass   |
| test_model_risk_report_mrc.py::TestBuildRAG       | test_model_risk_report_mrc.py::TestBuildRAG       | TestBuildRAGIsing_backup                        | Pass   |
| test_model_risk_report_mrc.py::TestBuildRAG       | test_model_risk_report_mrc.py::TestBuildRAG       | TestBuildRAGRoles                               | Pass   |
| test_model_risk_report_mrc.py::TestBuildRAG       | test_model_risk_report_mrc.py::TestBuildRAG       | TestBuildRAGHeaders                             | Pass   |
| test_model_risk_report_mrc.py::TestBuildRAG       | test_model_risk_report_mrc.py::TestBuildRAG       | TestBuildRAGWith_activities                     | Pass   |
| test_model_risk_report_mrc.py::TestBuildMediation | test_model_risk_report_mrc.py::TestBuildMediation | TestBuildMediationall_completed                 | Pass   |
| test_model_risk_report_mrc.py::TestBuildMediation | test_model_risk_report_mrc.py::TestBuildMediation | TestBuildMediationempty                         | Pass   |
| test_model_risk_report_mrc.py::TestBuildMediation | test_model_risk_report_mrc.py::TestBuildMediation | TestBuildMediationoverdue_highlighted           | Pass   |
| test_model_risk_report_mrc.py::TestBuildMediation | test_model_risk_report_mrc.py::TestBuildMediation | TestBuildMediationpriority_sorting              | Pass   |
| test_model_risk_report_mrc.py::TestBuildMediation | test_model_risk_report_mrc.py::TestBuildMediation | TestBuildMediationheaders                       | Pass   |
| test_model_risk_report_pdf.py::TestCreatePDF      | test_model_risk_report_pdf.py::TestCreatePDF      | TestCreatePDFReport_fallback                    | Pass   |
| test_model_risk_report_pdf.py::TestCreatePDF      | test_model_risk_report_pdf.py::TestCreatePDF      | TestCreatePDFReport_dir                         | Pass   |
| test_model_risk_report_pdf.py::TestCreatePDF      | test_model_risk_report_pdf.py::TestCreatePDF      | TestCreatePDFReport                             | Pass   |
| test_model_risk_report_pdf.py::TestCreatePDF      | test_model_risk_report_pdf.py::TestCreatePDF      | TestCreatePDFReportempty_data                   | Pass   |
| test_model_risk_report_pdf.py::TestCreatePDF      | test_model_risk_report_pdf.py::TestCreatePDF      | TestCreatePDFReports_summary                    | Pass   |
| test_model_risk_report_pdf.py::TestCreatePDF      | test_model_risk_report_pdf.py::TestCreatePDF      | TestCreatePDFReports_collect                    | Pass   |
| test_model_risk_report_pdf.py::TestBuildCase      | test_model_risk_report_pdf.py::TestBuildCase      | TestBuildCaseLog_single_event                   | Pass   |
| test_model_risk_report_pdf.py::TestBuildCase      | test_model_risk_report_pdf.py::TestBuildCase      | TestBuildCasezero_max_score                     | Pass   |
| test_model_risk_report_pdf.py::TestBuildCase      | test_model_risk_report_pdf.py::TestBuildCase      | TestBuildCaseno_coverage                        | Pass   |
| test_model_risk_report_pdf.py::TestBuildCase      | test_model_risk_report_pdf.py::TestBuildCase      | TestBuildCasewith_long_name                     | Pass   |
| test_model_risk_report_pdf.py::TestBuildCase      | test_model_risk_report_pdf.py::TestBuildCase      | TestBuildCasewith_no_name                       | Pass   |
| test_model_risk_report_pdf.py::TestBuildCase      | test_model_risk_report_pdf.py::TestBuildCase      | TestBuildCaseactions_with_missing_fields        | Pass   |
| test_model_risk_report_pdf.py::TestBuildCase      | test_model_risk_report_pdf.py::TestBuildCase      | TestBuildCasesingle_models_risk_ratings         | Pass   |
| test_model_risk_report_pdf.py::TestBuildCase      | test_model_risk_report_pdf.py::TestBuildCase      | TestBuildCaseno_activities_no_triggers          | Pass   |
| test_model_risk_report_pdf.py::TestBuildCase      | test_model_risk_report_pdf.py::TestBuildCase      | TestBuildCaseescalation_null_due_date           | Pass   |
| test_model_risk_report_pdf.py::TestBuildCase      | test_model_risk_report_pdf.py::TestBuildCase      | TestBuildCaseescalation_question_no_short_label | Pass   |
| test_model_risk_report_pdf.py::TestSection        | test_model_risk_report_pdf.py::TestSection        | TestSectionBuildHeaders_exported                | Pass   |
| test_model_risk_report_pdf.py::TestSection        | test_model_risk_report_pdf.py::TestSection        | TestSectionList                                 | Pass   |
| test_model_risk_report_styles.py::TestDownload    | test_model_risk_report_styles.py::TestDownload    | TestDownloadLink_keys                           | Pass   |
| test_model_risk_report_styles.py::TestDownload    | test_model_risk_report_styles.py::TestDownload    | TestDownloadLinks_files                         | Pass   |
| test_model_risk_report_styles.py::TestDownload    | test_model_risk_report_styles.py::TestDownload    | TestDownloadLinks_files_missing_dir             | Pass   |
| test_model_risk_report_styles.py::TestDownload    | test_model_risk_report_styles.py::TestDownload    | TestDownloadLinks_sensitivity_generators        | Pass   |
| test_model_risk_report_styles.py::TestDownload    | test_model_risk_report_styles.py::TestDownload    | TestDownloadLinks_sensitivity_generators_no_dir | Pass   |
| test_model_risk_report_styles.py::TestDownload    | test_model_risk_report_styles.py::TestDownload    | TestDownloadLinks_log                           | Pass   |
| test_model_risk_report_styles.py::TestDownload    | test_model_risk_report_styles.py::TestDownload    | TestDownloadLinks_log_dict_fallback             | Pass   |
| test_model_risk_report_styles.py::TestDownload    | test_model_risk_report_styles.py::TestDownload    | TestDownloadLinks                               | Pass   |

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| Model | Module                                     | Test Name                          | Status |
|-------|--------------------------------------------|------------------------------------|--------|
|       | test_model_risk_report_styles.py:          | TestDataCoverage                   | Pass   |
|       | test_model_risk_report_styles.py:          | TestDataCoverageCorrupt            | Pass   |
|       | test_model_risk_report_styles.py:          | TestDataCoverageMissing            | Pass   |
|       | test_model_risk_report_styles.py:          | TestDataInventory                  | Pass   |
|       | test_model_risk_report_styles.py:          | TestDataInventoryBadJson           | Pass   |
|       | test_model_risk_report_styles.py:          | TestDataInventoryMissing           | Pass   |
|       | test_model_risk_report_styles.py:          | TestDataJudits                     | Pass   |
|       | test_model_risk_report_styles.py:          | TestDataJuditsCorrupt              | Pass   |
|       | test_model_risk_report_styles.py:          | TestDataJuditsMissing              | Pass   |
|       | test_model_risk_report_styles.py:          | TestDataMeetings                   | Pass   |
|       | test_model_risk_report_styles.py:          | TestDataMeetingsDictFallback       | Pass   |
|       | test_model_risk_report_styles.py:          | TestDataRadars                     | Pass   |
|       | test_model_risk_report_styles.py:          | TestStylesConstants                | Pass   |
|       | test_model_risk_report_styles.py:          | TestStylesKeys                     | Pass   |
|       | test_model_risk_report_styles.py:          | TestStylesColourMaps               | Pass   |
|       | test_model_risk_report_styles.py:          | TestSectionRuleAppendsFlowables    | Pass   |
|       | test_model_risk_report_styles.py:          | TestStyleStyleReturnsTableStyle    | Pass   |
|       | test_blotter_data.py::TestGaugeData:       | curve.points_present               | Pass   |
|       | test_blotter_data.py::TestGaugeData:       | each_gauge_has_required_fields     | Pass   |
|       | test_blotter_data.py::TestGaugeData:       | file_exists                        | Pass   |
|       | test_blotter_data.py::TestGaugeData:       | flood_trigger_ordering             | Pass   |
|       | test_blotter_data.py::TestGaugeData:       | gauge_count_at_least_40            | Pass   |
|       | test_blotter_data.py::TestGaugeData:       | gauge_ids_are_well_formed          | Pass   |
|       | test_blotter_data.py::TestGaugeData:       | has_hazard_curves_key              | Pass   |
|       | test_blotter_data.py::TestGaugeData:       | hazard_rate_ordering               | Pass   |
|       | test_blotter_data.py::TestGaugeData:       | hazard_rates_positive              | Pass   |
|       | test_blotter_data.py::TestGaugeData:       | term_structure_present             | Pass   |
|       | test_blotter_data.py::TestMarketStateData: | base_rates_not_empty               | Pass   |
|       | test_blotter_data.py::TestMarketStateData: | file_exists                        | Pass   |
|       | test_blotter_data.py::TestMarketStateData: | base_required_keys                 | Pass   |
|       | test_blotter_data.py::TestMarketStateData: | hazard_term_structure_has_triggers | Pass   |
|       | test_blotter_data.py::TestMarketStateData: | hazard_term_structure_not_empty    | Pass   |
|       | test_blotter_data.py::TestMarketStateData: | last_updated_present               | Pass   |
|       | test_blotter_data.py::TestMarketStateData: | yield_curve_has_all_tenors         | Pass   |
|       | test_blotter_data.py::TestMarketStateData: | yield_curve_rates_in_range         | Pass   |
|       | test_blotter_data.py::TestPRTradeData:     | each_trade_has_physical_swap_root  | Pass   |
|       | test_blotter_data.py::TestPRTradeData:     | each_trade_has_swap_id             | Pass   |
|       | test_blotter_data.py::TestPRTradeData:     | trades_dir_exists                  | Pass   |
|       | test_blotter_data.py::TestPRTradeData:     | swap_id_matches_filename           | Pass   |
|       | test_blotter_data.py::TestPRTradeData:     | trade_count_thames_central         | Pass   |
|       | test_blotter_data.py::TestPRTradeData:     | trade_statuses_are_valid           | Pass   |
|       | test_blotter_data.py::TestPRTradeData:     | trades_are_present                 | Pass   |
|       | test_blotter_data.py::TestPRTradeData:     | trades_have_gauge_set              | Pass   |
|       | test_id_consistency.py::TestClassifier:    | gauge_ids_match                    | Pass   |

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| Model | Module                                       | Test Name                                 | Status |
|-------|----------------------------------------------|-------------------------------------------|--------|
|       | test_id_consistency.py::TestClassifiers      | test_classifiers_exist                    | Pass   |
|       | test_id_consistency.py::TestClassifiers      | test_classifier_summary_exists            | Pass   |
|       | test_id_consistency.py::TestDataLineage      | test_dependency_graph_complete            | Pass   |
|       | test_id_consistency.py::TestDataLineage      | test_manifest_exists                      | Pass   |
|       | test_id_consistency.py::TestDataLineage      | test_no_stale_inputs                      | Pass   |
|       | test_id_consistency.py::TestDataLineage      | test_output_hashes_current                | Pass   |
|       | test_id_consistency.py::TestDeterministicIDs | test_ids_are_deterministic                | Skip   |
|       | test_id_consistency.py::TestGauges           | test_gauge_json_has_gauges                | Pass   |
|       | test_id_consistency.py::TestGauges           | test_gauges_match_gauges                  | Pass   |
|       | test_id_consistency.py::TestGauges           | test_hazard_curves_match_gauges           | Pass   |
|       | test_id_consistency.py::TestGauges           | test_hazard_curves_subset_of_gauges       | Pass   |
|       | test_id_consistency.py::TestGauges           | test_consider_gauges_exist_in_gauge_json  | Pass   |
|       | test_id_consistency.py::TestGauges           | test_consider_gauges_have_hazard_curves   | Pass   |
|       | test_id_consistency.py::TestPipeline         | test_pipeline_outputs_exist               | Pass   |
|       | test_id_consistency.py::TestPipeline         | test_classifiers_match_storm_data         | Pass   |
|       | test_id_consistency.py::TestPipeline         | test_no_empty_output_directories          | Pass   |
|       | test_id_consistency.py::TestProperties       | test_properties_ids_in_header             | Pass   |
|       | test_id_consistency.py::TestProperties       | test_properties_json_has_properties       | Pass   |
|       | test_id_consistency.py::TestProperties       | test_properties_files_match_properties    | Pass   |
|       | test_id_consistency.py::TestStorms           | test_gauges_storms_from_sequences         | Pass   |
|       | test_id_consistency.py::TestStorms           | test_properties_storms_from_sequences     | Pass   |
|       | test_lineage.py::TestCheckInputs             | test_consumer_has_no_hash_for_input       | Pass   |
|       | test_lineage.py::TestCheckInputs             | test_consumer_never_run                   | Pass   |
|       | test_lineage.py::TestCheckInputs             | test_fresh_when_hashes_match              | Pass   |
|       | test_lineage.py::TestCheckInputs             | test_no_hash_recorded_for_producer_output | Pass   |
|       | test_lineage.py::TestCheckInputs             | test_producer_never_run                   | Pass   |
|       | test_lineage.py::TestCheckInputs             | test_root_step_always_fresh               | Pass   |
|       | test_lineage.py::TestCheckInputs             | test_stale_when_hashes_differ             | Pass   |
|       | test_lineage.py::TestCheckInputs             | test_unknown_step                         | Pass   |
|       | test_lineage.py::TestCheckPipeline           | test_complete_pipeline                    | Pass   |
|       | test_lineage.py::TestCheckPipeline           | test_empty_directory_detected             | Pass   |
|       | test_lineage.py::TestCheckPipeline           | test_missing_file                         | Pass   |
|       | test_lineage.py::TestCheckPipeline           | test_partial_pipeline                     | Pass   |
|       | test_lineage.py::TestCheckStep               | test_step_present                         | Pass   |
|       | test_lineage.py::TestCheckStep               | test_missing_upstream                     | Pass   |
|       | test_lineage.py::TestGetCurrentRunID         | test_format                               | Pass   |
|       | test_lineage.py::TestGetFileLineage          | test_file_with_no_producer_entry          | Pass   |
|       | test_lineage.py::TestGetFileLineage          | test_known_file                           | Pass   |
|       | test_lineage.py::TestGetFileLineage          | test_unknown_file                         | Pass   |
|       | test_lineage.py::TestGetStaleDownstream      | test_gauges_affects_many                  | Pass   |
|       | test_lineage.py::TestGetStaleDownstream      | test_head_has_no_downstream               | Pass   |
|       | test_lineage.py::TestGetStaleDownstream      | test_mid_chain                            | Pass   |
|       | test_lineage.py::TestGetStepLineage          | test_mid_step                             | Pass   |
|       | test_lineage.py::TestGetStepLineage          | test_root_step                            | Pass   |

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| Model | Module                                          | Test Name                               | Status |
|-------|-------------------------------------------------|-----------------------------------------|--------|
|       | test_lineage.py::TestGetStepLineage             | test_step_not_in_manifest               | Pass   |
|       | test_lineage.py::TestHashArtifact               | test_hash_directory                     | Pass   |
|       | test_lineage.py::TestHashArtifact               | test_hash_file                          | Pass   |
|       | test_lineage.py::TestHashArtifact               | test_hash_missing                       | Pass   |
|       | test_lineage.py::TestHashDirectory              | test_hash_empty_directory               | Pass   |
|       | test_lineage.py::TestHashDirectory              | test_hash_multiple_files_deterministic  | Pass   |
|       | test_lineage.py::TestHashDirectory              | test_hash_respects_pattern              | Pass   |
|       | test_lineage.py::TestHashDirectory              | test_hash_single_file                   | Pass   |
|       | test_lineage.py::TestHashFile                   | test_hash_binary_file                   | Pass   |
|       | test_lineage.py::TestHashFile                   | test_hash_empty_file                    | Pass   |
|       | test_lineage.py::TestHashFile                   | test_hash_known_content                 | Pass   |
|       | test_lineage.py::TestInputToStep                | test_known_inputs                       | Pass   |
|       | test_lineage.py::TestLoadManifest               | test_load_corrupt_json_returns_skeleton | Pass   |
|       | test_lineage.py::TestLoadManifest               | test_load_missing_returns_skeleton      | Pass   |
|       | test_lineage.py::TestLoadManifest               | test_load_valid_manifest                | Pass   |
|       | test_lineage.py::TestOutputToStep               | test_known_outputs                      | Pass   |
|       | test_lineage.py::TestRecordStep                 | test_record_and_reload                  | Pass   |
|       | test_lineage.py::TestRecordStep                 | test_record_with_auto_run_id            | Pass   |
|       | test_lineage.py::TestRepairManifest             | test_repair_empty_dir_skipped           | Pass   |
|       | test_lineage.py::TestRepairManifest             | test_repair_from_scratch                | Pass   |
|       | test_lineage.py::TestRepairManifest             | test_repair_hashes_are_current          | Pass   |
|       | test_lineage.py::TestRepairManifest             | test_repair_idempotent                  | Pass   |
|       | test_lineage.py::TestRepairManifest             | test_repair_inputs_hashed_correctly     | Pass   |
|       | test_lineage.py::TestRepairManifest             | test_repair_preserves_existing_metadata | Pass   |
|       | test_lineage.py::TestRepairManifest             | test_repair_run_id_format               | Pass   |
|       | test_lineage.py::TestRepairManifest             | test_repair_skips_missing_outputs       | Pass   |
|       | test_lineage.py::TestRepairManifest             | test_repair_updates_runs_index          | Pass   |
|       | test_lineage.py::TestSaveManifest               | test_save_cleans_up_on_failure          | Pass   |
|       | test_lineage.py::TestSaveManifest               | test_save_creates_file                  | Pass   |
|       | test_lineage.py::TestSaveManifest               | test_save_overwrites_existing           | Pass   |
|       | test_lineage.py::TestTraceDataPoint             | test_gauge_id_chain                     | Pass   |
|       | test_lineage.py::TestTraceDataPoint             | test_property_id_chain                  | Pass   |
|       | test_lineage.py::TestTraceDataPoint             | test_unknown_data_type                  | Pass   |
|       | test_lineage.py::TestValidateFullChain          | test_inconsistent_chain                 | Pass   |
|       | test_lineage.py::TestValidateFullChain          | test_empty_manifest                     | Pass   |
|       | test_lineage.py::TestValidateFullChain          | test_stale_step_detected                | Pass   |
|       | test_lineage.py::TestWalkUpstream               | test_cycle_safety                       | Pass   |
|       | test_lineage.py::TestWalkUpstream               | test_deep_chain                         | Pass   |
|       | test_lineage.py::TestWalkUpstream               | test_mid_chain                          | Pass   |
|       | test_lineage.py::TestWalkUpstream               | test_root_has_no_upstream               | Pass   |
|       | test_lineage_prerequisites.py::TestOutputsExist | test_output_empty                       | Pass   |
|       | test_lineage_prerequisites.py::TestOutputsExist | test_output_missing                     | Pass   |
|       | test_lineage_prerequisites.py::TestOutputsExist | test_output_present_non_empty           | Pass   |
|       | test_lineage_prerequisites.py::TestOutputsExist | test_output_missing                     | Pass   |

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| Model | Module                                      | Test Name                                      | Status |
|-------|---------------------------------------------|------------------------------------------------|--------|
|       | test_storm_data.storms.py::TestStorms       | Storms file ids use valid prefix               | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file one storm exceeds alert            | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file has gauge responses                | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file has required fields                | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file is                                 | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file well formed                        | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file responses have required fields     | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file is                                 | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file categories present                 | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file storm count                        | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file exceeds base                       | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file is positive                        | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file unique                             | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file well formed                        | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file over multiple gauges               | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file sorted by severity descending      | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file forms key matches list length      | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file summary max trigger valid          | Pass   |
|       | test_storm_data.storms.py::TestStorms       | Storms file summary populated for all storms   | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader       | Leader                                         | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader       | Leader of Base gauges key also works           | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader       | Leader of Base all returns list                | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader       | Leader of Base missing file returns empty      | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader       | Leader of Base caching works                   | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader       | Leader of Base cache                           | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader       | Leader of Base reload                          | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader       | Leader of Base lists false                     | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader       | Leader of Base lists true                      | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader       | Leader of Base by id missing returns none      | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader       | Leader of Base by id returns entity            | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader       | Leader of Base list all returns summaries      | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader       | Leader of Base status returns dict             | Pass   |
|       | test_mortgage_loader.py::TestMortgageLeader | Leader of Base returns summaries               | Pass   |
|       | test_mortgage_loader.py::TestMortgageLeader | Leader of Base returns list                    | Pass   |
|       | test_mortgage_loader.py::TestMortgageLeader | Leader of Base by id lookup                    | Pass   |
|       | test_mortgage_loader.py::TestMortgageLeader | Leader of Base by id property nested structure | Pass   |
|       | test_mortgage_loader.py::TestMortgageLeader | Leader of Base loading lookups none            | Pass   |
|       | test_property_county.py::TestFindByCounty   | FindBy county case insensitive                 | Pass   |
|       | test_property_county.py::TestFindByCounty   | FindBy county empty portfolio                  | Pass   |
|       | test_property_county.py::TestFindByCounty   | FindBy county exact match                      | Pass   |
|       | test_property_county.py::TestFindByCounty   | FindBy county missing county field             | Pass   |
|       | test_property_county.py::TestFindByCounty   | FindBy county no match                         | Pass   |
|       | test_property_county.py::TestFindByCounty   | FindBy county partial match                    | Pass   |
|       | test_property_loader.py::TestPropertyLoader | Property loader filename                       | Pass   |

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| Model | Module                                           | Test Name                                 | Status |
|-------|--------------------------------------------------|-------------------------------------------|--------|
|       | test_property_loader.py::TestPropertyLoaderBasic | test_property_loader_returns_list         | Pass   |
|       | test_property_loader.py::TestPropertyLoaderBasic | test_property_loader_file_empty           | Pass   |
|       | test_property_loader.py::TestPropertyLoaderBasic | test_property_loader_lookup               | Pass   |
|       | test_property_loader.py::TestPropertyLoaderBasic | test_property_loader_has_property_id      | Pass   |
|       | test_registry.py::TestCreateRegistry             | test_creates_registry                     | Pass   |
|       | test_registry.py::TestCreateRegistry             | test_with_file_overrides                  | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_clear_all_caches                     | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_get_all_status                       | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_clear_and_validate_all_caches        | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_data_files_all_missing               | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_data_files_present                   | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_data_file_overrides                  | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_data_dir                             | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_get_available_loaders                | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_get_gauge_loader                     | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_get_loader_by_name                   | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_get_loader_unknown_name_returns_none | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_get_mortgage_loader                  | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_get_property_loader                  | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_get_storm_loader                     | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_get_timeseries_loader                | Pass   |
|       | test_registry.py::TestLoaderRegistry             | test_loader_cached_on_repeated_access     | Pass   |
|       | storm_loader.py::TestFindByBasin                 | test_atl_basin                            | Pass   |
|       | storm_loader.py::TestFindByBasin                 | test_case_insensitive                     | Pass   |
|       | storm_loader.py::TestFindByBasin                 | test_unknown_basin_returns_empty          | Pass   |
|       | storm_loader.py::TestFindByBasin                 | test_wpac_basin                           | Pass   |
|       | storm_loader.py::TestFindByCategory              | test_all_category_4                       | Pass   |
|       | storm_loader.py::TestFindByCategory              | test_category_3                           | Pass   |
|       | storm_loader.py::TestFindByCategory              | test_category_5                           | Pass   |
|       | storm_loader.py::TestFindByCategory              | test_no_matches_returns_empty             | Pass   |
|       | storm_loader.py::TestFindByName                  | test_case_insensitive                     | Pass   |
|       | storm_loader.py::TestFindByName                  | test_exact_match                          | Pass   |
|       | storm_loader.py::TestFindByName                  | test_no_match_returns_none                | Pass   |
|       | storm_loader.py::TestFindByName                  | test_partial_match                        | Pass   |
|       | storm_loader.py::TestFindBySeason                | test_missing_season_returns_empty         | Pass   |
|       | storm_loader.py::TestFindBySeason                | test_season_2017                          | Pass   |
|       | storm_loader.py::TestFindBySeason                | test_season_with_multiple                 | Pass   |
|       | storm_loader.py::TestGetEntityId                 | test_empty_entity_returns_none            | Pass   |
|       | storm_loader.py::TestGetEntityId                 | test_extract_event_id                     | Pass   |
|       | storm_loader.py::TestGetEntityId                 | test_missing_id_returns_none              | Pass   |
|       | storm_loader.py::TestGetEntitySummary            | test_missing_fields_return_none           | Pass   |
|       | storm_loader.py::TestGetEntitySummary            | test_returns_summary_dict                 | Pass   |
|       | storm_loader.py::TestGetImpact                   | test_missing_id_returns_none              | Pass   |
|       | storm_loader.py::TestGetImpact                   | test_returns_impact_dict                  | Pass   |

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| Model | Module                               | Test Name                                               | Status |
|-------|--------------------------------------|---------------------------------------------------------|--------|
|       | storm_loader.py::TestGetMajorH       | test_excludes_category_1_and_2                          | Pass   |
|       | storm_loader.py::TestGetMajorH       | test_includes_cat_3_4_5                                 | Pass   |
|       | storm_loader.py::TestGetMajorH       | test_returns_category_3_plus                            | Pass   |
|       | storm_loader.py::TestGetStorms       | test_each_storm_returned_at_most_once                   | Pass   |
|       | storm_loader.py::TestGetStorms       | test_empty_cat_in_remote_location                       | Pass   |
|       | storm_loader.py::TestGetStorms       | test_high_severity_by_storm                             | Pass   |
|       | storm_loader.py::TestGetStorms       | test_large_cradius_finds_more                           | Pass   |
|       | storm_loader.py::TestGetTrack        | test_missing_id_returns_none                            | Pass   |
|       | storm_loader.py::TestGetTrack        | test_returns_track_for_existing_storm                   | Pass   |
|       | storm_loader.py::TestGetTrack        | test_track_has_lat_lon                                  | Pass   |
|       | test_base_loader_coverage.py::Test   | test_get_entity_is_called                               | Pass   |
|       | test_base_loader_coverage.py::Test   | test_get_entity_summary_called                          | Pass   |
|       | test_base_loader_coverage.py::Test   | test_get_entity_aliases_delegates_to_get_entity_summary | Pass   |
|       | test_base_loader_coverage.py::Test   | test_get_entity_aliases_delegates_to_get_entity         | Pass   |
|       | test_base_loader_coverage.py::Test   | test_get_entity_aliases_keys_returns_class_attrs        | Pass   |
|       | test_base_loader_coverage.py::Test   | test_get_entity_aliases_delegates_to_get_status         | Pass   |
|       | test_base_loader_coverage.py::Test   | test_normalize_list_of_dicts_returns_empty              | Pass   |
|       | test_base_loader_coverage.py::Test   | test_normalize_list_of_dicts_returns_empty              | Pass   |
|       | test_base_loader_coverage.py::Test   | test_normalize_list_of_dicts_returns_empty              | Pass   |
|       | test_base_loader_coverage.py::Test   | test_normalize_list_of_dicts_returns_empty              | Pass   |
|       | test_lookups_gauge_property.py::Test | test_build_gauge_properties_empty                       | Pass   |
|       | test_lookups_gauge_property.py::Test | test_build_gauge_properties_missing_fields              | Pass   |
|       | test_lookups_gauge_property.py::Test | test_build_gauge_properties_skipped                     | Pass   |
|       | test_lookups_gauge_property.py::Test | test_build_gauge_properties_only                        | Pass   |
|       | test_lookups_gauge_property.py::Test | test_build_gauge_properties_precedence                  | Pass   |
|       | test_lookups_gauge_property.py::Test | test_build_properties_from_data_when_not_in_phc         | Pass   |
|       | test_lookups_gauge_property.py::Test | test_build_properties_from_data                         | Pass   |
|       | test_lookups_gauge_property.py::Test | test_build_properties_precedence_over_property_data     | Pass   |
|       | test_lookups_gauge_property.py::Test | test_classify_higher_risk_is_high                       | Pass   |
|       | test_lookups_gauge_property.py::Test | test_classify_higher_risk                               | Pass   |
|       | test_lookups_gauge_property.py::Test | test_classify_lower_risk_is_low                         | Pass   |
|       | test_lookups_gauge_property.py::Test | test_classify_lower_risk                                | Pass   |
|       | test_lookups_gauge_property.py::Test | test_classify_lower_risk                                | Pass   |
|       | test_lookups_gauge_property.py::Test | test_classify_lower_risk                                | Pass   |
|       | test_lookups_ids.py::TestAnalyze     | test_relationships_returns_zeros                        | Pass   |
|       | test_lookups_ids.py::TestAnalyze     | test_relationships_match_data                           | Pass   |
|       | test_lookups_ids.py::TestAnalyze     | test_relationships_gauges_counted                       | Pass   |
|       | test_lookups_ids.py::TestAnalyze     | test_relationships_properties_overlap                   | Pass   |
|       | test_lookups_ids.py::TestAnalyze     | test_relationships_required_keys                        | Pass   |
|       | test_lookups_ids.py::TestExtract     | test_extract_ids                                        | Pass   |
|       | test_lookups_ids.py::TestExtract     | test_missing_gauge_id_skipped                           | Pass   |
|       | test_lookups_ids.py::TestExtract     | test_missing_returns_empty_set                          | Pass   |
|       | test_lookups_ids.py::TestExtract     | test_extract_ids                                        | Pass   |

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| Model                                           | Module                                          | Test Name                                                   | Status |
|-------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------|--------|
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_time_series_loader_statistics_missing_gauge            | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_time_series_loader_statistics_no_numeric_level         | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_time_series_loader_statistics_with_data                | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_time_series_loader_timestamp                           | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_time_series_loader_status_found_returns_none           | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_time_series_loader_status_alias                        | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_time_series_loader_status_returns_dict                 | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_appends_entry                                          | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_bad_audit_path_does_not_raise                          | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_corrupted_file_restarted                               | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_creates_audit_file                                     | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_entry_has_required_keys                                | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_existing_file_appended                                 | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_multiple_calls_accumulate                              | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_case_insensitive                                       | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_depth_damage_maps_to_dd                                | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_distribution_maps_to_si                                | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_mkm_prefix_returned_as_is                              | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_shorthand_gev_resolved                                 | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_shorthand_intensity_resolved                           | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_shorthand_mortgage_resolved                            | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_shorthand_prs_resolved                                 | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_spatial_maps_to_sp                                     | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_unknown_shorthand_returned_as_is                       | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_bool_passed_through                                    | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_empty_params                                           | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_float_passed_through                                   | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_int_passed_through                                     | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_long_list_summarised                                   | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_none_passed_through                                    | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_object_converted_to_string                             | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_short_list_converted_to_str                            | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_string_passed_through                                  | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_tuple_treated_like_list                                | Pass   |
| depth_damage.py::TestCalculateDepthDamage       | depth_damage.py::TestCalculateDepthDamage       | test_calculate_depth_damage_severe_level_some_positive      | Pass   |
| depth_damage.py::TestCalculateDepthDamage       | depth_damage.py::TestCalculateDepthDamage       | test_calculate_depth_damage_severe_level_returns_zeros      | Pass   |
| depth_damage.py::TestCalculateDepthDamage       | depth_damage.py::TestCalculateDepthDamage       | test_calculate_depth_damage_capped_at_5m                    | Pass   |
| depth_damage.py::TestCalculateDepthDamage       | depth_damage.py::TestCalculateDepthDamage       | test_calculate_depth_damage_gauge_data_returns_zeros        | Pass   |
| depth_damage.py::TestCalculateDepthDamage       | depth_damage.py::TestCalculateDepthDamage       | test_calculate_depth_damage_gauge_data_returns_zeros        | Pass   |
| depth_damage.py::TestCalculateDepthDamage       | depth_damage.py::TestCalculateDepthDamage       | test_calculate_depth_damage_correct_length                  | Pass   |
| depth_damage.py::TestDepthDamage                | depth_damage.py::TestDepthDamage                | test_depth_damage_commercial_higher_damage_than_residential | Pass   |
| depth_damage.py::TestDepthDamage                | depth_damage.py::TestDepthDamage                | test_depth_damage_flood_level_reduces_damage                | Pass   |
| depth_damage.py::TestDepthDamage                | depth_damage.py::TestDepthDamage                | test_depth_damage_flood_depth_all_near_one                  | Pass   |
| depth_damage.py::TestDepthDamage                | depth_damage.py::TestDepthDamage                | test_depth_damage_flood_depth_clipped_to_0_1                | Pass   |

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| Model | Module                                  | Test Name                                           | Status |
|-------|-----------------------------------------|-----------------------------------------------------|--------|
|       | depth_damage.py::TestDepthDamage        | test_flood_returns_array_same_length                | Pass   |
|       | depth_damage.py::TestDepthDamage        | test_flood_returns_all_zero_damage                  | Pass   |
|       | depth_damage.py::TestScalarDepthDamage  | test_damage_max_depth_returns_one                   | Pass   |
|       | depth_damage.py::TestScalarDepthDamage  | test_damage_between_control_points_interpolated     | Pass   |
|       | depth_damage.py::TestScalarDepthDamage  | test_damage_between_0_and_1                         | Pass   |
|       | depth_damage.py::TestScalarDepthDamage  | test_damage_meter_damage                            | Pass   |
|       | depth_damage.py::TestScalarDepthDamage  | test_damage_max_depth_returns_one                   | Pass   |
|       | depth_damage.py::TestScalarDepthDamage  | test_damage_monotonically_increasing                | Pass   |
|       | depth_damage.py::TestScalarDepthDamage  | test_damage_negative_depth_returns_zero             | Pass   |
|       | depth_damage.py::TestScalarDepthDamage  | test_damage_meter_damage                            | Pass   |
|       | depth_damage.py::TestScalarDepthDamage  | test_damage_returns_float                           | Pass   |
|       | depth_damage.py::TestScalarDepthDamage  | test_damage_small_depth_returns_small_damage        | Pass   |
|       | depth_damage.py::TestScalarDepthDamage  | test_damage_zero_depth_returns_zero                 | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_init                                     | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_correlation_matrix_built                 | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_correlation_params                       | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_missing_elevation_warns_and_defaults     | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_missing_floor_level_defaults_to_zero     | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_missing_geometry_raises                  | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_missing_property_id_auto_assigned        | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_missing_property_type_warns_and_defaults | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_missing_value_raises                     | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_properties_copied                        | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_modeling_gauge_data                            | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_methods_risk_clusters_returns_df         | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_methods_spatial_concentration_returns_df | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_methods_advanced_metrics_returns_dict    | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_methods_direct_impacts_returns_array     | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_methods_flood_depths_returns_array       | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_methods_distance_delegate                | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel | test_model_methods_portfolio_impact_returns_dict    | Pass   |
|       | spatial.py::TestBuildCorrelationMatrix  | test_matrix_diagonal_is_one                         | Pass   |
|       | spatial.py::TestBuildCorrelationMatrix  | test_matrix_off_diagonal_between_zero_and_one       | Pass   |
|       | spatial.py::TestBuildCorrelationMatrix  | test_matrix_shape_is_n_by_n                         | Pass   |
|       | spatial.py::TestBuildCorrelationMatrix  | test_matrix_symmetric                               | Pass   |
|       | spatial.py::TestBuildSpatialIndex       | test_empty_dataframe_returns_none                   | Pass   |
|       | spatial.py::TestBuildSpatialIndex       | test_none_gauge_data_returns_none                   | Pass   |
|       | spatial.py::TestBuildSpatialIndex       | test_with_gauges_returns_kdtree                     | Pass   |
|       | spatial.py::TestHaversineDistance       | test_equatorial_degree_approx_111km                 | Pass   |
|       | spatial.py::TestHaversineDistance       | test_london_paris_approx_340km                      | Pass   |
|       | spatial.py::TestHaversineDistance       | test_north_south_degree_approx_111km                | Pass   |
|       | spatial.py::TestHaversineDistance       | test_returns_float                                  | Pass   |
|       | spatial.py::TestHaversineDistance       | test_same_point_is_zero                             | Pass   |
|       | spatial.py::TestHaversineDistance       | test_symmetry                                       | Pass   |

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| Model | Module                                                                           | Test Name                                              | Status |
|-------|----------------------------------------------------------------------------------|--------------------------------------------------------|--------|
|       | spatial.py::TestIDWInterpolate                                                   | test_all_zero_weights_returns_zero                     | Pass   |
|       | spatial.py::TestIDWInterpolate                                                   | test_closer_gauge_has_higher_weight                    | Pass   |
|       | spatial.py::TestIDWInterpolate                                                   | test_empty_list_returns_zero                           | Pass   |
|       | spatial.py::TestIDWInterpolate                                                   | test_equal_distance_gauges_averages_wse                | Pass   |
|       | spatial.py::TestIDWInterpolate                                                   | test_near_gauge_returns_wse_directly                   | Pass   |
|       | spatial.py::TestIDWInterpolate                                                   | test_power_one_changes_result                          | Pass   |
|       | spatial.py::TestIDWInterpolate                                                   | test_single_far_gauge                                  | Pass   |
|       | spatial.py::TestSpatialInterpolateWSE                                            | test_at_gauge_location_returns_wse_directly            | Fails  |
|       | spatial.py::TestSpatialInterpolateWSE                                            | test_empty_gauge_data_returns_zero                     | Pass   |
|       | spatial.py::TestSpatialInterpolateWSE                                            | test_idw_returns_weighted_value                        | Pass   |
|       | spatial.py::TestSpatialInterpolateWSE                                            | test_single_gauge                                      | Pass   |
|       | test_depth_damage_coverage.py::TestCubicGaugeGoodBaptisteNegativesToSynthesize   | testCubicGaugeGoodBaptisteNegativesToSynthesize        | Fails  |
|       | test_depth_damage_coverage.py::TestDepthDamageFunctionOfCFactor                  | testDepthDamageFunctionOfCFactor                       | Pass   |
|       | test_depth_damage_coverage.py::TestDepthDamageAdjustmentCoverage                 | testDepthDamageAdjustmentCoverage                      | Pass   |
|       | test_depth_damage_coverage.py::TestDepthDamageFunctionReturnsCorrectLength       | testDepthDamageFunctionReturnsCorrectLength            | Pass   |
|       | test_depth_damage_coverage.py::TestDepthDamageFunctionOfCFactor                  | testDepthDamageFunctionOfCFactor                       | Pass   |
|       | test_depth_damage_coverage.py::TestDepthDamagePredictionInterval                 | testDepthDamagePredictionInterval                      | Pass   |
|       | test_depth_damage_coverage.py::TestDepthDamageTypeDefaultCoverage0               | testDepthDamageTypeDefaultCoverage0                    | Pass   |
|       | test_depth_damage_coverage.py::TestDepthDepthsFunctionCoverage                   | testDepthDepthsFunctionCoverage                        | Pass   |
|       | test_depth_damage_coverage.py::TestScaledDepthDamageBallback                     | testScaledDepthDamageBallback                          | Pass   |
|       | test_depth_damage_coverage.py::TestSplatDepthDamageBallback                      | testSplatDepthDamageBallback                           | Pass   |
|       | test_depth_damage_coverage.py::TestSplatDepthDamageReturnsZeros                  | testSplatDepthDamageReturnsZeros                       | Pass   |
|       | test_depth_damage_coverage.py::TestSplatDepthFlippedData1m                       | testSplatDepthFlippedData1m                            | Pass   |
|       | test_depth_damage_coverage.py::TestSplatShallowFloodDepthApplied                 | testSplatShallowFloodDepthApplied                      | Pass   |
|       | test_depth_damage_coverage.py::TestSplatHeightFlippedReturnsZeros                | testSplatHeightFlippedReturnsZeros                     | Pass   |
|       | test_depth_damage_coverage.py::TestSplatInertiaFloppinessGetsZero                | testSplatInertiaFloppinessGetsZero                     | Pass   |
|       | test_depth_damage_coverage.py::TestSplatInertiaFloppinessProducesPositiveDepth   | testSplatInertiaFloppinessProducesPositiveDepth        | Fails  |
|       | test_depth_damage_coverage.py::TestSyntheticGaugeReturnsZeros                    | testSyntheticGaugeReturnsZeros                         | Pass   |
|       | test_spatial_coverage.py::TestBuildCorrelationMatrixOneCoverage                  | testBuildCorrelationMatrixOneCoverage                  | Pass   |
|       | test_spatial_coverage.py::TestBuildCorrelationMatrixCoverageIncreasesOffDiagonal | testBuildCorrelationMatrixCoverageIncreasesOffDiagonal | Fails  |
|       | test_spatial_coverage.py::TestBuildCorrelationMatrixCoverageIncreasesCorrelation | testBuildCorrelationMatrixCoverageIncreasesCorrelation | Fails  |
|       | test_spatial_coverage.py::TestBuildCorrelationMatrixPositivity                   | testBuildCorrelationMatrixPositivity                   | Pass   |
|       | test_spatial_coverage.py::TestBuildCorrelationMatrixSquareCovariance             | testBuildCorrelationMatrixSquareCovariance             | Pass   |
|       | test_spatial_coverage.py::TestBuildCorrelationMatrixCoverage                     | testBuildCorrelationMatrixCoverage                     | Pass   |
|       | test_spatial_coverage.py::TestBuildSpatialTreeCoverageRadians                    | testBuildSpatialTreeCoverageRadians                    | Pass   |
|       | test_spatial_coverage.py::TestBuildSpatialTreeCoverageTree                       | testBuildSpatialTreeCoverageTree                       | Pass   |
|       | test_spatial_coverage.py::TestBuildSpatialTreeCoverageFrameReturnsKdtree         | testBuildSpatialTreeCoverageFrameReturnsKdtree         | Pass   |
|       | test_spatial_coverage.py::TestIDWInterpolationPolygons                           | testIDWInterpolationPolygons                           | Pass   |
|       | test_spatial_coverage.py::TestIDWInterpolationEmptyGaugesZero                    | testIDWInterpolationEmptyGaugesZero                    | Pass   |
|       | test_spatial_coverage.py::TestIDWInterpolationSingleEdgeCases                    | testIDWInterpolationSingleEdgeCases                    | Pass   |
|       | test_spatial_coverage.py::TestIDWInterpolationWeighted                           | testIDWInterpolationWeighted                           | Pass   |
|       | test_spatial_coverage.py::TestIDWInterpolationPolyGaugeReturnsDirectly           | testIDWInterpolationPolyGaugeReturnsDirectly           | Pass   |
|       | test_spatial_coverage.py::TestSpatialInterpolationWeightedLocationsReturnsWSE    | testSpatialInterpolationWeightedLocationsReturnsWSE    | Pass   |
|       | test_spatial_coverage.py::TestSpatialInterpolationWSEEdgeCasesReturnsZero        | testSpatialInterpolationWSEEdgeCasesReturnsZero        | Pass   |

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| Model | Module                                                                                  | Test Name                                                            | Status |
|-------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------|--------|
|       | cli.py::TestMain                                                                        | test_exceedance_mode                                                 | Pass   |
|       | cli.py::TestMain                                                                        | test_json_default_mode                                               | Pass   |
|       | cli.py::TestMain                                                                        | test_return_period_json_mode                                         | Pass   |
|       | cli.py::TestMain                                                                        | test_return_period_mode                                              | Pass   |
|       | cli.py::TestMain                                                                        | test_samples_json_mode                                               | Pass   |
|       | cli.py::TestMain                                                                        | test_samples_mode                                                    | Pass   |
|       | cli.py::TestMain                                                                        | test_scenario_mode                                                   | Pass   |
|       | cli.py::TestMain                                                                        | test_seed_reproducible                                               | Pass   |
|       | prs_hazard_curve.py::TestCreateFlatHazardCurve                                          | test_flat_hazard_curve                                               | Pass   |
|       | prs_hazard_curve.py::TestCreateFlatHazardCurveAndQuote                                  | test_flat_hazard_curve_and_quote                                     | Pass   |
|       | prs_hazard_curve.py::TestCreateSensitivityFlatHazard                                    | test_sensitivity_flat_hazard                                         | Pass   |
|       | prs_hazard_curve.py::TestCreateSensitivityFlatHazardCounter                             | test_sensitivity_flat_hazard_counter                                 | Pass   |
|       | prs_hazard_curve.py::TestCreateSensitivityClusterFromHazardStart                        | test_sensitivity_cluster_from_hazard_start                           | Pass   |
|       | prs_hazard_curve.py::TestCreateSensitivityCurvesFromHazard                              | test_sensitivity_curves_from_hazard                                  | Pass   |
|       | prs_hazard_curve.py::TestCreateSensitivitySurvivalFrequencyHazardUsesFallbackHazardRate | test_sensitivity_survival_frequency_hazard_uses_fallback_hazard_rate | Pass   |
|       | prs_hazard_curve.py::TestCreateSensitivitySurvivalFrequencyHazardUsesFallbackHazardRate | test_sensitivity_survival_frequency_hazard_uses_fallback_hazard_rate | Pass   |
|       | prs_hazard_curve.py::TestLoadHazardCodesDataHasGauge                                    | test_load_hazard_codes_data_has_gauge                                | Pass   |
|       | prs_hazard_curve.py::TestLoadHazardCodesFromFile                                        | test_load_hazard_codes_from_file                                     | Pass   |
|       | prs_hazard_curve.py::TestPricePointAlertTrigger                                         | test_alert_trigger                                                   | Pass   |
|       | prs_hazard_curve.py::TestPricePointCustomNotional                                       | test_custom_notional                                                 | Pass   |
|       | prs_hazard_curve.py::TestPricePointFlatHazardCurve                                      | test_flat_hazard_curve                                               | Pass   |
|       | prs_hazard_curve.py::TestPricePointResultHasFairSpreadBps                               | test_result_has_fair_spread_bps                                      | Pass   |
|       | prs_hazard_curve.py::TestPricePointResultHasNPV                                         | test_result_has_npv                                                  | Pass   |
|       | prs_hazard_curve.py::TestPricePointResultHasSurvivalProbabilities                       | test_result_has_survival_probabilities                               | Pass   |
|       | prs_hazard_curve.py::TestPricePointReturnsDict                                          | test_returns_dict                                                    | Pass   |
|       | prs_hazard_curve.py::TestPricePointSevereTrigger                                        | test_severe_trigger                                                  | Pass   |
|       | prs_hazard_curve.py::TestPricePointTermStructureTrue                                    | test_term_structure_true                                             | Pass   |
|       | prs_hazard_curve.py::TestPrintPricePointResultsGaugeName                                | test_print_price_point_results_gauge_name                            | Pass   |
|       | prs_hazard_curve.py::TestPrintPricePointResultsWithoutError                             | test_print_price_point_results_without_error                         | Pass   |
|       | test_load_and_curves.py::TestCreateFlatHazard360DayAdditionalAccepted                   | test_flat_hazard_360_day_additional_accepted                         | Pass   |
|       | test_load_and_curves.py::TestCreateFlatHazard360DayAdditional                           | test_flat_hazard_360_day_additional                                  | Pass   |
|       | test_load_and_curves.py::TestCreateFlatHazardCurveAdditionalInput                       | test_flat_hazard_curve_additional_input                              | Pass   |
|       | test_load_and_curves.py::TestCreateFlatHazardCurveAdditionalFormula                     | test_flat_hazard_curve_additional_formula                            | Pass   |
|       | test_load_and_curves.py::TestCreateFlatHigh6HazardRateNearZeroSurvival                  | test_flat_high_6_hazard_rate_near_zero_survival                      | Pass   |
|       | test_load_and_curves.py::TestCreateSurvival360DayAdditionalProducesCurve                | test_survival_360_day_additional_produces_curve                      | Pass   |
|       | test_load_and_curves.py::TestCreateSimplifiedHazardAdditionalFormula                    | test_simplified_hazard_additional_formula                            | Pass   |
|       | test_load_and_curves.py::TestCreateSimplifiedHazardAdditionalPositiveHazard             | test_simplified_hazard_additional_positive_hazard                    | Pass   |
|       | test_load_and_curves.py::TestCreateSurvival6PointAdditionalRiskTs                       | test_survival_6_point_additional_risk_ts                             | Pass   |
|       | test_load_and_curves.py::TestCreateSurvival6PointAdditionalStructure                    | test_survival_6_point_additional_structure                           | Pass   |
|       | test_load_and_curves.py::TestCreateSurvivalCurveAdditionalPositive                      | test_survival_curve_additional_positive                              | Pass   |
|       | test_load_and_curves.py::TestLoadHazardPyjssFromObjectReturnsEmptyDict                  | test_load_hazard_pyjss_from_object_returns_empty_dict                | Pass   |
|       | test_load_and_curves.py::TestLoadHazardPyjssFromRaisesJsonDecodeError                   | test_load_hazard_pyjss_from_raises_json_decode_error                 | Pass   |
|       | test_load_and_curves.py::TestLoadHazardPyjssFromRaisesFileNotFound                      | test_load_hazard_pyjss_from_raises_file_not_found                    | Pass   |
|       | test_load_and_curves.py::TestLoadHazardPyjssFromRaisesFileStructure                     | test_load_hazard_pyjss_from_raises_file_structure                    | Pass   |

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| Model | Module                                      | Test Name                                    | Status |
|-------|---------------------------------------------|----------------------------------------------|--------|
|       | test_price_prs.py::TestPricePrsGauge        | test_annual_hazard_rate_matches_input        | Pass   |
|       | test_price_prs.py::TestPricePrsGauge        | test_gauge_in_result                         | Pass   |
|       | test_price_prs.py::TestPricePrsGauge        | test_gauge_name_in_result                    | Pass   |
|       | test_price_prs.py::TestPricePrsGauge        | test_severe_annual_hazard_rate_matches_input | Pass   |
|       | test_price_prs.py::TestPricePrsLegSignp     | test_signp_is_sum_of_legs                    | Pass   |
|       | test_price_prs.py::TestPricePrsLegSignp     | test_signpremium_leg_negative                | Pass   |
|       | test_price_prs.py::TestPricePrsLegSignp     | test_signprotection_leg_positive             | Pass   |
|       | test_price_prs.py::TestPricePrsRecovery     | test_zero_recovery_rate_stored               | Pass   |
|       | test_price_prs.py::TestPricePrsRecovery     | test_recovery_rate_affects_fair_spread       | Pass   |
|       | test_price_prs.py::TestPricePrsRiskFreeRate | test_right_risk_free_rate_changes_npv        | Pass   |
|       | test_price_prs.py::TestPricePrsRiskFreeRate | test_left_risk_free_rate_not_in_output       | Pass   |
|       | test_price_prs.py::TestPricePrsType         | test_fair_spread_is_float                    | Pass   |
|       | test_price_prs.py::TestPricePrsType         | test_fair_upfront_is_float                   | Pass   |
|       | test_price_prs.py::TestPricePrsType         | test_npv_is_float                            | Pass   |
|       | test_price_prs.py::TestPricePrsType         | test_running_spread_bps_identity             | Pass   |
|       | test_price_prs.py::TestPricePrsType         | test_survival_probs_all_in_zero_one          | Pass   |
|       | test_price_prs.py::TestPricePrsType         | test_tenor_1yr_gives_one_survival_entry      | Pass   |
|       | test_price_prs.py::TestPricePrsType         | test_use_term_structure_stored_false         | Pass   |
|       | test_price_prs.py::TestPricePrsType         | test_use_term_structure_stored_true          | Pass   |
|       | test_print_and_cli.py::TestMain             | test_custom_notional_and_spread              | Pass   |
|       | test_print_and_cli.py::TestMain             | test_default_gauge_selection_returns_zero    | Pass   |
|       | test_print_and_cli.py::TestMain             | test_explicit_gauge_returns_zero             | Pass   |
|       | test_print_and_cli.py::TestMain             | test_flat_flag_dispatches_flat_mode          | Pass   |
|       | test_print_and_cli.py::TestMain             | test_list_flag_logs_gauge_names              | Pass   |
|       | test_print_and_cli.py::TestMain             | test_list_flag_returns_zero                  | Pass   |
|       | test_print_and_cli.py::TestMain             | test_missing_gauge_logs_available_gauge      | Pass   |
|       | test_print_and_cli.py::TestMain             | test_missing_gauge_logs_error                | Pass   |
|       | test_print_and_cli.py::TestMain             | test_missing_gauge_returns_one               | Pass   |
|       | test_print_and_cli.py::TestMain             | test_trigger_alert_explicit                  | Pass   |
|       | test_print_and_cli.py::TestMain             | test_trigger_severe_explicit                 | Pass   |
|       | test_print_and_cli.py::TestMain             | test_trigger_warning_explicit                | Pass   |
|       | test_print_and_cli.py::TestPrint            | test_alerts_trigger_in_log                   | Pass   |
|       | test_print_and_cli.py::TestPrint            | test_results_corporate_years_in_log          | Pass   |
|       | test_print_and_cli.py::TestPrint            | test_structured_terms_header_in_log          | Pass   |
|       | test_print_and_cli.py::TestPrint            | test_pain_points_in_log                      | Pass   |
|       | test_print_and_cli.py::TestPrint            | test_gauges_in_log                           | Pass   |
|       | test_print_and_cli.py::TestPrint            | test_gauges_name_in_log                      | Pass   |
|       | test_print_and_cli.py::TestPrint            | test_results_corporate_in_log                | Pass   |
|       | test_print_and_cli.py::TestPrint            | test_premium_in_log                          | Pass   |
|       | test_print_and_cli.py::TestPrint            | test_protection_and_in_log                   | Pass   |
|       | test_print_and_cli.py::TestPrint            | test_resolve_corporate_in_log                | Pass   |
|       | test_print_and_cli.py::TestPrint            | test_severe_trigger_in_log                   | Pass   |
|       | test_print_and_cli.py::TestPrint            | test_results_corporate_label_in_log          | Pass   |
|       | test_print_and_cli.py::TestPrint            | test_triggers_in_log                         | Pass   |

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| Model | Module                                            | Test Name                                | Status |
|-------|---------------------------------------------------|------------------------------------------|--------|
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore100                 | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore100                 | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScoreReturns_float       | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore.concrete_lower     | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore.steel_lowest       | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore.timber_higher      | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore.wood_equals_timber | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore.multiplier         | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore100_multiplier      | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore105_multiplier      | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore_multiplier         | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore.construction_type  | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore.flood_level        | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore.flood_capped       | Pass   |
|       | test_risk_assessor_core.py::TestConstantRiskScore | testConstantRiskScore.flood_lowest_score | Pass   |
|       | test_risk_assessor_core.py::TestFloodRisk         | testFloodRiskHigh_threshold              | Pass   |
|       | test_risk_assessor_core.py::TestFloodRisk         | testFloodRiskLow_threshold               | Pass   |
|       | test_risk_assessor_core.py::TestFloodRisk         | testFloodRiskMedium_threshold            | Pass   |
|       | test_risk_assessor_core.py::TestFloodRisk         | testFloodRiskBelow_low_threshold         | Pass   |
|       | test_risk_assessor_core.py::TestFloodRisk         | testFloodRiskIn_range                    | Pass   |
|       | test_risk_assessor_core.py::TestFloodRisk         | testFloodRiskIn_range                    | Pass   |
|       | test_risk_assessor_core.py::TestFloodRisk         | testFloodRiskNot_in_range_with_depth     | Pass   |
|       | test_risk_assessor_core.py::TestFloodRisk         | testFloodRiskNegative_returns_unknown    | Pass   |
|       | test_risk_assessor_core.py::TestFloodRisk         | testFloodRiskZero_returns_unknown        | Pass   |
|       | test_risk_assessor_core.py::TestFloodRisk         | testFloodRiskHigh                        | Pass   |
|       | test_risk_assessor_core.py::TestFloodRisk         | testFloodRiskVery_low                    | Pass   |
|       | test_risk_assessor_core.py::TestLTVRisk           | testLTVRiskHigh_boundary                 | Pass   |
|       | test_risk_assessor_core.py::TestLTVRisk           | testLTVRiskLow_boundary                  | Pass   |
|       | test_risk_assessor_core.py::TestLTVRisk           | testLTVRiskModerate_boundary             | Pass   |
|       | test_risk_assessor_core.py::TestLTVRisk           | testLTVRiskCritical_ltv                  | Pass   |
|       | test_risk_assessor_core.py::TestLTVRisk           | testLTVRiskHigh_ltv                      | Pass   |
|       | test_risk_assessor_core.py::TestLTVRisk           | testLTVRiskLow_ltv                       | Pass   |
|       | test_risk_assessor_core.py::TestLTVRisk           | testLTVRiskModerate_ltv                  | Pass   |
|       | test_risk_assessor_core.py::TestLTVRisk           | testLTVRiskNone_returns_unknown          | Pass   |
|       | test_risk_assessor_core.py::TestLTVRisk           | testLTVRiskNormalises_95_pct             | Pass   |
|       | test_risk_assessor_core.py::TestLTVRisk           | testLTVRiskNormalises_percentage_input   | Pass   |
|       | test_risk_assessor_core.py::TestMessageRisk       | testMessageRisk_ltv_medium_flood         | Pass   |
|       | test_risk_assessor_core.py::TestMessageRisk       | testMessageRisk_flood_overrides          | Pass   |
|       | test_risk_assessor_core.py::TestMessageRisk       | testMessageRisk_ltv_medium_flood         | Pass   |
|       | test_risk_assessor_core.py::TestMessageRisk       | testMessageRisk_flood_risk               | Pass   |
|       | test_risk_assessor_core.py::TestMessageRisk       | testMessageRisk_flood_default            | Pass   |
|       | test_risk_assessor_core.py::TestMessageRisk       | testMessageRisk_l_risk                   | Pass   |
|       | test_risk_assessor_core.py::TestMessageRisk       | testMessageRisk_value_critical           | Pass   |
|       | test_risk_assessor_core.py::TestMessageRisk       | testMessageRisk_value_high               | Pass   |

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| Model | Module                                        | Test Name                                    | Status |
|-------|-----------------------------------------------|----------------------------------------------|--------|
|       | data_structures.py::TestGaugeRequest          | test_is_bool                                 | Pass   |
|       | data_structures.py::TestGaugeRequest          | test_has_required_keys                       | Pass   |
|       | data_structures.py::TestGaugeRequest          | test_is_timeseries_by_default                | Pass   |
|       | data_structures.py::TestGaugeRequest          | test_returns_dict                            | Pass   |
|       | data_structures.py::TestGaugeRequest          | test_value_is_rounded                        | Pass   |
|       | data_structures.py::TestIntensityProfile      | test_beta_value                              | Pass   |
|       | data_structures.py::TestIntensityProfile      | test_gaussian_value                          | Pass   |
|       | data_structures.py::TestIntensityProfile      | test_triangular_value                        | Pass   |
|       | data_structures.py::TestStormTotal            | test_custom_kernel                           | Pass   |
|       | data_structures.py::TestStormTotal            | test_decay_kernel_serialised_as_string       | Pass   |
|       | data_structures.py::TestStormTotal            | test_has_storm_id                            | Pass   |
|       | data_structures.py::TestStormTotal            | test_has_track                               | Pass   |
|       | data_structures.py::TestStormTotal            | test_intensity_profile_serialised_as_string  | Pass   |
|       | data_structures.py::TestStormTotal            | test_returns_dict                            | Pass   |
|       | data_structures.py::TestStormTotal            | test_track_has_coords                        | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_auto_generated_id_starts_with_storm     | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_decay_kernel_stored                     | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_decay_parameter_stored                  | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_default_num_track_points                | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_default_start_time_is_recent            | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_duration_hours_stored                   | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_explicit_id_preserved                   | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_explicit_name                           | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_explicit_start_time                     | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_footprint_km_stored                     | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_num_track_points                        | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_peak_intensity_stored                   | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_returns_storm_object                    | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_speed_is_positive                       | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_track_end_time_equals_duration          | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_track_longitude_moves_from_start_to_end | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_track_points_are_trackpoint_instances   | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_track_start_time_is_zero                | Pass   |
|       | storm_factory.py::TestCreateStorm             | test_track_time_increases                    | Pass   |
|       | storm_factory.py::TestIntensityProfile        | test_all_profiles_produce_valid_storm        | Pass   |
|       | storm_factory.py::TestIntensityProfile        | test_beta_profile_creates_storm              | Pass   |
|       | storm_factory.py::TestIntensityProfile        | test_gaussian_profile_peak_near_midpoint     | Pass   |
|       | storm_factory.py::TestIntensityProfile        | test_triangular_max_equals_peak_intensity    | Pass   |
|       | storm_factory.py::TestIntensityProfile        | test_triangular_profile_peak_near_midpoint   | Pass   |
|       | storm_factory.py::TestLoadGaugeFromPortfolio  | test_empty_portfolio_returns_empty_list      | Pass   |
|       | storm_factory.py::TestLoadGaugeFromPortfolio  | test_loads_gauge_from_portfolio              | Pass   |
|       | flood_classifier.py::TestFloodClassifierTrain | test_gauge_creates_joblib_file               | Pass   |
|       | flood_classifier.py::TestFloodClassifierTrain | test_gauge_has_metrics                       | Pass   |
|       | flood_classifier.py::TestFloodClassifierTrain | test_gauge_returns_dict                      | Pass   |

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| Model | Module                                           | Test Name                                     | Status |
|-------|--------------------------------------------------|-----------------------------------------------|--------|
|       | flood_classifier.py::TestFloodClassifierTrain    | test_train_gauge_single_class_skipped         | Pass   |
|       | flood_classifier.py::TestFloodClassifierTrain    | test_train_gauge_status_trained               | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_travavailable_gauges                     | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_travavailable_gauges_empty_dir           | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_load_summary                             | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_load_summary_cached                      | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_predict_caches_model                     | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_predict_high_level_higher_probability    | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_predict_in_range                         | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_predict_missing_model_raises             | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_predict_returns_float                    | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_predict_series_all_in_range              | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_predict_series_default_derivatives       | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_predict_series_returns_list              | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_predict_series_with_derivatives          | Pass   |
|       | flood_classifier.py::TestFloodPredictor          | test_predict_surface_returns_grid             | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_all_levels_positive                      | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_increases_length_approximately           | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_dates_are_sequential                     | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_different_seeds_different_output         | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_each_obs_has_date_and_level              | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_instant_gauge_data_uses_defaults         | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_historical_high_date_in_range            | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_historical_high_date_invalid_format      | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_historical_high_date_out_of_range        | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_historical_high_injected_when_not_placed | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_missing_flood_stages_uses_defaults       | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_partial_tidal_gauge_produces_output      | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_returns_list                             | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_series_reproducible                      | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_tidal_gauge_produces_output              | Pass   |
|       | synthetic.py::TestGenerateSyntheticTimeSeries    | test_tidal_spring_neap_modulation             | Pass   |
|       | test_hazard_pricing.py::TestCreateEngine         | test_create_engine_one                        | Pass   |
|       | test_hazard_pricing.py::TestCreateEngines        | test_create_engines_with_engines              | Pass   |
|       | test_hazard_pricing.py::TestCreateHazardData     | test_create_hazard_data_approach              | Pass   |
|       | test_hazard_pricing.py::TestCreateHazardData     | test_create_hazard_data_probability_decreases | Pass   |
|       | test_hazard_pricing.py::TestCreateHazardData     | test_create_hazard_data_approach_survival     | Pass   |
|       | test_hazard_pricing.py::TestHazardPointsToGrid   | test_points_to_grid_count                     | Pass   |
|       | test_hazard_pricing.py::TestHazardPointsToGrid   | test_points_to_grid_gives_low_survival        | Pass   |
|       | test_hazard_pricing.py::TestHazardPointsToGrid   | test_points_to_grid_credit_curve              | Pass   |
|       | test_hazard_pricing.py::TestHazardPointsToGrid   | test_points_to_grid_probability_decreases     | Pass   |
|       | test_hazard_pricing.py::TestHazardPointsToGrid   | test_points_to_grid_gives_survival_near_one   | Pass   |
|       | test_engine_coverage.py::TestPropertyBasis       | test_property_basis_trade_has_basis_info      | Pass   |
|       | test_engine_coverage.py::TestRemainderByEdgeCase | test_remainder_by_edge_case_returns_5         | Pass   |

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| Model | Module                                       | Test Name                                      | Status |
|-------|----------------------------------------------|------------------------------------------------|--------|
|       | insurance.py::TestGetAreaPremiumFactorsRange | test_factor_range_tuple                        | Pass   |
|       | insurance.py::TestGetAreaPremiumFactorsRange | test_factor_range_first_band                   | Pass   |
|       | insurance.py::TestInsuranceFactorsTable      | test_base_premium_factors_all_positive         | Pass   |
|       | insurance.py::TestInsuranceFactorsTable      | test_base_area_bands_are_sorted                | Pass   |
|       | insurance.py::TestInsuranceFactorsTable      | test_base_rate_range                           | Pass   |
|       | insurance.py::TestInsuranceFactorsTable      | test_construction_type_factors_present         | Pass   |
|       | insurance.py::TestInsuranceFactorsTable      | test_flood_risk_factors_increase_with_risk     | Pass   |
|       | insurance.py::TestInsuranceFactorsTable      | test_flood_risk_factors_present                | Pass   |
|       | insurance.py::TestInsuranceFactorsTable      | test_min_max_premium_bounds                    | Pass   |
|       | insurance.py::TestInsuranceFactorsTable      | test_property_type_factors_present             | Pass   |
|       | property_value.py::TestApplyValueFactor      | test_factor_applied                            | Pass   |
|       | property_value.py::TestApplyValueFactor      | test_factor_calculation                        | Pass   |
|       | property_value.py::TestApplyValueFactor      | test_factor_clamped_to_max                     | Pass   |
|       | property_value.py::TestApplyValueFactor      | test_factor_clamped_to_min                     | Pass   |
|       | property_value.py::TestApplyValueFactor      | test_factor_risk_discount_reduces_value        | Pass   |
|       | property_value.py::TestApplyValueFactor      | test_factor_applied                            | Pass   |
|       | property_value.py::TestApplyValueFactor      | test_factor_returns_float                      | Pass   |
|       | property_value.py::TestFactorTable           | test_age_band_factors_within_reasonable_bounds | Pass   |
|       | property_value.py::TestFactorTable           | test_base_area_ranges_all_positive             | Pass   |
|       | property_value.py::TestFactorTable           | test_base_price_per_sqm_all_positive           | Pass   |
|       | property_value.py::TestFactorTable           | test_condition_factors_ordered_by_severity     | Pass   |
|       | property_value.py::TestFactorTable           | test_epc_factors_a_highest                     | Pass   |
|       | property_value.py::TestFactorTable           | test_flood_risk_factors_decrease_with_risk     | Pass   |
|       | property_value.py::TestFactorTable           | test_min_max_value_bounds                      | Pass   |
|       | property_value.py::TestFactorTable           | test_proximity_zones_have_range_key            | Pass   |
|       | property_value.py::TestFactorTable           | test_rent_per_sqm_all_present                  | Pass   |
|       | property_value.py::TestFactorTable           | test_rental_yield_rates_all_present            | Pass   |
|       | property_value.py::TestGetAgeBand            | test_100_years_is_heritage                     | Pass   |
|       | property_value.py::TestGetAgeBand            | test_10_years_is_modern                        | Pass   |
|       | property_value.py::TestGetAgeBand            | test_25_years_is_established                   | Pass   |
|       | property_value.py::TestGetAgeBand            | test_50_years_is_period                        | Pass   |
|       | property_value.py::TestGetAgeBand            | test_default_current_year                      | Pass   |
|       | property_value.py::TestGetAgeBand            | test_established                               | Pass   |
|       | property_value.py::TestGetAgeBand            | test_exactly_24_years_is_modern                | Pass   |
|       | property_value.py::TestGetAgeBand            | test_exactly_49_years_is_established           | Pass   |
|       | property_value.py::TestGetAgeBand            | test_exactly_99_years_is_period                | Pass   |
|       | property_value.py::TestGetAgeBand            | test_exactly_9_years_is_new_build              | Pass   |
|       | property_value.py::TestGetAgeBand            | test_heritage_old                              | Pass   |
|       | property_value.py::TestGetAgeBand            | test_modern                                    | Pass   |
|       | property_value.py::TestGetAgeBand            | test_new_build                                 | Pass   |
|       | property_value.py::TestGetAgeBand            | test_period                                    | Pass   |
|       | property_value.py::TestGetProximityZone      | test_zone_high_risk_high                       | Pass   |
|       | property_value.py::TestGetProximityZone      | test_zone_high_risk_medium                     | Pass   |
|       | property_value.py::TestGetProximityZone      | test_zone_high_risk_very_high                  | Pass   |

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| Model | Module                                  | Test Name                                               | Status         |
|-------|-----------------------------------------|---------------------------------------------------------|----------------|
|       | property_value.py::TestGetProximityZone | testZone_low_risk_low                                   | Pass           |
|       | property_value.py::TestGetProximityZone | testZone_low_risk_very_low                              | Pass           |
|       | property_value.py::TestGetProximityZone | testZone_exactly_200m_is_medium_zone                    | Pass           |
|       | property_value.py::TestGetProximityZone | testZone_exactly_500m_is_far                            | Pass           |
|       | property_value.py::TestGetProximityZone | testZone_far_zone                                       | Pass           |
|       | property_value.py::TestGetProximityZone | testZone_medium_zone                                    | Pass           |
|       | property_value.py::TestGetProximityZone | testZone_medium_zone_upper_bound                        | Pass           |
|       | display.py::TestBlotterTradeExistence   | testblotter_glob_pattern_matches_prs_id_protocol        | Pass           |
|       | display.py::TestBlotterTradeExistence   | testblotter_returns_trades_with_fixture                 | Pass           |
|       | display.py::TestStormDisplayFormat      | test_all_files_include_storm_id_in_label                | Pass           |
|       | display.py::TestStormDisplayFormat      | test_all_files_use_sentinel_or_pipe_separator           | Pass           |
|       | display.py::TestStormDisplayFormat      | test_canonical_format_defined_in_config                 | Format         |
|       | display.py::TestStormDisplayFormat      | test_no_file_uses_old_precipitation_label               | Pass           |
|       | display.py::TestStormDisplayFormat      | test_no_file_uses_old_severe_levels_format              | Pass           |
|       | display.py::TestStormDisplayFormat      | test_port_stress_uses_sentinel_with_warning             | Pass           |
|       | display.py::TestStormDisplayFormat      | test_sp_table_uses_sentinel_pattern                     | Pass           |
|       | display.py::TestStormDisplayFormat      | test_storm_option_js_custom_var                         | Pass           |
|       | display.py::TestStormDisplayFormat      | test_storm_option_js_peak_variant                       | Pass           |
|       | display.py::TestStormDisplayFormat      | test_storm_option_js_warning_variant                    | Pass           |
|       | display.py::TestStormDisplayFormat      | test_trading_stress_uses_sentinel_with_peak             | Peak           |
|       | display.py::TestStormDisplayFormat      | test_visual_files_use_sentinel_not_hardcoded_format     | Code           |
|       | fixtures.py::TestProtocolUniformities   | test_all_entity_suffixes_have_same_length               | Pass           |
|       | fixtures.py::TestProtocolUniformities   | test_all_uuid4_suffixes_produce_lowercased_hex          | Pass           |
|       | fixtures.py::TestProtocolUniformities   | test_gauge_prop_mort_storm_share_same_suffix_derivation | Prefix         |
|       | fixtures.py::TestTestFixtureGaugeIDs    | testbackward_compat_gauge_ids_are_documented            | Doc            |
|       | fixtures.py::TestTestFixtureGaugeIDs    | testgauge_json_and_gaugehc_ids_are_consistent           | Pass           |
|       | fixtures.py::TestTestFixtureGaugeIDs    | testgaugehc_new_entries_follow_protocol                 | Pass           |
|       | fixtures.py::TestTestFixtureGaugeIDs    | testnew_conf_test_gauge_ids_match_protocol              | Pass           |
|       | fixtures.py::TestTestFixtureGaugeIDs    | teststress_storm_gauge_ids_are_in_known_set             | Set            |
|       | fixtures.py::TestTestFixtureStormIDs    | testport_stress_routes_storm_constants_match_protocol   | Match protocol |
|       | fixtures.py::TestTestFixtureStormIDs    | testsample_port_stress_storms_ids_match_protocol        | Pass           |
|       | fixtures.py::TestTestFixtureStormIDs    | testsample_stress_storms_storm_ids_match_protocol       | Pass           |
|       | fixtures.py::TestTestFixtureStormIDs    | teststress_routes_storm_constants_match_protocol        | Pass           |
|       | generators.py::TestGaugeIDGenerator     | testgenerator_does_not_use_sequential_format            | Format         |
|       | generators.py::TestGaugeIDGenerator     | testgenerator_source_uses_gauge_prefix                  | Pass           |
|       | generators.py::TestGaugeIDGenerator     | testgenerator_source_uses_uuid                          | Pass           |
|       | generators.py::TestGaugeIDGenerator     | testuuid4_produces_protocol_matching_ids                | Pass           |
|       | generators.py::TestMortgageIDGenerator  | testgenerator_does_not_use_sequential_format            | Format         |
|       | generators.py::TestMortgageIDGenerator  | testgenerator_source_uses_mort_prefix                   | Pass           |
|       | generators.py::TestMortgageIDGenerator  | testgenerator_source_uses_uuid                          | Pass           |
|       | generators.py::TestMortgageIDGenerator  | testuuid4_produces_protocol_matching_ids                | Pass           |
|       | generators.py::TestPRSTradeIDGenerator  | testthames_uses_prs_prefix                              | Pass           |
|       | generators.py::TestPRSTradeIDGenerator  | testthames_uses_uuid                                    | Pass           |
|       | generators.py::TestPRSTradeIDGenerator  | testuuid4_hex_upper_produces_protocol_matching_ids      | Pass           |

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| Model | Module                                                                           | Test Name                                                   | Status |
|-------|----------------------------------------------------------------------------------|-------------------------------------------------------------|--------|
|       | generators.py::TestPropertyIDGenerator                                           | test_generator_does_not_use_sequential_format               | Fail   |
|       | generators.py::TestPropertyIDGenerator                                           | test_generator_source_uses_prop_prefix                      | Pass   |
|       | generators.py::TestPropertyIDGenerator                                           | test_generator_source_uses_uuid                             | Pass   |
|       | generators.py::TestPropertyIDGenerator                                           | test_uuid4_produces_protocol_matching_id                    | Pass   |
|       | generators.py::TestStormIDGenerator                                              | test_multi_storm_make_storm_id_follows_protocol             | Pass   |
|       | generators.py::TestStormIDGenerator                                              | test_single_storm_generator_does_not_use_sequential_format  | Pass   |
|       | generators.py::TestStormIDGenerator                                              | test_single_storm_generator_uses_storm_prefix               | Pass   |
|       | generators.py::TestStormIDGenerator                                              | test_single_storm_generator_uses_uuid                       | Pass   |
|       | generators.py::TestStormIDGenerator                                              | test_uuid4_produces_protocol_matching_id                    | Pass   |
|       | patterns.py::TestGaugeIDPattern                                                  | test_accepts_valid_uuid_derived_ids                         | Pass   |
|       | patterns.py::TestGaugeIDPattern                                                  | test_rejects_non_hex_chars                                  | Pass   |
|       | patterns.py::TestGaugeIDPattern                                                  | test_rejects_sequential_integers                            | Pass   |
|       | patterns.py::TestGaugeIDPattern                                                  | test_rejects_uppercase_hex                                  | Pass   |
|       | patterns.py::TestGaugeIDPattern                                                  | test_rejects_wrong_length                                   | Pass   |
|       | patterns.py::TestMortgageIDPattern                                               | test_accepts_valid                                          | Pass   |
|       | patterns.py::TestMortgageIDPattern                                               | test_rejects_sequential                                     | Pass   |
|       | patterns.py::TestPRSTradeIDPattern                                               | test_accepts_lowercase                                      | Pass   |
|       | patterns.py::TestPRSTradeIDPattern                                               | test_accepts_uppercase                                      | Pass   |
|       | patterns.py::TestPRSTradeIDPattern                                               | test_rejects_non_hex_suffix                                 | Pass   |
|       | patterns.py::TestPropertyIDPattern                                               | test_accepts_valid                                          | Pass   |
|       | patterns.py::TestPropertyIDPattern                                               | test_rejects_sequential                                     | Pass   |
|       | patterns.py::TestPropertyIDPattern                                               | test_rejects_uppercase                                      | Pass   |
|       | patterns.py::TestStormIDPattern                                                  | test_accepts_valid                                          | Pass   |
|       | patterns.py::TestStormIDPattern                                                  | test_rejects_descriptive_labels                             | Pass   |
|       | patterns.py::TestStormIDPattern                                                  | test_rejects_sequential_integers                            | Pass   |
|       | patterns.py::TestStormIDPattern                                                  | test_rejects_uppercase                                      | Pass   |
|       | patterns.py::TestStormIDPattern                                                  | test_rejects_wrong_prefix_format                            | Pass   |
|       | test_book_common_coverage_b6.py::TestBulkIDPatternValidating                     | test_bulk_id_pattern_validating_logs_warning                | Pass   |
|       | test_book_common_coverage_b6.py::TestEmptyCounterpartyFileUsesFallback           | test_empty_counterparty_file_uses_fallback                  | Fail   |
|       | test_book_common_coverage_b6.py::TestHeaderCounterpartyFileUsesFallback          | test_header_counterparty_file_uses_fallback                 | Pass   |
|       | test_book_common_coverage_b6.py::TestShouldConfigureFallbackWithData             | test_should_configure_fallback_with_data                    | Pass   |
|       | test_book_coverage.py::TestEmptyReturnsRaisesValueError                          | test_empty_returns_raises_value_error                       | Pass   |
|       | test_book_coverage.py::TestFallbackEmptyCounterpartyFileUsesFallback             | test_fallback_empty_counterparty_file_uses_fallback         | Fail   |
|       | test_book_coverage.py::TestFallbackMissingCounterpartyFileUsesFallback           | test_fallback_missing_counterparty_file_uses_fallback       | Fail   |
|       | test_book_coverage.py::TestGenerateTradePairFailureReturnsEmptyList              | test_trade_pair_failure_returns_empty_list                  | Pass   |
|       | test_book_coverage.py::TestGenerateTradePairFailureReturnsSuccessFullOnly        | test_trade_pair_failure_returns_success_full_only           | Fail   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMappingFieldsExtracted                 | test_cbm_create_mapping_fields_extracted                    | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMappingFieldsPresent                   | test_cbm_create_mapping_fields_present                      | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMappingFieldsDict                      | test_cbm_create_mapping_fields_dict                         | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMappingFieldsFlatDict                  | test_cbm_create_mapping_fields_flat_dict                    | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMappingFieldsValueError                | test_cbm_create_mapping_fields_value_error                  | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMappingFieldsExcluded                  | test_cbm_create_mapping_fields_excluded                     | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMappingFieldsRequiredFieldsReturnsList | test_cbm_create_mapping_fields_required_fields_returns_list | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMappingFieldsReturnsDict               | test_cbm_create_mapping_fields_returns_dict                 | Pass   |

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| Model | Module                                       | Test Name                                                      | Status |
|-------|----------------------------------------------|----------------------------------------------------------------|--------|
|       | ctpy_mapping.py::TestCounterparty            | test_CDMV_validate_counterparties_adds_error                   | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMV_validate_returns_errors                              | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMV_validate_counterparty_role_adds_error                | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMV_validate_counterparty_role_adds_error                | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMV_validate_party_id_adds_error                         | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMV_validate_party_name_adds_error                       | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMV_validate_party_ref_adds_error                        | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMV_validate_party_role_adds_error                       | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMV_validate_no_errors                                   | Pass   |
|       | class_attrs.py::TestClassAttribute           | test_metadata_categories_contains_expected_keys                | Pass   |
|       | class_attrs.py::TestClassAttribute           | test_metadata_categories_station_has_required_subkeys          | Pass   |
|       | class_attrs.py::TestClassAttribute           | test_schema_version_is_string                                  | Pass   |
|       | class_attrs.py::TestGaugeHistoricalData      | test_daily_available_flag_is_bool                              | Pass   |
|       | directory_main.py::TestMain                  | test_main_skips_bad_file_without_raising                       | Pass   |
|       | directory_main.py::TestMain                  | test_main_with_csv_files_runs_without_error                    | Pass   |
|       | directory_main.py::TestMain                  | test_main_with_no_csv_files_runs_without_error                 | Pass   |
|       | filter.py::TestFilterByYears                 | test_empty_flows_returns_empty                                 | Pass   |
|       | filter.py::TestFilterByYears                 | test_filter_1_year_keeps_only_recent                           | Pass   |
|       | filter.py::TestFilterByYears                 | test_filter_keeps_recent                                       | Pass   |
|       | filter.py::TestFilterByYears                 | test_none_years_returns_all                                    | Pass   |
|       | filter.py::TestFilterByYearsBoundary         | test_all_dates_within_range_kept                               | Pass   |
|       | filter.py::TestFilterByYearsBoundary         | test_zero_years_returns_only_most_recent_year                  | Pass   |
|       | generate.py::TestGenerate                    | test_daily_flows_filtered_by_years                             | Pass   |
|       | generate.py::TestGenerate                    | test_explicit_gauge_id_overrides_metadata                      | Pass   |
|       | generate.py::TestGenerate                    | test_output_file_written                                       | Pass   |
|       | generate.py::TestGenerate                    | test_returns_dict_with_required_keys                           | Pass   |
|       | generate.py::TestGenerateUsesInstanceDefault | test_instance_default_years_overrides_instance_default         | Pass   |
|       | generate.py::TestGenerateUsesInstanceDefault | test_instance_default_years_used_when_no_years_provided        | Pass   |
|       | generate.py::TestGenerateWithNoOutputPath    | test_generate_with_no_output_path_none_uses_config_input_dir   | Pass   |
|       | metadata.py::TestGetNrfaMetadataFor          | test_for_returns_expected_keys                                 | Pass   |
|       | metadata.py::TestGetNrfaMetadataFor          | test_for_formats_data_quality_notes                            | Pass   |
|       | metadata.py::TestGetNrfaMetadataFor          | test_for_checks_expected_keys_present                          | Pass   |
|       | metadata.py::TestGetNrfaMetadataFor          | test_for_checks_missing_keys_always_none                       | Pass   |
|       | metadata.py::TestGetNrfaMetadataFor          | test_for_checks_missing_keys                                   | Pass   |
|       | metadata.py::TestGetNrfaMetadataFor          | test_for_checks_missing_keys_from_station_metadata             | Pass   |
|       | metadata.py::TestGetNrfaMetadataFor          | test_for_checks_missing_keys_m3_per_s                          | Pass   |
|       | metadata.py::TestGetNrfaMetadataFor          | test_for_checks_missing_keys_quality_notes_when_no_description | Pass   |
|       | metadata.py::TestGetNrfaMetadataFor          | test_for_checks_missing_keys_source_returns_none               | Pass   |
|       | metadata.py::TestGetNrfaMetadataFor          | test_for_checks_missing_keys_return_none                       | Pass   |
|       | parse_csv.py::TestParseNrfaCsv               | test_flow_values_are_floats                                    | Pass   |
|       | parse_csv.py::TestParseNrfaCsv               | test_flows_have_date_and_flow_cumeecs                          | Pass   |
|       | parse_csv.py::TestParseNrfaCsv               | test_invalid_rows_skipped                                      | Pass   |
|       | parse_csv.py::TestParseNrfaCsv               | test_metadata_contains_station_name                            | Pass   |
|       | parse_csv.py::TestParseNrfaCsv               | test_metadata_station_id                                       | Pass   |

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| Model | Module                                           | Test Name                                                         | Status |
|-------|--------------------------------------------------|-------------------------------------------------------------------|--------|
|       | parse_csv.py::TestParseNrfaCsv                   | test_returns_metadata_and_flows                                   | Pass   |
|       | parse_csv.py::TestParseNrfaCsv                   | test_short_row_skipped                                            | Pass   |
|       | parse_csv.py::TestParseNrfaCsv                   | test_metadata_categories_parsed                                   | Pass   |
|       | parse_csv.py::TestParseNrfaCsv                   | test_metadata_with_non_numeric_value_skipped                      | Pass   |
|       | parse_csv.py::TestParseNrfaCsv                   | test_metadata_row_with_no_value_uses_empty_string                 | Pass   |
|       | parse_csv.py::TestParseNrfaCsv                   | test_metadata_row_with_three_columns_stores_value                 | Pass   |
|       | process_directory.py::TestProcessDirectory       | test_empty_directory_returns_empty_list                           | Pass   |
|       | process_directory.py::TestProcessDirectory       | test_nonmatching_files_skipped                                    | Pass   |
|       | process_directory.py::TestProcessDirectory       | test_output_dir_none_uses_input_dir                               | Pass   |
|       | process_directory.py::TestProcessDirectory       | test_output_files_created                                         | Pass   |
|       | process_directory.py::TestProcessDirectory       | test_process_directory_error_continues                            | Pass   |
|       | process_directory.py::TestProcessDirectory       | test_processes_matching_files                                     | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_annual_maxima_present                                        | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_basic_stats_keys                                             | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_empty_returns_empty_dict                                     | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_extreme_events_above_p99                                     | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_mean_correct                                                 | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_min_max_correct                                              | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_monthly_means_present                                        | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_percentiles_present                                          | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_single_flow_no_std_dev_error                                 | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_added_percentile_keys_present                                | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_added_maxima_events_capped_at_ten                            | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_added_events_top_10_capped                                   | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_flow_date_correctly_identified                               | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_flow_equals_p95                                              | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_flow_equals_p5                                               | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_flow_start_and_end_match_data                                | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_flow_year_annual_maxima_std_dev_zero                         | Zero   |
|       | statistics.py::TestCalculateStatistics           | test_flow_years_approximate                                       | Pass   |
|       | statistics.py::TestCalculateStatistics           | test_flow_std_dev_non_zero                                        | Pass   |
|       | test_block1.coverage_base.py::TestBaseDMCoverage | test_base_dm_coverage_required_fields_default                     | Pass   |
|       | test_block1.coverage_base.py::TestBaseDMCoverage | test_base_dm_coverage_include_info_leaf_returns_dict              | Pass   |
|       | test_block1.coverage_base.py::TestBaseDMCoverage | test_base_dm_coverage_include_info_not_found_returns_none         | Pass   |
|       | test_block1.coverage_base.py::TestBaseDMCoverage | test_base_dm_coverage_include_info_partial_path_returns_none      | None   |
|       | test_block1.coverage_base.py::TestBaseDMCoverage | test_base_dm_coverage_include_info_returns_none_for_non_dict_leaf | Dict   |
|       | test_block1.coverage_base.py::TestBaseDMCoverage | test_base_dm_coverage_fields_nested_metadata_keys                 | Pass   |
|       | test_block1.coverage_base.py::TestBaseDMCoverage | test_base_dm_coverage_fields_prs                                  | Pass   |
|       | test_block1.coverage_base.py::TestBaseDMCoverage | test_base_dm_coverage_fields_returns_nested_path                  | Pass   |
|       | test_block1.coverage_base.py::TestBaseDMCoverage | test_base_dm_coverage_fields_skips_metadata_keys                  | Pass   |
|       | test_block1.coverage_base.py::TestBaseDMCoverage | test_base_dm_coverage                                             | Pass   |
|       | test_block1.coverage_base.py::TestBaseDMCoverage | test_base_dm_coverage_returns_validation_error                    | Pass   |
|       | test_block1.coverage_base.py::TestBaseDMCoverage | test_base_dm_coverage_menu_value_produces_sensor_error            | Error  |
|       | test_block1.coverage_base.py::TestBaseDMCoverage | test_base_dm_coverage_invalid_menu_values                         | Pass   |

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| Model | Module                                               | Test Name                                             | Status |
|-------|------------------------------------------------------|-------------------------------------------------------|--------|
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorTotalGenerated           | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorValidJson                | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorWithRequiredKeys         | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorReturnsEmpty             | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorUnspecifiedFunction      | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorUnspecifiedFunction      | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorHasCsa                   | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorPartiesListHasTwoParties | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorHasCounterpartySet       | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorAgreementIsTrue          | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorHasAccounts              | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorContactInfo              | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorHasUid                   | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorHasName                  | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorNaturalPerson            | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorHasCreditRating          | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | GenerateCounterpartyGeneratorIds                      | Pass   |
|       | test_counterparty_coverage_b6.py::TestVerboseLogging | VerboseLoggingNoPerCounterpartyLogs                   | Pass   |
|       | test_counterparty_coverage_b6.py::TestVerboseLogging | VerboseLoggingPerCounterpartyLogs                     | Pass   |
|       | test_counterparty_coverage_b6.py::TestVerboseLogging | VerboseLoggingSummary                                 | Pass   |
|       | business_days.py::TestBusinessDays                   | BusinessDaysChronologicalOrder                        | Pass   |
|       | business_days.py::TestBusinessDays                   | BusinessDaysEndsBeforeStart                           | Pass   |
|       | business_days.py::TestBusinessDays                   | BusinessDaysLargeCount                                | Pass   |
|       | business_days.py::TestBusinessDays                   | BusinessDaysLastDayIsDayBeforeStart                   | Pass   |
|       | business_days.py::TestBusinessDays                   | BusinessDaysNoWeekends                                | Pass   |
|       | business_days.py::TestBusinessDays                   | BusinessDaysReturnsCorrectCount                       | Pass   |
|       | business_days.py::TestBusinessDays                   | BusinessDaysSingleDay                                 | Pass   |
|       | business_days.py::TestBusinessDays                   | BusinessDaysStartOnMondaySkipsWeekend                 | Pass   |
|       | business_days.py::TestBusinessDays                   | BusinessDaysStartOnWednesday                          | Pass   |
|       | business_days.py::TestSimulation                     | SimulationBusinessDaysIsThreeMonths                   | Pass   |
|       | business_days.py::TestSimulation                     | SimulationDailyVolIsSmall                             | Pass   |
|       | business_days.py::TestSimulation                     | SimulationFloorIsPositive                             | Pass   |
|       | business_days.py::TestSimulation                     | SimulationMoveAtLeastTenDailyVols                     | Pass   |
|       | business_days.py::TestSimulation                     | SimulationMoveBoundsVol                               | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | GenerateHistoryFilesProduceNoFile                     | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | GenerateHistoryFilesHaveDate                          | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | GenerateHistoryFilesHaveTenors                        | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | GenerateHistoryFilesKeyedByGauge                      | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | GenerateHistoryFilesKeyedByTrigger                    | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | GenerateHistoryFilesGeneratedAt                       | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | GenerateHistoryFilesSnapshots                         | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | GenerateHistoryFilesInMetadata                        | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | GenerateHistoryFilesProduceMultipleEntries            | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | GenerateHistoryFilesHasMetadataAndHistory             | Pass   |

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| Model | Module                                                | Test Name                                | Status |
|-------|-------------------------------------------------------|------------------------------------------|--------|
|       | history_files.py::TestGenerateHazard                  | testSingleHazardFileCreatesFile          | Pass   |
|       | history_files.py::TestGenerateTrades                  | testHistoryFileProduceNoFile             | Pass   |
|       | history_files.py::TestGenerateTrades                  | testHistoryFileEntryStructure            | Pass   |
|       | history_files.py::TestGenerateTrades                  | testHazardFileGeneratedAt                | Pass   |
|       | history_files.py::TestGenerateTrades                  | testHazardFileNumSnapshots               | Pass   |
|       | history_files.py::TestGenerateTrades                  | testHistoryFileBuildsHistoryVector       | Pass   |
|       | history_files.py::TestGenerateTrades                  | testHistoryFileTradesInSnapshot          | Pass   |
|       | history_files.py::TestGenerateTrades                  | testHistoryFileMetadataAndTrades         | Pass   |
|       | history_files.py::TestGenerateTrades                  | testHistoryFileMissingSwapIdIsSkipped    | Pass   |
|       | history_files.py::TestGenerateTrades                  | testHazardFileSnapshotCreatesFile        | Pass   |
|       | history_files.py::TestGenerateTrades                  | testHistoryFileKeyedBySwapId             | Pass   |
|       | history_files.py::TestGenerateTrades                  | testHistoryFileMetadataFields            | Pass   |
|       | gaugets.py::TestGaugeTimeSeriesDefault                | testDefaultSimulationHoursIs168          | Pass   |
|       | gaugets.py::TestGaugeTimeSeriesDefault                | testDefaultGenerateDefaultArgumentIs168  | Pass   |
|       | gaugets.py::TestGaugeTimeSeriesDefault                | testDefaultGaugetsFunctionDefaultIs168   | Pass   |
|       | gaugets.py::TestGaugeTimeSeriesDefault                | testDefaultHourMaxWithinSimulationWindow | Pass   |
|       | gaugets.py::TestGaugeTimeSeriesDefault                | testDefaultHourMinLessThanMax            | Pass   |
|       | gaugets.py::TestGaugeTimeSeriesDefault                | testDefault68ProduceMoreReadingsThan60   | Pass   |
|       | gaugets.py::TestGaugeTimeSeriesDefault                | testConfigureKnownParam                  | Pass   |
|       | gaugets.py::TestGaugeTimeSeriesDefault                | testConfigureUnknownParamIgnored         | Pass   |
|       | gaugets.py::TestGaugeTimeSeriesDefault                | testExplicit168HoursProduce168Readings   | Pass   |
|       | gaugets.py::TestGaugeTimeSeriesDefault                | testExplicit60HoursStores60InMetadata    | Pass   |
|       | gaugets.py::TestGaugeTimeSeriesDefault                | testMissingGaugeFileRaises               | Pass   |
|       | gaugets.py::TestGaugeTimeSeriesDefault                | testSimulationHoursStoredInOutput        | Pass   |
|       | gaugets.py::TestGaugeTimeSeriesDefault                | testStaleGaugeFilesRemoved               | Pass   |
|       | gaugets.py::TestGaugetsDateTimeEncoder                | testEncodesDatetime                      | Pass   |
|       | gaugets.py::TestGaugetsDateTimeEncoder                | testEncodesNumpyFloat32                  | Pass   |
|       | gaugets.py::TestGaugetsDateTimeEncoder                | testEncodesNumpyInteger                  | Pass   |
|       | gaugets.py::TestGaugetsDateTimeEncoder                | testEncodesNumpyNdarray                  | Pass   |
|       | gaugets.py::TestGaugetsDateTimeEncoder                | testFallbackRaisesForUnknown             | Pass   |
|       | gaugets.py::TestGaugetsMain                           | testMainBlock                            | Pass   |
|       | gaugets.py::TestGaugetsRandomDefault                  | testRandomDefaultPeakHourMaxWithinWindow | Pass   |
|       | gaugets.py::TestGaugetsRandomDefault                  | testRandomDefaultSimulationHoursIs168    | Pass   |
|       | gaugets.py::TestGenerateGaugets                       | testGenerateGaugetsCallsGenerator        | Pass   |
|       | test_gauge_random_coverage.py::TestFieldNoneDefault   | testFieldNoneDefaultReturnsEmptyString   | Pass   |
|       | test_gauge_random_coverage.py::TestFieldNoneDefault   | testFieldNoneDefaultReturnsEmptyString   | Pass   |
|       | test_gauge_random_coverage.py::TestFieldNoneDefault   | testFieldNoneDefaultReturnedAsIs         | Pass   |
|       | test_gauge_random_coverage.py::TestFieldNoneDefault   | testFieldNoneDefaultReturnsText          | Pass   |
|       | test_gauge_random_coverage.py::TestFieldNoneDefault   | testFieldNoneDefaultReturnsEmpty         | Pass   |
|       | test_gauge_random_coverage.py::TestFieldNameFallback  | testFieldNameFallbackBackToArea          | Pass   |
|       | test_gauge_random_coverage.py::TestFieldNameFallback  | testFieldNameFallbackBackToArea          | Pass   |
|       | test_gauge_random_coverage.py::TestIntegerDefault     | testIntegerDefaultReturnsInt             | Pass   |
|       | test_gauge_random_coverage.py::TestMarkOptionsDefault | testMarkOptionsDefaultReturnsEmpty       | Pass   |
|       | test_gauge_random_coverage.py::TestMarkOptionsDefault | testMarkOptionsDefaultOptionsCycles      | Pass   |

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| Model                                                  | Module                                                 | Test Name                                                     | Status    |
|--------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------|-----------|
| test_gauge_random_coverage.py::TestMarkOptionsDefault  | test_gauge_random_coverage.py::TestMarkOptionsDefault  | test_mark_options_default                                     | Pass      |
| test_gaugehd_module.py::TestGenerateAllChangeHist      | test_gaugehd_module.py::TestGenerateAllChangeHist      | test_generate_all_change_histories                            | Pass      |
| test_gaugehd_module.py::TestGenerateAllChangeHist      | test_gaugehd_module.py::TestGenerateAllChangeHist      | test_generate_all_change_histories_error                      | Continues |
| test_gaugehd_module.py::TestLoadGaugePortfolio         | test_gaugehd_module.py::TestLoadGaugePortfolio         | test_load_gauge_portfolio_file_not_found                      | Pass      |
| test_gaugehd_module.py::TestLoadGaugePortfolio         | test_gaugehd_module.py::TestLoadGaugePortfolio         | test_load_gauge_portfolio_success                             | Pass      |
| test_gaugehd_module.py::TestProcessNrfaDirectory       | test_gaugehd_module.py::TestProcessNrfaDirectory       | test_process_nrfa_directory_bad_csv_content                   | Pass      |
| test_gaugehd_module.py::TestProcessNrfaDirectory       | test_gaugehd_module.py::TestProcessNrfaDirectory       | test_process_nrfa_directory_basic                             | Pass      |
| test_gaugehd_module.py::TestProcessNrfaDirectory       | test_gaugehd_module.py::TestProcessNrfaDirectory       | test_process_nrfa_directory_default_output                    | Pass      |
| test_gaugehd_module.py::TestProcessNrfaDirectory       | test_gaugehd_module.py::TestProcessNrfaDirectory       | test_process_nrfa_directory_empty_dir                         | Pass      |
| test_gaugehd_runner.py::TestGenerateEmptyPortfolio     | test_gaugehd_runner.py::TestGenerateEmptyPortfolio     | test_generate_empty_portfolio_returns_empty                   | Pass      |
| test_gaugehd_runner.py::TestGenerateFromPackets        | test_gaugehd_runner.py::TestGenerateFromPackets        | test_generate_from_packets_gauge_id_from_nested_header        | Pass      |
| test_gaugehd_runner.py::TestGenerateFromPackets        | test_gaugehd_runner.py::TestGenerateFromPackets        | test_generate_from_packets_no_header_uses_unknown             | Pass      |
| test_gaugehd_runner.py::TestGenerateFromPackets        | test_gaugehd_runner.py::TestGenerateFromPackets        | test_generate_from_packets_and_bad_gauges                     | Pass      |
| test_gaugehd_runner.py::TestMain                       | test_gaugehd_runner.py::TestMain                       | test_main_default_synthetic                                   | Pass      |
| test_gaugehd_runner.py::TestMain                       | test_gaugehd_runner.py::TestMain                       | test_main_no_catchment_leaves_config_unchanged                | Changed   |
| test_gaugehd_runner.py::TestMain                       | test_gaugehd_runner.py::TestMain                       | test_main_short_catchment_flag                                | Pass      |
| test_gaugehd_runner.py::TestMain                       | test_gaugehd_runner.py::TestMain                       | test_main_short_years_flag                                    | Pass      |
| test_gaugehd_runner.py::TestMain                       | test_gaugehd_runner.py::TestMain                       | test_main_with_catchment                                      | Pass      |
| test_gaugehd_runner.py::TestMain                       | test_gaugehd_runner.py::TestMain                       | test_main_with_nrfa_dir                                       | Pass      |
| test_gaugehd_runner.py::TestMain                       | test_gaugehd_runner.py::TestMain                       | test_main_with_years                                          | Pass      |
| test_gaugehd_runner.py::TestProcessNrfaDirectory       | test_gaugehd_runner.py::TestProcessNrfaDirectory       | test_process_nrfa_directory_from_filename                     | Pass      |
| test_gaugehd_runner.py::TestProcessNrfaDirectory       | test_gaugehd_runner.py::TestProcessNrfaDirectory       | test_process_nrfa_directory_from_dir                          | Pass      |
| test_gaugehd_runner.py::TestStateFileCleanUp           | test_gaugehd_runner.py::TestStateFileCleanUp           | test_state_file_clean_up_remove_non_matching_files            | Pass      |
| test_gaugehd_runner.py::TestStateFileCleanUp           | test_gaugehd_runner.py::TestStateFileCleanUp           | test_state_file_clean_up_multiple_stale_files                 | Pass      |
| test_gaugehd_runner.py::TestStateFileCleanUp           | test_gaugehd_runner.py::TestStateFileCleanUp           | test_state_file_clean_up_stale_gauge_files                    | Pass      |
| test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_generate_historical_daily_statistics_from_levels         | Pass      |
| test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_generate_historical_daily_statistics_via_class           | Pass      |
| test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_generate_historical_daily                                | Pass      |
| test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_generate_historical_daily_years                          | Pass      |
| test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_generate_historical_daily_years_explicit                 | Pass      |
| test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_generate_historical_daily_years_none                     | Pass      |
| test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_generate_historical_daily_gauge_portfolio_explicit_years | Pass      |
| test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_generate_historical_daily_gauge_portfolio_via_class      | Pass      |
| test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_generate_historical_daily_nrfa_uses_default_years        | Pass      |
| test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_generate_historical_daily_nrfa_via_class                 | Pass      |
| test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_generate_historical_daily_synthetic_timeseries_via_class | Pass      |
| test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_gaugehd_synthetic.py::TestGenerateHistoricalDaily | test_generate_historical_daily_via_class                      | Pass      |
| test_gaugehd_synthetic.py::TestGenerateFromGauges      | test_gaugehd_synthetic.py::TestGenerateFromGauges      | test_generate_from_gauges_portfolio                           | Pass      |
| test_gaugehd_synthetic.py::TestGenerateFromGauges      | test_gaugehd_synthetic.py::TestGenerateFromGauges      | test_generate_from_gauges_portfolio_generated                 | Pass      |
| test_gaugehd_synthetic.py::TestGenerateFromGauges      | test_gaugehd_synthetic.py::TestGenerateFromGauges      | test_generate_from_gauges_portfolio_directed                  | Pass      |
| test_gaugehd_synthetic.py::TestGenerateFromGauges      | test_gaugehd_synthetic.py::TestGenerateFromGauges      | test_generate_from_gauges_portfolio_data                      | Pass      |
| test_gaugehd_synthetic.py::TestGenerateFromGauges      | test_gaugehd_synthetic.py::TestGenerateFromGauges      | test_generate_from_gauges_portfolio_predefined                | Pass      |

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| Model | Module                                                                           | Test Name                                        | Status |
|-------|----------------------------------------------------------------------------------|--------------------------------------------------|--------|
|       | test_gaugehd_synthetic.py::TestGenerateFromGaugePortfolio                        | TestGenerateFromGaugePortfolio                   | Pass   |
|       | test_gaugehd_synthetic.py::TestGenerateFromGaugePortfolio                        | TestGenerateFromGaugePortfolio                   | Pass   |
|       | test_gaugets_random_coverage.py::TestDuplicateNoneParams                         | TestDuplicateNoneParams                          | Pass   |
|       | test_gaugets_random_coverage.py::TestDuplicateNoneParams                         | TestDuplicateNoneParams                          | Pass   |
|       | test_gaugets_random_coverage.py::TestDefaultParamsDefaults                       | TestDefaultParamsDefaults                        | Pass   |
|       | test_gaugets_random_coverage.py::TestDefaultParamsDefaults                       | TestDefaultParamsDefaults                        | Pass   |
|       | test_gaugets_random_coverage.py::TestStringSimulationStrings                     | TestStringSimulationStrings                      | Pass   |
|       | test_gaugets_random_coverage.py::TestGaugerReadingReadingsDefaultFields          | TestGaugerReadingReadingsDefaultFields           | Pass   |
|       | test_gaugets_random_coverage.py::TestGaugerReadingReadingsDefaultFields          | TestGaugerReadingReadingsDefaultFields           | Pass   |
|       | test_gaugets_random_coverage.py::TestGenerateTimeSeriesAlerts                    | TestGenerateTimeSeriesAlerts                     | Pass   |
|       | test_gaugets_random_coverage.py::TestGenerateTimeSeriesAlerts                    | TestGenerateTimeSeriesAlerts                     | Pass   |
|       | test_gaugets_random_coverage.py::TestFilterByYear5_year_filter                   | TestFilterByYear5_year_filter                    | Pass   |
|       | test_nrfa_parsing.py::TestFilterByYearempty_list_returns_empty                   | TestFilterByYearempty_list_returns_empty         | Pass   |
|       | test_nrfa_parsing.py::TestFilterByYearlarge_years_returns_all                    | TestFilterByYearlarge_years_returns_all          | Pass   |
|       | test_nrfa_parsing.py::TestFilterByYearone_years_returns_all                      | TestFilterByYearone_years_returns_all            | Pass   |
|       | test_nrfa_parsing.py::TestGenerateFromNrfOutputPath                              | TestGenerateFromNrfOutputPath                    | Pass   |
|       | test_nrfa_parsing.py::TestGenerateFromNrfJsonFile                                | TestGenerateFromNrfJsonFile                      | Pass   |
|       | test_nrfa_parsing.py::TestGenerateFromNrfIdOverride                              | TestGenerateFromNrfIdOverride                    | Pass   |
|       | test_nrfa_parsing.py::TestGenerateFromNrfReturnsDict                             | TestGenerateFromNrfReturnsDict                   | Pass   |
|       | test_nrfa_parsing.py::TestGenerateFromNrfVersionInOutput                         | TestGenerateFromNrfVersionInOutput               | Pass   |
|       | test_nrfa_parsing.py::TestGenerateFromNrfMetadataExtracted                       | TestGenerateFromNrfMetadataExtracted             | Pass   |
|       | test_nrfa_parsing.py::TestParseNrfEmptyFile                                      | TestParseNrfEmptyFile                            | Pass   |
|       | test_nrfa_parsing.py::TestParseNrfCsfFlowsHaveDateAndFlow                        | TestParseNrfCsfFlowsHaveDateAndFlow              | Pass   |
|       | test_nrfa_parsing.py::TestParseNrfCsfMetadataContainsStationId                   | TestParseNrfCsfMetadataContainsStationId         | Pass   |
|       | test_nrfa_parsing.py::TestParseNrfCsfMetadataContainsStationName                 | TestParseNrfCsfMetadataContainsStationName       | Pass   |
|       | test_nrfa_parsing.py::TestParseNrfCsfReturnsTuple                                | TestParseNrfCsfReturnsTuple                      | Pass   |
|       | test_nrfa_parsing.py::TestParseNrfCsfSkipsShortRows                              | TestParseNrfCsfSkipsShortRows                    | Pass   |
|       | test_stress_storms_core.py::TestGenerateHugeIdPresentInAllResponses              | TestGenerateHugeIdPresentInAllResponses          | Pass   |
|       | test_stress_storms_core.py::TestGenerateMultipleGaugesHaveCorrectIds             | TestGenerateMultipleGaugesHaveCorrectIds         | Pass   |
|       | test_stress_storms_core.py::TestGenerateStressStormsWhenGaugesEmpty              | TestGenerateStressStormsWhenGaugesEmpty          | Pass   |
|       | test_stress_storms_core.py::TestGenerateIndexStressGaugeIdsAlert                 | TestGenerateIndexStressGaugeIdsAlert             | Pass   |
|       | test_stress_storms_core.py::TestGenerateIndexStressRequiredTopLevelKeys          | TestGenerateIndexStressRequiredTopLevelKeys      | Pass   |
|       | test_stress_storms_core.py::TestGenerateStressStormFileHasGaugeResponses         | TestGenerateStressStormFileHasGaugeResponses     | Pass   |
|       | test_stress_storms_core.py::TestGenerateStressStormPrefixUsesConfigConstant      | TestGenerateStressStormPrefixUsesConfigConstant  | Pass   |
|       | test_stress_storms_core.py::TestGenerateSubStressStormsExcluded                  | TestGenerateSubStressStormsExcluded              | Pass   |
|       | test_stress_storms_core.py::TestGenerateSubStressStormsMatchesListLength         | TestGenerateSubStressStormsMatchesListLength     | Pass   |
|       | test_stress_storms_core.py::TestGenerateStressStormsAndStormFiles                | TestGenerateStressStormsAndStormFiles            | Pass   |
|       | test_stress_storms_core.py::TestReturnReturnsDictWithTotalStorms                 | TestReturnReturnsDictWithTotalStorms             | Pass   |
|       | test_stress_storms_core.py::TestReturnReturnsElapsedS                            | TestReturnReturnsElapsedS                        | Pass   |
|       | test_stress_storms_coverage.py::TestCorruptGaugeFileGaugeFilesNoStorms           | TestCorruptGaugeFileGaugeFilesNoStorms           | Pass   |
|       | test_stress_storms_coverage.py::TestCorruptGaugeFileGaugeFileSkipped             | TestCorruptGaugeFileGaugeFileSkipped             | Pass   |
|       | test_stress_storms_coverage.py::TestCorruptStormsJsonDoesNotCrash                | TestCorruptStormsJsonDoesNotCrash                | Pass   |
|       | test_stress_storms_coverage.py::TestDenegateIsCategoryWarningPathReturnsBaseline | TestDenegateIsCategoryWarningPathReturnsBaseline | Pass   |
|       | test_stress_storms_coverage.py::TestDenegateIsCategoryWarningPathReturnsModerate | TestDenegateIsCategoryWarningPathReturnsModerate | Pass   |

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| Model | Module                                 | Test Name                                            | Status |
|-------|----------------------------------------|------------------------------------------------------|--------|
|       | business.days.py::TestBusinessDays     | test_all_days_before_start                           | Pass   |
|       | business.days.py::TestBusinessDays     | test_returns_ascending_order                         | Pass   |
|       | business.days.py::TestBusinessDays     | test_returns_business_days_only                      | Pass   |
|       | business.days.py::TestBusinessDays     | test_returns_correct_count                           | Pass   |
|       | business.days.py::TestBusinessDays     | test_single_day                                      | Pass   |
|       | business.days.py::TestBusinessDays     | test_skips_weekends                                  | Pass   |
|       | business.days.py::TestBusinessDays     | test_zero_days                                       | Pass   |
|       | eod_series.py::TestGenerateHistory     | test_EODSeries_days_are_weekdays                     | Pass   |
|       | eod_series.py::TestGenerateHistory     | test_EODSeries_existing_eod_files                    | Pass   |
|       | eod_series.py::TestGenerateHistory     | test_EODSeries_eod_dir                               | Pass   |
|       | eod_series.py::TestGenerateHistory     | test_EODSeries_trades_returns_zero                   | Pass   |
|       | eod_series.py::TestGenerateHistory     | test_EODSeries_instrument_failure_is_skipped         | Pass   |
|       | eod_series.py::TestGenerateHistory     | test_EODSeries_trades_adds_more_snapshot             | Pass   |
|       | eod_series.py::TestGenerateHistory     | test_EODSeries_integer                               | Pass   |
|       | eod_series.py::TestGenerateHistory     | test_EODSeries_reproducible                          | Pass   |
|       | eod_series.py::TestGenerateHistory     | test_EODSeries_trade_generates_snapshots             | Pass   |
|       | eod_series.py::TestGenerateHistory     | test_EODSeries_trade_writes_eod_files                | Pass   |
|       | eod_series.py::TestGenerateHistory     | test_EODSeries_trade_writes_history_files            | Pass   |
|       | hazard_history.py::TestGenerateHistory | test_features_history_output_file                    | Pass   |
|       | hazard_history.py::TestGenerateHistory | test_HistoryHistory_entries_have_date                | Pass   |
|       | hazard_history.py::TestGenerateHistory | test_HistoryHistory_lists_gauges                     | Pass   |
|       | hazard_history.py::TestGenerateHistory | test_HistoryHistory_gauges                           | Pass   |
|       | hazard_history.py::TestGenerateHistory | test_HistoryHistory_returns_early                    | Pass   |
|       | hazard_history.py::TestGenerateHistory | test_HistoryHistory_history                          | Pass   |
|       | hazard_history.py::TestGenerateHistory | test_HistoryHistory_metadata                         | Pass   |
|       | hazard_history.py::TestGenerateHistory | test_HistoryHistory_not_numerically                  | Pass   |
|       | hazard_history.py::TestGenerateHistory | test_HistoryHistory_has_two_history_entries          | Pass   |
|       | pnl_history.py::TestGenerateTrades     | test_PnlHistory_output_file                          | Pass   |
|       | pnl_history.py::TestGenerateTrades     | test_PnlHistory_entry_has_required_keys              | Pass   |
|       | pnl_history.py::TestGenerateTrades     | test_PnlHistory_multiple_eods_accumulate             | Pass   |
|       | pnl_history.py::TestGenerateTrades     | test_PnlHistory_multiple_trades_in_same_eod          | Pass   |
|       | pnl_history.py::TestGenerateTrades     | test_PnlHistory_files_returns_early                  | Pass   |
|       | pnl_history.py::TestGenerateTrades     | test_PnlHistory_snapshots_in_metadata                | Pass   |
|       | pnl_history.py::TestGenerateTrades     | test_PnlHistory_has_metadata                         | Pass   |
|       | pnl_history.py::TestGenerateTrades     | test_PnlHistory_has_trades                           | Pass   |
|       | pnl_history.py::TestGenerateTrades     | test_PnlHistory_on_without_swap_id_skipped           | Pass   |
|       | pnl_history.py::TestGenerateTrades     | test_PnlHistory_has_history                          | Pass   |
|       | pnl_history.py::TestGenerateTrades     | test_PnlHistory_metadata_preserved                   | Pass   |
|       | test_mortgage_builder.py::TestBuilder  | test_empty_schema_returns_empty_dict                 | Pass   |
|       | test_mortgage_builder.py::TestBuilder  | test_existing_dict_without_type_recurse              | Pass   |
|       | test_mortgage_builder.py::TestBuilder  | test_empty_dict_schema_returns_empty_dict            | Pass   |
|       | test_mortgage_builder.py::TestBuilder  | test_empty_schema_returns_empty_dict                 | Pass   |
|       | test_mortgage_builder.py::TestBuilder  | test_remove_value_from_generate_field_value_excluded | Pass   |
|       | test_mortgage_builder.py::TestBuilder  | test_remove_reserved_key_description_skipped         | Pass   |

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| Model | Module                                                    | Test Name                              | Status |
|-------|-----------------------------------------------------------|----------------------------------------|--------|
|       | test_mortgage.builder.py::TestBuilder                     | test_reserved_key_options_skipped      | Pass   |
|       | test_mortgage.builder.py::TestBuilder                     | test_reserved_key_type_skipped         | Pass   |
|       | test_mortgage.builder.py::TestBuilder                     | test_reserved_key_values_skipped       | Pass   |
|       | test_mortgage.builder.py::TestQualityConsistencyChecker   | test_initial_is_capped_95pct           | Pass   |
|       | test_mortgage.builder.py::TestQualityConsistencyChecker   | test_initial_unchanged                 | Pass   |
|       | test_mortgage.builder.py::TestQualityDelegator            | test_choose_module_when_method_present | Pass   |
|       | test_mortgage.builder.py::TestQualityConsistencyChecker   | test_full_95k                          | Pass   |
|       | test_mortgage.builder.py::TestQualityConsistencyChecker   | test_full_60k                          | Pass   |
|       | test_mortgage.builder.py::TestQualityConsistencyChecker   | test_unchanged                         | Pass   |
|       | test_mortgage.builder.py::TestSetSpecificMortgageValues   | test_dict_updated                      | Pass   |
|       | test_mortgage.builder.py::TestSetSpecificMortgageValues   | test_header_key_when_absent            | Pass   |
|       | test_mortgage.builder.py::TestSetSpecificMortgageValues   | test_mortgage_key_when_absent          | Pass   |
|       | test_mortgage.builder.py::TestSetSpecificMortgageValues   | test_id_key_defaults_empty_string      | Pass   |
|       | test_mortgage.builder.py::TestSetSpecificMortgageValues   | test_valid                             | Pass   |
|       | test_mortgage.builder.py::TestSetSpecificMortgageValues   | test_fields                            | Pass   |
|       | test_mortgage.builder.py::TestSetSpecificMortgageValues   | test_valid                             | Pass   |
|       | test_mortgage.builder.py::TestSetSpecificMortgageValues   | test_invalid_in_header                 | Pass   |
|       | test_mortgage.coverage.py::TestEstimate                   | test_error_is_reraised                 | Pass   |
|       | test_mortgage.coverage.py::TestSingleMortgage             | test_refail_returns_empty_list         | Pass   |
|       | test_mortgage.coverage.py::TestSingleMortgage             | test_is_counted_and_skipped            | Pass   |
|       | test_mortgage.encoder_extractor.py::TestDatetimeEncoder   | test_isoformat                         | Pass   |
|       | test_mortgage.encoder_extractor.py::TestDatetimeEncoder   | test_datetime                          | Pass   |
|       | test_mortgage.encoder_extractor.py::TestDatetimeEncoder   | test_float_encoded                     | Pass   |
|       | test_mortgage.encoder_extractor.py::TestDatetimeEncoder   | test_float_encoded_as_float            | Pass   |
|       | test_mortgage.encoder_extractor.py::TestDatetimeEncoder   | test_int32_encoded                     | Pass   |
|       | test_mortgage.encoder_extractor.py::TestDatetimeEncoder   | test_int64_encoded_as_int              | Pass   |
|       | test_mortgage.encoder_extractor.py::TestDatetimeEncoder   | test_int_encoded_as_list               | Pass   |
|       | test_mortgage.encoder_extractor.py::TestDatetimeEncoder   | test_raises_type_error                 | Pass   |
|       | test_mortgage.encoder_extractor.py::TestDictExtractor     | test_dict_extracted                    | Pass   |
|       | test_mortgage.encoder_extractor.py::TestDictExtractor     | test_dict_extracted                    | Pass   |
|       | test_mortgage.encoder_extractor.py::TestDictExtractor     | test_info                              | Pass   |
|       | test_mortgage.encoder_extractor.py::TestMissingProperty   | test_init_construction_year            | Pass   |
|       | test_mortgage.encoder_extractor.py::TestMissingProperty   | test_defaults_property_id              | Pass   |
|       | test_mortgage.encoder_extractor.py::TestMissingProperty   | test_defaults_postcode_and_coords      | Pass   |
|       | test_mortgage.encoder_extractor.py::TestMissingProperty   | test_header_returns_defaults           | Pass   |
|       | test_mortgage.encoder_extractor.py::TestMissingProperty   | test_defaults_risk                     | Pass   |
|       | test_mortgage.encoder_extractor.py::TestMissingProperty   | test_defaults_value                    | Pass   |
|       | test_mortgage.encoder_extractor.py::TestPropertyExtractor | test_info                              | Pass   |
|       | test_mortgage.encoder_extractor.py::TestPropertyExtractor | test_id_extracted_from_header          | Pass   |
|       | test_mortgage.encoder_extractor.py::TestPropertyExtractor | test_type_extracted                    | Pass   |
|       | test_mortgage.encoder_extractor.py::TestPropertyExtractor | test_value_extracted                   | Pass   |
|       | test_mortgage.generate.py::TestGeneratePropertyList       | test_returns_zero_mortgages            | Pass   |
|       | test_mortgage.generate.py::TestGeneratePropertyFile       | test_file_not_found                    | Pass   |
|       | test_mortgage.generate.py::TestGenerateBalance            | test_does_not_exceed_or_equal_loan     | Pass   |

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| Model | Module                                | Test Name                                        | Status |
|-------|---------------------------------------|--------------------------------------------------|--------|
|       | test_mortgage_random_values.py        | TestOverriddenDecimalValue                       | Pass   |
|       | test_mortgage_random_values.py        | TestOriginalDendailyValue                        | Pass   |
|       | test_mortgage_random_values.py        | TestOriginalDoinalValue                          | Pass   |
|       | test_mortgage_random_values.py        | TestOriginalDoinalAmount                         | Pass   |
|       | test_mortgage_random_values.py        | TestOriginalDctinalValue                         | Pass   |
|       | test_mortgage_random_values.py        | TestGetstandOriginalValue                        | Pass   |
|       | test_mortgage_random_values.py        | TestPurchaseDealValue                            | Pass   |
|       | test_mortgage_random_values.py        | TestApplicationValue                             | Pass   |
|       | test_mortgage_random_values.py        | TestLoanpurposeValue                             | Pass   |
|       | test_mortgage_random_values.py        | TestLoanpurposeResidential                       | Pass   |
|       | test_mortgage_random_values.py        | TestMortgageTypeValue                            | Pass   |
|       | test_mortgage_random_values.py        | TestOccupancyTypeValue                           | Pass   |
|       | test_mortgage_random_values.py        | TestOriginalMortType                             | Pass   |
|       | test_mortgage_random_values.py        | TestPaymentFrequency                             | Pass   |
|       | test_mortgage_random_values.py        | TestCepaymentMonthly                             | Pass   |
|       | test_mortgage_random_values.py        | TestCepaymentMonthlyResidential                  | Pass   |
|       | test_mortgage_random_values.py        | TestUnknownFieldNoOptions                        | Pass   |
|       | test_mortgage_random_values.py        | TestUnknownFieldWithOptions                      | Pass   |
|       | generators.py::TestCounterparty       | test_all_counterparties_have_party_id            | Pass   |
|       | generators.py::TestCounterparty       | test_cdm_structure                               | Pass   |
|       | generators.py::TestCounterparty       | test_custom_count                                | Pass   |
|       | generators.py::TestCounterparty       | test_generates_counterparty_file                 | Pass   |
|       | generators.py::TestGaugeHistorical    | test_generates_per_gauge_files                   | Pass   |
|       | generators.py::TestGaugeHistorical    | test_daily_output_has_daily_observations         | Pass   |
|       | generators.py::TestGaugeHistorical    | test_daily_statistics_computed                   | Pass   |
|       | generators.py::TestGaugeTimeSeries    | test_generates_per_gauge_files                   | Pass   |
|       | generators.py::TestGaugeTimeSeries    | test_output_file_has_readings                    | Pass   |
|       | generators.py::TestGaugeTimeSeries    | test_requires_gauge_portfolio                    | Pass   |
|       | generators.py::TestNostressFlag       | test_explicit_stressm_flag_runs_despite_nostress | Pass   |
|       | generators.py::TestNostressFlag       | test_nostress_flag_exists_in_parser              | Pass   |
|       | generators.py::TestNostressFlag       | test_nostress_skips_stressm_when_all             | Pass   |
|       | generators.py::TestPropertyTimeSeries | test_generates_per_property_files                | Pass   |
|       | generators.py::TestPropertyTimeSeries | test_per_property_files_have_flood_data          | Pass   |
|       | generators.py::TestPropertyTimeSeries | test_portfolio_summary_has_statistics            | Pass   |
|       | integration.py::TestDependencyChain   | test_gaugehd_references_gauges                   | Pass   |
|       | integration.py::TestDependencyChain   | test_gaugets_reads_gauge                         | Pass   |
|       | integration.py::TestDependencyChain   | test_hazard_reads_gauge_and_storms               | Pass   |
|       | integration.py::TestDependencyChain   | test_mortgage_reads_property                     | Pass   |
|       | integration.py::TestDependencyChain   | test_propertyhc_references_properties            | Pass   |
|       | integration.py::TestDependencyChain   | test_propertyts_uses_nearest_gauges              | Pass   |
|       | integration.py::TestEndToEndPipeline  | test_full_pipeline_generates_all_output          | Pass   |
|       | integration.py::TestEndToEndPipeline  | test_gauge_count_propagates                      | Pass   |
|       | integration.py::TestEndToEndPipeline  | test_pipeline_output_file_integrity              | Pass   |
|       | integration.py::TestEndToEndPipeline  | test_property_count_propagates                   | Pass   |

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| Model | Module                                                       | Test Name                                                                  | Status |
|-------|--------------------------------------------------------------|----------------------------------------------------------------------------|--------|
|       | property_energy.py::TestCalculateActualEnergy                | testActualEnergyArea.gives.larger.energy                                   | Pass   |
|       | property_energy.py::TestCalculateActualEnergy                | testActualEnergyArea.positive.value                                        | Pass   |
|       | property_energy.py::TestCalculateActualEnergy                | testActualEnergyArea.default.area.when.missing                             | Pass   |
|       | property_energy.py::TestCalculateActualEnergy                | testActualEnergyArea.floats                                                | Pass   |
|       | property_energy.py::TestCalculateActualEnergy                | testActualEnergyArea.tends.to.give.larger.en                               | Pass   |
|       | property_energy.py::TestCalculateActualEnergy                | testActualEnergyArea.positive.value                                        | Pass   |
|       | property_energy.py::TestCalculateActualEnergy                | testActualEnergyArea.default.area.when.missing                             | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.calculation                                                 | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.at                                                          | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.s_positive.value                                            | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.defaults.when.missing                                       | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.returns.zero                                           | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.less.than.remaining.energy                             | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.negative.when.electricity.exceeds.total                | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.returns.non.negative                                   | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.defaults.when.missing                                  | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.fraction.of.total                                      | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.positive.value                                         | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.default.energy.when.missing                            | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.returns.zero                                           | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.returns.positive                                       | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.returns.zero                                           | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.returns.zero                                           | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.returns.positive                                       | Pass   |
|       | property_energy.py::TestCalculateEnergyBill                  | testEnergyBill.sage.returns.default.area.when.missing                      | Pass   |
|       | test_geometry.py::TestEnsureOffset                           | testEnsureOffset.close.point.pushed.away                                   | Pass   |
|       | test_geometry.py::TestEnsureOffset                           | testEnsureOffset.far.point.unchanged                                       | Pass   |
|       | test_geometry.py::TestEnsureOffset                           | testEnsureOffset.returns.tuple.of.two.floats                               | Pass   |
|       | test_geometry.py::TestHaversine                              | testApproximateOneDegreeLat                                                | Pass   |
|       | test_geometry.py::TestHaversine                              | testApproximateOneDegreeLonAtEquator                                       | Pass   |
|       | test_geometry.py::TestHaversine                              | testSamePointReturnsZero                                                   | Pass   |
|       | test_geometry.py::TestHaversine                              | testSymmetry                                                               | Pass   |
|       | test_geometry.py::TestMinRiverDistance                       | testMinRiverDistance.farthest.point.returns.larger.distance                | Pass   |
|       | test_geometry.py::TestMinRiverDistance                       | testMinRiverDistance.point.on.gauge.returns.small.distance                 | Pass   |
|       | test_geometry.py::TestMinRiverDistance                       | testMinRiverDistance.returns.float                                         | Pass   |
|       | test_geometry.py::TestNearestOnSegment                       | testNearestOnSegment.generate.segment.returns.start                        | Pass   |
|       | test_geometry.py::TestNearestOnSegment                       | testNearestOnSegment.distance.is.non.negative                              | Pass   |
|       | test_geometry.py::TestNearestOnSegment                       | testNearestOnSegment.midpoint.projects.to.midpoint                         | Pass   |
|       | test_geometry.py::TestNearestOnSegment                       | testNearestOnSegment.point.before.start.clamps.to.start                    | Pass   |
|       | test_geometry.py::TestNearestOnSegment                       | testNearestOnSegment.point.past.end.clamps.to.end                          | Pass   |
|       | test_geometry.py::TestPerpendicularOffset                    | testPerpendicularOffset.generate.segment.falls.back.to.previous.lat.offset | Pass   |
|       | test_geometry.py::TestPerpendicularOffset                    | testPerpendicularOffset.moves.point                                        | Pass   |
|       | test_geometry.py::TestPerpendicularOffset                    | testPerpendicularOffset.returns.two.floats                                 | Pass   |
|       | test_location_coverage_b6.py::TestGetBuildAreaBoundaryValues | testGetBuildAreaBoundaryValues                                             | Pass   |

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| Model | Module                                                       | Test Name                                                   | Status |
|-------|--------------------------------------------------------------|-------------------------------------------------------------|--------|
|       | test_location_coverage.b6.py::TestGenerateLocation           | test_house_3_bedrooms_house_paths_1_or_2                    | Pass   |
|       | test_location_coverage.b6.py::TestGenerateLocation           | test_house_4_bedrooms_house_paths_2_or_3                    | Pass   |
|       | test_location_coverage.b6.py::TestGenerateLocation           | test_house_5_bedrooms_house_paths_2_or_3                    | Pass   |
|       | test_locations.py::TestGenerateLocations                     | test_location_has_required_keys                             | Pass   |
|       | test_locations.py::TestGenerateLocations                     | test_location_elevation_is_non_negative                     | Pass   |
|       | test_locations.py::TestGenerateLocations                     | test_location_fallback_used_when_fewer_than_3_gauge_points  | Pass   |
|       | test_locations.py::TestGenerateLocations                     | test_location_fallback_used_when_no_gauge_points            | Pass   |
|       | test_locations.py::TestGenerateLocations                     | test_location_primary_idx_zero_uses_gauges_012              | Pass   |
|       | test_locations.py::TestGenerateLocations                     | test_location_reference_gauge_indices_sorted                | Pass   |
|       | test_locations.py::TestGenerateLocations                     | test_location_reference_gauge_indices_three_elements        | Pass   |
|       | test_locations.py::TestGenerateLocations                     | test_location_returns_correct_count                         | Pass   |
|       | test_locations.py::TestGenerateLocations                     | test_location_fallback_no_fallback_elevation_use_random     | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorms               | test_cluster_max_depth_is_from_highest_storm                | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorms               | test_cluster_produces_three_flood_events                    | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorms               | test_cluster_three_distinct_storm_ids_in_events             | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorms               | test_cluster_duplicates_storms_create_flood_events          | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorms               | test_cluster_duplicates_storms_not_in_flood_events          | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorms               | test_cluster_all_events_have_reading                        | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorms               | test_cluster_four_events_recorded                           | Pass   |
|       | test_multi_storm_doublet.py::TestDoubletStorms               | test_doublet_each_event_has_independent_damage              | Pass   |
|       | test_multi_storm_doublet.py::TestDoubletStorms               | test_doublet_events_have_distinct_storm_ids                 | Pass   |
|       | test_multi_storm_doublet.py::TestDoubletStorms               | test_doublet_produces_two_flood_events                      | Pass   |
|       | test_multi_storm_doublet.py::TestDoubletStorms               | test_doublet_enumerates_all_doublet_floods                  | Pass   |
|       | test_multi_storm_portfolio.py::TestPortfolioStorms           | test_portfolio_with_multiple_storms_generates_zero_floods   | Pass   |
|       | test_multi_storm_portfolio.py::TestPortfolioStorms           | test_portfolio_with_multiple_storms_generates_all_sequences | Pass   |
|       | test_multi_storm_portfolio.py::TestPortfolioStorms           | test_portfolio_with_multiple_storms_generates_all_sequences | Pass   |
|       | test_property_random_coverage.py::TestGenerateRandomCoverage | test_generate_random_coverage_exception_falls_back_to_type  | Pass   |
|       | test_property_random_coverage.py::TestGenerateRandomCoverage | test_integer_falls_back_to_int                              | Pass   |
|       | test_property_random_coverage.py::TestGenerateRandomCoverage | test_number_falls_back_to_float                             | Pass   |
|       | test_property_random_coverage.py::TestGenerateRandomCoverage | test_string_with_options_picks_from_list                    | Pass   |
|       | test_property_random_coverage.py::TestGenerateRandomCoverage | test_string_with_options_returns_empty                      | Pass   |
|       | test_property_values.py::TestGauging                         | test_gauging_gauge_json_does_not_crash                      | Pass   |
|       | test_property_values.py::TestGauging                         | test_gauging_map_empty_when_no_gauge_json                   | Pass   |
|       | test_property_values.py::TestGauging                         | test_gauging_map_loaded_when_gauge_json_present             | Pass   |
|       | test_property_values.py::TestGauging                         | test_gauging_map_works_with_gauges_key                      | Pass   |
|       | test_property_values.py::TestLogging                         | test_log_debug_does_not_raise                               | Pass   |
|       | test_property_values.py::TestLogging                         | test_log_error_does_not_raise                               | Pass   |
|       | test_property_values.py::TestLogging                         | test_log_info_does_not_raise                                | Pass   |
|       | test_property_values.py::TestLogging                         | test_log_access_falls_back_gracefully                       | Pass   |
|       | test_property_values.py::TestLogging                         | test_log_warning_does_not_raise                             | Pass   |

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| Model | Module                                        | Test Name                                 | Status |
|-------|-----------------------------------------------|-------------------------------------------|--------|
|       | storm_intensity.py::TestSampleStorm           | testUndefinedCategories[catastrophic]     | Pass   |
|       | storm_intensity.py::TestSampleStorm           | testUndefinedCategories[extreme]          | Pass   |
|       | storm_intensity.py::TestSampleStorm           | testUndefinedCategories[moderate]         | Pass   |
|       | storm_intensity.py::TestSampleStorm           | testUndefinedCategories[severe]           | Pass   |
|       | storm_intensity.py::TestSampleStorm           | testSequenceTypes.sequence_type           | Pass   |
|       | storm_intensity.py::TestSampleStorm           | testUnknownCategoryUsesDefaultWeight      | Pass   |
|       | storm_intensity.py::TestSampleStorm           | testIncreasingPathSubsequentStorm         | Pass   |
|       | storm_intensity.py::TestSampleStorm           | testIntensityStormDownwardVariation       | Pass   |
|       | storm_intensity.py::TestSampleStorm           | testIntensityStormReturnsPositive         | Pass   |
|       | storm_intensity.py::TestSampleStorm           | testIntensityWithin20PercentRange         | Pass   |
|       | storm_intensity.py::TestSampleStorm           | testSubsequentStormMostlyUp               | Pass   |
|       | storm_intensity.py::TestSampleStorm           | testSubsequentStormPositive               | Pass   |
|       | storm_intensity.py::TestSampleStorm           | testIntensityLowBaseClampedTo01           | Pass   |
|       | storm_intensity.py::TestShouldGenerate        | testAtSevereDefinedCategories             | Pass   |
|       | storm_intensity.py::TestShouldGenerate        | testAtCatastrophicHighProbability         | Pass   |
|       | storm_intensity.py::TestShouldGenerate        | testAtSevereLowProbability                | Pass   |
|       | storm_intensity.py::TestShouldGenerate        | testReturnsCool                           | Pass   |
|       | storm_intensity.py::TestShouldGenerate        | testUnknownCategoryDefaultsTo015          | Pass   |
|       | test_batch_train.py::TestBatchTrain           | testClassifierReadyTrained                | Pass   |
|       | test_batch_train.py::TestBatchTrain           | testClassifiersPrintedAfterFirstGauge     | Pass   |
|       | test_batch_train.py::TestBatchTrain           | testClassifyingGaugeJson                  | Pass   |
|       | test_batch_train.py::TestBatchTrain           | testClassifyingSequences                  | Pass   |
|       | test_batch_train.py::TestBatchTrain           | testClassifierIdGauges                    | Pass   |
|       | test_batch_train.py::TestBatchTrain           | testClassifierTraining                    | Pass   |
|       | test_batch_train.py::TestBatchTrain           | testClassifierSuccessfulTraining          | Pass   |
|       | test_batch_train.py::TestBatchTrain           | testClassifierFailureIsCaught             | Pass   |
|       | test_batch_train.py::TestUpdateTestingStatus  | testNewSummary                            | Pass   |
|       | test_batch_train.py::TestUpdateTestingStatus  | testNewSummaryIntoExisting                | Pass   |
|       | test_batch_train.py::TestUpdateTestingStatus  | testPlaceNewExistingGauge                 | Pass   |
|       | test_batch_train.py::TestUpdateTestingStatus  | testSpentGaugeNotCountedAsTrained         | Pass   |
|       | test_classifier.py::TestPortfolioClassifier   | testPortfolioClassifierSummaryUnchanged   | Pass   |
|       | test_classifier.py::TestPortfolioClassifier   | testPortfolioClassifierTrainsAllGauges    | Pass   |
|       | test_classifier.py::TestPrintClassifierReady  | testTrainedNoMetricsPrinted               | Pass   |
|       | test_classifier.py::TestPrintClassifierReady  | testTrainedStatus                         | Pass   |
|       | test_classifier.py::TestPrintClassifierReady  | testTrainedShowsFeatureImportance         | Pass   |
|       | test_classifier.py::TestPrintClassifierReady  | testTrainedShowsLabelInfo                 | Pass   |
|       | test_classifier.py::TestPrintClassifierReady  | testTrainedStatusPrintsMetrics            | Pass   |
|       | test_classifier.py::TestPrintGaugeProgression | testGaugeInMarkerInOutput                 | Pass   |
|       | test_classifier.py::TestPrintGaugeProgression | testGaugeSequenceTypesReported            | Pass   |
|       | test_classifier.py::TestPrintGaugeProgression | testEmptyCategorySkipped                  | Pass   |
|       | test_classifier.py::TestPrintGaugeProgression | testGaugeIdContainsGaugeId                | Pass   |
|       | test_classifier.py::TestPrintGaugeProgression | testSevereMarkerInOutput                  | Pass   |
|       | test_classifier.py::TestPrintGaugeProgression | testWarningMarkerInOutput                 | Pass   |
|       | test_classifier.py::TestTrainGauge            | testFailClassifierAlertWhenNoSevereEvents | Pass   |

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| Model | Module                                 | Test Name                                                            | Status |
|-------|----------------------------------------|----------------------------------------------------------------------|--------|
|       | test_classifier.py::TestTrainGauge     | test_model_classifier_written                                        | Pass   |
|       | test_classifier.py::TestTrainGauge     | test_model_classifier_gauge_id                                       | Pass   |
|       | test_classifier.py::TestTrainGauge     | test_model_classifier_label_threshold                                | Pass   |
|       | test_classifier.py::TestTrainGauge     | test_model_classifier_result_dict                                    | Pass   |
|       | test_classifier.py::TestTrainGauge     | test_model_classifier_branch                                         | Pass   |
|       | test_gauge_parser.py::TestLoadGauge    | test_load_base_line_file_skipped                                     | Pass   |
|       | test_gauge_parser.py::TestLoadGauge    | test_load_base_line_directory                                        | Pass   |
|       | test_gauge_parser.py::TestLoadGauge    | test_load_base_line_directory                                        | Pass   |
|       | test_gauge_parser.py::TestLoadGauge    | test_load_base_line_mean_level_skipped                               | Pass   |
|       | test_gauge_parser.py::TestLoadGauge    | test_load_base_line_gauges                                           | Pass   |
|       | test_gauge_parser.py::TestLoadGauge    | test_load_base_line_gaugehd_files                                    | Pass   |
|       | test_gauge_parser.py::TestParseGauge   | test_parse_base_line_without_monthly_means                           | Pass   |
|       | test_gauge_parser.py::TestParseGauge   | test_parse_base_line                                                 | Pass   |
|       | test_gauge_parser.py::TestParseGauge   | test_parse_base_line_uses_heuristic                                  | Pass   |
|       | test_gauge_parser.py::TestSeasonalBase | test_seasonal_base_monthly_means                                     | Pass   |
|       | test_gauge_parser.py::TestSeasonalBase | test_seasonal_base_falls_back                                        | Pass   |
|       | test_gauge_parser.py::TestSeasonalBase | test_seasonal_base_monthly_means                                     | Pass   |
|       | test_gauge_parser.py::TestSeasonalBase | test_seasonal_base_monthly_means                                     | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_gauge_ids                                      | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_json_written                                   | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_num_gauges                                     | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_schema_version                                 | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_sequence_count                                 | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_sequences_file_loadable                                | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_sequences_json_written                                 | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_json_written                                   | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_non_negative                                   | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_positive                                       | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_gauges                                         | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_sequences                                      | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_dict                                           | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_warning                                        | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_present                                        | Pass   |
|       | test_generate.py::TestGenerateSummary  | test_generate_summary_alert                                          | Pass   |
|       | test_generate.py::TestMissingGauge     | test_missing_gauge_crash_when_gauge_json_absent                      | Pass   |
|       | test_generate.py::TestMissingGauge     | test_missing_gauge_zero_without_gauge_json                           | Pass   |
|       | test_generate.py::TestMissingGauge     | test_missing_gauge_sequences_still_written_without_gauge_json        | Pass   |
|       | test_generate.py::TestNoValidGauges    | test_no_valid_gauges_returns_gracefully_when_all_gauges_invalid      | Pass   |
|       | test_generate.py::TestNoValidGauges    | test_no_valid_gauges_sequences_still_written_when_all_gauges_invalid | Pass   |
|       | test_generate.py::TestVerboseProgress  | test_verbose_progress_false_no_mid_loop_output_at_small_count        | Pass   |
|       | test_generate.py::TestVerboseProgress  | test_verbose_progress_run_completes                                  | Pass   |
|       | test_helpers.py::TestBuildSummary      | test_counts_default_to_zero                                          | Pass   |
|       | test_helpers.py::TestBuildSummary      | test_num_gauges                                                      | Pass   |
|       | test_helpers.py::TestBuildSummary      | test_num_sequences                                                   | Pass   |

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| Model | Module                                               | Test Name                                               | Status |
|-------|------------------------------------------------------|---------------------------------------------------------|--------|
|       | test_helpers.py::TestBuildSummary                    | test_returns_required_keys                              | Pass   |
|       | test_helpers.py::TestExtractGauges                   | test_dict_format                                        | Pass   |
|       | test_helpers.py::TestExtractGauges                   | test_empty_list                                         | Pass   |
|       | test_helpers.py::TestExtractGauges                   | test_list_format                                        | Pass   |
|       | test_helpers.py::TestExtractGauges                   | test_missing_key_returns_empty                          | Pass   |
|       | test_helpers.py::TestParseGauges                     | test_base_level_derived_from_alert                      | Pass   |
|       | test_helpers.py::TestParseGauges                     | test_flood_alert                                        | Pass   |
|       | test_helpers.py::TestParseGauges                     | test_flood_warning                                      | Pass   |
|       | test_helpers.py::TestParseGauges                     | test_gauge_id                                           | Pass   |
|       | test_helpers.py::TestParseGauges                     | test_lat_lon                                            | Pass   |
|       | test_helpers.py::TestParseGauges                     | test_missing_alert_returns_none                         | Pass   |
|       | test_helpers.py::TestParseGauges                     | test_missing_gauge_id_returns_none                      | Pass   |
|       | test_helpers.py::TestParseGauges                     | test_missing_lat_returns_none                           | Pass   |
|       | test_helpers.py::TestParseGauges                     | test_severe_warning                                     | Pass   |
|       | test_helpers.py::TestParseGauges                     | test_valid_record_returns_dict                          | Pass   |
|       | test_helpers.py::TestParseGauges                     | test_warning_defaults_when_absent                       | Pass   |
|       | test_records.py::TestIntensityCategoryBatchGenerator | test_batch_generator_populates_category                 | Pass   |
|       | test_records.py::TestIntensityCategoryBatchGenerator | test_batch_generator_missing_category_defaults_to_empty | Pass   |
|       | test_records.py::TestIntensityCategoryBatchGenerator | test_batch_generator_round_trip                         | Pass   |
|       | test_records.py::TestIntensityCategoryBatchGenerator | test_batch_generator_sequences_have_category            | Pass   |
|       | test_records.py::TestIntensityCategoryBatchGenerator | test_batch_generator_includes_intensity_category        | Pass   |
|       | test_records.py::TestSequenceRetrieval               | test_alert_flags_length_matches_num_gauges              | Pass   |
|       | test_records.py::TestSequenceRetrieval               | test_alert_have_intensity_category                      | Pass   |
|       | test_records.py::TestSequenceRetrieval               | test_alert_have_sequence_id                             | Pass   |
|       | test_records.py::TestSequenceRetrieval               | test_alert_have_sequence_type                           | Pass   |
|       | test_records.py::TestSequenceRetrieval               | test_alert_have_total_precip                            | Pass   |
|       | test_records.py::TestSequenceRetrieval               | test_peak_hours_in_valid_range                          | Pass   |
|       | test_records.py::TestSequenceRetrieval               | test_peak_hours_length_matches_num_gauges               | Pass   |
|       | test_records.py::TestSequenceRetrieval               | test_peak_hours_length_matches_num_gauges               | Pass   |
|       | test_records.py::TestSequenceRetrieval               | test_peak_hours_positive                                | Pass   |
|       | test_records.py::TestSequenceRetrieval               | test_severe_flags_length_matches_num_gauges             | Pass   |
|       | test_records.py::TestSequenceRetrieval               | test_severe_implies_warning                             | Pass   |
|       | test_records.py::TestSequenceRetrieval               | test_starting_flags_length_matches_num_gauges           | Pass   |
|       | test_records.py::TestSequenceRetrieval               | test_starting_implies_alert                             | Pass   |
|       | test_single_gauge.py::TestSingleGaugeMonitor         | test_monitor_message_lists_available_ids                | Pass   |
|       | test_single_gauge.py::TestSingleGaugeMonitor         | test_invalid_gauge_id_raises                            | Pass   |
|       | test_single_gauge.py::TestSingleGaugeMonitor         | test_monitor_file_gauge_ids                             | Pass   |
|       | test_single_gauge.py::TestSingleGaugeMonitor         | test_monitor_file_peaks_length_one                      | Pass   |
|       | test_single_gauge.py::TestSingleGaugeMonitor         | test_monitor_file_schema_version                        | Pass   |
|       | test_single_gauge.py::TestSingleGaugeMonitor         | test_monitor_file_sequence_count                        | Pass   |
|       | test_single_gauge.py::TestSingleGaugeMonitor         | test_monitor_output_file_written                        | Pass   |
|       | test_single_gauge.py::TestSingleGaugeMonitor         | test_monitor_num_gauges_is_one                          | Pass   |
|       | test_single_gauge.py::TestSingleGaugeClassifier      | test_with_classifier_result_dict                        | Pass   |
|       | test_single_gauge.py::TestSingleGaugeClassifier      | test_with_classifier_train_classifier                   | Pass   |

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| Model | Module                             | Test Name                                 | Status |
|-------|------------------------------------|-------------------------------------------|--------|
|       | test_sequence_response.compound.py | TestCompoundingStormGrowthForShortGap     | Pass   |
|       | test_sequence_response.compound.py | TestFastDrainpageDiscontinuity            | Pass   |
|       | test_sequence_response.compound.py | TestFirstGapnonbaseLineinrecovery         | Pass   |
|       | test_sequence_response.compound.py | TestFirstOnpeakrisingdistorm1majority     | Pass   |
|       | test_sequence_response.compound.py | TestPeakAmplitudeStormPeaks               | Pass   |
|       | test_sequence_response.compound.py | TestPeakIndicesStartOfstormindices        | Pass   |
|       | test_sequence_response.compound.py | TestResponsePeakPerastorm                 | Pass   |
|       | test_sequence_response.hydro.py    | TestAntecedentwetnessIncreaseResponsepeak | Pass   |
|       | test_sequence_response.hydro.py    | TestDrainageSwindowGaugedisplacement      | Pass   |
|       | test_sequence_response.hydro.py    | TestHighertensityGaigherResponse          | Pass   |
|       | test_sequence_response.hydro.py    | TestIsolatedSequentialGaugesforfirststorm | Pass   |
|       | test_sequence_response.hydro.py    | TestLevelabovebaseGaugedistorm            | Pass   |
|       | test_sequence_response.hydro.py    | TestNonnegativeLeaveGaugeResponse         | Pass   |
|       | test_sequence_response.hydro.py    | TestNonpeteSequentialGaugeresponse        | Pass   |
|       | test_sequence_response.hydro.py    | TestOutputShapeGaugeResponse              | Pass   |
|       | test_sequence_response.hydro.py    | TestRecoverySequenceAlignByhour_168       | Pass   |
|       | test_sequence_response.hydro.py    | TestHybridgenModRunSeries                 | Pass   |
|       | test_sequence_response.hydro.py    | TestHybridgenModRfrSeriesoutput           | Pass   |
|       | test_sequence_response.hydro.py    | TesthydraulicmodelRunSeries               | Pass   |
|       | test_sequence_response.hydro.py    | TestHydrologyModelBaseSeries              | Pass   |
|       | test_sequence_response.hydro.py    | TestHydrologyModelRunSeries               | Pass   |
|       | test_sequence_response.hydro.py    | TestHydrologyModelwithnostrain            | Pass   |
|       | test_sequence_response.hydro.py    | TestHydrolstartModelPreip_noquickflow     | Pass   |
|       | test_sequence_response.hydro.py    | TestGroundwaterAddStimulatesfromsoil      | Pass   |
|       | test_sequence_response.hydro.py    | TestHeightlogProbRepeatedaszero           | Pass   |
|       | test_sequence_response.hydro.py    | TestHydrologyModelSepoilmoisture          | Pass   |
|       | test_sequence_response.hydro.py    | TestHydrologyModelStopickflow             | Pass   |
|       | test_sequence_response.hydro.py    | TestHydrologyModelTrapswithoutrain        | Pass   |
|       | test_sequence_response.hydro.py    | TestHydrologyModelStopickflow             | Pass   |
|       | test_sequence_response.hydro.py    | TestModerateSinglestormpeak               | Pass   |
|       | test_sequence_response.hydro.py    | TestMajorProportionofstorms               | Pass   |
|       | test_sequence_response.hydro.py    | TestMakeGainSeries                        | Pass   |
|       | test_sequence_response.hydro.py    | TestMultiTrapSeries                       | Pass   |
|       | test_sequence_response.hydro.py    | TestMetalPrepSpringserved                 | Pass   |
|       | test_spatial_correlation_matrix.py | TestCholesky_cache_key_rounding           | Pass   |
|       | test_spatial_correlation_matrix.py | TestCholesky_cached                       | Pass   |
|       | test_spatial_correlation_matrix.py | TestCholesky_lowertriangular              | Pass   |
|       | test_spatial_correlation_matrix.py | TestCholesky_reconstruction               | Pass   |
|       | test_spatial_correlation_matrix.py | TestCholesky_reconstruction               | Pass   |
|       | test_spatial_correlation_matrix.py | TestCovarianceMatrix                      | Pass   |
|       | test_spatial_correlation_matrix.py | TestCorrelatingAtHigherCorrelation        | Pass   |
|       | test_spatial_correlation_matrix.py | TestCorrelatedAsOffdiagonal               | Pass   |
|       | test_spatial_correlation_matrix.py | TestOffdiagonalNaninopeninterval          | Pass   |
|       | test_spatial_correlation_matrix.py | TestPositiveDefinite                      | Pass   |

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| Model | Module                             | Test Name                                             | Status |
|-------|------------------------------------|-------------------------------------------------------|--------|
|       | test_spatial_correlation_matrix.py | TestSymmetricMatrix                                   | Pass   |
|       | test_spatial_correlation_matrix.py | TestSemiDefiniteMatrix                                | Pass   |
|       | test_spatial_correlation_matrix.py | TestHighDensityLongerRange                            | Pass   |
|       | test_spatial_correlation_matrix.py | TestHighDensityNoReduction                            | Pass   |
|       | test_spatial_correlation_matrix.py | TestEffectiveRangeDensityEqualsBase                   | Pass   |
|       | test_spatial_correlation_matrix.py | TestEffectiveRangeWithRho                             | Pass   |
|       | test_spatial_coverage_b6.py        | TestCirclesFallbackCallbackAddsJitter                 | Pass   |
|       | test_spatial_coverage_b6.py        | TestFromGaugesRangeToPortfolioFile                    | Pass   |
|       | test_spatial_coverage_b6.py        | TestFromSampleWithNoRange                             | Pass   |
|       | test_spatial_multipliers.py        | TestApplyToSequenceNegative                           | Pass   |
|       | test_spatial_multipliers.py        | TestApplyToSequenceMeanPreservedPerHour               | Pass   |
|       | test_spatial_multipliers.py        | TestApplyToSequenceShape                              | Pass   |
|       | test_spatial_multipliers.py        | TestApplyToSequenceVariationPresent                   | Pass   |
|       | test_spatial_multipliers.py        | TestApplyToSequencePrecipitationConserved             | Pass   |
|       | test_spatial_multipliers.py        | TestApplyToSequencePrecipHoursRemainZero              | Pass   |
|       | test_spatial_multipliers.py        | TestSampleMultipliersPositive                         | Pass   |
|       | test_spatial_multipliers.py        | TestSampleMultipliersCorrelationConsistentWithModel   | Pass   |
|       | test_spatial_multipliers.py        | TestSampleMultipliersDensityLowerVariance             | Pass   |
|       | test_spatial_multipliers.py        | TestSampleMultipliersOne                              | Pass   |
|       | test_spatial_multipliers.py        | TestSampleMultipliersLognormalMean                    | Pass   |
|       | test_spatial_multipliers.py        | TestSampleMultipliersReproducibleWithSeed             | Pass   |
|       | test_spatial_multipliers.py        | TestSampleMultipliersShape                            | Pass   |
|       | test_spatial_params_distance.py    | TestConfigFileConstantsValid                          | Pass   |
|       | test_spatial_params_distance.py    | TestConfigFileStructure                               | Pass   |
|       | test_spatial_params_distance.py    | TestConfigFileTripParams                              | Pass   |
|       | test_spatial_params_distance.py    | TestSaveFileCreatesFile                               | Pass   |
|       | test_spatial_params_distance.py    | TestDiagonalAllZero                                   | Pass   |
|       | test_spatial_params_distance.py    | TestDistancesMatrixKnownDistance                      | Pass   |
|       | test_spatial_params_distance.py    | TestDistancesMatrixFixedDiagonal                      | Pass   |
|       | test_spatial_params_distance.py    | TestDistanceMatrix                                    | Pass   |
|       | test_spatial_params_distance.py    | TestSymmetricMatrix                                   | Pass   |
|       | test_spatial_params_distance.py    | TestDistanceMatrixDistancePlausible                   | Pass   |
|       | test_spatial_params_distance.py    | TestDistanceMatrixGauges                              | Pass   |
|       | test_spatial_params_distance.py    | TestFromGaugesPortfolioFile                           | Pass   |
|       | test_spatial_params_distance.py    | TestFromGaugesPortfolioFile                           | Pass   |
|       | test_spatial_params_distance.py    | TestFromGaugesPortfolioFile                           | Pass   |
|       | test_spatial_params_distance.py    | TestDefaultCorrelationParams                          | Pass   |
|       | test_spatial_params_distance.py    | TestSampleCorrelationParams                           | Pass   |
|       | test_spatial_params_distance.py    | TestSampleCorrelationParams                           | Pass   |
|       | test_import_isolation.py           | TestBookRequirementsExportLibWhenQuantLibPresent      | Pass   |
|       | test_import_isolation.py           | TestBookRequirementsQuantLibImportableWithoutQuantLib | Pass   |
|       | test_import_isolation.py           | TestBookRequirementsNameLibCentralAreasCount          | Pass   |
|       | test_import_isolation.py           | TestBookRequirementsQuantLibRequiresQuantLib          | Pass   |
|       | test_import_isolation.py           | TestBookRequirementsNameLibInsideBlotterBlock         | Pass   |

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| Model | Module                                                 | Test Name                                                | Status |
|-------|--------------------------------------------------------|----------------------------------------------------------|--------|
|       | cli.py::TestOptionalLoading                            | test_gauge_id_filters_data                               | Pass   |
|       | cli.py::TestOptionalLoading                            | test_timeseries_file_is_loaded                           | Pass   |
|       | cli.py::TestReportGeneration                           | test_monitoring_without_timeseries_hits                  | Pass   |
|       | cli.py::TestReportGeneration                           | test_report_type_analysis                                | Pass   |
|       | cli.py::TestReportGeneration                           | test_report_type_basic                                   | Pass   |
|       | cli.py::TestReportGeneration                           | test_report_type_full_hits_else_branch                   | Pass   |
|       | cli.py::TestReportGeneration                           | test_report_type_monitoring_with_timeseries              | Pass   |
|       | cli.py::TestReportGeneration                           | test_success_logging_executes                            | Pass   |
|       | convenience.py::TestEdgeCases                          | test_completely_empty_gauge_data                         | Pass   |
|       | convenience.py::TestEdgeCases                          | test_gauge_data_with_hazard_curve_and_timeseries         | Pass   |
|       | convenience.py::TestEdgeCases                          | test_none_flood_gauge_header                             | Pass   |
|       | convenience.py::TestEdgeCases                          | test_output_is_non_empty_bytes                           | Pass   |
|       | convenience.py::TestEdgeCases                          | test_report_with_all_pages                               | Pass   |
|       | convenience.py::TestEdgeCases                          | test_single_page_report                                  | Pass   |
|       | convenience.py::TestFindGaugeByAdiction                | test_adiction_without_known_key_raises                   | Pass   |
|       | convenience.py::TestFindGaugeByAffiliation             | test_affiliation_gauges_key_first_gauge                  | Pass   |
|       | convenience.py::TestFindGaugeByAffiliation             | test_affiliation_gauges_key_last_gauge                   | Pass   |
|       | convenience.py::TestFindGaugeByAggregation             | test_aggregation_not_found_raises                        | Pass   |
|       | convenience.py::TestFindGaugeByAlert                   | test_alert_no_input_raises                               | Pass   |
|       | convenience.py::TestFindGaugeByAlert                   | test_alert_input_used_directly                           | Pass   |
|       | convenience.py::TestFindGaugeByAnalysis                | test_analysis_entry_missing_flood_gauge_key_skipped      | Pass   |
|       | convenience.py::TestFindGaugeByAnalysis                | test_analysis_entry_with_none_skipped                    | Pass   |
|       | convenience.py::TestFindGaugeByAscription              | test_ascription_flood_gauge_dict_returns_itself          | Pass   |
|       | convenience.py::TestFindGaugeByAscription              | test_ascription_input_raises                             | Pass   |
|       | convenience.py::TestGenerateGaugeReportModulePenchills | test_generate_analytically                               | Pass   |
|       | convenience.py::TestGenerateGaugeReportModulePenchills | test_generate_analytically_returns_path                  | Pass   |
|       | convenience.py::TestGenerateGaugeReportModulePenchills | test_generate_analytically_handles_exception             | Pass   |
|       | convenience.py::TestGenerateGaugeReportModulePenchills | test_generate_analytically_open_succeeds                 | Pass   |
|       | convenience.py::TestGenerateGaugeReportModulePenchills | test_generate_basic                                      | Pass   |
|       | convenience.py::TestGenerateGaugeReportMonitoringType  | test_generate_monitoring_type_with_timeseries_called     | Pass   |
|       | convenience.py::TestGenerateGaugeReportMonitoringType  | test_generate_monitoring_type_with_timeseries_falls_back | Pass   |
|       | convenience.py::TestGenerateGaugeReportModulePenchills | test_generate_report                                     | Pass   |
|       | convenience.py::TestGenerateGaugeReportModulePenchills | test_generate_report                                     | Pass   |
|       | elements.py::TestGenerateElement                       | test_empty_page_list_returns_empty                       | Pass   |
|       | elements.py::TestGenerateElement                       | test_exception_in_page_continues                         | Pass   |
|       | elements.py::TestGenerateElement                       | test_first_page_no_page_break                            | Pass   |
|       | elements.py::TestGenerateElement                       | test_mixed_valid_and_unknown_pages                       | Pass   |
|       | elements.py::TestGenerateElement                       | test_multiple_pages_have_page_break                      | Pass   |
|       | elements.py::TestGenerateElement                       | test_none_timeseries_passed_to_page                      | Pass   |
|       | elements.py::TestGenerateElement                       | test_returns_list                                        | Pass   |
|       | elements.py::TestGenerateElement                       | test_timeseries_passed_to_page                           | Pass   |
|       | elements.py::TestGenerateElement                       | test_unknown_page_skipped                                | Pass   |
|       | elements.py::TestGenerateReport                        | test_all_unknown_pages_still_creates_pdf                 | Pass   |
|       | elements.py::TestGenerateReport                        | test_auto_select_called_when_pages_none                  | Pass   |

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| Model | Module                                         | Test Name                                           | Status |
|-------|------------------------------------------------|-----------------------------------------------------|--------|
|       | elements.py::TestGenerateReport                | test_auto_select_not_called_when_pages_provided     | Pass   |
|       | elements.py::TestGenerateReport                | test_custom_output_filename                         | Pass   |
|       | elements.py::TestGenerateReport                | test_custom_pages_to_include                        | Pass   |
|       | elements.py::TestGenerateReport                | test_empty_gauge_data                               | Pass   |
|       | elements.py::TestGenerateReport                | test_pdf_exists                                     | Pass   |
|       | elements.py::TestGenerateReport                | test_pdf_is_bytes                                   | Pass   |
|       | elements.py::TestGenerateReport                | test_pdf_non_empty                                  | Pass   |
|       | elements.py::TestGenerateReport                | test_pdf_suffix                                     | Pass   |
|       | elements.py::TestGenerateReport                | test_returns_path                                   | Pass   |
|       | elements.py::TestGenerateReport                | test_with_timeseries_data                           | Pass   |
|       | init.py::TestAutoSelectPages                   | test_all_pages_exist                                | Pass   |
|       | init.py::TestAutoSelectPages                   | test_always_includes_gauge_info_pages               | Pass   |
|       | init.py::TestAutoSelectPages                   | test_always_includes_operational_pages              | Pass   |
|       | init.py::TestAutoSelectPages                   | test_always_includes_summary_pages                  | Pass   |
|       | init.py::TestAutoSelectPages                   | test_returns_list                                   | Pass   |
|       | init.py::TestAutoSelectPages                   | test_with_hazard_curve_includes_analysis_pages      | Pass   |
|       | init.py::TestAutoSelectPages                   | test_with_timeseries_includes_analysis_pages        | Pass   |
|       | init.py::TestAutoSelectPages                   | test_without_timeseries_or_hazard_no_analysis_pages | Pass   |
|       | init.py::TestGaugeReportGenerator              | test_categories_dict_initialized                    | Pass   |
|       | init.py::TestGaugeReportGenerator              | test_explicit_output_dir_path                       | Pass   |
|       | init.py::TestGaugeReportGenerator              | test_explicit_output_dir_string                     | Pass   |
|       | init.py::TestGaugeReportGenerator              | test_output_dir_uses_config                         | Pass   |
|       | init.py::TestGaugeReportGenerator              | test_output_dir_created                             | Pass   |
|       | init.py::TestGaugeReportGenerator              | test_pages_dict_initialized                         | Pass   |
|       | init.py::TestGenerateFilename                  | test_contains_gauge_id                              | Pass   |
|       | init.py::TestGenerateFilename                  | test_ends_with_pdf                                  | Pass   |
|       | init.py::TestGenerateFilename                  | test_missing_gauge_id_uses_unknown                  | Pass   |
|       | init.py::TestGenerateFilename                  | test_partial_path_missing_gauge_id                  | Pass   |
|       | init.py::TestGenerateFilename                  | test_partial_path_missing_header                    | Pass   |
|       | init.py::TestGenerateFilename                  | test_returns_string                                 | Pass   |
|       | init.py::TestGenerateFilename                  | test_timestamp_in_filename                          | Pass   |
|       | specialised.py::TestHeaderFooter               | test_header_footer_callable                         | Pass   |
|       | specialised.py::TestHeaderFooter               | test_header_footer_draws_confidential_footer        | Pass   |
|       | specialised.py::TestHeaderFooter               | test_header_footer_draws_header_strings             | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_analysis_report_custom_filename       | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_analysis_report_exists                | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_analysis_report_with_timeseries       | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_basic_report_custom_filename          | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_basic_report_exists                   | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_basic_report_non_empty                | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_basic_with_timeseries                 | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_monitoring_report_custom_filename     | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_monitoring_report_exists              | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_monitoring_report_no_timeseries       | Pass   |

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| Model | Module                                                | Test Name                                             | Status |
|-------|-------------------------------------------------------|-------------------------------------------------------|--------|
|       | specialised.py::TestUtilityMethods                    | test_get_page_categories_contains_expected_keys       | Fail   |
|       | specialised.py::TestUtilityMethods                    | test_get_page_categories_returns_copy                 | Pass   |
|       | specialised.py::TestUtilityMethods                    | test_get_page_categories_returns_dict                 | Pass   |
|       | specialised.py::TestUtilityMethods                    | test_list_available_pages_contains_expected           | Fail   |
|       | specialised.py::TestUtilityMethods                    | test_list_available_pages_non_empty                   | Pass   |
|       | specialised.py::TestUtilityMethods                    | test_list_available_pages_returns_list                | Pass   |
|       | specialised.py::TestUtilityMethods                    | test_validate_pages_all_invalid                       | Pass   |
|       | specialised.py::TestUtilityMethods                    | test_validate_pages_all_valid                         | Pass   |
|       | specialised.py::TestUtilityMethods                    | test_validate_pages_empty_list                        | Pass   |
|       | specialised.py::TestUtilityMethods                    | test_validate_pages_mixed                             | Pass   |
|       | metrics.py::TestGaugeMetrics                          | test_level_range                                      | Pass   |
|       | metrics.py::TestGaugeMetrics                          | test_max_level                                        | Pass   |
|       | metrics.py::TestGaugeMetrics                          | test_mean_level                                       | Pass   |
|       | metrics.py::TestGaugeMetrics                          | test_min_level                                        | Pass   |
|       | metrics.py::TestGaugeMetrics                          | test_rate_of_rise                                     | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_title_overview_page_does_not_crash               | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_empty_overview_page_does_not_crash               | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_title_overview_inv_output                        | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_title_overview_wide_section_header               | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_title_overview_narrow_section_header             | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_title_overview_card_missing                      | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_title_overview_card_present                      | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_title_overview_hide_page                         | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_title_overview_installation_page_calculated      | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_title_overview_hold_page_does_not_crash          | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_title_overview_fetch_one_list_page               | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_title_overview_varying_page_present              | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_title_overview_unknown_gauge_fallback            | Pass   |
|       | page_01_title_overview.py::TestGaugeTitleOverviewPage | test_title_overview_with_overview_page_does_not_crash | Pass   |
|       | page_02_sensor_details.py::TestGaugeSensorDetailsPage | test_sensor_details_page_does_not_crash               | Pass   |
|       | page_02_sensor_details.py::TestGaugeSensorDetailsPage | test_sensor_details_page_subtitle                     | Pass   |
|       | page_02_sensor_details.py::TestGaugeSensorDetailsPage | test_sensor_details_page_on_section                   | Pass   |
|       | page_02_sensor_details.py::TestGaugeSensorDetailsPage | test_sensor_details_page_missing                      | Pass   |
|       | page_02_sensor_details.py::TestGaugeSensorDetailsPage | test_sensor_details_page_present                      | Pass   |
|       | page_02_sensor_details.py::TestGaugeSensorDetailsPage | test_sensor_details_page                              | Pass   |
|       | page_02_sensor_details.py::TestGaugeSensorDetailsPage | test_sensor_details_page_statistics_section           | Pass   |
|       | page_02_sensor_details.py::TestGaugeSensorDetailsPage | test_sensor_details_page_information_section          | Pass   |
|       | page_02_sensor_details.py::TestGaugeSensorDetailsPage | test_sensor_details_page_configuration_section        | Pass   |
|       | page_02_sensor_details.py::TestGaugeSensorDetailsPage | test_sensor_details_page_does_not_crash               | Pass   |
|       | page_02_sensor_details.py::TestGaugeSensorDetailsPage | test_sensor_details_page                              | Pass   |
|       | page_02_sensor_details.py::TestGaugeSensorDetailsPage | test_sensor_details_page                              | Pass   |
|       | page_02_sensor_details.py::TestGaugeSensorDetailsPage | test_sensor_details_page_does_not_crash               | Pass   |
|       | page_03_location.py::TestGaugeLocationPage            | test_location_page_close_to_thames                    | Pass   |
|       | page_03_location.py::TestGaugeLocationPage            | test_location_page_zero_on_thames                     | Pass   |

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| Model | Module                                | Test Name                                             | Status |
|-------|---------------------------------------|-------------------------------------------------------|--------|
|       | page_03_location.py::TestGauge        | testEmptyPageData_does_not_crash                      | Pass   |
|       | page_03_location.py::TestGauge        | testEmptyPageRisk_section                             | Pass   |
|       | page_03_location.py::TestGauge        | testEmptyPageGraphic_coordinates_section              | Pass   |
|       | page_03_location.py::TestGauge        | testEmptyPageTables                                   | Pass   |
|       | page_03_location.py::TestGauge        | testEmptyPageCentral_london                           | Pass   |
|       | page_03_location.py::TestGauge        | testEmptyPageNorth_london                             | Pass   |
|       | page_03_location.py::TestGauge        | testEmptyPageSouth_london                             | Pass   |
|       | page_03_location.py::TestGauge        | testEmptyPageThames_info_does_not_crash               | Pass   |
|       | page_03_location.py::TestGauge        | testEmptyPageZoning_summary_section                   | Pass   |
|       | page_03_location.py::TestGauge        | testEmptyPageList                                     | Pass   |
|       | page_03_location.py::TestGauge        | testEmptyPageCategory_in_location_profile             | Pass   |
|       | page_03_location.py::TestGauge        | testEmptyPageScore_formatted_as_out_of_10             | Pass   |
|       | page_03_location.py::TestGauge        | testEmptyPageContext_section                          | Pass   |
|       | page_03_location.py::TestGauge        | testEmptyPagePresent                                  | Pass   |
|       | page_04_measurements.py::TestGauge    | testEmptyPageData_does_not_crash                      | Pass   |
|       | page_04_measurements.py::TestGauge    | testEmptyPageList_does_not_crash                      | Pass   |
|       | page_04_measurements.py::TestGauge    | testEmptyPageWithMissing                              | Pass   |
|       | page_04_measurements.py::TestGauge    | testEmptyPageWithPresent                              | Pass   |
|       | page_04_measurements.py::TestGauge    | testEmptyPageTables                                   | Pass   |
|       | page_04_measurements.py::TestGauge    | testEmptyPageHistoricalStatistics_section             | Pass   |
|       | page_04_measurements.py::TestGauge    | testEmptyPageConfiguration_section                    | Pass   |
|       | page_04_measurements.py::TestGauge    | testEmptyPageShows_message                            | Pass   |
|       | page_04_measurements.py::TestGauge    | testEmptyPageStatus_section                           | Pass   |
|       | page_04_measurements.py::TestGauge    | testEmptyPageList                                     | Pass   |
|       | page_04_measurements.py::TestGauge    | testEmptyPagePresent                                  | Pass   |
|       | page_04_measurements.py::TestGauge    | testEmptyPageHas_extra_table                          | Pass   |
|       | page_04_measurements.py::TestGauge    | testEmptyPageShows_current_readings                   | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testFloodStageLevel_at_alert                          | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testFloodStageLevel_at_severe                         | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testFloodStageLevel_at_warning                        | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testFloodStageLevel_below_alert_is_normal             | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testEmptyStagePage_does_not_crash                     | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testEmptyStagePageMissing                             | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testEmptyStagePagePresent                             | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testEmptyStagePage                                    | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testEmptyStagePageExceedance_section                  | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testEmptyStagePageShows_message                       | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testEmptyStagePage                                    | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testEmptyStagePageAnalysis_section                    | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testEmptyStagePageDifferences_computed                | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testEmptyStagePagePresent                             | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testEmptyStagePagePolys_section                       | Pass   |
|       | page_05_flood_stages.py::TestGauge    | testEmptyStagePageSeries_shows_current_status_section | Pass   |
|       | page_06_risk_assessment.py::TestGauge | testDistanceToThames_label                            | Pass   |

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| Model | Module                                                                | Test Name                                               | Status |
|-------|-----------------------------------------------------------------------|---------------------------------------------------------|--------|
|       | page_06_risk_assessment.py::TestDistanceToThreat                      | Distance highest risk                                   | Pass   |
|       | page_06_risk_assessment.py::TestDistanceToThreat                      | Distance missing key                                    | Pass   |
|       | page_06_risk_assessment.py::TestDistanceToThreat                      | Distance gauge data returns error                       | Pass   |
|       | page_06_risk_assessment.py::TestFrequencyExceedances                  | Frequency Exceedances high                              | Pass   |
|       | page_06_risk_assessment.py::TestFrequencyExceedances                  | Frequency Exceedance                                    | Pass   |
|       | page_06_risk_assessment.py::TestFrequencyExceedances                  | Frequency Exceedances moderate                          | Pass   |
|       | page_06_risk_assessment.py::TestFrequencyExceedances                  | Frequency Exceedances                                   | Pass   |
|       | page_06_risk_assessment.py::TestGenerateRiskAssessmentPageBasic       | Generate Risk Assessment Page Basic                     | Pass   |
|       | page_06_risk_assessment.py::TestGenerateRiskAssessmentPagePlaceholder | Generate Risk Assessment Page Placeholder               | Pass   |
|       | page_06_risk_assessment.py::TestGenerateRiskAssessmentPageBasic       | Generate Risk Assessment Page Basic                     | Pass   |
|       | page_06_risk_assessment.py::TestHistoricalHigh                        | Historical High severe label                            | Pass   |
|       | page_06_risk_assessment.py::TestHistoricalHigh                        | Historical High severe level                            | Pass   |
|       | page_06_risk_assessment.py::TestLastKnownData                         | Last Known Data string gracefully skipped               | Pass   |
|       | page_06_risk_assessment.py::TestLastKnownData                         | Last Known Data recent activity                         | Pass   |
|       | page_06_risk_assessment.py::TestLastKnownData                         | Last Known Data activity                                | Pass   |
|       | page_06_risk_assessment.py::TestLastKnownData                         | Last Known Data activity within 1 year                  | Pass   |
|       | page_06_risk_assessment.py::TestRiskScoreBrackethigh                  | Risk Score Bracket high label                           | Pass   |
|       | page_06_risk_assessment.py::TestRiskScoreBracketlow                   | Risk Score Bracket low label                            | Pass   |
|       | page_06_risk_assessment.py::TestRiskScoreBracketrisk                  | Risk Score Bracket risk label                           | Pass   |
|       | page_06_risk_assessment.py::TestRiskScoreBracketscore                 | Risk Score Bracket score formatted                      | Pass   |
|       | page_06_risk_assessment.py::TestRiskScoreBrackethigh                  | Risk Score Bracket high risk label                      | Pass   |
|       | page_06_risk_assessment.py::TestThresholdAndAlert                     | Threshold And Alert high exceeds severe by less than 1m | Pass   |
|       | page_06_risk_assessment.py::TestThresholdAndAlert                     | Threshold And Alert high exceeds severe by more than 1m | Pass   |
|       | page_06_risk_assessment.py::TestThresholdAndAlert                     | Threshold And Alert high within severe                  | Pass   |
|       | page_06_risk_assessment.py::TestThresholdAndAlert                     | Threshold And Alert range                               | Pass   |
|       | page_06_risk_assessment.py::TestThresholdAndAlert                     | Threshold And Alert range                               | Pass   |
|       | page_06_risk_assessment.py::TestThresholdAndAlert                     | Threshold And Alert range                               | Pass   |
|       | page_06_risk_assessment.py::TestThresholdAndAlert                     | Threshold And Alert status                              | Pass   |
|       | page_06_risk_assessment.py::TestThresholdAndAlert                     | Threshold And Alert status                              | Pass   |
|       | page_06_risk_assessment.py::TestThresholdAndAlert                     | Threshold And Alert status                              | Pass   |
|       | page_06_risk_assessment.py::TestThresholdAndAlert                     | Threshold And Alert status with no level                | Pass   |
|       | page_06_risk_assessment.py::TestThresholdAndAlert                     | Threshold And Alert status with no readings             | Pass   |
|       | page_06_risk_assessment.py::TestThresholdAndAlert                     | Threshold And Alert status                              | Pass   |
|       | page_07_data_summary.py::TestGenerateDataSummaryPage                  | Generate Data Summary Page                              | Pass   |
|       | page_07_data_summary.py::TestGenerateDataSummaryPage                  | Generate Data Summary Page does not crash               | Pass   |
|       | page_07_data_summary.py::TestGenerateDataSummaryPage                  | Generate Data Summary Page significance                 | Pass   |
|       | page_07_data_summary.py::TestGenerateDataSummaryPage                  | Generate Data Summary Page significance                 | Pass   |
|       | page_07_data_summary.py::TestGenerateDataSummaryPage                  | Generate Data Summary Page exceedances conclusion       | Pass   |
|       | page_07_data_summary.py::TestGenerateDataSummaryPage                  | Generate Data Summary Page significance                 | Pass   |
|       | page_07_data_summary.py::TestGenerateDataSummaryPage                  | Generate Data Summary Page summary section              | Pass   |
|       | page_07_data_summary.py::TestGenerateDataSummaryPage                  | Generate Data Summary Page missing                      | Pass   |
|       | page_07_data_summary.py::TestGenerateDataSummaryPage                  | Generate Data Summary Page present                      | Pass   |
|       | page_07_data_summary.py::TestGenerateDataSummaryPage                  | Generate Data Summary Page missing                      | Pass   |
|       | page_07_data_summary.py::TestGenerateDataSummaryPage                  | Generate Data Summary Page present                      | Pass   |

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| Model                                                                           | Module                                                                          | Test Name                                         | Status |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------|--------|
| GaugeDataCables                                                                 | page_07_data_summary.py::TestGaugeDataSummaryPage                               | GaugeDataSummaryPage                              | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeHighRiskConclusion                            | GaugeHighRiskConclusion                           | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeDataSummaryPage_severe_significance           | GaugeDataSummaryPage_severe_significance          | Fail   |
|                                                                                 | page_07_data_summary.py::TestGaugeDataSummaryPage_range_significance            | GaugeDataSummaryPage_range_significance           | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeDataSummaryPage_hazard_section                | GaugeDataSummaryPage_hazard_section               | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeLowRiskConclusion                             | GaugeLowRiskConclusion                            | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeDataSummaryPage_conclusion                    | GaugeDataSummaryPage_conclusion                   | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeOperationalStatusConclusion                   | GaugeOperationalStatusConclusion                  | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeOperationalStatusSummarySection               | GaugeOperationalStatusSummarySection              | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeDeploymentDesignsSection                      | GaugeDeploymentDesignsSection                     | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeDataSummaryPage                               | GaugeDataSummaryPage                              | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeDataSeriesDailyConclusion                     | GaugeDataSeriesDailyConclusion                    | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeDataSeriesWeeklyConclusion                    | GaugeDataSeriesWeeklyConclusion                   | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeDataSeriesMonthlyConclusion                   | GaugeDataSeriesMonthlyConclusion                  | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeDataSeriesYearlyConclusion                    | GaugeDataSeriesYearlyConclusion                   | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeDataSeriesWarningConclusion                   | GaugeDataSeriesWarningConclusion                  | Pass   |
|                                                                                 | page_07_data_summary.py::TestGaugeDataSummaryPage                               | GaugeDataSummaryPage                              | Pass   |
|                                                                                 | page_08_flood_history.py::TestGaugeFloodHistoryPage_no_dates_fallback           | GaugeFloodHistoryPage_no_dates_fallback           | Pass   |
|                                                                                 | page_08_flood_history.py::TestGaugeFloodHistoryPage_with_thresholds             | GaugeFloodHistoryPage_with_thresholds             | Pass   |
|                                                                                 | page_08_flood_history.py::TestGaugeFloodHistoryPage_responses_shows_message     | GaugeFloodHistoryPage_responses_shows_message     | Pass   |
|                                                                                 | page_08_flood_history.py::TestGaugeFloodHistoryPage_data_shows_no_data_message  | GaugeFloodHistoryPage_data_shows_no_data_message  | Pass   |
|                                                                                 | page_08_flood_history.py::TestGaugeFloodHistoryPage_no_exceedances              | GaugeFloodHistoryPage_no_exceedances              | Pass   |
|                                                                                 | page_08_flood_history.py::TestGaugeFloodHistoryPage_with_exceedances            | GaugeFloodHistoryPage_with_exceedances            | Pass   |
|                                                                                 | page_08_flood_history.py::TestGaugeFloodHistoryPage                             | GaugeFloodHistoryPage                             | Pass   |
|                                                                                 | page_08_flood_history.py::TestGaugeFloodHistoryPage_daily_returns_more_elements | GaugeFloodHistoryPage_daily_returns_more_elements | Fail   |
|                                                                                 | page_08_flood_history.py::TestGaugeFloodHistoryPage_responses_includes_table    | GaugeFloodHistoryPage_responses_includes_table    | Pass   |
|                                                                                 | page_08_flood_history.py::TestGaugeFloodHistoryPage_severity_classification     | GaugeFloodHistoryPage_severity_classification     | Pass   |
|                                                                                 | page_08_flood_history.py::TestGaugeFloodHistoryPage_table_top_10                | GaugeFloodHistoryPage_table_top_10                | Pass   |
|                                                                                 | page_09_hazard_curves.py::TestGaugeHazardCurvesPeriodLevelsNoExtraTable         | GaugeHazardCurvesPeriodLevelsNoExtraTable         | Fail   |
|                                                                                 | page_09_hazard_curves.py::TestGaugeHazardCurvesPageShowsFallback                | GaugeHazardCurvesPageShowsFallback                | Pass   |
| page_09_hazard_curves.py::TestGaugeHazardCurvesPage                             | GaugeHazardCurvesPage                                                           | Pass                                              |        |
| page_09_hazard_curves.py::TestGaugeHazardCurvesPage_basic                       | GaugeHazardCurvesPage_basic                                                     | Pass                                              |        |
| page_09_hazard_curves.py::TestGaugeHazardCurvesPage_period_levels_renders_table | GaugeHazardCurvesPage_period_levels_renders_table                               | Fail                                              |        |
| page_10_prs_pricing.py::TestGaugePRSPricingPageCurveShowsFallback               | GaugePRSPricingPageCurveShowsFallback                                           | Pass                                              |        |
| page_10_prs_pricing.py::TestGaugePRSPricingPage                                 | GaugePRSPricingPage                                                             | Pass                                              |        |
| page_10_prs_pricing.py::TestGaugePRSPricingPage_rates_renders_spreads           | GaugePRSPricingPage_rates_renders_spreads                                       | Pass                                              |        |
| page_10_prs_pricing.py::TestGaugePRSPricingPage_rates_renders_na                | GaugePRSPricingPage_rates_renders_na                                            | Pass                                              |        |
| page_11_current_risk.py::TestGaugeCurrentRiskHistoricalDailyShowsFallback       | GaugeCurrentRiskHistoricalDailyShowsFallback                                    | Fail                                              |        |
| page_11_current_risk.py::TestGaugeCurrentRiskHistoricalDataShowsFallback        | GaugeCurrentRiskHistoricalDataShowsFallback                                     | Pass                                              |        |
| page_11_current_risk.py::TestGaugeCurrentRiskPage                               | GaugeCurrentRiskPage                                                            | Pass                                              |        |
| page_11_current_risk.py::TestGaugeCurrentRiskPage_includes_content              | GaugeCurrentRiskPage_includes_content                                           | Skip                                              |        |
| page_11_current_risk.py::TestGaugeCurrentRiskPage_returns_elements              | GaugeCurrentRiskPage_returns_elements                                           | Skip                                              |        |
| page_12_trading.py::TestBuildBlockSectionMakesReturnsMessage                    | test_BuildBlockSectionMakesReturnsMessage                                       | Pass                                              |        |
| page_12_trading.py::TestBuildBlockSectionInverTradeDirection                    | test_BuildBlockSectionInverTradeDirection                                       | Pass                                              |        |
| page_12_trading.py::TestBuildBlockSectionTriggerCapitalized                     | test_BuildBlockSectionTriggerCapitalized                                        | Pass                                              |        |

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| Model | Module                                           | Test Name                                       | Status |
|-------|--------------------------------------------------|-------------------------------------------------|--------|
|       | page_12_trading.py::TestBuildBlotter             | test_Settlement.trades_returns_table            | Pass   |
|       | page_12_trading.py::TestBuildBlotter             | test_Settlement.trades_shows_count              | Pass   |
|       | page_12_trading.py::TestBuildMarket              | test_Governance.state_returns_message           | Pass   |
|       | page_12_trading.py::TestBuildMarket              | test_Governance.state_handles_none_rates        | Pass   |
|       | page_12_trading.py::TestBuildMarket              | test_Governance.adjustments_returns_extra_table | Pass   |
|       | page_12_trading.py::TestBuildMarket              | test_Governance.state_returns_table             | Pass   |
|       | page_12_trading.py::TestDeriveTenor              | test_both_missing_returns_na                    | Pass   |
|       | page_12_trading.py::TestDeriveTenor              | test_five_year_tenor                            | Pass   |
|       | page_12_trading.py::TestDeriveTenor              | test_invalid_date_format_returns_na             | Pass   |
|       | page_12_trading.py::TestDeriveTenor              | test_missing_end_returns_na                     | Pass   |
|       | page_12_trading.py::TestDeriveTenor              | test_missing_start_returns_na                   | Pass   |
|       | page_12_trading.py::TestDeriveTenor              | test_valid_dates_returns_years                  | Pass   |
|       | page_12_trading.py::TestGenerateElectron         | test_return_gauge_id_key_returns_message        | Pass   |
|       | page_12_trading.py::TestGenerateElectron         | test_returns_list                               | Pass   |
|       | page_12_trading.py::TestGenerateElectron         | test_with_gauge_id_returns_trading_header       | Pass   |
|       | page_12_trading.py::TestLoadGauges               | test_get_reports_dir_raises_fallback            | Pass   |
|       | page_12_trading.py::TestLoadGauges               | test_reports_dir_not_exist_returns_empty        | Pass   |
|       | page_12_trading.py::TestLoadGauges               | test_trade_matching_gauge_id_appended           | Pass   |
|       | page_12_trading.py::TestLoadGauges               | test_trade_not_matching_gauge_skipped           | Pass   |
|       | page_12_trading.py::TestLoadMarket               | test_market_state_exception_returns_empty       | Pass   |
|       | test_base_page_format.py::TestFormatCoordinate   | test_coordinate_six_decimals                    | Pass   |
|       | test_base_page_format.py::TestFormatCoordinate   | test_negative_coordinate                        | Pass   |
|       | test_base_page_format.py::TestFormatCoordinate   | test_none_delegates_to_format_value             | Pass   |
|       | test_base_page_format.py::TestFormatDivision     | test_division_replacements                      | Pass   |
|       | test_base_page_format.py::TestFormatName         | test_name_split                                 | Pass   |
|       | test_base_page_format.py::TestFormatNumeric      | test_numeric_delegates                          | Pass   |
|       | test_base_page_format.py::TestFormatPositive     | test_positive_returns_times                     | Pass   |
|       | test_base_page_format.py::TestFormatFrequency    | test_frequency_never                            | Pass   |
|       | test_base_page_format.py::TestFormatInteger      | test_integer_value                              | Pass   |
|       | test_base_page_format.py::TestFormatNonnumeric   | test_nonnumeric_delegates_to_format_value       | Pass   |
|       | test_base_page_format.py::TestFormatNonnumeric   | test_nonnumeric_value_with_custom_unit          | Pass   |
|       | test_base_page_format.py::TestFormatNonnumeric   | test_nonnumeric_value_with_default_unit         | Pass   |
|       | test_base_page_format.py::TestFormatString       | test_empty_string_returns_not_specified         | Pass   |
|       | test_base_page_format.py::TestFormatString       | test_false_returns_no                           | Pass   |
|       | test_base_page_format.py::TestFormatString       | test_float_whole_number_strips_decimal          | Pass   |
|       | test_base_page_format.py::TestFormatString       | test_float_with_decimals_formatted              | Pass   |
|       | test_base_page_format.py::TestFormatString       | test_integer_returns_str                        | Pass   |
|       | test_base_page_format.py::TestFormatString       | test_none_returns_not_specified                 | Pass   |
|       | test_base_page_format.py::TestFormatString       | test_normal_string_returned                     | Pass   |
|       | test_base_page_format.py::TestFormatString       | test_value_returns_yes                          | Pass   |
|       | test_current_risk.py::TestCurrentRiskEmptyIdFile | test_observed_key                               | Pass   |
|       | test_current_risk.py::TestCurrentRiskEmptyIdFile | test_listsEdgeCases                             | Pass   |
|       | test_current_risk.py::TestCurrentRiskFullPath    | test_returnsEdgeCases                           | Pass   |
|       | test_current_risk.py::TestCurrentRiskGeneral     | test_general_data_in_list_of_lists              | Pass   |

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| Model | Module                                                                            | Test Name                                                                         | Status |
|-------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------|
|       | test_current_risk.py::TestCurrentRiskGeneratorDateInExceedanceCases               | test_current_risk.py::TestCurrentRiskGeneratorDateInExceedanceCases               | Pass   |
|       | test_current_risk.py::TestCurrentRiskGeneratorMonthInExceedanceCases_all_level    | test_current_risk.py::TestCurrentRiskGeneratorMonthInExceedanceCases_all_level    | Pass   |
|       | test_current_risk.py::TestCurrentRiskGeneratorFloodInExceedanceCases              | test_current_risk.py::TestCurrentRiskGeneratorFloodInExceedanceCases              | Pass   |
|       | test_current_risk.py::TestCurrentRiskGeneratorFloodInExceedanceCases_skipped      | test_current_risk.py::TestCurrentRiskGeneratorFloodInExceedanceCases_skipped      | Pass   |
|       | test_flood_history.py::TestBuildFloodHistoryClassification                        | test_flood_history.py::TestBuildFloodHistoryClassification                        | Pass   |
|       | test_flood_history.py::TestBuildFloodHistoryDateInExceedanceSkipped               | test_flood_history.py::TestBuildFloodHistoryDateInExceedanceSkipped               | Pass   |
|       | test_flood_history.py::TestBuildFloodHistoryDayEventUpdatesPeak                   | test_flood_history.py::TestBuildFloodHistoryDayEventUpdatesPeak                   | Pass   |
|       | test_flood_history.py::TestBuildFloodHistoryDayRangeDateFormat                    | test_flood_history.py::TestBuildFloodHistoryDayRangeDateFormat                    | Pass   |
|       | test_flood_history.py::TestBuildFloodHistoryEventsReturnsMessage                  | test_flood_history.py::TestBuildFloodHistoryEventsReturnsMessage                  | Pass   |
|       | test_flood_history.py::TestBuildFloodHistoryClassification                        | test_flood_history.py::TestBuildFloodHistoryClassification                        | Pass   |
|       | test_flood_history.py::TestBuildFloodHistoryDayEventDateFormat                    | test_flood_history.py::TestBuildFloodHistoryDayEventDateFormat                    | Pass   |
|       | test_flood_history.py::TestBuildFloodHistorySeparateEvents                        | test_flood_history.py::TestBuildFloodHistorySeparateEvents                        | Pass   |
|       | test_flood_history.py::TestBuildFloodHistoryClassification                        | test_flood_history.py::TestBuildFloodHistoryClassification                        | Pass   |
|       | test_gauge_generator_extras.py::TestAutoOpenCalled                                | test_gauge_generator_extras.py::TestAutoOpenCalled                                | Pass   |
|       | test_gauge_generator_extras.py::TestAutoOpenFalseDoesNotRaise                     | test_gauge_generator_extras.py::TestAutoOpenFalseDoesNotRaise                     | Pass   |
|       | test_gauge_generator_extras.py::TestAutoOpenSuccess                               | test_gauge_generator_extras.py::TestAutoOpenSuccess                               | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugesByLin_flood_gauges_list             | test_gauge_generator_extras.py::TestFindGaugesByLin_flood_gauges_list             | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugesByLin_gauges_key                    | test_gauge_generator_extras.py::TestFindGaugesByLin_gauges_key                    | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugesByLin                               | test_gauge_generator_extras.py::TestFindGaugesByLin                               | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugesByLin_gauge_object                  | test_gauge_generator_extras.py::TestFindGaugesByLin_gauge_object                  | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugesByLin_raises_value_error            | test_gauge_generator_extras.py::TestFindGaugesByLin_raises_value_error            | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugesByLin_raises_value_error            | test_gauge_generator_extras.py::TestFindGaugesByLin_raises_value_error            | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugesByLin_list_raises_value_error       | test_gauge_generator_extras.py::TestFindGaugesByLin_list_raises_value_error       | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugesByLin_raises_value_error            | test_gauge_generator_extras.py::TestFindGaugesByLin_raises_value_error            | Pass   |
|       | test_generator_direct.py::TestFindGaugesByLin_gauges_key_found                    | test_generator_direct.py::TestFindGaugesByLin_gauges_key_found                    | Pass   |
|       | test_generator_direct.py::TestFindGaugesByLin_dict_structure_raises               | test_generator_direct.py::TestFindGaugesByLin_dict_structure_raises               | Pass   |
|       | test_generator_direct.py::TestFindGaugesByLin_type_raises                         | test_generator_direct.py::TestFindGaugesByLin_type_raises                         | Pass   |
|       | test_generator_direct.py::TestFindGaugesByLin_input_found                         | test_generator_direct.py::TestFindGaugesByLin_input_found                         | Pass   |
|       | test_generator_direct.py::TestFindGaugesByLin_dmed_gauge_entry_skipped            | test_generator_direct.py::TestFindGaugesByLin_dmed_gauge_entry_skipped            | Pass   |
|       | test_generator_direct.py::TestFindGaugesByLin_found_raises_value_error            | test_generator_direct.py::TestFindGaugesByLin_found_raises_value_error            | Pass   |
|       | test_generator_direct.py::TestFindGaugesByLin_gauge_dict_returns_self             | test_generator_direct.py::TestFindGaugesByLin_gauge_dict_returns_self             | Pass   |
|       | test_generator_direct.py::TestGaugeGeneratorDirectCallReportDirect                | test_generator_direct.py::TestGaugeGeneratorDirectCallReportDirect                | Pass   |
|       | test_generator_direct.py::TestGaugeGeneratorDirectCallReportFnAnalysis            | test_generator_direct.py::TestGaugeGeneratorDirectCallReportFnAnalysis            | Pass   |
|       | test_generator_direct.py::TestGaugeGeneratorDirectCallReportFnAutoOpen            | test_generator_direct.py::TestGaugeGeneratorDirectCallReportFnAutoOpen            | True   |
|       | test_generator_direct.py::TestGaugeGeneratorDirectCallReportFnBasic               | test_generator_direct.py::TestGaugeGeneratorDirectCallReportFnBasic               | Pass   |
|       | test_generator_direct.py::TestGaugeGeneratorDirectCallReportFnFallback            | test_generator_direct.py::TestGaugeGeneratorDirectCallReportFnFallback            | Pass   |
|       | test_generator_direct.py::TestGaugeGeneratorDirectCallReportFnMonitoring          | test_generator_direct.py::TestGaugeGeneratorDirectCallReportFnMonitoring          | Pass   |
|       | test_generator_direct.py::TestGaugeGeneratorDirectCallReportNoPagesUsesAutoSelect | test_generator_direct.py::TestGaugeGeneratorDirectCallReportNoPagesUsesAutoSelect | Select |
|       | test_generator_direct.py::TestGaugeGeneratorDirectCallReturnsFalse                | test_generator_direct.py::TestGaugeGeneratorDirectCallReturnsFalse                | Pass   |
|       | test_generator_direct.py::TestGaugeGeneratorDirectCallNoAnalysisData              | test_generator_direct.py::TestGaugeGeneratorDirectCallNoAnalysisData              | Pass   |
|       | test_generator_direct.py::TestGaugeGeneratorDirectCallWithHazard                  | test_generator_direct.py::TestGaugeGeneratorDirectCallWithHazard                  | Pass   |
|       | test_generator_direct.py::TestGaugeGeneratorDirectCallReport                      | test_generator_direct.py::TestGaugeGeneratorDirectCallReport                      | Pass   |
|       | test_generator_direct.py::TestGaugeGeneratorDirectCallExceptionContinue           | test_generator_direct.py::TestGaugeGeneratorDirectCallExceptionContinue           | Pass   |
|       | test_generator_direct.py::TestGaugeGeneratorDirectCallSkipsUnknownPages           | test_generator_direct.py::TestGaugeGeneratorDirectCallSkipsUnknownPages           | Pass   |

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| Model | Module                                                      | Test Name                                                              | Status |
|-------|-------------------------------------------------------------|------------------------------------------------------------------------|--------|
|       | test_generator_direct.py::TestGenerateReport                | test_generator_direct.py::TestGenerateReportWithNoGaugeId              | Pass   |
|       | test_generator_direct.py::TestGenerateReport                | test_generator_direct.py::TestGenerateReportWithGaugeId                | Pass   |
|       | test_generator_direct.py::TestGenerateReport                | test_generator_direct.py::TestGenerateReportWithMonitoringReport       | Pass   |
|       | test_generator_direct.py::TestGenerateReport                | test_generator_direct.py::TestGenerateReportWithCategories             | Pass   |
|       | test_generator_direct.py::TestGenerateReport                | test_generator_direct.py::TestGenerateReportWithAvailablePages         | Pass   |
|       | test_generator_direct.py::TestGenerateReport                | test_generator_direct.py::TestGenerateReportWithDissectConfig          | Pass   |
|       | test_generator_direct.py::TestGenerateReport                | test_generator_direct.py::TestGenerateReportWithInvalidId              | Pass   |
|       | test_generator_direct.py::TestGenerateReport                | test_generator_direct.py::TestGenerateReportWithInvalidId              | Pass   |
|       | test_integrator.py::TestGenerateReport                      | test_integrator.py::TestGenerateReportForKeyStructureWorks             | Pass   |
|       | test_integrator.py::TestGenerateReport                      | test_integrator.py::TestGenerateReportForGaugeReturnsPath              | Pass   |
|       | test_integrator.py::TestGenerateReport                      | test_integrator.py::TestGenerateReportForGaugeNotFoundReturnsNone      | Pass   |
|       | test_integrator.py::TestGenerateReport                      | test_integrator.py::TestGenerateReportForGaugeFileReturnsNone          | Pass   |
|       | test_integrator.py::TestGenerateReport                      | test_integrator.py::TestGenerateReportForGaugeContainsGaugeId          | Pass   |
|       | test_integrator.py::TestGenerateReport                      | test_integrator.py::TestGenerateReportForGaugeCreated                  | Pass   |
|       | test_integrator.py::TestGenerateReport                      | test_integrator.py::TestGenerateReportForGaugeFileCreated              | Pass   |
|       | test_integrator.py::TestGetAvailableGauges                  | test_integrator.py::TestGetAvailableGaugesUsingFileReturnsEmptyList    | Pass   |
|       | test_integrator.py::TestGetAvailableGauges                  | test_integrator.py::TestGetAvailableGaugesReturnsAllIds                | Pass   |
|       | test_integrator.py::TestGetAvailableGauges                  | test_integrator.py::TestGetAvailableGaugesReturnsCorrectId             | Pass   |
|       | test_integrator.py::TestGetAvailableGauges                  | test_integrator.py::TestGetAvailableGaugesReturnsList                  | Pass   |
|       | test_integrator.py::TestValidateGauge                       | test_integrator.py::TestValidateGaugeExistsKeyStructure                | Pass   |
|       | test_integrator.py::TestValidateGauge                       | test_integrator.py::TestValidateGaugeExistingGaugeReturnsTrue          | Pass   |
|       | test_integrator.py::TestValidateGauge                       | test_integrator.py::TestValidateGaugeExistingFileReturnsFalse          | Pass   |
|       | test_integrator.py::TestValidateGauge                       | test_integrator.py::TestValidateGaugeNonexistentGaugeReturnsFalse      | Pass   |
|       | test_integrator_exception.py::TestGenerateReport            | test_integrator_exception.py::TestGenerateReportForGaugeException      | Pass   |
|       | test_integrator_exception.py::TestGenerateReport            | test_integrator_exception.py::TestGenerateReportForGaugeDescription    | Pass   |
|       | test_integrator_exception.py::TestGetAvailableGauges        | test_integrator_exception.py::TestGetAvailableGaugesReturnsEmptyList   | Pass   |
|       | test_integrator_exception.py::TestGetAvailableGauges        | test_integrator_exception.py::TestGetAvailableGaugesReturnsEmptyList   | Pass   |
|       | test_integrator_exception.py::TestValidateGauge             | test_integrator_exception.py::TestValidateGaugeExistsReturnsFalse      | Pass   |
|       | test_integrator_exception.py::TestValidateGauge             | test_integrator_exception.py::TestValidateGaugeExistsReturnsFalse      | Pass   |
|       | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels            | Pass   |
|       | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabelsWhenPresent | Pass   |
|       | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels            | Pass   |
|       | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels            | Pass   |
|       | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels            | Pass   |
|       | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabelsGuard       | Pass   |
|       | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels            | Pass   |
|       | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels | test_percentile_gauge.py::TestButtonToToggleGaugeRiskLabels            | Pass   |
|       | test_report_generation.py::TestGenerateReport               | test_report_generation.py::TestGenerateReportType                      | Pass   |
|       | test_report_generation.py::TestGenerateReport               | test_report_generation.py::TestGenerateReportFailureDoesNotRaise       | Pass   |
|       | test_report_generation.py::TestGenerateReport               | test_report_generation.py::TestGenerateReportCallsOpenPdf              | Pass   |
|       | test_report_generation.py::TestGenerateReport               | test_report_generation.py::TestGenerateReportType                      | Pass   |
|       | test_report_generation.py::TestGenerateReport               | test_report_generation.py::TestGenerateReportTypeWithTimestamp         | Pass   |
|       | test_report_generation.py::TestGenerateReport               | test_report_generation.py::TestGenerateReportPath                      | Pass   |
|       | test_report_generation.py::TestGenerateReport               | test_report_generation.py::TestGenerateReportTypeFallsBack             | Pass   |

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| Model | Module                                         | Test Name                                              | Status           |
|-------|------------------------------------------------|--------------------------------------------------------|------------------|
|       | mortgage_pages.py::TestGenerator               | test_generate_report_function_creates_pdf              | Pass             |
|       | mortgage_pages.py::TestGenerator               | test_generate_report_function_returns_path             | Pass             |
|       | mortgage_pages.py::TestMortgageReportGenerator | test_get_elements_returns_list                         | Pass             |
|       | mortgage_pages.py::TestMortgageReportGenerator | test_get_filename_fallback                             | Pass             |
|       | mortgage_pages.py::TestMortgageReportGenerator | test_get_filename_with_id                              | Pass             |
|       | mortgage_pages.py::TestMortgageReportGenerator | test_generate_report_creates_pdf                       | Pass             |
|       | mortgage_pages.py::TestMortgageReportGenerator | test_generate_report_returns_path                      | Pass             |
|       | mortgage_pages.py::TestMortgageReportGenerator | test_output_dir_created                                | Pass             |
|       | mortgage_pages.py::TestMortgageReportGenerator | test_page_order_set                                    | Pass             |
|       | mortgage_pages.py::TestMortgageReportGenerator | test_page_generator_initialised                        | Pass             |
|       | mortgage_pages.py::TestMortgageTitlePage       | test_title_page_non_empty                              | Pass             |
|       | mortgage_pages.py::TestMortgageTitlePage       | test_title_page_property_data_does_not_crash           | Pass             |
|       | mortgage_pages.py::TestMortgageTitlePage       | test_title_page_mortgage_id                            | Pass             |
|       | mortgage_pages.py::TestMortgageTitlePage       | test_title_page_mortgage_data_returns_elements         | Pass             |
|       | mortgage_pages.py::TestMortgageTitlePage       | test_title_page_returns_list                           | Pass             |
|       | mortgage_pages.py::TestMortgageTitlePage       | test_title_page_elements_included                      | Pass             |
|       | test_mortgage_generator_coverage               | test_test_mortgage_generator_default_output_dir        | Fail             |
|       | test_mortgage_generator_coverage               | test_test_mortgage_generator_no_file_name_raises_error | Distribute_error |
|       | test_mortgage_generator_coverage               | test_test_mortgage_generator_no_file_name_fails        | Fail             |
|       | test_mortgage_generator_coverage               | test_test_mortgage_generator_page_exception            | Pass             |
|       | test_mortgage_generator_coverage               | test_test_mortgage_generator_skip_skipped_pages        | Pass             |
|       | gauge-pdf.py::TestGaugePdfGenerator            | test_basic_gauge_report_generates_pdf                  | Pass             |
|       | gauge-pdf.py::TestGaugePdfGenerator            | test_full_gauge_report_generates_pdf                   | Pass             |
|       | gauge-pdf.py::TestGaugePdfGenerator            | test_gauge_output_dir_created_if_missing               | Pass             |
|       | gauge-pdf.py::TestGaugePdfGenerator            | test_gauge_pdf_filename_contains_gauge                 | Pass             |
|       | gauge-pdf.py::TestGaugePdfGenerator            | test_gauge_pdf_has_reasonable_size                     | Pass             |
|       | gauge-pdf.py::TestGaugePdfGenerator            | test_gauge_pdf_starts_with_pdf_header                  | Pass             |
|       | gauge-pdf.py::TestGaugePdfGenerator            | test_gauge_report_with_minimal_data                    | Pass             |
|       | gauge-pdf.py::TestGaugeReportGenerator         | test_generator_initialises                             | Pass             |
|       | gauge-pdf.py::TestGaugeReportGenerator         | test_list_available_pages                              | Pass             |
|       | gauge-pdf.py::TestGaugeReportGenerator         | test_validate_pages                                    | Pass             |
|       | property-pdf.py::TestPropertyPdfGenerator      | test_full_property_report_generates_pdf                | Pass             |
|       | property-pdf.py::TestPropertyPdfGenerator      | test_mortgage_focused_report_generates_pdf             | Pass             |
|       | property-pdf.py::TestPropertyPdfGenerator      | test_output_dir_created_if_missing                     | Pass             |
|       | property-pdf.py::TestPropertyPdfGenerator      | test_pdf_filename_contains_property_id                 | Pass             |
|       | property-pdf.py::TestPropertyPdfGenerator      | test_pdf_has_reasonable_size                           | Pass             |
|       | property-pdf.py::TestPropertyPdfGenerator      | test_pdf_starts_with_pdf_header                        | Pass             |
|       | property-pdf.py::TestPropertyPdfGenerator      | test_property_only_report_generates_pdf                | Pass             |
|       | property-pdf.py::TestPropertyPdfGenerator      | test_report_with_minimal_property_data                 | Pass             |
|       | property-pdf.py::TestPropertyPdfGenerator      | test_risk_focused_report_generates_pdf                 | Pass             |
|       | property-pdf.py::TestPropertyReportGenerator   | test_generator_select_pages_with_mortgage              | Pass             |
|       | property-pdf.py::TestPropertyReportGenerator   | test_generator_select_pages_without_mortgage           | Pass             |
|       | property-pdf.py::TestPropertyReportGenerator   | test_generator_initialises                             | Pass             |
|       | property-pdf.py::TestPropertyReportGenerator   | test_list_available_pages                              | Pass             |

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| Model | Module                                        | Test Name                                        | Status |
|-------|-----------------------------------------------|--------------------------------------------------|--------|
|       | property_pdf.py::TestPropertyReport           | test_validate_pages_valid                        | Pass   |
|       | server_flow.py::TestServerReport              | test_gauge_report_via_route_function             | Pass   |
|       | server_flow.py::TestServerReport              | test_gauge_report_with_real_data                 | Pass   |
|       | server_flow.py::TestServerReport              | test_property_report_via_route_function          | Pass   |
|       | server_flow.py::TestServerReport              | test_property_report_with_real_data              | Pass   |
|       | test_data_loader.py::TestCountDir             | test_counts_matching_files                       | Pass   |
|       | test_data_loader.py::TestCountDir             | test_empty_directory                             | Pass   |
|       | test_data_loader.py::TestCountDir             | test_returns_zero_for_no_matches                 | Pass   |
|       | test_data_loader.py::TestCountDir             | test_returns_zero_for_none_dir                   | Pass   |
|       | test_data_loader.py::TestCountDir             | test_returns_zero_for_nonexistent_dir            | Pass   |
|       | test_data_loader.py::TestLoad                 | test_loads_list_json                             | Pass   |
|       | test_data_loader.py::TestLoad                 | test_loads_nested_json                           | Pass   |
|       | test_data_loader.py::TestLoad                 | test_loads_valid_json                            | Pass   |
|       | test_data_loader.py::TestLoad                 | test_returns_empty_dict_for_directory            | Pass   |
|       | test_data_loader.py::TestLoad                 | test_returns_empty_dict_for_invalid_json         | Pass   |
|       | test_data_loader.py::TestLoad                 | test_returns_empty_dict_for_missing_file         | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_classifier_count                            | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_eod_count_defaults_zero_on_import_error     | Error  |
|       | test_data_loader.py::TestLoadAll              | test_gaugehd_baselines_loaded                    | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_gaugehd_count                               | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_gaugets_count                               | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_loads_counterparties                        | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_loads_gaugehc                               | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_loads_gauges                                | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_loads_mortgages                             | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_loads_properties                            | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_loads_propertyhc                            | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_loads_seq_summary                           | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_loads_stress_storms_index                   | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_loads_stress_storms_legacy_fallback         | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_missing_files_return_empty                  | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_propertyts_count                            | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_stress_storms_empty_when_neither_exists     | Pass   |
|       | test_data_loader.py::TestLoadAll              | test_trade_count_defaults_zero_on_import_error   | Error  |
|       | test_data_loader.py::TestLoadAll              | test_training_summary                            | Pass   |
|       | test_data_loader.py::TestLoadGaugehdBaselines | test_gaugehd_baselines_summer_mean               | Pass   |
|       | test_data_loader.py::TestLoadGaugehdBaselines | test_gaugehd_baselines_winter_mean               | Pass   |
|       | test_data_loader.py::TestLoadGaugehdBaselines | test_gaugehd_baselines_missing_monthly_means     | Pass   |
|       | test_data_loader.py::TestLoadGaugehdBaselines | test_gaugehd_baselines                           | Pass   |
|       | test_data_loader.py::TestLoadGaugehdBaselines | test_gaugehd_baselines_empty_for_nonexistent_dir | Pass   |
|       | test_data_loader.py::TestLoadGaugehdBaselines | test_gaugehd_baselines_files_without_mean_level  | Pass   |
|       | test_data_loader.py::TestLoadGaugehdBaselines | test_gaugehd_baselines_invalid_files             | Pass   |
|       | test_generator.py::TestEdgeCases              | test_gauge_with_dash_values                      | Pass   |
|       | test_generator.py::TestEdgeCases              | test_mortgage_with_none_fields                   | Pass   |

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| Model | Module                                      | Test Name                             | Status |
|-------|---------------------------------------------|---------------------------------------|--------|
|       | test_generator.py::TestEdgeCases            | test_property_with_none_value         | Pass   |
|       | test_generator.py::TestGenerate             | test_catchment_name_from_input_dir    | Pass   |
|       | test_generator.py::TestGenerate             | test_generate_with_empty_data         | Pass   |
|       | test_generator.py::TestGenerate             | test_generate_with_minimal_data       | Pass   |
|       | test_generator.py::TestGenerate             | test_output_path_creates_parent_dirs  | Pass   |
|       | test_generator.py::TestGenerate             | test_pdf_has_multiple_pages           | Pass   |
|       | test_generator.py::TestGenerate             | test_pdf_starts_with_magic_bytes      | Pass   |
|       | test_generator.py::TestGenerate             | test_produces_pdf_file                | Pass   |
|       | test_generator.py::TestGenerate             | test_returns_path                     | Pass   |
|       | test_generator.py::TestInit                 | test_input_dir_accepts_string         | Pass   |
|       | test_generator.py::TestInit                 | test_input_dir_set                    | Pass   |
|       | test_generator.py::TestInit                 | test_output_path_explicit             | Pass   |
|       | test_generator.py::TestInit                 | test_styles_initialised               | Pass   |
|       | test_generator.py::TestMixinIntegration     | test_has_kv_table                     | Pass   |
|       | test_generator.py::TestMixinIntegration     | test_has_load_all                     | Pass   |
|       | test_generator.py::TestMixinIntegration     | test_has_make_table                   | Pass   |
|       | test_generator.py::TestMixinIntegration     | test_has_section_methods              | Pass   |
|       | test_generator.py::TestMixinIntegration     | test_load_all_and_sections_compatible | Pass   |
|       | test_sections.gauges.py::TestSectionHeading | test_heading                          | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_baselines_message         | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_computed                  | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_returns_list              | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_sectional_averages        | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_when_baselines            | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_gauges_message            | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_multiple_gauges           | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_returns_list              | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_section_heading           | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_summary_line_counts       | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_table_present             | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_alternative_key_format    | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_as_non_dict_gauge_curve   | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_curves                    | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_curves_message            | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_returns_list              | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_with_curves               | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_in_summary                | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_mortgages_message         | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_heading                   | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_returns_list              | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_present                   | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_properties_message        | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_heading_present           | Pass   |
|       | test_sections.gauges.py::TestSectionGlobal  | test_global_property_count_in_summary | Pass   |

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| Model | Module                                         | Test Name                          | Status |
|-------|------------------------------------------------|------------------------------------|--------|
|       | test_sections.gauges.py::TestSectionProperties | test_returns_list                  | Pass   |
|       | test_sections.gauges.py::TestSectionProperties | test_table_with_properties         | Pass   |
|       | test_sections.gauges.py::TestSectionProperties | test_heading                       | Pass   |
|       | test_sections.gauges.py::TestSectionProperties | test_table_present                 | Pass   |
|       | test_sections.gauges.py::TestSectionProperties | test_returns_list                  | Pass   |
|       | test_sections.gauges.py::TestSectionProperties | test_with_full_summary             | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_heading                       | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_table                         | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_returns_list                  | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_alternative_cdm_format        | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_empty_message                 | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_loading_parties               | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_returns_list                  | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_table_present                 | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_headingts                     | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_properties_section_when_empty | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_properties_section_added      | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_returns_list                  | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_data_size_computed            | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_heading                       | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_table_present                 | Pass   |
|       | test_sections.portfolio.py::TestSectionBlot    | test_returns_list                  | Pass   |
|       | test_styles.py::TestAvailWidth                 | test_avail_width_value             | Pass   |
|       | test_styles.py::TestColourConstants            | test_blue_is_hex_colour            | Pass   |
|       | test_styles.py::TestColourConstants            | test_dark_blue                     | Pass   |
|       | test_styles.py::TestColourConstants            | test_grey                          | Pass   |
|       | test_styles.py::TestColourConstants            | test_light_amber                   | Pass   |
|       | test_styles.py::TestColourConstants            | test_light_blue                    | Pass   |
|       | test_styles.py::TestColourConstants            | test_light_green                   | Pass   |
|       | test_styles.py::TestColourConstants            | test_light_red                     | Pass   |
|       | test_styles.py::TestKvTable                    | test_custom_label_width            | Pass   |
|       | test_styles.py::TestKvTable                    | test_default_label_width           | Pass   |
|       | test_styles.py::TestKvTable                    | test_multiple_pairs                | Pass   |
|       | test_styles.py::TestKvTable                    | test_returns_table                 | Pass   |
|       | test_styles.py::TestKvTable                    | test_value_converted_to_string     | Pass   |
|       | test_styles.py::TestMakeTable                  | test_col_widths_propagated         | Pass   |
|       | test_styles.py::TestMakeTable                  | test_custom_alt_color              | Pass   |
|       | test_styles.py::TestMakeTable                  | test_default_alt_color_is_blue     | Pass   |
|       | test_styles.py::TestMakeTable                  | test_empty_rows                    | Pass   |
|       | test_styles.py::TestMakeTable                  | test_repeat_rows_set               | Pass   |
|       | test_styles.py::TestMakeTable                  | test_returns_table                 | Pass   |
|       | test_styles.py::TestMakeTable                  | test_table_has_correct_row_count   | Pass   |
|       | test_styles.py::TestSetupStyles                | test_body_style_font_size          | Pass   |
|       | test_styles.py::TestSetupStyles                | test_kv_style_font_size            | Pass   |

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| Model | Module                                     | Test Name                                    | Status |
|-------|--------------------------------------------|----------------------------------------------|--------|
|       | test_styles.py::TestSetupStyles            | test_kv_style_left_indent                    | Pass   |
|       | test_styles.py::TestSetupStyles            | test_section_style_font_size                 | Pass   |
|       | test_styles.py::TestSetupStyles            | test_section_style_text_color                | Pass   |
|       | test_styles.py::TestSetupStyles            | test_subsection_style_font_size              | Pass   |
|       | test_styles.py::TestSetupStyles            | test_subsection_style_text_color             | Pass   |
|       | test_styles.py::TestSetupStyles            | test_subtitle_style_font_size                | Pass   |
|       | test_styles.py::TestSetupStyles            | test_subtitle_style_text_color               | Pass   |
|       | test_styles.py::TestSetupStyles            | test_title_style_font_name                   | Pass   |
|       | test_styles.py::TestSetupStyles            | test_title_style_font_size                   | Pass   |
|       | test_styles.py::TestSetupStyles            | test_title_style_text_color                  | Pass   |
|       | test_styles.py::TestTableStyleConstants    | test_table_rows_amber                        | Pass   |
|       | test_styles.py::TestTableStyleConstants    | test_table_rows_blue                         | Pass   |
|       | test_styles.py::TestTableStyleConstants    | test_table_rows_green                        | Pass   |
|       | test_styles.py::TestTableStyleConstants    | test_table_style_first_command_is_background | Pass   |
|       | test_styles.py::TestTableStyleConstants    | test_table_style_has_grid                    | Pass   |
|       | test_styles.py::TestTableStyleConstants    | test_table_style_is_list                     | Pass   |
|       | analysis.py::TestSinglePropertyAnalysis    | test_find_nearest_gauge                      | Pass   |
|       | analysis.py::TestSinglePropertyAnalysis    | test_full_analysis_high_risk_property        | Pass   |
|       | analysis.py::TestSinglePropertyAnalysis    | test_full_analysis_low_risk_property         | Pass   |
|       | analysis.py::TestSinglePropertyAnalysis    | test_load_gauge_data                         | Pass   |
|       | analysis.py::TestSinglePropertyAnalysis    | test_load_property_data                      | Pass   |
|       | calculations.py::TestDamageRatio           | test_commercial_higher_damage                | Pass   |
|       | calculations.py::TestDamageRatio           | test_damage_capped                           | Pass   |
|       | calculations.py::TestDamageRatio           | test_damage_increases_with_depth             | Pass   |
|       | calculations.py::TestDamageRatio           | test_no_damage_no_flood                      | Pass   |
|       | calculations.py::TestDistanceCalculation   | test_haversine_known_distance                | Pass   |
|       | calculations.py::TestDistanceCalculation   | test_haversine_same_point                    | Pass   |
|       | calculations.py::TestDistanceCalculation   | test_haversine_teddington_to_richmond        | Pass   |
|       | calculations.py::TestFloodDepthCalculation | test_flood_above_trigger                     | Pass   |
|       | calculations.py::TestFloodDepthCalculation | test_flood_attenuates_with_distance          | Pass   |
|       | calculations.py::TestFloodDepthCalculation | test_high_elevation_no_flood                 | Pass   |
|       | calculations.py::TestFloodDepthCalculation | test_no_flood_below_trigger                  | Pass   |
|       | calculations.py::TestRiskClassification    | test_high_risk                               | Pass   |
|       | calculations.py::TestRiskClassification    | test_low_risk                                | Pass   |
|       | calculations.py::TestRiskClassification    | test_medium_risk                             | Pass   |
|       | calculations.py::TestRiskClassification    | test_minimal_risk                            | Pass   |
|       | formatters.py::TestConstants               | test_margin_positive                         | Pass   |
|       | formatters.py::TestConstants               | test_page_dimensions                         | Pass   |
|       | formatters.py::TestFmtGbp                  | test_large_value                             | Pass   |
|       | formatters.py::TestFmtGbp                  | test_negative                                | Pass   |
|       | formatters.py::TestFmtGbp                  | test_non_numeric_fallback                    | Pass   |
|       | formatters.py::TestFmtGbp                  | test_none_fallback                           | Pass   |
|       | formatters.py::TestFmtGbp                  | test_positive_float                          | Pass   |
|       | formatters.py::TestFmtGbp                  | test_zero                                    | Pass   |

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| Model | Module                                 | Test Name                                    | Status |
|-------|----------------------------------------|----------------------------------------------|--------|
|       | formatters.py::TestLayouts             | test.body_style.alignment                    | Pass   |
|       | formatters.py::TestLayouts             | test.body_style.bold                         | Pass   |
|       | formatters.py::TestLayouts             | test.body_style.default                      | Pass   |
|       | formatters.py::TestLayouts             | test.build_header_footer.callable            | Pass   |
|       | formatters.py::TestLayouts             | test.white_hdr_style_returns_paragraph_style | Pass   |
|       | formatters.py::TestSeqTypeColor        | test.case_insensitive                        | Pass   |
|       | formatters.py::TestSeqTypeColor        | test.cluster                                 | Pass   |
|       | formatters.py::TestSeqTypeColor        | test.doublet                                 | Pass   |
|       | formatters.py::TestSeqTypeColor        | test.isolated                                | Pass   |
|       | formatters.py::TestSeqTypeColor        | test.none_returns_white                      | Pass   |
|       | formatters.py::TestSeqTypeColor        | test.persistent                              | Pass   |
|       | formatters.py::TestSeqTypeColor        | test.unknown_returns_white                   | Pass   |
|       | formatters.py::TestSetupStyles         | test.required_keys                           | Pass   |
|       | formatters.py::TestSetupStyles         | test.returns_stylesheet                      | Pass   |
|       | formatters.py::TestSetupStyles         | test.styles_are_paragraph_styles             | Pass   |
|       | generator.py::TestClaimReportGenerator | test.claim_ref_contains_prop_id_suffix       | Pass   |
|       | generator.py::TestClaimReportGenerator | test.generate_empty_sequence_lookup          | Pass   |
|       | generator.py::TestClaimReportGenerator | test.generate_returns_bytes                  | Pass   |
|       | generator.py::TestClaimReportGenerator | test.generate_without_mortgage               | Pass   |
|       | generator.py::TestClaimReportGenerator | test.minimal_prop_data                       | Pass   |
|       | generator.py::TestClaimReportGenerator | test.output_is_valid_pdf                     | Pass   |
|       | generator.py::TestClaimReportGenerator | test.pdf_size_reasonable                     | Pass   |
|       | generator.py::TestCompatShim           | test.shim_class_is_same                      | Pass   |
|       | generator.py::TestCompatShim           | test.shim_imports_generator                  | Pass   |
|       | generator.py::TestPublicAPI            | test.all_exports                             | Pass   |
|       | generator.py::TestPublicAPI            | test.import_from_package                     | Pass   |
|       | pages1.3.py::TestPage1Cover            | test.contains_claim_ref_paragraph            | Pass   |
|       | pages1.3.py::TestPage1Cover            | test.no_flood_events_still_renders           | Pass   |
|       | pages1.3.py::TestPage1Cover            | test.non_float_coordinates                   | Pass   |
|       | pages1.3.py::TestPage1Cover            | test.returns_list                            | Pass   |
|       | pages1.3.py::TestPage1Cover            | test.stats_reflect_events                    | Pass   |
|       | pages1.3.py::TestPage1Cover            | test.without_mortgage                        | Pass   |
|       | pages1.3.py::TestPage2Chronology       | test.events_with_no_sequence_id              | Pass   |
|       | pages1.3.py::TestPage2Chronology       | test.no_flood_events                         | Pass   |
|       | pages1.3.py::TestPage2Chronology       | test.returns_list                            | Pass   |
|       | pages1.3.py::TestPage2Chronology       | test.stats_bar_present                       | Pass   |
|       | pages1.3.py::TestPage2Chronology       | test.table_included                          | Pass   |
|       | pages1.3.py::TestPage2Chronology       | test.unknown_sequence_id_handled             | Pass   |
|       | pages1.3.py::TestPage3Damage           | test.contains_tables                         | Pass   |
|       | pages1.3.py::TestPage3Damage           | test.high_damage_rows_highlighted            | Pass   |
|       | pages1.3.py::TestPage3Damage           | test.multiple_sequences_grouped              | Pass   |
|       | pages1.3.py::TestPage3Damage           | test.no_flood_events                         | Pass   |
|       | pages1.3.py::TestPage3Damage           | test.returns_list                            | Pass   |
|       | pages1.3.py::TestPage3Damage           | test.zero_property_value                     | Pass   |

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| Model | Module                                           | Test Name                                  | Status |
|-------|--------------------------------------------------|--------------------------------------------|--------|
|       | pages4.5.py::TestPage4Mortgage                   | test_contains_tables                       | Pass   |
|       | pages4.5.py::TestPage4Mortgage                   | test_negative_equity_highlights_row        | Pass   |
|       | pages4.5.py::TestPage4Mortgage                   | test_no_flood_events_with_mortgage         | Pass   |
|       | pages4.5.py::TestPage4Mortgage                   | test_with_mortgage                         | Pass   |
|       | pages4.5.py::TestPage4Mortgage                   | test_without_mortgage                      | Pass   |
|       | pages4.5.py::TestPage5Deterministic              | test_contains_reference_table              | Pass   |
|       | pages4.5.py::TestPage5Deterministic              | test_damage_calculation                    | Pass   |
|       | pages4.5.py::TestPage5Deterministic              | test_negative_equity_flag                  | Pass   |
|       | pages4.5.py::TestPage5Deterministic              | test_returns_list                          | Pass   |
|       | pages4.5.py::TestPage5Deterministic              | test_signature_block_present               | Pass   |
|       | pages4.5.py::TestPage5Deterministic              | test_without_mortgage                      | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_building_page_calculated              | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_empty_property_does_not_crash         | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_get_property_id_missing_nested_key    | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_get_property_id_present               | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_has_tables                            | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_has_title_paragraph                   | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_proper_pid_in_output                  | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_proper_page_overview_section_header   | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_proper_page_summary_section_header    | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_returns_list                          | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_unknown_property_id_fallback          | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_with_no_page_data_does_not_crash      | Pass   |
|       | page_02_location.py::TestLocationPage            | test_address_subsection_header             | Pass   |
|       | page_02_location.py::TestLocationPage            | test_administrative_subsection_header      | Pass   |
|       | page_02_location.py::TestLocationPage            | test_area_classification_subsection_header | Pass   |
|       | page_02_location.py::TestLocationPage            | test_density_high                          | Pass   |
|       | page_02_location.py::TestLocationPage            | test_density_low                           | Pass   |
|       | page_02_location.py::TestLocationPage            | test_density_low_medium                    | Pass   |
|       | page_02_location.py::TestLocationPage            | test_density_medium                        | Pass   |
|       | page_02_location.py::TestLocationPage            | test_density_very_high                     | Pass   |
|       | page_02_location.py::TestLocationPage            | test_empty_property_does_not_crash         | Pass   |
|       | page_02_location.py::TestLocationPage            | test_geographic_subsection_header          | Pass   |
|       | page_02_location.py::TestLocationPage            | test_has_section_header                    | Pass   |
|       | page_02_location.py::TestLocationPage            | test_has_tables                            | Pass   |
|       | page_02_location.py::TestLocationPage            | test_no_coordinates_skips_geo_rows         | Pass   |
|       | page_02_location.py::TestLocationPage            | test_returns_list                          | Pass   |
|       | page_02_location.py::TestLocationPage            | test_with_mortgage_data_does_not_crash     | Pass   |
|       | page_03_attributes.py::TestAttributesPage        | test_contemporary                          | Pass   |
|       | page_03_attributes.py::TestAttributesPage        | test_historic                              | Pass   |
|       | page_03_attributes.py::TestAttributesPage        | test_mature                                | Pass   |
|       | page_03_attributes.py::TestAttributesPage        | test_modern                                | Pass   |
|       | page_03_attributes.py::TestAttributesPage        | test_basic_info_subsection                 | Pass   |
|       | page_03_attributes.py::TestAttributesPage        | test_building_age_subsection               | Pass   |

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| Model | Module                                                                  | Test Name           | Status |
|-------|-------------------------------------------------------------------------|---------------------|--------|
|       | page_03_attributes.py::TestAttributeCondition                           | fair_recommendation | Pass   |
|       | page_03_attributes.py::TestAttributeCondition                           | good_recommendation | Pass   |
|       | page_03_attributes.py::TestAttributeCondition                           | poor_recommendation | Pass   |
|       | page_03_attributes.py::TestAttributeCondition                           | unknown_falls_back  | Pass   |
|       | page_03_attributes.py::TestAttributeEmptyPropertyDoesNotCrash           |                     | Pass   |
|       | page_03_attributes.py::TestAttributeExternalFeaturesSubsection          |                     | Pass   |
|       | page_03_attributes.py::TestAttributePageSectionHeader                   |                     | Pass   |
|       | page_03_attributes.py::TestAttributePageTables                          |                     | Pass   |
|       | page_03_attributes.py::TestAttributePageConstructionYearSkipsAge        |                     | Pass   |
|       | page_03_attributes.py::TestAttributePhysicalDimensionsSubsection        |                     | Pass   |
|       | page_03_attributes.py::TestAttributeReturnsList                         |                     | Pass   |
|       | page_03_attributes.py::TestAttributePageCompositionSubsection           |                     | Pass   |
|       | page_03_attributes.py::TestAttributeWithMortgageDataDoesNotCrash        |                     | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeriesReturnsShortList       |                     | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeriesTitle                  |                     | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeriesNameDoesNotCrash       |                     | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeriesMessageIncluded        |                     | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeriesReturnsShortList       |                     | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeriesTable                  |                     | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeriesMixedStatuses          |                     | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeriesWithThresholds         |                     | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeriesReturnsMoreElements    |                     | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeriesTableIncluded          |                     | Pass   |
|       | page_04_construction.py::TestConstructionPageBasicsYearShown            |                     | Pass   |
|       | page_04_construction.py::TestConstructionPagePropertyDoesNotCrash       |                     | Pass   |
|       | page_04_construction.py::TestConstructionPageTableIsShown               |                     | Pass   |
|       | page_04_construction.py::TestConstructionPageTableHeader                |                     | Pass   |
|       | page_04_construction.py::TestConstructionPageBasics                     |                     | Pass   |
|       | page_04_construction.py::TestConstructionPageButtonOnDataReturnsMessage |                     | Pass   |
|       | page_04_construction.py::TestConstructionPageButtonOnYear               |                     | Pass   |
|       | page_04_construction.py::TestConstructionPageBasicsWithData             |                     | Pass   |
|       | page_04_construction.py::TestConstructionPageTableIsShown               |                     | Pass   |
|       | page_04_construction.py::TestConstructionTypeAnalysisOn                 |                     | Pass   |
|       | page_04_construction.py::TestConstructionTypeAnalysisConstruction       |                     | Pass   |
|       | page_04_construction.py::TestConstructionTypeAnalysisOnType             |                     | Pass   |
|       | page_04_construction.py::TestConstructionTypeAnalysisConstruction       |                     | Pass   |
|       | page_04_construction.py::TestConstructionTypeAnalysisConstruction       |                     | Pass   |
|       | page_04_construction.py::TestConstructionTypeAnalysisConstructionType   |                     | Pass   |
|       | page_04_construction.py::TestFoundationPadForAdaptation                 |                     | Pass   |
|       | page_04_construction.py::TestFoundationTypeAndAnalysis                  |                     | Pass   |
|       | page_04_construction.py::TestFoundationTypeAndAnalysis                  |                     | Pass   |
|       | page_04_construction.py::TestFoundationTypeAndAnalysis                  |                     | Pass   |
|       | page_04_construction.py::TestFoundationTypeAndAnalysis                  |                     | Pass   |

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| Model | Module                                          | Test Name                                           | Status |
|-------|-------------------------------------------------|-----------------------------------------------------|--------|
|       | page_05_risk_assessment.py::TestRiskProximity   | TestRiskProximity1000m                              | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity   | TestRiskProximitycloser_1000m                       | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity   | TestRiskProximity300closer_1000m                    | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity   | TestRiskProximityBrands_100m                        | Pass   |
|       | page_05_risk_assessment.py::TestRiskAssessment  | TestRiskAssessmenthigh                              | Pass   |
|       | page_05_risk_assessment.py::TestRiskAssessment  | TestRiskAssessmentrisk                              | Pass   |
|       | page_05_risk_assessment.py::TestRiskAssessment  | TestRiskAssessmentkeyword                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskAssessment  | TestRiskAssessmentrisk                              | Pass   |
|       | page_05_risk_assessment.py::TestRiskAssessment  | TestRiskAssessmentrisk                              | Pass   |
|       | page_05_risk_assessment.py::TestRiskAssessment  | TestRiskAssessmentrisk                              | Pass   |
|       | page_06_financial.py::TestFinancialPage         | testPageBasicproperty_does_not_crash                | Pass   |
|       | page_06_financial.py::TestFinancialPage         | testPageBasicfinancial_header                       | Pass   |
|       | page_06_financial.py::TestFinancialPage         | testPageBasic                                       | Pass   |
|       | page_06_financial.py::TestFinancialPage         | testPageBasicarea_skips_value_per_sqm               | Pass   |
|       | page_06_financial.py::TestFinancialPage         | testPageBasicproperty_value                         | Pass   |
|       | page_06_financial.py::TestFinancialPage         | testPageBasicvaluation_data                         | Pass   |
|       | page_06_financial.py::TestFinancialPage         | testPageBasicvaluation_field_with_value_in_name     | Pass   |
|       | page_06_financial.py::TestFinancialPage         | testPageBasicvaluation_field_without_value_in_name  | Pass   |
|       | page_06_financial.py::TestFinancialPage         | testPageBasicbins_list                              | Pass   |
|       | page_06_financial.py::TestFinancialPage         | testPageBasicbins_non_empty_list                    | Pass   |
|       | page_06_financial.py::TestFinancialPage         | testPageBasicper_sqm_shown_when_area_present        | Pass   |
|       | page_06_financial.py::TestInvestmentRating      | testRatingaverage_2_to_4pct                         | Pass   |
|       | page_06_financial.py::TestInvestmentRating      | testRatinglent_over_8pct                            | Pass   |
|       | page_06_financial.py::TestInvestmentRating      | testRating4_to_6pct                                 | Pass   |
|       | page_06_financial.py::TestInvestmentRating      | testRating6_to_8pct                                 | Pass   |
|       | page_06_financial.py::TestInvestmentRating      | testRatingunder_2pct                                | Pass   |
|       | page_06_financial.py::TestMarketSegmentAnalysis | testSegmentAnalysis1_under_250k                     | Pass   |
|       | page_06_financial.py::TestMarketSegmentAnalysis | testSegmentAnalysis250k_to_500k                     | Pass   |
|       | page_06_financial.py::TestMarketSegmentAnalysis | testSegmentAnalysisover_1m                          | Pass   |
|       | page_06_financial.py::TestMarketSegmentAnalysis | testSegmentAnalysisidle_500k_to_1m                  | Pass   |
|       | page_06_financial.py::TestTransaction           | testMonthlyrent_and_annual_rent                     | Pass   |
|       | page_06_financial.py::TestTransaction           | testMonthlytransaction_data                         | Pass   |
|       | page_06_financial.py::TestTransaction           | testMonthlyfield_computed                           | Pass   |
|       | page_06_financial.py::TestTransaction           | testSaledate_shown                                  | Pass   |
|       | page_06_financial.py::TestTransaction           | testSaleprice_and_appreciation                      | Pass   |
|       | page_06_financial.py::TestTransaction           | testSalesection_rendered                            | Pass   |
|       | page_07_protection.py::TestProtectionPage       | testPageNaturalMeasurerecommendation                | Pass   |
|       | page_07_protection.py::TestProtectionPage       | testPageNaturalMeasuressection_rendered             | Pass   |
|       | page_07_protection.py::TestProtectionPage       | testPageNoDataproperty_returns_list                 | Pass   |
|       | page_07_protection.py::TestProtectionPage       | testPageNoDataaction_measures_dict_shows_fallback   | Pass   |
|       | page_07_protection.py::TestProtectionPage       | testPageNoDataon_measures_shows_fallback            | Pass   |
|       | page_07_protection.py::TestProtectionPage       | testPageProtectionfewer_recommendations             | Pass   |
|       | page_07_protection.py::TestProtectionPage       | testPageRateprotection_recommendations_returns_dict | Pass   |
|       | page_07_protection.py::TestProtectionPage       | testPageRegcriticalmeasures_in_recommendations      | Pass   |

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| Model | Module                                                 | Test Name                                     | Status |
|-------|--------------------------------------------------------|-----------------------------------------------|--------|
|       | page_09_history.py::TestHistoryPage                    | test_empty_flood_events                       | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_empty_property_does_not_crash            | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_has_history_header                       | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_no_flood_history                         | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_returns_list                             | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_with_environmental_issues                | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_with_fire_incidents                      | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_with_ground_conditions                   | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_with_mortgage_data                       | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_avg_yield_average                        | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_avg_yield_below_average                  | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_avg_yield_excellent                      | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_avg_yield_good                           | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_avg_yield_poor                           | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_empty_property_does_not_crash            | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_empty_transactions                       | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_has_transactions_header                  | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_appreciation             | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_few_vacancies            | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_many_vacancies           | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_marketing_above_30       | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_marketing_above_90       | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_never_rented             | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_no_vacancy               | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_yield                    | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_transactions_action_history              | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_transactions_list                        | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_transactions_extended_marketing          | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_transactions_quick_sale                  | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_transactions_very_long_marketing         | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_transactions_mortgage_data               | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_transactions_rental_data                 | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_transactions_pages_data                  | Pass   |
|       | page_11_mortgage_overview.py::TestMortgageOverviewPage | test_application_submission                   | Pass   |
|       | page_11_mortgage_overview.py::TestMortgageOverviewPage | test_mortgage_reasons_when_default            | Pass   |
|       | page_11_mortgage_overview.py::TestMortgageOverviewPage | test_mortgage_reasons_when_not_default        | Pass   |
|       | page_11_mortgage_overview.py::TestMortgageOverviewPage | test_empty_mortgage_page_does_not_crash       | Pass   |
|       | page_11_mortgage_overview.py::TestMortgageOverviewPage | test_mortgage_page_accepted                   | Pass   |
|       | page_11_mortgage_overview.py::TestMortgageOverviewPage | test_mortgage_page_overview                   | Pass   |
|       | page_11_mortgage_overview.py::TestMortgageOverviewPage | test_mortgage_financial_terms_subsection      | Pass   |
|       | page_11_mortgage_overview.py::TestMortgageOverviewPage | test_mortgage_oriented_from_loan_and_purchase | Pass   |
|       | page_11_mortgage_overview.py::TestMortgageOverviewPage | test_mortgage_features_subsection             | Pass   |
|       | page_11_mortgage_overview.py::TestMortgageOverviewPage | test_mortgage_evidence_ligation_subsection    | Pass   |
|       | page_11_mortgage_overview.py::TestMortgageOverviewPage | test_mortgage_page_data_returns_message       | Pass   |

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| Model | Module                         | Test Name                                   | Status |
|-------|--------------------------------|---------------------------------------------|--------|
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage returns message    | Pass   |
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage data               | Pass   |
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage present            | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage field               | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage subsection          | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                     | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                     | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage risk                | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                     | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                     | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                     | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage subsection          | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage of_purchase_calcul  | Failed |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                     | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage risk                | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                     | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                     | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                     | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                     | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                     | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage returns message     | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage returns message     | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage with data           | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage present             | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage subsection            | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage                       | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage                       | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage burden                | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage                       | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage                       | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage                       | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage                       | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage dict_does_not_crash   | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage                       | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage reduces_net_cost      | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage terms_returns_message | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage data_returns_message  | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage setup_subsection      | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage                       | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage with data             | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage present               | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage percentage_of_loan    | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage overview_subsection   | Pass   |
|       | page_11c_regulatory.py::       | TestContentedSections together              | Pass   |
|       | page_11c_regulatory.py::       | TestContentedSection handled_gracefully     | Pass   |
|       | page_11c_regulatory.py::       | TestContentedSection mortgage_structure     | Pass   |

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| Model | Module                                                 | Test Name                                   | Status |
|-------|--------------------------------------------------------|---------------------------------------------|--------|
|       | page_11c_regulatory.py::TestContent                    | testRegulatoryFlag                          | Pass   |
|       | page_11c_regulatory.py::TestContent                    | testRegulatorypurpose_false                 | Pass   |
|       | page_11c_regulatory.py::TestContent                    | testRegulatorypurpose_true                  | Pass   |
|       | page_11c_regulatory.py::TestContent                    | testRegulatorysection_returns_elements      | Pass   |
|       | page_11c_regulatory.py::TestContent                    | testRegulatoryreference_number              | Pass   |
|       | page_11c_regulatory.py::TestContent                    | testRegulatoryregulatory_flags              | Pass   |
|       | page_11c_regulatory.py::TestHmdaSection                | testHmda_minimal_fields                     | Pass   |
|       | page_11c_regulatory.py::TestHmdaSection                | testHmda_rate_spread_formatted              | Pass   |
|       | page_11c_regulatory.py::TestHmdaSection                | testHmda_section_creates_table              | Pass   |
|       | page_11c_regulatory.py::TestMcotSection                | testMcot_advise_rejection_reason            | Pass   |
|       | page_11c_regulatory.py::TestMcotSection                | testMcot_all_fields_returns_list            | Pass   |
|       | page_11c_regulatory.py::TestMcotSection                | testMcot_section_creates_tables             | Pass   |
|       | page_11c_regulatory.py::TestMcotSection                | testMcot_with_minimal_fields                | Pass   |
|       | page_11c_regulatory.py::TestNoMsgPage                  | testNoMsg Mortgage message                  | Pass   |
|       | page_11c_regulatory.py::TestNoMsgPage                  | testNoMsg Mortgage returns_list             | Pass   |
|       | page_11c_regulatory.py::TestNoRegulatoryData           | testRegulatory_returns_message              | Pass   |
|       | page_11c_regulatory.py::TestNoRegulatoryData           | testRegulatory_data_list                    | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasics                 | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasicsShown            | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasicsHeader           | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasics                 | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasicsShown            | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasicsDate_shown       | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasicsShown            | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasicsShown            | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasics returns_message | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasics returns_message | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasics balance_shown   | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasicsShown            | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasicsShown            | Pass   |
|       | page_12_current_status.py::TestCurrentStatusPageBasics | testCurrentStatusPageBasics returns_list    | Pass   |
|       | page_12_current_status.py::TestLFEBrnClass             | testLFEBrnClass_percentage_not_decimal      | Pass   |
|       | page_12_current_status.py::TestLFEBrnClass             | testLFEBrnClass_lower_60_low_medium         | Pass   |
|       | page_12_current_status.py::TestLFEBrnClass             | testLFEBrnClass_lower_70_medium             | Pass   |
|       | page_12_current_status.py::TestLFEBrnClass             | testLFEBrnClass_lower_80_medium_high        | Pass   |
|       | page_12_current_status.py::TestLFEBrnClass             | testLFEBrnClass_lower_90_high_risk          | Pass   |
|       | page_12_current_status.py::TestLFEBrnClass             | testLFEBrnClass_under_60_low_risk           | Pass   |
|       | page_12_current_status.py::TestPresentHoliday          | testPresentHoliday_fields_shown             | Pass   |
|       | page_12_current_status.py::TestPresentHoliday          | testPresentHoliday_taken_shown              | Pass   |
|       | page_12_current_status.py::TestPresentHoliday          | testPresentHoliday_date_shown               | Pass   |
|       | page_12_current_status.py::TestPresentHoliday          | testPresentHoliday_data_no_section          | Pass   |
|       | page_12_current_status.py::TestPresentHoliday          | testPresentHoliday_shown                    | Pass   |
|       | page_12_current_status.py::TestPresentPersonalBalance  | testPresentPersonalBalance_shown            | Pass   |
|       | page_12_current_status.py::TestPresentPersonalBalance  | testPresentPersonalBalance_poor             | Pass   |

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| Model | Module                                 | Test Name                                         | Status |
|-------|----------------------------------------|---------------------------------------------------|--------|
|       | page_12_current_status.py::TestPerfor  | testPerforissedPayments concerning                | Pass   |
|       | page_12_current_status.py::TestPerfor  | testPerforissedPayments concerning                | Pass   |
|       | page_12_current_status.py::TestPerfor  | testPerforissedPayments concerning                | Pass   |
|       | page_12_current_status.py::TestPerfor  | testPerforissedPayments concerning                | Pass   |
|       | page_12_current_status.py::TestPerfor  | testPerforissedPayments concerning                | Pass   |
|       | page_12_current_status.py::TestPerfor  | testPerforissedPayments concerning                | Pass   |
|       | page_12_current_status.py::TestPerfor  | testPerforissedPayments concerning                | Pass   |
|       | page_13_risk_analysis.py::TestCredit   | testCreditRiskBranded credit_800_plus             | Pass   |
|       | page_13_risk_analysis.py::TestCredit   | testCreditRiskBranded credit_580_to_669           | Pass   |
|       | page_13_risk_analysis.py::TestCredit   | testCreditRiskBranded credit_670_to_739           | Pass   |
|       | page_13_risk_analysis.py::TestCredit   | testCreditRiskBranded credit_under_580            | Pass   |
|       | page_13_risk_analysis.py::TestCredit   | testCreditRiskBranded credit_740_to_799           | Pass   |
|       | page_13_risk_analysis.py::TestDTI      | testDTIStandards dti_under_28                     | Pass   |
|       | page_13_risk_analysis.py::TestDTI      | testDTIStandards dti_36_to_43                     | Pass   |
|       | page_13_risk_analysis.py::TestDTI      | testDTIStandards dti_28_to_36                     | Pass   |
|       | page_13_risk_analysis.py::TestDTI      | testDTIStandards payment_dti_unknown              | Pass   |
|       | page_13_risk_analysis.py::TestDTI      | testDTIStandards high_dti_over_43                 | Pass   |
|       | page_13_risk_analysis.py::TestHedge    | testHedgeMethod date_monitoring_high_score        | Pass   |
|       | page_13_risk_analysis.py::TestHedge    | testHedgeMethod date_monitoring_low_score         | Pass   |
|       | page_13_risk_analysis.py::TestHedge    | testHedgeMethod date_monitoring_medium_score      | Pass   |
|       | page_13_risk_analysis.py::TestHedge    | testHedgeMethod date_recommendations_high_score   | Pass   |
|       | page_13_risk_analysis.py::TestHedge    | testHedgeMethod date_recommendations_low_score    | Pass   |
|       | page_13_risk_analysis.py::TestHedge    | testHedgeMethod date_recommendations_medium_score | Pass   |
|       | page_13_risk_analysis.py::TestHedge    | testHedgeMethod identify_key_factors_high_risk    | Pass   |
|       | page_13_risk_analysis.py::TestHedge    | testHedgeMethod identify_key_factors_medium_risk  | Pass   |
|       | page_13_risk_analysis.py::TestInterest | testInterestRateRate 5_to_7                       | Pass   |
|       | page_13_risk_analysis.py::TestInterest | testInterestRateRate over_7                       | Pass   |
|       | page_13_risk_analysis.py::TestInterest | testInterestRateRate under_3                      | Pass   |
|       | page_13_risk_analysis.py::TestInterest | testInterestRateRate 3_to_5                       | Pass   |
|       | page_13_risk_analysis.py::TestInterest | testInterestRateRate unknown                      | Pass   |
|       | page_13_risk_analysis.py::TestLTV      | testLTVRiskHigh ltv_90_to_95                      | Pass   |
|       | page_13_risk_analysis.py::TestLTV      | testLTVRiskLow ltv_under_70                       | Pass   |
|       | page_13_risk_analysis.py::TestLTV      | testLTVRiskMedium ltv_70_to_80                    | Pass   |
|       | page_13_risk_analysis.py::TestLTV      | testLTVRiskMedium ltv_80_to_90                    | Pass   |
|       | page_13_risk_analysis.py::TestLTV      | testLTVRiskVeryHigh ltv_over_95                   | Pass   |
|       | page_13_risk_analysis.py::TestOverall  | testOverallRisk scenario                          | Pass   |
|       | page_13_risk_analysis.py::TestOverall  | testOverallRisk scenario                          | Pass   |
|       | page_13_risk_analysis.py::TestOverall  | testOverallRisk only_high_flood_risk              | Pass   |
|       | page_13_risk_analysis.py::TestOverall  | testOverallRisk only_very_low_flood_risk          | Pass   |
|       | page_13_risk_analysis.py::TestPayment  | testPaymentPerforissedPayments                    | Pass   |
|       | page_13_risk_analysis.py::TestPayment  | testPaymentPerforissedPayments over_3             | Pass   |
|       | page_13_risk_analysis.py::TestPayment  | testPaymentPerforissedPayments good               | Pass   |
|       | page_13_risk_analysis.py::TestPayment  | testPaymentPerforissedPayments 1_to_3             | Pass   |
|       | page_13_risk_analysis.py::TestRisk     | testRiskAnalysis PagePass with mortgage           | Pass   |

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| Model | Module                                                                    | Test Name                                       | Status |
|-------|---------------------------------------------------------------------------|-------------------------------------------------|--------|
|       | page_13_risk_analysis.py::TestRiskAnalysisPageBasics                      | test_analyze_page_does_not_crash                | Pass   |
|       | page_13_risk_analysis.py::TestRiskAnalysisPageHeader                      | test_analyze_page_header                        | Pass   |
|       | page_13_risk_analysis.py::TestRiskAnalysisPageBasics                      | test_analyze_page_basics                        | Pass   |
|       | page_13_risk_analysis.py::TestRiskAnalysisPageBasics                      | test_analyze_page_basics_mode                   | Pass   |
|       | page_13_risk_analysis.py::TestRiskAnalysisPageBasics                      | test_analyze_page_basics                        | Pass   |
|       | page_14_borrower_profile.py::TestAgeRiskBranching_retirement_over_55      | test_age_branching_retirement_over_55           | Pass   |
|       | page_14_borrower_profile.py::TestAgeRiskBranching_under_25                | test_age_branching_under_25                     | Pass   |
|       | page_14_borrower_profile.py::TestAgeRiskBranching_years_25_to_55          | test_age_branching_years_25_to_55               | Pass   |
|       | page_14_borrower_profile.py::TestAgeRiskBranching_coaching_retirement     | test_age_branching_coaching_retirement          | Pass   |
|       | page_14_borrower_profile.py::TestAgeRiskBranching_risky_career            | test_age_branching_risky_career                 | Pass   |
|       | page_14_borrower_profile.py::TestAgeRiskBranching_risky_years             | test_age_branching_risky_years                  | Pass   |
|       | page_14_borrower_profile.py::TestAssetPortfolioRisk_high_for_poor_score   | test_asset_portfolio_high_for_poor_score        | Pass   |
|       | page_14_borrower_profile.py::TestAssetPortfolioRisk_low_for_high_score    | test_asset_portfolio_low_for_high_score         | Pass   |
|       | page_14_borrower_profile.py::TestAssetPortfolioRisk_medium_for_good_score | test_asset_portfolio_medium_for_good_score      | Pass   |
|       | page_14_borrower_profile.py::TestAssetPortfolioRisk_medium_for_fair_score | test_asset_portfolio_medium_for_fair_score      | Pass   |
|       | page_14_borrower_profile.py::TestAssetEmploymentRisk_established_2_to_5   | test_asset_employment_established_2_to_5        | Pass   |
|       | page_14_borrower_profile.py::TestAssetEmploymentRisk_recent_under_2       | test_asset_employment_recent_under_2            | Pass   |
|       | page_14_borrower_profile.py::TestAssetEmploymentRisk_stable_5_plus        | test_asset_employment_stable_5_plus             | Pass   |
|       | page_14_borrower_profile.py::TestAssetIncomeRisk_low_for_high_income      | test_asset_income_low_for_high_income           | Pass   |
|       | page_14_borrower_profile.py::TestAssetIncomeRisk_low_medium_adequate      | test_asset_income_low_medium_adequate           | Pass   |
|       | page_14_borrower_profile.py::TestAssetIncomeRisk_medium_limited           | test_asset_income_medium_limited                | Pass   |
|       | page_14_borrower_profile.py::TestAssetsBondsandRisk                       | test_assets_bonds_and_risk                      | Pass   |
|       | page_14_borrower_profile.py::TestBorrowerProfilePageShown                 | test_borrower_profile_page_shown                | Pass   |
|       | page_14_borrower_profile.py::TestBorrowerProfilePageHeader                | test_borrower_profile_page_header               | Pass   |
|       | page_14_borrower_profile.py::TestBorrowerProfilePageBasics                | test_borrower_profile_page_basics               | Pass   |
|       | page_14_borrower_profile.py::TestBorrowerProfilePageValidation            | test_borrower_profile_page_validation           | Pass   |
|       | page_14_borrower_profile.py::TestBorrowerProfilePageStatusShown           | test_borrower_profile_page_status_shown         | Pass   |
|       | page_14_borrower_profile.py::TestBorrowerProfilePageBasics                | test_borrower_profile_page_basics               | Pass   |
|       | page_14_borrower_profile.py::TestBorrowerProfilePageBasics_returns_early  | test_borrower_profile_page_basics_returns_early | Pass   |
|       | page_14_borrower_profile.py::TestBorrowerProfilePageBasics_message        | test_borrower_profile_page_basics_message       | Pass   |
|       | page_14_borrower_profile.py::TestBorrowerProfilePageBasics                | test_borrower_profile_page_basics               | Pass   |
|       | page_14_borrower_profile.py::TestBorrowerProfilePageBasics                | test_borrower_profile_page_basics               | Pass   |
|       | page_14_borrower_profile.py::TestBorrowerProfilePageBasics_list           | test_borrower_profile_page_basics_list          | Pass   |
|       | page_14_borrower_profile.py::TestCreditScoreBand800plus                   | test_credit_score_band_800_plus                 | Pass   |
|       | page_14_borrower_profile.py::TestCreditScoreBand669to739                  | test_credit_score_band_669_to_739               | Pass   |
|       | page_14_borrower_profile.py::TestCreditScoreBand580to669                  | test_credit_score_band_580_to_669               | Pass   |
|       | page_14_borrower_profile.py::TestCreditScoreBand740to799                  | test_credit_score_band_740_to_799               | Pass   |
|       | page_14_borrower_profile.py::TestEmploymentStabilityBranchear             | test_employment_stability_branch_ear            | Pass   |
|       | page_14_borrower_profile.py::TestEmploymentStabilityBranches              | test_employment_stability_branches              | Pass   |
|       | page_14_borrower_profile.py::TestEmploymentStabilityBranches              | test_employment_stability_branches              | Pass   |
|       | page_14_borrower_profile.py::TestEmploymentStabilityBranches              | test_employment_stability_branches              | Pass   |
|       | page_14_borrower_profile.py::TestLoanAnalysisDe_30k_to_50k                | test_loan_analysis_de_30k_to_50k                | Pass   |
|       | page_14_borrower_profile.py::TestLoanAnalysisDe_50k_to_100k               | test_loan_analysis_de_50k_to_100k               | Pass   |

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| Model                         | Module                             | Test Name                                              | Status |
|-------------------------------|------------------------------------|--------------------------------------------------------|--------|
| Test                          | page_14.borrower_profile.py::Test  | test_high_analysis_over_100k                           | Pass   |
|                               | page_14.borrower_profile.py::Test  | test_min_analysis_is_bene_has                          | Pass   |
|                               | page_14.borrower_profile.py::Test  | test_min_analysis_is_bene_with_secondary               | Pass   |
|                               | page_15.data_summary.py::Test      | test_analyze_data_generated_percentage_computed        | Pass   |
|                               | page_15.data_summary.py::Test      | test_analyze_data_generated_excluded                   | Pass   |
|                               | page_15.data_summary.py::Test      | test_analyze_data_generated_included                   | Pass   |
|                               | page_15.data_summary.py::Test      | test_analyze_data_generated_section_present            | Pass   |
|                               | page_15.data_summary.py::Test      | test_analyze_data_generated_tenness                    | Pass   |
|                               | page_15.data_summary.py::Test      | test_analyze_data_generated_present_excluded           | Pass   |
|                               | page_15.data_summary.py::Test      | test_assets_data_and_top_sections_shown                | Pass   |
|                               | page_15.data_summary.py::Test      | test_assets_data_quality_completeness_high             | Pass   |
|                               | page_15.data_summary.py::Test      | test_assets_data_quality_completeness_low              | Pass   |
|                               | page_15.data_summary.py::Test      | test_assets_data_quality_completeness_medium           | Pass   |
|                               | page_15.data_summary.py::Test      | test_assets_data_quality_over_90                       | Pass   |
|                               | page_15.data_summary.py::Test      | test_assets_data_quality_75                            | Pass   |
|                               | page_15.data_summary.py::Test      | test_assets_data_quality_90                            | Pass   |
|                               | page_15.data_summary.py::Test      | test_assets_data_quality_40_to_60                      | Pass   |
|                               | page_15.data_summary.py::Test      | test_assets_data_quality_40                            | Pass   |
|                               | page_15.data_summary.py::Test      | test_both_empty_PageBasics                             | Pass   |
|                               | page_15.data_summary.py::Test      | test_empty_PageBasics_does_not_crash                   | Pass   |
|                               | page_15.data_summary.py::Test      | test_shows_data_PageBasics                             | Pass   |
|                               | page_15.data_summary.py::Test      | test_shows tab PageBasics                              | Pass   |
|                               | page_15.data_summary.py::Test      | test_summary_PageBasics_property                       | Pass   |
|                               | page_15.data_summary.py::Test      | test_returns PageBasics                                | Pass   |
|                               | page_15.data_summary.py::Test      | test_returns PageEmpty                                 | Pass   |
|                               | page_15.data_summary.py::Test      | test_summary_PageBasics_data                           | Pass   |
|                               | page_15.data_summary.py::Test      | test_generate_sections_150_to_175_gets_enhancement_rec | Pass   |
|                               | page_15.data_summary.py::Test      | test_generate_sections_maintains_recommendation        | Pass   |
|                               | page_15.data_summary.py::Test      | test_generate_51_recommendations                       | Pass   |
|                               | page_15.data_summary.py::Test      | test_generate_sections_150_gets_recommendation         | Pass   |
|                               | page_15.data_summary.py::Test      | test_generate_recommendations                          | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_timestamp                     | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_extracted                     | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_extracted                     | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_extracted                     | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_extracted                     | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_extracted                     | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_extracted                     | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_extracted                     | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_extracted                     | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_extracted                     | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_extracted                     | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_extracted                     | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_extracted                     | Pass   |
|                               | page_15.data_summary.py::Test      | test_has_report_metadata_extracted                     | Pass   |
| page_15.data_summary.py::Test | test_has_report_metadata_extracted | Pass                                                   |        |
| page_15.data_summary.py::Test | test_has_report_metadata_extracted | Pass                                                   |        |
| cli.py::Test                  | ErrorHandling                      | test.bad_property_file.exits_with_1                    | Pass   |
| cli.py::Test                  | ErrorHandling                      | test.invalid.json.exits_with_1                         | Pass   |
| cli.py::Test                  | ListCategories                     | test.list.categories_does_not_open_property_file       | Pass   |
| cli.py::Test                  | ListCategories                     | test.list.categories_runs_without_error                | Pass   |

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| Model | Module                                                                            | Test Name                                                                  | Status |
|-------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------|
|       | cli.py::TestListPages                                                             | test_list_pages_does_not_open_property_file                                | Fail   |
|       | cli.py::TestListPages                                                             | test_list_pages_runs_without_error                                         | Pass   |
|       | cli.py::TestOptionalLoading                                                       | test_mortgage_file_is_loaded                                               | Pass   |
|       | cli.py::TestOptionalLoading                                                       | test_property_id_filters_without_mortgage                                  | Pass   |
|       | cli.py::TestOptionalLoading                                                       | test_property_id_with_mortgage_file                                        | Pass   |
|       | cli.py::TestReportGeneration                                                      | test_else_branch_triggers_except                                           | Pass   |
|       | cli.py::TestReportGeneration                                                      | test_report_type_mortgage_focused_with_mortgage                            | Fail   |
|       | cli.py::TestReportGeneration                                                      | test_report_type_mortgage_focused_without_mortgage_falls_through           | Pass   |
|       | cli.py::TestReportGeneration                                                      | test_report_type_property_only                                             | Pass   |
|       | cli.py::TestReportGeneration                                                      | test_report_type_risk_focused                                              | Pass   |
|       | cli.py::TestReportGeneration                                                      | test_success_logging_executes                                              | Pass   |
|       | convenience.py::TestGeneratePropertyReportConceptFindsDoesNotRaise                | test_generate_property_report_concept_finds_does_not_raise                 | Pass   |
|       | convenience.py::TestGeneratePropertyReportConceptFindsFileOpen                    | test_generate_property_report_concept_finds_file_open                      | Pass   |
|       | convenience.py::TestGeneratePropertyReportConceptTrueCallsFunction                | test_generate_property_report_concept_true_calls_function                  | Pass   |
|       | convenience.py::TestGeneratePropertyReportConceptFalseSubfunctionToGenerateReport | test_generate_property_report_concept_false_subfunction_to_generate_report | Fail   |
|       | convenience.py::TestGeneratePropertyReportMortgageFocusedTypeWithMortgage         | test_generate_property_report_mortgage_focused_type_with_mortgage          | Pass   |
|       | convenience.py::TestGeneratePropertyReportMortgageFocusedWithoutMortgageFallsBack | test_generate_property_report_mortgage_focused_without_mortgage_falls_back | Fail   |
|       | convenience.py::TestGeneratePropertyReportOnlyTypeFunction                        | test_generate_property_report_only_type_function                           | Pass   |
|       | convenience.py::TestGeneratePropertyReportsPathConvenienceFunction                | test_generate_property_reports_path_convenience_function                   | Pass   |
|       | convenience.py::TestGeneratePropertyRiskFocusedSearchTypeFunction                 | test_generate_property_risk_focused_search_type_function                   | Pass   |
|       | convenience.py::TestGeneratePropertyUnknownTypeFallsBackToGenerateReport          | test_generate_property_unknown_type_falls_back_to_generate_report          | Fail   |
|       | elements.py::TestGenerateElementsEmptyPageList                                    | test_empty_page_list                                                       | Pass   |
|       | elements.py::TestGenerateElementsExceptionInPageModuleContinues                   | test_exception_in_page_module_continues                                    | Pass   |
|       | elements.py::TestGenerateElementsFirstPageNoPageBreak                             | test_first_page_no_page_break                                              | Pass   |
|       | elements.py::TestGenerateElementsMixedValidAndUnknownPages                        | test_mixed_valid_and_unknown_pages                                         | Pass   |
|       | elements.py::TestGenerateElementsMortgageDataPassedToPage                         | test_mortgage_data_passed_to_page                                          | Pass   |
|       | elements.py::TestGenerateElementsMultiplePagesHavePageBreaks                      | test_multiple_pages_have_page_breaks                                       | Pass   |
|       | elements.py::TestGenerateElementsReturnsList                                      | test_returns_list                                                          | Pass   |
|       | elements.py::TestGenerateElementsUnknownPageSkipped                               | test_unknown_page_skipped                                                  | Pass   |
|       | elements.py::TestGenerateReportsAllUnknownPagesStillCreatesPdf                    | test_all_unknown_pages_still_creates_pdf                                   | Pass   |
|       | elements.py::TestGenerateReportsAutoSelectCalledWhenPagesNone                     | test_auto_select_called_when_pages_none                                    | Pass   |
|       | elements.py::TestGenerateReportsAutoSelectNotCalledWhenPagesProvided              | test_auto_select_not_called_when_pages_provided                            | Fail   |
|       | elements.py::TestGenerateReportsCustomOutputFilename                              | test_custom_output_filename                                                | Pass   |
|       | elements.py::TestGenerateReportsCustomPagesToInclude                              | test_custom_pages_to_include                                               | Pass   |
|       | elements.py::TestGenerateReportsEmptyPropertyData                                 | test_empty_property_data                                                   | Pass   |
|       | elements.py::TestGenerateReportsFullDataReport                                    | test_full_data_report                                                      | Pass   |
|       | elements.py::TestGenerateReportsPdfExists                                         | test_pdf_exists                                                            | Pass   |
|       | elements.py::TestGenerateReportsPdfNonEmpty                                       | test_pdf_non_empty                                                         | Pass   |
|       | elements.py::TestGenerateReportsPdfSuffix                                         | test_pdf_suffix                                                            | Pass   |
|       | elements.py::TestGenerateReportsReturnsPath                                       | test_returns_path                                                          | Pass   |
|       | elements.py::TestGenerateReportsWithMortgageData                                  | test_with_mortgage_data                                                    | Pass   |
|       | finders.py::TestEdgeCasesCompletelyEmptyPropertyData                              | test_completely_empty_property_data                                        | Pass   |
|       | finders.py::TestEdgeCasesNegativeValueProperty                                    | test_negative_value_property                                               | Pass   |
|       | finders.py::TestEdgeCasesPdfContentIsBytes                                        | test_pdf_content_is_bytes                                                  | Pass   |

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| Model | Module                                   | Test Name                                                    | Status |
|-------|------------------------------------------|--------------------------------------------------------------|--------|
|       | finders.py::TestEdgeCases                | test_property_data_with_none_header                          | Pass   |
|       | finders.py::TestEdgeCases                | test_report_with_all_pages                                   | Pass   |
|       | finders.py::TestEdgeCases                | test_zero_value_property                                     | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_find_mortgage_with_mortgages_found                      | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_find_mortgage_with_mortgages_not_found_returns_none     | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_find_mortgage_without_mortgages_key_returns_data_itself | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_find_mortgage_with_empty_mortgages_list_returns_none    | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_find_mortgage_with_invalid_type_returns_data_as_is      | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_find_mortgage_with_invalid_input_found                  | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_find_mortgage_with_invalid_input_not_found_returns_none | Pass   |
|       | finders.py::TestFindPropertyById         | test_dict_without_list_key_wraps_itself                      | Pass   |
|       | finders.py::TestFindPropertyById         | test_empty_properties_list_raises                            | Pass   |
|       | finders.py::TestFindPropertyById         | test_invalid_type_raises_value_error                         | Pass   |
|       | finders.py::TestFindPropertyById         | test_list_input_found                                        | Pass   |
|       | finders.py::TestFindPropertyById         | test_list_not_found_raises                                   | Pass   |
|       | finders.py::TestFindPropertyById         | test_none_raises_value_error                                 | Pass   |
|       | finders.py::TestFindPropertyById         | test_not_found_raises_value_error                            | Pass   |
|       | finders.py::TestFindPropertyById         | test_portfolio_key_found                                     | Pass   |
|       | finders.py::TestFindPropertyById         | test_properties_key_found                                    | Pass   |
|       | finders.py::TestOpenPdfFile              | test_darwin_success                                          | Pass   |
|       | finders.py::TestOpenPdfFile              | test_generic_exception_returns_false                         | Pass   |
|       | finders.py::TestOpenPdfFile              | test_linux_success                                           | Pass   |
|       | finders.py::TestOpenPdfFile              | test_subprocess_called_process_error_returns_false           | Pass   |
|       | finders.py::TestOpenPdfFile              | test_unknown_platform_uses_webbrowser                        | Pass   |
|       | finders.py::TestOpenPdfFile              | test_windows_case_insensitive                                | Pass   |
|       | finders.py::TestOpenPdfFile              | test_windows_success                                         | Pass   |
|       | init.py::TestAutoSelectPages             | test_always_includes_analysis_pages                          | Pass   |
|       | init.py::TestAutoSelectPages             | test_always_includes_property_pages                          | Pass   |
|       | init.py::TestAutoSelectPages             | test_empty_property_data                                     | Pass   |
|       | init.py::TestAutoSelectPages             | test_with_mortgage_includes_mortgage_pages                   | Pass   |
|       | init.py::TestAutoSelectPages             | test_without_mortgage_excludes_mortgage_pages                | Pass   |
|       | init.py::TestGenerateFilename            | test_contains_property_id                                    | Pass   |
|       | init.py::TestGenerateFilename            | test_ends_with_pdf                                           | Pass   |
|       | init.py::TestGenerateFilename            | test_missing_property_id_uses_unknown                        | Pass   |
|       | init.py::TestGenerateFilename            | test_partial_path_missing_header                             | Pass   |
|       | init.py::TestGenerateFilename            | test_partial_path_missing_property_id                        | Pass   |
|       | init.py::TestGenerateFilename            | test_returns_string                                          | Pass   |
|       | init.py::TestGenerateFilename            | test_timestamp_in_filename                                   | Pass   |
|       | init.py::TestPropertyReportGenerator     | test_report_categories_dict_initialized                      | Pass   |
|       | init.py::TestPropertyReportGenerator     | test_report_explicit_output_dir_path                         | Pass   |
|       | init.py::TestPropertyReportGenerator     | test_report_explicit_output_dir_string                       | Pass   |
|       | init.py::TestPropertyReportGenerator     | test_report_output_dir_uses_config                           | Pass   |
|       | init.py::TestPropertyReportGenerator     | test_report_output_dir_created                               | Pass   |
|       | init.py::TestPropertyReportGenerator     | test_report_property_pages_dict_initialized                  | Pass   |

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| Model | Module                                                                           | Test Name                                                   | Status   |
|-------|----------------------------------------------------------------------------------|-------------------------------------------------------------|----------|
|       | specialised.py::TestHeaderFooter                                                 | test_header_footer_callable                                 | Pass     |
|       | specialised.py::TestHeaderFooter                                                 | test_header_footer_draws_confidential                       | Pass     |
|       | specialised.py::TestSpecialisedReportGenerator                                   | test_generate_mortgage_focused_report_custom_filename       | Fail     |
|       | specialised.py::TestSpecialisedReportGenerator                                   | test_generate_mortgage_focused_report_exists                | Pass     |
|       | specialised.py::TestSpecialisedReportGenerator                                   | test_generate_property_only_report_custom_filename          | Fail     |
|       | specialised.py::TestSpecialisedReportGenerator                                   | test_generate_property_only_report_exists                   | Pass     |
|       | specialised.py::TestSpecialisedReportGenerator                                   | test_generate_risk_focused_report_custom_filename           | Fail     |
|       | specialised.py::TestSpecialisedReportGenerator                                   | test_generate_risk_focused_report_with_mortgage             | Pass     |
|       | specialised.py::TestSpecialisedReportGenerator                                   | test_generate_risk_focused_report_without_mortgage          | Pass     |
|       | specialised.py::TestSpecialisedReportGenerator                                   | test_Mortgage_focused_includes_mortgage_content_pages       | Pass     |
|       | specialised.py::TestSpecialisedReportGenerator                                   | test_Method_focused_with_mortgage_include_current_status    | Pass     |
|       | specialised.py::TestSpecialisedReportGenerator                                   | test_Method_focused_without_mortgage_exclude_current_status | Fail     |
|       | specialised.py::TestUtilityMethods                                               | test_get_page_categories_returns_copy                       | Pass     |
|       | specialised.py::TestUtilityMethods                                               | test_get_page_categories_returns_dict                       | Pass     |
|       | specialised.py::TestUtilityMethods                                               | test_list_available_pages_contains_all_expected             | Expected |
|       | specialised.py::TestUtilityMethods                                               | test_list_available_pages_returns_list                      | Pass     |
|       | specialised.py::TestUtilityMethods                                               | test_validate_pages_all_invalid                             | Pass     |
|       | specialised.py::TestUtilityMethods                                               | test_validate_pages_all_valid                               | Pass     |
|       | specialised.py::TestUtilityMethods                                               | test_validate_pages_empty_list                              | Pass     |
|       | specialised.py::TestUtilityMethods                                               | test_validate_pages_mixed                                   | Pass     |
|       | test_base_page_coverage.py::TestAbstractConcreteSubClass                         | returns_empty_list                                          | Pass     |
|       | test_base_page_coverage.py::TestFormatCurrencyNoCurrency                         | returns_not_specified                                       | Pass     |
|       | test_base_page_coverage.py::TestFormattingCurrencyNone                           | delegates                                                   | Pass     |
|       | test_base_page_coverage.py::TestFormatPercentageNoPercentage                     | returns_not_specified                                       | Fail     |
|       | test_base_page_coverage.py::TestFormattingPercentageNone                         | delegates                                                   | Pass     |
|       | test_base_page_coverage.py::TestFormatEmptyString                                | returns_not_specified                                       | Pass     |
|       | test_base_page_coverage.py::TestFormatMalformedString                            | returns_not_specified                                       | Fail     |
|       | test_base_page_coverage.py::TestFormatValue                                      | returns_not_specified                                       | Pass     |
|       | test_base_page_coverage.py::TestPropertyBasePageTitleSet                         | Exists                                                      | Pass     |
|       | test_generator_direct.py::TestGenerateReportExceptionDoesNotRaise                |                                                             | Pass     |
|       | test_generator_direct.py::TestGenerateReportFalseSkips                           |                                                             | Pass     |
|       | test_generator_direct.py::TestGenerateReportTrueTries                            |                                                             | Pass     |
|       | test_generator_direct.py::TestGenerateReportRecursiveType                        |                                                             | Pass     |
|       | test_generator_direct.py::TestGenerateReportPropertyOnlyType                     |                                                             | Pass     |
|       | test_generator_direct.py::TestGenerateReportRiskOnlyType                         |                                                             | Pass     |
|       | test_generator_direct.py::TestGenerateReportUnknownTypeFallsBack                 |                                                             | Pass     |
|       | test_generator_direct.py::TestProduceReportSelectPageDirWithMortgage             |                                                             | Pass     |
|       | test_generator_direct.py::TestProduceReportSelectPageDirWithoutMortgage          |                                                             | Pass     |
|       | test_generator_direct.py::TestProduceReportGenerateDirExceptionContinues         |                                                             | Fail     |
|       | test_generator_direct.py::TestProduceReportGenerateDirSkipsUnknown               |                                                             | Pass     |
|       | test_generator_direct.py::TestProduceReportGenerateDirUnknownId                  |                                                             | Pass     |
|       | test_generator_direct.py::TestProduceReportGenerateDirFocusedReport              |                                                             | Pass     |
|       | test_generator_direct.py::TestProduceReportGenerateDirPropertyOnlyReport         |                                                             | Pass     |
|       | test_generator_direct.py::TestProduceReportGenerateDirRiskOfLossReportNoMortgage |                                                             | Pass     |

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| Model | Module                                                                    | Test Name                                                  | Status |
|-------|---------------------------------------------------------------------------|------------------------------------------------------------|--------|
|       | test_generator_direct.py::TestProjectGeneratorGeneral                     | test_report_with_mortgage                                  | Pass   |
|       | test_generator_direct.py::TestProjectGeneratorDirect                      | test_report_direct                                         | Pass   |
|       | test_generator_direct.py::TestProjectGeneratorPages                       | test_report_available_pages                                | Pass   |
|       | test_generator_direct.py::TestProjectGeneratorBaseconfig                  | test_report_baseconfig                                     | Pass   |
|       | test_generator_direct.py::TestProjectGeneratorDirect                      | test_report_direct                                         | Pass   |
|       | test_helpers.py::TestFindMortgageByDirect                                 | test_find_mortgages_found                                  | Pass   |
|       | test_helpers.py::TestFindMortgageByDirect                                 | test_find_mortgages_not_found_returns_none                 | Pass   |
|       | test_helpers.py::TestFindMortgageByProperty                               | test_find_type_returns_data                                | Pass   |
|       | test_helpers.py::TestFindMortgageByProperty                               | test_find_type_not_found                                   | Pass   |
|       | test_helpers.py::TestFindPropertyByDict                                   | test_dict_without_properties_wraps_in_list                 | Pass   |
|       | test_helpers.py::TestFindPropertyByDict                                   | test_invalid_type_raises                                   | Pass   |
|       | test_helpers.py::TestFindPropertyByDict                                   | test_list_input                                            | Pass   |
|       | test_helpers.py::TestFindPropertyByDict                                   | test_not_found_raises                                      | Pass   |
|       | test_helpers.py::TestFindPropertyByDict                                   | test_portfolio_key                                         | Pass   |
|       | test_helpers.py::TestFindPropertyByDict                                   | test_properties_key                                        | Pass   |
|       | test_helpers.py::TestOpenPdfFile                                          | test_darwin_success                                        | Pass   |
|       | test_helpers.py::TestOpenPdfFile                                          | test_generic_exception_returns_false                       | Pass   |
|       | test_helpers.py::TestOpenPdfFile                                          | test_linux_success                                         | Pass   |
|       | test_helpers.py::TestOpenPdfFile                                          | test_subprocess_failure_returns_false                      | Pass   |
|       | test_helpers.py::TestOpenPdfFile                                          | test_unknown_platform_uses_webbrowser                      | Pass   |
|       | test_helpers.py::TestOpenPdfFile                                          | test_windows_success                                       | Pass   |
|       | test_integrator.py::TestGenerateReportFromProperty                        | test_report_from_property_returns_path                     | Pass   |
|       | test_integrator.py::TestGenerateReportFromPropertyFile                    | test_report_from_property_file_returns_none                | Pass   |
|       | test_integrator.py::TestGenerateReportFromPropertyFile                    | test_report_from_property_file_still_generates             | Pass   |
|       | test_integrator.py::TestGenerateReportFromProperty                        | test_report_from_property_contains_property_id             | Pass   |
|       | test_integrator.py::TestGenerateReportFromProperty                        | test_report_from_property_created                          | Pass   |
|       | test_integrator.py::TestGenerateReportFromPropertyFile                    | test_report_from_property_file_created                     | Pass   |
|       | test_integrator.py::TestGenerateReportFromProperty                        | test_report_from_property_not_found_returns_none           | Pass   |
|       | test_integrator_exception.py::TestGenerateReportFromPropertyFileException | test_report_from_property_file_exception_returns_none      | Pass   |
|       | test_integrator_exception.py::TestGenerateReportFromPropertyFileException | test_report_from_property_file_exception_returns_exception | Pass   |
|       | generation.py::TestGenerateRiskReportAnalysisType                         | test_report_analysis_type                                  | Pass   |
|       | generation.py::TestGenerateRiskReportBasicType                            | test_report_basic_type                                     | Pass   |
|       | generation.py::TestGenerateRiskReportDetailedType                         | test_report_detailed_type                                  | Pass   |
|       | generation.py::TestGenerateRiskReportSummaryType                          | test_report_summary_type                                   | Pass   |
|       | generation.py::TestGenerateRiskReportUnknownTypeFallback                  | test_report_unknown_type_fallback                          | Pass   |
|       | generation.py::TestRiskReportEdgesCase                                    | test_edges_case_select_pages_returns_list                  | Pass   |
|       | generation.py::TestRiskReportEdgesCase                                    | test_edges_case_empty_gauge_data                           | Pass   |
|       | generation.py::TestRiskReportEdgesCase                                    | test_edges_case_generate_elements_exception_continues      | Pass   |
|       | generation.py::TestRiskReportEdgesCase                                    | test_edges_case_generate_elements_skips_unknown            | Pass   |
|       | generation.py::TestRiskReportEdgesCase                                    | test_edges_case_get_page_categories                        | Pass   |
|       | generation.py::TestRiskReportEdgesCase                                    | test_edges_case_high_risk_portfolio_report                 | Pass   |
|       | generation.py::TestRiskReportEdgesCase                                    | test_edges_case_list_available_pages                       | Pass   |
|       | generation.py::TestRiskReportEdgesCase                                    | test_edges_case_minimal_data_report                        | Pass   |
|       | generation.py::TestRiskReportEdgesCase                                    | test_edges_case_missing_mortgage_data                      | Pass   |

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| Model | Module                                  | Test Name                                       | Status |
|-------|-----------------------------------------|-------------------------------------------------|--------|
|       | generation.py::TestRiskReportEdgeCases  | test_output_dir_uses_config                     | Pass   |
|       | generation.py::TestRiskReportEdgeCases  | test_validate_pages_valid_and_invalid           | Pass   |
|       | generation.py::TestRiskReportGeneration | test_analysis_report_generation                 | Pass   |
|       | generation.py::TestRiskReportGeneration | test_basic_report_generation                    | Pass   |
|       | generation.py::TestRiskReportGeneration | test_custom_filename                            | Pass   |
|       | generation.py::TestRiskReportGeneration | test_custom_page_selection                      | Pass   |
|       | generation.py::TestRiskReportGeneration | test_detailed_report_generation                 | Pass   |
|       | generation.py::TestRiskReportGeneration | test_summary_report_generation                  | Pass   |
|       | cli.py::TestErrorHandling               | test_bad_flood_file_exits_with_1                | Pass   |
|       | cli.py::TestErrorHandling               | test_invalid_json_exits_with_1                  | Pass   |
|       | cli.py::TestListCategories              | test_list_categories_iterates_all_categories    | Pass   |
|       | cli.py::TestListCategories              | test_list_categories_runs_without_error         | Pass   |
|       | cli.py::TestListPages                   | test_list_pages_does_not_open_flood_file        | Pass   |
|       | cli.py::TestListPages                   | test_list_pages_runs_without_error              | Pass   |
|       | cli.py::TestReportGeneration            | test_logging_after_generation                   | Pass   |
|       | cli.py::TestReportGeneration            | test_report_type_analysis                       | Pass   |
|       | cli.py::TestReportGeneration            | test_report_type_basic                          | Pass   |
|       | cli.py::TestReportGeneration            | test_report_type_detailed                       | Pass   |
|       | cli.py::TestReportGeneration            | test_report_type_else_branch                    | Pass   |
|       | cli.py::TestReportGeneration            | test_report_type_summary                        | Pass   |
|       | convenience.py::TestEdgeCases           | test_all_unknown_pages_still_creates_pdf        | Pass   |
|       | convenience.py::TestEdgeCases           | test_completely_empty_flood_data                | Pass   |
|       | convenience.py::TestEdgeCases           | test_large_property_risk_dataset                | Pass   |
|       | convenience.py::TestEdgeCases           | test_missing_summary_key                        | Pass   |
|       | convenience.py::TestEdgeCases           | test_negative_values_in_summary                 | Pass   |
|       | convenience.py::TestEdgeCases           | test_none_gauge_data_key                        | Pass   |
|       | convenience.py::TestEdgeCases           | test_report_with_all_pages                      | Pass   |
|       | convenience.py::TestEdgeCases           | test_single_page_report                         | Pass   |
|       | convenience.py::TestEdgeCases           | test_zero_properties_at_risk                    | Pass   |
|       | convenience.py::TestGenerateRiskReport  | test_basic_type_returns_path                    | Pass   |
|       | convenience.py::TestGenerateRiskReport  | test_basic_type_returns_ipath                   | Pass   |
|       | convenience.py::TestGenerateRiskReport  | test_defective_report_type_is_basic             | Pass   |
|       | convenience.py::TestGenerateRiskReport  | test_defective_report_type_is_path              | Pass   |
|       | convenience.py::TestGenerateRiskReport  | test_output_dir_is_string                       | Pass   |
|       | convenience.py::TestGenerateRiskReport  | test_pdf_created_in_empty_bytes_on_disk         | Pass   |
|       | convenience.py::TestGenerateRiskReport  | test_summary_type_returns_path                  | Pass   |
|       | convenience.py::TestGenerateRiskReport  | test_unknown_type_falls_back_to_generate_report | Pass   |
|       | elements.py::TestGenerateElements       | test_empty_page_list                            | Pass   |
|       | elements.py::TestGenerateElements       | test_exception_in_page_continues                | Pass   |
|       | elements.py::TestGenerateElements       | test_first_page_no_page_break                   | Pass   |
|       | elements.py::TestGenerateElements       | test_mixed_valid_and_unknown_pages              | Pass   |
|       | elements.py::TestGenerateElements       | test_multiple_pages_inserts_page_breaks         | Pass   |
|       | elements.py::TestGenerateElements       | test_returns_list                               | Pass   |
|       | elements.py::TestGenerateElements       | test_unknown_page_skipped                       | Pass   |

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| Model | Module                                         | Test Name                                          | Status |
|-------|------------------------------------------------|----------------------------------------------------|--------|
|       | elements.py::TestGenerateReport                | test_auto_select_pages_when_none                   | Pass   |
|       | elements.py::TestGenerateReport                | test_auto_select_skipped_when_provided             | Pass   |
|       | elements.py::TestGenerateReport                | test_custom_output_filename                        | Pass   |
|       | elements.py::TestGenerateReport                | test_custom_pages_to_include                       | Pass   |
|       | elements.py::TestGenerateReport                | test_empty_flood_data                              | Pass   |
|       | elements.py::TestGenerateReport                | test_full_data_report                              | Pass   |
|       | elements.py::TestGenerateReport                | test_pdf_exists                                    | Pass   |
|       | elements.py::TestGenerateReport                | test_pdf_non_empty                                 | Pass   |
|       | elements.py::TestGenerateReport                | test_pdf_suffix                                    | Pass   |
|       | elements.py::TestGenerateReport                | test_returns_path                                  | Pass   |
|       | init.py::TestAutoSelectPages                   | test_all_pages_exist                               | Pass   |
|       | init.py::TestAutoSelectPages                   | test_always_includes_appendix                      | Pass   |
|       | init.py::TestAutoSelectPages                   | test_empty_flood_data                              | Pass   |
|       | init.py::TestAutoSelectPages                   | test_includes_analysis_pages                       | Pass   |
|       | init.py::TestAutoSelectPages                   | test_includes_overview_pages                       | Pass   |
|       | init.py::TestAutoSelectPages                   | test_returns_list                                  | Pass   |
|       | init.py::TestGenerateFilename                  | test_contains_flood_risk_report                    | Pass   |
|       | init.py::TestGenerateFilename                  | test_contains_timestamp                            | Pass   |
|       | init.py::TestGenerateFilename                  | test_ends_with_pdf                                 | Pass   |
|       | init.py::TestGenerateFilename                  | test_returns_string                                | Pass   |
|       | init.py::TestGenerateFilename                  | test_two_calls_produce_string_names                | Pass   |
|       | init.py::TestRiskReportGenerator               | test_categories_dict_initialized                   | Pass   |
|       | init.py::TestRiskReportGenerator               | test_explicit_output_dir_path                      | Pass   |
|       | init.py::TestRiskReportGenerator               | test_explicit_output_dir_string                    | Pass   |
|       | init.py::TestRiskReportGenerator               | test_no_output_dir_uses_config                     | Pass   |
|       | init.py::TestRiskReportGenerator               | test_output_dir_created                            | Pass   |
|       | init.py::TestRiskReportGenerator               | test_pages_dict_initialized                        | Pass   |
|       | specialised.py::TestHeaderFooter               | test_header_footer_callable                        | Pass   |
|       | specialised.py::TestHeaderFooter               | test_header_footer_draws_strings                   | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_method_calls_detailed_page_counts             | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_analysis_report_custom_filename      | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_analysis_report_exists               | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_basic_report_exists                  | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_basic_report_non_empty               | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_detailed_report_exists               | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_detailed_report_includes_report_page | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_summary_report_custom_filename       | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator | test_generate_summary_report_exists                | Pass   |
|       | specialised.py::TestUtilityMethods             | test_get_page_categories_contains_expected_keys    | Pass   |
|       | specialised.py::TestUtilityMethods             | test_get_page_categories_returns_copy              | Pass   |
|       | specialised.py::TestUtilityMethods             | test_get_page_categories_returns_dict              | Pass   |
|       | specialised.py::TestUtilityMethods             | test_list_available_pages_contains_all_expected    | Pass   |
|       | specialised.py::TestUtilityMethods             | test_list_available_pages_non_empty                | Pass   |
|       | specialised.py::TestUtilityMethods             | test_list_available_pages_returns_list             | Pass   |

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| Model | Module                                   | Test Name                            | Status |
|-------|------------------------------------------|--------------------------------------|--------|
|       | specialised.py::TestUtilityMethods       | test_validate_pages_all_invalid      | Pass   |
|       | specialised.py::TestUtilityMethods       | test_validate_pages_all_valid        | Pass   |
|       | specialised.py::TestUtilityMethods       | test_validate_pages_empty_list       | Pass   |
|       | specialised.py::TestUtilityMethods       | test_validate_pages_mixed            | Pass   |
|       | utilities.py::TestConvenienceFunctions   | test_generate_risk_report_basic      | Pass   |
|       | utilities.py::TestConvenienceFunctions   | test_generate_risk_report_default    | Pass   |
|       | utilities.py::TestConvenienceFunctions   | test_generate_risk_report_detailed   | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_count_non_numeric        | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_count_numeric            | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_currency_non_numeric     | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_currency_numeric         | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_depth_non_numeric        | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_depth_numeric            | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_field_name_camel_case    | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_field_name_replacements  | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_percentage_non_numeric   | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_percentage_numeric       | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_value_bool_false         | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_value_bool_true          | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_value_empty_string       | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_value_float_decimal      | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_value_float_integer      | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_value_int                | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_value_non_empty_string   | Pass   |
|       | utilities.py::TestRiskBasePageFormatters | test_format_value_none               | Pass   |
|       | utilities.py::TestRiskReportUtilities    | test_get_page_categories             | Pass   |
|       | utilities.py::TestRiskReportUtilities    | test_list_available_pages            | Pass   |
|       | utilities.py::TestRiskReportUtilities    | test_validate_pages_invalid          | Pass   |
|       | utilities.py::TestRiskReportUtilities    | test_validate_pages_valid            | Pass   |
|       | utilities.py::TestRiskReportWorkflow     | test_complete_workflow_all_types     | Pass   |
|       | utilities.py::TestRiskReportWorkflow     | test_complete_workflow_basic         | Pass   |
|       | eod_pages.py::TestEODChartsPages         | test_insufficient_data_message       | Pass   |
|       | eod_pages.py::TestEODChartsPages         | test_returns_list                    | Pass   |
|       | eod_pages.py::TestEODChartsPages         | test_single_entry_insufficient       | Pass   |
|       | eod_pages.py::TestEODChartsPages         | test_sufficient_data_creates_content | Pass   |
|       | eod_pages.py::TestEODPnLDetails          | test_many_market_movers_top_20       | Pass   |
|       | eod_pages.py::TestEODPnLDetails          | test_market_move_pnl_creates_table   | Pass   |
|       | eod_pages.py::TestEODPnLDetails          | test_new_trade_pnl_creates_table     | Pass   |
|       | eod_pages.py::TestEODPnLDetails          | test_no_positions_shows_messages     | Pass   |
|       | eod_pages.py::TestEODPnLDetails          | test_returns_list                    | Pass   |
|       | eod_pages.py::TestEODPositionsPage       | test_long_gauge_id_truncated         | Pass   |
|       | eod_pages.py::TestEODPositionsPage       | test_multiple_gauges_grouped         | Pass   |
|       | eod_pages.py::TestEODPositionsPage       | test_no_positions_shows_message      | Pass   |
|       | eod_pages.py::TestEODPositionsPage       | test_returns_list                    | Pass   |

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| Model | Module                                                 | Test Name                                    | Status |
|-------|--------------------------------------------------------|----------------------------------------------|--------|
|       | eod_pages.py::TestEODPositionsPage                     | single_position                              | Pass   |
|       | eod_pages.py::TestEODPositionsPage                     | with_positions_creates_table                 | Pass   |
|       | eod_pages.py::TestEODPositionsPage                     | zero_pnl_renders_dash                        | Pass   |
|       | eod_pages.py::TestEODReportGenerate                    | no_id_in_filename                            | Pass   |
|       | eod_pages.py::TestEODReportGenerate                    | generate_empty_snapshot                      | Pass   |
|       | eod_pages.py::TestEODReportGenerate                    | generate_report_returns_path                 | Pass   |
|       | eod_pages.py::TestEODReportGenerate                    | generate_report_without_pnl_series           | Pass   |
|       | eod_pages.py::TestEODReportGenerate                    | no_output_dir_uses_config                    | Pass   |
|       | eod_pages.py::TestEODReportGenerate                    | output_dir_created                           | Pass   |
|       | eod_pages.py::TestEODReportGenerate                    | pdf_not_empty                                | Pass   |
|       | eod_pages.py::TestEODSummaryPage                       | date_appears_in_paragraph                    | Pass   |
|       | eod_pages.py::TestEODSummaryPage                       | empty_summary                                | Pass   |
|       | eod_pages.py::TestEODSummaryPage                       | has_paragraph_and_table                      | Pass   |
|       | eod_pages.py::TestEODSummaryPage                       | negative_pnl_snapshot                        | Pass   |
|       | eod_pages.py::TestEODSummaryPage                       | returns_list                                 | Pass   |
|       | eod_pages.py::TestPnLDetailZeroPnL                     | zero_pnl_returns_dash                        | Pass   |
|       | test_eod_charts_coverage.py::TestDailyEmpChartFlyout   | returns_drawing                              | Pass   |
|       | test_eod_charts_coverage.py::TestDailyPnLChartNegative | negative_renders                             | Pass   |
|       | test_eod_charts_coverage.py::TestDailyRegChartNegative | pnl_renders                                  | Pass   |
|       | test_eod_charts_coverage.py::TestByConsolidation       | key_returns_drawing                          | Pass   |
|       | test_eod_charts_coverage.py::TestByGrowthChart         | renders_correctly                            | Pass   |
|       | test_eod_charts_coverage.py::TestByGrowthChart         | returns_drawing                              | Pass   |
|       | test_eod_charts_coverage.py::TestByGrowthChart         | returns_drawing                              | Pass   |
|       | test_eod_charts_coverage.py::TestByGrowthChart         | returns_drawing                              | Pass   |
|       | test_eod_charts_coverage.py::TestByGrowthChart         | returns_drawing                              | Pass   |
|       | open_pdf.py::TestOpenPdfFile                           | test_called_process_error_returns_false      | Pass   |
|       | open_pdf.py::TestOpenPdfFile                           | test_darwin_calls_open                       | Pass   |
|       | open_pdf.py::TestOpenPdfFile                           | test_generic_exception_returns_false         | Pass   |
|       | open_pdf.py::TestOpenPdfFile                           | test_linux_calls_xdg_open                    | Pass   |
|       | open_pdf.py::TestOpenPdfFile                           | test_unknown_platform_uses_webbrowser        | Pass   |
|       | open_pdf.py::TestOpenPdfFile                           | test_windows_calls_start                     | Pass   |
|       | catchment.py::TestCatchmentException                   | test_invalid_catchment_selector_exception    | Pass   |
|       | catchment.py::TestCatchmentException                   | test_invalid_catchment_exception_returns_500 | Pass   |
|       | catchment.py::TestServeCatchmentSelector               | returns_404_when_file_missing                | Pass   |
|       | catchment.py::TestServeCatchmentSelector               | returns_html_when_file_exists                | Pass   |
|       | catchment.py::TestServeCatchmentSelector               | returns_json_error_or_html                   | Pass   |
|       | catchment.py::TestSetCatchment                         | test_case_insensitive_thames                 | Pass   |
|       | catchment.py::TestSetCatchment                         | test_missing_catchment_id_returns_400        | Pass   |
|       | catchment.py::TestSetCatchment                         | test_no_json_returns_400                     | Pass   |
|       | catchment.py::TestSetCatchment                         | test_null_catchment_id_returns_400           | Pass   |
|       | catchment.py::TestSetCatchment                         | test_options_returns_200                     | Pass   |
|       | catchment.py::TestSetCatchment                         | test_unknown_catchment_returns_400           | Pass   |
|       | catchment.py::TestSetCatchment                         | test_valid_thames_returns_success            | Pass   |
|       | counterparty.py::TestCounterpartyError                 | test_get_counterparty_error_returns_500      | Pass   |
|       | counterparty.py::TestCounterpartyError                 | test_list_counterparties_error_returns_500   | Pass   |

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| Model | Module                                      | Test Name                                    | Status |
|-------|---------------------------------------------|----------------------------------------------|--------|
|       | counterparty.py::TestGetCounterparty        | test_found_returns_200                       | Pass   |
|       | counterparty.py::TestGetCounterparty        | test_found_second_counterparty               | Pass   |
|       | counterparty.py::TestGetCounterparty        | test_not_found_returns_404                   | Pass   |
|       | counterparty.py::TestListCounterparties     | test_no_file_returns_empty_list              | Pass   |
|       | counterparty.py::TestListCounterparties     | test_summary_fields                          | Pass   |
|       | counterparty.py::TestListCounterparties     | test_with_data_returns_summary               | Pass   |
|       | cross_entity.py::TestCrossEntityConsistency | test_all_gauges_have_timeseries              | Pass   |
|       | cross_entity.py::TestCrossEntityConsistency | test_all_gauges_hazard_no_500                | Pass   |
|       | cross_entity.py::TestCrossEntityConsistency | test_all_properties_have_flood_data          | Pass   |
|       | cross_entity.py::TestInteractivity          | test_modeling_menus_renders                  | Pass   |
|       | cross_entity.py::TestInteractivity          | test_gauging_hazard_curve_renders            | Pass   |
|       | cross_entity.py::TestInteractivity          | test_gauging_pdf_panel_renders               | Pass   |
|       | cross_entity.py::TestInteractivity          | test_gauging_storm_analysis_renders          | Pass   |
|       | cross_entity.py::TestInteractivity          | test_modeling_notifications_renders          | Pass   |
|       | cross_entity.py::TestInteractivity          | test_property_hazard_curve_renders           | Pass   |
|       | cross_entity.py::TestInteractivity          | test_property_pdf_panel_renders              | Pass   |
|       | cross_entity.py::TestInteractivity          | test_property_storm_analysis_renders         | Pass   |
|       | cross_entity.py::TestInteractivity          | test_property_storm_js_guards_empty_events   | Pass   |
|       | gauge.py::TestDataFormatValidation          | test_gaugehc_file_structure                  | Pass   |
|       | gauge.py::TestDataFormatValidation          | test_gaugets_file_structure                  | Pass   |
|       | gauge.py::TestDataFormatValidation          | test_propertyhc_file_structure               | Pass   |
|       | gauge.py::TestDataFormatValidation          | test_propertyts_file_structure               | Pass   |
|       | gauge.py::TestGaugeHazardRoutes             | test_gauge_hazard_existing                   | Pass   |
|       | gauge.py::TestGaugeHazardRoutes             | test_gauge_hazard_gev_params                 | Pass   |
|       | gauge.py::TestGaugeHazardRoutes             | test_gauge_hazard_nonexistent_gauge          | Pass   |
|       | gauge.py::TestGaugeHazardRoutes             | test_gauge_hazard_not_found                  | Pass   |
|       | gauge.py::TestGaugeHazardRoutes             | test_gauge_storms_existing                   | Pass   |
|       | gauge.py::TestGaugeIDConsistency            | test_gaugehc_ids_match_gauge_portfolio       | Pass   |
|       | gauge.py::TestGaugeIDConsistency            | test_gaugets_ids_match_gauge_portfolio       | Pass   |
|       | gauge.py::TestGaugeIDConsistency            | test_gaugets_internal_id_matches_filename    | Pass   |
|       | property.py::TestMissingDataHandling        | test_property_hazard_no_file                 | Pass   |
|       | property.py::TestMissingDataHandling        | test_propertyhc_summary_no_file              | Pass   |
|       | property.py::TestMissingDataHandling        | test_propertyts_summary_no_dir               | Pass   |
|       | property.py::TestPropertyHCRoutes           | test_property_hazard_existing                | Pass   |
|       | property.py::TestPropertyHCRoutes           | test_property_hazard_nonexistent             | Pass   |
|       | property.py::TestPropertyHCRoutes           | test_property_hazard_not_in_curves           | Pass   |
|       | property.py::TestPropertyHCRoutes           | test_propertyhc_basis                        | Pass   |
|       | property.py::TestPropertyHCRoutes           | test_propertyhc_summary                      | Pass   |
|       | property.py::TestPropertyIDConsistency      | test_propertyhc_ids_match_property_portfolio | Pass   |
|       | property.py::TestPropertyIDConsistency      | test_propertyhc_ids_subset_of_propertyts     | Pass   |
|       | property.py::TestPropertyIDConsistency      | test_propertyts_ids_match_property_portfolio | Pass   |
|       | property.py::TestPropertyTSRoutes           | test_list_flood_storms                       | Pass   |
|       | property.py::TestPropertyTSRoutes           | test_list_flood_storms_only_returns_damaging | Pass   |
|       | property.py::TestPropertyTSRoutes           | test_property_floods_existing                | Pass   |

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| Model | Module                                   | Test Name                                         | Status |
|-------|------------------------------------------|---------------------------------------------------|--------|
|       | property.py::TestPropertyTSRouter        | test_property_floods_few_events                   | Pass   |
|       | property.py::TestPropertyTSRouter        | test_property_floods_not_found                    | Pass   |
|       | property.py::TestPropertyTSRouter        | test_property_storms_existing                     | Pass   |
|       | property.py::TestPropertyTSRouter        | test_property_storms_not_found                    | Pass   |
|       | property.py::TestPropertyTSRouter        | test_property_storms_zero_floods                  | Pass   |
|       | property.py::TestPropertyTSRouter        | test_propertyts_summary                           | Pass   |
|       | financial.py::TestPortfolioImpactNoData  | pts_dir_returns_404                               | Pass   |
|       | financial.py::TestPortfolioImpactOptions | options_returns_200                               | Pass   |
|       | financial.py::TestPortfolioImpactStorm   | unknown_storm_returns_404                         | Pass   |
|       | financial.py::TestPortfolioImpactStorm   | mortgage_enrichment_present                       | Pass   |
|       | financial.py::TestPortfolioImpactStorm   | property_entry_has_damage_fields                  | Pass   |
|       | financial.py::TestPortfolioImpactStorm   | response_has_portfolio_key                        | Pass   |
|       | financial.py::TestPortfolioImpactStorm   | response_has_properties_list                      | Pass   |
|       | financial.py::TestPortfolioImpactStorm   | storm_id_in_response                              | Pass   |
|       | financial.py::TestPortfolioImpactStorm   | valid_storm_returns_200                           | Pass   |
|       | test_crud.py::TestCrudGetGauge           | test_get_gauge_not_found                          | Pass   |
|       | test_crud.py::TestCrudGetGauge           | test_get_gauge_success                            | Pass   |
|       | test_crud.py::TestCrudListGauge          | test_list_gauges_error_returns_500                | Pass   |
|       | test_crud.py::TestCrudListGauge          | test_list_gauges_legacy_route                     | Pass   |
|       | test_crud.py::TestCrudListGauge          | test_list_gauges_options                          | Pass   |
|       | test_crud.py::TestCrudListGauge          | test_list_gauges_returns_200                      | Pass   |
|       | test_hazard.py::TestHazardError          | test_get_gauge_hazard_corrupted_file_returns_500  | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | price_prs_exception_returns_500                   | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | price_prs_gauge_not_in_hazard                     | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | price_prs_no_hazard_file                          | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | price_prs_no_json_body                            | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | price_prs_null_json_body                          | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | price_prs_severe_trigger                          | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | price_prs_success                                 | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | price_prs_with_defaults                           | Pass   |
|       | test_history.py::TestHistoryError        | test_get_gauge_history_corrupted_file_returns_500 | Pass   |
|       | test_history.py::TestHistoryError        | test_get_gauge_statistics_exception_returns_500   | Pass   |
|       | test_history.py::TestHistoryError        | test_get_gaugets_exception_returns_500            | Pass   |
|       | test_reports.py::TestReportsError        | test_gaugehd_load_error_warning                   | Pass   |
|       | test_reports.py::TestReportsError        | test_general_exception_returns_500                | Pass   |
|       | test_reports.py::TestReportsError        | test_hazard_load_error_warning                    | Pass   |
|       | test_reports.py::TestReportsError        | test_import_error_returns_500                     | Pass   |
|       | test_reports.py::TestReportsError        | test_timeseries_load_success_with_storm_responses | Pass   |
|       | test_reports.py::TestReportsError        | test_timeseries_load_warning                      | Pass   |
|       | test_storms.py::TestStormsError          | test_gaugets_dir_corrupted_file_continues         | Pass   |
|       | test_storms.py::TestStormsError          | test_no_gaugets_dir                               | Pass   |
|       | test_storms.py::TestStormsError          | test_no_storm_sequences_file                      | Pass   |
|       | test_storms.py::TestStormsError          | test_storm_sequences_corrupted_continues          | Pass   |
|       | test_storms.py::TestStormsError          | test_storms_exception_returns_500                 | Pass   |

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| Model | Module                              | Test Name                                              | Status |
|-------|-------------------------------------|--------------------------------------------------------|--------|
|       | gauges_hazard.py::TestGetGauge      | test_hazard_gauge_not_found_returns_404                | Pass   |
|       | gauges_hazard.py::TestGetGauge      | test_hazard_curve_for_gauge_returns_404                | Pass   |
|       | gauges_hazard.py::TestGetGauge      | test_hazard_hazard_file_returns_404                    | Pass   |
|       | gauges_hazard.py::TestGetGauge      | test_hazard_response_has_annual_flood_probs            | Pass   |
|       | gauges_hazard.py::TestGetGauge      | test_hazard_response_has_curve_points                  | Pass   |
|       | gauges_hazard.py::TestGetGauge      | test_hazard_response_has_flood_stages                  | Pass   |
|       | gauges_hazard.py::TestGetGauge      | test_hazard_response_has_gauge_id                      | Pass   |
|       | gauges_hazard.py::TestGetGauge      | test_hazard_response_has_gev_parameters                | Pass   |
|       | gauges_hazard.py::TestGetGauge      | test_hazard_response_has_term_structures               | Pass   |
|       | gauges_hazard.py::TestGetGauge      | test_hazard_valid_hazard_returns_200                   | Pass   |
|       | gauges_hazard.py::TestPricePrs      | test_gauge_not_in_hazard_returns_404                   | Pass   |
|       | gauges_hazard.py::TestPricePrs      | test_invalid_trigger_level_returns_400                 | Pass   |
|       | gauges_hazard.py::TestPricePrs      | test_missing_gauge_id_returns_400                      | Pass   |
|       | gauges_hazard.py::TestPricePrs      | test_no_hazard_file_returns_404                        | Pass   |
|       | gauges_hazard.py::TestPricePrs      | test_no_json_returns_400                               | Pass   |
|       | gauges_hazard.py::TestPricePrs      | test_options_returns_ok                                | Pass   |
|       | gauges_history.py::TestGaugeHist    | test_historys_response_has_count_and_gauges_keys       | Pass   |
|       | gauges_history.py::TestGaugeHist    | test_historys_returns_200_with_no_gaugehd_files        | Pass   |
|       | gauges_history.py::TestGaugeHist    | test_historys_returns_correct_count_when_files_present | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_days_larger_than_available_returns_all   | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_days_param_limits_observations           | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_days_param_zero_returns_all              | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_no_history_file_returns_404              | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_response_has_daily_observations          | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_response_has_flood_stages                | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_response_has_gauge_metadata              | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_response_has_statistics                  | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_valid_history_returns_200                | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_gauge_not_found_returns_404              | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_invalid_gauge_returns_success            | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_gauge_not_found_returns_404              | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_response_has_count_and_readings          | Pass   |
|       | gauges_history.py::TestGetGaugeHist | test_historys_valid_gauge_returns_success              | Pass   |
|       | gauges_reports.py::TestGenerate     | test_generate_report_gauge_not_found_returns_404       | Pass   |
|       | gauges_reports.py::TestGenerate     | test_generate_report_route_alias                       | Pass   |
|       | gauges_reports.py::TestGenerate     | test_generate_report_gauge_id_returns_400              | Pass   |
|       | gauges_reports.py::TestGenerate     | test_generate_report_no_json_returns_400               | Pass   |
|       | gauges_reports.py::TestGenerate     | test_generate_report_no_json_body_returns_400          | Pass   |
|       | gauges_reports.py::TestGenerate     | test_generate_report_options_returns_ok                | Pass   |
|       | gauges_reports.py::TestGenerate     | test_generate_report_response_has_file_path            | Pass   |
|       | gauges_reports.py::TestGenerate     | test_generate_report_response_has_pdf_base64           | Pass   |
|       | gauges_reports.py::TestGenerate     | test_generate_report_valid_gauge_generates_report      | Pass   |
|       | gauges_reports.py::TestGenerate     | test_generate_report_gaugehd_file_present              | Pass   |
|       | gauges_reports.py::TestGenerate     | test_generate_report_hazard_file_present               | Pass   |

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| Model | Module                           | Test Name                                     | Status |
|-------|----------------------------------|-----------------------------------------------|--------|
|       | gauges.storms.py::TestGetGauge   | test_gauge_not_found_returns_404              | Pass   |
|       | gauges.storms.py::TestGetGauge   | test_response_has_flood_simulation            | Pass   |
|       | gauges.storms.py::TestGetGauge   | test_response_has_flood_stages                | Pass   |
|       | gauges.storms.py::TestGetGauge   | test_response_has_gauge_id                    | Pass   |
|       | gauges.storms.py::TestGetGauge   | test_response_has_storm_responses             | Pass   |
|       | gauges.storms.py::TestGetGauge   | test_storm_enrichment_from_storms_json        | Pass   |
|       | gauges.storms.py::TestGetGauge   | test_invalid_gauge_returns_200                | Pass   |
|       | gauges.storms.py::TestGetGauge   | test_with_severe_exceeded_counts_gauges       | Pass   |
|       | audit.py::TestGetAuditTrail      | test_after_log_entry_appears                  | Pass   |
|       | audit.py::TestGetAuditTrail      | test_empty_trail_returns_200                  | Pass   |
|       | audit.py::TestGetAuditTrail      | test_filter_by_event_type                     | Pass   |
|       | audit.py::TestGetAuditTrail      | test_filter_by_model_id                       | Pass   |
|       | audit.py::TestGetAuditTrail      | test_filter_by_model_id_no_match              | Pass   |
|       | audit.py::TestGetAuditTrail      | test_has_returned_field                       | Pass   |
|       | audit.py::TestGetAuditTrail      | test_limit_parameter                          | Pass   |
|       | audit.py::TestGetRiskRating      | test_model_not_found_returns_404              | Pass   |
|       | audit.py::TestGetRiskRating      | test_response_has_risk_rating                 | Pass   |
|       | audit.py::TestGetRiskRating      | test_risk_rating_has_component_scores         | Pass   |
|       | audit.py::TestGetRiskRating      | test_valid_model_returns_success              | Pass   |
|       | audit.py::TestLogModelUsage      | test_defaults_applied_when_no_json            | Pass   |
|       | audit.py::TestLogModelUsage      | test_entry_has_expected_fields                | Pass   |
|       | audit.py::TestLogModelUsage      | test_model_not_found_returns_404              | Pass   |
|       | audit.py::TestLogModelUsage      | test_parameters_stored                        | Pass   |
|       | audit.py::TestLogModelUsage      | test_valid_model_returns_200                  | Pass   |
|       | audit.py::TestOverrideRiskRating | test_clear_override_with_null                 | Pass   |
|       | audit.py::TestOverrideRiskRating | test_invalid_rating_returns_400               | Pass   |
|       | audit.py::TestOverrideRiskRating | test_model_not_found_returns_404              | Pass   |
|       | audit.py::TestOverrideRiskRating | test_no_reason_field_returns_400              | Pass   |
|       | audit.py::TestOverrideRiskRating | test_no_reason_returns_400                    | Pass   |
|       | audit.py::TestOverrideRiskRating | test_override_stored_in_model                 | Pass   |
|       | audit.py::TestOverrideRiskRating | test_response_has_audit_entry                 | Pass   |
|       | audit.py::TestOverrideRiskRating | test_valid_override_returns_success           | Pass   |
|       | audit.py::TestPdfEndpoints       | test_documentation_pdf_known_model_no_file    | Pass   |
|       | audit.py::TestPdfEndpoints       | test_documentation_pdf_unknown_model_404      | Pass   |
|       | audit.py::TestPdfEndpoints       | test_mrc_tor_pdf_responds                     | Pass   |
|       | audit.py::TestPdfEndpoints       | test_parameter_inventory_pdf_responds         | Pass   |
|       | audit.py::TestPdfEndpoints       | test_test_results_pdf_known_model_no_file     | Pass   |
|       | audit.py::TestPdfEndpoints       | test_test_results_pdf_unknown_model_404       | Pass   |
|       | audit.py::TestUpdateValidation   | test_all_valid_statuses                       | Pass   |
|       | audit.py::TestUpdateValidation   | test_invalid_status_returns_400               | Pass   |
|       | audit.py::TestUpdateValidation   | test_model_not_found_returns_404              | Pass   |
|       | audit.py::TestUpdateValidation   | test_question_not_found_returns_404           | Pass   |
|       | audit.py::TestUpdateValidation   | test_risk_rating_recalculated                 | Pass   |
|       | audit.py::TestUpdateValidation   | test_invalid_update_has_model_and_audit_entry | Pass   |

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| Model | Module                                     | Test Name                                       | Status |
|-------|--------------------------------------------|-------------------------------------------------|--------|
|       | audit.py::TestUpdateValidation             | test_invalid_update_returns_success             | Pass   |
|       | audit_reports.py::TestDownloadAuditReport  | test_report_not_found_returns_404               | Pass   |
|       | audit_reports.py::TestDownloadAuditReport  | test_report_pdf_served_as_attachment            | Pass   |
|       | audit_reports.py::TestDownloadAuditReport  | test_report_traversal_returns_403               | Pass   |
|       | audit_reports.py::TestDownloadAuditReport  | test_report_pdf_served_inline                   | Pass   |
|       | audit_reports.py::TestListAuditReports     | test_coverage_available_false_when_no_index     | Pass   |
|       | audit_reports.py::TestListAuditReports     | test_coverage_available_true_when_index_present | Pass   |
|       | audit_reports.py::TestListAuditReports     | test_empty_dir_returns_empty_list               | Pass   |
|       | audit_reports.py::TestListAuditReports     | test_file_has_size_and_modified                 | Pass   |
|       | audit_reports.py::TestListAuditReports     | test_file_type_label_json                       | Pass   |
|       | audit_reports.py::TestListAuditReports     | test_file_type_label_pdf                        | Pass   |
|       | audit_reports.py::TestListAuditReports     | test_file_type_label_unknown                    | Pass   |
|       | audit_reports.py::TestListAuditReports     | test_no_dir_returns_empty_list                  | Pass   |
|       | audit_reports.py::TestListAuditReports     | test_pdf_file_listed                            | Pass   |
|       | audit_reports.py::TestListAuditReports     | test_subdirectories_not_listed                  | Pass   |
|       | audit_reports.py::TestServeCoverageIndex   | test_index_coverage_dir_returns_404             | Pass   |
|       | audit_reports.py::TestServeCoverageIndex   | test_with_coverage_index_returns_200            | Pass   |
|       | bibliography.py::TestAddBibliographyRef    | test_ref_appears_in_list                        | Pass   |
|       | bibliography.py::TestAddBibliographyRef    | test_ref_assigns_id                             | Pass   |
|       | bibliography.py::TestAddBibliographyRef    | test_ref_no_json_body_uses_defaults             | Pass   |
|       | bibliography.py::TestAddBibliographyRef    | test_ref_returns_200                            | Pass   |
|       | bibliography.py::TestAddBibliographyRef    | test_ref_returns_reference                      | Pass   |
|       | bibliography.py::TestAddBibliographyRef    | test_ref_same_category_twice_no_duplicates      | Pass   |
|       | bibliography.py::TestAddBibliographyRef    | test_ref_stores_all_fields                      | Pass   |
|       | bibliography.py::TestAddBibliographyRef    | test_ref_with_category_updates_category         | Pass   |
|       | bibliography.py::TestDeleteBibliographyRef | test_ref_nonexistent_still_returns_success      | Pass   |
|       | bibliography.py::TestDeleteBibliographyRef | test_ref_returns_success                        | Pass   |
|       | bibliography.py::TestDeleteBibliographyRef | test_refed_ref_not_in_list                      | Pass   |
|       | bibliography.py::TestGetBibliographyRef    | test_empty_references                           | Pass   |
|       | bibliography.py::TestGetBibliographyRef    | test_has_categories_key                         | Pass   |
|       | bibliography.py::TestGetBibliographyRef    | test_returns_200                                | Pass   |
|       | bibliography.py::TestGetBibliographyRef    | test_returns_success_status                     | Pass   |
|       | bibliography.py::TestUpdateBibliographyRef | test_ref_not_found_returns_404                  | Pass   |
|       | bibliography.py::TestUpdateBibliographyRef | test_update_ref_changes_field                   | Pass   |
|       | bibliography.py::TestUpdateBibliographyRef | test_update_ref_returns_success                 | Pass   |
|       | bibliography.py::TestUpdateBibliographyRef | test_update_ref_with_new_category_adds_to_list  | Pass   |
|       | compliance.py::TestGetBcbs239              | test_no_data_returns_404                        | Pass   |
|       | compliance.py::TestGetBcbs239              | test_principles_count                           | Pass   |
|       | compliance.py::TestGetBcbs239              | test_response_has_assessment                    | Pass   |
|       | compliance.py::TestGetBcbs239              | test_with_data_returns_200                      | Pass   |
|       | compliance.py::TestGetBcbs239              | test_bcbs239_pdf_responds                       | Pass   |
|       | compliance.py::TestUpdateBcbs239           | test_assessment_date_updated                    | Pass   |
|       | compliance.py::TestUpdateBcbs239           | test_invalid_file_returns_404                   | Pass   |
|       | compliance.py::TestUpdateBcbs239           | test_principle_not_found_returns_404            | Pass   |

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| Model | Module                               | Test Name                                          | Status |
|-------|--------------------------------------|----------------------------------------------------|--------|
|       | compliance.py::TestUpdateBcbs        | 239Principle 1 derives non-compliant status        | Pass   |
|       | compliance.py::TestUpdateBcbs        | 239Principle 4 derives fully compliant status      | Pass   |
|       | compliance.py::TestUpdateBcbs        | 239Principle update_evidence_field                 | Pass   |
|       | compliance.py::TestUpdateBcbs        | 239Principle update_gaps_and_remediation           | Pass   |
|       | compliance.py::TestUpdateBcbs        | 239Principle update_returns_200                    | Pass   |
|       | compliance.py::TestUpdateRaci        | Activity not found returns 404                     | Pass   |
|       | compliance.py::TestUpdateRaci        | Activity invalid role_id returns 400               | Pass   |
|       | compliance.py::TestUpdateRaci        | Activity not updated                               | Pass   |
|       | compliance.py::TestUpdateRaci        | Activity invalid raci_code allowed                 | Pass   |
|       | compliance.py::TestUpdateRaci        | Activity invalid activity update returns success   | Pass   |
|       | compliance.py::TestUpdateRaci        | Escalation tier threshold updated                  | Pass   |
|       | compliance.py::TestUpdateRaci        | Escalation response has audit entry                | Pass   |
|       | compliance.py::TestUpdateRaci        | Escalation response not found returns 404          | Pass   |
|       | compliance.py::TestUpdateRaci        | Escalation index escalation update returns success | Pass   |
|       | compliance.py::TestUpdateRaci        | File backup field updated                          | Pass   |
|       | compliance.py::TestUpdateRaci        | File extra raci file returns 404                   | Pass   |
|       | compliance.py::TestUpdateRaci        | File response has audit entry                      | Pass   |
|       | compliance.py::TestUpdateRaci        | File not found returns 404                         | Pass   |
|       | compliance.py::TestUpdateRaci        | File invalid role update returns success           | Pass   |
|       | documents.py::TestDeleteDocument     | test_deleted_doc_not_in_list                       | Pass   |
|       | documents.py::TestDeleteDocument     | test_not_found_returns_404                         | Pass   |
|       | documents.py::TestDeleteDocument     | test_valid_delete_returns_success                  | Pass   |
|       | documents.py::TestDeleteSaveModel    | test_delete_save_failure_returns_500               | Pass   |
|       | documents.py::TestDownloadDocument   | test_not_found_returns_404                         | Pass   |
|       | documents.py::TestDownloadDocument   | test_valid_download_returns_200                    | Pass   |
|       | documents.py::TestDownloadFile       | test_missing_on_disk_returns_404                   | Pass   |
|       | documents.py::TestGetDocuments       | test_empty_list_returns_success                    | Pass   |
|       | documents.py::TestUploadDocument     | test_empty_filename_returns_400                    | Pass   |
|       | documents.py::TestUploadDocument     | test_no_file_returns_400                           | Pass   |
|       | documents.py::TestUploadDocument     | test_upload_appears_in_list                        | Pass   |
|       | documents.py::TestUploadDocument     | test_upload_response_has_id                        | Pass   |
|       | documents.py::TestUploadDocument     | test_upload_stores_description                     | Pass   |
|       | documents.py::TestUploadDocument     | test_valid_upload_returns_200                      | Pass   |
|       | documents.py::TestUploadSaveModel    | test_upload_save_failure_returns_500               | Pass   |
|       | inventory.py::TestInventorySummary   | test_get_models_includes_risk_fields               | Pass   |
|       | inventory.py::TestRiskRating         | test_get_risk_rating_calculation                   | Pass   |
|       | inventory.py::TestRiskRating         | test_risk_rating_clear_override                    | Pass   |
|       | inventory.py::TestRiskRating         | test_risk_rating_override                          | Pass   |
|       | inventory.py::TestRiskRating         | test_risk_rating_override_invalid_rating           | Pass   |
|       | inventory.py::TestRiskRating         | test_risk_rating_override_requires_reason          | Pass   |
|       | inventory.py::TestValidationQuestion | test_update_triggers_risk_recalculation            | Pass   |
|       | inventory.py::TestValidationQuestion | test_update_validation_question                    | Pass   |
|       | inventory.py::TestValidationQuestion | test_update_validation_question_invalid_status     | Pass   |
|       | inventory.py::TestValidationQuestion | test_update_validation_question_not_found          | Pass   |

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| Model | Module                               | Test Name                                           | Status  |
|-------|--------------------------------------|-----------------------------------------------------|---------|
|       | models.py::TestGetEditableField      | test_has_fields                                     | Pass    |
|       | models.py::TestGetEditableField      | test_returns_200                                    | Pass    |
|       | models.py::TestGetEditableField      | test_returns_success_status                         | Pass    |
|       | models.py::TestGetModelDetail        | test_model_has_validation_questions                 | Pass    |
|       | models.py::TestGetModelDetail        | test_model_not_found_returns_404                    | Pass    |
|       | models.py::TestGetModelDetail        | test_response_has_audit_entries                     | Pass    |
|       | models.py::TestGetModelDetail        | test_valid_model_returns_success                    | Pass    |
|       | models.py::TestGetModels             | test_category_distribution                          | Pass    |
|       | models.py::TestGetModels             | test_model_has_review_status                        | Pass    |
|       | models.py::TestGetModels             | test_model_has_risk_rating                          | Pass    |
|       | models.py::TestGetModels             | test_model_has_validation_coverage                  | Pass    |
|       | models.py::TestGetModels             | test_response_has_metadata                          | Pass    |
|       | models.py::TestGetModels             | test_returns_200                                    | Pass    |
|       | models.py::TestGetModels             | test_returns_models_list                            | Pass    |
|       | models.py::TestGetModels             | test_returns_success_status                         | Pass    |
|       | models.py::TestGetModels             | test_tier_distribution                              | Pass    |
|       | models.py::TestUpdateModelField      | test_empty_value_returns_400                        | Pass    |
|       | models.py::TestUpdateModelField      | test_invalid_choice_value_returns_400               | Pass    |
|       | models.py::TestUpdateModelField      | test_invalid_field_returns_400                      | Pass    |
|       | models.py::TestUpdateModelField      | test_model_not_found_returns_404                    | Pass    |
|       | models.py::TestUpdateModelField      | test_no_reason_returns_400                          | Pass    |
|       | models.py::TestUpdateModelField      | test_response_has_audit_entry                       | Pass    |
|       | models.py::TestUpdateModelField      | test_update_lifecycle_stage                         | Pass    |
|       | models.py::TestUpdateModelField      | test_update_validation_status                       | Pass    |
|       | models.py::TestUpdateModelField      | test_valid_choice_field_update_returns_success      | Success |
|       | models.py::TestUpdateModelField      | test_valid_text_field_update_returns_success        | Success |
|       | mrc_crud_actions.py::TestActions     | test_add_action_auto_id                             | Pass    |
|       | mrc_crud_actions.py::TestActions     | test_add_action_missing_meeting_returns_404         | 404     |
|       | mrc_crud_actions.py::TestActions     | test_add_action_returns_200                         | Pass    |
|       | mrc_crud_actions.py::TestActions     | test_delete_action_missing_meeting_returns_404      | 404     |
|       | mrc_crud_actions.py::TestActions     | test_delete_action_returns_200                      | Pass    |
|       | mrc_crud_actions.py::TestActions     | test_update_action_not_found_returns_404            | 404     |
|       | mrc_crud_actions.py::TestActions     | test_update_action_returns_200                      | Pass    |
|       | mrc_crud_actions.py::TestParticipant | test_add_participant_auto_id                        | Pass    |
|       | mrc_crud_actions.py::TestParticipant | test_add_participant_missing_meeting_returns_404    | 404     |
|       | mrc_crud_actions.py::TestParticipant | test_add_participant_returns_200                    | Pass    |
|       | mrc_crud_actions.py::TestParticipant | test_delete_participant_missing_meeting_returns_404 | 404     |
|       | mrc_crud_actions.py::TestParticipant | test_delete_participant_returns_200                 | Pass    |
|       | mrc_crud_actions.py::TestParticipant | test_update_participant_not_found_returns_404       | 404     |
|       | mrc_crud_actions.py::TestParticipant | test_update_participant_returns_200                 | Pass    |
|       | mrc_crud_agenda.py::TestAgenda       | test_add_agenda_item_increments_count               | Pass    |
|       | mrc_crud_agenda.py::TestAgenda       | test_add_agenda_item_returns_200                    | Pass    |
|       | mrc_crud_agenda.py::TestAgenda       | test_add_agenda_missing_meeting_returns_404         | 404     |
|       | mrc_crud_agenda.py::TestAgenda       | test_delete_agenda_item_returns_200                 | Pass    |

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| Model                                      | Module                                                | Test Name | Status |
|--------------------------------------------|-------------------------------------------------------|-----------|--------|
| mrc_crud_agenda.py::TestAgenda             | test_delete_agenda_missing_meeting_returns_404        | Pass      | 404    |
| mrc_crud_agenda.py::TestAgenda             | test_update_agenda_item_not_found_returns_404         | Pass      | 404    |
| mrc_crud_agenda.py::TestAgenda             | test_update_agenda_item_returns_200                   | Pass      |        |
| mrc_crud_agenda.py::TestDecision           | test_add_decision_auto_ids                            | Pass      |        |
| mrc_crud_agenda.py::TestDecision           | test_add_decision_missing_meeting_returns_404         | Pass      | 404    |
| mrc_crud_agenda.py::TestDecision           | test_add_decision_returns_200                         | Pass      |        |
| mrc_crud_agenda.py::TestDecision           | test_delete_decision_missing_meeting_returns_404      | Pass      | 404    |
| mrc_crud_agenda.py::TestDecision           | test_delete_decision_returns_200                      | Pass      |        |
| mrc_crud_agenda.py::TestDecision           | test_update_decision_not_found_returns_404            | Pass      | 404    |
| mrc_crud_agenda.py::TestDecision           | test_update_decision_returns_200                      | Pass      |        |
| mrc_crud_agenda.py::TestMinutes            | test_update_minutes_missing_meeting_returns_404       | Pass      | 404    |
| mrc_crud_agenda.py::TestMinutes            | test_update_minutes_returns_200                       | Pass      |        |
| mrc_crud_edge.py::TestMalformedAction      | test_add_action_with_malformed_existing_id            | Pass      |        |
| mrc_crud_edge.py::TestMalformedDecision    | test_add_decision_with_malformed_existing_id          | Pass      |        |
| mrc_crud_edge.py::TestMalformedParticipant | test_add_participant_with_malformed_existing_id       | Pass      |        |
| mrc_crud_edge.py::TestMinutesItem          | test_add_minute_item_missing_meeting_returns_404      | Pass      | 404    |
| mrc_crud_edge.py::TestMinutesItem          | test_add_minute_item_returns_200                      | Pass      |        |
| mrc_crud_edge.py::TestMinutesItem          | test_delete_minute_item_legacy_format_returns_400     | Pass      | 400    |
| mrc_crud_edge.py::TestMinutesItem          | test_delete_minute_item_missing_meeting_returns_404   | Pass      | 404    |
| mrc_crud_edge.py::TestMinutesItem          | test_delete_minute_item_returns_200                   | Pass      |        |
| mrc_crud_edge.py::TestMinutesItem          | test_update_minute_item_legacy_format_returns_400     | Pass      | 400    |
| mrc_crud_edge.py::TestMinutesItem          | test_update_minute_item_not_found_returns_404         | Pass      | 404    |
| mrc_crud_edge.py::TestMinutesItem          | test_update_minute_item_returns_200                   | Pass      |        |
| mrc_crud_edge.py::TestMinutesString        | test_convert_string_to_minutes_when_minutes_is_string | Pass      |        |
| mrc_crud_edge.py::TestUpdateMeeting        | test_update_action_missing_meeting_returns_404        | Pass      | 404    |
| mrc_crud_edge.py::TestUpdateMeeting        | test_update_decision_missing_meeting_returns_404      | Pass      | 404    |
| mrc_crud_edge.py::TestUpdateMeeting        | test_update_minute_item_missing_meeting_returns_404   | Pass      | 404    |
| mrc_crud_edge.py::TestUpdateMeeting        | test_update_participant_missing_meeting_returns_404   | Pass      | 404    |
| mrc_pdf.py::TestBuildMeetingPdf            | test_all_sections                                     | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf            | test_buffer_not_empty                                 | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf            | test_meeting_with_actions                             | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf            | test_meeting_with_agenda                              | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf            | test_meeting_with_decisions                           | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf            | test_meeting_with_minutes                             | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf            | test_meeting_with_no_optional_fields                  | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf            | test_meeting_with_participants                        | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf            | test_minimal_meeting                                  | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf            | test_returns_bytesio                                  | Pass      |        |
| mrc_pdf.py::TestGetMeetingPdf              | test_meeting_not_found_returns_404                    | Pass      |        |
| mrc_pdf.py::TestGetMeetingPdf              | test_valid_meeting_content_type_pdf                   | Pass      |        |
| mrc_pdf.py::TestGetMeetingPdf              | test_valid_meeting_returns_bytes                      | Pass      |        |
| mrc_pdf.py::TestGetMeetingPdf              | test_valid_meeting_returns_pdf                        | Pass      |        |
| raci.py::TestRACIMatrix                    | test_get_raci_matrix                                  | Pass      |        |
| raci.py::TestRACIMatrix                    | test_get_raci_matrix_file_missing                     | Pass      |        |

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| Model | Module                                        | Test Name                                       | Status |
|-------|-----------------------------------------------|-------------------------------------------------|--------|
|       | test_helpers_data.py::TestLoadBibliography    | test_loads_valid_data                           | Pass   |
|       | test_helpers_data.py::TestLoadBibliography    | test_loads_returns_none_when_missing            | Pass   |
|       | test_helpers_data.py::TestLoadBibliography    | test_loads_bibliography                         | Pass   |
|       | test_helpers_data.py::TestLoadBibliography    | test_loads_returns_default_when_missing         | Pass   |
|       | test_helpers_data.py::TestLoadDocuments       | test_loads_documents                            | Pass   |
|       | test_helpers_data.py::TestLoadDocuments       | test_loads_returns_empty_list_when_missing      | Pass   |
|       | test_helpers_data.py::TestLoadReports         | test_loads_valid_data                           | Pass   |
|       | test_helpers_data.py::TestLoadReports         | test_loads_returns_none_when_missing            | Pass   |
|       | test_helpers_data.py::TestSaveBibliography    | test_save_returns_false_on_oserror              | Pass   |
|       | test_helpers_data.py::TestSaveBibliography    | test_save_saves_data                            | Pass   |
|       | test_helpers_data.py::TestSaveBibliography    | test_save_returns_false_on_oserror              | Pass   |
|       | test_helpers_data.py::TestSaveBibliography    | test_save_saves_bibliography                    | Pass   |
|       | test_helpers_data.py::TestSaveDocuments       | test_save_returns_false_on_oserror              | Pass   |
|       | test_helpers_data.py::TestSaveDocuments       | test_save_saves_documents                       | Pass   |
|       | test_helpers_data.py::TestSaveReports         | test_save_returns_false_on_oserror              | Pass   |
|       | test_helpers_data.py::TestSaveReports         | test_save_saves_data                            | Pass   |
|       | test_helpers_io.py::TestFindMeeting           | test_finds_existing_meeting                     | Pass   |
|       | test_helpers_io.py::TestFindMeeting           | test_returns_error_for_missing_meeting          | Pass   |
|       | test_helpers_io.py::TestLoadAudits            | test_loads_entries                              | Pass   |
|       | test_helpers_io.py::TestLoadAudits            | test_returns_empty_list_on_bad_json             | Pass   |
|       | test_helpers_io.py::TestLoadAudits            | test_returns_empty_list_when_missing            | Pass   |
|       | test_helpers_io.py::TestLoadInventory         | test_loads_valid_file                           | Pass   |
|       | test_helpers_io.py::TestLoadInventory         | test_returns_none_on_bad_json                   | Pass   |
|       | test_helpers_io.py::TestLoadInventory         | test_returns_none_when_missing                  | Pass   |
|       | test_helpers_io.py::TestLoadMeetings          | test_loads_list                                 | Pass   |
|       | test_helpers_io.py::TestLoadMeetings          | test_returns_empty_list_when_missing            | Pass   |
|       | test_helpers_io.py::TestSaveAndRespond        | test_saves_and_returns_response                 | Pass   |
|       | test_helpers_io.py::TestSaveAuditLog          | test_returns_false_on_oserror                   | Pass   |
|       | test_helpers_io.py::TestSaveAuditLog          | test_saves_entries                              | Pass   |
|       | test_helpers_io.py::TestSaveInventory         | test_returns_false_on_oserror                   | Pass   |
|       | test_helpers_io.py::TestSaveInventory         | test_saves_data                                 | Pass   |
|       | test_helpers_io.py::TestSaveMeetings          | test_returns_false_on_oserror                   | Pass   |
|       | test_helpers_io.py::TestSaveMeetings          | test_saves_meetings                             | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating | test_calculating_additional_rating_middle_range | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating | test_calculating_gives_half_score               | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating | test_calculating_review_gives_full_score        | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating | test_calculating_assumptions_penalised          | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating | test_calculating_coverage_acceptable            | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating | test_calculating_limitations_penalised          | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating | test_calculating_coverage_unacceptable          | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating | test_calculating_review_date_gives_zero         | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating | test_calculating_review_gives_zero_score        | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating | test_calculating_dict_with_required_keys        | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating | test_calculating_is_float                       | Pass   |

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| Model | Module                                                                     | Test Name                                            | Status |
|-------|----------------------------------------------------------------------------|------------------------------------------------------|--------|
|       | test_helpers_risk.py::TestVqSummary                                        | test_all_addressed                                   | Pass   |
|       | test_helpers_risk.py::TestVqSummary                                        | test_empty_questions                                 | Pass   |
|       | test_helpers_risk.py::TestVqSummary                                        | test_none_addressed                                  | Pass   |
|       | test_helpers_risk.py::TestVqSummary                                        | test_some_not_applicable                             | Pass   |
|       | test_inventory_integrity.py::TestAPIEndpoints                              | test_returns_200                                     | Pass   |
|       | test_inventory_integrity.py::TestAPIEndpoints                              | test_endpoint_succeeds                               | Pass   |
|       | test_inventory_integrity.py::TestAPIEndpoints                              | test_model_count_matches_file                        | Pass   |
|       | test_inventory_integrity.py::TestInventoryFileExists                       | test_file_exists                                     | Pass   |
|       | test_inventory_integrity.py::TestInventoryFileIsValid                      | test_valid_json                                      | Pass   |
|       | test_inventory_integrity.py::TestInventoryFileHasAtLeastOneModel           | test_has_at_least_one_model                          | Pass   |
|       | test_inventory_integrity.py::TestInventoryFileHasData                      | test_has_data                                        | Pass   |
|       | test_inventory_integrity.py::TestInventoryFileHasChain                     | test_has_chain                                       | Pass   |
|       | test_inventory_integrity.py::TestInventoryFileHasKey                       | test_has_key                                         | Pass   |
|       | test_inventory_integrity.py::TestModelIDs                                  | test_models_have_model_id                            | Pass   |
|       | test_inventory_integrity.py::TestModelIDs                                  | test_model_ids_are_unique                            | Pass   |
|       | test_inventory_integrity.py::TestModelIDs                                  | test_model_ids_follow_naming_convention              | Pass   |
|       | test_inventory_integrity.py::TestOverallRiskRatingStringIsValid            | test_overall_risk_rating_string_is_valid             | Pass   |
|       | test_inventory_integrity.py::TestOverallRiskRatingStringHasComponentScores | test_overall_risk_rating_string_has_component_scores | Pass   |
|       | test_inventory_integrity.py::TestOverallRiskRatingStringHasRequiredKeys    | test_overall_risk_rating_string_has_required_keys    | Pass   |
|       | test_inventory_integrity.py::TestRequiredFieldsList[overall_risk_rating]   | test_required_fields_list[overall_risk_rating]       | Pass   |
|       | test_inventory_integrity.py::TestRequiredFieldsList[test_coverage]         | test_required_fields_list[test_coverage]             | Pass   |
|       | test_inventory_integrity.py::TestRequiredFieldsList[assumptions]           | test_required_fields_list[assumptions]               | Pass   |
|       | test_inventory_integrity.py::TestRequiredFieldsList[downstream_models]     | test_required_fields_list[downstream_models]         | Pass   |
|       | test_inventory_integrity.py::TestRequiredFieldsList[limitations]           | test_required_fields_list[limitations]               | Pass   |
|       | test_inventory_integrity.py::TestRequiredFieldsList[upstream_models]       | test_required_fields_list[upstream_models]           | Pass   |
|       | test_inventory_integrity.py::TestRequiredFieldsList[category]              | test_required_fields_list[category]                  | Pass   |
|       | test_inventory_integrity.py::TestRequiredFieldsList[description]           | test_required_fields_list[description]               | Pass   |
|       | test_inventory_integrity.py::TestRequiredFieldsList[name]                  | test_required_fields_list[name]                      | Pass   |
|       | test_inventory_integrity.py::TestRequiredFieldsList[owner]                 | test_required_fields_list[owner]                     | Pass   |
|       | test_inventory_integrity.py::TestRequiredFieldsList[status]                | test_required_fields_list[status]                    | Pass   |
|       | test_inventory_integrity.py::TestRequiredFieldsList[version]               | test_required_fields_list[version]                   | Pass   |
|       | test_inventory_integrity.py::TestTestCoverageStorage                       | test_test_coverage_storage_booleans_are_bool         | Pass   |
|       | test_inventory_integrity.py::TestTestCoverageStorage                       | test_test_coverage_storage_has_required_keys         | Pass   |
|       | test_inventory_integrity.py::TestTestCoverageStorage                       | test_test_coverage_storage_has_test_file             | Pass   |
|       | test_lineage_data_routes.py::TestDataLineage                               | test_data_lineage                                    | Pass   |
|       | test_lineage_data_routes.py::TestDataLineageHealth                         | test_data_lineage_health                             | Pass   |
|       | test_lineage_data_routes.py::TestDataLineageAge                            | test_data_lineage_age                                | Pass   |
|       | test_lineage_data_routes.py::TestDataLineageStatus                         | test_data_lineage_status                             | Pass   |
|       | test_lineage_data_routes.py::TestDataLineageManifest                       | test_data_lineage_manifest                           | Pass   |
|       | test_lineage_data_routes.py::TestDataLineageGetRoute                       | test_data_lineage_get_route                          | Pass   |
|       | test_lineage_data_routes.py::TestDataLineageGetRoute                       | test_data_lineage_get_route_type                     | Pass   |
|       | test_lineage_data_routes.py::TestDataLineageGetRoute                       | test_data_lineage_get_route_params                   | Pass   |
|       | test_lineage_data_routes.py::TestDataLineageGetRoute                       | test_data_lineage_get_route_auth                     | Pass   |
|       | test_lineage_data_routes.py::TestDataLineageGetRoute                       | test_data_lineage_get_route_permissions              | Pass   |

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| Model | Module                                                                                   | Test Name                                                       | Status |
|-------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------|
|       | test_lineage_data_routes.py::TestTraceLineageBundle                                      | TestTraceLineageBundle                                          | Pass   |
|       | test_lineage_data_routes.py::TestValidLineageRoute                                       | TestValidLineageRoute                                           | Pass   |
|       | test_lineage_data_routes.py::TestUnitParamsOnly                                          | TestUnitParamsOnly                                              | Pass   |
|       | test_lineage_field.py::TestFieldLineageLookup_found                                      | TestFieldLineageLookup_found                                    | Pass   |
|       | test_lineage_field.py::TestFieldLineageLookup_not_found                                  | TestFieldLineageLookup_not_found                                | Pass   |
|       | test_lineage_field.py::TestFieldLineageLookup_wrong_report                               | TestFieldLineageLookup_wrong_report                             | Pass   |
|       | test_lineage_field.py::TestFieldLineageLookup_wrong_section                              | TestFieldLineageLookup_wrong_section                            | Pass   |
|       | test_lineage_field.py::TestFieldLineageLookup_params_400                                 | TestFieldLineageLookup_params_400                               | Pass   |
|       | test_lineage_field.py::TestFieldLineageLookup_registry_404                               | TestFieldLineageLookup_registry_404                             | Pass   |
|       | test_lineage_field.py::TestFieldLineageLookup_by_cdm_path                                | TestFieldLineageLookup_by_cdm_path                              | Pass   |
|       | test_lineage_field.py::TestFieldLineageLookup_by_source_field                            | TestFieldLineageLookup_by_source_field                          | Pass   |
|       | test_lineage_field.py::TestFieldLineageLookup_case_insensitive                           | TestFieldLineageLookup_case_insensitive                         | Pass   |
|       | test_lineage_field.py::TestFieldLineageLookup_finds_matches                              | TestFieldLineageLookup_finds_matches                            | Pass   |
|       | test_lineage_field.py::TestFieldLineageLookup_no_results                                 | TestFieldLineageLookup_no_results                               | Pass   |
|       | test_lineage_field.py::TestFieldLineageLookup_result_structure                           | TestFieldLineageLookup_result_structure                         | Pass   |
|       | test_lineage_field.py::TestFieldLineageLookup_takes_priority_over_partial_exact          | TestFieldLineageLookup_takes_priority_over_partial_exact        | Pass   |
|       | test_lineage_field.py::TestGetFieldHistoryFilter_by_report                               | TestGetFieldHistoryFilter_by_report                             | Pass   |
|       | test_lineage_field.py::TestGetFieldHistoryFilter_unknown_report_404                      | TestGetFieldHistoryFilter_unknown_report_404                    | Pass   |
|       | test_lineage_field.py::TestGetFieldHistoryFilter_registry                                | TestGetFieldHistoryFilter_registry                              | Pass   |
|       | test_lineage_field.py::TestGetFieldHistoryFilter_registry_404                            | TestGetFieldHistoryFilter_registry_404                          | Pass   |
|       | test_lineage_field.py::TestGetFieldHistorySummary_contents                               | TestGetFieldHistorySummary_contents                             | Pass   |
|       | test_lineage_helpers.py::TestCheckStaleness_missing                                      | TestCheckStaleness_missing                                      | Pass   |
|       | test_lineage_helpers.py::TestCheckStaleness_directory_step_fresh                         | TestCheckStaleness_directory_step_fresh                         | Pass   |
|       | test_lineage_helpers.py::TestCheckStaleness_directory_step_stale                         | TestCheckStaleness_directory_step_stale                         | Pass   |
|       | test_lineage_helpers.py::TestCheckStaleness_directory_is_missing                         | TestCheckStaleness_directory_is_missing                         | Pass   |
|       | test_lineage_helpers.py::TestCheckStaleness_fresh_file                                   | TestCheckStaleness_fresh_file                                   | Pass   |
|       | test_lineage_helpers.py::TestCheckStaleness_hidden_files_ignored_in_directory            | TestCheckStaleness_hidden_files_ignored_in_directory            | Pass   |
|       | test_lineage_helpers.py::TestCheckStaleness_manifest_last_run_not_overwritten_by_runtime | TestCheckStaleness_manifest_last_run_not_overwritten_by_runtime | Pass   |
|       | test_lineage_helpers.py::TestCheckStaleness_manifest_last_run_used                       | TestCheckStaleness_manifest_last_run_used                       | Pass   |
|       | test_lineage_helpers.py::TestCheckStaleness_stale_file                                   | TestCheckStaleness_stale_file                                   | Pass   |
|       | test_lineage_helpers.py::TestTraceDirectTrace_found                                      | TestTraceDirectTrace_found                                      | Pass   |
|       | test_lineage_helpers.py::TestTraceFallbackScan_finds_in_outputs                          | TestTraceFallbackScan_finds_in_outputs                          | Pass   |
|       | test_lineage_helpers.py::TestTraceFallbackScan_no_match                                  | TestTraceFallbackScan_no_match                                  | Pass   |
|       | test_lineage_helpers.py::TestTraceFallbackScan_with_non_string_output                    | TestTraceFallbackScan_with_non_string_output                    | Pass   |
|       | test_lineage_helpers.py::TestTraceNoneLineage_returns_empty                              | TestTraceNoneLineage_returns_empty                              | Pass   |
|       | test_lineage_helpers.py::TestTraceTraceId_not_found                                      | TestTraceTraceId_not_found                                      | Pass   |
|       | test_lineage_helpers.py::TestTraceTraceType_not_found                                    | TestTraceTraceType_not_found                                    | Pass   |
|       | test_lineage_staleness.py::TestStalenessAllRefresh_full_health                           | TestStalenessAllRefresh_full_health                             | Pass   |
|       | test_lineage_staleness.py::TestStalenessAllMissing_zero_health                           | TestStalenessAllMissing_zero_health                             | Pass   |
|       | test_lineage_staleness.py::TestStalenessAllReads_of                                      | TestStalenessAllReads_of                                        | Pass   |
|       | test_lineage_staleness.py::TestStalenessPartial health                                   | TestStalenessPartial health                                     | Pass   |
|       | test_lineage_staleness.py::TestStalenessRate_does_not_count_as_fresh                     | TestStalenessRate_does_not_count_as_fresh                       | Pass   |
|       | test_models_coverage.py::TestAuditLogTruncation_truncated_to_10000                       | TestAuditLogTruncation_truncated_to_10000                       | Pass   |
|       | test_models_coverage.py::TestGetWorldDemographics_by_report                              | TestGetWorldDemographics_by_report                              | Pass   |

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| Model | Module                          | Test Name                                            | Status |
|-------|---------------------------------|------------------------------------------------------|--------|
|       | test_models_coverage.py::       | TestGetModelsNotInventory returns_404                | Pass   |
|       | test_models_coverage.py::       | TestGetModelsNotScheduled review_status              | Pass   |
|       | test_models_coverage.py::       | TestGetModelsOverdueReview status                    | Pass   |
|       | test_models_coverage.py::       | TestUpdateModelInventory returns_404                 | Pass   |
|       | test_models_coverage.py::       | TestUpdateModelFailFuture returns_500                | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestAddAgenda item                                   | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestAddAgenda unknown meeting returns_404            | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestAddAgenda multiple items increments number       | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestDeleteAgenda item rennumbers                     | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestDeleteAgenda unknown meeting returns_404         | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestUpdateAgenda item                                | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestUpdateAgenda unknown meeting returns_404         | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestUpdateAgenda missing item returns_404            | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestAddAgenda item                                   | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestUpdateAgenda item legacy format returns_400      | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestUpdateAgenda item legacy format returns_400      | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestUpdateAgenda item not found returns_404          | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestUpdateAgenda minutes text                        | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestUpdateAgenda minutes unknown meeting returns_404 | Pass   |
|       | test_mrc_decisions_actions.py:: | TestAddAction auto generates id                      | Pass   |
|       | test_mrc_decisions_actions.py:: | TestAddAction unknown meeting returns_404            | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDeleteAction                                     | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDeleteAction unknown meeting returns_404         | Pass   |
|       | test_mrc_decisions_actions.py:: | TestUpdateAction status                              | Pass   |
|       | test_mrc_decisions_actions.py:: | TestUpdateAction missing action returns_404          | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDecision auto generates id                       | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDecision unknown meeting returns_404             | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDeleteDecision                                   | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDeleteDecision unknown meeting returns_404       | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDecision increments id                           | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDecision                                         | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDecision missing decision returns_404            | Pass   |
|       | test_mrc_decisions_actions.py:: | TestParticipant auto generates id                    | Pass   |
|       | test_mrc_decisions_actions.py:: | TestParticipant unknown meeting returns_404          | Pass   |
|       | test_mrc_decisions_actions.py:: | TestParticipant                                      | Pass   |
|       | test_mrc_decisions_actions.py:: | TestParticipant unknown meeting returns_404          | Pass   |
|       | test_mrc_decisions_actions.py:: | TestParticipant missing participant returns_404      | Pass   |
|       | test_mrc_decisions_actions.py:: | TestParticipant                                      | Pass   |
|       | test_mrc_meetings.py::          | TestCreateMeeting auto generates id                  | Pass   |
|       | test_mrc_meetings.py::          | TestCreateMeeting returns meeting                    | Pass   |
|       | test_mrc_meetings.py::          | TestCreateMeeting attendees become participant       | Pass   |
|       | test_mrc_meetings.py::          | TestCreateMeeting title used                         | Pass   |
|       | test_mrc_meetings.py::          | TestCreateMeeting default chair populated            | Pass   |
|       | test_mrc_meetings.py::          | TestCreateMeeting default participants created       | Pass   |

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| Model | Module                                                    | Test Name                                           | Status    |
|-------|-----------------------------------------------------------|-----------------------------------------------------|-----------|
|       | test_mrc_meetings.py::TestCreateMeeting                   | test_create_meeting_json_body_uses_defaults         | Pass      |
|       | test_mrc_meetings.py::TestGetMeeting                      | test_get_created_meeting                            | Pass      |
|       | test_mrc_meetings.py::TestGetMeeting                      | test_get_unknown_id_returns_404                     | Pass      |
|       | test_mrc_meetings.py::TestListMeetings                    | test_created_meeting_appears_in_list                | Pass      |
|       | test_mrc_meetings.py::TestListMeetings                    | test_empty_list_returns_success                     | Pass      |
|       | test_mrc_meetings.py::TestListMeetings                    | test_list_summary_has_required_keys                 | Pass      |
|       | test_mrc_meetings.py::TestListMeetings                    | test_sorted_by_date_desc                            | Pass      |
|       | test_mrc_meetings.py::TestMeetingDocument                 | test_document_missing_document_returns_404          | Pass      |
|       | test_mrc_meetings.py::TestMeetingDocument                 | test_upload_empty_filename_returns_400              | Pass      |
|       | test_mrc_meetings.py::TestMeetingDocument                 | test_upload_no_file_returns_400                     | Pass      |
|       | test_mrc_meetings.py::TestMeetingDocument                 | test_upload_not_found_meeting_returns_404           | Pass      |
|       | test_mrc_meetings.py::TestUpdateMeeting                   | test_update_sets_updated_at                         | Pass      |
|       | test_mrc_meetings.py::TestUpdateMeeting                   | test_update_title                                   | Pass      |
|       | test_mrc_meetings.py::TestUpdateMeeting                   | test_update_unknown_id_returns_404                  | Pass      |
|       | test_report_routes.py::TestAppendOutput                   | test_append_line                                    | Pass      |
|       | test_report_routes.py::TestAppendOutput                   | test_clear_then_append                              | Pass      |
|       | test_report_routes.py::TestAppendOutput                   | test_trims_buffer_at_2000                           | Pass      |
|       | test_report_routes.py::TestRunAndSave                     | test_append_started_header                          | Pass      |
|       | test_report_routes.py::TestRunAndSave                     | test_error_line_appended_on_exception               | Pass      |
|       | test_report_routes.py::TestRunAndSave                     | test_handles_exception_gracefully                   | Pass      |
|       | test_report_routes.py::TestRunAndSave                     | test_sets_running_false_on_completion               | Pass      |
|       | test_report_routes.py::TestRunTestReport                  | test_already_running_message                        | Pass      |
|       | test_report_routes.py::TestRunTestReport                  | test_already_running_returns_202                    | Pass      |
|       | test_report_routes.py::TestRunTestReport                  | test_response_has_started_status                    | Pass      |
|       | test_report_routes.py::TestRunTestReport                  | test_starts_run_returns_202                         | Pass      |
|       | test_report_routes.py::TestStreamProc                     | test_calls_proc_wait                                | Pass      |
|       | test_report_routes.py::TestStreamProc                     | test_reads_stdout_into_buffer                       | Pass      |
|       | test_report_routes.py::TestStreamProc                     | test_strips_newlines                                | Pass      |
|       | test_report_routes.py::TestTestReportOutput               | test_has_lines_field                                | Pass      |
|       | test_report_routes.py::TestTestReportOutput               | test_has_running_field                              | Pass      |
|       | test_report_routes.py::TestTestReportOutput               | test_includes_appended                              | Pass      |
|       | test_report_routes.py::TestTestReportOutput               | test_not_running_initially                          | Pass      |
|       | test_report_routes.py::TestTestReportOutput               | test_returns_200                                    | Pass      |
|       | test_report_routes.py::TestTestReportOutput               | test_returns_not_found_when_no_file                 | Pass      |
|       | test_report_routes.py::TestTestReportOutput               | test_returns_ok_when_file_exists                    | Pass      |
|       | test_report_routes.py::TestTestReportOutput               | test_returns_summary_fields                         | Pass      |
|       | test_report_routes.py::TestTestReportOutput               | test_running_field_is_bool                          | Pass      |
|       | test_cluster_and_isolation.py::TestClusterIsolationStages | test_cluster_isolation_stages_tv_computed_per Storm | Storm     |
|       | test_cluster_and_isolation.py::TestClusterIsolationStages | test_individual_storms_individually_queryable       | Queryable |
|       | test_cluster_and_isolation.py::TestClusterIsolationStages | test_most_damaged_storm_has_most_damaged            | Damaged   |
|       | test_cluster_and_isolation.py::TestClusterIsolationStages | test_unknown_storm_from_sequence_returns_404        | 404       |
|       | test_cluster_and_isolation.py::TestNegativeEquityFunction | test_negative_equity_function_computed_correctly    | Pass      |
|       | test_cluster_and_isolation.py::TestNegativeEquityFunction | test_negative_equity_function_set_on_property       | Pass      |
|       | test_cluster_and_isolation.py::TestNegativeDamageFunction | test_negative_damage_function_returns_100_pct       | Pass      |

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| Model | Module                                                                        | Test Name                                                     | Status |
|-------|-------------------------------------------------------------------------------|---------------------------------------------------------------|--------|
|       | test_cluster_and_isolation.py::TestNegativeEquityFromSignsenegative-equity    | test_negative_equality_from_signsenegative_equality           | Pass   |
|       | test_cluster_and_isolation.py::TestSequenceIsolationNoContamination           | test_sequence_isolation_no_contamination                      | Pass   |
|       | test_cluster_and_isolation.py::TestSequenceIsolationStormReturnsOnlyItsDamage | test_sequence_isolation_storm_returns_only_its_damage         | Pass   |
|       | test_cluster_and_isolation.py::TestSequenceIsolationReturnsOnlySeqB.Damage    | test_sequence_isolation_returns_only_seqB_damage              | Pass   |
|       | test_doublet_storms.py::TestDoubletFirstStormDamageIsIndependentOfSecond      | test_doublet_first_storm_damage_is_independent_of_second      | Pass   |
|       | test_doublet_storms.py::TestDoubletFirstStormReturnsOnlyFloodedProperties     | test_doublet_first_storm_returns_only_flooded_properties      | Pass   |
|       | test_doublet_storms.py::TestDoubletFirstStormTotalsDifferBetweenStorms        | test_doublet_first_storm_totals_differ_between_storms         | Pass   |
|       | test_doublet_storms.py::TestDoubletSecondStormLargerDamageForProperties       | test_doublet_second_storm_larger_damage_for_properties        | Pass   |
|       | test_doublet_storms.py::TestDoubletSecondStormReturnsBothProperties           | test_doublet_second_storm_returns_both_properties             | Pass   |
|       | test_doublet_storms.py::TestDoubletStormInitialStormResponseMatchesQuery      | test_doublet_storm_initial_storm_response_matches_query       | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointIsMaxDepth                     | test_sequence_endpoint_is_max_depth                           | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointPerPropertyReported            | test_sequence_endpoint_per_property_reported                  | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointInSequenceReported             | test_sequence_endpoint_in_sequence_reported                   | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointDoubleCounted                  | test_sequence_endpoint_double_counted                         | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointMaxDepthEvent                  | test_sequence_endpoint_max_depth_event                        | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointIncludedViaSecondStorm         | test_sequence_endpoint_included_via_second_storm              | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointIncludedFromDifferentSequence  | test_sequence_endpoint_included_from_different_sequence       | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointDamageLessThanSumOfStorms      | test_sequence_endpoint_damage_less_than_sum_of_storms         | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointOptions                        | test_sequence_endpoint_options                                | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointReturns200                     | test_sequence_endpoint_returns_200                            | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointIdInResponse                   | test_sequence_endpoint_id_in_response                         | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointSequenceReturns404             | test_sequence_endpoint_sequence_returns_404                   | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointNegativeEquityFromWorstStorm   | test_sequence_endpoint_negative_equality_from_worst_storm     | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointMultipleFloodEventsFour        | test_sequence_endpoint_multiple_flood_events_four             | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpointNegativeEquityStorm            | test_sequence_endpoint_negative_equality_storm                | Pass   |
|       | properties.py::TestGenerateMortgageReportExceptionReturns500                  | test_generate_mortgage_report_exception_returns_500           | Pass   |
|       | properties.py::TestGenerateMortgageReportSuccess                              | test_generate_mortgage_report_success                         | Pass   |
|       | properties.py::TestGenerateMortgageReportMissingPropertyIdReturns400          | test_generate_mortgage_report_missing_property_id_returns_400 | Pass   |
|       | properties.py::TestGenerateMortgageReportMortgageNotFoundReturns404           | test_generate_mortgage_report_mortgage_not_found_returns_404  | Pass   |
|       | properties.py::TestGenerateMortgageReportJsonBodyReturns400                   | test_generate_mortgage_report_json_body_returns_400           | Pass   |
|       | properties.py::TestGenerateMortgageReportJsonReturnsError                     | test_generate_mortgage_report_json_returns_error              | Pass   |
|       | properties.py::TestGenerateMortgageReportOptionsReturnsOk                     | test_generate_mortgage_report_options_returns_ok              | Pass   |
|       | properties.py::TestGenerateMortgageReportPropertyNotFoundReturns404           | test_generate_mortgage_report_property_not_found_returns_404  | Pass   |
|       | properties.py::TestGenerateMortgageReportMortgageNotFoundReturns500           | test_generate_mortgage_report_mortgage_not_found_returns_500  | Pass   |
|       | properties.py::TestGeneratePropertyReportExceptionReturns500                  | test_generate_report_exception_returns_500                    | Pass   |
|       | properties.py::TestGeneratePropertyReportSuccess                              | test_generate_report_success                                  | Pass   |
|       | properties.py::TestGeneratePropertyReportMissingPropertyIdReturns400          | test_generate_report_missing_property_id_returns_400          | Pass   |
|       | properties.py::TestGeneratePropertyReportJsonBodyReturns400                   | test_generate_report_json_body_returns_400                    | Pass   |
|       | properties.py::TestGeneratePropertyReportJsonReturnsError                     | test_generate_report_json_returns_error                       | Pass   |
|       | properties.py::TestGeneratePropertyReportOptionsReturnsOk                     | test_generate_report_options_returns_ok                       | Pass   |
|       | properties.py::TestGeneratePropertyReportPropertyNotFoundReturns404           | test_generate_report_property_not_found_returns_404           | Pass   |
|       | properties.py::TestGenerateReportExceptionReturns500                          | test_report_exception_returns_500                             | Pass   |
|       | properties.py::TestGetPropertyFoundReturnsProperty                            | test_found_returns_property                                   | Pass   |
|       | properties.py::TestGetPropertyNotFoundReturns404                              | test_not_found_returns_404                                    | Pass   |

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| Model | Module                              | Test Name                                  | Status |
|-------|-------------------------------------|--------------------------------------------|--------|
|       | properties.py::TestListMortgages    | test_list_mortgages_empty                  | Pass   |
|       | properties.py::TestListMortgages    | test_list_mortgages_exception_returns_500  | Pass   |
|       | properties.py::TestListMortgages    | test_list_mortgages_success                | Pass   |
|       | properties.py::TestListProperties   | test_loader_exception_returns_500          | Pass   |
|       | properties.py::TestListProperties   | test_success_empty_list                    | Pass   |
|       | properties.py::TestListProperties   | test_success_with_properties               | Pass   |
|       | properties.py::TestNoJsonBody       | test_mortgage_report_null_json_returns_400 | Pass   |
|       | properties.py::TestNoJsonBody       | test_property_report_null_json_returns_400 | Pass   |
|       | properties.py::TestPropertyMortgage | test_found_returns_mortgage                | Pass   |
|       | properties.py::TestPropertyMortgage | test_loader_exception_returns_500          | Pass   |
|       | properties.py::TestPropertyMortgage | test_not_found_returns_404                 | Pass   |
|       | properties.py::TestPropertyMortgage | test_options_returns_ok                    | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_data_contains_hazard_curve            | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_data_contains_nearest_gauges          | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_data_contains_property_id             | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_data_contains_summary                 | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_known_property_returns_200            | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_known_property_status_success         | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_no_data_returns_404                   | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_no_data_status_error                  | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_options_returns_200                   | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_options_status_ok                     | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_response_has_data_key                 | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_second_property_also_works            | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_unknown_property_message_contains_id  | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_unknown_property_returns_404          | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_unknown_property_status_error         | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_entry_has_avg_basis_bps         | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_entry_has_flood_count           | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_entry_has_nearest_gauges        | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_entry_has_property_id           | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_table_count_matches             | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_table_length                    | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_table_sorted_desc_by_basis      | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_nearest_gauge_has_distance_km         | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_nearest_gauge_has_gauge_id            | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_no_data_returns_404                   | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_no_data_status_error                  | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_options_returns_200                   | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_options_status_ok                     | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_response_has_basis_table              | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_response_has_count                    | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_returns_200_with_data                 | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_status_is_success                     | Pass   |

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| Model | Module                                    | Test Name                                     | Status |
|-------|-------------------------------------------|-----------------------------------------------|--------|
|       | propertyhc.py::TestPropertyhcBasisEmpty   | test_empty_basis_table_is_list                | Pass   |
|       | propertyhc.py::TestPropertyhcBasisEmpty   | test_empty_count_is_zero                      | Pass   |
|       | propertyhc.py::TestPropertyhcBasisEmpty   | test_empty_returns_200                        | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_distribution_avg_basis_bps         | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_distribution_avg_flood_count       | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_distribution_max_depth_m           | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_distribution_num_properties        | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_data_message_mentions_command      | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_data_returns_404                   | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_data_status_is_error               | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_options_returns_200                | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_options_status_ok                  | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_response_has_data_key              | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_response_has_distribution          | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_response_has_metadata              | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_response_has_summary               | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_returns_200_with_data              | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_status_is_success                  | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_hyptcurves_avg_flood_count_is_zero | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_hyptcurves_num_properties_is_zero  | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmpty  | test_empty_hyptcurves_returns_200             | Pass   |
|       | claim.py::TestClaimReportGeneratorError   | test_generator_error_has_message              | Pass   |
|       | claim.py::TestClaimReportGeneratorError   | test_generator_error_returns_500              | Pass   |
|       | claim.py::TestClaimReportGeneratorError   | test_generator_error_status_is_error          | Pass   |
|       | claim.py::TestClaimReportHappyPath        | test_content_length_header_set                | Pass   |
|       | claim.py::TestClaimReportHappyPath        | test_filename_contains_prop_id                | Pass   |
|       | claim.py::TestClaimReportHappyPath        | test_response_body_is_pdf_bytes               | Pass   |
|       | claim.py::TestClaimReportHappyPath        | test_response_has_content_disposition         | Pass   |
|       | claim.py::TestClaimReportHappyPath        | test_response_is_pdf_content_type             | Pass   |
|       | claim.py::TestClaimReportHappyPath        | test_success_returns_200                      | Pass   |
|       | claim.py::TestClaimReportHelperEstimation | test_sequences_file_handled                   | Pass   |
|       | claim.py::TestClaimReportHelperEstimation | test_missing_mortgage_json_still_generates    | Pass   |
|       | claim.py::TestClaimReportHelperEstimation | test_missing_property_json_still_generates    | Pass   |
|       | claim.py::TestClaimReportHelperEstimation | test_missing_sequences_file_still_generates   | Pass   |
|       | claim.py::TestClaimReportNoDir            | test_no_dir_message_mentions_propertyts       | Pass   |
|       | claim.py::TestClaimReportNoDir            | test_no_dir_returns_404                       | Pass   |
|       | claim.py::TestClaimReportNoDir            | test_no_dir_status_error                      | Pass   |
|       | claim.py::TestClaimReportNoFloodEvent     | test_no_floods_message_mentions_events        | Pass   |
|       | claim.py::TestClaimReportNoFloodEvent     | test_no_floods_returns_404                    | Pass   |
|       | claim.py::TestClaimReportNoFloodEvent     | test_no_floods_status_error                   | Pass   |
|       | claim.py::TestClaimReportNotFound         | test_unknown_property_message_contains_id     | Pass   |
|       | claim.py::TestClaimReportNotFound         | test_unknown_property_returns_404             | Pass   |
|       | claim.py::TestClaimReportNotFound         | test_unknown_property_status_error            | Pass   |
|       | core_enrichment.py::TestListFloodStagnant | test_stagnant_error_continues                 | Pass   |

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| Model | Module                                  | Test Name                                             | Status |
|-------|-----------------------------------------|-------------------------------------------------------|--------|
|       | core_enrichment.py::TestListFloodStorms | test_storms_data_json_read_error_continues            | Pass   |
|       | core_enrichment.py::TestListFloodStorms | test_storms_enrich_flooding_enrichment                | Pass   |
|       | core_enrichment.py::TestListFloodStorms | test_storms_enrich_json_read_error_continues          | Pass   |
|       | core_enrichment.py::TestListFloodStorms | test_storms_enrichment_in_valued_ids_skipped          | Pass   |
|       | core_enrichment.py::TestListFloodStorms | test_storms_enrich_with_empty_storm_id_skipped        | Pass   |
|       | core_enrichment.py::TestListFloodStorms | test_storms_enrich_data_from_storms_json              | Pass   |
|       | core_enrichment.py::TestListFloodStorms | test_storms_enrich_severity_classification_all_levels | Pass   |
|       | core_enrichment.py::TestListFloodStorms | test_storms_enrich_depth_skipped                      | Pass   |
|       | core_storms.py::TestListFloodStorms     | test_no_storms_file_returns_404                       | Pass   |
|       | core_storms.py::TestListFloodStorms     | test_options_returns_ok                               | Pass   |
|       | core_storms.py::TestListFloodStorms     | test_storm_has_expected_fields                        | Pass   |
|       | core_storms.py::TestListFloodStorms     | test_storm_name_from_intensity_category               | Pass   |
|       | core_storms.py::TestListFloodStorms     | test_storms_list_count                                | Pass   |
|       | core_storms.py::TestListFloodStorms     | test_storms_sorted_by_gauges_severe                   | Pass   |
|       | core_storms.py::TestListFloodStorms     | test_with_data_returns_success                        | Pass   |
|       | core_storms.py::TestPropertyStorms      | test_missing_file_without_gauge_json                  | Pass   |
|       | core_storms.py::TestPropertyStorms      | test_missing_file_without_storm_sequences_json        | Pass   |
|       | core_summary.py::TestPropertyStorms     | test_existing_property_returns_success                | Pass   |
|       | core_summary.py::TestPropertyStorms     | test_include_readings_false_strips_readings           | Pass   |
|       | core_summary.py::TestPropertyStorms     | test_include_readings_true_keeps_reading              | Pass   |
|       | core_summary.py::TestPropertyStorms     | test_missing_property_returns_404                     | Pass   |
|       | core_summary.py::TestPropertyStorms     | test_options_returns_ok                               | Pass   |
|       | core_summary.py::TestPropertyStorms     | test_existing_property_returns_success                | Pass   |
|       | core_summary.py::TestPropertyStorms     | test_flood_events_tagged_with_sequence_type           | Pass   |
|       | core_summary.py::TestPropertyStorms     | test_missing_property_returns_404                     | Pass   |
|       | core_summary.py::TestPropertyStorms     | test_nearest_gauges_enriched_with_flood_stages        | Pass   |
|       | core_summary.py::TestPropertyStorms     | test_options_returns_ok                               | Pass   |
|       | core_summary.py::TestPropertyStorms     | test_response_has_flood_events                        | Pass   |
|       | core_summary.py::TestPropertyStorms     | test_response_has_nearest_gauges                      | Pass   |
|       | core_summary.py::TestPropertyStorms     | test_response_has_property_info                       | Pass   |
|       | core_summary.py::TestSummary            | test_no_data_returns_404                              | Pass   |
|       | core_summary.py::TestSummary            | test_options_returns_ok                               | Pass   |
|       | core_summary.py::TestSummary            | test_with_data_returns_success                        | Pass   |
|       | financial.py::TestPortfolioImpact       | test_known_storm_returns_success                      | Pass   |
|       | financial.py::TestPortfolioImpact       | test_mortgaged_property_has_ltv_fields                | Pass   |
|       | financial.py::TestPortfolioImpact       | test_no_pts_dir_returns_404                           | Pass   |
|       | financial.py::TestPortfolioImpact       | test_options_returns_ok                               | Pass   |
|       | financial.py::TestPortfolioImpact       | test_portfolio_damage_pct_non_negative                | Pass   |
|       | financial.py::TestPortfolioImpact       | test_portfolio_totals_count                           | Pass   |
|       | financial.py::TestPortfolioImpact       | test_properties_sorted_by_damage_descending           | Pass   |
|       | financial.py::TestPortfolioImpact       | test_property_entry_has_financial_fields              | Pass   |
|       | financial.py::TestPortfolioImpact       | test_response_has_portfolio                           | Pass   |
|       | financial.py::TestPortfolioImpact       | test_response_has_properties_list                     | Pass   |
|       | financial.py::TestPortfolioImpact       | test_response_has_storm_id                            | Pass   |

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| Model | Module                                      | Test Name                                           | Status |
|-------|---------------------------------------------|-----------------------------------------------------|--------|
|       | financial.py::TestPortfolioImpact           | test_unknown_storm_returns_404                      | Pass   |
|       | financial.py::TestSequencePortfolioImpact   | test_unknown_sequence_returns_success               | Pass   |
|       | financial.py::TestSequencePortfolioImpact   | test_no_pts_dir_returns_404                         | Pass   |
|       | financial.py::TestSequencePortfolioImpact   | test_options_returns_ok                             | Pass   |
|       | financial.py::TestSequencePortfolioImpact   | test_properties_sorted_by_damage_descending         | Fail   |
|       | financial.py::TestSequencePortfolioImpact   | test_property_entry_has_financial_field             | Pass   |
|       | financial.py::TestSequencePortfolioImpact   | test_property_entry_has_num_sequence_flow           | Fail   |
|       | financial.py::TestSequencePortfolioImpact   | test_property_entry_has_worst_storm_id              | Pass   |
|       | financial.py::TestSequencePortfolioImpact   | test_response_has_damage_model                      | Pass   |
|       | financial.py::TestSequencePortfolioImpact   | test_response_has_num_storms                        | Pass   |
|       | financial.py::TestSequencePortfolioImpact   | test_response_has_portfolio                         | Pass   |
|       | financial.py::TestSequencePortfolioImpact   | test_response_has_properties_list                   | Pass   |
|       | financial.py::TestSequencePortfolioImpact   | test_response_has_sequence_id                       | Pass   |
|       | financial.py::TestSequencePortfolioImpact   | test_unknown_sequence_returns_404                   | Pass   |
|       | risk.py::TestPortfolioVarErrors             | test_missing_sequences_message_mentions_doublet     | Fail   |
|       | risk.py::TestPortfolioVarErrors             | test_missing_sequences_returns_404                  | Pass   |
|       | risk.py::TestPortfolioVarErrors             | test_no_pts_dir_returns_404                         | Pass   |
|       | risk.py::TestPortfolioVarErrors             | test_no_pts_dir_status_error                        | Pass   |
|       | risk.py::TestPortfolioVarErrors             | test_options_returns_ok                             | Pass   |
|       | risk.py::TestPortfolioVarMissingFiles       | test_missing_mortgage_json_returns_200              | Pass   |
|       | risk.py::TestPortfolioVarMissingFiles       | test_missing_mortgage_json_zero_portfolio_mortgages | Fail   |
|       | risk.py::TestPortfolioVarMissingFiles       | test_missing_property_json_returns_200              | Pass   |
|       | risk.py::TestPortfolioVarMissingFiles       | test_missing_property_json_status_success           | Pass   |
|       | risk.py::TestPortfolioVarMissingFiles       | test_missing_property_json_zero_portfolio_value     | Fail   |
|       | risk.py::TestPortfolioVarMortgageImpairment | test_mortgage_impairment_max_is_positive            | Pass   |
|       | risk.py::TestPortfolioVarMortgageImpairment | test_mortgage_impairment_value                      | Pass   |
|       | risk.py::TestPortfolioVarMortgageImpairment | test_single_storm_has_mortgage_impairment           | Pass   |
|       | risk.py::TestPortfolioVarMultipleStorms     | test_prob_loss_pct_is_50                            | Pass   |
|       | risk.py::TestPortfolioVarMultipleStorms     | test_storm_count_is_two                             | Pass   |
|       | risk.py::TestPortfolioVarMultipleStorms     | test_storms_with_damage_is_one                      | Pass   |
|       | risk.py::TestPortfolioVarMultipleStorms     | test_storms_zero_damage_is_one                      | Pass   |
|       | risk.py::TestPortfolioVarMultipleStorms     | test_mail_storms_contains_the_damaged_storm         | Fail   |
|       | risk.py::TestPortfolioVarMultipleStorms     | test_mail_storms_limited_to_50                      | Pass   |
|       | risk.py::TestPortfolioVarNumerics           | test_no_mortgage_impairment_when_above_water        | Fail   |
|       | risk.py::TestPortfolioVarNumerics           | test_prob_loss_pct_is_100                           | Pass   |
|       | risk.py::TestPortfolioVarNumerics           | test_property_damage_cond_mean_positive             | Pass   |
|       | risk.py::TestPortfolioVarNumerics           | test_property_damage_max_is_positive                | Pass   |
|       | risk.py::TestPortfolioVarNumerics           | test_storm_count_is_one                             | Pass   |
|       | risk.py::TestPortfolioVarNumerics           | test_storms_with_damage_is_one                      | Pass   |
|       | risk.py::TestPortfolioVarNumerics           | test_storms_zero_plus_with_damage_equals_total      | Fail   |
|       | risk.py::TestPortfolioVarNumerics           | test_total_portfolio_mortgages                      | Pass   |
|       | risk.py::TestPortfolioVarNumerics           | test_total_portfolio_value                          | Pass   |
|       | risk.py::TestPortfolioVarSuccess            | test_histogram_bin_has_lo_hi_count                  | Pass   |
|       | risk.py::TestPortfolioVarSuccess            | test_mort_histogram_has_50_bins                     | Pass   |

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| Model | Module                                          | Test Name                                             | Status |
|-------|-------------------------------------------------|-------------------------------------------------------|--------|
|       | routes.py::TestGaugeRoutes                      | test_get_gauge_not_found                              | Pass   |
|       | routes.py::TestGaugeRoutes                      | test_get_gauge_statistics                             | Pass   |
|       | routes.py::TestGaugeRoutes                      | test_get_gaugets                                      | Pass   |
|       | routes.py::TestGaugeRoutes                      | test_legacy_list_gauges                               | Pass   |
|       | routes.py::TestGaugeRoutes                      | test_list_gauges                                      | Pass   |
|       | routes.py::TestHealthRoutes                     | test_detailed_health_check                            | Pass   |
|       | routes.py::TestHealthRoutes                     | test_health_check                                     | Pass   |
|       | routes.py::TestHealthRoutes                     | test_root_redirects                                   | Pass   |
|       | routes.py::TestPropertyRoutes                   | test_generate_report_missing_body                     | Pass   |
|       | routes.py::TestPropertyRoutes                   | test_generate_report_missing_property_id              | Pass   |
|       | routes.py::TestPropertyRoutes                   | test_generate_report_property_not_found               | Pass   |
|       | routes.py::TestPropertyRoutes                   | test_get_property                                     | Pass   |
|       | routes.py::TestPropertyRoutes                   | test_get_property_not_found                           | Pass   |
|       | routes.py::TestPropertyRoutes                   | test_legacy_generate_property_report                  | Pass   |
|       | routes.py::TestPropertyRoutes                   | test_list_properties                                  | Pass   |
|       | routes.py::TestPropertyRoutes                   | test_options_preflight                                | Pass   |
|       | routes.py::TestServerMain                       | test_main_calls_app_run                               | Pass   |
|       | consistency.py::TestStormIdConsistency          | test_new_storm_from_list_is_runnable                  | Pass   |
|       | consistency.py::TestStormIdConsistency          | test_worst_storm_auto_selection_is_runnable           | Pass   |
|       | consistency.py::TestStressRunInputs             | test_storm_id_exact_match_required                    | Pass   |
|       | consistency.py::TestStressRunInputs             | test_storm_list_has_all_run_inputs                    | Pass   |
|       | hydrograph.py::TestActualHydrograph             | test_peak_function decays after peak                  | Pass   |
|       | hydrograph.py::TestActualHydrograph             | test_peak_function reaches peak                       | Pass   |
|       | hydrograph.py::TestActualHydrograph             | test_peak_function returns 168 hours                  | Pass   |
|       | hydrograph.py::TestActualHydrograph             | test_peak_function starts at base                     | Pass   |
|       | hydrograph.py::TestActualHydrograph             | test_peak_anchor matches peak position                | Pass   |
|       | hydrograph.py::TestGaugetsHydrograph            | test_phillips_synthesis when no gaugets               | Pass   |
|       | hydrograph.py::TestGaugetsHydrograph            | test_phillips_level field fallback                    | Pass   |
|       | hydrograph.py::TestGaugetsHydrograph            | test_phillips_winter level field                      | Pass   |
|       | ordering.py::TestPeakOrdering                   | test_severe_storm_alpha_is_first                      | Pass   |
|       | ordering.py::TestPeakOrdering                   | test_storms_sorted_worst_first                        | Pass   |
|       | ordering.py::TestPeakOrdering                   | test_worst_storm_is_first                             | Pass   |
|       | presentation.py::TestPresentationConsistency    | test_consistency use same storms endpoint             | Pass   |
|       | presentation.py::TestPresentationConsistency    | test_stress_gauge stress auto selects worst case      | Pass   |
|       | presentation.py::TestPresentationConsistency    | test_stress_gauge stress tab uses all required fields | Pass   |
|       | presentation.py::TestPresentationConsistency    | test_historical tab uses all required fields          | Pass   |
|       | presentation.py::TestPresentationConsistency    | test_peak_level_min in trading stress label           | Pass   |
|       | presentation.py::TestPresentationConsistency    | test_trading stress auto selects worst case           | Pass   |
|       | presentation.py::TestPresentationConsistency    | test_trading stress tab uses all required fields      | Pass   |
|       | response_fields.py::TestStressRunResponseFields | test_alert_warning_severe levels                      | Pass   |
|       | response_fields.py::TestStressRunResponseFields | test_hydrograph_source_info                           | Pass   |
|       | response_fields.py::TestStressRunResponseFields | test_storm_name_and_category                          | Pass   |
|       | response_fields.py::TestStressRunResponseFields | test_count_is_168                                     | Pass   |
|       | response_fields.py::TestStressRunResponseFields | test_probability_surface_has_full_168h                | Pass   |

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| Model | Module                                         | Test Name                                     | Status |
|-------|------------------------------------------------|-----------------------------------------------|--------|
|       | response_fields.py::TestStressStor             | testNoGaugeId.response_has_count              | Pass   |
|       | response_fields.py::TestStressStor             | testNoGaugeId_returns_200_not_400             | Pass   |
|       | response_fields.py::TestStressStor             | testNoGaugeId_returns_all_storms              | Pass   |
|       | response_fields.py::TestStressStor             | testNoGaugeId_returns_success                 | Pass   |
|       | response_fields.py::TestStressStor             | testNoGaugeId_still_filters                   | Pass   |
|       | scenario_fields.py::TestScenarioVisites        | testFieldslist_has_base_level_m               | Pass   |
|       | scenario_fields.py::TestScenarioVisites        | testFieldslist_has_duration_hours             | Pass   |
|       | scenario_fields.py::TestScenarioVisites        | testFieldslist_has_effective_precipitation_mm | Pass   |
|       | scenario_fields.py::TestScenarioVisites        | testFieldslist_has_gauges_severe              | Pass   |
|       | scenario_fields.py::TestScenarioVisites        | testFieldslist_has_intensity_category         | Pass   |
|       | scenario_fields.py::TestScenarioVisites        | testFieldslist_has_level_change_m             | Pass   |
|       | scenario_fields.py::TestScenarioVisites        | testFieldslist_has_max_trigger                | Pass   |
|       | scenario_fields.py::TestScenarioVisites        | testFieldslist_has_name                       | Pass   |
|       | scenario_fields.py::TestScenarioVisites        | testFieldslist_has_peak_level_m               | Pass   |
|       | scenario_fields.py::TestScenarioVisites        | testFieldslist_has_peak_position              | Pass   |
|       | test_health_coverage.py::TestState             | testapple.touch.icon                          | Pass   |
|       | test_health_coverage.py::TestState             | testapple.touch.icon.precomposed              | Pass   |
|       | test_health_coverage.py::TestState             | testfont.avicon.ico                           | Pass   |
|       | test_prs_basic.py::TestCommitTrade             | test_commit_with_cashflows                    | Pass   |
|       | test_prs_basic.py::TestCommitTrade             | test_commit_with_explicit_maturity            | Pass   |
|       | test_prs_basic.py::TestCommitTrade             | test_empty_json_returns_400                   | Pass   |
|       | test_prs_basic.py::TestCommitTrade             | test_minimal_valid_commit                     | Pass   |
|       | test_prs_basic.py::TestCommitTrade             | test_no_json_returns_error                    | Pass   |
|       | test_prs_basic.py::TestGetTradePDF             | test_existing_pdf_served                      | Pass   |
|       | test_prs_basic.py::TestGetTradePDF             | test_missing_pdf_returns_404                  | Pass   |
|       | test_prs_basic.py::TestListTrades              | test_empty_dir_returns_zero_count             | Pass   |
|       | test_prs_basic.py::TestListTrades              | test_returns_trades_list_key                  | Pass   |
|       | test_prs_basic.py::TestListTrades              | testWithFiles_returns_trades_from_json_files  | Pass   |
|       | test_prs_generate.py::TestCommitTrade          | testCloseOut_path_exercised                   | Pass   |
|       | test_prs_generate.py::TestCommitTrade          | testValidationErrors_fails_returns_400        | Pass   |
|       | test_prs_generate.py::TestGenerateTradePDF     | testTradePDF_payable_direction                | Pass   |
|       | test_prs_generate.py::TestGenerateTradePDF     | testTradePDF_no_cashflows                     | Pass   |
|       | test_prs_generate.py::TestGenerateTradePDF     | testTradePDF_with_cashflows                   | Pass   |
|       | test_prs_generate.py::TestGenerateTradePDF     | testTradePDF_with_close_out                   | Pass   |
|       | test_prs_generate.py::TestGenerateTradePDF     | testTradePDF_with_close_out_date              | Pass   |
|       | test_prs_generate.py::TestListTrades           | testEmptyTrade_file_returns_500               | Pass   |
|       | test_classifiers_routes.py::TestBatchTrainings | test_eta_fallback_from_elapsed                | Pass   |
|       | test_classifiers_routes.py::TestBatchTrainings | test_eta_none_when_zero_completed             | Pass   |
|       | test_classifiers_routes.py::TestBatchTrainings | test_all_already_trained                      | Pass   |
|       | test_classifiers_routes.py::TestBatchTrainings | test_all_already_running                      | Pass   |
|       | test_classifiers_routes.py::TestBatchTrainings | test_all_complete_includes_results            | Pass   |
|       | test_classifiers_routes.py::TestBatchTrainings | test_all_starts                               | Pass   |
|       | test_classifiers_routes.py::TestBatchTrainings | test_all_status_fields                        | Pass   |
|       | test_classifiers_routes.py::TestBatchTrainings | test_all_status_idle                          | Pass   |

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| Model | Module                                                            | Test Name                                               | Status |
|-------|-------------------------------------------------------------------|---------------------------------------------------------|--------|
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_all_trained                     | Pass   |
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_error                           | Pass   |
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_no_models                       | Pass   |
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_partial                         | Pass   |
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_cached                          | Pass   |
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_fields                          | Pass   |
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_by_longitude                    | Pass   |
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_models_initially                | Pass   |
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_all_gauges                      | Pass   |
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_success                         | Pass   |
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_gauge_shows_metrics             | Pass   |
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_empty                           | Pass   |
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_error                           | Pass   |
|       | test_classifiers_routes.py::TestClassifierPredict                 | test_classifier_predict_moves_models                    | Pass   |
|       | test_classifiers_routes.py::TestRouteBatching                     | test_route_batching_cancelled_mid_run                   | Pass   |
|       | test_classifiers_routes.py::TestRouteBatching                     | test_route_batching_with_failure                        | Pass   |
|       | test_classifiers_routes.py::TestRouteBatching                     | test_route_batching_successful_batch                    | Pass   |
|       | test_classifiers_routes.py::TestSummaryEdgeCases                  | test_summary_edge_case_all_when_no_trained              | Pass   |
|       | test_classifiers_routes.py::TestSummaryEdgeCases                  | test_summary_edge_case_fields_present                   | Pass   |
|       | test_classifiers_routes.py::TestSummaryEdgeCases                  | test_summary_edge_case_trained_avg_auc                  | Pass   |
|       | test_classifiers_routes.py::TestSummaryEdgeCases                  | test_summary_edge_case_error                            | Pass   |
|       | test_classifiers_routes.py::TestSummaryEdgeCases                  | test_summary_edge_case_without_auc                      | Pass   |
|       | test_classifiers_routes.py::TestTimeGauges                        | test_time_gauges_gauge_seconds                          | Pass   |
|       | test_classifiers_routes.py::TestTimeGauges                        | test_time_gauges_gauge_seconds_empty                    | Pass   |
|       | test_classifiers_routes.py::TestTimeGauges                        | test_time_gauges_timings_corrupt_json                   | Pass   |
|       | test_classifiers_routes.py::TestTimeGauges                        | test_time_gauges_timings_missing_file                   | Pass   |
|       | test_classifiers_routes.py::TestTimeGauges                        | test_time_gauges_timings_valid                          | Pass   |
|       | test_classifiers_routes.py::TestTimeGauges                        | test_time_gauges_timings_truncates_to_20                | Pass   |
|       | test_port_stress_coverage.py::TestAssembleThresholdClassification | test_assemble_threshold_classification                  | Pass   |
|       | test_port_stress_coverage.py::TestExcludeBadGauges                | test_exclude_bad_gauges_excluded_from_stress            | Pass   |
|       | test_port_stress_coverage.py::TestEmptyFormIdReturns400           | test_empty_form_id_returns_400                          | Pass   |
|       | test_port_stress_coverage.py::TestHydrographFallback              | test_hydrograph_fallback_falls_back                     | Pass   |
|       | test_port_stress_coverage.py::TestHydrographFallback              | test_hydrograph_fallback_enough_readings                | Pass   |
|       | test_port_stress_coverage.py::TestPortExceptionRetries500         | test_port_exception_retries_500                         | Pass   |
|       | test_port_stress_coverage.py::TestPortExceptionRetries500         | test_port_exception_retries_exception_defaults_to_zero  | Pass   |
|       | test_port_stress_coverage.py::TestStormNameFallback               | test_storm_name_fallback_name_uses_storm_id             | Pass   |
|       | test_port_stress_coverage.py::TestStormNameFallback               | test_storm_name_fallback_with_fallback_loads_prediction | Pass   |
|       | test_port_stress_infra.py::TestPortStressErrorReturns500          | test_port_stress_error_returns_500                      | Pass   |
|       | test_port_stress_infra.py::TestPortStressErrorReturns404          | test_port_stress_error_returns_file_returns_404         | Pass   |
|       | test_port_stress_infra.py::TestPortStressErrorReturnsError        | test_port_stress_error_returns_error                    | Pass   |
|       | test_port_stress_infra.py::TestPortStressPredict                  | test_port_stress_predict_called_per_gauge               | Pass   |
|       | test_port_stress_infra.py::TestPortStressPredict                  | test_port_stress_predict_cache_hit                      | Pass   |
|       | test_port_stress_infra.py::TestPortStressPredict                  | test_port_stress_predict_falls_back                     | Pass   |
|       | test_port_stress_infra.py::TestStormCacheAndReload                | test_storm_cache_and_reload_when_mtime_unchanged        | Pass   |

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| Model                                          | Module                                         | Test Name                                                                                              | Status |
|------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------|
| test_port_stress.infra.py::TestStressor        | test_stress_infra.py::TestStressor             | test_stress_infra.py::TestStressor::TestStressorHasAdditionalTimeChange                                | Pass   |
| test_port_stress.run.py::TestPortStressPpl     | test_port_stress_run.py::TestPortStressPpl     | test_port_stress_run.py::TestPortStressPpl::TestPortStressPplCultivationNotional                       | Pass   |
| test_port_stress.run.py::TestPortStressPpl     | test_port_stress_run.py::TestPortStressPpl     | test_port_stress_run.py::TestPortStressPpl::TestPortStressPplCalculationsSumOfGauges                   | Pass   |
| test_port_stress.run.py::TestPortStressPpl     | test_port_stress_run.py::TestPortStressPpl     | test_port_stress_run.py::TestPortStressPpl::TestPortStressPplCalegativeNotional                        | Pass   |
| test_port_stress.run.py::TestPortStressPpl     | test_port_stress_run.py::TestPortStressPpl     | test_port_stress_run.py::TestPortStressPpl::TestPortStressPplGreachesP_flood_to_100                    | Pass   |
| test_port_stress.run.py::TestPortStressPpl     | test_port_stress_run.py::TestPortStressPpl     | test_port_stress_run.py::TestPortStressPpl::TestPortStressPplCalculationA                              | Pass   |
| test_port_stress.run.py::TestPortStressPpl     | test_port_stress_run.py::TestPortStressPpl     | test_port_stress_run.py::TestPortStressPpl::TestPortStressPplCedGaugeHasZeroP_flood                    | Pass   |
| test_port_stress.run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl::TestRunStressPplGaugeResultsFields                          | Pass   |
| test_port_stress.run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl::TestRunStressPplRunGaugesSevereList                         | Pass   |
| test_port_stress.run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl::TestRunStressPplRunGaugesWarningList                        | Pass   |
| test_port_stress.run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl::TestRunStressPplRunHasGaugeResults                          | Pass   |
| test_port_stress.run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl::TestRunStressPplRunHasPortfolioTotals                       | Pass   |
| test_port_stress.run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl::TestRunStressPplPortfolioInvalidStorm                       | Pass   |
| test_port_stress.run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl::TestRunStressPplPortfolioMissingStormId                     | Pass   |
| test_port_stress.run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl      | test_port_stress_run.py::TestRunStressPpl::TestRunStressPplPortfolioReturnsSuccess                     | Pass   |
| test_port_stress.storms.py::TestGetPortfolio   | test_port_stress_storms.py::TestGetPortfolio   | test_port_stress_storms.py::TestGetPortfolio::TestGetPortfolioStormsHasStorms                          | Pass   |
| test_port_stress.storms.py::TestGetPortfolio   | test_port_stress_storms.py::TestGetPortfolio   | test_port_stress_storms.py::TestGetPortfolio::TestGetPortfolioStormsSuccess                            | Pass   |
| test_port_stress.storms.py::TestGetPortfolio   | test_port_stress_storms.py::TestGetPortfolio   | test_port_stress_storms.py::TestGetPortfolio::TestGetPortfolioStormsHasRequiredFields                  | Pass   |
| test_port_stress.storms.py::TestGetPortfolio   | test_port_stress_storms.py::TestGetPortfolio   | test_port_stress_storms.py::TestGetPortfolio::TestGetPortfolioStormsNoData                             | Pass   |
| test_port_stress.storms.py::TestGetPortfolio   | test_port_stress_storms.py::TestGetPortfolio   | test_port_stress_storms.py::TestGetPortfolio::TestGetPortfolioStormsSecondarySort                      | Pass   |
| test_port_stress.storms.py::TestGetPortfolio   | test_port_stress_storms.py::TestGetPortfolio   | test_port_stress_storms.py::TestGetPortfolio::TestGetPortfolioStormsSortedBySeverity                   | Pass   |
| test_port_stress.storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName::TestPortfolioNameEqualsStormId                          | Pass   |
| test_port_stress.storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName::TestPortfolioNameMainDisplayLabelNotId                  | Pass   |
| test_port_stress.storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName::TestPortfolioNameSevereBeforeWarning                    | Pass   |
| test_port_stress.storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName::TestPortfolioNameWarningBeforeAlert                     | Pass   |
| test_port_stress.storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName::TestPortfolioNameTradingGradesHaveTradeDetails          | Pass   |
| test_port_stress.storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName::TestPortfolioNameTradingFields                          | Pass   |
| test_port_stress.storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName  | test_port_stress_storms.py::TestPortfolioName::TestPortfolioNameImpactsForModerateWithEqualSevereCount | Pass   |
| test_stress.infra.py::TestGetPredictor         | test_stress_infra.py::TestGetPredictor         | test_stress_infra.py::TestGetPredictor::TestGetPredictorFallsBackToLegacyDir                           | Pass   |
| test_stress.infra.py::TestGetPredictor         | test_stress_infra.py::TestGetPredictor         | test_stress_infra.py::TestGetPredictor::TestGetPredictorLoadsFromStressmDir                            | Pass   |
| test_stress.infra.py::TestGetPredictor         | test_stress_infra.py::TestGetPredictor         | test_stress_infra.py::TestGetPredictor::TestGetPredictorWarnsWhenNoModels                              | Pass   |
| test_stress.infra.py::TestHydrograph           | test_stress_infra.py::TestHydrograph           | test_stress_infra.py::TestHydrograph::TestHydrographRemainsExactlyZero                                 | Pass   |
| test_stress.infra.py::TestInvalidatedPredictor | test_stress_infra.py::TestInvalidatedPredictor | test_stress_infra.py::TestInvalidatedPredictor::TestInvalidatedPredictorClearsCache                    | Pass   |
| test_stress.infra.py::TestLegacyFallback       | test_stress_infra.py::TestLegacyFallback       | test_stress_infra.py::TestLegacyFallback::TestLegacyFallbackCacheReuse                                 | Pass   |
| test_stress.infra.py::TestLegacyFallback       | test_stress_infra.py::TestLegacyFallback       | test_stress_infra.py::TestLegacyFallback::TestLegacyFallbackFileLoadedWhenNoIndex                      | Pass   |
| test_stress.infra.py::TestLegacyFallback       | test_stress_infra.py::TestLegacyFallback       | test_stress_infra.py::TestLegacyFallback::TestLegacyFallbackStatOSErrorReturnsCache                    | Pass   |
| test_stress.infra.py::TestLoadStorm            | test_stress_infra.py::TestLoadStorm            | test_stress_infra.py::TestLoadStorm::TestLoadStormLoadFromLegacyFile                                   | Pass   |
| test_stress.infra.py::TestLoadStorm            | test_stress_infra.py::TestLoadStorm            | test_stress_infra.py::TestLoadStorm::TestLoadStormStatOSErrorReturnsCached                             | Pass   |
| test_stress.infra.py::TestLoadStorm            | test_stress_infra.py::TestLoadStorm            | test_stress_infra.py::TestLoadStorm::TestLoadStormStatOSErrorReturnsNoneWhenNoCache                    | Pass   |
| test_stress.infra.py::TestRunStress            | test_stress_infra.py::TestRunStress            | test_stress_infra.py::TestRunStress::TestRunStressEmptyGaugeIdReturns400                               | Pass   |
| test_stress.infra.py::TestRunStress            | test_stress_infra.py::TestRunStress            | test_stress_infra.py::TestRunStress::TestRunStressNoOpenTradesReturns404                               | Pass   |
| test_stress.infra.py::TestRunStress            | test_stress_infra.py::TestRunStress            | test_stress_infra.py::TestRunStress::TestRunStressNullBodyReturns400                                   | Pass   |
| test_stress.infra.py::TestRunStress            | test_stress_infra.py::TestRunStress            | test_stress_infra.py::TestRunStress::TestRunStressPredictorNotFoundReturns404                          | Pass   |
| test_stress.infra.py::TestRunStress            | test_stress_infra.py::TestRunStress            | test_stress_infra.py::TestRunStress::TestRunStressUsesGaugetsHydrograph                                | Pass   |

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| Model | Module                                                      | Test Name                                                                        | Status |
|-------|-------------------------------------------------------------|----------------------------------------------------------------------------------|--------|
|       | test_stress_infra.py::TestStressCache                       | test_stress_infra.py::TestStressCachePredictor_cache_hit                         | Pass   |
|       | test_stress_infra.py::TestStressCache                       | test_stress_infra.py::TestStressCachePredictor_returns_none_when_model_missing   | Pass   |
|       | test_stress_infra.py::TestStressErrorHandler                | test_stress_infra.py::TestStressErrorHandlerEngine_error_returns_500             | Pass   |
|       | test_stress_infra.py::TestStressErrorHandler                | test_stress_infra.py::TestStressErrorHandlerGauges_engine_error_returns_500      | Pass   |
|       | test_stress_infra.py::TestStressErrorHandler                | test_stress_infra.py::TestStressErrorHandlerStorms_engine_error_returns_500      | Pass   |
|       | test_stress_infra.py::TestStressHydrograph                  | test_stress_infra.py::TestStressHydrographPeak_at_zero                           | Pass   |
|       | test_stress_infra.py::TestStressHydrograph                  | test_stress_infra.py::TestStressHydrographRemaining_zero                         | Pass   |
|       | test_stress_run.py::TestRunStress                           | test_stress_run.py::TestRunStressHourly_fields                                   | Pass   |
|       | test_stress_run.py::TestRunStress                           | test_stress_run.py::TestRunStressInvalid_storm                                   | Pass   |
|       | test_stress_run.py::TestRunStress                           | test_stress_run.py::TestRunStressKnock_out                                       | Pass   |
|       | test_stress_run.py::TestRunStress                           | test_stress_run.py::TestRunStressMissing_params                                  | Pass   |
|       | test_stress_run.py::TestRunStress                           | test_stress_run.py::TestRunStressNo_body                                         | Pass   |
|       | test_stress_run.py::TestRunStress                           | test_stress_run.py::TestRunStressPortfolio_summary                               | Pass   |
|       | test_stress_run.py::TestRunStress                           | test_stress_run.py::TestRunStressProbability_surface                             | Pass   |
|       | test_stress_run.py::TestRunStress                           | test_stress_run.py::TestRunStressReturns_hourly                                  | Pass   |
|       | test_stress_run.py::TestRunStress                           | test_stress_run.py::TestRunStressTrade_summary                                   | Pass   |
|       | test_stress_run.py::TestRunStress                           | test_stress_run.py::TestRunStressSurface_caps_at_severe                          | Pass   |
|       | test_stress_run.py::TestRunStress                           | test_stress_run.py::TestRunStressSurface_shows_full_168h                         | Pass   |
|       | test_stress_scenario_coverage.py::TestHydrographAdd         | test_stress_scenario_coverage.py::TestHydrographAddGauge_falls_back              | Pass   |
|       | test_stress_scenario_coverage.py::TestHydrographAdd         | test_stress_scenario_coverage.py::TestHydrographAddGauge_added_to_storm_hours    | Pass   |
|       | test_stress_scenario_coverage.py::TestModelAUF              | test_stress_scenario_coverage.py::TestModelAUFNone_when_summary_no_match         | Pass   |
|       | test_stress_scenario_coverage.py::TestModelAUF              | test_stress_scenario_coverage.py::TestModelAUFNone_when_summary_raises           | Pass   |
|       | test_stress_scenario_coverage.py::TestModelAUF              | test_stress_scenario_coverage.py::TestModelAUFPresent_when_summary_has_gauge     | Pass   |
|       | test_stress_scenario_coverage.py::TestModelAUF              | test_stress_scenario_coverage.py::TestModelAUFExcludes_untrained_gauge           | Pass   |
|       | test_stress_scenario_coverage.py::TestNgaugesWithNoAtGauges | test_stress_scenario_coverage.py::TestNgaugesWithNoAtGauges_returns_404          | Pass   |
|       | test_stress_scenario_coverage.py::TestPredictorNotFound     | test_stress_scenario_coverage.py::TestPredictorNotFound_returns_404              | Pass   |
|       | test_stress_scenario_coverage.py::TestStressStormsNotFound  | test_stress_scenario_coverage.py::TestStressStormsNotFound_file                  | Pass   |
|       | test_stress_storms.py::TestGetStorms                        | test_stress_storms.py::TestGetStormsHas_peak_level                               | Pass   |
|       | test_stress_storms.py::TestGetStorms                        | test_stress_storms.py::TestGetStormsHas_storm_fields                             | Pass   |
|       | test_stress_storms.py::TestGetStorms                        | test_stress_storms.py::TestGetStormsNo_data                                      | Pass   |
|       | test_stress_storms.py::TestGetStorms                        | test_stress_storms.py::TestGetStormsNo_gauge_id_returns_all_storms               | Pass   |
|       | test_stress_storms.py::TestGetStorms                        | test_stress_storms.py::TestGetStormsNo_matching_gauge                            | Pass   |
|       | test_stress_storms.py::TestGetStorms                        | test_stress_storms.py::TestGetStormsReturns_list                                 | Pass   |
|       | test_stress_storms.py::TestGetStorms                        | test_stress_storms.py::TestGetStormsSorted_by_peak                               | Pass   |
|       | test_stress_storms.py::TestGetStorms                        | test_stress_storms.py::TestGetStormsGauges_has_required_fields                   | Pass   |
|       | test_stress_storms.py::TestGetStorms                        | test_stress_storms.py::TestGetStormsGauges_has_trade_counts                      | Pass   |
|       | test_stress_storms.py::TestGetStorms                        | test_stress_storms.py::TestGetStormsGauges_returns_list                          | Pass   |
|       | test_stress_storms.py::TestGetStorms                        | test_stress_storms.py::TestGetStormsGauges_sorted_by_longitude                   | Pass   |
|       | test_training_status.py::TestClassifyStatus                 | test_training_status.py::TestClassifyStatusFailed_from_job_dict                  | Pass   |
|       | test_training_status.py::TestClassifyStatus                 | test_training_status.py::TestClassifyStatusFailed_no_error_key                   | Pass   |
|       | test_training_status.py::TestClassifyStatus                 | test_training_status.py::TestClassifyStatusJoblib_takes_precedence_over_job_dict | Pass   |
|       | test_training_status.py::TestClassifyStatus                 | test_training_status.py::TestClassifyStatusNot_trained_when_no_joblib_exists     | Pass   |
|       | test_training_status.py::TestClassifyStatus                 | test_training_status.py::TestClassifyStatusReady_from_job_dict                   | Pass   |
|       | test_training_status.py::TestClassifyStatus                 | test_training_status.py::TestClassifyStatusReady_when_joblib_exists              | Pass   |

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| Model                          | Module                                          | Test Name                                                           | Status                         |
|--------------------------------|-------------------------------------------------|---------------------------------------------------------------------|--------------------------------|
| test_training                  | test_training_status.py::TestClassifierStatus   | test_training_in_progress                                           | Pass                           |
|                                | test_training_status.py::TestClassifierStatus   | test_training_unknown_gauge_not_trained                             | Pass                           |
|                                | test_training_status.py::TestEdgeCase           | test_gauge_json_dict_format                                         | Pass                           |
|                                | test_training_status.py::TestEdgeCase           | test_train_does_not_start_if_joblib_created_between_check_and_train | Tested between check and train |
|                                | test_training_status.py::TestEdgeCase           | test_training_jobs_isolated_between_gauges                          | Pass                           |
|                                | test_training_summary.py::TestUpdater           | test_all_tasks_skipped_avg_auc_is_zero                              | Pass                           |
|                                | test_training_summary.py::TestUpdater           | test_checking_new_summary                                           | Pass                           |
|                                | test_training_summary.py::TestUpdater           | test_checking_for_existing_summary                                  | Pass                           |
|                                | test_training_summary.py::TestUpdater           | test_replace_existing_gauge_entry                                   | Pass                           |
|                                | test_training_summary.py::TestUpdater           | test_skipped_gauge_not_counted_in_auc                               | Pass                           |
|                                | test_training_train.py::TestTrainClassifier     | test_train_already_in_progress                                      | Pass                           |
|                                | test_training_train.py::TestTrainClassifier     | test_train_already_trained_returns_ready                            | Pass                           |
|                                | test_training_train.py::TestTrainClassifier     | test_train_corrupt_gauge_json_500                                   | Pass                           |
|                                | test_training_train.py::TestTrainClassifier     | test_train_gauge_not_found_404                                      | Pass                           |
|                                | test_training_train.py::TestTrainClassifier     | test_train_no_gauge_json_404                                        | Pass                           |
|                                | test_training_train.py::TestTrainClassifier     | test_train_no_storm_sequences_404                                   | Pass                           |
|                                | test_training_train.py::TestTrainClassifier     | test_train_previously_failed_can_restart                            | Pass                           |
|                                | test_training_train.py::TestTrainClassifier     | test_train_starts_background_thread                                 | Pass                           |
|                                | test_training_train.py::TestTrainClassifier     | test_train_gauge_failure_updates_job_to_failed                      | Pass                           |
|                                | test_training_train.py::TestTrainClassifier     | test_train_gauge_success_updates_job_to_ready                       | Pass                           |
|                                | visualization.py::TestServeVisualization        | test_get_returns_json_error                                         | Pass                           |
|                                | visualization.py::TestServeVisualization        | test_no_gauge_data_returns_error                                    | Pass                           |
|                                | visualization.py::TestServeVisualization        | test_no_gauge_data_returns_error_status                             | Pass                           |
|                                | visualization.py::TestServeVisualization        | test_no_kubernetes_exception_returns_500                            | Pass                           |
|                                | visualization.py::TestServeVisualization        | test_no_hmc_data_is_insensitive                                     | Pass                           |
|                                | visualization.py::TestServeVisualization        | test_no_hmc_data_redirects                                          | Pass                           |
|                                | visualization.py::TestServeVisualization        | test_no_kubernetes_attachment_returns_400                           | Pass                           |
|                                | visualization.py::TestServeVisualization        | test_no_supported_attachment_message                                | Pass                           |
|                                | visualization.py::TestServeVisualization        | test_no_support_error_returns_503                                   | Pass                           |
|                                | visualization.py::TestServeVisualization        | test_no_success_returns_200                                         | Pass                           |
|                                | base_loader.py::TestBaseLoader                  | test_caching_prevents_reload                                        | Pass                           |
|                                | base_loader.py::TestBaseLoader                  | test_clear_cache                                                    | Pass                           |
|                                | base_loader.py::TestBaseLoader                  | test_force_reload_bypasses_cache                                    | Pass                           |
| base_loader.py::TestBaseLoader | test_invalidate_cache                           | Pass                                                                |                                |
| base_loader.py::TestBaseLoader | test_abstract_methods_enforced                  | Pass                                                                |                                |
| base_loader.py::TestBaseLoader | test_subclass_must_implement_get_entity         | Pass                                                                |                                |
| base_loader.py::TestBaseLoader | test_subclass_must_implement_get_entity_summary | Summary                                                             |                                |
| base_loader.py::TestBaseLoader | test_loading_exists_check                       | Skip                                                                |                                |
| base_loader.py::TestBaseLoader | test_loading_alternative_container_key          | Pass                                                                |                                |
| base_loader.py::TestBaseLoader | test_loading_dict_with_container_key            | Pass                                                                |                                |
| base_loader.py::TestBaseLoader | test_loading_list_format                        | Pass                                                                |                                |
| base_loader.py::TestBaseLoader | test_loading_missing_file                       | Pass                                                                |                                |
| base_loader.py::TestBaseLoader | test_loading_single_object                      | Pass                                                                |                                |
| base_loader.py::TestBaseLoader | test_queries_count                              | Pass                                                                |                                |

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| Model | Module                                 | Test Name                                              | Status |
|-------|----------------------------------------|--------------------------------------------------------|--------|
|       | base_loader.py::TestBaseLoader         | TestBaseLoaderExists                                   | Pass   |
|       | base_loader.py::TestBaseLoader         | TestBaseLoaderFindById                                 | Pass   |
|       | base_loader.py::TestBaseLoader         | TestBaseLoaderFindByIdNotFound                         | Pass   |
|       | base_loader.py::TestBaseLoader         | TestBaseLoaderGetIds                                   | Skip   |
|       | base_loader.py::TestBaseLoader         | TestBaseLoaderGetStatus                                | Skip   |
|       | base_loader.py::TestBaseLoader         | TestBaseLoaderListAll                                  | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderEntityIdExtraction                      | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderEntitySummary                           | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderLoadGauges                              | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetActiveGauges                         | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetCoordinates                          | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetFloodStages                          | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetGaugeName                            | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetGaugeNameNotFound                    | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetGaugesById                           | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetGaugesByName                         | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetGaugesByNameCaseInsensitive          | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetGaugesByOwner                        | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetGaugesByStatusActive                 | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetGaugesByStatusInactive               | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetGaugesByTypeRiver                    | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetGaugesByTypeTidal                    | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetSpatialGaugesInRadius                | Pass   |
|       | gauge_loader.py::TestGaugeLoader       | TestGaugeLoaderGetSpatialGaugesInRadiusNoneFound       | Pass   |
|       | loader_registry.py::TestCreateRegistry | TestCreateRegistryCreateRegistry                       | Pass   |
|       | loader_registry.py::TestCreateRegistry | TestCreateRegistryCreateRegistryWithOverrides          | Pass   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryCreateRegistry                       | Pass   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryGetAvailableLoaders                  | Pass   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryClearAllCaches                       | Pass   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryGetDifferentLoadersSeparateInstances | Pass   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryGetCachedInstancesCached             | Pass   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryFileOverrides                        | Pass   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryGetLoaderByName                      | Pass   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryGetLoaderUnknownType                 | Pass   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryGetAllDataFiles                      | Pass   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryGetAllStatus                         | Skip   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryGetTypeGaugeLoader                   | Pass   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryGetTypeAverageLoader                 | Pass   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryGetTypeProportionsLoader             | Pass   |
|       | loader_registry.py::TestLoaderRegistry | TestLoaderRegistryGetTypeAssessmentsLoader             | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | TestMortgageLoaderGetMortgages                         | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | TestMortgageLoaderGetMortgagesByLender                 | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | TestMortgageLoaderGetMortgagesWithMortgages            | Pass   |

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| Model | Module                                     | Test Name                                     | Status |
|-------|--------------------------------------------|-----------------------------------------------|--------|
|       | mortgage_loader.py::TestMortgageLoader     | testBagetAnnualExposure                       | Pass   |
|       | mortgage_loader.py::TestMortgageLoader     | testBagetBasicExtraction                      | Pass   |
|       | mortgage_loader.py::TestMortgageLoader     | testBagetBasicSummary                         | Pass   |
|       | mortgage_loader.py::TestMortgageLoader     | testBagetBasicMortgages                       | Pass   |
|       | mortgage_loader.py::TestMortgageLoader     | testBagetById                                 | Pass   |
|       | mortgage_loader.py::TestMortgageLoader     | testBagetByIdLender                           | Pass   |
|       | mortgage_loader.py::TestMortgageLoader     | testBagetByIdLender_case_insensitive          | Pass   |
|       | mortgage_loader.py::TestMortgageLoader     | testBagetByIdPropertyId                       | Pass   |
|       | mortgage_loader.py::TestMortgageLoader     | testBagetByIdPropertyId_not_found             | Pass   |
|       | mortgage_loader.py::TestMortgageLoader     | testBagetByIdStatusActive                     | Pass   |
|       | mortgage_loader.py::TestMortgageLoader     | testBagetByIdStatusDelinquent                 | Pass   |
|       | mortgage_loader.py::TestMortgageLoader     | testPortageLinkToProperty                     | Pass   |
|       | mortgage_loader.py::TestMortgageLoader     | testPropertyWithMortgage                      | Pass   |
|       | property_loader.py::TestPropertyLoader     | testBagetAnnualValue                          | Pass   |
|       | property_loader.py::TestPropertyLoader     | testBagetAnnualByState                        | Pass   |
|       | property_loader.py::TestPropertyLoader     | testBagetBasicExtraction                      | Pass   |
|       | property_loader.py::TestPropertyLoader     | testBagetBasicSummary                         | Pass   |
|       | property_loader.py::TestPropertyLoader     | testBagetBasicProperties                      | Pass   |
|       | property_loader.py::TestPropertyLoader     | testBagetCoordinates                          | Pass   |
|       | property_loader.py::TestPropertyLoader     | testBagetCoordinates_not_found                | Pass   |
|       | property_loader.py::TestPropertyLoader     | testBagetLocation                             | Pass   |
|       | property_loader.py::TestPropertyLoader     | testBagetLocation_not_found                   | Pass   |
|       | property_loader.py::TestPropertyLoader     | testBagetPropertiesInRadius                   | Pass   |
|       | property_loader.py::TestPropertyLoader     | testByIdByAddress                             | Pass   |
|       | property_loader.py::TestPropertyLoader     | testByIdByAddress_case_insensitive            | Pass   |
|       | property_loader.py::TestPropertyLoader     | testByIdByAddress_not_found                   | Pass   |
|       | property_loader.py::TestPropertyLoader     | testByIdById                                  | Pass   |
|       | property_loader.py::TestPropertyLoader     | testByIdByState                               | Pass   |
|       | property_loader.py::TestPropertyLoader     | testByIdByState_case_insensitive              | Pass   |
|       | property_loader.py::TestPropertyLoader     | testByIdByZip                                 | Pass   |
|       | property_loader.py::TestPropertyLoader     | testByIdByZip_multiple                        | Pass   |
|       | timeseries_loader.py::TestTimeseriesLoader | testChangeBetweenSeriesForPortfolioGauge      | Pass   |
|       | timeseries_loader.py::TestTimeseriesLoader | testChangeBetweenSeriesGaugeIdsMatchPortfolio | Pass   |
|       | timeseries_loader.py::TestTimeseriesLoader | testLoadBySeriesTimestamp                     | Pass   |
|       | timeseries_loader.py::TestTimeseriesLoader | testLoadBySeriesSummary                       | Pass   |
|       | timeseries_loader.py::TestTimeseriesLoader | testLoadBySeries                              | Pass   |
|       | timeseries_loader.py::TestTimeseriesLoader | testLoggerGaugeQueueing                       | Pass   |
|       | timeseries_loader.py::TestTimeseriesLoader | testLoggerPeakQueueing                        | Pass   |
|       | timeseries_loader.py::TestTimeseriesLoader | testLoggerGaugesForGauge                      | Pass   |
|       | timeseries_loader.py::TestTimeseriesLoader | testLoggerGaugesForGaugeByName                | Pass   |
|       | timeseries_loader.py::TestTimeseriesLoader | testLoggerGaugesForGaugeNotFound              | Pass   |
|       | timeseries_loader.py::TestTimeseriesLoader | testLoggerStalableGaugeIds                    | Pass   |
|       | timeseries_loader.py::TestTimeseriesLoader | testLoggerStatistics                          | Pass   |
|       | timeseries_loader.py::TestTimeseriesLoader | testLoggerStatisticsNotFound                  | Pass   |

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| Model | Module                                                | Test Name                                        | Status |
|-------|-------------------------------------------------------|--------------------------------------------------|--------|
|       | timeseries_loader.py::TestTimeseriesLoaderEndOfSeries | test_loader_end_of_series_range                  | Skip   |
|       | timeseries_loader.py::TestTimeseriesLoaderEndOfSeries | test_loader_end_of_series_range_for_gauge        | Skip   |
|       | test_b10_quick_wins.py::TestAuditFunctions            | test_functions_truncated_above_10000             | Pass   |
|       | test_b10_quick_wins.py::TestEOBTestDreadZooNormal     | test_dread_zoo_normal_new_trade                  | Pass   |
|       | test_b10_quick_wins.py::TestGaugeTestDreadZooNormal   | test_dread_zoo_normal_exists                     | Pass   |
|       | test_b10_quick_wins.py::TestInsuranceArea             | test_insurance_area_fallthrough                  | Pass   |
|       | test_b10_quick_wins.py::TestIntensityIntensity        | test_intensity_intensity_module_runs             | Pass   |
|       | test_b10_quick_wins.py::TestIntensityGetPattern       | test_get_pattern_unknown_type_returns_pass       | Pass   |
|       | test_b10_quick_wins.py::TestJSONGetGauge              | test_get_gauge_on_unknown_key                    | Pass   |
|       | test_b10_quick_wins.py::TestJSONGetGauge              | test_get_gauge_known_key                         | Pass   |
|       | test_b10_quick_wins.py::TestLtvPercentage             | test_ltv_percentage_percentage                   | Pass   |
|       | test_b10_quick_wins.py::TestPropertyManipulation      | test_manipulation_typeerror                      | Pass   |
|       | test_b10_quick_wins.py::TestPropertyManipulation      | test_manipulation_encoded_as_float               | Pass   |
|       | test_b10_quick_wins.py::TestServiceBlock              | test_block_main_callable                         | Pass   |
|       | test_b10_quick_wins.py::TestStorageBlock              | test_block_linear_fallback                       | Pass   |
|       | test_b10_quick_wins.py::TestTradingStatistics         | test_trading_statistics_returns_dimensions       | Pass   |
|       | test_b10_quick_wins.py::TestVisualizationFactory      | test_create_visualization_returns_instance       | Pass   |
|       | data_structures.py::TestDataValidation                | test_validation_result_fields                    | Pass   |
|       | data_structures.py::TestDataValidation                | test_validation_result_tuple                     | Pass   |
|       | data_structures.py::TestDataValidation                | test_validation_result_unpacking                 | Pass   |
|       | data_structures.py::TestDataValidation                | test_validation_result_fields                    | Pass   |
|       | data_structures.py::TestLoadedData                    | test_can_set_gauge_data                          | Pass   |
|       | data_structures.py::TestLoadedData                    | test_can_set_multiple_fields                     | Pass   |
|       | data_structures.py::TestLoadedData                    | test_counts_default_to_zero                      | Pass   |
|       | data_structures.py::TestLoadedData                    | test_defaults_are_none                           | Pass   |
|       | data_structures.py::TestLoadedData                    | test_is_dataclass                                | Pass   |
|       | data_structures.py::TestLoadedData                    | test_lookup_defaults_are_none                    | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_data_summary_contains_errors_count          | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_data_summary_contains_warnings_count        | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_data_summary_loaded_flag_false_when_no_file | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_data_summary_loaded_flag_true_for_gauge     | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_get_data_summary_empty_before_load          | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_get_data_summary_returns_dict               | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_get_validation_summary_empty_before_load    | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_get_validation_summary_is_copy              | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_get_validation_summary_returns_dict         | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_is_data_complete_depends_only_on_gauge      | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_is_data_complete_false_before_load          | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_is_data_complete_false_when_no_gauge        | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_is_data_complete_true_with_gauge            | Pass   |
|       | helpers.py::TestDataLoaderHelper                      | test_validation_summary_has_gauge_key            | Pass   |
|       | helpers.py::TestPrintLoadingSummary                   | test_summary_with_full_data_does_not_crash       | Pass   |
|       | helpers.py::TestPrintLoadingSummary                   | test_summary_with_no_data_does_not_crash         | Pass   |
|       | helpers.py::TestPrintLoadingSummary                   | test_summary_with_partial_data_does_not_crash    | Pass   |

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| Model | Module                                | Test Name                                                 | Status |
|-------|---------------------------------------|-----------------------------------------------------------|--------|
|       | init.py::TestDataLoaderInit           | test_init_with_input_dir                                  | Pass   |
|       | init.py::TestDataLoaderInit           | test_init_with_string_path_converts_to_path               | Pass   |
|       | init.py::TestDataLoaderInit           | test_loaded_data_initialised                              | Pass   |
|       | init.py::TestDataLoaderInit           | test_private_loaders_created                              | Pass   |
|       | init.py::TestDataLoaderInit           | test_validation_results_empty_at_init                     | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_counterparty_data_loaded                             | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_empty_dir_returns_loaded_data                        | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_gauge_data_count_matches_records                     | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_gauge_data_has_items_key                             | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_gaugehd_dir_counted                                  | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_gaugets_dir_counted                                  | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_gaugets_dir_multiple_files_counted                   | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_hazard_curves_loaded                                 | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_lookups_built_after_load                             | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_no_gaugehd_dir_count_stays_zero                      | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_no_gaugets_dir_count_stays_zero                      | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_no_propertyts_dir_count_stays_zero                   | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_property_hazard_curves_loaded                        | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_property_hazard_summary_fields_accessible            | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_propertyts_dir_counted                               | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_storm_data_list_format                               | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_storm_data_loaded_dict_format                        | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_storm_data_unknown_format                            | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_storm_data_with_items_key                            | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_with_gauge_data_loaded                               | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_with_mortgage_data_loaded                            | Pass   |
|       | load_all.py::TestDataLoaderLoadAll    | test_with_property_data_loaded                            | Pass   |
|       | loaders.py::TestLoadCounterpartyData  | test_corrupt_counterparty_file_leaves_data_none           | Pass   |
|       | loaders.py::TestLoadCounterpartyData  | test_missing_counterparty_file_leaves_data_none           | Pass   |
|       | loaders.py::TestLoadCounterpartyData  | test_valid_counterparty_file_loads                        | Pass   |
|       | loaders.py::TestLoadHazardCurves      | test_corrupt_hazard_file_leaves_hazard_data_none          | Pass   |
|       | loaders.py::TestLoadHazardCurves      | test_corrupt_property_hazard_file_leaves_hazard_data_none | Pass   |
|       | loaders.py::TestLoadHazardCurves      | test_missing_hazard_file_leaves_hazard_data_none          | Pass   |
|       | loaders.py::TestLoadHazardCurves      | test_missing_property_hazard_file_leaves_hazard_data_none | Pass   |
|       | loaders.py::TestLoadHazardCurves      | test_valid_hazard_file_populates_hazard_data              | Pass   |
|       | loaders.py::TestLoadStormData         | test_corrupt_storm_file_leaves_storm_data_none            | Pass   |
|       | loaders.py::TestLoadStormData         | test_missing_storm_file_leaves_storm_data_none            | Pass   |
|       | loaders.py::TestLoadStormData         | test_storms_dict_with_storms_key                          | Pass   |
|       | loaders.py::TestLoadStormData         | test_storms_list_format                                   | Pass   |
|       | loaders.py::TestLoadStormData         | test_storms_other_type                                    | Pass   |
|       | validation.py::TestLoadWithValidation | test_empty_data_list_records_invalid_records              | Pass   |
|       | validation.py::TestLoadWithValidation | test_empty_data_list_returns_none                         | Pass   |
|       | validation.py::TestLoadWithValidation | test_exception_in_loader_returns_none                     | Pass   |
|       | validation.py::TestLoadWithValidation | test_exception_records_error_in_validation                | Pass   |

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| Model | Module                                                | Test Name                                            | Status |
|-------|-------------------------------------------------------|------------------------------------------------------|--------|
|       | validation.py::TestLoadWithValidation                 | test_validation_return_from_loader_returns_none      | Pass   |
|       | validation.py::TestLoadWithValidation                 | test_validation_successful_load_records_valid_result | Pass   |
|       | validation.py::TestLoadWithValidation                 | test_validation_successful_load_returns_data_dict    | Pass   |
|       | validation.py::TestLoadWithValidation                 | test_validation_summary_count_matches_items          | Pass   |
|       | validation.py::TestLoadWithValidation                 | test_validation_summary_has_type_field               | Pass   |
|       | gauge_panel.py::TestGaugeGraphHistogramConfiguration  | test_gauge_histogram_configure_panel_height          | Pass   |
|       | gauge_panel.py::TestGaugeGraphHistogramConfiguration  | test_gauge_histogram_configure_panel_width           | Pass   |
|       | gauge_panel.py::TestGaugeGraphHistogramConfiguration  | test_gauge_histogram_configure_statistics            | Pass   |
|       | gauge_panel.py::TestGaugeHazardStructureConfiguration | test_gauge_hazard_configure_panel_height             | Pass   |
|       | gauge_panel.py::TestGaugeHazardStructureConfiguration | test_gauge_hazard_configure_panel_width              | Pass   |
|       | gauge_panel.py::TestGaugeHazardStructureConfiguration | test_gauge_hazard_configure_statistics               | Pass   |
|       | gauge_panel.py::TestGaugeStormAnalyticsConfiguration  | test_gauge_storm_analyse_configure_panel_height      | Pass   |
|       | gauge_panel.py::TestGaugeStormAnalyticsConfiguration  | test_gauge_storm_analyse_configure_panel_width       | Pass   |
|       | gauge_panel.py::TestGaugeStormAnalyticsConfiguration  | test_gauge_storm_analyse_configure_statistics        | Pass   |
|       | gauge_panel.py::TestHistoricalTable                   | test_historical_fetches_gauge_history                | Pass   |
|       | gauge_panel.py::TestHistoricalTable                   | test_historical_fetches_stress_storms                | Pass   |
|       | gauge_panel.py::TestHistoricalTable                   | test_historical_has_storm_scenarios                  | Pass   |
|       | gauge_panel.py::TestHistoricalTable                   | test_historical_has_time_series_chart                | Pass   |
|       | gauge_panel.py::TestHistoricalTable                   | test_historical_renders                              | Pass   |
|       | gauge_panel.py::TestHistoricalTable                   | test_historical_renders_storm_list                   | Pass   |
|       | gauge_panel.py::TestHistoricalTable                   | test_historical_shows_flood_stages                   | Pass   |
|       | gauge_panel.py::TestHistoricalTable                   | test_historical_shows_trigger_colors                 | Pass   |
|       | gauge_panel.py::TestHistoricalTable                   | test_historical_storm_click_navigates_to_stress      | Pass   |
|       | gauge_panel.py::TestHistoricalTable                   | test_historical_uses_event_listeners                 | Pass   |
|       | gauge_panel.py::TestStressTestTable                   | test_stress_fetches_storms                           | Pass   |
|       | gauge_panel.py::TestStressTestTable                   | test_stress_has_cash_price_concept                   | Pass   |
|       | gauge_panel.py::TestStressTestTable                   | test_stress_has_chart_tabs                           | Pass   |
|       | gauge_panel.py::TestStressTestTable                   | test_stress_has_storm_selector                       | Pass   |
|       | gauge_panel.py::TestStressTestTable                   | test_stress_renders                                  | Pass   |
|       | gauge_panel.py::TestStressTestTable                   | test_stress_runs_scenario                            | Pass   |
|       | gauge_panel.py::TestStressTestTable                   | test_stress_shows_p_flood                            | Pass   |
|       | gauge_panel.py::TestStressTestTable                   | test_stress_storm_hint_pickup                        | Pass   |
|       | gauge_panel.py::TestStressTestTable                   | test_stress_trade_table                              | Pass   |
|       | gauge_panel.py::TestStressTestTable                   | test_stress_uses_event_listeners_for_tabs            | Pass   |
|       | historical_detail.py::TestHistoricalTableAlert        | test_historical_alert_colour_amber                   | Pass   |
|       | historical_detail.py::TestHistoricalTableAlert        | test_historical_alert_daily_observations_field       | Pass   |
|       | historical_detail.py::TestHistoricalTableAlert        | test_historical_alert_renders_historical_chart       | Pass   |
|       | historical_detail.py::TestHistoricalTableAlert        | test_historical_alert_severe_colour_red              | Pass   |
|       | historical_detail.py::TestHistoricalTableAlert        | test_historical_alert_shows_flood_alert_threshold    | Pass   |
|       | historical_detail.py::TestHistoricalTableAlert        | test_historical_alert_shows_flood_warning_threshold  | Pass   |
|       | historical_detail.py::TestHistoricalTableAlert        | test_historical_alert_shows_severe_flood_threshold   | Pass   |
|       | historical_detail.py::TestHistoricalTableAlert        | test_historical_alert_subsamples_observations        | Pass   |
|       | historical_detail.py::TestHistoricalTableAlert        | test_historical_alert_warning_colour_orange          | Pass   |
|       | historical_detail.py::TestHistoricalTableAlert        | test_historical_alert_data_cached_by_gauge_id        | Pass   |

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| Model | Module                                                                     | Test Name                                             | Status |
|-------|----------------------------------------------------------------------------|-------------------------------------------------------|--------|
|       | historical_detail.py::TestHistoricalDataCaching                            | test_data_caching_history_data                        | Pass   |
|       | historical_detail.py::TestHistoricalDataCaching                            | test_data_cached_data_on_revisit                      | Pass   |
|       | historical_detail.py::TestHistoricalStatsBar                               | test_stats_bar_setup_function                         | Pass   |
|       | historical_detail.py::TestHistoricalStatsBar                               | test_stats_bar_flood_exceedances                      | Pass   |
|       | historical_detail.py::TestHistoricalStatsBar                               | test_stats_bar_mean_level                             | Pass   |
|       | historical_detail.py::TestHistoricalStatsBar                               | test_stats_bar_record_years                           | Pass   |
|       | historical_detail.py::TestHistoricalTabRenderers                           | test_historical_tab_renderers                         | Pass   |
|       | historical_detail.py::TestHistoricalTabRenderers                           | test_historical_tab_safe_length                       | Pass   |
|       | historical_detail.py::TestHistoryEndpoint                                  | test_endpoint_fetches_gauge_history                   | Pass   |
|       | historical_detail.py::TestHistoryEndpoint                                  | test_endpoint_gauge_id_variable                       | Pass   |
|       | historical_detail.py::TestStormChiefNavigation                             | test_navigation_handler_defined                       | Pass   |
|       | historical_detail.py::TestStormChiefNavigation                             | test_navigation_handler_iife_safe                     | Pass   |
|       | historical_detail.py::TestStormChiefNavigation                             | test_navigation_goes_to_tab_5                         | Pass   |
|       | historical_detail.py::TestStormChiefNavigation                             | test_navigation_set_stress_storm_hint                 | Pass   |
|       | historical_detail.py::TestStormChiefNavigation                             | test_navigation_hint_set_before_switch_tab            | Pass   |
|       | historical_detail.py::TestStormChiefNavigation                             | test_navigation_id_passed_to_navigate                 | Pass   |
|       | historical_detail.py::TestStormChiefConsistencyWithStipulation             | test_consistency_with_stipulation_mm_consistent       | Pass   |
|       | historical_detail.py::TestStormChiefConsistencyWithStipulation             | test_consistency_with_stipulation_severe              | Pass   |
|       | historical_detail.py::TestStormChiefConsistencyWithStipulation             | test_consistency_with_stipulation_gross               | Pass   |
|       | historical_detail.py::TestStormChiefConsistencyWithStipulation             | test_consistency_with_stipulation_peak                | Pass   |
|       | historical_detail.py::TestStormChiefConsistencyWithStipulation             | test_consistency_with_stipulation_fab                 | Pass   |
|       | historical_detail.py::TestStormListOrders                                  | test_orders_overflow_count                            | Pass   |
|       | historical_detail.py::TestStormListOrders                                  | test_orders_top_30_storms                             | Pass   |
|       | historical_detail.py::TestStormListOrders                                  | test_orders_sorted_by_peak_level                      | Pass   |
|       | historical_detail.py::TestStormPresentationEffectivePrecipitation          | test_status_effective_precipitation                   | Pass   |
|       | historical_detail.py::TestStormPresentationGaugesSevere                    | test_status_gauges_severe                             | Pass   |
|       | historical_detail.py::TestStormPresentationIntensityCategory               | test_status_intensity_category                        | Pass   |
|       | historical_detail.py::TestStormPresentationPeakTriggerForColour            | test_status_peak_trigger_for_colour                   | Pass   |
|       | historical_detail.py::TestStormPresentationPeakLevelM                      | test_status_peak_level_m                              | Pass   |
|       | historical_detail.py::TestStormPresentationStormID                         | test_status_storm_id                                  | Pass   |
|       | historical_detail.py::TestStormPresentationStormNameField                  | test_status_storm_name_field                          | Pass   |
|       | historical_detail.py::TestStormScenarioEndpointUseGaugeStormsEndpoint      | test_scenario_endpoint_use_gauge_storms_endpoint      | Pass   |
|       | historical_detail.py::TestStormScenarioEndpointStormScenarios              | test_scenario_endpoint_storm_scenarios                | Pass   |
|       | historical_detail.py::TestStormScenarioEndpointStormHistoryRendersInPanel  | test_scenario_endpoint_storm_history_renders_in_panel | Pass   |
|       | historical_detail.py::TestStormScenarioEndpointTradingStressStormsEndpoint | test_scenario_endpoint_trading_stress_storms_endpoint | Pass   |
|       | prs_detail.py::TestAnalyticalPrice                                         | test_cashflow_computation                             | Pass   |
|       | prs_detail.py::TestAnalyticalPrice                                         | test_survival_interpolation                           | Pass   |
|       | prs_detail.py::TestAnalyticalPrice                                         | test_yield_rate_interpolation                         | Pass   |
|       | prs_detail.py::TestMaturityConversion                                      | test_compute_maturity_date                            | Pass   |
|       | prs_detail.py::TestMaturityConversion                                      | test_current_roll_date                                | Pass   |
|       | prs_detail.py::TestMaturityConversion                                      | test_last_friday_of_month                             | Pass   |
|       | prs_detail.py::TestMaturityConversion                                      | test_maturity_popup                                   | Pass   |
|       | prs_detail.py::TestPRSControls                                             | test_controls_function                                | Pass   |
|       | prs_detail.py::TestPRSControls                                             | test_counterparty_dropdown                            | Pass   |

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| Model | Module                                   | Test Name                                     | Status |
|-------|------------------------------------------|-----------------------------------------------|--------|
|       | prs_detail.py::TestPRSControls           | test.direction_toggle                         | Pass   |
|       | prs_detail.py::TestPRSControls           | test.maturity_selector                        | Pass   |
|       | prs_detail.py::TestPRSControls           | test.notional_input                           | Pass   |
|       | prs_detail.py::TestPRSControls           | test.trigger_selector                         | Pass   |
|       | prs_detail.py::TestPRSTabRenderers       | test.prs_tab_renderers                        | Pass   |
|       | prs_detail.py::TestPRSTabRenderers       | test.prs_yield_curve_fetch                    | Pass   |
|       | prs_detail.py::TestTradeCommit           | test.commit_endpoint                          | Pass   |
|       | prs_detail.py::TestTradeCommit           | test.payer_flag_in_commit                     | Pass   |
|       | stress_detail.py::TestChartSubTabs       | test.event_listeners_not_inline               | Pass   |
|       | stress_detail.py::TestChartSubTabs       | test.flood_probability_tab                    | Pass   |
|       | stress_detail.py::TestChartSubTabs       | test.stress_pnl_tab                           | Pass   |
|       | stress_detail.py::TestChartSubTabs       | test.surface_tab                              | Pass   |
|       | stress_detail.py::TestKnockOutDisplay    | test.first_trigger_hour                       | Pass   |
|       | stress_detail.py::TestKnockOutDisplay    | test.ko_annotation_line                       | Pass   |
|       | stress_detail.py::TestKnockOutDisplay    | test.ko_label                                 | Pass   |
|       | stress_detail.py::TestKnockOutDisplay    | test.triggered_hour_reference                 | Pass   |
|       | stress_detail.py::TestStormSelector      | test.auto_selects_worst_storm                 | Pass   |
|       | stress_detail.py::TestStormSelector      | test.storm_dropdown                           | Pass   |
|       | stress_detail.py::TestStressEndpoints    | test.run_endpoint                             | Pass   |
|       | stress_detail.py::TestStressEndpoints    | test.storms_endpoint                          | Pass   |
|       | stress_detail.py::TestStressTabRenderers | test.stress_tab_renderers                     | Pass   |
|       | stress_detail.py::TestSurfaceTable       | test.surface_ko_trim                          | Pass   |
|       | stress_detail.py::TestSurfaceTable       | test.surface_reads_probability_surface        | Pass   |
|       | stress_detail.py::TestSurfaceTable       | test.surface_renders_table                    | Pass   |
|       | stress_detail.py::TestSurfaceTable       | test.surface_trigger_band_shading             | Pass   |
|       | stress_detail.py::TestSurfaceTable       | test.surface_wrap_element                     | Pass   |
|       | js_safety.py::TestAuditReportsController | test.audit_terminal_refresh_exposed_on_window | Fail   |
|       | js_safety.py::TestAuditReportsController | test.no_json_download_button                  | Pass   |
|       | js_safety.py::TestAuditReportsController | test.poll_output_endpoint                     | Pass   |
|       | js_safety.py::TestAuditReportsController | test.refresh_audit_tab.button                 | Pass   |
|       | js_safety.py::TestAuditReportsController | test.run_test_suite_exposed_on_window         | Pass   |
|       | js_safety.py::TestAuditReportsController | test.run_test_suite_function                  | Pass   |
|       | js_safety.py::TestAuditReportsController | test.terminal_overlay_id                      | Pass   |
|       | js_safety.py::TestAuditReportsController | test.toggle_failure_detail_exposed_on_window  | Fail   |
|       | js_safety.py::TestFullCompilation        | test.full_js_contains_iife                    | Pass   |
|       | js_safety.py::TestFullCompilation        | test.full_js_contains_regulatory_button       | Pass   |
|       | js_safety.py::TestFullCompilation        | test.full_js_contains_render_audit_report     | Pass   |
|       | js_safety.py::TestFullCompilation        | test.full_js_contains_script_tags             | Pass   |
|       | js_safety.py::TestFullCompilation        | test.full_js_contains_show_mg_panel           | Pass   |
|       | js_safety.py::TestFullCompilation        | test.full_js_contains_terminal_popup          | Pass   |
|       | js_safety.py::TestFullCompilation        | test.full_js_is_str                           | Pass   |
|       | js_safety.py::TestFullCompilation        | test.full_js_non_empty                        | Pass   |
|       | js_safety.py::TestFullCompilation        | test.full_js_passes_node_syntax_check         | Pass   |
|       | js_safety.py::TestGetJsReturnsString     | test.all_modules_non_empty                    | Pass   |

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| Model | Module                                                                      | Test Name                                             | Status |
|-------|-----------------------------------------------------------------------------|-------------------------------------------------------|--------|
|       | js_safety.py::TestGetJsReturnsString                                        | test_all_modules_return_str                           | Pass   |
|       | js_safety.py::TestNoLiteralNewlineInJsStrings                               | test_all_modules_no_newline_in_js_strings             | Pass   |
|       | js_safety.py::TestNoLiteralNewlineInJsStringsReportsNoNewlineInJsStrings    | test_all_modules_reports_no_newline_in_js_strings     | Pass   |
|       | js_safety.py::TestNoLiteralNewlineInJsStringsCompiledJsNoNewlineInJsStrings | test_all_modules_compiled_js_no_newline_in_js_strings | Pass   |
|       | test_backend_handler.py::TestAddChildToMapCallsAddChild                     | test_add_to_map_calls_add_child                       | Pass   |
|       | test_backend_handler.py::TestBackendHandlerInit                             | test_backend_handler_init                             | Pass   |
|       | test_backend_handler.py::TestBackendHandlerInitUsesConfig                   | test_backend_handler_init_uses_config                 | Pass   |
|       | test_backend_handler.py::TestBackendHandlerInitCopies                       | test_backend_handler_init_copies                      | Pass   |
|       | test_backend_handler.py::TestConfigureNoArgsNoChange                        | test_configure_no_args_no_change                      | Pass   |
|       | test_backend_handler.py::TestConfigureUpdatesEndpoints                      | test_configure_updates_endpoints                      | Pass   |
|       | test_backend_handler.py::TestConfigureUpdatesServerUrl                      | test_configure_updates_server_url                     | Pass   |
|       | test_backend_handler.py::TestGetTestContainsTimeout                         | test_test_contains_timeout                            | Pass   |
|       | test_backend_handler.py::TestGetTestReturnsScriptTag                        | test_test_returns_script_tag                          | Pass   |
|       | test_backend_handler.py::TestGetTestGetStatisticsKeys                       | test_test_get_statistics_keys                         | Pass   |
|       | test_backend_handler.py::TestGetTestGetStatisticsSingleEndpoint             | test_test_get_statistics_single_endpoint              | Pass   |
|       | test_context_menus.py::TestAddChildToMapCallsAddChild                       | test_add_to_map_calls_add_child                       | Pass   |
|       | test_context_menus.py::TestConfigureBothMenus                               | test_configure_both_menus                             | Pass   |
|       | test_context_menus.py::TestConfigureGaugeMenu                               | test_configure_gauge_menu                             | Pass   |
|       | test_context_menus.py::TestConfigureNoArgsNoChange                          | test_configure_no_args_no_change                      | Pass   |
|       | test_context_menus.py::TestConfigurePropertyMenu                            | test_configure_property_menu                          | Pass   |
|       | test_context_menus.py::TestContextMenuShowMenus                             | test_context_menu_show_menus                          | Pass   |
|       | test_context_menus.py::TestContextMenuDefaultInit                           | test_context_menu_default_init                        | Pass   |
|       | test_context_menus.py::TestGetStatisticsCustomStatistics                    | test_get_statistics_custom_statistics                 | Pass   |
|       | test_context_menus.py::TestGetStatisticsDefaultStatistics                   | test_get_statistics_default_statistics                | Pass   |
|       | test_manager.py::TestConfigure test_configure_menus                         | test_configure_menus                                  | Pass   |
|       | test_manager.py::TestConfigure test_configure_notification_position         | test_configure_notification_position                  | Pass   |
|       | test_manager.py::TestConfigure test_configure_notification_timeout          | test_configure_notification_timeout                   | Pass   |
|       | test_manager.py::TestConfigure test_configure_returns_self                  | test_configure_returns_self                           | Pass   |
|       | test_manager.py::TestConfigure test_configure_server_url                    | test_configure_server_url                             | Pass   |
|       | test_manager.py::TestGetStatistics test_returns_dict_with_expected_keys     | test_test_returns_dict_with_expected_keys             | Pass   |
|       | test_manager.py::TestGetStatistics test_statistics_are_dicts                | test_test_statistics_are_dicts                        | Pass   |
|       | test_manager.py::TestSetupMap test_adds_all_components                      | test_setup_map_adds_all_components                    | Pass   |
|       | test_manager.py::TestSetupMap test_activate_right_element_added             | test_setup_map_activate_right_element_added           | Pass   |
|       | test_manager.py::TestSetupMap test_activate_returns_self                    | test_setup_map_activate_returns_self                  | Pass   |
|       | test_notifications.py::TestAddNotificationToMap                             | test_add_to_map                                       | Pass   |
|       | test_notifications.py::TestAddNotificationToMapNotificationSystem           | test_add_to_map_notification_system                   | Pass   |
|       | test_notifications.py::TestAddToMapAddToMapCallsAddChild                    | test_add_to_map_calls_add_child                       | Pass   |
|       | test_notifications.py::TestBuildPositionReturnsPositionFunction             | test_build_position_returns_position_function         | Pass   |
|       | test_notifications.py::TestBuildTemplateContainsAllTypes                    | test_build_template_contains_all_types                | Pass   |
|       | test_notifications.py::TestBuildTemplateInheritsTemplateValues              | test_build_template_inherits_template_values          | Pass   |
|       | test_notifications.py::TestBuildTemplateReturnsValidJson                    | test_build_template_returns_valid_json                | Pass   |
|       | test_notifications.py::TestConfigure test_configure_max_visible             | test_configure_max_visible                            | Pass   |
|       | test_notifications.py::TestConfigure test_configure_no_args_no_change       | test_configure_no_args_no_change                      | Pass   |
|       | test_notifications.py::TestConfigure test_configure_position                | test_configure_position                               | Pass   |

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| Model | Module                                                              | Test Name                                                  | Status |
|-------|---------------------------------------------------------------------|------------------------------------------------------------|--------|
|       | test_notifications.py::TestConfigure                                | test_configure_timeout                                     | Pass   |
|       | test_notifications.py::TestCreateNotificationSystem                 | test_create_notification_system                            | Pass   |
|       | test_notifications.py::TestCreateNotificationSystem                 | test_create_notification_timeout                           | Pass   |
|       | test_notifications.py::TestCreateNotificationSystem                 | test_create_notification_disposition_defaults_to_top_right | Pass   |
|       | test_notifications.py::TestCreateNotificationSystem                 | test_create_notification_disposition                       | Pass   |
|       | test_notifications.py::TestGetStatistics                            | test_get_statistics_returns_expected_keys                  | Pass   |
|       | test_notifications.py::TestGetStatistics                            | test_get_statistics_reflect_config                         | Pass   |
|       | test_startup.py::TestGetPreloaderJS                                 | test_get_preloader_js_matches_get_js                       | Pass   |
|       | test_startup.py::TestGetPreloaderJS                                 | test_get_preloader_js_returns_string                       | Pass   |
|       | test_startup.py::TestStartupPreloaderAddToMap                       | test_preloader_add_to_map_calls_add_child                  | Pass   |
|       | test_startup.py::TestStartupPreloaderAddToMap                       | test_preloader_add_to_map_output_contains_script           | Pass   |
|       | test_td_main_map_coverage.py::TestMainMapES01CallScriptAddChild     | test_main_map_es01_call_script_add_child                   | Pass   |
|       | test_td_main_map_coverage.py::TestMainMapES01GetStatistics          | test_main_map_es01_get_statistics                          | Pass   |
|       | test_td_main_map_coverage.py::TestMainMapES01GetStatisticsDict      | test_main_map_es01_get_statistics_dict                     | Pass   |
|       | property_panel.py::TestPropertyHeight                               | test_configure_config_height                               | Pass   |
|       | property_panel.py::TestPropertyHeight                               | test_configure_config_height_does_not_change               | Pass   |
|       | property_panel.py::TestPropertyHeight                               | test_configure_config_height                               | Pass   |
|       | property_panel.py::TestPropertyHeight                               | test_configure_config_width                                | Pass   |
|       | property_panel.py::TestPropertyHeight                               | test_configure_config_use                                  | Pass   |
|       | property_panel.py::TestPropertyHeight                               | test_configure_config_user_returns_dict                    | Pass   |
|       | sp_sim_detail.py::TestScrubberRange                                 | test_hour_label_present                                    | Pass   |
|       | sp_sim_detail.py::TestScrubberRange                                 | test_scrubber_id_present                                   | Pass   |
|       | sp_sim_detail.py::TestScrubberRange                                 | test_scrubber_max_from_config                              | Pass   |
|       | sp_sim_detail.py::TestScrubberRange                                 | test_scrubber_max_not_59                                   | Pass   |
|       | sp_sim_detail.py::TestScrubberRange                                 | test_scrubber_min_is_zero                                  | Pass   |
|       | sp_sim_detail.py::TestScrubberRange                                 | test_scrubber_type_is_range                                | Pass   |
|       | sp_sim_detail.py::TestSimFunction                                   | test_get_js_returns_nonempty                               | Pass   |
|       | sp_sim_detail.py::TestSimFunction                                   | test_spSeekTo_defined                                      | Pass   |
|       | sp_sim_detail.py::TestSimFunction                                   | test_spSetSpeed_defined                                    | Pass   |
|       | sp_sim_detail.py::TestSimFunction                                   | test_spStopAnim_defined                                    | Pass   |
|       | sp_table_detail.py::TestPanelTitle                                  | test_panel_title_id_present                                | Pass   |
|       | sp_table_detail.py::TestPanelTitle                                  | test_panel_title_prefix                                    | Pass   |
|       | sp_table_detail.py::TestPanelTitle                                  | test_title_reads_selected_option_text                      | Pass   |
|       | sp_table_detail.py::TestPanelTitle                                  | test_title_uses_label_variable_not_bare_id_concat          | Pass   |
|       | sp_table_detail.py::TestStormList                                   | test_loads_from_propertyts_storms                          | Pass   |
|       | sp_table_detail.py::TestStormList                                   | test_loads_storms_message                                  | Pass   |
|       | sp_table_detail.py::TestStormList                                   | test_loads_preloaded_cache                                 | Pass   |
|       | sp_table_detail.py::TestTableStructure                              | test_reload_portfolio_impact_defined                       | Pass   |
|       | sp_table_detail.py::TestTableStructure                              | test_reload_portfolio_impact_endpoint                      | Pass   |
|       | sp_table_detail.py::TestTableStructure                              | test_resort_columns_defined                                | Pass   |
|       | sp_table_detail.py::TestTableStructure                              | test_summary_cards_present                                 | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelRefreshMainMap | test_cross_iife_exports_panel_refresh_main_map             | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsApplyFilter         | test_cross_iife_exports_apply_filter                       | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsCellClick           | test_cross_iife_exports_cell_click                         | Pass   |

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| Model | Module                                               | Test Name                                                                         | Status |
|-------|------------------------------------------------------|-----------------------------------------------------------------------------------|--------|
|       | test_cross_iife_exports.py::TestCrossIIFEExports     | test_cross_iife_exports.py::TestCrossIIFEExports.window                           | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExports     | test_cross_iife_exports.py::TestCrossIIFEExports.01_exported                      | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExports     | test_cross_iife_exports.py::TestCrossIIFEExports.window                           | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExports     | test_cross_iife_exports.py::TestCrossIIFEExports.window_hint                      | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExports     | test_cross_iife_exports.py::TestCrossIIFEExports.show_hide                        | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExports     | test_cross_iife_exports.py::TestCrossIIFEExports.blotter_view                     | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExports     | test_cross_iife_exports.py::TestCrossIIFEExports.exists                           | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExports     | test_cross_iife_exports.py::TestCrossIIFEExports.display_flex                     | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExports     | test_cross_iife_exports.py::TestCrossIIFEExports.show_panel                       | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExports     | test_cross_iife_exports.py::TestCrossIIFEExports.exists                           | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExports     | test_cross_iife_exports.py::TestCrossIIFEExports.exists                           | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExports     | test_cross_iife_exports.py::TestCrossIIFEExports.window_preload_done              | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExports     | test_cross_iife_exports.py::TestCrossIIFEExports.open                             | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.on_window[preGovAudit]       | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.on_window[preGovBib]         | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.on_window[preStorms]         | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.on_window[tdPreBlotter]      | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.on_window[tdPreEodHistory]   | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.on_window[tdPreGauges]       | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.on_window[tdPreGovDocs]      | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.on_window[tdPreMarket]       | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.on_window[tdPrePortStorms]   | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.on_window[tdPreStressStorms] | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.on_window[tdPreYieldCurve]   | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.declarations_for_cache       | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.for_td_preload_done          | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.slope_initialised            | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.slope_set_after_start        | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.consume_after_consume        | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.pre_blotter                  | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.consume_after_consume        | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.pre_gov_docs                 | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.consume_after_consume        | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.pre_port_storms              | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.pre_read_assignment          | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.preload_done                 | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.consume_after_consume        | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.pre_storms                   | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.preload_done                 | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.pre_blotter                  | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.pre_gov_docs                 | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.pre_port_storms              | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.preload_done_read            | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnership | test_cross_iife_ownership.py::TestCrossIIFEOwnership.pre_storms                   | Pass   |

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| Model | Module                                                                        | Test Name                                                  | Status |
|-------|-------------------------------------------------------------------------------|------------------------------------------------------------|--------|
|       | test_cross.iife_ownership.py::TestContextTradingNoDeskRebars                  | test_context_trading_no_desk_rebars_preload_done           | Pass   |
|       | test_cross.iife_syntax.py::TestContextTradingMarketSubmodulesSyntax           | test_context_trading_market_submodules_syntax              | Pass   |
|       | test_cross.iife_syntax.py::TestContextTradingSupervisorPreloaderSyntax        | test_context_trading_supervisor_preloader_syntax           | Pass   |
|       | test_cross.iife_syntax.py::TestContextTradingPortfolioSyntax                  | test_context_trading_portfolio_syntax                      | Pass   |
|       | test_cross.iife_syntax.py::TestContextTradingAggregateSubmoduleSyntax         | test_context_trading_aggregate_submodule_syntax            | Pass   |
|       | test_cross.iife_syntax.py::TestContextTradingBiblotterSubmoduleSyntax         | test_context_trading_biblotter_submodule_syntax            | Pass   |
|       | test_cross.iife_syntax.py::TestContextTradingCurvesSubmoduleSyntax            | test_context_trading_curves_submodule_syntax               | Pass   |
|       | test_cross.iife_syntax.py::TestContextTradingDeskJsSyntax                     | test_context_trading_desk_js_syntax                        | Pass   |
|       | test_cross.iife_syntax.py::TestContextTradingGoodSubmoduleSyntax              | test_context_trading_good_submodule_syntax                 | Pass   |
|       | test_cross.iife_syntax.py::TestContextTradingGuifs01SubmoduleSyntax           | test_context_trading_guifs01_submodule_syntax              | Pass   |
|       | test_cross.iife_syntax.py::TestContextTradingMarketSubmoduleSyntax            | test_context_trading_market_submodule_syntax               | Pass   |
|       | test_cross.iife_syntax.py::TestContextTradingPortStressSubmoduleSyntax        | test_context_trading_port_stress_submodule_syntax          | Pass   |
|       | test_cross.iife_syntax.py::TestContextTradingRustStressSubmoduleSyntax        | test_context_trading_rust_stress_submodule_syntax          | Pass   |
|       | test_cross.iife_syntax.py::TestContextTradingGenerates                        | test_context_trading_generates                             | Pass   |
|       | test_cross.iife_syntax.py::TestContextFlattenGenerates                        | test_context_flatten_generates                             | Pass   |
|       | test_cross.iife_syntax.py::TestContextFullPresetGenerates                     | test_context_full_preset_generates                         | Pass   |
|       | test_cross.iife_syntax.py::TestContextFeedGeneration                          | test_context_feed_generation                               | Pass   |
|       | test_cross.iife_syntax.py::TestContextFsPhyGeneration                         | test_context_fs_phy_generation                             | Pass   |
|       | test_cross.iife_syntax.py::TestContextFormalGenCitebibliographyGenerates      | test_context_formal_gen_citebibliography_generates         | Pass   |
|       | test_cross.iife_syntax.py::TestContextFormalGenCitedatsGenerates              | test_context_formal_gen_citedats_generates                 | Pass   |
|       | test_cross.iife_syntax.py::TestContextFormalGenCiteGenerates                  | test_context_formal_gen_cite_generates                     | Pass   |
|       | test_cross.iife_syntax.py::TestContextFormalGenCiteSelfnessGenerates          | test_context_formal_gen_cite_selfness_generates            | Pass   |
|       | test_cross.iife_syntax.py::TestContextFormalGenCiteGenerates                  | test_context_formal_gen_cite_generates                     | Pass   |
|       | test_cross.iife_syntax.py::TestContextFormalGenCiteGenerates                  | test_context_formal_gen_cite_generates                     | Pass   |
|       | test_cross.iife_syntax.py::TestContextFormalGenCiteGenerates                  | test_context_formal_gen_cite_generates                     | Pass   |
|       | test_cross.iife_syntax.py::TestContextFormalGenCiteGenerates                  | test_context_formal_gen_cite_generates                     | Pass   |
|       | test_cross.iife_syntax.py::TestContextFormalGenCiteGenerates                  | test_context_formal_gen_cite_generates                     | Pass   |
|       | test_cross.iife_syntax.py::TestContextFormalGenCiteGenerates                  | test_context_formal_gen_cite_generates                     | Pass   |
|       | test_cross.iife_syntax.py::TestContextFormalGenCiteGenerates                  | test_context_formal_gen_cite_generates                     | Pass   |
|       | test_startup_preloader.py::TestContextApplyDocumentsCacheFunctionExists       | test_context_apply_documents_cache_function_exists         | Pass   |
|       | test_startup_preloader.py::TestContextCheckDocumentsCache                     | test_context_check_documents_cache                         | Pass   |
|       | test_startup_preloader.py::TestContextClearDocumentsCacheAfterUse             | test_context_clear_documents_cache_after_use               | Pass   |
|       | test_startup_preloader.py::TestHealthHealth_200_at_api_v1                     | test_health_health_200_at_api_v1                           | Pass   |
|       | test_startup_preloader.py::TestHealthHealth_200_at_root                       | test_health_health_200_at_root                             | Pass   |
|       | test_startup_preloader.py::TestHealthHealth_check_function_serves_both        | test_health_health_check_function_serves_both              | Pass   |
|       | test_startup_preloader.py::TestHealthHealth_py_has_api_v1_health_route        | test_health_health_py_has_api_v1_health_route              | Pass   |
|       | test_startup_preloader.py::TestHealthHealth_py_has_plain_health_route         | test_health_health_py_has_plain_health_route               | Pass   |
|       | test_startup_preloader.py::TestHealthHealth_returns_ok_status                 | test_health_health_returns_ok_status                       | Pass   |
|       | test_startup_preloader.py::TestInitSetupMapAndViningLoaderBeforeNotifications | test_init_setup_map_and_vining_loader_before_notifications | Fail   |
|       | test_startup_preloader.py::TestInitStartupMppelVinerImportedInInit            | test_init_startup_mppel_viner_imported_in_init             | Pass   |
|       | test_startup_preloader.py::TestInitStartupMppelVinerInstantiatedInInit        | test_init_startup_mppel_viner_instantiated_in_init         | Fail   |
|       | test_startup_preloader.py::TestStatData fifteen keys have stat handler        | test_stat_data_fifteen_keys_have_stat_handler              | Pass   |
|       | test_startup_preloader.py::TestStatDetailsStatUsesTradesLength                | test_stat_details_stat_uses_trades_length                  | Pass   |
|       | test_startup_preloader.py::TestStatDetailsStoryStatUsesHistoryLength          | test_stat_details_story_stat_uses_history_length           | Pass   |
|       | test_startup_preloader.py::TestStatDangleStatsStatUsesCount                   | test_stat_dangle_stats_stat_uses_count                     | Pass   |
|       | test_startup_preloader.py::TestStatDangleStatsStatUsesGaugesLength            | test_stat_dangle_stats_stat_uses_gauges_length             | Pass   |

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| Model | Module                                                                          | Test Name                                                                       | Status |
|-------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------|
|       | test_startup_preloader.py::TestStartupPreloaderStatUsesEventsLength             | test_startup_preloader.py::TestStartupPreloaderStatUsesEventsLength             | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderStatUsesReferencesLength         | test_startup_preloader.py::TestStartupPreloaderStatUsesReferencesLength         | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderStatUsesDocumentsLength          | test_startup_preloader.py::TestStartupPreloaderStatUsesDocumentsLength          | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderStatUsesObjectKeysGauges         | test_startup_preloader.py::TestStartupPreloaderStatUsesObjectKeysGauges         | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderStatUsesMortgagesLength          | test_startup_preloader.py::TestStartupPreloaderStatUsesMortgagesLength          | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderStatsStatUsesStormsLength        | test_startup_preloader.py::TestStartupPreloaderStatsStatUsesStormsLength        | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderStatsStatUsesStormsLength        | test_startup_preloader.py::TestStartupPreloaderStatsStatUsesStormsLength        | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderStatsStatUsesTotalProperties     | test_startup_preloader.py::TestStartupPreloaderStatsStatUsesTotalProperties     | Fail   |
|       | test_startup_preloader.py::TestStartupPreloaderStatsStatUsesStormsLength        | test_startup_preloader.py::TestStartupPreloaderStatsStatUsesStormsLength        | Fail   |
|       | test_startup_preloader.py::TestStartupPreloaderStatUsesObjectKeys               | test_startup_preloader.py::TestStartupPreloaderStatUsesObjectKeys               | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderBehaviorDefined                  | test_startup_preloader.py::TestStartupPreloaderBehaviorDefined                  | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderEndpointsPresent                 | test_startup_preloader.py::TestStartupPreloaderEndpointsPresent                 | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderContentLoaded                    | test_startup_preloader.py::TestStartupPreloaderContentLoaded                    | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderLabelIsHazardCurve               | test_startup_preloader.py::TestStartupPreloaderLabelIsHazardCurve               | Pass   |
|       | test_startup_preloader.py::TestStartupPreloadDone                               | test_startup_preloader.py::TestStartupPreloadDone                               | Pass   |
|       | test_startup_preloader.py::TestStartupShowsProgressPopup                        | test_startup_preloader.py::TestStartupShowsProgressPopup                        | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderClassDefined                     | test_startup_preloader.py::TestStartupPreloaderClassDefined                     | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCacheVagueHist                    | test_startup_preloader.py::TestStartupDeclaresCacheVagueHist                    | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCacheVagueAudit                   | test_startup_preloader.py::TestStartupDeclaresCacheVagueAudit                   | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCacheVagueBib                     | test_startup_preloader.py::TestStartupDeclaresCacheVagueBib                     | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCacheVaguePropertyTs              | test_startup_preloader.py::TestStartupDeclaresCacheVaguePropertyTs              | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCacheVagueVars                    | test_startup_preloader.py::TestStartupDeclaresCacheVagueVars                    | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCacheVagueBlotter                 | test_startup_preloader.py::TestStartupDeclaresCacheVagueBlotter                 | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCacheVagueHistory                 | test_startup_preloader.py::TestStartupDeclaresCacheVagueHistory                 | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCacheVagueGauges                  | test_startup_preloader.py::TestStartupDeclaresCacheVagueGauges                  | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCacheVagueGovDocs                 | test_startup_preloader.py::TestStartupDeclaresCacheVagueGovDocs                 | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCacheVagueMarket                  | test_startup_preloader.py::TestStartupDeclaresCacheVagueMarket                  | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCacheVaguePortStorms              | test_startup_preloader.py::TestStartupDeclaresCacheVaguePortStorms              | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCacheVagueStressStorms            | test_startup_preloader.py::TestStartupDeclaresCacheVagueStressStorms            | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCacheVagueYieldCurve              | test_startup_preloader.py::TestStartupDeclaresCacheVagueYieldCurve              | Pass   |
|       | test_startup_preloader.py::TestStartupApplyFormListFunctionExists               | test_startup_preloader.py::TestStartupApplyFormListFunctionExists               | Pass   |
|       | test_startup_preloader.py::TestStartupChecksCacheStormsCache                    | test_startup_preloader.py::TestStartupChecksCacheStormsCache                    | Pass   |
|       | test_startup_preloader.py::TestStartupClearsCacheAfterUse                       | test_startup_preloader.py::TestStartupClearsCacheAfterUse                       | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderDeclarationRedeclareTdblotter    | test_startup_preloader.py::TestStartupPreloaderDeclarationRedeclareTdblotter    | Fail   |
|       | test_startup_preloader.py::TestStartupPreloaderDeclarationRedeclareTdmarket     | test_startup_preloader.py::TestStartupPreloaderDeclarationRedeclareTdmarket     | Fail   |
|       | test_startup_preloader.py::TestStartupPreloaderDeclarationRedeclareTdpload_done | test_startup_preloader.py::TestStartupPreloaderDeclarationRedeclareTdpload_done | Fail   |
|       | blotter.js.py::TestBlotterSignedNotionalInPnlBar                                | blotter.js.py::TestBlotterSignedNotionalInPnlBar                                | Pass   |
|       | blotter.js.py::TestBlotterSignedNotionalPayerNegative                           | blotter.js.py::TestBlotterSignedNotionalPayerNegative                           | Pass   |
|       | blotter.js.py::TestMarketPnLDisplayBlotterHasFromMarketInPnlBar                 | blotter.js.py::TestMarketPnLDisplayBlotterHasFromMarketInPnlBar                 | Pass   |
|       | blotter.js.py::TestMarketPnLDisplayBlotterHasMarketPnlColumn                    | blotter.js.py::TestMarketPnLDisplayBlotterHasMarketPnlColumn                    | Pass   |
|       | blotter.js.py::TestMarketPnLDisplayCommitShowsPnlImpact                         | blotter.js.py::TestMarketPnLDisplayCommitShowsPnlImpact                         | Pass   |
|       | blotter.js.py::TestMarketPnLDisplayHazardColumnUpDownArrows                     | blotter.js.py::TestMarketPnLDisplayHazardColumnUpDownArrows                     | Pass   |
|       | classifiers.js.py::TestClassifiersEndpointStatusEndpoint                        | classifiers.js.py::TestClassifiersEndpointStatusEndpoint                        | Pass   |
|       | classifiers.js.py::TestClassifiersEndpointSummaryEndpoint                       | classifiers.js.py::TestClassifiersEndpointSummaryEndpoint                       | Pass   |

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| Model | Module                                         | Test Name                                   | Status |
|-------|------------------------------------------------|---------------------------------------------|--------|
|       | classifiers.js.py::TestClassifiersEndpoints    | test_train_all_endpoint                     | Pass   |
|       | classifiers.js.py::TestClassifiersEndpoints    | test_train_endpoint                         | Pass   |
|       | classifiers.js.py::TestClassifiersIntegrations | test_classifiers_cleanup                    | Pass   |
|       | classifiers.js.py::TestClassifiersIntegrations | test_classifiers_in_views_array             | Pass   |
|       | classifiers.js.py::TestClassifiersIntegrations | test_classifiers_switch_handler             | Pass   |
|       | classifiers.js.py::TestClassifiersIntegrations | test_classifiers_tab_in_array               | Pass   |
|       | classifiers.js.py::TestClassifiersIntegrations | test_classifiers_view_created               | Pass   |
|       | classifiers.js.py::TestClassifiersStatus       | test_classifiers_renders                    | Pass   |
|       | classifiers.js.py::TestClassifiersStatus       | test_has_batch_training                     | Pass   |
|       | classifiers.js.py::TestClassifiersStatus       | test_has_cleanup                            | Pass   |
|       | classifiers.js.py::TestClassifiersStatus       | test_has_create_view                        | Pass   |
|       | classifiers.js.py::TestClassifiersStatus       | test_has_feature_chart                      | Pass   |
|       | classifiers.js.py::TestClassifiersStatus       | test_has_load_data                          | Pass   |
|       | classifiers.js.py::TestClassifiersStatus       | test_has_metric_card                        | Pass   |
|       | classifiers.js.py::TestClassifiersStatus       | test_has_progress_bar                       | Pass   |
|       | classifiers.js.py::TestClassifiersStatus       | test_has_render_detail                      | Pass   |
|       | classifiers.js.py::TestClassifiersStatus       | test_has_render_summary_table               | Pass   |
|       | classifiers.js.py::TestClassifiersStatus       | test_has_single_training                    | Pass   |
|       | classifiers.js.py::TestClassifiersStatus       | test_has_state_vars                         | Pass   |
|       | market.js.py::TestCommitButtonTests            | test_commit_button_calls_commit_function    | Pass   |
|       | market.js.py::TestCommitButtonTests            | test_commit_button_in_html                  | Pass   |
|       | market.js.py::TestCommitButtonTests            | test_commit_button_not_hidden               | Pass   |
|       | market.js.py::TestCommitButtonTests            | test_commit_function_exists                 | Pass   |
|       | market.js.py::TestCommitButtonTests            | test_commit_posts_to_endpoint               | Pass   |
|       | market.js.py::TestCommitButtonTests            | test_commit_refreshes_blotter               | Pass   |
|       | market.js.py::TestCurvesTabRedesign            | test_curves_all_gauges_chart                | Pass   |
|       | market.js.py::TestCurvesTabRedesign            | test_curves_has_trigger_selector            | Pass   |
|       | market.js.py::TestCurvesTabRedesign            | test_curves_hsl_colours                     | Pass   |
|       | market.js.py::TestCurvesTabRedesign            | test_curves_renders                         | Pass   |
|       | market.js.py::TestCurvesTabRedesign            | test_curves_strips_thames_prefix            | Pass   |
|       | market.js.py::TestCurvesTabRedesign            | test_curves_uses_market_state               | Pass   |
|       | market.js.py::TestMapClickThrough              | test_map_hides_trading_desk                 | Pass   |
|       | market.js.py::TestMapClickThrough              | test_map_new_trade_link                     | Pass   |
|       | market.js.py::TestMapClickThrough              | test_map_view_hazard_curve                  | Pass   |
|       | panel.py::TestInteractivityManager             | test_manager_exports_trading_desk           | Pass   |
|       | panel.py::TestInteractivityManager             | test_manager_has_trading_desk               | Pass   |
|       | panel.py::TestPreloadDoneScoping               | test_preloader_sets_window_preload_done     | Pass   |
|       | panel.py::TestPreloadDoneScoping               | test_startup_uses_window_preload_done_in_js | Pass   |
|       | panel.py::TestPreloadDoneScoping               | test_startup_uses_window_preload_done_set   | Pass   |
|       | panel.py::TestPreloadDoneScoping               | test_tradingdesk_reads_window_preload_done  | Pass   |
|       | panel.py::TestTradingDeskPanelTests            | test_add_to_map                             | Pass   |
|       | panel.py::TestTradingDeskPanelTests            | test_blotter_submodule                      | Pass   |
|       | panel.py::TestTradingDeskPanelTests            | test_eod_submodule                          | Pass   |
|       | panel.py::TestTradingDeskPanelTests            | test_map_submodule                          | Pass   |

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| Model | Module                            | Test Name                                    | Status |
|-------|-----------------------------------|----------------------------------------------|--------|
|       | panel.py::TestTradingDeskPanel    | Test_market_submodule                        | Pass   |
|       | panel.py::TestTradingDeskPanel    | Test_no_unescaped_braces                     | Pass   |
|       | panel.py::TestTradingDeskPanel    | Test_panel_contains_all_tabs                 | Pass   |
|       | panel.py::TestTradingDeskPanel    | Test_panel_contains_global_export            | Pass   |
|       | panel.py::TestTradingDeskPanel    | Test_panel_contains_pi_symbol                | Pass   |
|       | panel.py::TestTradingDeskPanel    | Test_panel_renders                           | Pass   |
|       | panel.py::TestTradingDeskPanel    | Test_risk_submodule                          | Pass   |
|       | port_stress_js.py::TestPortStress | Test_portfolio_run_endpoint                  | Pass   |
|       | port_stress_js.py::TestPortStress | Test_portfolio_storms_endpoint               | Pass   |
|       | port_stress_js.py::TestPortStress | Test_post_method_used_for_run                | Pass   |
|       | port_stress_js.py::TestPortStress | Test_gauge_pnl_calls_switch_tab_stress       | Pass   |
|       | port_stress_js.py::TestPortStress | Test_gauge_pnl_full_detail_button            | Pass   |
|       | port_stress_js.py::TestPortStress | Test_gauge_pnl_has_gauge_table               | Pass   |
|       | port_stress_js.py::TestPortStress | Test_gauge_pnl_render_function               | Pass   |
|       | port_stress_js.py::TestPortStress | Test_gauge_pnl_sets_gauge_hint               | Pass   |
|       | port_stress_js.py::TestPortStress | Test_gauge_pnl_sets_storm_hint               | Pass   |
|       | port_stress_js.py::TestPortStress | Test_convertion_conflicts_with_stress_tab    | Pass   |
|       | port_stress_js.py::TestPortStress | Test_stress_prefix_on_content                | Pass   |
|       | port_stress_js.py::TestPortStress | Test_stress_prefix_on_storm_sel              | Pass   |
|       | port_stress_js.py::TestPortStress | Test_stress_prefix_on_tabs                   | Pass   |
|       | port_stress_js.py::TestPortStress | Test_full_desk_syntax_valid_with_port_stress | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_data_called         | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_artifacts                  | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_tab_in_full_js      | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_view_appended       | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_view_created        | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_in_switch_tab       | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_in_tabs_array       | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_in_views_array      | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_setup_in_hide_panel        | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_inside_iife                | Pass   |
|       | port_stress_js.py::TestPortStress | Test_topflood_canvas_id                      | Pass   |
|       | port_stress_js.py::TestPortStress | Test_topflood_render_function                | Pass   |
|       | port_stress_js.py::TestPortStress | Test_topflood_threshold_colours              | Pass   |
|       | port_stress_js.py::TestPortStress | Test_topflood_uses_chartjs                   | Pass   |
|       | port_stress_js.py::TestPortStress | Test_topflood_canvas_id                      | Pass   |
|       | port_stress_js.py::TestPortStress | Test_topflood_fmtgbp                         | Pass   |
|       | port_stress_js.py::TestPortStress | Test_topflood_render_function                | Pass   |
|       | port_stress_js.py::TestPortStress | Test_topflood_uses_horizontal_bars           | Pass   |
|       | port_stress_js.py::TestPortStress | Test_severity_gauge_pnl_link                 | Pass   |
|       | port_stress_js.py::TestPortStress | Test_severity_has_four_groups                | Pass   |
|       | port_stress_js.py::TestPortStress | Test_severity_render_function                | Pass   |
|       | port_stress_js.py::TestPortStress | Test_severity_sets_selected_gauge            | Pass   |
|       | port_stress_js.py::TestPortStress | Test_severity_severe_colour                  | Pass   |

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| Model | Module                                    | Test Name                                         | Status |
|-------|-------------------------------------------|---------------------------------------------------|--------|
|       | port_stress.js.py::TestPortStressSubmodel | test_stress_has_all_sub_tabs                      | Pass   |
|       | port_stress.js.py::TestPortStressSubmodel | test_stress_has_cleanup                           | Pass   |
|       | port_stress.js.py::TestPortStressSubmodel | test_stress_has_create_view                       | Pass   |
|       | port_stress.js.py::TestPortStressSubmodel | test_stress_has_load_data                         | Pass   |
|       | port_stress.js.py::TestPortStressSubmodel | test_stress_has_storm_changed                     | Pass   |
|       | port_stress.js.py::TestPortStressSubmodel | test_stress_has_switch_subtab                     | Pass   |
|       | port_stress.js.py::TestPortStressSubmodel | test_stress_renders                               | Pass   |
|       | port_stress.js.py::TestPortStressSubmodel | test_stress_state_vars                            | Pass   |
|       | stress.js.py::TestStressGaugeFunction     | test_full_js_contains_populate_gauge_dropdown     | Pass   |
|       | stress.js.py::TestStressGaugeFunction     | test_full_js_contains_stress_gauge_changed        | Pass   |
|       | stress.js.py::TestStressGaugeFunction     | test_full_js_no_unclosed_template_literals        | Pass   |
|       | stress.js.py::TestStressGaugeFunction     | test_gauge_selector_element_id_present            | Pass   |
|       | stress.js.py::TestStressGaugeFunction     | test_populate_gauge_dropdown_function_defined     | Pass   |
|       | stress.js.py::TestStressGaugeFunction     | test_populate_storm_dropdown_function_defined     | Pass   |
|       | stress.js.py::TestStressGaugeFunction     | test_storm_selector_element_id_present            | Pass   |
|       | stress.js.py::TestStressGaugeFunction     | test_stress_functions_not_before_iife_closure     | Pass   |
|       | stress.js.py::TestStressGaugeFunction     | test_stress_gauge_changed_binds_to_dropdown       | Pass   |
|       | stress.js.py::TestStressGaugeFunction     | test_stress_gauge_changed_function_defined        | Pass   |
|       | stress.js.py::TestStressGaugeFunction     | test_stress_storm_changed_binds_to_dropdown       | Pass   |
|       | stress.js.py::TestStressGaugeFunction     | test_stress_storm_changed_function_defined        | Pass   |
|       | stress.js.py::TestStressGaugeHint         | test_fallback_only_when_hint_not_matched          | Pass   |
|       | stress.js.py::TestStressGaugeHint         | test_hint_applied_before_fallback                 | Pass   |
|       | stress.js.py::TestStressGaugeHint         | test_hint_consumed_after_use                      | Pass   |
|       | stress.js.py::TestStressGaugeHint         | test_hint_fallback_to_blotter_filter              | Pass   |
|       | stress.js.py::TestStressGaugeHint         | test_load_stress_data_accepts_gauge_hint          | Pass   |
|       | stress.js.py::TestStressGaugeHint         | test_load_stress_data_stores_hint                 | Pass   |
|       | stress.js.py::TestStressGaugeHint         | test_populate_gauge_dropdown_reads_hint           | Pass   |
|       | stress.js.py::TestStressGaugeHint         | test_stress_gauge_hint_state_variable             | Pass   |
|       | stress.js.py::TestStressGaugeHint         | test_switch_tab_stress_blotter_filter_fallback    | Pass   |
|       | stress.js.py::TestStressGaugeHint         | test_switch_tab_stress_passes_gauge_hint          | Pass   |
|       | stress.js.py::TestStressGaugeHint         | test_switch_tab_stress_uses_blotter_data_fallback | Pass   |
|       | stress.js.py::TestStressKnockOut          | test_stress_ko_annotation_on_charts               | Pass   |
|       | stress.js.py::TestStressKnockOut          | test_stress_ko_status_in_table                    | Pass   |
|       | stress.js.py::TestStressKnockOut          | test_stress_stats_bar_ko_info                     | Pass   |
|       | stress.js.py::TestStressTabSubmodel       | test_stress_chart_canvas                          | Pass   |
|       | stress.js.py::TestStressTabSubmodel       | test_stress_chart_tabs                            | Pass   |
|       | stress.js.py::TestStressTabSubmodel       | test_stress_charts                                | Pass   |
|       | stress.js.py::TestStressTabSubmodel       | test_stress_cleanup                               | Pass   |
|       | stress.js.py::TestStressTabSubmodel       | test_stress_gauge_dropdown                        | Pass   |
|       | stress.js.py::TestStressTabSubmodel       | test_stress_gauges_endpoint                       | Pass   |
|       | stress.js.py::TestStressTabSubmodel       | test_stress_has_create_and_load                   | Pass   |
|       | stress.js.py::TestStressTabSubmodel       | test_stress_ko_annotation                         | Pass   |
|       | stress.js.py::TestStressTabSubmodel       | test_stress_link_property_damage                  | Pass   |
|       | stress.js.py::TestStressTabSubmodel       | test_stress_link_storm_simulation                 | Pass   |

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| Model | Module                              | Test Name                     | Status |
|-------|-------------------------------------|-------------------------------|--------|
|       | stress.js.py::TestStressTabSubmodel | test_stress_renders           | Pass   |
|       | stress.js.py::TestStressTabSubmodel | test_stress_stats_bar         | Pass   |
|       | stress.js.py::TestStressTabSubmodel | test_stress_storm_dropdown    | Pass   |
|       | stress.js.py::TestStressTabSubmodel | test_stress_surface_table     | Pass   |
|       | stress.js.py::TestStressTabSubmodel | test_stress_trade_table       | Pass   |
|       | stress.js.py::TestStressTabSubmodel | test_surface_table_scrollable | Pass   |

### 3 Flood Risk (Depth-Damage) (MKM-DD-001)

Table 4: Test Results Summary — Flood Risk (Depth-Damage) (MKM-DD-001)

|             |       |
|-------------|-------|
| Total Tests | 29    |
| Passed      | 29    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.00s |

Table 5: Detailed Test Results — Flood Risk (Depth-Damage)

| Test Description                                                                      | Acceptance Criteria                                                       | Result | Time   |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------|--------|
| At distance = characteristic length, factor must equal $1/e \approx 0.368$ .          | <code>abs(f - 1.0 / math.e) &lt; 0.001</code>                             | Pass   | 0.000s |
| At 5x the decay length, attenuation should be $\leq 1\%$ ( $e^{-5} \approx 0.0067$ ). | <code>f &lt; 0.01</code>                                                  | Pass   | 0.000s |
| Negative distance is clamped to zero; must return 1.0.                                | <code>compute_attenuation(-100) == 1.0</code>                             | Pass   | 0.000s |
| At zero distance, attenuation factor must be 1.0 (no decay).                          | <code>compute_attenuation(0) == 1.0</code>                                | Pass   | 0.000s |
| Zero decay length means instant decay; must return 0.0 for any distance $\neq 0$ .    | <code>compute_attenuation(100, 0) == 0.0</code>                           | Pass   | 0.000s |
| Every hour from 0..n must be present in output, in order.                             | <code>hours == list(range(len(gauge_readings)))</code>                    | Pass   | 0.000s |
| Flood depth must be non-negative at every time step.                                  | <code>r['depth_m'] &gt;= 0.0</code>                                       | Pass   | 0.000s |
| Each hydrograph entry must contain hour, wse_m, depth_m, and flooded.                 | <code>'hour' in r; 'wse_m' in r (+2 more)</code>                          | Pass   | 0.000s |
| Empty gauge readings must return empty property hydrograph.                           | <code>build_property_hydrograph([], 8.0, 1.0, 0.9, 6.0, 0.3) == []</code> | Pass   | 0.000s |
| When floor level exceeds peak WSE, no time step should be flooded.                    | <code>r['flooded'] is False</code>                                        | Pass   | 0.000s |
| With travel time of 5 hours, first 5 hours must show zero flood depth.                | <code>r['depth_m'] == 0.0; r['flooded'] is False</code>                   | Pass   | 0.000s |
| Property hydrograph must have same number of time steps as gauge input.               | <code>len(result) == len(gauge_readings)</code>                           | Pass   | 0.000s |

| Test Description                                                            | Acceptance Criteria                                                     | Result | Time   |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------|--------|--------|
| Low-elevation property at gauge location must flood at peak water level.    | <code>max(peak_depths) &gt; 0</code>                                    | Pass   | 0.000s |
| Slope from 10m to 8m over 1000m should be 0.002 (2m/1000m).                 | <code>abs(s - 0.002) &lt; 1e-6</code>                                   | Pass   | 0.000s |
| Flat terrain (equal elevations) must return MIN_SLOPE, not zero.            | <code>s == MIN_SLOPE</code>                                             | Pass   | 0.000s |
| Slope must be the same regardless of flow direction (uses absolute value).  | <code>s1 == s2</code>                                                   | Pass   | 0.000s |
| Zero horizontal distance must return MIN_SLOPE to avoid division by zero.   | <code>t == 0.0; s == MIN_SLOPE</code>                                   | Pass   | 0.000s |
| Monotonicity: deeper water must produce higher velocity at constant slope.  | <code>v2 &gt; v1</code>                                                 | Pass   | 0.000s |
| Negative depth is physically invalid; must return zero velocity.            | <code>compute_manning_velocity(-1.0, 0.005) == 0.0</code>               | Pass   | 0.000s |
| Negative slope (downhill in opposite direction) must use absolute value.    | <code>v_pos == v_neg</code>                                             | Pass   | 0.000s |
| Manning velocity at d=1.0m, S=0.005, n=0.04 should be ~1.77 m/s.            | <code>1.5 &lt; v &lt; 2.0</code>                                        | Pass   | 0.000s |
| Slopes below MIN_SLOPE must be clamped to MIN_SLOPE to avoid                | <code>v == v_min</code>                                                 | Pass   | 0.000s |
| Monotonicity: steeper slope must produce higher velocity at constant depth. | <code>v2 &gt; v1</code>                                                 | Pass   | 0.000s |
| Zero depth must produce zero velocity (no flow without water).              | <code>compute_manning_velocity(0.0, 0.005) == 0.0</code>                | Pass   | 0.000s |
| Zero roughness coefficient causes division by zero; must return 0.          | <code>compute_manning_velocity(1.0, 0.005, roughness=0.0) == 0.0</code> | Pass   | 0.000s |
| Travel time for 500m at d=1.0m, S=0.005 should be well under 1 hour.        | <code>0 &lt; t &lt; 1.0</code>                                          | Pass   | 0.000s |
| Monotonicity: greater distance must produce longer travel time.             | <code>t2 &gt; t1</code>                                                 | Pass   | 0.000s |
| Zero depth produces zero velocity, so travel time must be infinite.         | <code>t == float('inf')</code>                                          | Pass   | 0.000s |
| Zero distance must produce zero travel time.                                | <code>t == 0.0; s == MIN_SLOPE</code>                                   | Pass   | 0.000s |

## 4 GEV Hazard Model (MKM-GH-001)

Table 6: Test Results Summary — GEV Hazard Model (MKM-GH-001)

|             |       |
|-------------|-------|
| Total Tests | 16    |
| Passed      | 16    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.27s |

Table 7: Detailed Test Results — GEV Hazard Model

| Test Description                             | Acceptance Criteria                                                                                                                                                           | Result | Time   |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Exceedance probability in range              | <code>0.0 &lt;= prob &lt;= 1.0</code>                                                                                                                                         | Pass   | 0.000s |
| Fit returns three floats                     | <code>isinstance(shape, float);</code><br><code>isinstance(loc, float) (+1 more)</code>                                                                                       | Pass   | 0.018s |
| Force gumbel sets shape zero                 | <code>shape == 0.0</code>                                                                                                                                                     | Pass   | 0.001s |
| Higher threshold lower probability           | <code>p_high &lt; p_low</code>                                                                                                                                                | Pass   | 0.000s |
| Return level increases with period           | <code>r1_100 &gt; r1_10</code>                                                                                                                                                | Pass   | 0.000s |
| Compute all responses returns dict           | <code>isinstance(responses,</code><br><code>dict); len(responses) ==</code><br><code>len(synthetic_flat_gauges) (+1</code><br><code>more)</code>                              | Pass   | 0.000s |
| Compute response returns gauge re-<br>sponse | <code>isinstance(response, GaugeResponse);</code><br><code>response.gauge_id == gauge_id (+1</code><br><code>more)</code>                                                     | Pass   | 0.000s |
| Response threshold flags                     | <code>response.exceeded_alert;</code><br><code>response.exceeded_warning</code>                                                                                               | Pass   | 0.000s |
| Annual flood probabilities in range          | <code>0.0 &lt;= curve.annual_flood_prob.alert</code><br><code>&lt;= 1.0; 0.0 &lt;=</code><br><code>curve.annual_flood_prob.warning</code><br><code>&lt;= 1.0 (+1 more)</code> | Pass   | 0.082s |
| Build returns dict of curves                 | <code>isinstance(curves, dict); len(curves)</code><br><code>== len(synthetic_flat_gauges) (+2</code><br><code>more)</code>                                                    | Pass   | 0.083s |
| Each curve is gauge hazard curve             | <code>isinstance(curve, GaugeHazardCurve);</code><br><code>curve.gauge_id == gauge_id</code>                                                                                  | Pass   | 0.090s |
| Cumulative prob increasing                   | <code>probs[i] &gt;= probs[i - 1]</code>                                                                                                                                      | Pass   | 0.000s |
| Probabilities sum to one                     | <code>abs(point.survival_prob +</code><br><code>point.prob.at_least_one - 1.0) ...</code>                                                                                     | Pass   | 0.000s |
| Returns correct length                       | <code>len(ts) == 5</code>                                                                                                                                                     | Pass   | 0.000s |



| Test Description              | Acceptance Criteria                                | Result | Time   |
|-------------------------------|----------------------------------------------------|--------|--------|
| Returns term structure points | <code>isinstance(point, TermStructurePoint)</code> | Pass   | 0.000s |
| Survival prob decreasing      | <code>survivals[i] &lt;= survivals[i - 1]</code>   | Pass   | 0.000s |

## 5 Mortgage Pricer (MKM-MP-001)

Table 8: Test Results Summary — Mortgage Pricer (MKM-MP-001)

|             |       |
|-------------|-------|
| Total Tests | 67    |
| Passed      | 67    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.05s |

Table 9: Detailed Test Results — Mortgage Pricer

| Test Description                                                | Acceptance Criteria                                                                                               | Result | Time   |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------|--------|
| Default ids assigned                                            | <code>results[0]["mortgage_id"] == "MORTGAGE_0";</code><br><code>results[0]["property_id"] == "PROPERTY_0"</code> | Pass   | 0.001s |
| Error isolation                                                 | <code>len(results) == 3;</code> "error" not in <code>results[0]</code> (+2 more)                                  | Pass   | 0.001s |
| Prices multiple mortgages                                       | <code>len(results) == 2;</code><br><code>results[0]["mortgage_id"] == "M1"</code> (+1 more)                       | Pass   | 0.001s |
| Custom tax rate                                                 | <code>MortgagePricer(tax_rate=0.40).tax_rate == 0.40</code>                                                       | Pass   | 0.000s |
| Default tax rate                                                | <code>MortgagePricer().tax_rate == 0.20</code>                                                                    | Pass   | 0.000s |
| Custom tax rate                                                 | <code>spread_high_tax &gt; spread_low_tax</code>                                                                  | Pass   | 0.000s |
| Floor at 10bps                                                  | <code>spread &gt;= 0.001</code>                                                                                   | Pass   | 0.000s |
| Higher affordability higher spread                              | <code>spread_low_income &gt; spread_high_income</code>                                                            | Pass   | 0.000s |
| Negative income fallback                                        | <code>spread == 0.15</code>                                                                                       | Pass   | 0.000s |
| Spread bounded by schedule                                      | <code>spread &gt;= 0.001; spread &lt; 0.05</code>                                                                 | Pass   | 0.000s |
| Zero income fallback                                            | <code>spread == 0.15</code>                                                                                       | Pass   | 0.000s |
| debug=True in the credit spread calculation.                    | <code>"credit_spread" in result</code>                                                                            | Pass   | 0.001s |
| debug=True exercises logging branches without error.            | <code>result["mortgage_value"] &gt; 0</code>                                                                      | Pass   | 0.001s |
| Zero income with debug=True exercises zero-income debug branch. | <code>result["credit_spread"] &gt; 0</code>                                                                       | Pass   | 0.001s |
| flood_risk_category=None with debug=True.                       | <code>result["mortgage_value"] &gt; 0</code>                                                                      | Pass   | 0.001s |

| Test Description                                                             | Acceptance Criteria                                                                                                                | Result | Time   |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| When interest_rate=0, monthly_rate==0 → loan_amount/n_periods formula.       | result["mortgage_value"] > 0                                                                                                       | Pass   | 0.001s |
| Batch passes flood risk                                                      | results[0]["flood_risk_factor"] == 1.40; results[1]["flood_risk_factor"] == 1.00 (+1 more)                                         | Pass   | 0.001s |
| Case insensitive                                                             | MortgagePricer.calculate_flood_risk_impact("high") =...;<br>MortgagePricer.calculate_flood_risk_impact("VERY LOW") ==... (+1 more) | Pass   | 0.000s |
| Flood factor in result                                                       | result["flood_risk_factor"] == 1.20                                                                                                | Pass   | 0.001s |
| Flood risk increases spread                                                  | high["credit_spread"] > low["credit_spread"];<br>high["flood_risk_factor"] > low["flood_risk_factor"] (+1 more)                    | Pass   | 0.001s |
| Flood risk multipliers[high-1.4] [category='High', expected=1.4]             | MortgagePricer.calculate_flood_risk_impact("High") == e...                                                                         | Pass   | 0.000s |
| Flood risk multipliers[low-1.05] [category='Low', expected=1.05]             | MortgagePricer.calculate_flood_risk_impact("Low") == e...                                                                          | Pass   | 0.000s |
| Flood risk multipliers[medium-1.2] [category='Medium', expected=1.2]         | MortgagePricer.calculate_flood_risk_impact("Medium") == e...                                                                       | Pass   | 0.000s |
| Flood risk multipliers[very high-1.75] [category='Very High', expected=1.75] | MortgagePricer.calculate_flood_risk_impact("Very High") == e...                                                                    | Pass   | 0.000s |
| Flood risk multipliers[very low-1.0] [category='Very Low', expected=1.0]     | MortgagePricer.calculate_flood_risk_impact("Very Low") == e...                                                                     | Pass   | 0.000s |
| No flood risk defaults to unity                                              | result["flood_risk_factor"] == 1.0                                                                                                 | Pass   | 0.001s |
| None returns unity                                                           | MortgagePricer.calculate_flood_risk_impact("None") == 1.0                                                                          | Pass   | 0.000s |
| Unknown category returns unity                                               | MortgagePricer.calculate_flood_risk_impact("Unknown") == 1.0;<br>MortgagePricer.calculate_flood_risk_impact("Extreme") == 1.0      | Pass   | 0.000s |
| Ltv thresholds[0.5-1.0] [ltv=0.5, expected=1.0]                              | pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...                                                                       | Pass   | 0.000s |
| Ltv thresholds[0.8-1.0] [ltv=0.8, expected=1.0]                              | pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...                                                                       | Pass   | 0.000s |
| Ltv thresholds[0.85-1.1] [ltv=0.85, expected=1.1]                            | pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...                                                                       | Pass   | 0.000s |

| Test Description                                  | Acceptance Criteria                                                                                 | Result | Time   |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| Ltv thresholds[0.9-1.1] [ltv=0.9, expected=1.1]   | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Ltv thresholds[0.91-1.3] [ltv=0.91, expected=1.3] | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Ltv thresholds[0.95-1.3] [ltv=0.95, expected=1.3] | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Ltv thresholds[0.96-1.5] [ltv=0.96, expected=1.5] | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Ltv thresholds[1.0-1.5] [ltv=1.0, expected=1.5]   | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Zero property value                               | <code>pricer.calculate_loan_to_value_impact(100.0, 0) == 1.5</code>                                 | Pass   | 0.000s |
| Higher rate higher payment                        | <code>high &gt; low</code>                                                                          | Pass   | 0.000s |
| Shorter term higher payment                       | <code>short &gt; long</code>                                                                        | Pass   | 0.000s |
| £200k at 3.5% over 25yr<br>£1,001.25/month.       | <code>995 &lt; payment &lt; 1010</code>                                                             | Pass   | 0.000s |
| Zero-rate degenerate case: $M = P / n$ .          | <code>payment == pytest.approx(1000.0)</code>                                                       | Pass   | 0.000s |
| Each mortgage has property id                     | <code>mapping.get("property_id") is not None</code>                                                 | Pass   | 0.004s |
| Processing stats                                  | <code>stats["successful_mortgages"] == 5;</code><br><code>stats["failed_mortgages"] == 0</code>     | Pass   | 0.004s |
| Output file exists                                | <code>result["file_path"].exists()</code>                                                           | Pass   | 0.005s |
| Output file is valid json                         | <code>isinstance(data, dict)</code>                                                                 | Pass   | 0.005s |
| Generate returns expected keys                    | <code>"data" in result;</code> <code>"file_path" in result (+1 more)</code>                         | Pass   | 0.003s |
| Mortgage count matches properties                 | <code>len(result["data"]["mortgages"]) == 5</code>                                                  | Pass   | 0.003s |
| Mortgage id format                                | <code>re.match(r"~MORT-[a-f0-9]{8}\$", mid)</code>                                                  | Pass   | 0.003s |
| Mortgage ids unique                               | <code>len(ids) == len(set(ids))</code>                                                              | Pass   | 0.002s |
| All errors returns error                          | <code>"error" in calculate_portfolio_metrics(results)</code>                                        | Pass   | 0.000s |
| High risk count                                   | <code>calculate_portfolio_metrics(results)["high_risk_mortgages"]</code>                            | Pass   | 0.000s |
| Mixed results exclude errors                      | <code>metrics["total_mortgages"] == 1;</code><br><code>metrics["error_count"] == 1</code>           | Pass   | 0.001s |
| Valid results                                     | <code>metrics["total_mortgages"] == 2;</code><br><code>metrics["error_count"] == 0 (+3 more)</code> | Pass   | 0.001s |
| Balance converges to zero                         | <code>result["outstanding_balance"][-1] &lt; 1.0</code>                                             | Pass   | 0.001s |
| Expected losses non negative                      | <code>all(e1 &gt;= 0 for e1 in result["expected_losses"])</code>                                    | Pass   | 0.001s |

| Test Description                    | Acceptance Criteria                                                                                                      | Result | Time   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Fair value less than par            | <code>result["mortgage_value"] &lt; base.mortgage_params["loan_amo...; result["discount_to_par"] &gt; 0 (+1 more)</code> | Pass   | 0.001s |
| Hazard rates positive               | <code>result["hazard_rates"][0] == 0.0; all(result["hazard_rates"][i] &gt; 0 for i in range(1, len(r...</code>           | Pass   | 0.001s |
| High ltv increases spread           | <code>high_ltv["credit_spread"] &gt; low_ltv["credit_spread"]; high_ltv["ltv_factor"] &gt; low_ltv["ltv_factor"]</code>  | Pass   | 0.001s |
| Higher haircut more loss            | <code>high["pv_losses"] &gt; low["pv_losses"]</code>                                                                     | Pass   | 0.001s |
| Input validation clamps             | <code>result["mortgage_value"] &gt; 0</code>                                                                             | Pass   | 0.000s |
| Ltv factor in result                | <code>"ltv_factor" in result; result["ltv_factor"] == 1.0</code>                                                         | Pass   | 0.001s |
| Raises on invalid types             | —                                                                                                                        | Pass   | 0.000s |
| Recovery haircut zero low ltv       | <code>all(lgd == 0 for lgd in result["lgds"])</code>                                                                     | Pass   | 0.001s |
| Survival first period less than one | <code>result["survival_probs"][1] &lt; 1.0</code>                                                                        | Pass   | 0.001s |
| Survival monotonically decreasing   | <code>surv[i] &lt;= surv[i - 1]</code>                                                                                   | Pass   | 0.001s |
| Positive rate exceeds principal     | <code>cost &gt; 200.000</code>                                                                                           | Pass   | 0.000s |
| Zero rate equals principal          | <code>cost == pytest.approx(100_000.0)</code>                                                                            | Pass   | 0.000s |



## 6 PRS Pricing Model (MKM-PR-001)

Table 10: Test Results Summary — PRS Pricing Model (MKM-PR-001)

|             |       |
|-------------|-------|
| Total Tests | 61    |
| Passed      | 61    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 6.47s |

Table 11: Detailed Test Results — PRS Pricing Model

| Test Description                                 | Acceptance Criteria                                                                                 | Result | Time   |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| Basis waterfall in generated output              | <code>bw1["model.uncertainty_bp"] == 2.0;</code><br><code>bw1["terrain_bp"] == 2.0 (+6 more)</code> | Pass   | 0.534s |
| Composition flat penalty                         | <code>result["composition_bp"] ==</code><br><code>COMPOSITION_BASIS_BPS["Flat"]</code>              | Pass   | 0.000s |
| Composition pre2000 penalty                      | <code>result["composition_bp"] == 1.0</code>                                                        | Pass   | 0.000s |
| Distance basis capped                            | <code>result["distance_bp"] ==</code><br><code>DISTANCE_MAX_BPS</code>                              | Pass   | 0.000s |
| Distance basis scales linearly                   | <code>result["distance_bp"] ==</code><br><code>pytest.approx(1.0, abs=0.1)</code>                   | Pass   | 0.000s |
| Elevation above gauge reduces spread             | <code>result["elevation_bp"] ==</code><br><code>pytest.approx(-1.0, abs=0.1)</code>                 | Pass   | 0.000s |
| Elevation below gauge no benefit                 | <code>result["elevation_bp"] == 0.0</code>                                                          | Pass   | 0.000s |
| Elevation capped at 3bp                          | <code>result["elevation_bp"] ==</code><br><code>-ELEVATION_MAX_BENEFIT_BPS</code>                   | Pass   | 0.000s |
| Model uncertainty is fixed                       | <code>result["model.uncertainty_bp"] ==</code><br><code>MODEL_UNCERTAINTY_BPS</code>                | Pass   | 0.000s |
| Terrain zone1 low risk                           | <code>result["terrain_bp"] == 0.0</code>                                                            | Pass   | 0.000s |
| Terrain zone3 high risk                          | <code>result["terrain_bp"] ==</code><br><code>TERRAIN_BASIS_BPS["Zone 3a"]</code>                   | Pass   | 0.000s |
| Total basis sums components                      | <code>result["total_basis_bp"] ==</code><br><code>pytest.approx(expected_total,...</code>           | Pass   | 0.000s |
| 1Y from Feb 2026 = last Friday May 2027.         | <code>mat == last_friday_of_month(2027, 5)</code>                                                   | Pass   | 0.000s |
| 3Y from Feb 2026 = last Friday May 2029.         | <code>mat == last_friday_of_month(2029, 5);</code><br><code>mat.month == 5 (+1 more)</code>         | Pass   | 0.000s |
| Jan 2027 uses Aug 2026 roll → Nov 2029 maturity. | <code>mat == last_friday_of_month(2029, 11)</code>                                                  | Pass   | 0.000s |

| Test Description                                                             | Acceptance Criteria                                                                                              | Result | Time   |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------|--------|
| 3Y from Sep 2026 = last Friday Nov 2029 (Aug roll).                          | <code>mat == last_friday_of_month(2029, 11);</code><br><code>mat.month == 11</code>                              | Pass   | 0.000s |
| 5Y from Feb 2026 = last Friday May 2031.                                     | <code>mat == last_friday_of_month(2031, 5)</code>                                                                | Pass   | 0.000s |
| 5Y from Sep 2026 = last Friday Nov 2031.                                     | <code>mat == last_friday_of_month(2031, 11)</code>                                                               | Pass   | 0.000s |
| All maturities should fall on a Friday.                                      | <code>mat.weekday() == 4</code>                                                                                  | Pass   | 0.000s |
| 3Y maturity jumps from May to Nov at the Aug roll.                           | <code>may_mat.month == 5; nov_mat.month == 11 (+1 more)</code>                                                   | Pass   | 0.000s |
| Aug onwards uses Aug roll.                                                   | <code>roll == last_friday_of_month(2026, 8);</code><br><code>roll.month == 8</code>                              | Pass   | 0.000s |
| December uses the same year's Aug roll.                                      | <code>roll.month == 8; roll.year == 2026</code>                                                                  | Pass   | 0.000s |
| Feb-Jul uses Feb roll.                                                       | <code>roll == last_friday_of_month(2026, 2);</code><br><code>roll.month == 2</code>                              | Pass   | 0.000s |
| January uses the previous year's Aug roll.                                   | <code>roll == last_friday_of_month(2026, 8);</code><br><code>roll.month == 8 (+1 more)</code>                    | Pass   | 0.000s |
| July is the last month of the Feb roll period.                               | <code>roll.month == 2; roll.year == 2026</code>                                                                  | Pass   | 0.000s |
| March is still in the Feb roll period.                                       | <code>roll.month == 2; roll.year == 2026</code>                                                                  | Pass   | 0.000s |
| <code>compute_maturity_date</code> with no <code>ref_date</code> uses today. | <code>isinstance(mat, date); mat.weekday() == 4</code>                                                           | Pass   | 0.000s |
| <code>current_roll_date(None)</code> uses today and returns a valid date.    | <code>isinstance(roll, date);</code><br><code>roll.weekday() == 4</code>                                         | Pass   | 0.000s |
| <code>maturity_table</code> with no <code>ref_date</code> uses today.        | <code>len(table) == 5;</code><br><code>all(isinstance(e['maturity_date'], date) for e in table)</code>           | Pass   | 0.000s |
| Empty propertyts dir                                                         | <code>stats["total_properties"] == 0;</code><br><code>stats["properties_with_gev"] == 0</code>                   | Pass   | 0.002s |
| Generator should still work without gauge hazard curves (empty basis).       | <code>stats["properties_with_gev"] == 2;</code><br><code>stats["properties_with_floor"] == 1</code><br>(+1 more) | Pass   | 0.527s |
| No propertyts dir                                                            | —                                                                                                                | Pass   | 0.000s |
| Aug 2026: 31st is Monday, last Friday = 28th.                                | <code>last_friday_of_month(2026, 8) == date(2026, 8, 28)</code>                                                  | Pass   | 0.000s |
| Feb 2026: 28th is Saturday, last Friday = 27th.                              | <code>last_friday_of_month(2026, 2) == date(2026, 2, 27)</code>                                                  | Pass   | 0.000s |
| May 2029: 31st is Thursday, last Friday = 25th.                              | <code>last_friday_of_month(2029, 5) == date(2029, 5, 25)</code>                                                  | Pass   | 0.000s |

| Test Description                                       | Acceptance Criteria                                                                                  | Result | Time   |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------|--------|
| Nov 2029: 30th is Friday.                              | <code>last_friday_of_month(2029, 11) == date(2029, 11, 30)</code>                                    | Pass   | 0.000s |
| Result should always be a Friday (weekday=4).          | <code>d.weekday() == 4</code>                                                                        | Pass   | 0.000s |
| Actual years should be strictly increasing.            | <code>years[i] &gt; years[i - 1]</code>                                                              | Pass   | 0.000s |
| All maturities should be in May for the Feb roll.      | <code>e['maturity_date'].month == 5</code>                                                           | Pass   | 0.000s |
| All maturities should be in November for the Aug roll. | <code>e['maturity_date'].month == 11</code>                                                          | Pass   | 0.000s |
| Maturity string should be formatted as 'DD Mon YYYY'.  | <code>len(e['maturity_str']) &gt; 8; 'May' in e['maturity_str'] or 'Nov' in e['maturity_str']</code> | Pass   | 0.000s |
| Default table has 5 entries (1Y through 5Y).           | <code>len(table) == 5</code>                                                                         | Pass   | 0.000s |
| Tenors should be 1, 2, 3, 4, 5.                        | <code>tenors == [1, 2, 3, 4, 5]</code>                                                               | Pass   | 0.000s |
| Positive spread                                        | <code>spread &gt; 0; 50 &lt; spread &lt; 500</code>                                                  | Pass   | 0.000s |
| Spread increases with hazard                           | <code>spread_high &gt; spread_low</code>                                                             | Pass   | 0.000s |
| Spread values at different tenors                      | <code>s &gt; 0</code>                                                                                | Pass   | 0.000s |
| Zero hazard rate                                       | <code>spread == 0.0</code>                                                                           | Pass   | 0.000s |
| Basis calculation                                      | <code>len(nearest) &gt; 0; ng0["gauge_id"] == "GAUGE-001" (+5 more)</code>                           | Pass   | 0.534s |
| Depth thresholds                                       | <code>"any_flood" in thresholds; "moderate" in thresholds (+4 more)</code>                           | Pass   | 0.534s |
| Generate produces output                               | <code>output_path.exists(); "metadata" in data (+2 more)</code>                                      | Pass   | 0.540s |
| Gev params structure                                   | <code>"shape" in gev; "loc" in gev (+2 more)</code>                                                  | Pass   | 0.541s |
| High transmission rate                                 | <code>prop3["summary"]["flood.transmission_rate"] == 1.0</code>                                      | Pass   | 0.532s |
| Output structure                                       | <code>"PROP-001" in curves; "PROP-002" in curves (+21 more)</code>                                   | Pass   | 0.532s |
| Properties with gev                                    | <code>stats["properties_with_gev"] == 2; stats["properties_with_floor"] == 1 (+1 more)</code>        | Pass   | 0.595s |
| Survival decreases with tenor                          | <code>survival[i] &lt;= survival[i - 1]</code>                                                       | Pass   | 0.530s |
| Term structure                                         | <code>ts["tenors"] == TENORS; threshold_name in ts (+3 more)</code>                                  | Pass   | 0.533s |
| Recovery rates by trigger                              | <code>len(spreads) &gt; 0</code>                                                                     | Pass   | 0.538s |
| Recovery reduces spread                                | <code>spread_with_recovery &lt; spread_no_recovery</code>                                            | Pass   | 0.000s |

| Test Description                                        | Acceptance Criteria                                           | Result | Time   |
|---------------------------------------------------------|---------------------------------------------------------------|--------|--------|
| All tenors should round-trip through maturity and back. | <code>tenor_from_maturity(mat, ref) == t</code>               | Pass   | 0.000s |
| A date not in the table returns the closest tenor.      | <code>result in [1, 2, 3, 4, 5]</code>                        | Pass   | 0.000s |
| Exact maturity date returns correct tenor.              | <code>tenor_from_maturity(mat, date(2026, 2, 24)) == 3</code> | Pass   | 0.000s |

## 7 Property Valuation (MKM-PV-001)

Table 12: Test Results Summary — Property Valuation (MKM-PV-001)

|             |        |
|-------------|--------|
| Total Tests | 55     |
| Passed      | 55     |
| Failed      | 0      |
| Skipped     | 0      |
| Duration    | 19.07s |

Table 13: Detailed Test Results — Property Valuation

| Test Description                                                 | Acceptance Criteria                                                       | Result | Time   |
|------------------------------------------------------------------|---------------------------------------------------------------------------|--------|--------|
| Line 541: non-dict schema → returns {}.                          | <code>result == {}</code>                                                 | Pass   | 0.000s |
| Line 544-545: 'type'/'options'/'description' fields are skipped. | <code>"type" not in result; "options" not in result</code>                | Pass   | 0.000s |
| Calculate insurance premium flood risk high                      | <code>premium &gt; 0</code>                                               | Pass   | 0.097s |
| Line 214: area=None → computes it first.                         | <code>isinstance(premium, float); premium &gt; 0</code>                   | Pass   | 0.098s |
| Line 211: property_value=None → computes it first.               | <code>isinstance(premium, float); premium &gt; 0</code>                   | Pass   | 0.273s |
| Calculate monthly rent all types                                 | <code>800 &lt;= rent &lt;= 15.000</code>                                  | Pass   | 0.000s |
| Lines 165-166: area=None → computes it first.                    | <code>isinstance(rent, float); rent &gt; 0</code>                         | Pass   | 0.000s |
| Lines 162-163: property_value=None → computes it first.          | <code>isinstance(rent, float); rent &gt; 0</code>                         | Pass   | 0.098s |
| Calculate monthly rent positive                                  | <code>isinstance(rent, float); 800 &lt;= rent &lt;= 15.000</code>         | Pass   | 0.000s |
| Calculate property area all types                                | <code>area &gt; 0</code>                                                  | Pass   | 0.000s |
| Lines 128-129: invalid date format → random years_ago.           | <code>isinstance(price, float)</code>                                     | Pass   | 0.000s |
| Lines 120-121: property_value=None → computes it first.          | <code>isinstance(price, float); price &gt; 0</code>                       | Pass   | 0.098s |
| Calculate sale price positive                                    | <code>isinstance(price, float); 50.000 &lt;= price &lt;= 8.000.000</code> | Pass   | 0.000s |



| Test Description                                                  | Acceptance Criteria                                         | Result | Time   |
|-------------------------------------------------------------------|-------------------------------------------------------------|--------|--------|
| Lines 135-137: sale_date 1-2 years ago.                           | isinstance(price, float)                                    | Pass   | 0.000s |
| Lines 139-140: sale_date > 2 years ago.                           | isinstance(price, float)                                    | Pass   | 0.000s |
| Lines 124-128: sale_date within 1 year.                           | isinstance(price, float)                                    | Pass   | 0.000s |
| Line 60-61: datetime → isoformat string.                          | "2024-01-01" in result                                      | Pass   | 0.000s |
| Lines 64-65: np.float64 → float.                                  | "3.14" in result                                            | Pass   | 0.000s |
| Lines 62-63: np.integer → int.                                    | result == "42"                                              | Pass   | 0.000s |
| Lines 66-67: np.ndarray → list.                                   | result == [1, 2, 3]                                         | Pass   | 0.000s |
| Line 68: super().default() raises TypeError for unknown type.     | —                                                           | Pass   | 0.000s |
| Lines 510-517: each fallback location has lat/lon/name/elevation. | "lat" in loc; "lon" in loc (+2 more)                        | Pass   | 0.000s |
| Lines 499-518: fallback generates right number of locations.      | len(locs) == 5                                              | Pass   | 0.001s |
| Lines 505-506: uses params.get_elevation if available.            | all(loc["elevation"] == 10.0 for loc in locs)               | Pass   | 0.000s |
| Lines 608-612: generate_properties returns result dict.           | isinstance(result, dict); "data" in result (+1 more)        | Pass   | 0.923s |
| Lines 610-612: generate_properties creates output file.           | result["file_path"].exists()                                | Pass   | 0.590s |
| Calculate mortgage financials returns dict                        | isinstance(result, dict); "loan_amount" in result (+2 more) | Pass   | 0.000s |
| Generate financial data keys                                      | "mortgage_id" in data; "loan_amount" in data (+1 more)      | Pass   | 0.000s |
| Interest rate range                                               | 0.01 <= data["interest_rate"] <= 0.15                       | Pass   | 0.000s |
| Loan amount less than value                                       | data["loan_amount"] <= info["property_value"] * 1.05        | Pass   | 0.000s |
| Mortgage id format                                                | re.match(r"~MORT-[a-f0-9]{8}\$", data["mortgage_id"])       | Pass   | 0.000s |
| Quality consistency check                                         | isinstance(corrected, dict)                                 | Pass   | 0.000s |
| Calculate insurance premium positive                              | isinstance(premium, (int, float)); premium > 0              | Pass   | 0.100s |
| Calculate property area positive                                  | isinstance(area, (int, float)); area > 0                    | Pass   | 0.000s |
| Calculate property value in range                                 | 50_000 <= value <= 10_000_000                               | Pass   | 1.056s |
| Generate bedrooms detached                                        | isinstance(beds, int); 1 <= beds <= 7                       | Pass   | 0.000s |
| Generate bedrooms flat                                            | isinstance(beds, int); 1 <= beds <= 4                       | Pass   | 0.000s |

| Test Description                                                 | Acceptance Criteria                                                                   | Result | Time   |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|--------|
| Generate postcode format                                         | <code>isinstance(postcode, str);<br/>len(postcode) &gt;= 5</code>                     | Pass   | 0.000s |
| Generate field value decimal                                     | <code>isinstance(val, (int, float))</code>                                            | Pass   | 0.196s |
| Generate field value text                                        | <code>isinstance(val, str)</code>                                                     | Pass   | 0.099s |
| Coordinates in thames range                                      | <code>51.0 &lt;= loc["lat"] &lt;= 52.0;    -1.0 &lt;=<br/>loc["lon"] &lt;= 1.0</code> | Pass   | 2.964s |
| Processing stats                                                 | <code>stats["successful_properties"] == 5;<br/>stats["failed_properties"] == 0</code> | Pass   | 1.535s |
| Output file exists                                               | <code>result["file_path"].exists()</code>                                             | Pass   | 1.523s |
| Output file is valid json                                        | <code>isinstance(data, dict)</code>                                                   | Pass   | 1.479s |
| Generate correct count                                           | <code>len(result["data"]["properties"]) ==<br/>5</code>                               | Pass   | 1.517s |
| Generate returns expected keys                                   | <code>"data" in result;        "file_path" in<br/>result (+1 more)</code>             | Pass   | 1.506s |
| Property id format                                               | <code>re.match(r"~PROP-[a-f0-9]{8}\$", pid)</code>                                    | Pass   | 1.521s |
| Property ids unique                                              | <code>len(ids) == len(set(ids))</code>                                                | Pass   | 3.189s |
| Construction year in range                                       | <code>1600 &lt;= year &lt;= 2026</code>                                               | Pass   | 0.000s |
| Property id format                                               | <code>re.match(r"~PROP-[a-f0-9]{8}\$",<br/>meta["property_id"])</code>                | Pass   | 0.101s |
| Returns dict with required keys                                  | <code>"property_id" in meta; "property_type"<br/>in meta</code>                       | Pass   | 0.103s |
| Lines 73-75: generate_grid_reference returns a non-empty string. | <code>isinstance(ref, str); len(ref) &gt; 4</code>                                    | Pass   | 0.000s |
| Lines 50-52: generate_owner_name returns 'Firstname Lastname'.   | <code>len(parts) == 2;    all(p[0].isupper()<br/>for p in parts)</code>               | Pass   | 0.000s |
| Lines 43-45: generate_past_date returns YYYY-MM-DD string.       | <code>isinstance(result, str);    len(parts)<br/>== 3</code>                          | Pass   | 0.000s |
| Custom days_range is respected (result is a past date).          | <code>dt &lt; datetime.now()</code>                                                   | Pass   | 0.000s |

## 8 Storm-Gauge Response Model (MKM-SG-001)

Table 14: Test Results Summary — Storm-Gauge Response Model (MKM-SG-001)

|             |       |
|-------------|-------|
| Total Tests | 81    |
| Passed      | 81    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.04s |

Table 15: Detailed Test Results — Storm-Gauge Response Model

| Test Description                      | Acceptance Criteria                                                                           | Result | Time   |
|---------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| Gauge ids match                       | <code>{r.gauge_id for r in responses} ==</code><br><code>{gauge.gauge_id, gauge...</code>     | Pass   | 0.000s |
| Near gauge higher response            | <code>near_resp.peak_level &gt;=</code><br><code>far_resp.peak_level</code>                   | Pass   | 0.000s |
| Returns one response per gauge        | <code>len(responses) == 2</code>                                                              | Pass   | 0.000s |
| All kernels decrease with distance    | <code>near &gt;= far</code>                                                                   | Pass   | 0.000s |
| All kernels in 0 1 range              | <code>0.0 &lt;= decay &lt;= 1.0</code>                                                        | Pass   | 0.000s |
| Exponential at half                   | <code>decay ==</code><br><code>pytest.approx(math.exp(-0.5),</code><br><code>rel=1e-4)</code> | Pass   | 0.000s |
| Exponential at one lambda             | <code>decay ==</code><br><code>pytest.approx(math.exp(-1.0),</code><br><code>rel=1e-4)</code> | Pass   | 0.000s |
| Gaussian at one sigma                 | <code>decay ==</code><br><code>pytest.approx(math.exp(-0.5),</code><br><code>rel=1e-4)</code> | Pass   | 0.000s |
| Gaussian at two sigma                 | <code>decay ==</code><br><code>pytest.approx(math.exp(-2.0),</code><br><code>rel=1e-4)</code> | Pass   | 0.000s |
| Linear beyond cutoff is zero          | <code>decay == 0.0</code>                                                                     | Pass   | 0.000s |
| Linear midpoint                       | <code>decay == pytest.approx(0.75,</code><br><code>abs=0.001)</code>                          | Pass   | 0.000s |
| Zero distance returns one all kernels | <code>model.compute_decay(0, 100, kernel,</code><br><code>1.0) == 1.0</code>                  | Pass   | 0.000s |
| After end uses last point             | <code>intensity &gt;= 0</code>                                                                | Pass   | 0.000s |
| At peak time is nonzero               | <code>intensity &gt; 0</code>                                                                 | Pass   | 0.000s |
| Before start uses first point         | <code>intensity &gt;= 0</code>                                                                | Pass   | 0.000s |
| Empty track returns zero              | <code>result == 0.0</code>                                                                    | Pass   | 0.000s |

| Test Description                                                                 | Acceptance Criteria                                                                                                                                         | Result | Time   |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Far from track is lower                                                          | <code>near &gt; far</code>                                                                                                                                  | Pass   | 0.000s |
| Accumulation nonnegative                                                         | <code>resp.accumulation &gt;= 0</code>                                                                                                                      | Pass   | 0.000s |
| Duration nonnegative                                                             | <code>resp.duration_above_alert &gt;= 0;</code><br><code>resp.duration_above_warning &gt;= 0</code> (+1 more)                                               | Pass   | 0.000s |
| Flooded flag correct                                                             | <code>resp.flooded == (resp.peak_level &gt;= gauge.flood_alert)</code>                                                                                      | Pass   | 0.000s |
| A storm passing directly over with peak=100 should flood the gauge.              | <code>resp.peak_level &gt; gauge.base_level</code>                                                                                                          | Pass   | 0.000s |
| Peak level at least base                                                         | <code>resp.peak_level &gt;= gauge.base_level</code>                                                                                                         | Pass   | 0.000s |
| Peak level is max of series                                                      | <code>resp.peak_level ==</code><br><code>pytest.approx(max(levels))</code>                                                                                  | Pass   | 0.000s |
| Response ids match                                                               | <code>resp.gauge_id ==</code><br><code>gauge.gauge_id;</code> <code>resp.storm_id ==</code><br><code>simple_storm.storm_id</code>                           | Pass   | 0.000s |
| Timeseries length                                                                | <code>len(resp.level_timeseries)</code><br><code>== expected_steps;</code><br><code>len(resp.intensity_timeseries) ==</code><br><code>expected_steps</code> | Pass   | 0.000s |
| To dict roundtrip                                                                | <code>d['gauge_id'] == gauge.gauge_id;</code><br><code>'level_timeseries' in d</code> (+1 more)                                                             | Pass   | 0.000s |
| To dict without timeseries                                                       | <code>'level_timeseries' not in d</code>                                                                                                                    | Pass   | 0.000s |
| Encodes datetime                                                                 | <code>"2026-01-01" in result</code>                                                                                                                         | Pass   | 0.000s |
| Line 80: np.float32 is np.floating but NOT a Python float subclass in NumPy 2.x, | <code>"2.5" in result</code>                                                                                                                                | Pass   | 0.000s |
| Encodes numpy floating                                                           | <code>"3.14" in result</code>                                                                                                                               | Pass   | 0.000s |
| Encodes numpy integer                                                            | <code>"42" in result</code>                                                                                                                                 | Pass   | 0.000s |
| Encodes numpy ndarray                                                            | <code>"1.0" in result</code>                                                                                                                                | Pass   | 0.000s |
| Fallback raises for unknown                                                      | —                                                                                                                                                           | Pass   | 0.000s |
| Distance is positive                                                             | <code>dist &gt; 0</code>                                                                                                                                    | Pass   | 0.000s |
| Returns nearest of three                                                         | <code>nearest.intensity == 80.0;</code> <code>dist ==</code><br><code>pytest.approx(0.0, abs=1.0)</code>                                                    | Pass   | 0.000s |
| Single track point                                                               | <code>nearest is track[0]</code>                                                                                                                            | Pass   | 0.000s |
| Close to thames high risk                                                        | <code>result["FloodRiskScore"] &gt;= 7;</code><br><code>result["FloodRiskCategory"] in</code><br><code>["High", "Very High"]</code>                         | Pass   | 0.000s |
| Far from thames lower risk                                                       | <code>result["FloodRiskScore"] &lt;= 6;</code><br><code>result["FloodRiskCategory"] in</code><br><code>["Low", "Medium"]</code>                             | Pass   | 0.000s |
| Flood stages ordering                                                            | <code>alert &lt; warning &lt; severe</code>                                                                                                                 | Pass   | 0.001s |

| Test Description                                                                                                      | Acceptance Criteria                                                                                                                                                                     | Result | Time   |
|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Processing stats                                                                                                      | <code>stats["successful_gauges"] == 5;</code><br><code>stats["failed_gauges"] == 0</code>                                                                                               | Pass   | 0.001s |
| Lines 264-266: exception during <code>json.dump</code> is logged and re-raised.                                       | —                                                                                                                                                                                       | Pass   | 0.000s |
| Line 245: <code>end_time</code> is set by the first <code>generate()</code> ; the second <code>generate()</code> call | —                                                                                                                                                                                       | Pass   | 0.001s |
| Line 290: <code>non-dict section_schema</code> returns <code>{}</code> .                                              | <code>result == {}</code>                                                                                                                                                               | Pass   | 0.000s |
| Line 294: <code>'type', 'options', 'description', 'values'</code> keys are skipped.                                   | <code>skip not in result</code>                                                                                                                                                         | Pass   | 0.000s |
| Lines 159-162: <code>count i max_gauges</code> → capped to <code>max_gauges</code> .                                  | <code>len(result["data"]["flood_gauges"])</code><br><code>== max_available</code>                                                                                                       | Pass   | 0.007s |
| Line 155-156: <code>count i 1</code> raises <code>ValueError</code> .                                                 | —                                                                                                                                                                                       | Pass   | 0.000s |
| Lines 354-355: <code>generate_gauges()</code> convenience function.                                                   | <code>"data" in result;</code><br><code>len(result["data"]["flood_gauges"])</code><br><code>== 3</code>                                                                                 | Pass   | 0.001s |
| Lines 212-215: exception in <code>_generate_single_gauge</code> increments <code>failed_gauges</code> .               | <code>result["processing_stats"]["failed_gauges"]</code><br><code>== 1</code>                                                                                                           | Pass   | 0.000s |
| Lines 122-125: <code>log()</code> dispatches to correct logger method.                                                | —                                                                                                                                                                                       | Pass   | 0.000s |
| Lines 309-327: missing <code>FloodGauge/Header/Location/FloodStages</code> are created.                               | <code>gauge_data["FloodGauge"]["Header"]["GaugeID"]</code><br><code>== "GAUGE-t...;</code><br><code>gauge_data["FloodGauge"]["Location"]["GaugeLatitude"]</code><br><code>== ...</code> | Pass   | 0.000s |
| Lines 99-100: <code>verbose=False</code> sets logger to <code>WARNING</code> .                                        | <code>logging.getLogger("port.src.gauge.gauge")</code><br><code>== login...</code>                                                                                                      | Pass   | 0.000s |
| Output file exists                                                                                                    | <code>result["file_path"].exists()</code>                                                                                                                                               | Pass   | 0.001s |
| Output file is valid json                                                                                             | <code>"flood_gauges" in data</code>                                                                                                                                                     | Pass   | 0.003s |
| Output json has correct count                                                                                         | <code>len(data["flood_gauges"]) == 5</code>                                                                                                                                             | Pass   | 0.004s |
| Lines 378-381: <code>_main__</code> block calls <code>generate_gauges</code> and logs results.                        | <code>(tmp_path / 'gauge.json').exists()</code>                                                                                                                                         | Pass   | 0.005s |
| Gauge id format                                                                                                       | <code>re.match(r"~GAUGE-[a-f0-9]{8}\$",</code><br><code>gid)</code>                                                                                                                     | Pass   | 0.001s |
| Gauge ids unique                                                                                                      | <code>len(ids) == len(set(ids))</code>                                                                                                                                                  | Pass   | 0.002s |
| Generate correct count                                                                                                | <code>len(result["data"]["flood_gauges"])</code><br><code>== 5</code>                                                                                                                   | Pass   | 0.001s |
| Generate returns expected keys                                                                                        | <code>"data" in result;</code> <code>"file_path" in</code><br><code>result (+1 more)</code>                                                                                             | Pass   | 0.001s |



| Test Description                  | Acceptance Criteria                                                                                                     | Result | Time   |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Locations have coordinates        | "lat" in loc; "lon" in loc (+1 more)                                                                                    | Pass   | 0.001s |
| Date field returns date string    | isinstance(val, str);<br>re.match(r"\d{4}-\d{2}-\d{2}\$",<br>val)                                                       | Pass   | 0.000s |
| Decimal field returns float       | isinstance(val, (int, float))                                                                                           | Pass   | 0.000s |
| Gauge name includes area          | "TestArea" in val                                                                                                       | Pass   | 0.000s |
| Menu field returns valid option   | isinstance(val, str)                                                                                                    | Pass   | 0.000s |
| Text field returns string         | isinstance(val, str); len(val) > 0                                                                                      | Pass   | 0.000s |
| Catchment name stored             | meta["catchment_name"] == "Thames"                                                                                      | Pass   | 0.000s |
| Flood stage ordering              | meta["flood_alert"] <<br>meta["flood_warning"];<br>meta["flood_warning"] <<br>meta["severe_flood_warning"] (+1<br>more) | Pass   | 0.000s |
| Gauge id format                   | re.match(r"^GAUGE-[a-f0-9]{8}\$",<br>meta["gauge_id"])                                                                  | Pass   | 0.000s |
| Historical high level range       | 5.0 <= meta["historical_high_level"]<br><= 8.0                                                                          | Pass   | 0.000s |
| Install date format               | re.match(r"\d{4}-\d{2}-\d{2}\$",<br>meta["install_date"])                                                               | Pass   | 0.000s |
| Returns dict with required keys   | key in meta                                                                                                             | Pass   | 0.000s |
| Equatorial degree approx 111km    | 110 < d < 113                                                                                                           | Pass   | 0.000s |
| London to paris approx 340km      | 320 < d < 360                                                                                                           | Pass   | 0.000s |
| Same point is zero                | model.haversine_km(0.0, 0.0, 0.0,<br>0.0) == pytest.approx(0...                                                         | Pass   | 0.000s |
| Symmetric                         | d1 == pytest.approx(d2, rel=1e-6)                                                                                       | Pass   | 0.000s |
| Monotone increasing               | all(levels[i] < levels[i + 1] for i<br>in range(len(levels) ...                                                         | Pass   | 0.000s |
| Negative intensity returns zero   | model.intensity_to_level(-10.0,<br>gauge) == 0.0                                                                        | Pass   | 0.000s |
| Positive intensity positive level | model.intensity_to_level(50.0, gauge)<br>> 0.0                                                                          | Pass   | 0.000s |
| Sensitivity scales output         | model.intensity_to_level(50, g2) ==<br>pytest.approx( ...                                                               | Pass   | 0.000s |
| Uses historical high when set     | lv_with != lv_without                                                                                                   | Pass   | 0.000s |
| Zero intensity returns zero       | model.intensity_to_level(0.0, gauge)<br>== 0.0                                                                          | Pass   | 0.000s |

## 9 Storm Intensity Distribution (MKM-SI-001)

Table 16: Test Results Summary — Storm Intensity Distribution (MKM-SI-001)

|             |       |
|-------------|-------|
| Total Tests | 105   |
| Passed      | 105   |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.24s |

Table 17: Detailed Test Results — Storm Intensity Distribution

| Test Description                              | Acceptance Criteria                                                                                                                                                                         | Result | Time   |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Custom parameters                             | <code>d.params.base_mean == 50;</code>                                                                                                                                                      | Pass   | 0.000s |
|                                               | <code>d.params.tail.threshold == 70</code>                                                                                                                                                  |        |        |
| Default parameters                            | <code>d.params.base_mean == 45.0;</code>                                                                                                                                                    | Pass   | 0.000s |
|                                               | <code>d.params.base_std == 15.0 (+4 more)</code>                                                                                                                                            |        |        |
| From params object                            | <code>d.params.base_mean == 55</code>                                                                                                                                                       | Pass   | 0.000s |
| From scenario baseline                        | <code>d is not None;</code> <code>d.params.name ==</code><br><code>"baseline"</code>                                                                                                        | Pass   | 0.000s |
| From scenario unknown raises                  | —                                                                                                                                                                                           | Pass   | 0.000s |
| Seed gives deterministic output               | <code>s1 == s2</code>                                                                                                                                                                       | Pass   | 0.000s |
| Tail prob between 0 and 1                     | <code>0.0 &lt; d._tail_prob &lt; 1.0</code>                                                                                                                                                 | Pass   | 0.000s |
| All categories defined                        | <code>set(DURATION_PARAMS.keys()) ==</code><br><code>expected</code>                                                                                                                        | Pass   | 0.000s |
| Catastrophic extended                         | <code>high == 240</code>                                                                                                                                                                    | Pass   | 0.000s |
| Each tier's max roughly doubles the previous. | <code>1.5 &lt;= ratio &lt;= 3.0</code>                                                                                                                                                      | Pass   | 0.000s |
| Extreme extended                              | <code>DURATION_PARAMS["extreme"] == (36,</code><br><code>72, 144)</code>                                                                                                                    | Pass   | 0.000s |
| Moderate extended                             | <code>low == 12; mode == 30 (+1 more)</code>                                                                                                                                                | Pass   | 0.000s |
| Severe extended                               | <code>DURATION_PARAMS["severe"] == (24, 48,</code><br><code>96)</code>                                                                                                                      | Pass   | 0.000s |
| Gap types                                     | <code>GapType.SHORT.value == "short";</code><br><code>GapType.MEDIUM.value == "medium" (+1</code><br><code>more)</code>                                                                     | Pass   | 0.000s |
| Sequence gap mapping                          | <code>SEQUENCE_GAP_TYPE[SequenceType.DOUBLET]</code><br><code>== GapType.MEDIUM;</code><br><code>SEQUENCE_GAP_TYPE[SequenceType.CLUSTER]</code><br><code>== GapType.MEDIUM (+2 more)</code> | Pass   | 0.000s |

| Test Description            | Acceptance Criteria                                                                                                                  | Result | Time   |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Sequence types              | SequenceType.ISOLATED.value == "isolated";<br>SequenceType.DOUBLET.value == "doublet" (+2 more)                                      | Pass   | 0.000s |
| Storm count ranges          | SEQUENCE_TYPE.STORM.COUNTS[SequenceType.ISOLATED] == (1, 1);<br>SEQUENCE_TYPE.STORM.COUNTS[SequenceType.DOUBLET] == (2, 2) (+2 more) | Pass   | 0.000s |
| Above max is zero           | d.exceedance_probability(d.params.max_intensity + 10) == 0.0                                                                         | Pass   | 0.000s |
| Above threshold uses pareto | 0 < p < 1                                                                                                                            | Pass   | 0.000s |
| At max intensity is zero    | d.exceedance_probability(d.params.max_intensity) == 0.0                                                                              | Pass   | 0.000s |
| At min intensity is one     | d.exceedance_probability(d.params.min_intensity) == 1.0                                                                              | Pass   | 0.000s |
| At threshold continuous     | abs(p_just_below - p_just_above) < 0.05                                                                                              | Pass   | 0.000s |
| Below min is one            | d.exceedance_probability(d.params.min_intensity - 5) == 1.0                                                                          | Pass   | 0.000s |
| Below threshold uses normal | 0 < p < 1                                                                                                                            | Pass   | 0.000s |
| Monotonically decreasing    | all(probs[i] > probs[i + 1] for i in range(len(probs) - 1))                                                                          | Pass   | 0.000s |
| Constants                   | EVENT_WINDOW_HOURS == 168;<br>MIN_DRAINAGE_WINDOW_HOURS == 12<br>(+1 more)                                                           | Pass   | 0.000s |
| Exact boundary              | fits_in_window([78.0, 78.0], [0.0])                                                                                                  | Pass   | 0.000s |
| Exceeds window              | not fits_in_window([72.0, 72.0], [48.0])                                                                                             | Pass   | 0.000s |
| Simple doublet fits         | fits_in_window([24.0, 24.0], [48.0])                                                                                                 | Pass   | 0.000s |
| Single storm                | fits_in_window([100.0], [])                                                                                                          | Pass   | 0.000s |
| Long gaps                   | GAP_PARAMS[GapType.LONG] == (48, 96, 144)                                                                                            | Pass   | 0.000s |
| Medium gaps                 | GAP_PARAMS[GapType.MEDIUM] == (24, 48, 72)                                                                                           | Pass   | 0.000s |
| Short gaps                  | GAP_PARAMS[GapType.SHORT] == (6, 18, 36)                                                                                             | Pass   | 0.000s |
| Doublet medium              | get_gap_range(SequenceType.DOUBLET) == (24, 48, 72)                                                                                  | Pass   | 0.000s |
| Isolated zero               | get_gap_range(SequenceType.ISOLATED) == (0, 0, 0)                                                                                    | Pass   | 0.000s |

| Test Description                                                                          | Acceptance Criteria                                                                                                                   | Result | Time   |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Get duration range                                                                        | <code>get_duration_range("moderate") == (12, 30, 60)</code>                                                                           | Pass   | 0.000s |
| Get max duration                                                                          | <code>get_max_duration("catastrophic") == 240; get_max_duration("minimal") == 12</code>                                               | Pass   | 0.000s |
| Ids unique                                                                                | <code>len(ids) == 100</code>                                                                                                          | Pass   | 0.000s |
| Sequence id format                                                                        | <code>sid.startswith("STORM-"); len(sid) == 14</code>                                                                                 | Pass   | 0.000s |
| Sequence ids unique                                                                       | <code>len(ids) == 100</code>                                                                                                          | Pass   | 0.000s |
| Storm id format                                                                           | <code>sid.startswith("STORM-"); len(sid) == 14</code>                                                                                 | Pass   | 0.000s |
| One prob returns min                                                                      | <code>d.inverse_exceedance(1) == d.params.min_intensity</code>                                                                        | Pass   | 0.000s |
| Result in bounds                                                                          | <code>d.params.min_intensity &lt;= v &lt;= d.params.max_intensity; d.params.min_intensity &lt;= v &lt;= d.params.max_intensity</code> | Pass   | 0.000s |
| Roundtrip above threshold                                                                 | <code>abs(recovered - target) &lt; 3.0</code>                                                                                         | Pass   | 0.000s |
| Roundtrip below threshold                                                                 | <code>abs(recovered - target) &lt; 2.0</code>                                                                                         | Pass   | 0.000s |
| Zero prob returns max                                                                     | <code>d.inverse_exceedance(0) == d.params.max_intensity</code>                                                                        | Pass   | 0.000s |
| Fat tail higher exceedance                                                                | <code>p_fat &gt; p_thin</code>                                                                                                        | Pass   | 0.000s |
| Pareto above threshold between 0 and 1                                                    | <code>0 &lt; p &lt; 1</code>                                                                                                          | Pass   | 0.000s |
| Pareto at threshold is one                                                                | <code>d.pareto_exceedance(d.params.tail_threshold) == 1.0</code>                                                                      | Pass   | 0.000s |
| Pareto below threshold is one                                                             | <code>d.pareto_exceedance(d.params.tail_threshold - 5) == 1.0</code>                                                                  | Pass   | 0.000s |
| 100yr greater than 10yr                                                                   | <code>i100 &gt; i10</code>                                                                                                            | Pass   | 0.000s |
| Ordering 10 50 100                                                                        | <code>i10 &lt; i50 &lt; i100</code>                                                                                                   | Pass   | 0.000s |
| Result in bounds                                                                          | <code>d.params.min_intensity &lt;= v &lt;= d.params.max_intensity; d.params.min_intensity &lt;= v &lt;= d.params.max_intensity</code> | Pass   | 0.000s |
| Storms per year affects result                                                            | <code>many &gt; few</code>                                                                                                            | Pass   | 0.000s |
| Invalid category                                                                          | —                                                                                                                                     | Pass   | 0.000s |
| Mode should be near the most frequent value in a large sample.                            | <code>abs(median - mode) &lt; 15</code>                                                                                               | Pass   | 0.009s |
| Line 50: <code>rng=None</code> → creates <code>np.random.RandomState()</code> internally. | <code>low &lt;= d &lt;= high</code>                                                                                                   | Pass   | 0.000s |
| Reproducibility                                                                           | <code>d1 == d2</code>                                                                                                                 | Pass   | 0.000s |

| Test Description                                                                                  | Acceptance Criteria                                   | Result | Time   |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------|--------|--------|
| All sampled durations fall within [min, max] for the category. [category='baseline']              | <code>low &lt;= d &lt;= high</code>                   | Pass   | 0.000s |
| All sampled durations fall within [min, max] for the category. [category='catastrophic']          | <code>low &lt;= d &lt;= high</code>                   | Pass   | 0.000s |
| All sampled durations fall within [min, max] for the category. [category='extreme']               | <code>low &lt;= d &lt;= high</code>                   | Pass   | 0.000s |
| All sampled durations fall within [min, max] for the category. [category='minimal']               | <code>low &lt;= d &lt;= high</code>                   | Pass   | 0.000s |
| All sampled durations fall within [min, max] for the category. [category='moderate']              | <code>low &lt;= d &lt;= high</code>                   | Pass   | 0.000s |
| All sampled durations fall within [min, max] for the category. [category='severe']                | <code>low &lt;= d &lt;= high</code>                   | Pass   | 0.000s |
| Doublet gaps should fall in medium range (24-72h).                                                | <code>all(24 &lt;= g &lt;= 72 for g in gaps)</code>   | Pass   | 0.000s |
| Isolated returns zero                                                                             | <code>sample_gap(SequenceType.ISOLATED) == 0.0</code> | Pass   | 0.000s |
| Line 49: <code>rng=None</code> → creates <code>np.random.RandomState()</code> internally.         | <code>low &lt;= g &lt;= high</code>                   | Pass   | 0.000s |
| Persistent gaps should fall in short range (6-36h).                                               | <code>all(6 &lt;= g &lt;= 36 for g in gaps)</code>    | Pass   | 0.000s |
| Reproducibility                                                                                   | <code>g1 == g2</code>                                 | Pass   | 0.000s |
| Within<br>bounds[sequencetype.cluster]<br>[seq_type=iSequenceType.CLUSTER:<br>'cluster']          | <code>low &lt;= g &lt;= high</code>                   | Pass   | 0.000s |
| Within<br>bounds[sequencetype.doublet]<br>[seq_type=iSequenceType.DOUBLET:<br>'doublet']          | <code>low &lt;= g &lt;= high</code>                   | Pass   | 0.000s |
| Within<br>bounds[sequencetype.persistent]<br>[seq_type=iSequenceType.PERSISTENT:<br>'persistent'] | <code>low &lt;= g &lt;= high</code>                   | Pass   | 0.000s |
| Different seeds different output                                                                  | <code>d1.sample(10) != d2.sample(10)</code>           | Pass   | 0.000s |



| Test Description                  | Acceptance Criteria                                                                                                                                               | Result | Time   |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Sample n returns correct length   | <code>len(samples) == 100</code>                                                                                                                                  | Pass   | 0.000s |
| Sample single                     | <code>d.params.min_intensity &lt;= v &lt;=</code><br><code>d.params.max_intensity</code>                                                                          | Pass   | 0.000s |
| Samples within bounds             | <code>all(d.params.min_intensity &lt;= s &lt;=</code><br><code>d.params.max_intensity...</code>                                                                   | Pass   | 0.001s |
| Seeded reproducibility            | <code>s1 == s2</code>                                                                                                                                             | Pass   | 0.000s |
| Create basic                      | <code>storm.storm_id == "STORM-abc12345";</code><br><code>storm.duration.hours == 24.0</code>                                                                     | Pass   | 0.000s |
| From dict round trip              | <code>restored.storm_id ==</code><br><code>original.storm_id;</code><br><code>restored.duration.hours ==</code><br><code>original.duration.hours (+6 more)</code> | Pass   | 0.000s |
| Precipitation rounding            | <code>d["precipitation_mm"] == 35.12;</code><br><code>d["intensity_factor"] == 1.234568 (+1</code><br><code>more)</code>                                          | Pass   | 0.000s |
| To dict                           | <code>d["storm_id"] == "STORM-test1234";</code><br><code>d["start_time.hours"] == 48.0 (+6</code><br><code>more)</code>                                           | Pass   | 0.000s |
| Describe returns string           | <code>isinstance(desc, str); "base_mean" in</code><br><code>desc.lower() or "Base mean" in desc</code>                                                            | Pass   | 0.073s |
| Params roundtrip                  | <code>recovered.base_mean == 52;</code><br><code>recovered.name == "custom"</code>                                                                                | Pass   | 0.000s |
| Tail probability matches internal | <code>d.to_dict()['tail_probability'] ==</code><br><code>pytest.approx(d._tail...</code>                                                                          | Pass   | 0.000s |
| To dict keys                      | <code>'parameters' in result;</code><br><code>'tail_probability' in result</code>                                                                                 | Pass   | 0.000s |
| Antecedent defaults               | <code>seq.antecedent.soil_moisture ==</code><br><code>"normal"; seq.antecedent.groundwater</code><br><code>== "normal"</code>                                     | Pass   | 0.000s |
| Create doublet                    | <code>seq.num_storms == 2; len(seq.storms)</code><br><code>== 2 (+2 more)</code>                                                                                  | Pass   | 0.000s |
| From dict round trip              | <code>restored.storm_id ==</code><br><code>original.storm_id;</code><br><code>restored.duration.hours ==</code><br><code>original.duration.hours (+6 more)</code> | Pass   | 0.000s |
| Isolated no gaps                  | <code>len(seq.inter_storm_gaps_hours) == 0</code>                                                                                                                 | Pass   | 0.000s |
| Spatial correlation defaults      | <code>seq.spatial_correlation_enabled</code><br><code>is False;</code><br><code>seq.spatial_correlation_range_km</code><br><code>is None</code>                   | Pass   | 0.000s |

| Test Description              | Acceptance Criteria                                                                                          | Result | Time   |
|-------------------------------|--------------------------------------------------------------------------------------------------------------|--------|--------|
| Spatial correlation enabled   | <code>d["spatial.correlation.enabled"]<br/>is True;<br/>d["spatial.correlation.range_km"]<br/>== 43.2</code> | Pass   | 0.000s |
| To dict                       | <code>d["storm_id"] == "STORM-test1234";<br/>d["start_time.hours"] == 48.0</code> (+6 more)                  | Pass   | 0.000s |
| Exceedance probs decreasing   | <code>stats['prob.exceed_60'] &gt;=<br/>stats['prob.exceed_70'] &gt;= ...</code>                             | Pass   | 0.000s |
| Fat tail higher p99 than thin | <code>fat['p99'] &gt; thin['p99']</code>                                                                     | Pass   | 0.146s |
| Mean in reasonable range      | <code>30 &lt; stats['mean'] &lt; 70</code>                                                                   | Pass   | 0.000s |
| Percentiles ordered           | <code>stats['p10'] &lt;= stats['p25'] &lt;=<br/>stats['p50'] &lt;= \{\} ...</code>                           | Pass   | 0.000s |
| Required keys present         | <code>k in stats</code>                                                                                      | Pass   | 0.000s |
| Duration mismatch             | <code>any("total.duration.hours" in e for e<br/>in errors)</code>                                            | Pass   | 0.000s |
| Exceeds precip window         | <code>any("MAX_PRECIP_END_HOUR" in e for e<br/>in errors)</code>                                             | Pass   | 0.000s |
| Invalid intensity category    | <code>any("invalid intensity" in e for e<br/>in errors)</code>                                               | Pass   | 0.000s |
| Invalid sequence type         | <code>any("Invalid sequence.type" in e for<br/>e in errors)</code>                                           | Pass   | 0.000s |
| Overlapping storms            | <code>any("starts at" in e and "before<br/>storm" in e for e in err...</code>                                | Pass   | 0.000s |
| Storm count mismatch          | <code>any("num.storms" in e for e in<br/>errors)</code>                                                      | Pass   | 0.000s |
| Too many storms               | <code>any("not in [1, 5]" in e for e in<br/>errors)</code>                                                   | Pass   | 0.000s |
| Valid doublet                 | <code>errors == []</code>                                                                                    | Pass   | 0.000s |

## 10 Storm Sequence Generator (MKM-SS-001)

Table 18: Test Results Summary — Storm Sequence Generator (MKM-SS-001)

|             |       |
|-------------|-------|
| Total Tests | 52    |
| Passed      | 52    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.21s |

Table 19: Detailed Test Results — Storm Sequence Generator

| Test Description                                               | Acceptance Criteria                                                                                                                | Result | Time   |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Duration must not exceed MAX_PRECIP_END_HOUR (156).            | <code>result.storms[0].duration_hours &lt;= MAX_PRECIP_END_HOUR</code>                                                             | Pass   | 0.000s |
| Isolated has one storm                                         | <code>isinstance(result, StormSequence);</code><br><code>result.num_storms == 1 (+2 more)</code>                                   | Pass   | 0.000s |
| All storm sequences                                            | <code>isinstance(seq, StormSequence)</code>                                                                                        | Pass   | 0.000s |
| Catchment propagated                                           | <code>seq.catchment_id == "rhine";</code><br><code>storm.catchment_id == "rhine"</code>                                            | Pass   | 0.001s |
| Correct count                                                  | <code>len(small.batch) == 200</code>                                                                                               | Pass   | 0.000s |
| Different seeds differ                                         | <code>types_a != types_b</code>                                                                                                    | Pass   | 0.002s |
| Force sequence type doublet                                    | <code>doublets &gt; 50; seq.sequence_type in ("doublet", "isolated")</code>                                                        | Pass   | 0.002s |
| Force sequence type isolated                                   | <code>seq.sequence_type == "isolated";</code><br><code>seq.num_storms == 1</code>                                                  | Pass   | 0.001s |
| Reproducible with same seed                                    | <code>[s.sequence_id for s in a] != [s.sequence_id for s in b];</code><br><code>sa.num_storms == sb.num_storms</code><br>(+1 more) | Pass   | 0.003s |
| Returns list                                                   | <code>isinstance(small.batch, list)</code>                                                                                         | Pass   | 0.000s |
| Custom weight heavy on extreme shifts the distribution.        | <code>all(c == "extreme" for c in cats)</code>                                                                                     | Pass   | 0.007s |
| All four default weight categories appear in the batch.        | <code>cat in categories</code>                                                                                                     | Pass   | 0.000s |
| Multi-storm sequences appear at a meaningful rate for severe+. | <code>multi_frac &gt; 0.15</code>                                                                                                  | Pass   | 0.000s |
| Creates parent dirs                                            | <code>out.exists()</code>                                                                                                          | Pass   | 0.007s |
| Load missing file raises                                       | —                                                                                                                                  | Pass   | 0.000s |
| Load returns correct count                                     | <code>len(loaded) == len(small.batch)</code>                                                                                       | Pass   | 0.020s |
| Load wrong schema raises                                       | —                                                                                                                                  | Pass   | 0.002s |

| Test Description                                   | Acceptance Criteria                                                                                                                | Result | Time   |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Loaded sequences still valid                       | <code>result["invalid"] == 0</code>                                                                                                | Pass   | 0.019s |
| Round trip num storms                              | <code>orig.num_storms == back.num_storms</code>                                                                                    | Pass   | 0.020s |
| Round trip precipitation                           | <code>abs(orig.total_precipitation_mm -<br/>back.total_precipitatio...</code>                                                      | Pass   | 0.020s |
| Round trip sequence type                           | <code>orig.sequence_type ==<br/>back.sequence_type</code>                                                                          | Pass   | 0.019s |
| Round trip storm fields                            | <code>os_.storm_index == bs_.storm_index;<br/>abs(os_.duration_hours -<br/>bs_.duration_hours) &lt; 0.01</code> (+1 more)          | Pass   | 0.021s |
| Round trip total duration                          | <code>abs(orig.total_duration_hours -<br/>back.total_duration_hours...</code>                                                      | Pass   | 0.020s |
| Save creates file                                  | <code>out.exists()</code>                                                                                                          | Pass   | 0.007s |
| Schema version in file                             | <code>raw["schema_version"] ==<br/>SCHEMA_VERSION</code>                                                                           | Pass   | 0.021s |
| Creates file                                       | <code>out.exists()</code>                                                                                                          | Pass   | 0.000s |
| Duration stats present                             | <code>d["min"] &gt; 0; d["max"] &lt;= 156</code> (+1 more)                                                                         | Pass   | 0.001s |
| Num sequences matches                              | <code>data["num_sequences"] ==<br/>len(small.batch)</code>                                                                         | Pass   | 0.001s |
| Precipitation stats present                        | <code>p["min"] &gt; 0; p["max"] &gt;= p["min"]</code> (+1 more)                                                                    | Pass   | 0.002s |
| Schema version                                     | <code>data["schema_version"] ==<br/>SCHEMA_VERSION</code>                                                                          | Pass   | 0.002s |
| Type counts sum to total                           | <code>sum(data["sequence_type_counts"].values())<br/>== len(small.b...</code>                                                      | Pass   | 0.001s |
| Type fractions sum to one                          | <code>abs(total_frac - 1.0) &lt; 0.001</code>                                                                                      | Pass   | 0.001s |
| Aggregate stats                                    | <code>result.max_intensity_factor &gt;=<br/>result.avg_intensity_factor;<br/>result.total_precipitation_mm &gt; 0</code> (+1 more) | Pass   | 0.000s |
| Catchment id in storms                             | <code>result.storms[0].catchment_id ==<br/>"rhine"</code>                                                                          | Pass   | 0.000s |
| Cluster sequence                                   | <code>result.sequence_type in<br/>(SequenceType.CLUSTER.value, Sequ...</code>                                                      | Pass   | 0.000s |
| force_sequence_type='doublet'<br>always doublet.   | → <code>result.sequence_type ==<br/>SequenceType.DOUBLET.value;<br/>result.num_storms == 2</code>                                  | Pass   | 0.000s |
| force_sequence_type='isolated'<br>always isolated. | → <code>result.sequence_type ==<br/>SequenceType.ISOLATED.value;<br/>result.num_storms == 1</code>                                 | Pass   | 0.000s |

| Test Description                                                      | Acceptance Criteria                                                                                                                  | Result | Time   |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Minimal intensity (5% sequence probability) mostly produces isolated. | <code>isolated &gt;= 10</code>                                                                                                       | Pass   | 0.000s |
| Generate multiple sequences without error.                            | <code>isinstance(result, StormSequence);</code><br><code>result.num_storms &gt;= 1</code>                                            | Pass   | 0.000s |
| Persistent sequence                                                   | <code>result.sequence_type in</code><br><code>(SequenceType.PERSISTENT.value,</code><br><code>S...</code>                            | Pass   | 0.000s |
| Returns storm sequence                                                | <code>isinstance(result, StormSequence)</code>                                                                                       | Pass   | 0.000s |
| All sequences must have total_duration ≤ 168h.                        | <code>result.total_duration_hours &lt;= 168.0</code>                                                                                 | Pass   | 0.000s |
| Storms have correct fields                                            | <code>storm.duration_hours &gt; 0;</code><br><code>storm.intensity_factor &gt; 0</code> (+2 more)                                    | Pass   | 0.000s |
| Custom catchment                                                      | <code>gen.catchment_id == "rhine";</code><br><code>gen.base_precipitation ==</code><br><code>CATCHMENT_BASE_PRECIP["rhine"]</code>   | Pass   | 0.000s |
| Default catchment                                                     | <code>gen.catchment_id == "thames";</code><br><code>gen.base_precipitation ==</code><br><code>CATCHMENT_BASE_PRECIP["thames"]</code> | Pass   | 0.000s |
| Same seed → same sequence.                                            | <code>seq1.num_storms == seq2.num_storms;</code><br><code>seq1.sequence_type ==</code><br><code>seq2.sequence_type</code>            | Pass   | 0.000s |
| Unknown catchment defaults to 35                                      | <code>gen.base_precipitation == 35.0</code>                                                                                          | Pass   | 0.000s |
| Zero validation errors across the whole batch.                        | <code>result["invalid"] == 0</code>                                                                                                  | Pass   | 0.000s |
| Medium batch all valid                                                | <code>result["invalid"] == 0</code>                                                                                                  | Pass   | 0.001s |
| No negative durations                                                 | <code>storm.duration_hours &gt; 0</code>                                                                                             | Pass   | 0.000s |
| No negative precipitation                                             | <code>seq.total_precipitation_mm &gt; 0;</code><br><code>storm.precipitation_mm &gt; 0</code>                                        | Pass   | 0.000s |
| Last storm precipitation must end by hour 156 (1e-9 FP tolerance).    | <code>last_end &lt;= MAX_PRECIP_END_HOUR +</code><br><code>_FP_TOL</code>                                                            | Pass   | 0.000s |



## 11 Trading Desk (MKM-TD-001)

Table 20: Test Results Summary — Trading Desk (MKM-TD-001)

|             |       |
|-------------|-------|
| Total Tests | 278   |
| Passed      | 278   |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 2.33s |

Table 21: Detailed Test Results — Trading Desk

| Test Description                                                                  | Acceptance Criteria                                                | Result | Time   |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------|--------|--------|
| All buttons must use font-size:11px;font-weight:600 matching header tabs.         | count >= 4                                                         | Pass   | 0.000s |
| All buttons must use padding:4px 14px matching header tab buttons.                | count >= 4                                                         | Pass   | 0.000s |
| All four buttons must sit inside a single flex container (group).                 | 'display:flex;border:1px solid #ddd;border-radius:4px;ove...       | Pass   | 0.000s |
| Commit button must be blue (#1976d2) — primary action like active header tab.     | '#1976d2' in snippet; 'color:white' in snippet                     | Pass   | 0.000s |
| History button must be grey (#f5f5f5) — inactive like unselected header tabs.     | '#f5f5f5' in snippet; '#1976d2' not in snippet                     | Pass   | 0.000s |
| Individual buttons must NOT have border-radius — container provides the rounding. | 'border-radius' not in snippet                                     | Pass   | 0.000s |
| PL Hist button must be grey (#f5f5f5) — inactive like unselected header tabs.     | '#f5f5f5' in snippet; '#1976d2' not in snippet                     | Pass   | 0.000s |
| Reset button must be grey (#f5f5f5) — inactive.                                   | '#f5f5f5' in snippet                                               | Pass   | 0.000s |
| Westminster, Chelsea and Lambeth all have trades — all must appear.               | self.WESTMINSTER_ID in active; self.CHELSEA_ID in active (+1 more) | Pass   | 0.022s |
| No trades at all → empty gauge_ids list → all right-click                         | resp.status_code == 200; data['gauge_ids'] == []                   | Pass   | 0.001s |
| A gauge that has no PRS trades must not appear in active-gauges,                  | self.NO_TRADE_ID not in data['gauge_ids']                          | Pass   | 0.022s |

| Test Description                                                           | Acceptance Criteria                                                                                                            | Result | Time   |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Lambeth Bridge (GAUGE-9042bd95) must be in active-gauges                   | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)                                    | Pass   | 0.018s |
| After closing PRS-TEST-LAMBETH, Lambeth must leave active-gauges           | <code>self.LAMBETH_ID in</code><br><code>before['gauge_ids']; self.LAMBETH_ID</code><br><code>not in after['gauge_ids']</code> | Pass   | 0.044s |
| Closing Lambeth's trade must not affect Westminster or Chelsea.            | <code>self.WESTMINSTER_ID in</code><br><code>after['gauge_ids']; self.CHELSEA_ID in</code><br><code>after['gauge_ids']</code>  | Pass   | 0.036s |
| gauge_ids are returned in sorted order for stable JS dict lookup.          | <code>ids == sorted(ids)</code>                                                                                                | Pass   | 0.020s |
| Basis delta should be positive.                                            | <code>result['basis_delta_bps'] &gt; 0;</code><br><code>result['basis_dv01_gbp'] &gt; 0</code>                                 | Pass   | 0.000s |
| With constant basis, basis delta ~ gauge delta.                            | <code>abs(basis_delta['basis_delta_bps'] -</code><br><code>...</code>                                                          | Pass   | 0.000s |
| Closed trades excluded from blotter by default.                            | <code>'PRS-TEST-001' not in swap_ids</code>                                                                                    | Pass   | 0.038s |
| include_closed=true returns closed trades too.                             | <code>'PRS-TEST-001' in swap_ids</code>                                                                                        | Pass   | 0.039s |
| Hits /api/v1/trading/blotter.                                              | <code>'/api/v1/trading/blotter' in</code><br><code>blotter_js</code>                                                           | Pass   | 0.000s |
| Close-out hits /trading/close/.                                            | <code>'/trading/close/' in blotter_js</code>                                                                                   | Pass   | 0.000s |
| Corrupt JSON in prs dir is skipped via inner exception (lines 134-135).    | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>                                              | Pass   | 0.045s |
| Blotter returns 500 when engine raises (lines 94-96).                      | <code>resp.status_code == 500;</code><br><code>data['status'] == 'error'</code>                                                | Pass   | 0.004s |
| Close trade returns 500 when engine raises (lines 251-253).                | <code>resp.status_code == 500;</code><br><code>data['status'] == 'error'</code>                                                | Pass   | 0.001s |
| PDF regen exception is caught; trade close still succeeds (lines 236-237). | <code>resp.status_code == 200;</code><br><code>data['swap_id'] == 'PRS-TEST-001'</code><br>(+1 more)                           | Pass   | 0.019s |
| Trade with empty GaugeName gets name injected from gauge locations.        | <code>resp.status_code == 200;</code><br><code>noname[0].get('gauge_name', '')</code><br><code>!= ''</code>                    | Pass   | 0.054s |
| Close response includes final P&L.                                         | <code>'final_pnl' in data</code>                                                                                               | Pass   | 0.029s |
| After close, trade status is set to Closed in marks file.                  | <code>marks_file.exists();</code><br><code>marks['PRS-TEST-002']['trade_status']</code><br><code>== 'Closed' (+2 more)</code>  | Pass   | 0.030s |
| Closing non-existent trade returns 404.                                    | <code>resp.status_code == 404</code>                                                                                           | Pass   | 0.021s |

| Test Description                                                                     | Acceptance Criteria                                                                                      | Result | Time   |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------|--------|
| Close-out without close-out_spread_bps returns 400.                                  | <code>resp.status_code == 400</code>                                                                     | Pass   | 0.000s |
| Settlement amount reflects the negotiated closeout spread, not fair spread.          | <code>data['status'] == 'success';</code><br><code>data['close_spread_bps'] == 400.0</code><br>(+2 more) | Pass   | 0.028s |
| Closing a trade at a negotiated spread returns success with close spread.            | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+3 more)              | Pass   | 0.029s |
| Button must call <code>tdCommitMarket()</code> .                                     | <code>'tdCommitMarket()' in market_js</code>                                                             | Pass   | 0.000s |
| td-commit-btn ID present in output.                                                  | <code>'td-commit-btn' in market_js</code>                                                                | Pass   | 0.000s |
| Commit button must NOT have <code>display:none</code> in its own style.              | <code>'display:none' not in own.style</code>                                                             | Pass   | 0.000s |
| <code>tdCommitMarket</code> function must exist.                                     | <code>'tdCommitMarket' in market_js</code>                                                               | Pass   | 0.000s |
| Hits <code>/trading/hazard-term-structure/commit</code> .                            | <code>'/trading/hazard-term-structure/commit' in market_js</code>                                        | Pass   | 0.000s |
| Hits <code>/trading/yield-curve/commit</code> .                                      | <code>'/trading/yield-curve/commit' in market_js</code>                                                  | Pass   | 0.000s |
| Commit notification shows P&L impact.                                                | <code>'total_pnl_impact' in market_js;</code><br><code>'gross_pnl_impact' in market_js</code>            | Pass   | 0.000s |
| After commit, switches to blotter tab.                                               | <code>"switchTab('blotter')" in market_js</code>                                                         | Pass   | 0.000s |
| Higher spread means higher premium                                                   | <code>pvs.high["premium_leg_pv"] &gt; pvs.low["premium_leg_pv"]</code>                                   | Pass   | 0.000s |
| Positive hazard rate                                                                 | <code>pvs["premium_leg_pv"] &gt; 0;</code><br><code>pvs["protection_leg_pv"] &gt; 0</code> (+2 more)     | Pass   | 0.000s |
| Zero hazard rate                                                                     | <code>pvs["premium_leg_pv"] == 0;</code><br><code>pvs["protection_leg_pv"] == 0</code>                   | Pass   | 0.000s |
| Returns empty history when no snapshots exist.                                       | <code>resp.status_code == 200;</code><br><code>data['history'] == []</code>                              | Pass   | 0.001s |
| Missing gauge_id returns 400.                                                        | <code>resp.status_code == 400</code>                                                                     | Pass   | 0.000s |
| After EOD, returns term structure history for that gauge.                            | <code>resp.status_code == 200;</code><br><code>data['gauge_id'] == 'GAUGE-001'</code><br>(+1 more)       | Pass   | 0.040s |
| POST <code>hazard-term-structure/commit</code> returns 500 on error (lines 341-343). | <code>resp.status_code == 500</code>                                                                     | Pass   | 0.001s |

| Test Description                                                       | Acceptance Criteria                                                                                                   | Result | Time   |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------|--------|
| curve-history returns 500 on engine error (lines 82-84).               | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| GET hazard-term-structure returns 500 on engine error (lines 234-236). | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| GET yield-curve returns 500 on engine error (lines 104-106).           | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| pnl-series returns 500 on engine error (lines 36-38).                  | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| POST hazard-term-structure/reset returns 500 on error (lines 363-365). | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| POST yield-curve/reset returns 500 on engine error (lines 270-272).    | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| POST yield-curve returns 500 on engine error (lines 211-213).          | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| POST hazard-term-structure/commit with null body → 400 (line 282).     | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.000s |
| POST yield-curve/commit with null body → 400 (line 145).               | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.000s |
| POST hazard-term-structure with null body → 400 (line 245).            | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.000s |
| POST yield-curve with null body → 400 (line 115).                      | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.000s |
| Closed trades excluded                                                 | <code>pnl_engine.compute_daily_pnl(trades)["num_open_trades"] == 0</code>                                             | Pass   | 0.000s |
| Daily pnl without previous eod                                         | <code>result["num_open_trades"] == 2;</code><br><code>result["total_notional"] == 15_000_000 (+1 more)</code>         | Pass   | 0.000s |
| Pnl decomposition sums                                                 | <code>abs(pos["daily_pnl"] - (pos["new_trade_pnl"] + pos["marke...))</code>                                           | Pass   | 0.000s |
| Running pnl calculation                                                | <code>abs(pos1["running_pnl"] - 43.500) &lt; 1;</code><br><code>abs(pos2["running_pnl"] - (-7.125)) &lt; 1</code>     | Pass   | 0.000s |
| Test risk grid construction.                                           | <code>len(grid['buckets']) == 3;</code><br><code>len(grid['gauges']) == 2 (+3 more)</code>                            | Pass   | 0.000s |
| Receiver trade should have negative gauge DV01.                        | <code>enriched['gauge_fs01'] &lt; 0;</code><br><code>enriched['is_payer'] is False</code>                             | Pass   | 0.002s |
| Test trade enrichment with sample data.                                | <code>enriched['swap_id'] == 'PRS-TEST001';</code><br><code>enriched['gauge_id'] == 'GAUGE-TEST-001' (+4 more)</code> | Pass   | 0.002s |

| Test Description                                                           | Acceptance Criteria                                                                                    | Result | Time   |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------|--------|
| Payer and receiver DV01 should be equal magnitude, opposite sign.          | <code>abs(payer.enriched['gauge.fs01'] + rcv.enriched['gauge.fs...</code>                              | Pass   | 0.003s |
| <code>tdHazardDirtyKeys</code> tracks percentage hazard changes.           | <code>'tdHazardDirtyKeys' in market_js</code>                                                          | Pass   | 0.000s |
| Shows UNCOMMITTED label when dirty.                                        | <code>'UNCOMMITTED' in market_js</code>                                                                | Pass   | 0.000s |
| <code>tdYieldDirty</code> variable tracks yield curve changes.             | <code>'tdYieldDirty' in market_js</code>                                                               | Pass   | 0.000s |
| Also fetches blotter for live summary.                                     | <code>'/api/v1/trading/blotter' in eod_js</code>                                                       | Pass   | 0.000s |
| Fetches <code>/api/v1/trading/eod/history</code> .                         | <code>'/api/v1/trading/eod/history' in eod_js</code>                                                   | Pass   | 0.000s |
| Fetches <code>/api/v1/trading/pnl-series</code> .                          | <code>'/api/v1/trading/pnl-series' in eod_js</code>                                                    | Pass   | 0.000s |
| Hits <code>/api/v1/trading/eod</code> for submit.                          | <code>'/api/v1/trading/eod' in eod_js</code>                                                           | Pass   | 0.000s |
| <code>get_eod_history</code> returns 500 when engine raises (lines 88-90). | <code>resp.status_code == 500</code>                                                                   | Pass   | 0.001s |
| <code>get_eod_pdf</code> returns 500 on unexpected error (lines 107-109).  | <code>resp.status_code == 500</code>                                                                   | Pass   | 0.001s |
| <code>submit_eod</code> returns 500 when engine raises (lines 70-72).      | <code>resp.status_code == 500;</code><br><code>data['status'] == 'error'</code>                        | Pass   | 0.008s |
| <code>renderEodAttribution</code> function exists.                         | <code>'renderEodAttribution' in eod_js</code>                                                          | Pass   | 0.000s |
| <code>renderEodChart</code> function exists.                               | <code>'renderEodChart' in eod_js</code>                                                                | Pass   | 0.000s |
| <code>tdCleanupEodCharts</code> exists for chart cleanup.                  | <code>'tdCleanupEodCharts' in eod_js</code>                                                            | Pass   | 0.000s |
| <code>tdDownloadEodPdf</code> function exists.                             | <code>'tdDownloadEodPdf' in eod_js</code>                                                              | Pass   | 0.000s |
| <code>tdSubmitEod</code> function exists.                                  | <code>'tdSubmitEod' in eod_js</code>                                                                   | Pass   | 0.000s |
| Returns list of snapshots after submission.                                | <code>len(data['history']) == 1;</code><br><code>entry['eod_id'] == 'EOD-20260228'</code><br>(+3 more) | Pass   | 0.036s |
| Returns empty list when no snapshots exist.                                | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+2 more)            | Pass   | 0.001s |
| Can submit multiple EODs on different dates.                               | <code>data['count'] == 2</code>                                                                        | Pass   | 0.064s |
| POST with empty JSON body uses today's date (no date key).                 | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>                      | Pass   | 0.043s |



| Test Description                                                          | Acceptance Criteria                                                                                                                                           | Result | Time   |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| GET for existing PDF date serves the file.                                | <code>resp.status_code == 200;</code><br><code>resp.content_type ==</code><br><code>'application/pdf'</code>                                                  | Pass   | 0.003s |
| GET for non-existent date returns 404.                                    | <code>resp.status_code == 404</code>                                                                                                                          | Pass   | 0.000s |
| ImportError in PDF gen is caught; EOD snapshot still saved (lines 56-57). | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>                                                                             | Pass   | 0.030s |
| Exception in PDF gen is caught; EOD snapshot still saved (lines 58-59).   | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>                                                                             | Pass   | 0.043s |
| Attribution section container present.                                    | <code>'td-eod-attribution' in eod_js</code>                                                                                                                   | Pass   | 0.000s |
| P&L summary cards container present.                                      | <code>'td-eod-cards' in eod_js</code>                                                                                                                         | Pass   | 0.000s |
| Date picker for EOD date.                                                 | <code>'td-eod-date' in eod_js</code>                                                                                                                          | Pass   | 0.000s |
| EOD History section present.                                              | <code>'td-eod-history-wrap' in eod_js;</code> 'EOD History' in eod_js                                                                                         | Pass   | 0.000s |
| Submit EOD button present.                                                | <code>'td-eod-submit-btn' in eod_js;</code> 'EOD Submit' in eod_js                                                                                            | Pass   | 0.000s |
| All gauges from base should appear.                                       | <code>'GAUGE-001' in rates;</code> 'GAUGE-002' in rates                                                                                                       | Pass   | 0.001s |
| Adjusted gauges should be flagged.                                        | <code>rates['GAUGE-001']['is_adjusted']</code><br>is True;<br><code>rates['GAUGE-002']['is_adjusted']</code><br>is False                                      | Pass   | 0.002s |
| Adjusted rates should override base, non-adjusted should use base.        | <code>rates['GAUGE-001']['annual_hazard_rate_warning']</code><br>== 0.04;<br><code>rates['GAUGE-001']['annual_hazard_rate_alert']</code><br>== 0.04 (+1 more) | Pass   | 0.002s |
| Eod history                                                               | <code>len(history) == 2;</code> history[0]["date"]<br>== "2026-02-21"                                                                                         | Pass   | 0.002s |
| Generate snapshot                                                         | <code>snapshot["eod_id"] == "EOD-20260224";</code><br><code>snapshot["date"] == "2026-02-24"</code> (+3 more)                                                 | Pass   | 0.000s |
| Idempotent snapshot                                                       | <code>len(pnl_engine.get_eod_history()) ==</code><br>1                                                                                                        | Pass   | 0.003s |
| Pnl series                                                                | <code>len(series) == 2;</code> series[0]["date"]<br>== "2026-02-20" (+1 more)                                                                                 | Pass   | 0.003s |
| Snapshot persists                                                         | snapshot is not None;<br><code>snapshot["eod_id"] == "EOD-20260224"</code>                                                                                    | Pass   | 0.001s |

| Test Description                                                    | Acceptance Criteria                                                                           | Result | Time   |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| tdClearFilters function exists.                                     | 'tdClearFilters' in blotter_js                                                                | Pass   | 0.000s |
| tdApplyFilter exposed on window for external callers.               | 'tdApplyFilter' in blotter_js                                                                 | Pass   | 0.000s |
| tdFilterChanged function exists.                                    | 'tdFilterChanged' in blotter_js                                                               | Pass   | 0.000s |
| getFilteredTrades function exists.                                  | 'getFilteredTrades' in blotter_js                                                             | Pass   | 0.000s |
| tdRemoveFilter function exists.                                     | 'tdRemoveFilter' in blotter_js                                                                | Pass   | 0.000s |
| For small rates, 1bp bump in rate ~ = 1bp bump in spread.           | 0.5 < result['delta_spread.bps'] < 2.0                                                        | Pass   | 0.000s |
| DV01 scales with notional.                                          | abs(r2['dv01_gbp'] / r1['dv01_gbp'] - 2.0) < 0.01                                             | Pass   | 0.000s |
| DV01 should be positive for a buyer (spread increases with hazard). | result['dv01_gbp'] > 0;                                                                       | Pass   | 0.000s |
| At zero hazard rate, spread is at floor so delta may be minimal.    | result['delta_spread.bps'] > 0                                                                | Pass   | 0.000s |
| Balanced book                                                       | payer == rcv; len(trades) == 6                                                                | Pass   | 0.004s |
| Balanced notional                                                   | payer_not == rcv_not                                                                          | Pass   | 0.004s |
| Counterparties distributed                                          | len(ctpys) == 2                                                                               | Pass   | 0.004s |
| Files saved                                                         | len(list(output_dir.glob("PRS-*.json"))) == 6                                                 | Pass   | 0.004s |
| Payer trades below fair                                             | ps["Pricing"]["SpreadBps"] < ps["Pricing"]["FairSpreadBps"]                                   | Pass   | 0.004s |
| Receiver trades above fair                                          | ps["Pricing"]["SpreadBps"] > ps["Pricing"]["FairSpreadBps"]                                   | Pass   | 0.004s |
| Reproducible with seed                                              | t1["PhysicalSwap"]["Pricing"]["SpreadBps"] == t2["PhysicalSwap"]["Pricing"]["SpreadBps"]      | Pass   | 0.004s |
| Small mtm                                                           | abs(ps["Pricing"]["NPV"]) < ps["LegData"]["Notional"] * 0...                                  | Pass   | 0.004s |
| Valid cdm structure                                                 | ps["Header"]["SwapID"].startswith("PRS-"); isinstance(ps["LegData"]["Payer"], bool) (+2 more) | Pass   | 0.004s |
| Returns empty trades list when no PRS files exist.                  | resp.status_code == 200; data['status'] == 'success' (+1 more)                                | Pass   | 0.004s |
| Payer (long protection) trade should have positive gauge_fs01.      | len(payer_trades) > 0; t['gauge_fs01'] >= 0                                                   | Pass   | 0.030s |
| Receiver (short protection) trade should have negative gauge_fs01.  | len(rcv_trades) > 0; t['gauge_fs01'] <= 0                                                     | Pass   | 0.023s |
| Trades should have gauge_name from gauge locations.                 | 'Thames at Westminster' in names                                                              | Pass   | 0.031s |
| Each trade must have P&L decomposition fields.                      | 'daily_pnl' in trade; 'market_pnl' in trade (+2 more)                                         | Pass   | 0.025s |

| Test Description                                                             | Acceptance Criteria                                              | Result | Time   |
|------------------------------------------------------------------------------|------------------------------------------------------------------|--------|--------|
| Each trade must have prev_fair_spread_bps for curve-move indication.         | 'prev_fair_spread_bps' in trade                                  | Pass   | 0.026s |
| Each trade must have core identification and pricing fields.                 | 'swap_id' in trade; 'gauge_id' in trade (+5 more)                | Pass   | 0.026s |
| Summary must have all required aggregate fields.                             | 'num.trades' in summary; 'total_notional' in summary (+4 more)   | Pass   | 0.032s |
| Blotter returns enriched trades with status success.                         | resp.status_code == 200; data['status'] == 'success' (+1 more)   | Pass   | 0.027s |
| GET returns market_state with yield_curve and hazard_term_structure.         | resp.status_code == 200; data['status'] == 'success' (+2 more)   | Pass   | 0.003s |
| Response includes per-gauge effective rates.                                 | 'gauges' in data; len(gauge_ids) >= 1                            | Pass   | 0.003s |
| First load creates default yield curve.                                      | resp.status_code == 200; len(yc) > 0                             | Pass   | 0.002s |
| Bumping hazard curve higher should increase fair PRS spread.                 | spread_after > spread_before; spread_after - spread_before > 1.0 | Pass   | 0.003s |
| Bumping hazard curve should change trade MTM.                                | mtm_after > mtm_before                                           | Pass   | 0.003s |
| P&L from a 1bp parallel shift should approximately equal FS01.               | abs(mtm_change - fs01) < abs(fs01) * 0.20                        | Pass   | 0.003s |
| Full cycle: enrich → EOD snap → bump curve → re-enrich → daily P&L non-zero. | len(eod.snap['positions']) == 1; new_fair != prev_fair (+5 more) | Pass   | 0.005s |
| Shows down arrow for rate decrease.                                          | '\{\}\{u25bc' in blotter_js                                      | Pass   | 0.000s |
| Blotter compares to prev_fair_spread_bps.                                    | 'prev_fair_spread_bps' in blotter_js                             | Pass   | 0.000s |
| Shows up arrow for rate increase.                                            | '\{\}\{u25b2' in blotter_js                                      | Pass   | 0.000s |
| Commit returns affected_trades for the target gauge.                         | resp.status_code == 200; data['status'] == 'success' (+3 more)   | Pass   | 0.034s |
| Missing gauge_id returns 400.                                                | resp.status_code == 400                                          | Pass   | 0.000s |
| Missing rates returns 400.                                                   | resp.status_code == 400                                          | Pass   | 0.000s |
| Committing GAUGE-001 only affects its trades (2 trades).                     | data['affected_trades'] == 2                                     | Pass   | 0.032s |
| Reset all gauges.                                                            | resp.status_code == 200; 'all' in data['message']                | Pass   | 0.003s |

| Test Description                                                        | Acceptance Criteria                                                                                                   | Result | Time   |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------|--------|
| Reset one gauge.                                                        | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)                           | Pass   | 0.004s |
| GET returns per-gauge per-trigger structure.                            | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)                           | Pass   | 0.003s |
| Tenor outside 1-10 returns 400.                                         | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.000s |
| Missing gauge_id returns 400.                                           | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.000s |
| POST updates one gauge/trigger/tenor point.                             | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>                                     | Pass   | 0.005s |
| Verify the MTM formula.                                                 | <code>abs(mtm - 80_000) &lt; 1</code>                                                                                 | Pass   | 0.000s |
| Protection seller loses when spread widens.                             | <code>mtm &lt; 0</code>                                                                                               | Pass   | 0.000s |
| Protection buyer profits when spread widens.                            | <code>mtm &gt; 0</code>                                                                                               | Pass   | 0.000s |
| MTM is zero when fair = trade spread.                                   | <code>mtm == 0</code>                                                                                                 | Pass   | 0.000s |
| Blotter has market_pnl column.                                          | <code>'market_pnl' in blotter_js</code>                                                                               | Pass   | 0.000s |
| Blotter shows Mkt P&L label.                                            | <code>'Mkt P' in blotter_js</code>                                                                                    | Pass   | 0.000s |
| GET market-state returns 500 when engine raises (lines 44-46).          | <code>resp.status_code == 500;</code><br><code>data['status'] == 'error'</code>                                       | Pass   | 0.001s |
| POST market-state/reset returns 500 when engine raises (lines 119-121). | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| POST market-state with null JSON body → 400 (line 55).                  | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.000s |
| Base rates should match what's in gaugehc.json.                         | <code>g1['annual_hazard_rate_warning'] == 0.025;</code><br><code>g1['annual_hazard_rate_severe'] == 0.01</code>       | Pass   | 0.001s |
| First load should initialise from base curves.                          | <code>'base_rates' in state;</code><br><code>'gauge_adjustments' in state</code> (+2 more)                            | Pass   | 0.001s |
| Subsequent loads should return the saved state.                         | <code>state1['base_rates'] == state2['base_rates']</code>                                                             | Pass   | 0.003s |
| Reset all adjustments.                                                  | <code>len(state['gauge_adjustments']) == 0</code>                                                                     | Pass   | 0.005s |
| Reset a single gauge's adjustments.                                     | <code>'GAUGE-001' not in state['gauge_adjustments'];</code><br><code>'GAUGE-002' in state['gauge_adjustments']</code> | Pass   | 0.004s |
| Effective rate should use adjustment if present.                        | <code>rate == 0.025; rate == 0.04</code>                                                                              | Pass   | 0.003s |

| Test Description                                                                             | Acceptance Criteria                                                               | Result | Time   |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------|--------|
| Gauge without adjustment should return base rate.                                            | <code>rate == 0.03</code>                                                         | Pass   | 0.003s |
| Updating a rate should persist.                                                              | <code>adj['annual_hazard_rate_warning'] == 0.035</code>                           | Pass   | 0.002s |
| <code>tdCleanupMarketCharts</code> exists.                                                   | <code>'tdCleanupMarketCharts' in market_js</code>                                 | Pass   | 0.000s |
| <code>renderGaugeList</code> function exists.                                                | <code>'renderGaugeList' in market_js</code>                                       | Pass   | 0.000s |
| Chart canvas element present.                                                                | <code>'td-market-canvas' in market_js</code>                                      | Pass   | 0.000s |
| Curve mode selector with yield/hazard options.                                               | <code>'td-curve-mode' in market_js; 'Yield Curve' in market_js (+1 more)</code>   | Pass   | 0.000s |
| <code>commit_yield_curve</code> saves all tenors at once.                                    | <code>state['yield_curve']['1'] == 0.04; state['yield_curve']['3'] == 0.05</code> | Pass   | 0.002s |
| Tenor above maximum returns last key rate.                                                   | <code>rate == r6</code>                                                           | Pass   | 0.002s |
| Tenor below minimum returns first key rate.                                                  | <code>rate == r1</code>                                                           | Pass   | 0.001s |
| Exact tenor match from yield curve.                                                          | <code>isinstance(rate, float); rate &gt; 0</code>                                 | Pass   | 0.001s |
| Non-integer tenor triggers interpolation.                                                    | <code>min(r1, r2) &lt;= rate &lt;= max(r1, r2)</code>                             | Pass   | 0.001s |
| Gauge entries with no <code>gauge_id</code> are skipped.                                     | <code>'GAUGE-B' in base; '' not in base</code>                                    | Pass   | 0.001s |
| <code>gaugehc.json</code> with <code>'gauges'</code> key is handled.                         | <code>'GAUGE-A' in state.get('base_rates', {})</code>                             | Pass   | 0.002s |
| When <code>gaugehc.json</code> is missing, <code>base curves</code> returns empty dict.      | <code>isinstance(state, dict)</code>                                              | Pass   | 0.000s |
| <code>gaugehc.json</code> with unknown format returns empty <code>base_rates</code> .        | <code>state.get('base_rates', {}) == {}</code>                                    | Pass   | 0.002s |
| <code>reset_yield_curve</code> restores defaults.                                            | <code>state['yield_curve']['1'] == MarketStateManager.DEFAULT_Y...</code>         | Pass   | 0.004s |
| Notes parameter is stored in gauge adjustments.                                              | <code>'adjustment_notes' in adj; 'override' in adj['adjustment_notes']</code>     | Pass   | 0.003s |
| New Trade button must be grey (#f5f5f5) — non-primary action.                                | <code>'#f5f5f5' in snippet; '#1976d2' not in snippet</code>                       | Pass   | 0.000s |
| Button text must be 'New Trade'.                                                             | <code>'New Trade' in market_js</code>                                             | Pass   | 0.000s |
| <code>td-newtrade-btn</code> must exist in Market tab.                                       | <code>'td-newtrade-btn' in market_js</code>                                       | Pass   | 0.000s |
| Button must call <code>tdMarketNewTrade()</code> .                                           | <code>'tdMarketNewTrade' in market_js</code>                                      | Pass   | 0.000s |
| <code>tdMarketNewTrade</code> must hide trading desk and call <code>viewHazardCurve</code> . | <code>'viewHazardCurve' in market_js; 'tdSelectedGauge' in market_js</code>       | Pass   | 0.000s |
| New Trade button must appear to the LEFT of History in the DOM.                              | <code>nt_idx &lt; hist_idx</code>                                                 | Pass   | 0.000s |



| Test Description                                                           | Acceptance Criteria                                                          | Result | Time   |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------|--------|--------|
| New Trade button must not have its own border-radius (container clips it). | 'border-radius' not in snippet                                               | Pass   | 0.000s |
| Market pnl nonzero with eod mark                                           | abs(result["positions"][0]["market_pnl"] - 80.000.0) < 1                     | Pass   | 0.000s |
| Market pnl zero for missing eod trade                                      | abs(pos["market_pnl"]) < 0.01;<br>abs(pos["new_trade_pnl"]) < 0.01 (+1 more) | Pass   | 0.000s |
| New trade has inception pnl                                                | abs(pos["new_trade_pnl"] - expected) < 1; abs(pos["market_pnl"]) < 0.01      | Pass   | 0.000s |
| Pnl decomposition with prev eod                                            | abs(pos["daily_pnl"] - (pos["new_trade_pnl"] + pos["marke...                 | Pass   | 0.000s |
| Shows daily P&L metric.                                                    | 'dailyPnl' in blotter_js                                                     | Pass   | 0.000s |
| Shows From Market metric.                                                  | 'From Market' in blotter_js;<br>'fromMarket' in blotter_js                   | Pass   | 0.000s |
| Shows signed net notional.                                                 | 'netNotional' in blotter_js;<br>'fmtGBP(netNotional)' in blotter_js          | Pass   | 0.000s |
| Shows Running P&L.                                                         | 'Running' in blotter_js                                                      | Pass   | 0.000s |
| After EOD snap, next blotter P&L uses snap as baseline.                    | resp.status_code == 200; 'daily_pnl' in t (+1 more)                          | Pass   | 0.044s |
| After an EOD snap, series has one entry.                                   | len(data['series']) == 1; 'date' in entry (+4 more)                          | Pass   | 0.037s |
| Returns empty series when no EOD snapshots exist.                          | resp.status_code == 200;<br>data['status'] == 'success' (+1 more)            | Pass   | 0.001s |
| Summary runs                                                               | "Market-Making Book Summary" in out;<br>"Payer (short):" in out (+1 more)    | Pass   | 0.003s |
| tdResetCurve function exists.                                              | 'tdResetCurve' in market_js                                                  | Pass   | 0.000s |
| Hits /trading/hazard-term-structure/reset.                                 | '/trading/hazard-term-structure/reset' in market_js                          | Pass   | 0.000s |
| Hits /trading/yield-curve/reset.                                           | '/trading/yield-curve/reset' in market_js                                    | Pass   | 0.000s |
| Reset clears all adjustments.                                              | resp.status_code == 200;<br>data['status'] == 'success' (+1 more)            | Pass   | 0.026s |
| Reset clears one gauge only.                                               | resp.status_code == 200; 'GAUGE-001' in data['message']                      | Pass   | 0.003s |
| risk-grid returns 500 when engine raises (lines 44-46).                    | resp.status_code == 500;<br>data['status'] == 'error'                        | Pass   | 0.017s |
| trade-map returns 500 when engine raises (lines 197-199).                  | resp.status_code == 500                                                      | Pass   | 0.001s |

| Test Description                                              | Acceptance Criteria                                                                                                                         | Result | Time   |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Grid should contain entries for traded gauges.                | <code>len(grid) &gt; 0</code>                                                                                                               | Pass   | 0.026s |
| GET returns a risk grid structure.                            | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)                                                 | Pass   | 0.033s |
| Returns empty grid when no trades exist.                      | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>                                                           | Pass   | 0.004s |
| Longer tenor means more payments.                             | <code>a5 &gt; a1</code>                                                                                                                     | Pass   | 0.000s |
| Higher hazard rate means lower annuity (higher default prob). | <code>annuity_low &gt; annuity_high</code>                                                                                                  | Pass   | 0.000s |
| Annuity should be between 0 and tenor.                        | <code>0 &lt; annuity &lt; 5</code>                                                                                                          | Pass   | 0.000s |
| With no default risk, annuity equals risk-free annuity.       | <code>abs(annuity - expected) &lt; 1e-6</code>                                                                                              | Pass   | 0.000s |
| is_payer field used to determine sign.                        | <code>'is_payer' in blotter_js</code>                                                                                                       | Pass   | 0.000s |
| Payer notional should be negated.                             | <code>'is_payer' in blotter_js;</code> <code>'Math.abs'</code> in blotter_js                                                                | Pass   | 0.000s |
| tdToggleSort function exists.                                 | <code>'tdToggleSort' in blotter_js</code>                                                                                                   | Pass   | 0.000s |
| tdViewTrade function for click-through to PRS pricing.        | <code>'tdViewTrade' in blotter_js</code>                                                                                                    | Pass   | 0.000s |
| After knock-out, cash price must stay at full notional.       | <code>hourly[2]['per_trade'][0]['triggered'] is True;</code><br><code>hourly[2]['per_trade'][0]['cash_price'] == 1.000.000</code> (+2 more) | Pass   | 0.000s |
| Alert trade KOs before severe trade.                          | <code>'ALERT' in triggered;</code> <code>'SEVERE' in triggered</code> (+1 more)                                                             | Pass   | 0.000s |
| Trade must NOT knock out if water stays below trigger.        | <code>'T1' not in triggered</code>                                                                                                          | Pass   | 0.000s |
| Receiver (short protection) gets negative cash on KO.         | <code>hourly[1]['per_trade'][0]['cash_price'] == -1.000.000</code>                                                                          | Pass   | 0.000s |
| Trade must knock out when water $\geq$ trigger level.         | <code>'T1' in triggered;</code> <code>triggered['T1'] == 3</code>                                                                           | Pass   | 0.000s |
| Same-day snap overwrites previous.                            | <code>len(eod_files) == 1</code>                                                                                                            | Pass   | 0.058s |
| POST creates EOD snapshot.                                    | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)                                                 | Pass   | 0.044s |
| Response includes trade count and P&L summary.                | <code>'num_open_trades' in summary;</code><br><code>'total_notional' in summary</code> (+2 more)                                            | Pass   | 0.042s |

| Test Description                                                               | Acceptance Criteria                                                                                                       | Result | Time   |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------|--------|
| EOD-date.json created in eod/ directory.                                       | <code>snapshot.path.exists();</code><br><code>snapshot['eod_id'] == 'EOD-20260228'</code><br>(+1 more)                    | Pass   | 0.037s |
| Snapshot includes yield_curve and hazard_term_structure.                       | <code>'yield_curve' in ms;</code><br><code>'hazard_term_structure' in ms</code>                                           | Pass   | 0.040s |
| Snapshot includes per-trade positions.                                         | <code>len(positions) == 16;</code> <code>'swap_id' in p</code><br>(+2 more)                                               | Pass   | 0.038s |
| All levels should be $\geq$ base level (positive level_change).                | <code>lvl &gt;= base - 0.01</code>                                                                                        | Pass   | 0.000s |
| Custom num.hours should produce the right length.                              | <code>len(levels) == 100</code>                                                                                           | Pass   | 0.000s |
| Water levels should be non-decreasing before peak.                             | <code>levels[i] &gt;= levels[i - 1] - 0.01</code>                                                                         | Pass   | 0.000s |
| Peak should reach base + level_change.                                         | <code>peak &gt;= base + change * 0.9;</code> <code>peak &lt;= base + change + 0.1</code>                                  | Pass   | 0.000s |
| Water level should decrease after peak.                                        | <code>levels[-1] &lt; levels[peak_idx]</code>                                                                             | Pass   | 0.000s |
| Default hydrograph must return 168 hourly values (STORM_HOURS = 7-day window). | <code>len(levels) == 168</code>                                                                                           | Pass   | 0.000s |
| Very short storm peaks early.                                                  | <code>peak_idx &lt;= 10</code>                                                                                            | Pass   | 0.000s |
| Hour 0 should be at base level.                                                | <code>abs(levels[0] - 2.0) &lt; 0.01</code>                                                                               | Pass   | 0.000s |
| Zero duration should still produce valid levels.                               | <code>len(levels) == 168;</code> <code>levels[0] &gt; 2.0</code>                                                          | Pass   | 0.000s |
| gauge.fs01 column present.                                                     | <code>'gauge.fs01' in blotter_js</code>                                                                                   | Pass   | 0.000s |
| fntMaturity function exists.                                                   | <code>'fntMaturity' in blotter_js</code>                                                                                  | Pass   | 0.000s |
| Files saved                                                                    | <code>len(list(output_dir.glob("PRS-*.json")))</code><br><code>== 50</code>                                               | Pass   | 0.017s |
| Has calendar spreads                                                           | <code>calendar_spread_gauges &gt;= 5</code>                                                                               | Pass   | 0.014s |
| Reproducible with seed                                                         | <code>t1["PhysicalSwap"]["Pricing"]["SpreadBps"]</code><br>...                                                            | Pass   | 0.027s |
| Trade count                                                                    | <code>len(trades) == 50</code>                                                                                            | Pass   | 0.014s |
| Uses central gauges                                                            | <code>gauge_ids.issubset(expected_ids)</code>                                                                             | Pass   | 0.014s |
| Valid cdm structure                                                            | <code>ps["Header"]["SwapID"].startswith("PRS-");</code><br><code>ps["Pricing"]["RiskyAnnuity"] &gt; 0</code><br>(+1 more) | Pass   | 0.014s |
| Westminster is net short                                                       | <code>westminster_notional &lt; 0</code>                                                                                  | Pass   | 0.013s |
| Gauge positions include FS01 broken down by tenor.                             | <code>'fs01.by_tenor' in g;</code><br><code>isinstance(g['fs01.by_tenor'], dict)</code>                                   | Pass   | 0.027s |

| Test Description                                                         | Acceptance Criteria                                                                           | Result | Time   |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| Each gauge position has lat, lon, gauge_name.                            | 'lat' in g; 'lon' in g (+2 more)                                                              | Pass   | 0.028s |
| Each gauge has net_fs01.                                                 | 'net_fs01' in g                                                                               | Pass   | 0.026s |
| Each gauge has total and net notional.                                   | 'total_notional' in g; 'net_notional' in g (+1 more)                                          | Pass   | 0.031s |
| Gauge positions include daily P&L.                                       | 'daily_pnl' in g; 'running_pnl' in g                                                          | Pass   | 0.024s |
| Gauge positions show number of trades.                                   | gauges_by_id['GAUGE-001']['num_trades'] == 2;<br>gauges_by_id['GAUGE-002']['num_trades'] == 1 | Pass   | 0.032s |
| GET returns gauge positions.                                             | resp.status_code == 200;<br>data['status'] == 'success' (+1 more)                             | Pass   | 0.026s |
| Returns empty positions when no trades exist.                            | resp.status_code == 200;<br>len(data['gauges']) == 0                                          | Pass   | 0.004s |
| Gauge in gauge.json but not gaugehc.json gets coordinates (lines 83-85). | resp.status_code == 200                                                                       | Pass   | 0.046s |
| Property locations loaded from property.json (lines 97-101).             | resp.status_code == 200; 'gauges' in data                                                     | Pass   | 0.071s |
| Closed trades are excluded from trade map gauge aggregation (line 115).  | resp.status_code == 200;<br>'GAUGE-9042bd95' not in gauge_ids                                 | Pass   | 0.042s |
| Close trade                                                              | mark["trade_status"] == "Closed";<br>mark["close_spread_bps"] == 260.0 (+1 more)              | Pass   | 0.001s |
| Load empty marks                                                         | pnl_engine.load_trade_marks() == {}                                                           | Pass   | 0.000s |
| Update and load mark                                                     | pnl_engine.load_trade_marks()["PRS-001"]["trade_status"] == "Closed"<br>...                   | Pass   | 0.001s |
| Market making spreads are integers                                       | spread == round(spread)                                                                       | Pass   | 0.003s |
| Thames central spreads are integers                                      | spread == round(spread)                                                                       | Pass   | 0.014s |
| Updated state persists across requests.                                  | resp.status_code == 200;<br>data['market_state']['num_adjusted'] >= 1                         | Pass   | 0.037s |
| POST updates gauge rate successfully.                                    | resp.status_code == 200;<br>data['status'] == 'success' (+1 more)                             | Pass   | 0.025s |
| Missing gauge returns 400.                                               | resp.status_code == 400                                                                       | Pass   | 0.000s |
| No JSON body returns error (500 — Flask BadRequest caught).              | resp.status_code in (400, 500)                                                                | Pass   | 0.003s |

| Test Description                                      | Acceptance Criteria                                                  | Result | Time   |
|-------------------------------------------------------|----------------------------------------------------------------------|--------|--------|
| Update returns affected_trades and total_pnl_impact.  | 'affected_trades' in data;<br>'total_pnl_impact' in data             | Pass   | 0.025s |
| GET returns yield_curve dict.                         | resp.status_code == 200;<br>data['status'] == 'success' (+1<br>more) | Pass   | 0.003s |
| Default yield curve has standard tenor points.        | len(yc) > 0                                                          | Pass   | 0.003s |
| Tenor outside 1-30 returns 400.                       | resp.status_code == 400                                              | Pass   | 0.000s |
| POST without JSON body returns error.                 | resp.status_code in (400, 500)                                       | Pass   | 0.004s |
| POST updates one tenor point.                         | resp.status_code == 200;<br>data['status'] == 'success' (+1<br>more) | Pass   | 0.003s |
| Commit returns affected_trades and P&L impact fields. | resp.status_code == 200;<br>data['status'] == 'success' (+4<br>more) | Pass   | 0.032s |
| Commit should affect all open trades.                 | data['affected_trades'] == 16                                        | Pass   | 0.032s |
| Empty rates dict returns 400.                         | resp.status_code == 400                                              | Pass   | 0.000s |
| No JSON body returns error.                           | resp.status_code in (400, 500)                                       | Pass   | 0.001s |
| Reset returns to default shape.                       | resp.status_code == 200;<br>data['status'] == 'success' (+1<br>more) | Pass   | 0.008s |



## 12 CDM Schema Validation (CDM-ALL)

Table 22: Test Results Summary — CDM Schema Validation (CDM-ALL)

|             |       |
|-------------|-------|
| Total Tests | 76    |
| Passed      | 76    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.01s |

Table 23: Detailed Test Results — CDM Schema Validation

| Test Description                                                                 | Acceptance Criteria                                             | Result | Time   |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------|--------|--------|
| All cdm fields present                                                           | <code>gauge_mapping_summary.fields_missing == 0</code>          | Pass   | 0.000s |
| All cdm fields present                                                           | <code>mortgage_mapping_summary.fields_missing == 0</code>       | Pass   | 0.000s |
| All cdm fields present                                                           | <code>property_mapping_summary.fields_missing == 0</code>       | Pass   | 0.000s |
| All types valid                                                                  | <code>gauge_mapping_summary.fields_type_invalid == 0</code>     | Pass   | 0.000s |
| All types valid                                                                  | <code>mortgage_mapping_summary.fields_type_invalid == 0</code>  | Pass   | 0.000s |
| All types valid                                                                  | <code>property_mapping_summary.fields_type_invalid == 0</code>  | Pass   | 0.000s |
| All values valid                                                                 | <code>gauge_mapping_summary.fields_value_invalid == 0</code>    | Pass   | 0.000s |
| All values valid                                                                 | <code>mortgage_mapping_summary.fields_value_invalid == 0</code> | Pass   | 0.000s |
| All values valid                                                                 | <code>property_mapping_summary.fields_value_invalid == 0</code> | Pass   | 0.000s |
| Get required fields returns list[gauge cdm] [cdm_name='gauge_cdm']               | <code>isinstance(cdm.get_required_fields(), list)</code>        | Pass   | 0.000s |
| Get required fields re-<br>turns list[mortgage cdm]<br>[cdm_name='mortgage_cdm'] | <code>isinstance(cdm.get_required_fields(), list)</code>        | Pass   | 0.000s |
| Get required fields re-<br>turns list[property cdm]<br>[cdm_name='property_cdm'] | <code>isinstance(cdm.get_required_fields(), list)</code>        | Pass   | 0.000s |
| Get required fields returns list[prs cdm] [cdm_name='prs_cdm']                   | <code>isinstance(cdm.get_required_fields(), list)</code>        | Pass   | 0.000s |

| Test Description                                                       | Acceptance Criteria                                 | Result | Time   |
|------------------------------------------------------------------------|-----------------------------------------------------|--------|--------|
| Get required fields returns list[storm cdm] [cdm_name='storm_cdm']     | isinstance(cdm.get_required_fields(), list)         | Pass   | 0.000s |
| Get required fields returns list[stormts cdm] [cdm_name='stormts_cdm'] | isinstance(cdm.get_required_fields(), list)         | Pass   | 0.000s |
| List all fields returns paths[gauge cdm] [cdm_name='gauge_cdm']        | isinstance(fields, list); len(fields) > 0 (+2 more) | Pass   | 0.000s |
| List all fields returns paths[mortgage cdm] [cdm_name='mortgage_cdm']  | isinstance(fields, list); len(fields) > 0 (+2 more) | Pass   | 0.000s |
| List all fields returns paths[property cdm] [cdm_name='property_cdm']  | isinstance(fields, list); len(fields) > 0 (+2 more) | Pass   | 0.000s |
| List all fields returns paths[prs cdm] [cdm_name='prs_cdm']            | isinstance(fields, list); len(fields) > 0 (+2 more) | Pass   | 0.000s |
| List all fields returns paths[storm cdm] [cdm_name='storm_cdm']        | isinstance(fields, list); len(fields) > 0 (+2 more) | Pass   | 0.000s |
| List all fields returns paths[stormts cdm] [cdm_name='stormts_cdm']    | isinstance(fields, list); len(fields) > 0 (+2 more) | Pass   | 0.000s |
| Repr returns string[gauge cdm] [cdm_name='gauge_cdm']                  | isinstance(repr(cdm), str)                          | Pass   | 0.000s |
| Repr returns string[mortgage cdm] [cdm_name='mortgage_cdm']            | isinstance(repr(cdm), str)                          | Pass   | 0.000s |
| Repr returns string[property cdm] [cdm_name='property_cdm']            | isinstance(repr(cdm), str)                          | Pass   | 0.000s |
| Repr returns string[prs cdm] [cdm_name='prs_cdm']                      | isinstance(repr(cdm), str)                          | Pass   | 0.000s |
| Repr returns string[storm cdm] [cdm_name='storm_cdm']                  | isinstance(repr(cdm), str)                          | Pass   | 0.000s |
| Repr returns string[stormts cdm] [cdm_name='stormts_cdm']              | isinstance(repr(cdm), str)                          | Pass   | 0.000s |
| Schema returns non empty dict[gauge cdm] [cdm_name='gauge_cdm']        | isinstance(schema, dict); len(schema) > 0           | Pass   | 0.000s |
| Schema returns non empty dict[mortgage cdm] [cdm_name='mortgage_cdm']  | isinstance(schema, dict); len(schema) > 0           | Pass   | 0.000s |
| Schema returns non empty dict[property cdm] [cdm_name='property_cdm']  | isinstance(schema, dict); len(schema) > 0           | Pass   | 0.000s |
| Schema returns non empty dict[prs cdm] [cdm_name='prs_cdm']            | isinstance(schema, dict); len(schema) > 0           | Pass   | 0.000s |

| Test Description                                                    | Acceptance Criteria                                                              | Result | Time   |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|--------|
| Schema returns non empty dict[storm cdm] [cdm_name='storm_cdm']     | isinstance(schema, dict); len(schema) > 0                                        | Pass   | 0.000s |
| Schema returns non empty dict[stormts cdm] [cdm_name='stormts_cdm'] | isinstance(schema, dict); len(schema) > 0                                        | Pass   | 0.000s |
| Create mapping returns flat dict                                    | isinstance(mapping, dict); mapping.get("gauge_id") == "GAUGE-test1234" (+9 more) | Pass   | 0.000s |
| Create mapping with empty input                                     | isinstance(gauge_cdm.create_mapping({}), dict)                                   | Pass   | 0.000s |
| Get field info for gauge id                                         | info is not None; info.get("type") == "text"                                     | Pass   | 0.000s |
| Required fields include gauge id                                    | any("GaugeID" in f for f in gauge_cdm.get_required_fields())                     | Pass   | 0.000s |
| Schema has flood gauge root                                         | "FloodGauge" in gauge_cdm.schema                                                 | Pass   | 0.000s |
| Schema has header section                                           | "Header" in gauge_cdm.schema["FloodGauge"]                                       | Pass   | 0.000s |
| Validate empty data returns errors                                  | isinstance(gauge_cdm.validate({}), dict)                                         | Pass   | 0.000s |
| Validate valid gauge data                                           | len(gauge_cdm.validate(data)) == 0                                               | Pass   | 0.000s |
| Create mapping returns flat dict                                    | isinstance(mapping, dict); mapping.get("gauge_id") == "GAUGE-test1234" (+9 more) | Pass   | 0.000s |
| Required fields include mortgage id                                 | any("MortgageID" in f for f in mortgage_cdm.get_required...                      | Pass   | 0.000s |
| Schema has mortgage root                                            | len(list(mortgage_cdm.schema.keys())) > 0                                        | Pass   | 0.000s |
| Validate valid mortgage                                             | isinstance(mortgage_cdm.validate(data), dict)                                    | Pass   | 0.000s |
| Create mapping returns flat dict                                    | isinstance(mapping, dict); mapping.get("gauge_id") == "GAUGE-test1234" (+9 more) | Pass   | 0.000s |
| List all fields has gauge set                                       | any("GaugeSet" in f for f in prs_cdm.list_all_fields())                          | Pass   | 0.000s |
| Required fields include swap id                                     | any("SwapID" in f for f in prs_cdm.get_required_fields())                        | Pass   | 0.000s |
| Schema has root                                                     | isinstance(stormts_cdm.schema, dict); len(stormts_cdm.schema) > 0 (+2 more)      | Pass   | 0.000s |

| Test Description                                                           | Acceptance Criteria                                                                                                                | Result | Time   |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Create mapping returns flat dict                                           | <code>isinstance(mapping, dict);</code><br><code>mapping.get("gauge_id") ==</code><br><code>"GAUGE-test1234"</code> (+9 more)      | Pass   | 0.000s |
| Required fields not empty                                                  | <code>len(property_cdm.get_required_fields())</code><br><code>&gt; 0</code>                                                        | Pass   | 0.000s |
| Schema has property root                                                   | <code>len(list(property_cdm.schema.keys()))</code><br><code>&gt; 0</code>                                                          | Pass   | 0.000s |
| Validate valid property                                                    | <code>isinstance(property_cdm.validate(data), dict)</code>                                                                         | Pass   | 0.000s |
| Lines 255-256: exception in create_mapping → raises ValueError.            | —                                                                                                                                  | Pass   | 0.000s |
| Lines 213-256: create_mapping with all sections populated.                 | <code>mapping["storm_event_id"] ==</code><br><code>"STORM-001"; mapping["storm_name"] ==</code><br><code>"Eunice"</code> (+3 more) | Pass   | 0.000s |
| Create mapping returns flat dict                                           | <code>isinstance(mapping, dict);</code><br><code>mapping.get("gauge_id") ==</code><br><code>"GAUGE-test1234"</code> (+9 more)      | Pass   | 0.000s |
| Line 258-260: get_required_fields() returns list with StormEventID.        | <code>isinstance(fields, list);</code><br><code>any("StormEventID" in f for f in</code><br><code>fields)</code> (+2 more)          | Pass   | 0.000s |
| Required fields include storm id                                           | <code>any("StormEventID" in f for f in</code><br><code>storm_cdm.get_required_f...</code>                                          | Pass   | 0.000s |
| Schema has storm root                                                      | <code>len(list(storm_cdm.schema.keys())) &gt;</code><br><code>0</code>                                                             | Pass   | 0.000s |
| validate({}) returns errors for missing fields.                            | <code>isinstance(errors, dict);</code><br><code>isinstance(errors, dict)</code> (+1 more)                                          | Pass   | 0.000s |
| Lines 192-193: exception in validate → {'validation_error': [...]}.        | <code>isinstance(result, dict);</code><br><code>"validation_error" in result</code> (+2 more)                                      | Pass   | 0.000s |
| validate() with missing CatchmentID returns Header errors.                 | <code>"Header" in errors; any("CatchmentID"</code><br><code>in e for e in errors["Header"])</code> (+1 more)                       | Pass   | 0.000s |
| Lines 175-193: validate() with missing StormEventID returns Header errors. | <code>"Header" in errors;</code><br><code>any("StormEventID" in e for e in</code><br><code>errors["Header"])</code>                | Pass   | 0.000s |
| validate() with valid data returns empty dict.                             | <code>storm_cdm.validate(data) == {}</code>                                                                                        | Pass   | 0.000s |
| Lines 172-173: exception in create_mapping raises ValueError.              | —                                                                                                                                  | Pass   | 0.000s |

| Test Description                                                           | Acceptance Criteria                                                                           | Result | Time   |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| Lines 150-170: create_mapping with full Parameters section.                | mapping["storm_event_id"] ==<br>"STORM-001"; mapping["storm_name"] ==<br>"Eunice" (+3 more)   | Pass   | 0.000s |
| Create mapping returns flat dict                                           | isinstance(mapping, dict);<br>mapping.get("gauge_id") ==<br>"GAUGE-test1234" (+9 more)        | Pass   | 0.000s |
| Line 185: create_reading_mapping returns flat dict.                        | mapping["timestamp"]<br>== "2022-02-18T06:00";<br>mapping["rainfall_mm"] == 12.5<br>(+1 more) | Pass   | 0.000s |
| Lines 194-199: get_required_fields() returns list.                         | isinstance(fields, list);<br>any("StormEventID" in f for f in<br>fields) (+2 more)            | Pass   | 0.000s |
| Required fields include ts id                                              | any("TimeSeriesID" in f for f in<br>stormts_cdm.get_required...                               | Pass   | 0.000s |
| Schema has root                                                            | isinstance(stormts_cdm.schema, dict);<br>len(stormts_cdm.schema) > 0 (+2 more)                | Pass   | 0.000s |
| validate({}) returns errors for both required fields.                      | isinstance(errors, dict);<br>isinstance(errors, dict) (+1 more)                               | Pass   | 0.000s |
| Lines 137-138: exception in validate → {'validation_error': [...]}.        | isinstance(result, dict);<br>"validation_error" in result (+2<br>more)                        | Pass   | 0.000s |
| validate() with missing CatchmentID returns Header errors.                 | "Header" in errors; any("CatchmentID"<br>in e for e in errors["Header"]) (+1<br>more)         | Pass   | 0.000s |
| Lines 120-138: validate() with missing TimeSeriesID returns Header errors. | "Header" in errors;<br>any("TimeSeriesID" in e for e in<br>errors["Header"])                  | Pass   | 0.000s |
| validate() with valid data returns empty dict.                             | stormts_cdm.validate(data) == {}                                                              | Pass   | 0.000s |



## A Platform Infrastructure Tests

Table 24: Test Results Summary — Platform Infrastructure (PLATFORM)

|             |         |
|-------------|---------|
| Total Tests | 5599    |
| Passed      | 5585    |
| Failed      | 0       |
| Skipped     | 14      |
| Duration    | 169.91s |

Table 25: Detailed Test Results — Platform Infrastructure

| Test Description                                                             | Acceptance Criteria                                                                             | Result | Time   |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|--------|
| Test that config has CATCHMENT attribute.                                    | —                                                                                               | Pass   | 0.000s |
| Test that config ENDPOINTS includes health_check for backend handler.        | <code>cfg.ENDPOINTS['health_check'] == '/api/v1/health'; ep in cfg.ENDPOINTS</code>             | Pass   | 0.000s |
| Test that select_catchment.html exists.                                      | —                                                                                               | Pass   | 0.000s |
| Test that routes register correctly.                                         | —                                                                                               | Pass   | 0.050s |
| Existing synthetic fixture — confirms endpoint logic works.                  | <code>response.status_code == 200</code>                                                        | Pass   | 0.002s |
| Mounts the real model.inventory.json and confirms the endpoint does not 500. | <code>response.status_code == 200;</code><br><code>data["status"] == "success"</code> (+1 more) | Pass   | 0.040s |
| All models in the file are served by the endpoint.                           | <code>response.status_code == 200;</code><br><code>len(data["models"]) == expected.count</code> | Pass   | 0.042s |
| Concrete subclass returns empty list                                         | <code>result == []</code>                                                                       | Pass   | 0.000s |
| Get entity id called                                                         | <code>loader.find_by_id('A')</code> is not None                                                 | Pass   | 0.001s |
| Get entity summary called                                                    | <code>summaries[0]['id'] == 'B'</code>                                                          | Pass   | 0.001s |
| Add action auto generates id                                                 | <code>r.status_code == 200; actions[0]["id"] == "A-001"</code> (+1 more)                        | Pass   | 0.002s |
| Add action unknown meeting returns 404                                       | <code>r.status_code == 404</code>                                                               | Pass   | 0.002s |
| Delete action                                                                | <code>r.status_code == 200;</code><br><code>r.get_json()["meeting"]["actions"] == []</code>     | Pass   | 0.002s |
| Delete action unknown meeting returns 404                                    | <code>r.status_code == 404</code>                                                               | Pass   | 0.001s |

| Test Description                                                         | Acceptance Criteria                                                                                                            | Result | Time   |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Update action status                                                     | <code>r.status_code == 200;</code><br><code>r.get_json()["meeting"]["actions"][0]["status"]</code><br><code>== "Closed"</code> | Pass   | 0.003s |
| Update missing action returns 404                                        | <code>r.status_code == 404</code>                                                                                              | Pass   | 0.001s |
| Add action auto id                                                       | <code>"A-001" in ids; "A-002" in ids</code>                                                                                    | Pass   | 0.002s |
| Add action missing meeting returns 404                                   | <code>r.status_code == 404</code>                                                                                              | Pass   | 0.002s |
| Add action returns 200                                                   | <code>r.status_code == 200;</code><br><code>len(meeting["actions"]) == 2</code>                                                | Pass   | 0.002s |
| Delete action missing meeting returns 404                                | <code>r.status_code == 404</code>                                                                                              | Pass   | 0.002s |
| Delete action returns 200                                                | <code>r.status_code == 200;</code><br><code>len(r.get_json()["meeting"]["actions"])</code><br><code>== 0</code>                | Pass   | 0.002s |
| Update action not found returns 404                                      | <code>r.status_code == 404</code>                                                                                              | Pass   | 0.002s |
| Update action returns 200                                                | <code>r.status_code == 200; action["status"]</code><br><code>== "Closed"</code>                                                | Pass   | 0.002s |
| Production function: levels decay back toward base after peak.           | <code>levels[-1] &lt; max(levels)</code>                                                                                       | Pass   | 0.000s |
| Production function: peak base + level_change.                           | <code>max(levels) &gt;= base + change * 0.9</code>                                                                             | Pass   | 0.000s |
| _synthesize_hydrograph default returns 168 hours (STORM_HOURS constant). | <code>STORM_HOURS == 168; len(levels) == 168</code>                                                                            | Pass   | 0.000s |
| Production function: hour 0 = base_level.                                | <code>abs(levels[0] - 3.0) &lt; 0.01</code>                                                                                    | Pass   | 0.000s |
| Peak occurs near duration_hours * peak_position.                         | <code>abs(actual_peak_hour -</code><br><code>expected_peak_hour) &lt;= 2</code>                                                | Pass   | 0.000s |
| Add appears in list                                                      | <code>len(r.get_json()["references"]) == 1</code>                                                                              | Pass   | 0.003s |
| Add assigns id                                                           | <code>"id" in ref; len(ref["id"]) &gt; 0</code>                                                                                | Pass   | 0.002s |
| Add no json body uses defaults                                           | <code>r.status_code in (200, 400, 500)</code>                                                                                  | Pass   | 0.002s |
| Add returns 200                                                          | <code>r.status_code == 200</code>                                                                                              | Pass   | 0.002s |
| Add returns reference                                                    | <code>data["status"] == "success";</code><br><code>"reference" in data</code>                                                  | Pass   | 0.001s |
| Add same category twice no duplicate                                     | <code>r.get_json()["categories"].count("Risk")</code><br><code>== 1</code>                                                     | Pass   | 0.004s |
| Add stores all fields                                                    | <code>ref["author"] == "Jones";</code><br><code>ref["journal"] == "Nature"</code> (+1 more)                                    | Pass   | 0.002s |
| Add with category updates categories                                     | <code>"Flood Risk" in</code><br><code>r.get_json()["categories"]</code>                                                        | Pass   | 0.003s |

| Test Description                                                    | Acceptance Criteria                                                                                 | Result | Time   |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| Adds to map                                                         | —                                                                                                   | Pass   | 0.002s |
| Returns notification system                                         | <code>isinstance(result, NotificationSystem)</code>                                                 | Pass   | 0.002s |
| Add to map calls add child                                          | —                                                                                                   | Pass   | 0.004s |
| Add to map calls add child                                          | —                                                                                                   | Pass   | 0.002s |
| Add to map calls add child                                          | —                                                                                                   | Pass   | 0.002s |
| Approaching retirement over 55                                      | <code>isinstance(result, list)</code>                                                               | Pass   | 0.000s |
| Early career under 25                                               | <code>isinstance(result, list)</code>                                                               | Pass   | 0.000s |
| Prime years 25 to 55                                                | <code>isinstance(result, list)</code>                                                               | Pass   | 0.000s |
| Add agenda item                                                     | <code>r.status_code == 200; len(agenda) == 1</code><br>(+2 more)                                    | Pass   | 0.002s |
| Add agenda unknown meeting returns 404                              | <code>r.status_code == 404</code>                                                                   | Pass   | 0.002s |
| Add multiple items increments number                                | <code>[a["item"] for a in agenda] == [1, 2, 3]</code>                                               | Pass   | 0.007s |
| Delete agenda item rennumbers                                       | <code>len(agenda) == 2; [a["item"] for a in agenda] == [1, 2]</code> (+1 more)                      | Pass   | 0.007s |
| Delete agenda unknown meeting returns 404                           | <code>r.status_code == 404</code>                                                                   | Pass   | 0.001s |
| Update agenda item                                                  | <code>r.status_code == 200; item["title"] == "Updated"</code> (+1 more)                             | Pass   | 0.003s |
| Update agenda unknown meeting returns 404                           | <code>r.status_code == 404</code>                                                                   | Pass   | 0.002s |
| Update missing item returns 404                                     | <code>r.status_code == 404</code>                                                                   | Pass   | 0.001s |
| Add agenda item increments count                                    | <code>len(meeting["agenda"]) == 3</code>                                                            | Pass   | 0.005s |
| Add agenda item returns 200                                         | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>              | Pass   | 0.003s |
| Add agenda missing meeting returns 404                              | <code>r.status_code == 404</code>                                                                   | Pass   | 0.002s |
| Delete agenda item returns 200                                      | <code>r.status_code == 200;</code><br><code>len(meeting["agenda"]) == 1</code> (+1 more)            | Pass   | 0.003s |
| Delete agenda missing meeting returns 404                           | <code>r.status_code == 404</code>                                                                   | Pass   | 0.002s |
| Update agenda item not found returns 404                            | <code>r.status_code == 404</code>                                                                   | Pass   | 0.002s |
| Update agenda item returns 200                                      | <code>r.status_code == 200;</code><br><code>meeting["agenda"][0]["title"] == "Updated Title"</code> | Pass   | 0.002s |
| When peak_wl is alert but warning, threshold is 'alert' (line 294). | <code>resp.status_code == 200; g001 is not None</code> (+1 more)                                    | Pass   | 0.063s |

| Test Description                                                                 | Acceptance Criteria                                                                                                                                           | Result | Time   |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Empty report                                                                     | <code>result['num_clones'] == 0;</code><br><code>result['worst'] == []</code> (+1 more)                                                                       | Pass   | 0.000s |
| Fmt stats has python                                                             | <code>'python' in result['fmt_stats']</code>                                                                                                                  | Pass   | 0.000s |
| Line numbers extracted from 'start' field must be integers.                      | <code>isinstance(entry[1], int);</code><br><code>isinstance(entry[3], int)</code>                                                                             | Pass   | 0.000s |
| Regression: jscpd uses 'start' not 'startLine' — must not produce '?' in output. | <code>11 != '?'; 12 != '?'</code>                                                                                                                             | Pass   | 0.000s |
| Largest sorted descending                                                        | <code>lines == sorted(lines, reverse=True)</code>                                                                                                             | Pass   | 0.000s |
| Num clones correct                                                               | <code>result['num_clones'] == 2</code>                                                                                                                        | Pass   | 0.000s |
| Returns dict                                                                     | <code>isinstance(result, dict)</code>                                                                                                                         | Pass   | 0.000s |
| Total present                                                                    | <code>'total' in result;</code><br><code>result['total']['lines'] == 10000</code>                                                                             | Pass   | 0.000s |
| Worst populated                                                                  | <code>len(result['worst']) &gt; 0;</code><br><code>isinstance(fname, str)</code> (+2 more)                                                                    | Pass   | 0.000s |
| computePRSCashflows function exists.                                             | <code>'computePRSCashflows' in prs_js</code>                                                                                                                  | Pass   | 0.000s |
| interpolateSurvival function exists.                                             | <code>'interpolateSurvival' in prs_js</code>                                                                                                                  | Pass   | 0.000s |
| interpolateYieldRate function exists.                                            | <code>'interpolateYieldRate' in prs_js</code>                                                                                                                 | Pass   | 0.000s |
| Completeness percentage computed                                                 | <code>r["PropertyHeader"]["percentage"] ==</code><br><code>pytest.approx(66.67,...</code>                                                                     | Pass   | 0.000s |
| Empty section excluded                                                           | <code>"PropertyHeader" not in r</code>                                                                                                                        | Pass   | 0.000s |
| Mortgage data included                                                           | <code>"Mortgage Data" in r</code>                                                                                                                             | Pass   | 0.000s |
| Property header section present                                                  | <code>"PropertyHeader" in r</code>                                                                                                                            | Pass   | 0.000s |
| Returns dict                                                                     | <code>isinstance(r, dict);</code> <code>isinstance(r,</code><br><code>dict)</code>                                                                            | Pass   | 0.000s |
| Section not present excluded                                                     | <code>len(r) == 0</code>                                                                                                                                      | Pass   | 0.000s |
| All none returns zeros                                                           | <code>result["counts"]["properties"] == 0;</code><br><code>result["counts"]["mortgages"] == 0</code><br>(+1 more)                                             | Pass   | 0.000s |
| Counts match data                                                                | <code>result["counts"]["properties"] == 2;</code><br><code>result["counts"]["mortgages"] == 1</code><br>(+1 more)                                             | Pass   | 0.000s |
| Flood mortgages counted                                                          | <code>result["counts"]["flood_mortgages"]</code><br><code>== 1</code>                                                                                         | Pass   | 0.000s |
| Flood properties overlap                                                         | <code>result["counts"]["flood_properties"]</code><br><code>== 2;</code><br><code>result["overlaps"]["properties_with_flood_risk"]</code><br><code>== 1</code> | Pass   | 0.000s |

| Test Description                                                             | Acceptance Criteria                                                                           | Result | Time   |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| Returns required keys                                                        | "counts" in result; "overlaps" in result                                                      | Pass   | 0.000s |
| pts dir exists but property has no flood.events → 404.                       | client.get(_COMPOSITE_URL).status_code == 404                                                 | Pass   | 0.030s |
| pts dir exists but all flood_depths are zero → 404.                          | client.get(_COMPOSITE_URL).status_code == 404                                                 | Pass   | 0.030s |
| No pts dir returns 404                                                       | pts_client_no_data.get(_COMPOSITE_URL).status_code == 404                                     | Pass   | 0.000s |
| No pts dir status error                                                      | pts_client_no_data.get(_COMPOSITE_URL).get_json()["status"]                                   | Pass   | 0.000s |
| Options returns ok                                                           | pts_client_no_data.options(_COMPOSITE_URL).status_code == ...                                 | Pass   | 0.000s |
| Frame has gauges                                                             | "gauges" in frame0                                                                            | Pass   | 0.000s |
| Frame has hour                                                               | frame0["hour"] == 0                                                                           | Pass   | 0.000s |
| Frame has properties                                                         | "properties" in frame0                                                                        | Pass   | 0.000s |
| Frame has stats                                                              | "gauges_flooded" in stats; "properties_flooded" in stats (+1 more)                            | Pass   | 0.000s |
| Arrived true at arrival time                                                 | frames[4]["properties"][0]["arrived"] is False; frames[5]["properties"][0]["arrived"] is True | Pass   | 0.006s |
| Peak true at and after peak time                                             | frames[11]["properties"][0]["peak"] is False; frames[12]["properties"][0]["peak"] is True     | Pass   | 0.006s |
| hour i= len(readings) → flooded=False, depth_m=0, arrived=False, peak=False. | p["flooded"] is False; p["depth_m"] == 0 (+2 more)                                            | Pass   | 0.032s |
| Property does not have wse m                                                 | "wse_m" not in frames[0]["properties"][0]                                                     | Pass   | 0.000s |
| Property has peak field                                                      | "peak" in frames[0]["properties"][0]                                                          | Pass   | 0.000s |
| Property has required fields                                                 | f in p                                                                                        | Pass   | 0.000s |
| Frames list length                                                           | len(pts_env["client"].get(_COMPOSITE_URL).get_json())                                         | Pass   | 0.006s |
| N frames equals storm hours                                                  | pts_env["client"].get(_COMPOSITE_URL).get_json()["n_frames"]                                  | Pass   | 0.006s |
| N properties affected is one                                                 | pts_env["client"].get(_COMPOSITE_URL).get_json()["n_properties"]                              | Pass   | 0.008s |
| Returns 200                                                                  | pts_env["client"].get(_COMPOSITE_URL).status_code == 200                                      | Pass   | 0.005s |
| Status success                                                               | pts_env["client"].get(_COMPOSITE_URL).get_json()["status"]                                    | Pass   | 0.006s |
| Storm id is composite                                                        | pts_env["client"].get(_COMPOSITE_URL).get_json()["storm_id"]                                  | Pass   | 0.006s |



| Test Description                                                          | Acceptance Criteria                                                                                     | Result | Time   |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------|--------|
| Property with two storms: composite uses the one with higher flood_depth. | data["status"] == "success";<br>data["frames"][0]["properties"][0]["depth_m"]<br>== pytest.a...         | Pass   | 0.035s |
| No pts dir returns 404                                                    | r.status_code == 404                                                                                    | Pass   | 0.000s |
| No pts dir status error                                                   | pts_client.no_data.get(<br>f"/api/v1/propertyts/...                                                     | Pass   | 0.000s |
| Options returns ok                                                        | r.status_code == 200                                                                                    | Pass   | 0.000s |
| Unknown storm message mentions storm id                                   | "STORM-GHOST" in msg                                                                                    | Pass   | 0.032s |
| Unknown storm returns 404                                                 | client.get("/api/v1/propertyts/animate/STORM-GHOST")                                                    | Pass   | 0.030s |
| Unknown storm status error                                                | client.get(<br>"/api/v1/propertyts/animate/STORM...                                                     | Pass   | 0.030s |
| Frame has gauges                                                          | "gauges" in frame0;<br>len(frame0["gauges"]) == 1                                                       | Pass   | 0.000s |
| Frame has hour field                                                      | "hour" in frame0                                                                                        | Pass   | 0.000s |
| Frame has properties                                                      | "properties" in frame0;<br>len(frame0["properties"]) == 1                                               | Pass   | 0.000s |
| Frame has stats                                                           | "gauges_flooded" in stats;<br>"properties_flooded" in stats (+1 more)                                   | Pass   | 0.000s |
| Frame hour zero is first                                                  | frame0["hour"] == 0                                                                                     | Pass   | 0.000s |
| Hour values sequential                                                    | [f["hour"] for f in frames] ==<br>list(range(STORM_HOURS))                                              | Pass   | 0.033s |
| Gauge has required fields                                                 | f in gauge0                                                                                             | Pass   | 0.000s |
| Gauge id correct                                                          | gauge0["gauge_id"] == "GAUGE-001"                                                                       | Pass   | 0.000s |
| Gauge status alert when above alert                                       | gauge0["status"] == "alert"                                                                             | Pass   | 0.000s |
| Gauge status normal when below alert                                      | g["status"] == "normal"                                                                                 | Pass   | 0.034s |
| Gauge status severe                                                       | g["status"] == "severe"                                                                                 | Pass   | 0.032s |
| Gauge status warning                                                      | g["status"] == "warning"                                                                                | Pass   | 0.033s |
| Gauge readings using water_level_m key (not waterLevel) still work.       | g["water_level_m"] == 4.3                                                                               | Pass   | 0.033s |
| Gauge with no readings defaults to zero                                   | frames[10]["gauges"][0]["water_level_m"]<br>== 0                                                        | Pass   | 0.032s |
| Water level from readings                                                 | gauge0["water_level_m"] == 4.2                                                                          | Pass   | 0.000s |
| Arrived false before arrival time                                         | frames[0]["properties"][0]["arrived"]<br>is False;<br>frames[4]["properties"][0]["arrived"]<br>is False | Pass   | 0.000s |

| Test Description                                                                       | Acceptance Criteria                                                                                    | Result | Time   |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------|--------|
| Arrived true at and after arrival time                                                 | frames[5]["properties"][0]["arrived"]<br>is True;<br>frames[10]["properties"][0]["arrived"]<br>is True | Pass   | 0.000s |
| When hour $i = \text{len}(\text{readings})$ ,<br>flooded=False, depth_m=0,<br>wse_m=0. | p["flooded"] is False; p["depth_m"] ==<br>0 (+1 more)                                                  | Pass   | 0.031s |
| Property has required fields                                                           | f in p                                                                                                 | Pass   | 0.000s |
| Property id correct                                                                    | frames[0]["properties"][0]["property_id"]<br>== "PROP-001"                                             | Pass   | 0.000s |
| Gauges flooded counts non normal                                                       | stats["gauges_flooded"] == 1                                                                           | Pass   | 0.032s |
| Gauges flooded zero when normal                                                        | stats["gauges_flooded"] == 0                                                                           | Pass   | 0.033s |
| Properties flooded is positive early<br>on                                             | stats["properties_flooded"] >= 1                                                                       | Pass   | 0.033s |
| Total depth m non negative                                                             | all(f["stats"]["total_depth_m"] >= 0<br>for f in frames[:5])                                           | Pass   | 0.034s |
| Frames list length equals storm<br>hours                                               | len(data["frames"]) == STORM_HOURS                                                                     | Pass   | 0.006s |
| N frames equals storm hours                                                            | data["n_frames"] == STORM_HOURS                                                                        | Pass   | 0.006s |
| N properties affected is one                                                           | data["n_properties_affected"] == 1                                                                     | Pass   | 0.005s |
| Returns 200                                                                            | anim_client.get(f"/api/v1/propertyts/animal/STORM_ID").status_code == 200                              | Pass   | 0.005s |
| Status success                                                                         | data["status"] == "success"                                                                            | Pass   | 0.004s |
| Storm id in response                                                                   | data["storm_id"] == STORM_ID                                                                           | Pass   | 0.007s |
| Minimum spread applied for floor<br>property                                           | prob >= expected_min                                                                                   | Pass   | 0.100s |
| Return period finite for floor                                                         | rp is not None; rp > 0                                                                                 | Pass   | 0.102s |
| Appends line                                                                           | "hello world" in lines                                                                                 | Pass   | 0.000s |
| Clear then append                                                                      | lines == ["fresh line"]                                                                                | Pass   | 0.000s |
| Trims buffer at 2000                                                                   | length <= 2000                                                                                         | Pass   | 0.001s |
| All factors applied                                                                    | abs(result - round(expected, 2)) <<br>0.01                                                             | Pass   | 0.000s |
| Basic calculation                                                                      | result == 1250.0                                                                                       | Pass   | 0.000s |
| Claims factor increases premium                                                        | with_claims > base                                                                                     | Pass   | 0.000s |
| Clamped to max                                                                         | result == MAX_PREMIUM                                                                                  | Pass   | 0.000s |
| Clamped to min                                                                         | result == MIN_PREMIUM                                                                                  | Pass   | 0.000s |
| Contents premium added                                                                 | with_contents == base + 500.0                                                                          | Pass   | 0.000s |
| High flood risk increases premium                                                      | high_flood > low_flood                                                                                 | Pass   | 0.000s |
| Returns float                                                                          | isinstance(result, float)                                                                              | Pass   | 0.000s |
| Security factor reduces premium                                                        | with_security < without                                                                                | Pass   | 0.000s |
| All non negative                                                                       | np.all(result >= 0.0)                                                                                  | Pass   | 0.001s |

| Test Description                                                                  | Acceptance Criteria                                                               | Result | Time   |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------|--------|
| For each active hour, mean(gauge-precip) approximately equals catchment_precip.   | <code>result[:, h].mean() ==<br/>pytest.approx(precip[h], rel=1e-6)</code>        | Pass   | 0.001s |
| Output shape                                                                      | <code>result.shape ==<br/>(thames_model.n_gauges, 168)</code>                     | Pass   | 0.001s |
| Different gauges get different precipitation amounts.                             | <code>col.std() &gt; 0.0</code>                                                   | Pass   | 0.001s |
| Mean across gauges of total precip approximately equals original catchment total. | <code>per_gauge_total.mean() ==<br/>pytest.approx(catchment_total,<br/>...</code> | Pass   | 0.001s |
| Zero precip hours remain zero                                                     | <code>np.all(result[:, zero_hours] == 0.0)</code>                                 | Pass   | 0.001s |
| All factors applied                                                               | <code>abs(result - round(expected, 2)) &lt;<br/>0.01</code>                       | Pass   | 0.000s |
| Basic calculation                                                                 | <code>result == 800_000.0</code>                                                  | Pass   | 0.000s |
| Clamped to max                                                                    | <code>result == MAX_PROPERTY_VALUE</code>                                         | Pass   | 0.000s |
| Clamped to min                                                                    | <code>result == MIN_PROPERTY_VALUE</code>                                         | Pass   | 0.000s |
| Flood risk discount reduces value                                                 | <code>with_risk &lt; no_risk</code>                                               | Pass   | 0.000s |
| Noise factor applied                                                              | <code>result2 &gt; result1</code>                                                 | Pass   | 0.000s |
| Returns float                                                                     | <code>isinstance(result, float)</code>                                            | Pass   | 0.000s |
| Age risk approaching retirement                                                   | <code>"Medium" in r["Age Risk"]</code>                                            | Pass   | 0.000s |
| Age risk early career                                                             | <code>"Medium" in r["Age Risk"]</code>                                            | Pass   | 0.000s |
| Age risk prime years                                                              | <code>"Age Risk" in r; "Low" in r["Age<br/>Risk"]</code>                          | Pass   | 0.000s |
| Credit risk high for poor score                                                   | <code>"High" in r["Credit Risk"]</code>                                           | Pass   | 0.000s |
| Credit risk low for high score                                                    | <code>"Credit Risk" in r; "Low" in<br/>r["Credit Risk"]</code>                    | Pass   | 0.000s |
| Credit risk low medium for good score                                             | <code>"Low-Medium" in r["Credit Risk"]</code>                                     | Pass   | 0.000s |
| Credit risk medium for fair score                                                 | <code>"Medium" in r["Credit Risk"]</code>                                         | Pass   | 0.000s |
| Employment risk established 2 to 5                                                | <code>"Low-Medium" in r["Employment Risk"]</code>                                 | Pass   | 0.000s |
| Employment risk recent under 2                                                    | <code>"Medium-High" in r["Employment<br/>Risk"]</code>                            | Pass   | 0.000s |
| Employment risk stable 5 plus                                                     | <code>"Employment Risk" in r; "Low" in<br/>r["Employment Risk"]</code>            | Pass   | 0.000s |
| Income risk low for high income                                                   | <code>"Income Risk" in r; "Low" in<br/>r["Income Risk"]</code>                    | Pass   | 0.000s |
| Income risk low medium adequate                                                   | <code>"Low-Medium" in r["Income Risk"]</code>                                     | Pass   | 0.000s |
| Income risk medium limited                                                        | <code>"Medium" in r["Income Risk"]</code>                                         | Pass   | 0.000s |
| Returns dict                                                                      | <code>isinstance(result, dict)</code>                                             | Pass   | 0.000s |
| Best and worst sections shown                                                     | <code>"Best Data Section" in r; "Weakest<br/>Data Section" in r</code>            | Pass   | 0.000s |

| Test Description                                   | Acceptance Criteria                                                        | Result | Time   |
|----------------------------------------------------|----------------------------------------------------------------------------|--------|--------|
| Critical completeness high                         | "Critical Data Reliability" in r; "High" in r["Critical Data Reliability"] | Pass   | 0.000s |
| Critical completeness low                          | "Low" in r["Critical Data Reliability"]                                    | Pass   | 0.000s |
| Critical completeness medium                       | "Medium" in r["Critical Data Reliability"]                                 | Pass   | 0.000s |
| Excellent over 90                                  | "Excellent" in r["Overall Completeness"]                                   | Pass   | 0.000s |
| Fair 60 to 75                                      | "Fair" in r["Overall Completeness"]                                        | Pass   | 0.000s |
| Good 75 to 90                                      | "Good" in r["Overall Completeness"]                                        | Pass   | 0.000s |
| Limited 40 to 60                                   | "Limited" in r["Overall Completeness"]                                     | Pass   | 0.000s |
| Poor under 40                                      | "Poor" in r["Overall Completeness"]                                        | Pass   | 0.000s |
| Building 25 years → Contemporary category.         | isinstance(result, list)                                                   | Pass   | 0.001s |
| Building older than 100 years → Historic category. | isinstance(result, list)                                                   | Pass   | 0.001s |
| Building 50–100 years → Mature category.           | isinstance(result, list)                                                   | Pass   | 0.001s |
| Building 25–50 years → Modern category.            | isinstance(result, list)                                                   | Pass   | 0.001s |
| Basic info subsection                              | any("Basic Property Information" in t for t in texts)                      | Pass   | 0.001s |
| Building age subsection                            | any("Building Age" in t for t in texts)                                    | Pass   | 0.001s |
| Condition fair recommendation                      | isinstance(result, list)                                                   | Pass   | 0.001s |
| Condition good recommendation                      | isinstance(result, list)                                                   | Pass   | 0.001s |
| Condition poor recommendation                      | isinstance(result, list)                                                   | Pass   | 0.001s |
| Condition unknown falls back                       | isinstance(result, list)                                                   | Pass   | 0.001s |
| Empty property does not crash                      | isinstance(result, list)                                                   | Pass   | 0.000s |
| External features subsection                       | any("External Features" in t for t in texts)                               | Pass   | 0.001s |
| Has section header                                 | any("Property Attributes" in t for t in texts)                             | Pass   | 0.001s |
| Has tables                                         | any(isinstance(e, Table) for e in result)                                  | Pass   | 0.001s |
| No construction year skips age                     | isinstance(result, list)                                                   | Pass   | 0.001s |
| Physical dimensions subsection                     | any("Physical Dimensions" in t for t in texts)                             | Pass   | 0.001s |

| Test Description                                                                               | Acceptance Criteria                                                                               | Result | Time   |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------|--------|
| Returns list                                                                                   | <code>isinstance(result, list); len(result) &gt; 0</code>                                         | Pass   | 0.001s |
| Room composition subsection                                                                    | <code>any("Room Composition" in t for t in texts)</code>                                          | Pass   | 0.001s |
| With mortgage data does not crash                                                              | <code>isinstance(result, list)</code>                                                             | Pass   | 0.001s |
| Line 82: audit log $i$ 10000 entries gets trimmed.                                             | <code>r.status_code == 200; len(log) &lt;= 10000</code>                                           | Pass   | 0.069s |
| Line 57: inventory file empty/missing $\rightarrow$ 404.                                       | <code>r.status_code == 404</code>                                                                 | Pass   | 0.001s |
| Line 85: save failure $\rightarrow$ 500.                                                       | <code>r.status_code == 500</code>                                                                 | Pass   | 0.003s |
| Audit log truncated to 10000                                                                   | <code>r.status_code == 200; len(entries) == 10000 (+1 more)</code>                                | Pass   | 0.060s |
| <code>_auditTerminalRefreshNow</code> must be on window — called from onclick in innerHTML.    | <code>'window._auditTerminalRefreshNow' in audit_reports_js</code>                                | Pass   | 0.000s |
| JSON button was deliberately removed — must not reappear.                                      | <code>'test_failures_report.json' not in audit_reports_js</code>                                  | Pass   | 0.000s |
| Poll output endpoint                                                                           | <code>'test-report/output' in audit_reports_js</code>                                             | Pass   | 0.000s |
| Refresh audit tab button                                                                       | <code>'_auditTerminalRefreshNow' in audit_reports_js</code>                                       | Pass   | 0.000s |
| <code>_runTestSuite</code> must be on window — onclick attributes execute in global scope.     | <code>'window._runTestSuite' in audit_reports_js</code>                                           | Pass   | 0.000s |
| Run suite function                                                                             | <code>'_runTestSuite' in audit_reports_js</code>                                                  | Pass   | 0.000s |
| Terminal overlay id                                                                            | <code>'_audit-terminal-overlay' in audit_reports_js</code>                                        | Pass   | 0.000s |
| <code>_toggleFailureDetail</code> must be on window — called from onclick in innerHTML.        | <code>'window._toggleFailureDetail' in audit_reports_js</code>                                    | Pass   | 0.000s |
| When audit log exceeds 10 000 entries, it keeps only the last 10 000.                          | <code>len(entries) == 10_000; entries[0].get("seq") == 1 or entries[0].get("model_id") ...</code> | Pass   | 0.052s |
| <code>auto_open=True</code> enters the auto-open block (lines 324-325).                        | <code>report_path is not None; Path(report_path).exists()</code>                                  | Pass   | 0.017s |
| <code>auto_open=False</code> (default) — no open attempt, returns PDF path normally.           | <code>report_path is not None; Path(report_path).exists()</code>                                  | Pass   | 0.016s |
| <code>auto_open=True</code> with a successful <code>open_pdf_file</code> call (lines 324-325). | <code>report_path is not None; Path(report_path).exists()</code>                                  | Pass   | 0.017s |
| All pages exist                                                                                | <code>p in gen.pages</code>                                                                       | Pass   | 0.001s |



| Test Description                                                     | Acceptance Criteria                                                                         | Result | Time   |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| All pages exist                                                      | <code>p in gen.pages</code>                                                                 | Pass   | 0.001s |
| Always includes analysis pages                                       | <code>ap in pages</code>                                                                    | Pass   | 0.002s |
| Always includes appendix                                             | <code>"appendix" in pages</code>                                                            | Pass   | 0.001s |
| Always includes gauge info pages                                     | <code>gp in pages</code>                                                                    | Pass   | 0.001s |
| Always includes operational pages                                    | <code>op in pages</code>                                                                    | Pass   | 0.001s |
| Always includes property pages                                       | <code>pp in pages</code>                                                                    | Pass   | 0.002s |
| Always includes summary pages                                        | <code>sp in pages</code>                                                                    | Pass   | 0.001s |
| Empty flood data                                                     | <code>isinstance(pages, list)</code>                                                        | Pass   | 0.001s |
| Empty property data                                                  | <code>isinstance(pages, list)</code>                                                        | Pass   | 0.002s |
| Includes analysis pages                                              | <code>p in pages</code>                                                                     | Pass   | 0.001s |
| Includes overview pages                                              | <code>p in pages</code>                                                                     | Pass   | 0.001s |
| Returns list                                                         | <code>isinstance(pages, list); len(pages) &gt; 0</code>                                     | Pass   | 0.001s |
| Returns list                                                         | <code>isinstance(pages, list); len(pages) &gt; 0</code>                                     | Pass   | 0.001s |
| With hazard curve includes analysis pages                            | <code>any(p in pages for p in analysis_pages)</code>                                        | Pass   | 0.001s |
| With mortgage includes mortgage pages                                | <code>mp in pages</code>                                                                    | Pass   | 0.002s |
| With timeseries includes analysis pages                              | <code>any(p in pages for p in analysis_pages)</code>                                        | Pass   | 0.001s |
| Without mortgage excludes mortgage pages                             | <code>mp not in pages</code>                                                                | Pass   | 0.002s |
| Without timeseries or hazard no analysis pages                       | <code>not any(p in pages for p in analysis_pages)</code>                                    | Pass   | 0.001s |
| Avail width value                                                    | <code>AVAIL.WIDTH == pytest.approx(7.3 * inch)</code>                                       | Pass   | 0.000s |
| Lines 60-61: custom server_url and endpoints override config.        | <code>handler.server_url == 'http://test:9999'; handler.endpoints == {'a': '/api/a'}</code> | Pass   | 0.000s |
| Lines 60-61: default server_url and endpoints from config.           | <code>handler.server_url == config.SERVER_URL; handler.endpoints == config.ENDPOINTS</code> | Pass   | 0.000s |
| Endpoints dict is a copy, not a reference.                           | <code>'z' not in handler.endpoints</code>                                                   | Pass   | 0.000s |
| Line 154: BaseCDM.get_required_fields returns empty list by default. | <code>cdm.get_required_fields() == []</code>                                                | Pass   | 0.000s |
| get_field_info on a leaf field returns its definition.               | <code>info is not None; info["type"] == "text"</code>                                       | Pass   | 0.000s |

| Test Description                                                            | Acceptance Criteria                                                           | Result | Time   |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|--------|
| Line 116: field path not found returns None.                                | <code>mortgage_cdm.get_field_info("Mortgage.Nonexistent.Field")</code>        | Pass   | 0.000s |
| Line 116: partial path that resolves to non-dict returns None.              | <code>mortgage_cdm.get_field_info("Nonexistent")</code> is None               | Pass   | 0.000s |
| Line 118: <code>get_field_info</code> returns None when leaf is not a dict. | <code>cdm.get_field_info("Root.Field")</code> is None                         | Pass   | 0.000s |
| Line 133: metadata keys at nested level are also skipped.                   | "Root.Sub.Name" in fields; 'options' not in path.parts (+1 more)              | Pass   | 0.000s |
| list_all_fields on PRS CDM covers nested GaugeSet section.                  | any("GaugeSet" in f for f in fields); any("Triggers" in f for f in fields)    | Pass   | 0.000s |
| Line 133: list_all_fields recurses nested sections.                         | isinstance(fields, list); len(fields) > 10 (+2 more)                          | Pass   | 0.000s |
| Line 133: 'type', 'options', 'values', 'description', 'items' are skipped.  | len(fields) == 1; fields[0] == "Section.RealField"                            | Pass   | 0.000s |
| Line 157: <code>__repr__</code> returns class name.                         | "MortgageCDM" in repr(mortgage_cdm)                                           | Pass   | 0.000s |
| Verify data is cached after first load.                                     | len(result1) == len(result2) == 3                                             | Pass   | 0.001s |
| Test cache clearing.                                                        | loader._cache is not None; loader._cache is None (+1 more)                    | Pass   | 0.001s |
| Verify force_reload bypasses cache.                                         | len(result1) == 3; len(result2) == 0                                          | Pass   | 0.001s |
| Test cache invalidation.                                                    | loader._cache_valid is True; loader._cache_valid is False                     | Pass   | 0.001s |
| Verify abstract methods must be implemented.                                | —                                                                             | Pass   | 0.000s |
| Subclass must implement <code>get_entity_id</code> .                        | —                                                                             | Pass   | 0.000s |
| Subclass must implement <code>get_entity_summary</code> .                   | —                                                                             | Pass   | 0.000s |
| Test <code>file_exists</code> method.                                       | loader.file_exists() is True; loader2.file_exists() is False                  | Skip   | 0.000s |
| Load data with alternative container key (portfolio).                       | len(result) == 1                                                              | Pass   | 0.001s |
| Load data when JSON has container key.                                      | len(result) == 2                                                              | Pass   | 0.001s |
| Load data when JSON is a direct list.                                       | len(result) == 1; result[0]['PropertyHeader']['Header']['PropertyID'] == 'P1' | Pass   | 0.001s |
| Handle missing file gracefully.                                             | result == []                                                                  | Pass   | 0.000s |
| Load data when JSON is single object.                                       | len(result) == 1                                                              | Pass   | 0.001s |
| Test count method.                                                          | property_loader.count() == 3                                                  | Pass   | 0.001s |

| Test Description                                                        | Acceptance Criteria                                                                                                                                     | Result | Time   |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Test exists method.                                                     | <code>property_loader.exists("PROP-001")</code><br><code>is True;</code><br><code>property_loader.exists("NONEXISTENT")</code><br><code>is False</code> | Pass   | 0.001s |
| Test finding entity by ID.                                              | <code>result is not None;</code><br><code>result['PropertyHeader']['Header']['PropertyID']</code><br><code>== "PROP..."</code>                          | Pass   | 0.001s |
| Test finding non-existent entity.                                       | <code>result is None</code>                                                                                                                             | Pass   | 0.001s |
| Test getting all IDs.                                                   | <code>len(ids) == 3; "PROP-001" in ids (+2</code><br><code>more)</code>                                                                                 | Skip   | 0.000s |
| Test get_status method.                                                 | <code>status['entity_type'] == 'property';</code><br><code>status['file.exists'] is True (+1</code><br><code>more)</code>                               | Skip   | 0.000s |
| Test listing all entities.                                              | <code>len(result) == 3; all('propertyId' in</code><br><code>item for item in result)</code>                                                             | Pass   | 0.001s |
| Line 190: gauge_elevs defaults to [prop_elevation], elev_diff = 0.      | <code>result['elevation_diff_m'] == 0.0</code>                                                                                                          | Pass   | 0.000s |
| Lines 179, 190: empty nearest_gauges defaults distances and elevations. | <code>'total_basis_bp' in result;</code><br><code>'distance_bp' in result (+3 more)</code>                                                              | Pass   | 0.000s |
| All already trained                                                     | <code>result["trained"] == 0;</code><br><code>result["skipped"] == 3</code>                                                                             | Pass   | 0.002s |
| After the first gauge trains, ETA should appear in output.              | <code>"ETA" in captured</code>                                                                                                                          | Pass   | 0.202s |
| Missing gauge json                                                      | —                                                                                                                                                       | Pass   | 0.000s |
| Missing sequences                                                       | —                                                                                                                                                       | Pass   | 0.001s |
| No valid gauges                                                         | <code>result["trained"] == 0</code>                                                                                                                     | Pass   | 0.002s |
| One gauge already trained, one succeeds, one fails.                     | <code>result["trained"] == 1;</code><br><code>result["failed"] == 1 (+1 more)</code>                                                                    | Pass   | 0.139s |
| Successful training                                                     | <code>result["trained"] == 3;</code><br><code>result["failed"] == 0 (+2 more)</code>                                                                    | Pass   | 0.205s |
| Training failure is caught                                              | <code>result["trained"] == 0;</code><br><code>result["failed"] == 3 (+1 more)</code>                                                                    | Pass   | 0.214s |
| ETA computed from elapsed/completed when avg_per_gauge is 0.            | <code>data["eta.seconds"] is not None;</code><br><code>data["eta.seconds"] &gt;= 40</code>                                                              | Pass   | 0.000s |
| ETA is None when no gauges completed yet and no historical avg.         | <code>data["eta.seconds"] is None</code>                                                                                                                | Pass   | 0.000s |
| POST when every gauge has a .joblib returns 'complete'.                 | <code>data["status"] == "complete";</code><br><code>data["message"] == "All gauges</code><br><code>already trained"</code>                              | Pass   | 0.007s |

| Test Description                                                               | Acceptance Criteria                                                                                   | Result | Time   |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|--------|
| POST while batch is running returns current progress without starting new job. | <code>data["status"] == "running";</code><br><code>data["completed"] == 2 (+1 more)</code>            | Pass   | 0.000s |
| Complete status includes results list.                                         | <code>data["status"] == "complete";</code><br><code>len(data["results"]) == 2</code>                  | Pass   | 0.000s |
| POST returns running or complete.                                              | <code>data["status"] in ("running", "complete", "error")</code>                                       | Pass   | 0.002s |
| Status endpoint returns expected fields when batch is running.                 | <code>data["status"] == "running";</code><br><code>data["total"] == 10 (+4 more)</code>               | Pass   | 0.000s |
| Status endpoint returns idle when no batch is running.                         | <code>data["status"] == "idle"</code>                                                                 | Pass   | 0.001s |
| Pdf not generated returns 404                                                  | <code>r.status_code == 404;</code> "not yet generated" in <code>r.get_json()["message"]</code>        | Pass   | 0.000s |
| Save failure returns 500                                                       | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>                  | Pass   | 0.002s |
| Line 74: POST add when save fails.                                             | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>                  | Pass   | 0.002s |
| Line 112: DELETE when save fails.                                              | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>                  | Pass   | 0.002s |
| Line 100: PUT update when save fails.                                          | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>                  | Pass   | 0.002s |
| P&L bar should display net notional (signed sum).                              | <code>'netNotional' in js;</code><br><code>'fmtGBP(netNotional)' in js</code>                         | Pass   | 0.000s |
| Payer (Pay) should show negative notional.                                     | <code>'is_payer' in js; 'netNotional' in js</code>                                                    | Pass   | 0.000s |
| PRS-*.json glob must match PRS-{8hex} filenames (the production format).       | <code>fnmatch.fnmatch(name, 'PRS-*.json');</code><br><code>fnmatch.fnmatch(name, 'PRS-*.json')</code> | Pass   | 0.000s |
| When PRS trade files are present, GET /trading/blotter returns them.           | <code>len(found) == 1</code>                                                                          | Pass   | 0.001s |
| When QuantLib IS available, book must export its public API.                   | <code>isinstance(THAMES_CENTRAL_AREAS, list); len(THAMES_CENTRAL_AREAS) &gt; 0</code>                 | Pass   | 0.000s |
| port.src.book itself can be imported without QuantLib because                  | <code>hasattr(mod, "THAMES_CENTRAL_AREAS")</code>                                                     | Pass   | 0.002s |
| Thames Central book should cover exactly 10 gauge areas.                       | <code>len(THAMES_CENTRAL_AREAS) == 10</code>                                                          | Pass   | 0.000s |
| models.hazard.pricing has 'import QuantLib as ql' at module level.             | —                                                                                                     | Pass   | 0.001s |
| Family members shown                                                           | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Has borrower header                                                            | <code>any("Borrower" in t for t in texts)</code>                                                      | Pass   | 0.000s |

| Test Description                                                   | Acceptance Criteria                                                                                      | Result | Time   |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------|--------|
| Has table                                                          | <code>any(isinstance(e, Table) for e in result)</code>                                                   | Pass   | 0.000s |
| Income verification shown                                          | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Marital status shown                                               | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Nationality shown                                                  | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| No borrower details returns early                                  | <code>any("No borrower" in t for t in texts)</code>                                                      | Pass   | 0.000s |
| No mortgage returns message                                        | <code>any("No mortgage" in t for t in texts)</code>                                                      | Pass   | 0.000s |
| Residency status shown                                             | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Secondary income shown                                             | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| With mortgage returns list                                         | <code>isinstance(result, list) and len(result) &gt; 0</code>                                             | Pass   | 0.003s |
| All none returns empty dicts                                       | <code>result["mortgage_lookup"] == {};</code><br><code>result["gauge_flood_info"] == {}</code> (+1 more) | Pass   | 0.000s |
| All none returns four keys                                         | <code>set(result.keys()) == {"mortgage_lookup", "g..."</code>                                            | Pass   | 0.000s |
| With data populates lookups                                        | <code>"P1" in result["mortgage_lookup"]</code>                                                           | Pass   | 0.000s |
| Audit trail date range                                             | <code>len(story) &gt;= 6</code>                                                                          | Pass   | 0.000s |
| Audit trail empty                                                  | <code>len(story) &gt;= 2</code>                                                                          | Pass   | 0.000s |
| Audit trail no timestamps                                          | <code>len(story) &gt;= 4</code>                                                                          | Pass   | 0.000s |
| Audit trail renders                                                | <code>len(story) &gt; 0</code>                                                                           | Pass   | 0.000s |
| Bcbs239 all compliant                                              | <code>len(story) &gt;= 4</code>                                                                          | Pass   | 0.000s |
| Bcbs239 at risk                                                    | <code>len(story) &gt;= 5</code>                                                                          | Pass   | 0.000s |
| Bcbs239 no principles                                              | <code>len(story) &gt;= 2</code>                                                                          | Pass   | 0.000s |
| Bcbs239 renders                                                    | <code>len(story) &gt; 0</code>                                                                           | Pass   | 0.000s |
| No trades returns message                                          | <code>any("No open PRS trades" in t for t in texts)</code>                                               | Pass   | 0.000s |
| Receiver trade direction                                           | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Trigger capitalized                                                | <code>any(isinstance(e, Table) for e in result)</code>                                                   | Pass   | 0.000s |
| With trades returns table                                          | <code>len(tables) &gt;= 1</code>                                                                         | Pass   | 0.000s |
| With trades shows count                                            | <code>any("Total trades: 2" in t for t in texts)</code>                                                  | Pass   | 0.000s |
| Line 186: when CDM validation returns errors, a warning is logged. | <code>record is not None; "PhysicalSwap" in record</code> (+1 more)                                      | Pass   | 0.000s |
| Diagonal is one                                                    | —                                                                                                        | Pass   | 0.000s |
| Off diagonal between zero and one                                  | <code>np.all(off_diag &gt;= 0) and np.all(off_diag &lt;= 1)</code>                                       | Pass   | 0.000s |
| Shape is n by n                                                    | <code>mat.shape == (n, n)</code>                                                                         | Pass   | 0.000s |

| Test Description                                                          | Acceptance Criteria                                                                           | Result | Time   |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| Symmetric                                                                 | —                                                                                             | Pass   | 0.000s |
| Diagonal is one                                                           | —                                                                                             | Pass   | 0.000s |
| Higher base correlation increases off diagonal                            | <code>np.all(off_high &gt; off_low)</code>                                                    | Pass   | 0.000s |
| Larger correlation_distance → slower exponential decay → higher off-diag. | <code>np.all(off_far &gt;= off_near)</code>                                                   | Pass   | 0.000s |
| Off diagonal positive                                                     | <code>np.all(off &gt; 0)</code>                                                               | Pass   | 0.000s |
| Returns square matrix                                                     | <code>mat.shape == (3, 3)</code>                                                              | Pass   | 0.000s |
| Symmetric                                                                 | —                                                                                             | Pass   | 0.000s |
| Cover basic                                                               | <code>len(story) &gt; 0</code>                                                                | Pass   | 0.001s |
| Cover no bcbs                                                             | <code>len(story) &gt;= 3</code>                                                               | Pass   | 0.000s |
| Cover no tests                                                            | <code>len(story) &gt;= 3</code>                                                               | Pass   | 0.000s |
| Cover with bcbs                                                           | <code>'BCBS' in combined or len(story) &gt;= 5</code>                                         | Pass   | 0.001s |
| Cover with tests                                                          | <code>len(story) &gt;= 5</code>                                                               | Pass   | 0.000s |
| Documents empty                                                           | <code>len(story) &gt;= 2</code>                                                               | Pass   | 0.000s |
| Documents file list                                                       | <code>len(story) &gt;= 4</code>                                                               | Pass   | 0.000s |
| Documents renders                                                         | <code>len(story) &gt; 0</code>                                                                | Pass   | 0.000s |
| Exec summary all pass                                                     | <code>len(story) &gt; 0</code>                                                                | Pass   | 0.000s |
| Exec summary failures                                                     | <code>len(story) &gt; 0</code>                                                                | Pass   | 0.000s |
| Exec summary no coverage                                                  | <code>len(story) &gt; 0</code>                                                                | Pass   | 0.000s |
| Exec summary no tests                                                     | <code>len(story) &gt; 0</code>                                                                | Pass   | 0.001s |
| Exec summary renders                                                      | <code>len(story) &gt; 0</code>                                                                | Pass   | 0.000s |
| When flooded, arrival_time and peak_time are set.                         | <code>result["arrival_time_hrs"] is not None; result["peak_time_hrs"] is not None</code>      | Pass   | 0.001s |
| High WSE above property elevation -i flooded=True.                        | <code>result["flooded"] is True; result["flood_depth_m"] &gt; 0</code>                        | Pass   | 0.001s |
| Gaugets with empty readings -i no readings key (not flooded).             | <code>"readings" not in result</code>                                                         | Pass   | 0.001s |
| Low WSE -i flooded=False.                                                 | <code>result["flooded"] is False</code>                                                       | Pass   | 0.001s |
| _build_flood_event returns dict with required keys.                       | <code>key in result; "readings" in result</code>                                              | Pass   | 0.001s |
| Distance=0 -i travel_time=0 path.                                         | <code>result["travel_time_hrs"] == 0.0</code>                                                 | Pass   | 0.001s |
| Both none returns empty                                                   | <code>build_gauge_flood_info(None, None) == {}</code>                                         | Pass   | 0.000s |
| Gauge data supplements missing fields                                     | <code>"GAUGE-002" in result; result["GAUGE-002"]["gauge_name"] == "My Gauge" (+2 more)</code> | Pass   | 0.000s |
| Gauge without id skipped                                                  | <code>result == {}</code>                                                                     | Pass   | 0.000s |



| Test Description                                                  | Acceptance Criteria                                                                             | Result | Time   |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|--------|
| Hazard curves only                                                | "GAUGE-001" in result;<br>result["GAUGE-001"]["gauge_name"]<br>== "Thames at Chelsea" (+2 more) | Pass   | 0.000s |
| Hazard data takes precedence                                      | result["GAUGE-003"]["gauge_name"] ==<br>"From HC"                                               | Pass   | 0.000s |
| Line 247: FileNotFoundError when<br>gauge.json missing.           | —                                                                                               | Pass   | 0.002s |
| Line 235: FileNotFoundError when<br>storm_sequences.json missing. | —                                                                                               | Pass   | 0.000s |
| Verbose=False should skip log mes-<br>sages.                      | result['summary']['num_gauges'] == 1;<br>len(info_msgs) == 0                                    | Pass   | 0.021s |
| Lines 242, 254, 273, 280-281, 288:<br>verbose logging.            | result['summary']['num_gauges'] == 1;<br>result['summary']['num_storms'] == 20<br>(+3 more)     | Pass   | 0.023s |
| Inventory empty                                                   | len(story) >= 2                                                                                 | Pass   | 0.000s |
| Inventory no risk ratings                                         | len(story) >= 3                                                                                 | Pass   | 0.000s |
| Inventory rag colours                                             | len(story) >= 3                                                                                 | Pass   | 0.000s |
| Inventory renders                                                 | len(story) > 0                                                                                  | Pass   | 0.000s |
| Inventory risk ratings                                            | len(story) >= 4                                                                                 | Pass   | 0.000s |
| Mrc completed actions excluded                                    | len(story) >= 3                                                                                 | Pass   | 0.000s |
| Mrc no decisions                                                  | len(story) >= 3                                                                                 | Pass   | 0.000s |
| Mrc no meetings                                                   | len(story) >= 2                                                                                 | Pass   | 0.000s |
| Mrc open actions                                                  | len(story) >= 4                                                                                 | Pass   | 0.000s |
| Mrc renders                                                       | len(story) > 0                                                                                  | Pass   | 0.000s |
| Mrc with decisions                                                | len(story) >= 5                                                                                 | Pass   | 0.000s |
| No market state returns message                                   | any("No market hazard term<br>structure" in t for t in texts)                                   | Pass   | 0.000s |
| Some tenor slots missing → show<br>'N/A'.                         | any(isinstance(e, Table) for e in<br>result)                                                    | Pass   | 0.000s |
| With adjustments returns extra ta-<br>ble                         | len(tables) >= 2                                                                                | Pass   | 0.000s |
| With hts returns table                                            | len(tables) >= 1                                                                                | Pass   | 0.000s |
| All sections                                                      | len(content) > 1000                                                                             | Pass   | 0.003s |
| Buffer not empty                                                  | len(content) > 100                                                                              | Pass   | 0.003s |
| Meeting with actions                                              | isinstance(result, io.BytesIO)                                                                  | Pass   | 0.001s |
| Meeting with agenda                                               | isinstance(result, io.BytesIO)                                                                  | Pass   | 0.001s |
| Meeting with decisions                                            | isinstance(result, io.BytesIO)                                                                  | Pass   | 0.001s |
| Meeting with minutes                                              | isinstance(result, io.BytesIO)                                                                  | Pass   | 0.001s |
| Meeting with no date/time/loca-<br>tion/chair still works.        | isinstance(result, io.BytesIO)                                                                  | Pass   | 0.001s |
| Meeting with participants                                         | isinstance(result, io.BytesIO)                                                                  | Pass   | 0.003s |

| Test Description                                           | Acceptance Criteria                                                                                          | Result | Time   |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------|--------|
| Minimal meeting                                            | <code>isinstance(result, io.BytesIO)</code>                                                                  | Pass   | 0.001s |
| Returns bytesio                                            | <code>isinstance(result, io.BytesIO)</code>                                                                  | Pass   | 0.003s |
| Empty dict returns empty                                   | <code>build_mortgage_lookup({}) == {}</code>                                                                 | Pass   | 0.000s |
| Loan from loan details fallback                            | <code>result["PROP-002"]["OriginalLoan"] == 300.000</code>                                                   | Pass   | 0.000s |
| Mortgage without property id skipped                       | <code>build_mortgage_lookup(data) == {}</code>                                                               | Pass   | 0.000s |
| Multiple mortgages                                         | <code>set(result.keys()) == {"P1", "P2"}</code>                                                              | Pass   | 0.000s |
| No items key returns empty                                 | <code>build_mortgage_lookup({"other": []}) == {}</code>                                                      | Pass   | 0.000s |
| None returns empty                                         | <code>build_mortgage_lookup(None) == {}</code>                                                               | Pass   | 0.000s |
| Single mortgage                                            | <code>"PROP-001" in result;</code><br><code>result["PROP-001"]["mortgage_id"] == "MORT-001" (+3 more)</code> | Pass   | 0.000s |
| Basic risk info                                            | <code>"M1" in result["by_mortgage_id"]; "P1" in result["by_property_id"] (+3 more)</code>                    | Pass   | 0.000s |
| Missing mortgage or property id skipped                    | <code>result == {"by_mortgage_id": {}, "by_property_id": {}}</code>                                          | Pass   | 0.000s |
| No property flood info uses unknown                        | <code>result["by_mortgage_id"]["M2"]["flood_risk_level"] == "Un..."</code>                                   | Pass   | 0.000s |
| None mortgage returns empty structure                      | <code>result == {"by_mortgage_id": {}, "by_property_id": {}}</code>                                          | Pass   | 0.000s |
| Value at risk calculation                                  | <code>result["by_mortgage_id"]["M4"]["mortgage_value_at_risk"] == 0.000s</code><br>...                       | Pass   | 0.000s |
| 60 i= percentile i 80 -i 'Elevated'.                       | <code>any('Elevated' in t for t in texts)</code>                                                             | Pass   | 0.090s |
| FloodAlert and FloodWarning in metadata -i no crash.       | <code>len(elements) &gt; 0</code>                                                                            | Pass   | 0.095s |
| 80 i= percentile i 95 -i 'High'.                           | <code>any('High' in t for t in texts)</code>                                                                 | Pass   | 0.090s |
| percentile i 40 -i 'Low'.                                  | <code>any('Low' in t for t in texts)</code>                                                                  | Pass   | 0.189s |
| 40 i= percentile i 60 -i 'Normal'.                         | <code>any('Normal' in t for t in texts)</code>                                                               | Pass   | 0.089s |
| Single same_month_levels entry -i stdev guard -i no crash. | <code>len(elements) &gt; 0</code>                                                                            | Pass   | 0.085s |
| Stats table and distribution statistics header present.    | <code>any('Distribution Statistics' in t for t in texts)</code>                                              | Pass   | 0.089s |
| percentile i= 95 -i 'Very High'.                           | <code>any('Very High' in t for t in texts)</code>                                                            | Pass   | 0.090s |
| Returns position function                                  | <code>'getPositionStyles' in css;</code><br><code>'top-right' in css (+1 more)</code>                        | Pass   | 0.000s |
| All none returns empty                                     | <code>build_property_flood_info(None, None, None) == {}</code>                                               | Pass   | 0.000s |

| Test Description                                                  | Acceptance Criteria                                                                                 | Result | Time   |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| From property data when not in phc                                | "PROP-002" in result;<br>result["PROP-002"]["flood_zone"]<br>== "Zone 2" (+1 more)                  | Pass   | 0.000s |
| From property hazard data                                         | "PROP-001" in result;<br>result["PROP-001"]["annual_flood_prob"]<br>== pytest.approx(...) (+2 more) | Pass   | 0.000s |
| Phc takes precedence over property data                           | result["PROP-003"]["annual_flood_prob"]<br>== pytest.approx(...)                                    | Pass   | 0.000s |
| Raci escalation triggers dict                                     | len(story) >= 5                                                                                     | Pass   | 0.000s |
| Raci escalation triggers string                                   | len(story) >= 5                                                                                     | Pass   | 0.000s |
| Raci missing backup                                               | len(story) >= 3                                                                                     | Pass   | 0.000s |
| Raci no roles                                                     | len(story) >= 2                                                                                     | Pass   | 0.000s |
| Raci renders                                                      | len(story) > 0                                                                                      | Pass   | 0.000s |
| Raci with activities                                              | len(story) >= 5                                                                                     | Pass   | 0.000s |
| alert j= peak j warning -i 'Alert'.                               | len(elements) > 1                                                                                   | Pass   | 0.000s |
| ValueError/TypeError in date parsing (lines 179-180) -i continue. | len(elements) > 0                                                                                   | Pass   | 0.000s |
| Days within 3 days grouped; higher level updates peak.            | not any('No flood events' in t for t in texts)                                                      | Pass   | 0.000s |
| Consecutive days -i single event with range format.               | len(elements) > 1                                                                                   | Pass   | 0.000s |
| No days exceeding alert -i 'No flood events recorded'.            | any('No flood events' in t for t in texts)                                                          | Pass   | 0.000s |
| Peak i= severe.level -i 'Severe' row in table.                    | len(elements) > 1                                                                                   | Pass   | 0.000s |
| Single-day flood: date_str = start.strftime(...) only.            | len(elements) > 1                                                                                   | Pass   | 0.000s |
| Days i 3 days apart -i two separate flood events.                 | len(elements) > 1                                                                                   | Pass   | 0.000s |
| warning j= peak j severe -i 'Warning'.                            | len(elements) > 1                                                                                   | Pass   | 0.000s |
| Recommendations all clean                                         | len(story) >= 3                                                                                     | Pass   | 0.000s |
| Recommendations bcbs at risk                                      | len(story) >= 4                                                                                     | Pass   | 0.000s |
| Recommendations overdue remediation                               | len(story) >= 4                                                                                     | Pass   | 0.000s |
| Recommendations peer reviewer gap                                 | len(story) >= 4                                                                                     | Pass   | 0.000s |
| Recommendations raci backup gaps                                  | len(story) >= 4                                                                                     | Pass   | 0.000s |
| Recommendations renders                                           | len(story) > 0                                                                                      | Pass   | 0.000s |
| Recommendations failures                                          | len(story) >= 4                                                                                     | Pass   | 0.000s |
| Recommendations validation gap                                    | len(story) >= 4                                                                                     | Pass   | 0.000s |

| Test Description                                   | Acceptance Criteria                                                                           | Result | Time   |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| Remediation all completed                          | <code>len(story) &gt;= 4</code>                                                               | Pass   | 0.000s |
| Remediation empty                                  | <code>len(story) &gt;= 2</code>                                                               | Pass   | 0.000s |
| Remediation overdue highlighted                    | <code>len(story) &gt;= 4</code>                                                               | Pass   | 0.000s |
| Remediation priority sorting                       | <code>len(story) &gt;= 4</code>                                                               | Pass   | 0.000s |
| Remediation renders                                | <code>len(story) &gt; 0</code>                                                                | Pass   | 0.000s |
| Empty schema returns empty dict                    | <code>make_generator(tmp_path)..build_section({}, 0, {}) == {}</code>                         | Pass   | 0.000s |
| Nested dict without type recurses                  | <code>"SubSection" in result;</code><br><code>isinstance(result["SubSection"], dict)</code>   | Pass   | 0.000s |
| Non dict schema returns empty dict                 | <code>make_generator(tmp_path)..build_section("not a dict", 0, ...)</code>                    | Pass   | 0.000s |
| None schema returns empty dict                     | <code>make_generator(tmp_path)..build_section(None, 0, {}) == {}</code>                       | Pass   | 0.000s |
| None value from generate field value excluded      | <code>"AField" not in result</code>                                                           | Pass   | 0.000s |
| Reserved key description skipped                   | <code>"description" not in</code><br><code>make_generator(tmp_path)..build_sect...</code>     | Pass   | 0.000s |
| Reserved key options skipped                       | <code>"options" not in</code><br><code>make_generator(tmp_path)..build_section(...</code>     | Pass   | 0.000s |
| Reserved key type skipped                          | <code>"type" not in result</code>                                                             | Pass   | 0.000s |
| Reserved key values skipped                        | <code>"values" not in</code><br><code>make_generator(tmp_path)..build_section({...</code>     | Pass   | 0.000s |
| Empty dataframe returns none                       | <code>build_spatial_index(df) is None</code>                                                  | Pass   | 0.000s |
| None gauge data returns none                       | <code>build_spatial_index(None) is None</code>                                                | Pass   | 0.000s |
| With gauges returns kdtree                         | <code>isinstance(result, cKDTree)</code>                                                      | Pass   | 0.000s |
| Lines 43-45: coordinates are converted to radians. | <code>tree.data[0][1] ==</code><br><code>pytest.approx(np.pi, rel=1e-6)</code>                | Pass   | 0.000s |
| Single gauge should still produce a valid KDTree.  | <code>isinstance(tree, cKDTree); tree.n == 1</code>                                           | Pass   | 0.000s |
| Lines 42-46: non-empty DataFrame builds a cKDTree. | <code>isinstance(tree, cKDTree); tree.n == 3</code>                                           | Pass   | 0.000s |
| Counts default to zero                             | <code>s["alert_sequences"] == 0;</code><br><code>s["warning_sequences"] == 0</code> (+1 more) | Pass   | 0.000s |
| Num gauges                                         | <code>s["num_gauges"] == 4</code>                                                             | Pass   | 0.000s |
| Num sequences                                      | <code>s["num_sequences"] == 7</code>                                                          | Pass   | 0.001s |
| Returns required keys                              | <code>key in s</code>                                                                         | Pass   | 0.000s |
| Contains all types                                 | <code>t.key in parsed;</code> 'icon' in<br><code>parsed[t.key]</code> (+2 more)               | Pass   | 0.000s |
| Info template values                               | <code>parsed['info']['color'] == '#2196F3'</code>                                             | Pass   | 0.000s |

| Test Description                                                                            | Acceptance Criteria                                                                                                 | Result | Time   |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------|--------|
| Returns valid json                                                                          | <code>isinstance(parsed, dict)</code>                                                                               | Pass   | 0.000s |
| Testing no junit                                                                            | <code>len(story) &gt;= 3</code>                                                                                     | Pass   | 0.000s |
| Testing no sensitivity generators                                                           | <code>len(story) &gt;= 4</code>                                                                                     | Pass   | 0.000s |
| Testing per model coverage                                                                  | <code>len(story) &gt;= 6</code>                                                                                     | Pass   | 0.000s |
| Testing renders                                                                             | <code>len(story) &gt; 0</code>                                                                                      | Pass   | 0.000s |
| Testing with coverage                                                                       | <code>len(story) &gt;= 5</code>                                                                                     | Pass   | 0.000s |
| Testing with sensitivity generators                                                         | <code>len(story) &gt;= 6</code>                                                                                     | Pass   | 0.000s |
| Validation all reviewers assigned                                                           | <code>len(story) &gt;= 3</code>                                                                                     | Pass   | 0.000s |
| Validation high assumptions                                                                 | <code>len(story) &gt; 3</code>                                                                                      | Pass   | 0.000s |
| Validation no assumptions                                                                   | <code>len(story) &gt;= 3</code>                                                                                     | Pass   | 0.000s |
| Validation no questions                                                                     | <code>len(story) &gt;= 1</code>                                                                                     | Pass   | 0.000s |
| Validation peer reviewer gaps                                                               | <code>len(story) &gt;= 4</code>                                                                                     | Pass   | 0.000s |
| Validation renders                                                                          | <code>len(story) &gt; 0</code>                                                                                      | Pass   | 0.000s |
| Line 88: exceedance probs between 0 and 0.01 are capped to 0.01.                            | <code>pt.annual.exceedance_prob &gt;= 0.01</code> or<br><code>pt.annual.exceedance...</code>                        | Pass   | 0.022s |
| Line 61: <code>log()</code> does nothing when <code>verbose=False</code> .                  | <code>"should not appear" not in caplog.text</code>                                                                 | Pass   | 0.000s |
| Line 61: <code>log()</code> calls <code>logger.info</code> when <code>verbose=True</code> . | <code>"test message" in caplog.text</code>                                                                          | Pass   | 0.000s |
| Line 152: progress logged every 10 gauges.                                                  | <code>len(cores) == 11;</code> "10/11" in<br><code>caplog.text</code> or "Processed" in<br><code>caplog.text</code> | Pass   | 0.189s |
| All days before start                                                                       | <code>d &lt; today</code>                                                                                           | Pass   | 0.000s |
| Chronological order                                                                         | <code>days[i] &gt; days[i - 1]</code>                                                                               | Pass   | 0.000s |
| Ends before start                                                                           | <code>d &lt; start</code>                                                                                           | Pass   | 0.000s |
| Large count                                                                                 | <code>len(days) == 252; d.weekday() &lt; 5</code>                                                                   | Pass   | 0.000s |
| Last element should be the business day immediately before start.                           | <code>last &lt; start; last.weekday() &lt; 5</code>                                                                 | Pass   | 0.000s |
| No weekends                                                                                 | <code>d.weekday() &lt; 5</code>                                                                                     | Pass   | 0.000s |
| Returns ascending order                                                                     | <code>result == sorted(result)</code>                                                                               | Pass   | 0.000s |
| Returns business days only                                                                  | <code>d.weekday() &lt; 5</code>                                                                                     | Pass   | 0.000s |
| Returns correct count                                                                       | <code>len(days) == 63</code>                                                                                        | Pass   | 0.000s |
| Returns correct count                                                                       | <code>len(result) == 5</code>                                                                                       | Pass   | 0.000s |
| Single day                                                                                  | <code>len(days) == 1; days[0].weekday() &lt; 5</code>                                                               | Pass   | 0.000s |
| Single day                                                                                  | <code>len(result) == 1; result[0].weekday() &lt; 5</code>                                                           | Pass   | 0.000s |
| Skips weekends                                                                              | <code>wd not in returned_set</code>                                                                                 | Pass   | 0.000s |
| The day before Monday is Sunday — should skip back to Friday.                               | <code>days[0].weekday() == 4</code>                                                                                 | Pass   | 0.000s |
| 5 days back from Wednesday should contain no Saturday/Sunday.                               | <code>len(days) == 5; d.weekday() &lt; 5</code>                                                                     | Pass   | 0.000s |

| Test Description                                                         | Acceptance Criteria                                                                       | Result | Time   |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| Zero days                                                                | <code>result == []</code>                                                                 | Pass   | 0.000s |
| Test CORS headers on GET request.                                        | <code>response.status_code == 200</code>                                                  | Pass   | 0.000s |
| Test OPTIONS requests return 200.                                        | <code>response.status_code == 200</code>                                                  | Pass   | 0.001s |
| Larger area gives larger energy                                          | <code>sum(large) &gt; sum(small)</code>                                                   | Pass   | 0.000s |
| Returns positive value                                                   | <code>result &gt; 0; result &gt; 0 (+2 more)</code>                                       | Pass   | 0.000s |
| Uses default area when missing                                           | <code>result &gt; 0; result &gt; 0</code>                                                 | Pass   | 0.000s |
| Is float                                                                 | <code>isinstance(result, float);</code><br><code>isinstance(result, float)</code>         | Pass   | 0.000s |
| Larger area tends to give larger emissions                               | <code>sum(large) &gt; sum(small)</code>                                                   | Pass   | 0.000s |
| Returns positive value                                                   | <code>result &gt; 0; result &gt; 0 (+2 more)</code>                                       | Pass   | 0.000s |
| Uses default area when missing                                           | <code>result &gt; 0; result &gt; 0</code>                                                 | Pass   | 0.000s |
| Correct calculation                                                      | <code>abs(result - 2430.0) &lt; 0.01</code>                                               | Pass   | 0.000s |
| Is float                                                                 | <code>isinstance(result, float);</code><br><code>isinstance(result, float)</code>         | Pass   | 0.000s |
| Returns positive value                                                   | <code>result &gt; 0; result &gt; 0 (+2 more)</code>                                       | Pass   | 0.000s |
| Uses defaults when missing                                               | <code>result &gt;= 0; abs(result - 2430.0) &lt; 0.01</code>                               | Pass   | 0.000s |
| Zero usage returns zero                                                  | <code>result == 0.0</code>                                                                | Pass   | 0.000s |
| Above severe level some positive                                         | <code>np.any(depths &gt; 0)</code>                                                        | Pass   | 0.000s |
| Below severe level returns zeros                                         | —                                                                                         | Pass   | 0.000s |
| Depths capped at 5m                                                      | <code>np.all(depths &lt;= 5.0)</code>                                                     | Pass   | 0.000s |
| Empty gauge data returns zeros                                           | —                                                                                         | Pass   | 0.000s |
| No gauge data returns zeros                                              | —                                                                                         | Pass   | 0.000s |
| Returns correct length                                                   | <code>len(depths) == len(properties)</code>                                               | Pass   | 0.000s |
| Lines 50-57: non-None gauge_data calls calculate_synthetic_flood_depths. | <code>len(depths) == 3; isinstance(depths, np.ndarray)</code>                             | Pass   | 0.000s |
| Key remapping: 'flow_cumecs' → 'flow'.                                   | <code>"mean_flow" in result;</code><br><code>"mean_flow_cumecs" not in result</code>      | Pass   | 0.000s |
| Flow stats should include q5_flow and q95_flow.                          | <code>"q5_flow" in result; "q95_flow" in result</code>                                    | Pass   | 0.000s |
| Returns dict                                                             | <code>isinstance(result, dict);</code><br><code>isinstance(result, dict) (+1 more)</code> | Pass   | 0.000s |
| Less than remaining energy                                               | <code>result &lt;= 9001</code>                                                            | Pass   | 0.000s |
| No negative when electricity exceeds total                               | <code>result == 0</code>                                                                  | Pass   | 0.000s |
| Returns non negative                                                     | <code>result &gt;= 0</code>                                                               | Pass   | 0.000s |
| Uses defaults when missing                                               | <code>result &gt;= 0; abs(result - 2430.0) &lt; 0.01</code>                               | Pass   | 0.000s |



| Test Description                         | Acceptance Criteria                                                                                | Result | Time   |
|------------------------------------------|----------------------------------------------------------------------------------------------------|--------|--------|
| Fraction of total                        | <code>all(4500 &lt;= r &lt;= 12001 for r in results)</code>                                        | Pass   | 0.000s |
| Returns positive value                   | <code>result &gt; 0; result &gt; 0 (+2 more)</code>                                                | Pass   | 0.000s |
| Uses default energy when missing         | <code>result &gt; 0</code>                                                                         | Pass   | 0.000s |
| Key remapping: 'level_meters' → 'level'. | <code>"mean_level" in result;</code><br><code>"mean_level_meters" not in result</code>             | Pass   | 0.001s |
| Has min max                              | <code>"min_level" in result; "max_level" in result</code>                                          | Pass   | 0.001s |
| Returns dict                             | <code>isinstance(result, dict);</code><br><code>isinstance(result, dict) (+1 more)</code>          | Pass   | 0.001s |
| Buy to let type                          | <code>result["mortgage_type"] == "Buy-to-Let"</code>                                               | Pass   | 0.000s |
| High flood risk reduces ltv              | <code>high_risk["ltv_ratio"] &lt;= normal["ltv_ratio"] + 0.01</code>                               | Pass   | 0.000s |
| Loan amount positive                     | <code>result["loan_amount"] &gt; 0</code>                                                          | Pass   | 0.000s |
| Monthly payment positive                 | <code>result["monthly_payment"] &gt; 0</code>                                                      | Pass   | 0.000s |
| Required keys present                    | <code>key in result</code>                                                                         | Pass   | 0.000s |
| Returns dict                             | <code>isinstance(result, dict)</code>                                                              | Pass   | 0.000s |
| Second home type                         | <code>result["mortgage_type"] == "Second Home"</code>                                              | Pass   | 0.000s |
| Conditional rating middle range          | <code>result["calculated_rating"] in ("Conditional", "Unaccepta...</code>                          | Pass   | 0.000s |
| Due soon gives half score                | <code>result["component_scores"]["review_currency"] == 0.5</code>                                  | Pass   | 0.000s |
| Far review gives full score              | <code>result["component_scores"]["review_currency"] == 1.0</code>                                  | Pass   | 0.000s |
| High assumptions penalised               | <code>result["component_scores"]["assumption_risk"] == 0.0</code>                                  | Pass   | 0.000s |
| High coverage acceptable                 | <code>result["calculated_rating"] == "Acceptable"</code>                                           | Pass   | 0.000s |
| High limitations penalised               | <code>result["component_scores"]["limitation_risk"] == 0.0</code>                                  | Pass   | 0.000s |
| No coverage unacceptable                 | <code>result["calculated_rating"] == "Unacceptable"</code>                                         | Pass   | 0.000s |
| No review date gives zero                | <code>result["component_scores"]["review_currency"] == 0.0</code>                                  | Pass   | 0.000s |
| Overdue review gives zero score          | <code>result["component_scores"]["review_currency"] == 0.0</code>                                  | Pass   | 0.000s |
| Returns dict with required keys          | <code>"calculated_rating" in result;</code><br><code>"calculated_score" in result (+1 more)</code> | Pass   | 0.000s |

| Test Description                     | Acceptance Criteria                                                                                                             | Result | Time   |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Score is float                       | <code>isinstance(result["calculated_score"], float)</code>                                                                      | Pass   | 0.000s |
| Heat pump returns zero               | <code>result == 0</code>                                                                                                        | Pass   | 0.000s |
| Multiple returns positive            | <code>result &gt; 0</code>                                                                                                      | Pass   | 0.000s |
| No solar returns zero                | <code>result == 0</code>                                                                                                        | Pass   | 0.000s |
| No system returns zero               | <code>result == 0</code>                                                                                                        | Pass   | 0.000s |
| Solar pv returns positive            | <code>result &gt; 0</code>                                                                                                      | Pass   | 0.000s |
| Solar uses default area when missing | <code>result &gt; 0</code>                                                                                                      | Pass   | 0.000s |
| Annual maxima present                | <code>"annual_maxima" in stats;</code><br><code>stats["annual_maxima"]["years_count"]</code><br><code>== 2</code>               | Pass   | 0.000s |
| Basic stats keys                     | <code>key in stats</code>                                                                                                       | Pass   | 0.000s |
| Empty returns empty dict             | <code>GaugeHistoricalDaily().calculate_statistics()</code><br><code>== {}</code>                                                | Pass   | 0.000s |
| Extreme events above p99             | <code>"extreme_events" in stats;</code><br><code>stats["extreme_events"]["count"]</code><br><code>&gt;= 1</code>                | Pass   | 0.000s |
| Mean correct                         | <code>gen.calculate_statistics(flows)["mean_flow"]</code><br><code>== pytest.approx(5.0)</code>                                 | Pass   | 0.000s |
| Min max correct                      | <code>stats["min_flow"] ==</code><br><code>pytest.approx(5.0); stats["max_flow"]</code><br><code>== pytest.approx(100.0)</code> | Pass   | 0.000s |
| Monthly means present                | <code>"01" in stats["monthly_means"]; "02"</code><br><code>in stats["monthly_means"]</code>                                     | Pass   | 0.000s |
| Percentiles present                  | <code>"percentiles" in stats; p in</code><br><code>stats["percentiles"]</code>                                                  | Pass   | 0.000s |
| Single flow no std dev error         | <code>gen.calculate_statistics(flows)["std.dev"]</code><br><code>== 0</code>                                                    | Pass   | 0.000s |
| All percentile keys present          | <code>p in stats["percentiles"]</code>                                                                                          | Pass   | 0.000s |
| Annual maxima events capped at ten   | <code>len(stats["annual_maxima"]["events"])</code><br><code>&lt;= 10</code>                                                     | Pass   | 0.000s |
| Extreme events top 10 capped         | <code>len(stats["extreme_events"]["top_10"])</code><br><code>&lt;= 10</code>                                                    | Pass   | 0.000s |
| Max flow date correctly identified   | <code>stats["max_flow_date"] ==</code><br><code>"2020-01-02"; stats["max_flow"]</code><br><code>== pytest.approx(200.0)</code>  | Pass   | 0.000s |
| Q5 flow equals p95                   | <code>stats["q5_flow"] ==</code><br><code>stats["percentiles"]["p95"]</code>                                                    | Pass   | 0.000s |
| Q95 flow equals p5                   | <code>stats["q95_flow"] ==</code><br><code>stats["percentiles"]["p5"]</code>                                                    | Pass   | 0.000s |

| Test Description                       | Acceptance Criteria                                                                              | Result | Time   |
|----------------------------------------|--------------------------------------------------------------------------------------------------|--------|--------|
| Record start and end match data        | stats["record_start"] ==<br>"2018-05-01"; stats["record_end"]<br>== "2021-08-15"                 | Pass   | 0.000s |
| Single year annual maxima std dev zero | stats["annual_maxima"]["years_count"]<br>== 1;<br>stats["annual_maxima"]["std_dev"]<br>== 0      | Pass   | 0.000s |
| Total years approximate                | stats["total_days"] == 20;<br>stats["total_years"] == round(20<br>/ 365.25, 1)                   | Pass   | 0.000s |
| Two flow std dev non zero              | gen.calculate_statistics(flows)["std_dev"]<br>> 0                                                | Pass   | 0.000s |
| All same values                        | result["mean_level_meters"] == 3.0;<br>result["std_dev"] == 0.0                                  | Pass   | 0.000s |
| Empty returns empty dict               | calculate_timeseries_statistics([],<br>"level_meters") == {}                                     | Pass   | 0.000s |
| Flood exceedances zero when all below  | result["flood_exceedances"]["flood_alert"]<br>== 0                                               | Pass   | 0.000s |
| Flood stages adds exceedances          | "flood_exceedances" in result;<br>result["flood_exceedances"]["severe_warning"]["count"]<br>>= 1 | Pass   | 0.001s |
| Flow cumecs adds q flows               | "q95_flow" in result; "q5_flow" in<br>result                                                     | Pass   | 0.000s |
| Has annual maxima                      | "annual_maxima" in<br>result; "years_count" in<br>result["annual_maxima"]                        | Pass   | 0.001s |
| Has basic keys                         | "total_days" in result;<br>"mean_level_meters" in result (+2<br>more)                            | Pass   | 0.001s |
| Has extreme events                     | "extreme_events" in result                                                                       | Pass   | 0.001s |
| Has monthly means                      | "monthly_means" in result                                                                        | Pass   | 0.001s |
| Has percentiles                        | "percentiles" in result; "p5" in<br>result["percentiles"] (+1 more)                              | Pass   | 0.001s |
| Max date correct                       | result["max_level_meters_date"] ==<br>"2026-01-02"                                               | Pass   | 0.000s |
| No flood stages no exceedances key     | "flood_exceedances" not in result                                                                | Pass   | 0.001s |
| Record start end                       | result["record_start"] ==<br>"2020-01-01"; result["record_end"] ==<br>"2025-12-31"               | Pass   | 0.000s |
| Returns dict                           | isinstance(result, dict);<br>isinstance(result, dict) (+1 more)                                  | Pass   | 0.001s |

| Test Description                                                      | Acceptance Criteria                                                                                  | Result | Time   |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------|--------|
| Single observation no stdev error                                     | <code>result["std.dev"] == 0</code>                                                                  | Pass   | 0.000s |
| Total days correct                                                    | <code>result["total.days"] == 100</code>                                                             | Pass   | 0.000s |
| Explicit none params                                                  | <code>isinstance(level, float)</code>                                                                | Pass   | 0.000s |
| None params uses defaults                                             | <code>isinstance(level, float); level &gt; 0</code>                                                  | Pass   | 0.000s |
| Lines 54-56: internal error returns 500 JSON error.                   | <code>r.status_code in (404, 500);</code><br><code>data["status"] == "error"</code>                  | Pass   | 0.001s |
| Lines 108-113: exception in <code>set_catchment</code> returns 500.   | <code>r.status_code in (200, 500)</code>                                                             | Pass   | 0.001s |
| Uses <code>addEventListener</code> , not inline onclick for sub-tabs. | <code>'addEventListener' in stress_js;</code><br><code>'stress-ctab-0' in stress_js</code> (+2 more) | Pass   | 0.000s |
| Flood Probability sub-tab present.                                    | <code>'Flood Probability' in stress_js</code>                                                        | Pass   | 0.000s |
| Stress P&L sub-tab present.                                           | <code>'Stress P' in stress_js</code>                                                                 | Pass   | 0.000s |
| Surface sub-tab present.                                              | <code>'Surface' in stress_js</code>                                                                  | Pass   | 0.000s |
| Consumer has no hash for input                                        | not ok; <code>"no recorded hash" in issues[0].lower()</code>                                         | Pass   | 0.000s |
| Consumer never run                                                    | not ok; <code>"never run" in issues[0].lower()</code>                                                | Pass   | 0.000s |
| Fresh when hashes match                                               | ok; <code>issues == []</code>                                                                        | Pass   | 0.000s |
| No hash recorded for producer output                                  | not ok; <code>"No hash recorded" in issues[0]</code>                                                 | Pass   | 0.000s |
| Producer never run                                                    | not ok; <code>"never run" in issues[0].lower()</code>                                                | Pass   | 0.000s |
| Root steps (gauges, counterparties) have no inputs → always fresh.    | ok; <code>issues == []</code>                                                                        | Pass   | 0.000s |
| Stale when hashes differ                                              | not ok; <code>"stale" in issues[0].lower()</code>                                                    | Pass   | 0.000s |
| Unknown step                                                          | not ok; <code>"Unknown step" in issues[0]</code>                                                     | Pass   | 0.000s |
| All outputs present → complete.                                       | <code>result["complete"]; result["missing"] == []</code> (+1 more)                                   | Pass   | 0.003s |
| Empty directory is as broken as missing.                              | not <code>result["complete"]</code> ; <code>len(empty) == 1</code> (+2 more)                         | Pass   | 0.003s |
| Missing gauge.json → incomplete.                                      | not <code>result["complete"]</code> ; <code>"gauge.json" in missing_outputs</code> (+2 more)         | Pass   | 0.000s |
| Only root steps present → reports downstream as missing.              | not <code>result["complete"]</code> ;<br><code>result["present"] == 2</code> (+3 more)               | Pass   | 0.000s |
| All pipeline steps missing produces 'missing' statuses.               | <code>len(results) == 10</code> ; <code>r["status"] == "missing"</code> (+2 more)                    | Pass   | 0.000s |
| A directory step checks most recent file mtime.                       | <code>ghd_step["status"] == "fresh"</code>                                                           | Pass   | 0.001s |
| A directory with only old files is stale.                             | <code>ghd_step["status"] == "stale"</code>                                                           | Pass   | 0.001s |

| Test Description                                                                    | Acceptance Criteria                                                                                           | Result | Time   |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------|--------|
| An empty directory counts as missing.                                               | <code>ghd_step["status"] == "missing"</code>                                                                  | Pass   | 0.000s |
| A recently modified file is reported as 'fresh'.                                    | <code>gauge_step["status"] == "fresh";</code><br><code>gauge_step["last_run"] is not None</code><br>(+1 more) | Pass   | 0.000s |
| Dotfiles in directories are ignored.                                                | <code>ghd_step["status"] == "missing"</code>                                                                  | Pass   | 0.001s |
| If manifest has last_run AND file exists, manifest last_run is kept.                | <code>gauge_step["last_run"] ==</code><br><code>"2026-01-01T00:00:00"</code>                                  | Pass   | 0.000s |
| If lineage manifest has last_run, it is carried through.                            | <code>gauge_step["last_run"] ==</code><br><code>"2026-03-01T12:00:00"</code>                                  | Pass   | 0.000s |
| A file older than 72 hours is reported as 'stale'.                                  | <code>prop_step["status"] == "stale";</code><br><code>len(prop_step["issues"]) == 1</code> (+1 more)          | Pass   | 0.000s |
| All present                                                                         | <code>ok; missing == []</code>                                                                                | Pass   | 0.000s |
| Missing upstream                                                                    | <code>not ok; "gauges" in missing</code>                                                                      | Pass   | 0.000s |
| Nearby ranges round to same cache key.                                              | <code>L1 is L2</code>                                                                                         | Pass   | 0.000s |
| Second call at same range returns cached object.                                    | <code>L1 is L2</code>                                                                                         | Pass   | 0.000s |
| Cholesky lower triangular                                                           | <code>np.allclose(np.triu(L, k=1), 0.0)</code>                                                                | Pass   | 0.000s |
| $L @ L.T == C$ within numerical tolerance.                                          | <code>np.allclose(L @ L.T, C, atol=1e-10)</code>                                                              | Pass   | 0.001s |
| Thames cholesky reconstruction                                                      | <code>np.allclose(L @ L.T, C, atol=1e-8)</code>                                                               | Pass   | 0.004s |
| When <code>np.linalg.cholesky</code> raises <code>LinAlgError</code> on first call, | <code>L.shape == (model.n_gauges,</code><br><code>model.n_gauges); call_count[0] ==</code><br><code>2</code>  | Pass   | 0.000s |
| Generator error has message                                                         | <code>r.get_json().get("message")</code>                                                                      | Pass   | 0.005s |
| Generator error returns 500                                                         | <code>r.status_code == 500</code>                                                                             | Pass   | 0.008s |
| Generator error status is error                                                     | <code>r.get_json()["status"] == "error"</code>                                                                | Pass   | 0.005s |
| Claim ref is built from last 8 chars of property_id.                                | <code>len(result) &gt; 1000</code>                                                                            | Pass   | 0.016s |
| Generate empty sequence lookup                                                      | <code>isinstance(result, bytes)</code>                                                                        | Pass   | 0.021s |
| Generate returns bytes                                                              | <code>isinstance(result, bytes);</code><br><code>len(result) &gt; 0</code>                                    | Pass   | 0.021s |
| Generate without mortgage                                                           | <code>isinstance(result, bytes); result[:4]</code><br><code>== b'%PDF'</code>                                 | Pass   | 0.016s |
| Minimal prop_data with a single flood event should not raise.                       | <code>isinstance(result, bytes)</code>                                                                        | Pass   | 0.013s |
| Output is valid pdf                                                                 | <code>result[:4] == b'%PDF'</code>                                                                            | Pass   | 0.021s |
| Pdf size reasonable                                                                 | <code>len(result) &gt; 8.000</code>                                                                           | Pass   | 0.021s |

| Test Description                                                                    | Acceptance Criteria                                                                                                | Result | Time   |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------|--------|
| Content length header set                                                           | <code>r.headers.get("Content-Length") == str(len(fake_pdf))</code>                                                 | Pass   | 0.005s |
| Filename contains prop id                                                           | <code>PROP_ID in cd</code>                                                                                         | Pass   | 0.004s |
| Response body is pdf bytes                                                          | <code>r.data == fake_pdf</code>                                                                                    | Pass   | 0.003s |
| Response has content disposition                                                    | <code>"attachment" in cd; PROP_ID in cd</code>                                                                     | Pass   | 0.004s |
| Response is pdf content type                                                        | <code>r.content_type == "application/pdf"</code>                                                                   | Pass   | 0.004s |
| Success returns 200                                                                 | <code>r.status_code == 200</code>                                                                                  | Pass   | 0.004s |
| Corrupt storm_sequences.json triggers the except in load_sequence_lookup (L56-57).  | <code>r.status_code == 200; seq_lookup == {}</code>                                                                | Pass   | 0.030s |
| load_mortgage_record exception (L89-91) returns None; report still generated.       | <code>r.status_code == 200; mort_record is None</code>                                                             | Pass   | 0.002s |
| load_prop_record exception (L71-73) returns {}; report still generated.             | <code>r.status_code == 200; prop_record == {}</code>                                                               | Pass   | 0.002s |
| load_sequence_lookup exception (L56-57) returns empty dict; report still generated. | <code>r.status_code == 200</code>                                                                                  | Pass   | 0.002s |
| No dir message mentions propertyts                                                  | <code>"propertyts" in msg or "property" in msg or "not found" i...</code>                                          | Pass   | 0.000s |
| No dir returns 404                                                                  | <code>r.status_code == 404</code>                                                                                  | Pass   | 0.000s |
| No dir status error                                                                 | <code>r.get_json()["status"] == "error"</code>                                                                     | Pass   | 0.000s |
| No floods message mentions events                                                   | <code>"flood" in msg or "event" in msg</code>                                                                      | Pass   | 0.002s |
| No floods returns 404                                                               | <code>r.status_code == 404</code>                                                                                  | Pass   | 0.002s |
| No floods status error                                                              | <code>r.get_json()["status"] == "error"</code>                                                                     | Pass   | 0.002s |
| Unknown property message contains id                                                | <code>"PROP-GHOST" in r.get_json()["message"]</code>                                                               | Pass   | 0.000s |
| Unknown property returns 404                                                        | <code>r.status_code == 404</code>                                                                                  | Pass   | 0.000s |
| Unknown property status error                                                       | <code>r.get_json()["status"] == "error"</code>                                                                     | Pass   | 0.000s |
| Metadata categories contains expected keys                                          | <code>key in cats</code>                                                                                           | Pass   | 0.000s |
| Metadata categories station has required subkeys                                    | <code>key in station_keys</code>                                                                                   | Pass   | 0.000s |
| Schema version is string                                                            | <code>isinstance(GaugeHistoricalDaily.SCHEMA_VERSION, str); len(GaugeHistoricalDaily.SCHEMA_VERSION) &gt; 0</code> | Pass   | 0.000s |
| Classifiers that exist should reference current gauge IDs.                          | <code>len(valid) &gt; 0</code>                                                                                     | Pass   | 0.002s |



| Test Description                                                              | Acceptance Criteria                                                                                      | Result | Time   |
|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------|--------|
| At least one trained classifier must exist.                                   | <code>len(classifiers) &gt; 0</code>                                                                     | Pass   | 0.001s |
| <code>training_summary.json</code> must exist alongside classifiers.          | —                                                                                                        | Pass   | 0.000s |
| Job dict says 'failed' => status failed with error message.                   | <code>resp.status_code == 200;</code><br><code>data["status"] == "failed"</code> (+2 more)               | Pass   | 0.001s |
| Failed job with no 'error' key falls back to 'Unknown error'.                 | <code>data["status"] == "failed";</code><br><code>data["error"] == "Unknown error"</code>                | Pass   | 0.001s |
| If joblib exists on disk, status is 'ready' even if job dict says 'training'. | <code>data["status"] == "ready"</code>                                                                   | Pass   | 0.001s |
| No joblib file and no training job => not_trained.                            | <code>resp.status_code == 200;</code><br><code>data["status"] == "not_trained"</code><br>(+1 more)       | Pass   | 0.001s |
| Job dict says 'ready' (joblib might not exist yet) => status ready.           | <code>resp.status_code == 200;</code><br><code>data["status"] == "ready"</code>                          | Pass   | 0.001s |
| If a .joblib file exists for the gauge, status is 'ready'.                    | <code>resp.status_code == 200;</code><br><code>data["status"] == "ready"</code> (+1 more)                | Pass   | 0.001s |
| When a training job is in progress, status is 'training' with elapsed.        | <code>resp.status_code == 200;</code><br><code>data["status"] == "training"</code> (+4 more)             | Pass   | 0.001s |
| Unknown gauge with no joblib and no job => not_trained.                       | <code>resp.status_code == 200;</code><br><code>data["status"] == "not_trained"</code>                    | Pass   | 0.001s |
| Status endpoint                                                               | <code>"/api/v1/trading/stress/classifier-status"</code> in <code>cl_js</code>                            | Pass   | 0.000s |
| Summary endpoint                                                              | <code>"/api/v1/trading/classifiers/summary"</code> in <code>cl_js</code>                                 | Pass   | 0.000s |
| Train all endpoint                                                            | <code>"/api/v1/trading/classifiers/train-all"</code> in <code>cl_js</code>                               | Pass   | 0.000s |
| Train endpoint                                                                | <code>"/api/v1/trading/stress/train/"</code> in <code>cl_js</code>                                       | Pass   | 0.000s |
| <code>hidePanel</code> calls <code>clCleanupCharts</code> .                   | <code>'clCleanupCharts'</code> in <code>full_js</code>                                                   | Pass   | 0.000s |
| 'classifiers' included in <code>switchTab</code> views array.                 | <code>"classifiers"</code> in <code>full_js</code>                                                       | Pass   | 0.000s |
| <code>switchTab</code> calls <code>loadClassifiersData</code> .               | <code>'loadClassifiersData()'</code> in <code>full_js</code>                                             | Pass   | 0.000s |
| Tab button array includes classifiers.                                        | <code>"classifiers"</code> in <code>full_js</code> or <code>"classifiers"</code> in <code>full_js</code> | Pass   | 0.000s |
| <code>createClassifiersView()</code> called in panel setup.                   | <code>'createClassifiersView()'</code> in <code>full_js</code>                                           | Pass   | 0.000s |

| Test Description                                              | Acceptance Criteria                                                     | Result | Time   |
|---------------------------------------------------------------|-------------------------------------------------------------------------|--------|--------|
| All gauges have .joblib → ready=True.                         | data["ready"] is True;<br>data["trained"] == data["total"]<br>(+1 more) | Pass   | 0.003s |
| Internal error → 500 with error status.                       | resp.status_code == 500;<br>data["status"] == "error" (+1 more)         | Pass   | 0.007s |
| No .joblib files → ready=False.                               | resp.status_code == 200;<br>data["status"] == "success" (+4 more)       | Pass   | 0.003s |
| Some gauges trained → ready=False, correct counts.            | data["ready"] is False;<br>data["trained"] == 2 (+1 more)               | Pass   | 0.008s |
| classifiers.get_js() returns non-empty string.                | len(cl_js) > 0                                                          | Pass   | 0.000s |
| '_clStartBatchTraining' triggers train-all.                   | '_clStartBatchTraining' in cl_js                                        | Pass   | 0.000s |
| 'clCleanupCharts' destroys Chart.js instances.                | 'clCleanupCharts' in cl_js                                              | Pass   | 0.000s |
| 'createClassifiersView' builds the Tab 9 DOM.                 | 'createClassifiersView' in cl_js                                        | Pass   | 0.000s |
| '_clRenderFeatureChart' renders feature importance bar chart. | '_clRenderFeatureChart' in cl_js                                        | Pass   | 0.000s |
| 'loadClassifiersData' fetches summary on tab open.            | 'loadClassifiersData' in cl_js                                          | Pass   | 0.000s |
| '_clMetricCard' builds metric display cards.                  | '_clMetricCard' in cl_js                                                | Pass   | 0.000s |
| '_clUpdateProgress' updates the batch progress bar.           | '_clUpdateProgress' in cl_js                                            | Pass   | 0.000s |
| '_clRenderDetail' renders single-gauge detail panel.          | '_clRenderDetail' in cl_js                                              | Pass   | 0.000s |
| '_clRenderSummaryTable' renders the all-gauge table.          | '_clRenderSummaryTable' in cl_js                                        | Pass   | 0.000s |
| '_clStartSingleTraining' triggers per-gauge training.         | '_clStartSingleTraining' in cl_js                                       | Pass   | 0.000s |
| Module-level state variables must be declared.                | 'clSummary' in cl_js;<br>'clSelectedGaugeId' in cl_js (+2 more)         | Pass   | 0.000s |
| Avg auc computed                                              | data["num_trained"] == 1;<br>data["avg_auc_roc"] == 0.95                | Pass   | 0.004s |
| Gauge fields                                                  | "gauge_id" in g; "gauge_name" in g (+4 more)                            | Pass   | 0.004s |
| Gauges sorted by longitude                                    | lons == sorted(lons)                                                    | Pass   | 0.003s |

| Test Description                                                  | Acceptance Criteria                                                                         | Result | Time   |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| No .joblib files → all gauges show has_model=False.               | data["num_trained"] == 0;<br>g["has_model"] is False                                        | Pass   | 0.003s |
| Returns all gauges                                                | "gauges" in data; data["num_total"] > 0                                                     | Pass   | 0.003s |
| Returns success                                                   | resp.status_code == 200;<br>data["status"] == "success"                                     | Pass   | 0.003s |
| When training_summary.json + .joblib exist, metrics appear.       | len(trained) == 1; g["has_model"] is True (+4 more)                                         | Pass   | 0.004s |
| Above high threshold is high                                      | _classify_property_risk(0.5) == "High"                                                      | Pass   | 0.000s |
| High threshold                                                    | _classify_property_risk(0.03) == "High"                                                     | Pass   | 0.000s |
| Just below medium is low                                          | _classify_property_risk(0.009) == "Low"                                                     | Pass   | 0.000s |
| Medium threshold                                                  | _classify_property_risk(0.01) == "Medium"                                                   | Pass   | 0.000s |
| Tiny prob is low                                                  | _classify_property_risk(0.005) == "Low"                                                     | Pass   | 0.000s |
| Zero is negligible                                                | _classify_property_risk(0.0) == "Negligible"                                                | Pass   | 0.000s |
| No models to clear → removed=0.                                   | resp.status_code == 200;<br>data["status"] == "success" (+1 more)                           | Pass   | 0.001s |
| Internal error → 500.                                             | resp.status_code == 500;<br>data["status"] == "error"                                       | Pass   | 0.001s |
| Clears .joblib files and training_summary.json.                   | data["status"] == "success";<br>data["removed"] == 3 (+3 more)                              | Pass   | 0.003s |
| Closed trade should not appear in stress results.                 | resp.status_code == 200; t['swap_id'] != 'PRS-CLOSED-001'                                   | Pass   | 0.067s |
| post_damage_ltv is recomputed independently for each storm.       | ltvs["STORM-cl2"] > ltvs["STORM-cl1"]; ltvs["STORM-cl2"] > ltvs["STORM-cl3"]                | Pass   | 0.005s |
| Each cluster storm individually queryable                         | r.status_code == 200; data["status"] == "success"                                           | Pass   | 0.005s |
| STORM-cl2 (damage_ratio=0.15) causes more damage than cl1 or cl3. | damages["STORM-cl2"] > damages["STORM-cl1"];<br>damages["STORM-cl2"] > damages["STORM-cl3"] | Pass   | 0.005s |
| Querying a storm_id that didn't breach alert returns 404.         | r.status_code == 404                                                                        | Pass   | 0.004s |

| Test Description                                                  | Acceptance Criteria                                                    | Result | Time   |
|-------------------------------------------------------------------|------------------------------------------------------------------------|--------|--------|
| Summary max_depth_m reflects the worst storm in the cluster.      | result["summary"]["max_depth_m"] >= 0.0                                | Pass   | 0.001s |
| Cluster produces three flood events                               | len(data["flood_events"]) == 3                                         | Pass   | 0.004s |
| Cluster three distinct storm ids in events                        | storm_ids == {"STORM-c001a", "STORM-c001b", "STORM-c001c"}             | Pass   | 0.004s |
| The 'from port.src.book import ...' statement in cmd.port must be | blotter_idx != -1; book_idx != -1 (+1 more)                            | Pass   | 0.005s |
| Importing app.commands.port must not trigger QuantLib.            | hasattr(mod, "cmd.port"); hasattr(mod, "register_parser")              | Pass   | 0.001s |
| chain_result has expected keys.                                   | 'is_consistent' in chain; 'stale_steps' in chain (+1 more)             | Pass   | 0.002s |
| Every step in graph should have a step_details entry.             | graph.steps == detail_steps                                            | Pass   | 0.002s |
| Pipeline must have at least one step.                             | data['num_steps'] > 0                                                  | Pass   | 0.002s |
| collect_all() returns a dict with expected keys.                  | isinstance(data, dict); 'manifest' in data (+4 more)                   | Pass   | 0.016s |
| step_details is a list of dicts.                                  | isinstance(data['step_details'], list); isinstance(sd, dict) (+5 more) | Pass   | 0.002s |
| Blue is hex colour                                                | BLUE == colors.HexColor('#1565C0')                                     | Pass   | 0.000s |
| Dark blue                                                         | DARK_BLUE == colors.darkblue                                           | Pass   | 0.000s |
| Grey                                                              | GREY == colors.HexColor('#757575')                                     | Pass   | 0.000s |
| Light amber                                                       | LIGHT_AMBER == colors.HexColor('#FF8E1')                               | Pass   | 0.000s |
| Light blue                                                        | LIGHT_BLUE == colors.HexColor('#E3F2FD')                               | Pass   | 0.000s |
| Light green                                                       | LIGHT_GREEN == colors.HexColor('#E8F5E9')                              | Pass   | 0.000s |
| Light red                                                         | LIGHT_RED == colors.HexColor('#FFEBEE')                                | Pass   | 0.000s |
| Age 50 to 100                                                     | mid_age > young                                                        | Pass   | 0.000s |
| Age over 100                                                      | old > young                                                            | Pass   | 0.000s |
| Basic returns float                                               | isinstance(score, float); 0 < score <= 10                              | Pass   | 0.000s |
| Construction concrete lower                                       | concrete < brick                                                       | Pass   | 0.000s |
| Construction steel lowest                                         | steel < concrete                                                       | Pass   | 0.000s |
| Construction timber higher                                        | timber > brick                                                         | Pass   | 0.000s |
| Construction wood equals timber                                   | timber == wood                                                         | Pass   | 0.000s |
| High ltv multiplier                                               | high > low                                                             | Pass   | 0.000s |
| Ltv 60 to 80 multiplier                                           | mid > base                                                             | Pass   | 0.000s |
| Ltv 80 to 95 multiplier                                           | mid > base                                                             | Pass   | 0.000s |
| None ltv no multiplier                                            | 0 < score <= 10                                                        | Pass   | 0.000s |

| Test Description                                                                            | Acceptance Criteria                                                                                      | Result | Time   |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------|--------|
| Unknown construction type                                                                   | <code>0 &lt; score &lt;= 10</code>                                                                       | Pass   | 0.000s |
| Unknown flood level                                                                         | <code>0 &lt; score &lt;= 10</code>                                                                       | Pass   | 0.000s |
| Very high flood capped                                                                      | <code>score == 10.0</code>                                                                               | Pass   | 0.000s |
| Very low flood lowest score                                                                 | <code>vl &lt; vh</code>                                                                                  | Pass   | 0.000s |
| All sections together                                                                       | <code>isinstance(result, list); len(tables) &gt;= 2</code>                                               | Pass   | 0.000s |
| Feeding bad data (non-dict regulatory) exercises the except block.                          | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| mortgage_data without 'Mortgage' key → uses top-level.                                      | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Commit button must call <code>tdCommitMarket()</code> .                                     | <code>'tdCommitMarket()' in js</code> or <code>'tdCommitMarket(' in js</code>                            | Pass   | 0.000s |
| Market tab must contain a visible commit button with correct ID.                            | <code>'td-commit-btn' in js</code>                                                                       | Pass   | 0.000s |
| Commit button must NOT have <code>display:none</code> in its own style.                     | <code>'display:none' not in own.style</code>                                                             | Pass   | 0.000s |
| <code>tdCommitMarket</code> function must be defined.                                       | <code>'tdCommitMarket = function' in js</code> or <code>\{\} 'fu...</code>                               | Pass   | 0.000s |
| Commit function must POST to <code>hazard-term-structure/commit</code> .                    | <code>'/trading/hazard-term-structure/commit' in js</code>                                               | Pass   | 0.000s |
| After commit, must switch to blotter tab (which loads blotter data).                        | <code>"switchTab('blotter')" in js</code>                                                                | Pass   | 0.000s |
| Close out path exercised                                                                    | <code>r.status_code in (200, 400)</code>                                                                 | Pass   | 0.035s |
| Commit with cashflows — exercises PDF cashflow schedule section.                            | <code>r.status_code in (200, 400)</code>                                                                 | Pass   | 0.007s |
| Commit with <code>maturity_date</code> provided explicitly.                                 | <code>r.status_code in (200, 400)</code>                                                                 | Pass   | 0.007s |
| Empty json returns 400                                                                      | <code>r.status_code == 400;</code><br><code>r.get_json()["status"] == "error"</code>                     | Pass   | 0.001s |
| Commit with enough fields to pass CDM validation.                                           | <code>r.status_code in (200, 400);</code><br><code>r.get_json()["status"] in ("success", "error")</code> | Pass   | 0.007s |
| No json returns error                                                                       | <code>r.status_code in (400, 500);</code><br><code>r.get_json()["status"] == "error"</code>              | Pass   | 0.010s |
| Validation fails returns 400                                                                | <code>r.status_code == 400; data["status"] == "error" (+1 more)</code>                                   | Pass   | 0.027s |
| <code>commit_yield_curve</code> creates <code>yield_curve</code> from default when missing. | <code>updated['yield_curve']['1'] == 0.04;</code><br><code>updated['yield_curve']['2'] == 0.045</code>   | Pass   | 0.003s |
| Advised flag                                                                                | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Business purpose false                                                                      | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |

| Test Description                                                                                | Acceptance Criteria                                                                                                                           | Result | Time   |
|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Business purpose true                                                                           | <code>isinstance(result, list)</code>                                                                                                         | Pass   | 0.000s |
| Common section returns elements                                                                 | <code>isinstance(result, list); len(result) &gt; 2</code>                                                                                     | Pass   | 0.000s |
| Fca reference number                                                                            | <code>len(tables) &gt;= 1</code>                                                                                                              | Pass   | 0.000s |
| Multiple regulatory flags                                                                       | <code>isinstance(result, list)</code>                                                                                                         | Pass   | 0.000s |
| Contains header                                                                                 | <code>"Scenario Comparison" in result</code>                                                                                                  | Pass   | 0.394s |
| Contains scenario names                                                                         | <code>name in result</code>                                                                                                                   | Pass   | 0.363s |
| None uses all scenarios                                                                         | <code>len(result) &gt; 50</code>                                                                                                              | Pass   | 0.363s |
| Returns string                                                                                  | <code>isinstance(result, str)</code>                                                                                                          | Pass   | 0.364s |
| Single scenario                                                                                 | <code>first_name in result</code>                                                                                                             | Pass   | 0.073s |
| Two scenarios                                                                                   | <code>name in result</code>                                                                                                                   | Pass   | 0.146s |
| Shim class is same                                                                              | <code>New is Shim</code>                                                                                                                      | Pass   | 0.000s |
| Shim imports generator                                                                          | <code>ClaimReportGenerator is not None</code>                                                                                                 | Pass   | 0.002s |
| Line 159: <code>_create_summary</code> calls <code>get_entity_summary</code> .                  | <code>summary == {'id': 'Y', 'name': 'Ypsilon'}</code>                                                                                        | Pass   | 0.000s |
| Line 155: <code>_extract_id</code> calls <code>get_entity_id</code> .                           | <code>loader._extract_id(entity) == 'X'</code>                                                                                                | Pass   | 0.000s |
| Line 163: <code>_get_container_keys</code> returns <code>CONTAINER_KEYS</code> .                | <code>loader._get_container_keys() == ['items']</code>                                                                                        | Pass   | 0.000s |
| Line 167: <code>get_file_status</code> is alias for <code>get_status</code> .                   | <code>status['entity_name'] == 'stub'; status['exists'] is True (+1 more)</code>                                                              | Pass   | 0.000s |
| Persistent sequences (SHORT gaps) should show stronger compounding than doublets (MEDIUM gaps). | <code>persistent_excess &gt;= doublet_excess - 0.1</code>                                                                                     | Pass   | 0.046s |
| During the gap, level should drop from Storm 1 peak.                                            | <code>level.at_gap_end &lt; s1_peak</code>                                                                                                    | Pass   | 0.000s |
| For a short-gap persistent sequence, level does not return to baseline during gap.              | <code>level.at_gap_end &gt; default_gauge.base_level + 0.05</code>                                                                            | Pass   | 0.000s |
| Spec requirement: Storm 2 peak $\geq$ Storm 1 peak in $\geq 60\%$ of doublets.                  | <code>fraction &gt;= 0.60</code>                                                                                                              | Pass   | 0.036s |
| Consistent when all clear                                                                       | <code>dl_mod._compute_health(chain, lr) == 'CONSISTENT'</code>                                                                                | Pass   | 0.000s |
| Degraded on stale steps                                                                         | <code>dl_mod._compute_health(chain, lr) == 'DEGRADED'</code>                                                                                  | Pass   | 0.000s |
| Degraded on failures                                                                            | <code>dl_mod._compute_health(chain, lr) == 'DEGRADED'</code>                                                                                  | Pass   | 0.000s |
| Health colour mapping                                                                           | <code>dl_mod._health.colour('CONSISTENT') == dl_mod.GREEN;</code><br><code>dl_mod._health.colour('DEGRADED') == dl_mod.AMBER (+2 more)</code> | Pass   | 0.000s |



| Test Description                                                              | Acceptance Criteria                                                                  | Result | Time   |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|--------|
| Inconsistent on broken chain                                                  | <code>dl.mod..compute_health(chain, lr) == 'INCONSISTENT'</code>                     | Pass   | 0.000s |
| dist i 1.0 -i short-circuit to _build_flood_event directly.                   | <code>result is not None; result["storm_id"] == "STORM-001"</code>                   | Pass   | 0.001s |
| Multiple gauges in nearest -i IDW path used.                                  | <code>result is not None</code>                                                      | Pass   | 0.001s |
| gauge_responses has no entry for nearest gauge -i None.                       | <code>result is None</code>                                                          | Pass   | 0.001s |
| Normal IDW path returns dict with storm_id.                                   | <code>result is not None; result["storm_id"] == "STORM-001"</code>                   | Pass   | 0.001s |
| Line 126: annuity i= 0 returns annual_hazard_rate * 10000.                    | <code>result &gt; 0</code>                                                           | Pass   | 0.000s |
| Negative hazard returns zero                                                  | <code>result == 0.0</code>                                                           | Pass   | 0.000s |
| Verify compute_prs_spread uses yield_curve when provided.                     | <code>result &gt; 0</code>                                                           | Pass   | 0.000s |
| Zero hazard returns zero                                                      | <code>result == 0.0</code>                                                           | Pass   | 0.000s |
| Pre-wetting the catchment before the same storm raises the peak.              | <code>np.max(wet) &gt; np.max(dry)</code>                                            | Pass   | 0.001s |
| Level should be declining in the later part of the drainage window.           | <code>np.all(np.diff(tail) &lt;= 0.01)</code>                                        | Pass   | 0.004s |
| Extreme storm should produce higher peak than moderate.                       | <code>np.max(l_ext) &gt; np.max(l_mod)</code>                                        | Pass   | 0.001s |
| With identical first storm parameters, doublet produces i= isolated peak.     | <code>np.max(l_dbl) &gt;= np.max(l_iso)</code>                                       | Pass   | 0.001s |
| Severe storm should raise level above baseline.                               | <code>np.max(levels) &gt; default_gauge.base_level + 0.5</code>                      | Pass   | 0.000s |
| River level must always be i= base_level.                                     | <code>np.all(levels &gt;= default_gauge.base_level - 1e-6)</code>                    | Pass   | 0.005s |
| Spec constraint: no precipitation after hour 156.                             | <code>np.all(p[156:] == pytest.approx(0.0))</code>                                   | Pass   | 0.001s |
| Output shape                                                                  | <code>p.shape == (EVENT_WINDOW_HOURS,); levels.shape == (EVENT_WINDOW_HOURS,)</code> | Pass   | 0.000s |
| Level at hour 167 should be below the peak level (recovering, not diverging). | <code>final &lt;= peak + 0.01; final_excess &lt; 1.5</code>                          | Pass   | 0.005s |
| Peak above base                                                               | <code>peak &gt;= default_gauge.base_level</code>                                     | Pass   | 0.004s |
| Peak indices match storm indices                                              | <code>idx == storm.storm_index</code>                                                | Pass   | 0.002s |
| Returns one peak per storm                                                    | <code>len(peaks) == seq.num_storms</code>                                            | Pass   | 0.004s |
| Compliant when all pass                                                       | <code>dl.mod..compute_verdict(chain, lr) == 'COMPLIANT'</code>                       | Pass   | 0.000s |

| Test Description                                                      | Acceptance Criteria                                                                                                   | Result | Time   |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------|--------|
| Empty lineage results                                                 | <code>dl.mod..compute_verdict(chain, {}) == 'PARTIALLY COMPLIANT'</code>                                              | Pass   | 0.000s |
| Non compliant on broken chain                                         | <code>dl.mod..compute_verdict(chain, 1r) == 'NON-COMPLIANT'</code>                                                    | Pass   | 0.000s |
| Non compliant on majority failures                                    | <code>dl.mod..compute_verdict(chain, 1r) == 'NON-COMPLIANT'</code>                                                    | Pass   | 0.000s |
| Non compliant on missing steps                                        | <code>dl.mod..compute_verdict(chain, 1r) == 'NON-COMPLIANT'</code>                                                    | Pass   | 0.000s |
| Partially compliant on failures                                       | <code>dl.mod..compute_verdict(chain, 1r) == 'PARTIALLY COMPLIANT'</code>                                              | Pass   | 0.000s |
| No tests run cannot prove compliance.                                 | <code>dl.mod..compute_verdict(chain, 1r) == 'PARTIALLY COMPLIANT'</code>                                              | Pass   | 0.000s |
| Spatial correlation constants in config.port must be self-consistent. | <code>SPATIAL_CORR_ENABLED is True;</code><br><code>SPATIAL_CORR_MODEL_TYPE == "exponential" (+9 more)</code>         | Pass   | 0.000s |
| Config structure                                                      | <code>sc["enabled"] is True;</code><br><code>sc["model.type"] == "exponential" (+3 more)</code>                       | Pass   | 0.001s |
| Round trip params                                                     | <code>loaded.base_range_km == 35.0;</code><br><code>loaded.nugget == 0.08</code>                                      | Pass   | 0.001s |
| Save creates file                                                     | <code>path.exists()</code>                                                                                            | Pass   | 0.000s |
| Configure both menus                                                  | <code>handler.property_menu == pm;</code><br><code>handler.gauge_menu == gm</code>                                    | Pass   | 0.000s |
| Configure gauge menu                                                  | <code>handler.gauge_menu == new_gm</code>                                                                             | Pass   | 0.000s |
| Configure max visible                                                 | <code>ns.max_visible == 10</code>                                                                                     | Pass   | 0.000s |
| Configure menus                                                       | <code>mgr.context_menus.property_menu == custom_prop</code>                                                           | Pass   | 0.000s |
| Configure no args no change                                           | <code>handler.server_url == 'http://test:1234'</code>                                                                 | Pass   | 0.000s |
| Configure no args no change                                           | <code>handler.property_menu == DEFAULT_PROPERTY_MENU;</code><br><code>handler.gauge_menu == DEFAULT_GAUGE_MENU</code> | Pass   | 0.000s |
| Configure no args no change                                           | <code>ns.position == NotificationPosition.TOP_LEFT;</code><br><code>ns.timeout == 7000 (+1 more)</code>               | Pass   | 0.000s |
| Configure notification position                                       | <code>mgr.notifications.position == NotificationPosition.BOTTOM...</code>                                             | Pass   | 0.000s |
| Configure notification timeout                                        | <code>mgr.notifications.timeout == 2000</code>                                                                        | Pass   | 0.000s |
| Configure position                                                    | <code>ns.position == NotificationPosition.BOTTOM_LEFT</code>                                                          | Pass   | 0.000s |

| Test Description                     | Acceptance Criteria                                                                                   | Result | Time   |
|--------------------------------------|-------------------------------------------------------------------------------------------------------|--------|--------|
| Configure property menu              | <code>handler.property_menu == new_pm</code>                                                          | Pass   | 0.000s |
| Configure returns self               | <code>result is mgr</code>                                                                            | Pass   | 0.000s |
| Configure server url                 | <code>result is mgr; mgr.backend.server_url == 'http://new:9999'</code>                               | Pass   | 0.000s |
| Configure timeout                    | <code>ns.timeout == 3000</code>                                                                       | Pass   | 0.000s |
| Configure updates endpoints          | <code>handler.endpoints == {'a': '/b', 'c': '/d'}</code>                                              | Pass   | 0.000s |
| Configure updates server url         | <code>handler.server_url == 'http://new:2222'; handler.endpoints == {'a': '/b'}</code>                | Pass   | 0.000s |
| Margin positive                      | <code>MARGIN &gt; 0</code>                                                                            | Pass   | 0.000s |
| Page dimensions                      | <code>PAGE_W &gt; 0; PAGE_H &gt; 0 (+1 more)</code>                                                   | Pass   | 0.000s |
| Construction year shown              | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Empty property does not crash        | <code>isinstance(self._page().generate_elements(prop, list))</code>                                   | Pass   | 0.000s |
| Floor level shown                    | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Has section header                   | <code>any("Construction" in t for t in texts)</code>                                                  | Pass   | 0.000s |
| Has table                            | <code>any(isinstance(e, Table) for e in result)</code>                                                | Pass   | 0.000s |
| No construction data returns message | <code>any("No construction" in t for t in texts)</code>                                               | Pass   | 0.000s |
| No construction year                 | <code>isinstance(self._page().generate_elements(prop, list))</code>                                   | Pass   | 0.000s |
| Returns list with data               | <code>isinstance(result, list) and len(result) &gt; 0</code>                                          | Pass   | 0.000s |
| Site height shown                    | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Brick construction                   | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Concrete construction                | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Empty construction type              | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Masonry construction                 | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Steel construction                   | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Timber construction                  | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Unknown construction type            | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Custom menus                         | <code>handler.property_menu == pm; handler.gauge_menu == gm</code>                                    | Pass   | 0.000s |
| Default init                         | <code>handler.property_menu == DEFAULT_PROPERTY_MENU; handler.gauge_menu == DEFAULT_GAUGE_MENU</code> | Pass   | 0.000s |

| Test Description                                                                                                | Acceptance Criteria                                 | Result | Time   |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------|--------|
| Each startup cache variable must be initialised as <code>window.X = null</code> .<br>[var='_preGovAudit']       | <code>f'window.{var}' in src</code>                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as <code>window.X = null</code> .<br>[var='_preGovBib']         | <code>f'window.{var}' in src</code>                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as <code>window.X = null</code> .<br>[var='_preStorms']         | <code>f'window.{var}' in src</code>                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as <code>window.X = null</code> .<br>[var='_tdPreBlotter']      | <code>f'window.{var}' in src</code>                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as <code>window.X = null</code> .<br>[var='_tdPreEodHistory']   | <code>f'window.{var}' in src</code>                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as <code>window.X = null</code> .<br>[var='_tdPreGauges']       | <code>f'window.{var}' in src</code>                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as <code>window.X = null</code> .<br>[var='_tdPreGovDocs']      | <code>f'window.{var}' in src</code>                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as <code>window.X = null</code> .<br>[var='_tdPreMarket']       | <code>f'window.{var}' in src</code>                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as <code>window.X = null</code> .<br>[var='_tdPrePortStorms']   | <code>f'window.{var}' in src</code>                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as <code>window.X = null</code> .<br>[var='_tdPreStressStorms'] | <code>f'window.{var}' in src</code>                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as <code>window.X = null</code> .<br>[var='_tdPreYieldCurve']   | <code>f'window.{var}' in src</code>                 | Pass   | 0.000s |
| No startup cache var should have a bare var declaration.                                                        | <code>f'var {var}' not in src</code>                | Pass   | 0.000s |
| The preload-done flag must NOT be a var (regression: 2026-03-12 incident).                                      | <code>'var _tdPreloadDone' not in src</code>        | Pass   | 0.008s |
| Window td preload done initialised                                                                              | <code>'window._tdPreloadDone = false' in src</code> | Pass   | 0.000s |

| Test Description                                                              | Acceptance Criteria                     | Result | Time   |
|-------------------------------------------------------------------------------|-----------------------------------------|--------|--------|
| Window td preload done set after settle                                       | 'window._tdPreloadDone = true' in src   | Pass   | 0.000s |
| Blotter nulls cache after consume                                             | 'window._tdPreBlotter = null' in src    | Pass   | 0.000s |
| Blotter reads window pre blotter                                              | 'window._tdPreBlotter' in src           | Pass   | 0.000s |
| Gov docs nulls cache after consume                                            | 'window._tdPreGovDocs = null' in src    | Pass   | 0.000s |
| Gov docs reads window pre gov docs                                            | 'window._tdPreGovDocs' in src           | Pass   | 0.004s |
| Port stress nulls cache after consume                                         | 'window._tdPrePortStorms = null' in src | Pass   | 0.000s |
| Port stress reads window pre port storms                                      | 'window._tdPrePortStorms' in src        | Pass   | 0.001s |
| preloader.py must not assign bare _tdPreloadDone (creates unreliable global). | not bare_assigns                        | Pass   | 0.000s |
| trading/preloader.py fallback must also set window._tdPreloadDone.            | 'window._tdPreloadDone = true' in src   | Pass   | 0.002s |
| Storm portfolio nulls cache after consume                                     | 'window._preStorms = null' in src       | Pass   | 0.000s |
| Storm portfolio reads window pre storms                                       | 'window._preStorms' in src              | Pass   | 0.001s |
| showPanel() must read window._tdPreloadDone, not bare _tdPreloadDone.         | 'window._tdPreloadDone' in src          | Pass   | 0.000s |
| Blotter setup no bare pre blotter                                             | —                                       | Pass   | 0.001s |
| Gov docs no bare pre gov docs                                                 | —                                       | Pass   | 0.001s |
| Port stress no bare pre port storms                                           | —                                       | Pass   | 0.001s |
| preloader.py checks should all use window._tdPreloadDone.                     | not bad_lines                           | Pass   | 0.001s |
| Storm table no bare pre storms                                                | —                                       | Pass   | 0.001s |
| Tradingdesk no bare preload done                                              | —                                       | Pass   | 0.001s |
| Governance JS sub-modules must be syntactically valid.                        | —                                       | Pass   | 0.175s |
| startup.py JS must be syntactically valid.                                    | —                                       | Pass   | 0.056s |
| Storm portfolio syntax                                                        | —                                       | Pass   | 0.062s |
| Trading aggregate submodule syntax                                            | —                                       | Pass   | 0.057s |
| Trading blotter submodule syntax                                              | —                                       | Pass   | 0.061s |
| Trading curves submodule syntax                                               | —                                       | Pass   | 0.059s |
| Trading desk js syntax                                                        | —                                       | Pass   | 0.084s |
| Trading eod submodule syntax                                                  | —                                       | Pass   | 0.063s |
| Trading fs01 submodule syntax                                                 | —                                       | Pass   | 0.063s |

| Test Description                                                                                | Acceptance Criteria                                                                                                  | Result | Time   |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------|--------|
| Trading market submodule syntax                                                                 | —                                                                                                                    | Pass   | 0.060s |
| Trading port stress submodule syntax                                                            | —                                                                                                                    | Pass   | 0.060s |
| Trading stress submodule syntax                                                                 | —                                                                                                                    | Pass   | 0.064s |
| Aggregate generates                                                                             | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Blotter generates                                                                               | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Curves generates                                                                                | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Eod generates                                                                                   | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Fs01 generates                                                                                  | <code>len(js) &gt; 500</code>                                                                                        | Pass   | 0.000s |
| Governance bibliography generates                                                               | <code>len(js) &gt; 500</code>                                                                                        | Pass   | 0.000s |
| Governance docs generates                                                                       | <code>len(js) &gt; 500</code>                                                                                        | Pass   | 0.000s |
| Market generates                                                                                | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Port stress generates                                                                           | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Preloader generates                                                                             | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Startup generates                                                                               | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Storm sp table generates                                                                        | <code>len(js) &gt; 500</code>                                                                                        | Pass   | 0.000s |
| Stress generates                                                                                | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Trading desk generates                                                                          | <code>len(js) &gt; 50.000</code>                                                                                     | Pass   | 0.000s |
| blotter/actions.py must call <code>window.refreshMainMapFS01</code> after close-out.            | <code>'window.refreshMainMapFS01'</code> in <code>src</code>                                                         | Pass   | 0.015s |
| <code>tdApplyFilter</code> must be on window for FS01 cell click to work.                       | <code>'window.tdApplyFilter'</code> in <code>src</code>                                                              | Pass   | 0.008s |
| <code>tdRiskCellClick</code> must be on window for blotter filter drill-through.                | <code>'window.tdRiskCellClick'</code> in <code>src</code>                                                            | Pass   | 0.002s |
| <code>_tdPendingFilter</code> must be on window for context-menu → blotter filter.              | <code>'window._tdPendingFilter'</code> in <code>blotter_src</code>                                                   | Pass   | 0.003s |
| <code>refreshMainMapFS01</code> must be on window for blotter close-out to call it.             | <code>'window.refreshMainMapFS01'</code> in <code>td_map_src</code>                                                  | Pass   | 0.002s |
| <code>_stressStormHint</code> must be set via <code>window.*</code> for cross-panel navigation. | <code>'window._stressStormHint'</code> in <code>hist_src</code>                                                      | Pass   | 0.003s |
| Stress setup reads window hint                                                                  | <code>'window._stressStormHint'</code> in <code>stress_src</code>                                                    | Pass   | 0.001s |
| TradingDeskPanel must export <code>TradingDesk</code> and <code>showTradingDesk</code> .        | <code>'window.TradingDesk'</code> in <code>src</code> ;<br><code>'window.showTradingDesk'</code> in <code>src</code> | Pass   | 0.030s |
| Create panel creates blotter view                                                               | <code>'createBlotterView'</code> in <code>js</code>                                                                  | Pass   | 0.003s |
| Open panel function exists                                                                      | <code>'tdOpenPanel'</code> in <code>js</code>                                                                        | Pass   | 0.003s |
| Open panel sets display flex                                                                    | <code>"style.display = 'flex'"</code> in <code>js</code>                                                             | Pass   | 0.003s |



| Test Description                                                      | Acceptance Criteria                                                                | Result | Time   |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------|--------|--------|
| Pi button calls show panel                                            | 'showPanel()' in js                                                                | Pass   | 0.003s |
| Fallback preloader function must exist for first-open race condition. | '_tdRunPreload' in js                                                              | Pass   | 0.003s |
| Show panel function exists                                            | 'function showPanel()' in js                                                       | Pass   | 0.004s |
| Show panel reads window preload done                                  | 'window._tdPreloadDone' in js                                                      | Pass   | 0.003s |
| Panel must switch to blotter tab when opened.                         | "switchTab('blotter')" in js                                                       | Pass   | 0.003s |
| Generate risk report basic                                            | generate_risk_report(sample_portfolio_data, output_dir=ou...                       | Pass   | 0.017s |
| Generate risk report default                                          | generate_risk_report(sample_portfolio_data, output_dir=ou...                       | Pass   | 0.016s |
| Generate risk report detailed                                         | generate_risk_report(sample_portfolio_data, output_dir=ou...                       | Pass   | 0.022s |
| Diagonal is one                                                       | np.allclose(np.diag(C), 1.0)                                                       | Pass   | 0.000s |
| Longer range -i higher off-diagonal correlations.                     | C.long[mask].mean() ><br>C.short[mask].mean()                                      | Pass   | 0.000s |
| Higher nugget -i lower off-diagonal values.                           | C.high[mask].mean() <<br>C.low[mask].mean()                                        | Pass   | 0.000s |
| Off diagonal in open interval                                         | 0.0 < C[i, j] < 1.0                                                                | Pass   | 0.000s |
| Positive definite                                                     | np.all(eigenvalues > 0)                                                            | Pass   | 0.009s |
| Symmetric                                                             | np.allclose(C, C.T)                                                                | Pass   | 0.000s |
| Thames positive definite                                              | np.all(eigenvalues > 0)                                                            | Pass   | 0.005s |
| All gauge files corrupt → no storms (but file still written).         | result["total_storms"] == 0                                                        | Pass   | 0.002s |
| Lines 151-153: unreadable gauge file is skipped with warning.         | out.exists() and (out /<br>"_index.json").exists();<br>result["total_storms"] == 1 | Pass   | 0.003s |
| Lines 133-134: invalid JSON in storms.json logs warning, continues.   | out.exists() and (out /<br>"_index.json").exists();<br>result["total_storms"] == 1 | Pass   | 0.002s |
| Exactly at 119999 should be band 'A'.                                 | band == "A"                                                                        | Pass   | 0.000s |
| Line 159: property value j 120000 returns band 'A'.                   | band == "A"                                                                        | Pass   | 0.000s |
| Line 161: property value = 120000 returns band 'B'.                   | band == "B"                                                                        | Pass   | 0.000s |
| Property value 140000 returns band 'B'.                               | band == "B"                                                                        | Pass   | 0.000s |
| Exactly at 155999 should still be band 'B'.                           | band == "B"                                                                        | Pass   | 0.000s |

| Test Description                                             | Acceptance Criteria                                                                                                            | Result | Time   |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Counts matching files                                        | <code>_count_dir(tmp_path, 'GAUGE-*.json')</code><br><code>== 5</code>                                                         | Pass   | 0.001s |
| Empty directory                                              | <code>_count_dir(tmp_path, '*') == 0</code>                                                                                    | Pass   | 0.000s |
| Returns zero for no matches                                  | <code>_count_dir(tmp_path, '*.json') == 0</code>                                                                               | Pass   | 0.000s |
| Returns zero for none dir                                    | <code>_count_dir(None, '*.json') == 0</code>                                                                                   | Pass   | 0.000s |
| Returns zero for nonexistent dir                             | <code>_count_dir(tmp_path / 'nope',<br/> '*.json') == 0</code>                                                                 | Pass   | 0.000s |
| All counterparties have party id                             | <code>len(counterparties) &gt; 0;</code><br><code>ctpy["CounterpartySet"]["Party"]["PartyID"].startswith("C...</code>          | Pass   | 0.004s |
| Cdm structure                                                | <code>"Party" in ctpy;</code> <code>"PartyID" in</code><br><code>ctpy["Party"] (+1 more)</code>                                | Pass   | 0.001s |
| Custom count                                                 | <code>len(result["data"]) == 3</code>                                                                                          | Pass   | 0.001s |
| Generates counterparty file                                  | <code>(pipeline_dir /</code><br><code>"counterparty.json").exists()</code>                                                     | Pass   | 0.000s |
| Lines 263-269: ContactInformation fields mapped.             | <code>mapping.get("primary_email") ==</code><br><code>"test@bank.com"; mapping.get("city")</code><br><code>== "London"</code>  | Pass   | 0.000s |
| Lines 280-285: count fields in mapping.                      | <code>"counterparty_count" in mapping;</code><br><code>mapping["counterparty_count"] == 2</code>                               | Pass   | 0.000s |
| Minimal data (empty Counterparty-Set) returns dict.          | <code>isinstance(mapping, dict)</code>                                                                                         | Pass   | 0.000s |
| Lines 245-289: create_mapping with full data.                | <code>isinstance(mapping, dict);</code><br><code>mapping["party_id"] == "CTPY-001"</code><br><code>(+3 more)</code>            | Pass   | 0.000s |
| Lines 291-292: exception → raises ValueError.                | —                                                                                                                              | Pass   | 0.000s |
| Lines 288-289: None values are stripped from result.         | <code>None not in mapping.values()</code>                                                                                      | Pass   | 0.000s |
| Lines 294-301: get_required_fields() returns non-empty list. | <code>isinstance(fields, list); len(fields)</code><br><code>&gt; 0 (+2 more)</code>                                            | Pass   | 0.000s |
| Schema is dict                                               | <code>isinstance(cdm.schema, dict);</code><br><code>len(cdm.schema) &gt; 0</code>                                              | Pass   | 0.000s |
| Lines 201-202: empty Counterparties array → error.           | <code>"Counterparties" in errors</code>                                                                                        | Pass   | 0.000s |
| validate({}) returns errors for all required fields.         | <code>isinstance(errors, dict)</code>                                                                                          | Pass   | 0.000s |
| Lines 211-214: invalid role (not Party1/Party2) → error.     | <code>"Counterparties" in errors;</code><br><code>any("Invalid" in e for e in</code><br><code>errors["Counterparties"])</code> | Pass   | 0.000s |

| Test Description                                             | Acceptance Criteria                                                                            | Result | Time   |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------|--------|
| Lines 209-210: missing CounterpartyRole → error.             | "Counterparties" in errors;<br>any("CounterpartyRole" in e for<br>e in errors["Counterparti... | Pass   | 0.000s |
| Lines 188-189: missing PartyID → Party errors.               | "Party" in errors; any("PartyID" in e<br>for e in errors["Party"])                             | Pass   | 0.000s |
| Lines 191-192: missing PartyName → Party errors.             | "Party" in errors; any("PartyName" in<br>e for e in errors["Party"])                           | Pass   | 0.000s |
| Lines 218-219: missing PartyRef → error.                     | "Counterparties" in errors;<br>any("PartyRef" in e for e in<br>errors["Counterparties"])       | Pass   | 0.000s |
| Lines 222-225: two CPs with same role → error.               | "Counterparties" in errors                                                                     | Pass   | 0.000s |
| Valid counterparty data produces empty error dict.           | errors == {}                                                                                   | Pass   | 0.000s |
| Lines 74-75: json.JSONDecodeError in get_counterparty → 500. | r.status_code == 500; data["status"]<br>== "error"                                             | Pass   | 0.002s |
| Lines 58-59: json.JSONDecodeError → 500.                     | r.status_code == 500; data["status"]<br>== "error"                                             | Pass   | 0.003s |
| Creates output file                                          | output_file.exists()                                                                           | Pass   | 0.000s |
| Data is list                                                 | isinstance(result["data"], list);<br>len(result["data"]) == 3                                  | Pass   | 0.000s |
| Default count uses all counterparties                        | len(result["data"]) ==<br>len(_ALL_COUNTERPARTIES)                                             | Pass   | 0.002s |
| File path is pathlib path                                    | isinstance(result["file_path"], Path)                                                          | Pass   | 0.000s |
| Metadata has total generated                                 | result["metadata"]["total_generated"]<br>== 4                                                  | Pass   | 0.001s |
| Output file is valid json                                    | "counterparties" in data;<br>"generation_metadata" in data                                     | Pass   | 0.004s |
| Returns dict with required keys                              | "data" in result; "file_path" in<br>result (+1 more)                                           | Pass   | 0.000s |
| Zero count returns empty                                     | result["data"] == []                                                                           | Pass   | 0.000s |
| Consistent cover                                             | badge._cellvalues[1][0] ==<br>'CONSISTENT'                                                     | Pass   | 0.000s |
| Degraded cover on failures                                   | badge._cellvalues[1][0] == 'DEGRADED'                                                          | Pass   | 0.000s |
| Inconsistent cover on broken chain                           | badge._cellvalues[1][0] ==<br>'INCONSISTENT'                                                   | Pass   | 0.000s |
| Engines not none                                             | engine is not None                                                                             | Pass   | 0.000s |
| Returns dict with engines                                    | isinstance(engines, dict); "ISDA" in<br>engines (+2 more)                                      | Pass   | 0.000s |
| Default day counter                                          | curve is not None; curve is not None                                                           | Pass   | 0.000s |

| Test Description                                                 | Acceptance Criteria                                                                                | Result | Time   |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------|--------|
| Returns curve and quote                                          | curve is not None; quote is not None                                                               | Pass   | 0.000s |
| Actual360 day counter accepted                                   | curve is not None                                                                                  | Pass   | 0.000s |
| Quote is simple quote                                            | isinstance(quote, ql.SimpleQuote)                                                                  | Pass   | 0.000s |
| Quote value matches input                                        | quote.value() == pytest.approx(rate, rel=1e-10)                                                    | Pass   | 0.000s |
| Survival probability formula                                     | sp == pytest.approx(math.exp(-lam * 3), rel=0.02)                                                  | Pass   | 0.000s |
| Very high hazard rate near zero survival                         | s1 == pytest.approx(math.exp(-2.0), rel=0.01)                                                      | Pass   | 0.000s |
| Returns hazard curve                                             | curve is not None                                                                                  | Pass   | 0.000s |
| Survival probability decreases                                   | sp2 < sp1                                                                                          | Pass   | 0.000s |
| Zero hazard near full survival                                   | sp > 0.99                                                                                          | Pass   | 0.000s |
| Auto generates id                                                | "id" in data["meeting"];<br>len(data["meeting"]["id"]) > 0                                         | Pass   | 0.002s |
| Create returns meeting                                           | r.status_code == 200; data["status"] == "success" (+1 more)                                        | Pass   | 0.001s |
| Custom attendees become participants                             | "Alice Smith" in names                                                                             | Pass   | 0.003s |
| Custom title used                                                | data["meeting"]["title"] == "Quarterly Review"                                                     | Pass   | 0.002s |
| Default chair populated                                          | data["meeting"]["chair"] == "Johnny Mattimore"                                                     | Pass   | 0.001s |
| Default participants created                                     | len(participants) >= 1                                                                             | Pass   | 0.002s |
| No json body uses defaults                                       | r.status_code == 200;<br>r.get_json()["status"] == "success"                                       | Pass   | 0.002s |
| Custom max notifications                                         | ns.max_visible == 8                                                                                | Pass   | 0.000s |
| Custom timeout                                                   | ns.timeout == 1000                                                                                 | Pass   | 0.000s |
| Invalid position defaults to top right                           | ns.position.value == "top-right"                                                                   | Pass   | 0.000s |
| Valid position                                                   | ns.position.value == "bottom-left"                                                                 | Pass   | 0.000s |
| create_pdf_report(None) calls collect_all().                     | —                                                                                                  | Pass   | 0.016s |
| Creates audit dir                                                | audit.is_dir(); result.exists()                                                                    | Pass   | 0.016s |
| Creates pdf                                                      | result.exists(); result.stat().st_size > 0                                                         | Pass   | 0.016s |
| Creates pdf empty data                                           | result.exists()                                                                                    | Pass   | 0.007s |
| Main prints summary                                              | 'Generating Model Risk Governance Report' in captured; 'models in inventory' in captured (+5 more) | Pass   | 0.016s |
| Integration: main() with actual data loading against empty dirs. | out.exists()                                                                                       | Pass   | 0.010s |
| PDF file is created on disk.                                     | out.exists()                                                                                       | Pass   | 0.023s |

| Test Description                                               | Acceptance Criteria                                                                  | Result | Time   |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|--------|
| Generated file has PDF magic bytes.                            | <code>out.read_bytes()[:4] == b'%PDF'</code>                                         | Pass   | 0.023s |
| PDF has reasonable size (multi-page report).                   | <code>out.stat().st_size &gt; 5000</code>                                            | Pass   | 0.016s |
| PDF renders when no steps have been recorded.                  | <code>out.exists(); out.read_bytes()[:4] == b'%PDF'</code>                           | Pass   | 0.017s |
| PDF generates correctly when pipeline has stale/missing steps. | <code>out.exists(); out.read_bytes()[:4] == b'%PDF'</code> (+1 more)                 | Pass   | 0.023s |
| PDF renders when lineage_results is empty (tests not run).     | <code>out.exists(); out.stat().st_size &gt; 5000</code>                              | Pass   | 0.016s |
| <code>create_pdf_report()</code> returns a Path object.        | <code>isinstance(result, Path)</code>                                                | Pass   | 0.036s |
| Creates registry                                               | <code>create_registry(str(tmp_path)) is not None</code>                              | Pass   | 0.000s |
| With file overrides                                            | <code>registry.get_gauge_loader().count() == 2</code>                                | Pass   | 0.001s |
| Test create_registry helper.                                   | <code>registry.data_dir == populated_data_dir; registry is not None</code> (+1 more) | Pass   | 0.001s |
| Test create_registry with file overrides.                      | <code>loader.count() == 0</code>                                                     | Pass   | 0.001s |
| Auto generated id starts with storm                            | <code>storm.storm_id.startswith("STORM-")</code>                                     | Pass   | 0.000s |
| Decay kernel stored                                            | <code>storm.decay_kernel == DecayKernel.EXPONENTIAL</code>                           | Pass   | 0.000s |
| Decay parameter stored                                         | <code>storm.decay_parameter == pytest.approx(1.2)</code>                             | Pass   | 0.000s |
| Default num track points                                       | <code>len(storm.track) == 25</code>                                                  | Pass   | 0.000s |
| Default start time is recent                                   | <code>before &lt;= storm.start_time &lt;= after</code>                               | Pass   | 0.000s |
| Duration hours stored                                          | <code>storm.duration_hours == pytest.approx(36.0)</code>                             | Pass   | 0.000s |
| Explicit id preserved                                          | <code>storm.storm_id == "MY-STORM-001"</code>                                        | Pass   | 0.000s |
| Explicit name                                                  | <code>storm.name == "Ciara"</code>                                                   | Pass   | 0.000s |
| Explicit start time                                            | <code>storm.start_time == t</code>                                                   | Pass   | 0.000s |
| Footprint km stored                                            | <code>storm.footprint_km == pytest.approx(60.0)</code>                               | Pass   | 0.000s |
| Num track points                                               | <code>len(storm.track) == 10</code>                                                  | Pass   | 0.000s |
| Peak intensity stored                                          | <code>storm.peak_intensity == pytest.approx(75.0)</code>                             | Pass   | 0.000s |
| Returns storm object                                           | <code>isinstance(storm, Storm)</code>                                                | Pass   | 0.000s |
| Speed is positive                                              | <code>storm.speed_kmh &gt; 0</code>                                                  | Pass   | 0.000s |
| Track end time equals duration                                 | <code>storm.track[-1].time_hours == pytest.approx(24.0)</code>                       | Pass   | 0.000s |

| Test Description                                                           | Acceptance Criteria                                                                     | Result | Time   |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------|--------|
| Track longitude moves from start to end                                    | <code>lons[0] &lt; lons[-1]</code>                                                      | Pass   | 0.000s |
| Track points are trackpoint instances                                      | <code>isinstance(tp, TrackPoint)</code>                                                 | Pass   | 0.000s |
| Track start time is zero                                                   | <code>storm.track[0].time.hours ==<br/>pytest.approx(0.0)</code>                        | Pass   | 0.000s |
| Track time increases                                                       | <code>all(times[i] &lt;= times[i + 1] for i<br/>in range(len(times) - 1))</code>        | Pass   | 0.000s |
| Actual360 day counter produces curve                                       | <code>0.0 &lt; sp &lt;= 1.0</code>                                                      | Pass   | 0.000s |
| Implied hazard rate formula                                                | <code>sp5 == pytest.approx(expected_5yr,<br/>rel=0.01)</code>                           | Pass   | 0.000s |
| Near zero survival positive hazard                                         | <code>s1 &lt; 0.9</code>                                                                | Pass   | 0.000s |
| Single point high risk ts                                                  | <code>curve is not None</code>                                                          | Pass   | 0.000s |
| Single point term structure                                                | <code>curve is not None</code>                                                          | Pass   | 0.000s |
| Survival strictly positive                                                 | <code>sp &gt; 0.0</code>                                                                | Pass   | 0.000s |
| Custom day counter                                                         | <code>curve is not None</code>                                                          | Pass   | 0.000s |
| Default day counter                                                        | <code>curve is not None; curve is not None</code>                                       | Pass   | 0.000s |
| Full survival at start                                                     | <code>abs(sp - 1.0) &lt; 1e-6</code>                                                    | Pass   | 0.000s |
| Returns curve                                                              | <code>curve is not None</code>                                                          | Pass   | 0.000s |
| Line 83: <code>final_survival j= 0</code> → uses 0.05 default hazard rate. | <code>curve is not None</code>                                                          | Pass   | 0.000s |
| Line 83: <code>final_survival j= 1</code> → uses 0.05 default hazard rate. | <code>curve is not None</code>                                                          | Pass   | 0.000s |
| Excellent credit 800 plus                                                  | <code>isinstance(result, list)</code>                                                   | Pass   | 0.000s |
| Fair credit 580 to 669                                                     | <code>isinstance(result, list)</code>                                                   | Pass   | 0.000s |
| Good credit 670 to 739                                                     | <code>isinstance(result, list)</code>                                                   | Pass   | 0.000s |
| Poor credit under 580                                                      | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Very good credit 740 to 799                                                | <code>isinstance(result, list)</code>                                                   | Pass   | 0.000s |
| Excellent 800 plus                                                         | <code>isinstance(result, list)</code>                                                   | Pass   | 0.000s |
| Fair 580 to 669                                                            | <code>isinstance(result, list)</code>                                                   | Pass   | 0.000s |
| Good 670 to 739                                                            | <code>isinstance(result, list)</code>                                                   | Pass   | 0.000s |
| Poor under 580                                                             | <code>isinstance(result, list)</code>                                                   | Pass   | 0.000s |
| Very good 740 to 799                                                       | <code>isinstance(result, list)</code>                                                   | Pass   | 0.000s |
| Gauges returned by list endpoint should have valid timeseries responses.   | <code>response.status_code == 200;<br/>ts_response.status_code in [200,<br/>404]</code> | Pass   | 0.001s |
| Gauge hazard endpoint should never return 500 for listed gauges.           | <code>hz_response.status_code != 500</code>                                             | Pass   | 0.002s |



| Test Description                                                                | Acceptance Criteria                                                                                | Result | Time   |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------|--------|
| Properties returned by list endpoint should have flood data if property exists. | <code>response.status_code == 200;</code><br><code>flood_response.status_code in [200, 404]</code> | Pass   | 0.002s |
| crud.py lines 63-67: gauge not found -i 404.                                    | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code>               | Pass   | 0.140s |
| crud.py lines 59-72: valid gauge retrieval.                                     | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                           | Pass   | 0.028s |
| crud.py lines 51-53: loader error -i 500.                                       | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>               | Pass   | 0.029s |
| crud.py line 34: /list_gauges alias works.                                      | <code>r.status_code == 200;</code><br><code>r.get_json()["count"] == 1</code>                      | Pass   | 0.030s |
| crud.py lines 37-38: OPTIONS returns ok.                                        | <code>r.status_code == 200</code>                                                                  | Pass   | 0.027s |
| crud.py lines 40-50: successful gauge listing.                                  | <code>r.status_code == 200; data["status"] == "success" (+2 more)</code>                           | Pass   | 0.028s |
| hd present but daily_observations missing -i same paragraph.                    | <code>any('No historical daily data' in t for t in texts)</code>                                   | Pass   | 0.000s |
| daily_observations = [] -i falsy, caught by first guard.                        | <code>any('No historical daily data' in t for t in texts)</code>                                   | Pass   | 0.000s |
| Normal flow: valid obs -i chart + table elements returned.                      | <code>len(elements) &gt; 3</code>                                                                  | Pass   | 0.090s |
| ValueError/KeyError in date parse loop -i skipped.                              | <code>len(elements) &gt; 0</code>                                                                  | Pass   | 0.088s |
| Invalid date in latest observation -i current_month = None.                     | <code>len(elements) &gt; 0</code>                                                                  | Pass   | 0.087s |
| No obs in same month -i same_month_levels = all_levels.                         | <code>len(elements) &gt; 0</code>                                                                  | Pass   | 0.085s |
| No hd data -i 'No historical daily data' paragraph.                             | <code>any('No historical daily data' in t for t in texts)</code>                                   | Pass   | 0.000s |
| level is None -i skipped, does not crash.                                       | <code>len(elements) &gt; 0</code>                                                                  | Pass   | 0.089s |
| Boe base shown                                                                  | <code>isinstance(result, list)</code>                                                              | Pass   | 0.000s |
| Current payment shown                                                           | <code>isinstance(result, list)</code>                                                              | Pass   | 0.000s |
| Has status header                                                               | <code>any("Status" in t or "Mortgage" in t for t in texts)</code>                                  | Pass   | 0.000s |
| Has table                                                                       | <code>any(isinstance(e, Table) for e in result)</code>                                             | Pass   | 0.000s |
| Interest paid shown                                                             | <code>isinstance(result, list)</code>                                                              | Pass   | 0.000s |
| Last payment date shown                                                         | <code>isinstance(result, list)</code>                                                              | Pass   | 0.000s |
| Last status shown                                                               | <code>isinstance(result, list)</code>                                                              | Pass   | 0.000s |
| Lending rate shown                                                              | <code>isinstance(result, list)</code>                                                              | Pass   | 0.000s |

| Test Description                                              | Acceptance Criteria                                            | Result | Time   |
|---------------------------------------------------------------|----------------------------------------------------------------|--------|--------|
| No current status returns message                             | any("No current status" in t for t in texts)                   | Pass   | 0.000s |
| No mortgage returns message                                   | any("No mortgage" in t for t in texts)                         | Pass   | 0.000s |
| Outstanding balance shown                                     | isinstance(result, list)                                       | Pass   | 0.000s |
| Principal paid shown                                          | isinstance(result, list)                                       | Pass   | 0.000s |
| Total payments shown                                          | isinstance(result, list)                                       | Pass   | 0.000s |
| With mortgage returns list                                    | isinstance(result, list) and len(result) > 0                   | Pass   | 0.000s |
| Curves tab should show all gauges on one chart.               | 'tdRenderAllCurves' in js; 'tdCurveAllData' in js (+1 more)    | Pass   | 0.000s |
| Curves tab should have alert/warning/severe trigger selector. | 'td-curve-trigger' in js; 'severe' in js (+2 more)             | Pass   | 0.000s |
| Curves tab should use HSL for distinct line colours.          | 'hsl(' in js                                                   | Pass   | 0.000s |
| Curves sub-module should render without errors.               | len(js) > 0                                                    | Pass   | 0.000s |
| Legend should strip 'Thames' prefix for brevity.              | 'Thames' in js                                                 | Pass   | 0.000s |
| Curves tab should fetch from market-state endpoint.           | '/api/v1/trading/market-state' in js                           | Pass   | 0.000s |
| Good dti under 28                                             | isinstance(result, list)                                       | Pass   | 0.000s |
| High dti 36 to 43                                             | isinstance(result, list)                                       | Pass   | 0.000s |
| Moderate dti 28 to 36                                         | isinstance(result, list)                                       | Pass   | 0.000s |
| No payment dti unknown                                        | isinstance(result, list)                                       | Pass   | 0.000s |
| Very high dti over 43                                         | isinstance(result, list)                                       | Pass   | 0.000s |
| Empty daily pnl returns drawing                               | isinstance(result, Drawing)                                    | Pass   | 0.000s |
| Mixed positive negative renders                               | isinstance(result, Drawing)                                    | Pass   | 0.000s |
| Negative daily pnl renders                                    | isinstance(result, Drawing)                                    | Pass   | 0.000s |
| Commercial higher damage                                      | com_ratio >= res_ratio                                         | Pass   | 0.000s |
| Damage capped                                                 | calculate_damage_ratio(10.0) <= 1.0                            | Pass   | 0.000s |
| Damage increases with depth                                   | ratios == sorted(ratios)                                       | Pass   | 0.000s |
| No damage no flood                                            | calculate_damage_ratio(0.0) == 0.0                             | Pass   | 0.000s |
| GaugeHC file must have required structure.                    | 'hazard_curves' in data; not missing                           | Pass   | 0.001s |
| Gaugets files must have required structure.                   | 'flood_simulation' in data; 'readings' in sim (+1 more)        | Pass   | 0.002s |
| PropertyHC file must have required structure.                 | 'property_hazard_curves' in data; 'metadata' in data (+2 more) | Pass   | 0.001s |
| PropertyTS files must have required fields.                   | not missing; not missing_loc (+3 more)                         | Pass   | 0.003s |

| Test Description                                                   | Acceptance Criteria                                                                  | Result | Time   |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|--------|
| Every step in manifest must appear in the static dependency graph. | <code>len(missing) == 0</code>                                                       | Pass   | 0.000s |
| <code>data_lineage.json</code> must exist after any port run.      | <code>"steps" in data; len(data["steps"]) &gt; 0</code>                              | Pass   | 0.001s |
| No step should have stale inputs.                                  | <code>len(stale) == 0</code>                                                         | Pass   | 0.002s |
| Recorded output hashes must match current files on disk.           | —                                                                                    | Pass   | 1.739s |
| Data summary contains errors count                                 | <code>"errors" in dl.get_data_summary()["gauge"]</code>                              | Pass   | 0.002s |
| Data summary contains warnings count                               | <code>"warnings" in dl.get_data_summary()["gauge"]</code>                            | Pass   | 0.002s |
| Data summary loaded flag false when no file                        | <code>summary["gauge"]["loaded"] is False</code>                                     | Pass   | 0.001s |
| Data summary loaded flag true for gauge                            | <code>dl.get_data_summary()["gauge"]["loaded"] is True</code>                        | Pass   | 0.002s |
| Get data summary empty before load                                 | <code>DataLoader(input_dir=tmp_path).get_data_summary() == {}</code>                 | Pass   | 0.000s |
| Get data summary returns dict                                      | <code>isinstance(dl.get_data_summary(), dict)</code>                                 | Pass   | 0.002s |
| Get validation summary empty before load                           | <code>DataLoader(input_dir=tmp_path).get_validation_summary() == ...</code>          | Pass   | 0.000s |
| Get validation summary is copy                                     | <code>s1 is not s2</code>                                                            | Pass   | 0.005s |
| Get validation summary returns dict                                | <code>isinstance(dl.get_validation_summary(), dict)</code>                           | Pass   | 0.001s |
| Is data complete depends only on gauge                             | <code>dl.is_data_complete() is False</code>                                          | Pass   | 0.011s |
| Is data complete false before load                                 | <code>DataLoader(input_dir=tmp_path).is_data_complete() is False</code>              | Pass   | 0.000s |
| Is data complete false when no gauge                               | <code>dl.is_data_complete() is False</code>                                          | Pass   | 0.001s |
| Is data complete true with gauge                                   | <code>dl.is_data_complete() is True</code>                                           | Pass   | 0.026s |
| Validation summary has gauge key                                   | <code>"gauge" in dl.get_validation_summary()</code>                                  | Pass   | 0.005s |
| Init with input dir                                                | <code>dl.input_dir == tmp_path</code>                                                | Pass   | 0.000s |
| Init with string path converts to path                             | <code>isinstance(dl.input_dir, Path); dl.input_dir == tmp_path</code>                | Pass   | 0.000s |
| Loaded data initialised                                            | <code>isinstance(dl.loaded_data, LoadedData)</code>                                  | Pass   | 0.000s |
| Private loaders created                                            | <code>dl._gauge_loader is not None; dl._property_loader is not None (+1 more)</code> | Pass   | 0.000s |

| Test Description                          | Acceptance Criteria                                                                      | Result | Time   |
|-------------------------------------------|------------------------------------------------------------------------------------------|--------|--------|
| Validation results empty at init          | <code>dl.validation.results == {}</code>                                                 | Pass   | 0.000s |
| Counterparty data loaded                  | <code>result.counterparty_data is not None</code>                                        | Pass   | 0.003s |
| Empty dir returns loaded data             | <code>isinstance(result, LoadedData)</code>                                              | Pass   | 0.000s |
| Gauge data count matches records          | <code>result.gauge_data["count"] ==<br/>len(result.gauge_data["item...</code>            | Pass   | 0.002s |
| Gauge data has items key                  | <code>"items" in result.gauge_data</code>                                                | Pass   | 0.002s |
| Gaugehd dir counted                       | <code>result.gaugehd_count == 1</code>                                                   | Pass   | 0.002s |
| Gaugets dir counted                       | <code>result.gaugets_count == 1</code>                                                   | Pass   | 0.003s |
| Gaugets dir multiple files counted        | <code>result.gaugets_count == 3</code>                                                   | Pass   | 0.016s |
| Hazard curves loaded                      | <code>result.hazard_data is not None</code>                                              | Pass   | 0.004s |
| Lookups built after load                  | <code>result.mortgage_lookup is not None;<br/>result.gauge_flood_info is not None</code> | Pass   | 0.008s |
| No gaugehd dir count stays zero           | <code>result.gaugehd_count == 0</code>                                                   | Pass   | 0.006s |
| No gaugets dir count stays zero           | <code>result.gaugets_count == 0</code>                                                   | Pass   | 0.011s |
| No propertyts dir count stays zero        | <code>result.propertyts_count == 0</code>                                                | Pass   | 0.014s |
| Property hazard curves loaded             | <code>result.property_hazard_data is not<br/>None</code>                                 | Pass   | 0.003s |
| Property hazard summary fields accessible | <code>"summary" in<br/>result.property_hazard_data</code>                                | Pass   | 0.003s |
| Propertyts dir counted                    | <code>result.propertyts_count &gt;= 1</code>                                             | Pass   | 0.003s |
| Storm data list format                    | <code>result.storm_data is not None</code>                                               | Pass   | 0.003s |
| Storm data loaded dict format             | <code>result.storm_data is not None</code>                                               | Pass   | 0.003s |
| Storm data unknown format                 | <code>result.storm_data is not None</code>                                               | Pass   | 0.003s |
| Storm data with items key                 | <code>result.storm_data is not None</code>                                               | Pass   | 0.003s |
| With gauge data loaded                    | <code>result.gauge_data is not None</code>                                               | Pass   | 0.002s |
| With mortgage data loaded                 | <code>result.mortgage_data is not None</code>                                            | Pass   | 0.002s |
| With property data loaded                 | <code>result.property_data is not None</code>                                            | Pass   | 0.003s |
| Collect all keys                          | <code>expected_keys == set(result.keys())</code>                                         | Pass   | 0.004s |
| List audit files                          | <code>len(result) == 1; result[0]['name']<br/>== 'report.pdf' (+2 more)</code>           | Pass   | 0.001s |
| List audit files missing dir              | <code>result == []</code>                                                                | Pass   | 0.000s |
| List sensitivity generators               | <code>result == ['hazard']</code>                                                        | Pass   | 0.001s |
| List sensitivity generators no dir        | <code>result == []</code>                                                                | Pass   | 0.000s |
| Load audit log                            | <code>len(result) == 1</code>                                                            | Pass   | 0.001s |
| Load audit log dict fallback              | <code>result == []</code>                                                                | Pass   | 0.001s |
| Load bcbs                                 | <code>'principles' in result</code>                                                      | Pass   | 0.001s |
| Load coverage                             | <code>result == pytest.approx(82.5)</code>                                               | Pass   | 0.001s |
| Load coverage corrupt                     | <code>result is None</code>                                                              | Pass   | 0.001s |
| Load coverage missing                     | <code>result is None</code>                                                              | Pass   | 0.000s |
| Load inventory                            | <code>result['models'][0]['model_id'] ==<br/>'M-TEST'</code>                             | Pass   | 0.001s |

| Test Description                   | Acceptance Criteria                                                                                                | Result | Time    |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------|---------|
| Load inventory bad json            | <code>result == {}</code>                                                                                          | Pass   | 0.001s  |
| Load inventory missing             | <code>result == {}</code>                                                                                          | Pass   | 0.000s  |
| Load junit                         | <code>result['total'] == 100;</code><br><code>result['failed'] == 2 (+4 more)</code>                               | Pass   | 0.002s  |
| Load junit corrupt                 | <code>result['total'] == 0</code>                                                                                  | Pass   | 0.001s  |
| Load junit missing                 | <code>result['total'] == 0</code>                                                                                  | Pass   | 0.000s  |
| Load meetings                      | <code>len(result) == 1</code>                                                                                      | Pass   | 0.001s  |
| Load meetings dict fallback        | <code>result == []</code>                                                                                          | Pass   | 0.001s  |
| Load raci                          | <code>'roles' in result</code>                                                                                     | Pass   | 0.001s  |
| Both empty                         | <code>isinstance(self._page().generate_elements({}), list)</code>                                                  | Pass   | 0.000s  |
| Empty property does not crash      | <code>isinstance(self._page().generate_elements({}), list)</code>                                                  | Pass   | 0.000s  |
| Has data header                    | <code>any("Data" in t for t in texts)</code>                                                                       | Pass   | 0.001s  |
| Has table                          | <code>any(isinstance(e, Table) for e in result)</code>                                                             | Pass   | 0.001s  |
| No sections in property            | <code>isinstance(result, list)</code>                                                                              | Pass   | 0.000s  |
| Returns list                       | <code>isinstance(self._page().generate_elements(make_property()), list)</code><br><code>isinstance(r, list)</code> | Pass   | 0.001s; |
| Returns non empty                  | <code>len(self._page().generate_elements(make_property())) &gt; 0</code>                                           | Pass   | 0.001s  |
| With mortgage data                 | <code>isinstance(result, list) and len(result) &gt; 0</code>                                                       | Pass   | 0.001s  |
| Invalid result fields              | <code>result.is_valid is False;</code> "file not found" in <code>result.errors</code>                              | Pass   | 0.000s  |
| Is named tuple                     | <code>isinstance(result, tuple)</code>                                                                             | Pass   | 0.000s  |
| Tuple unpacking                    | <code>is_valid is True</code>                                                                                      | Pass   | 0.000s  |
| Valid result fields                | <code>result.is_valid is True;</code><br><code>result.warnings == ["minor issue"]</code><br>(+2 more)              | Pass   | 0.000s  |
| Datetime encoded as isoformat      | <code>"2024-06-15" in result</code>                                                                                | Pass   | 0.000s  |
| Nested numpy in dict               | <code>result["a"] == 1;</code> <code>abs(result["b"] - 2.5) &lt; 1e-6</code> (+1 more)                             | Pass   | 0.000s  |
| Numpy float32 encoded              | <code>abs(result - 1.5) &lt; 0.01</code>                                                                           | Pass   | 0.000s  |
| Numpy float64 encoded as float     | <code>abs(result - 3.14) &lt; 1e-6</code>                                                                          | Pass   | 0.000s  |
| Numpy int32 encoded                | <code>result == 7</code>                                                                                           | Pass   | 0.000s  |
| Numpy int64 encoded as int         | <code>result == 99; isinstance(result, int)</code>                                                                 | Pass   | 0.000s  |
| Numpy ndarray encoded as list      | <code>result == [1, 2, 3]</code>                                                                                   | Pass   | 0.000s  |
| Unknown type raises type error     | —                                                                                                                  | Pass   | 0.000s  |
| Line 59-60: datetime -j isoformat. | <code>"2024-06-01" in result</code>                                                                                | Pass   | 0.000s  |
| Line 63-64: np.floating -j float.  | <code>"2.71" in result</code>                                                                                      | Pass   | 0.000s  |

| Test Description                                      | Acceptance Criteria                                                                        | Result | Time   |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------|--------|--------|
| Line 61-62: np.integer -i int.                        | result == "99"                                                                             | Pass   | 0.000s |
| Line 65-66: np.ndarray -i list.                       | result == [10, 20]                                                                         | Pass   | 0.000s |
| Line 67: super().default() -i Type-Error for unknown. | —                                                                                          | Pass   | 0.000s |
| Exponential value                                     | DecayKernel.EXPONENTIAL.value == "exponential"                                             | Pass   | 0.000s |
| Gaussian value                                        | IntensityProfile.GAUSSIAN.value == "gaussian";<br>DecayKernel.GAUSSIAN.value == "gaussian" | Pass   | 0.000s |
| Linear value                                          | DecayKernel.LINEAR.value == "linear"                                                       | Pass   | 0.000s |
| Add decision auto generates id                        | r.status_code == 200;<br>decisions[0]["id"] == "D-001"                                     | Pass   | 0.002s |
| Add decision unknown meeting returns 404              | r.status_code == 404                                                                       | Pass   | 0.002s |
| Delete decision                                       | r.status_code == 200;<br>len(r.get_json()["meeting"]["decisions"]) == 1                    | Pass   | 0.004s |
| Delete decision unknown meeting returns 404           | r.status_code == 404                                                                       | Pass   | 0.001s |
| Second decision increments id                         | ids == ["D-001", "D-002"]                                                                  | Pass   | 0.003s |
| Update decision                                       | r.status_code == 200;<br>r.get_json()["meeting"]["decisions"][0]["description"] == ...     | Pass   | 0.002s |
| Update missing decision returns 404                   | r.status_code == 404                                                                       | Pass   | 0.001s |
| Add decision auto ids                                 | "D-001" in ids; "D-002" in ids                                                             | Pass   | 0.003s |
| Add decision missing meeting returns 404              | r.status_code == 404                                                                       | Pass   | 0.002s |
| Add decision returns 200                              | r.status_code == 200;<br>len(meeting["decisions"]) == 2                                    | Pass   | 0.002s |
| Delete decision missing meeting returns 404           | r.status_code == 404                                                                       | Pass   | 0.002s |
| Delete decision returns 200                           | r.status_code == 200;<br>len(r.get_json()["meeting"]["decisions"]) == 0                    | Pass   | 0.002s |
| Update decision not found returns 404                 | r.status_code == 404                                                                       | Pass   | 0.002s |
| Update decision returns 200                           | r.status_code == 200;<br>decisions[0]["description"] == "Updated decision"                 | Pass   | 0.002s |



| Test Description                                                | Acceptance Criteria                                                                    | Result | Time   |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------|--------|--------|
| Delete is idempotent — filtering out a missing ID is still 200. | <code>r.status_code == 200</code>                                                      | Pass   | 0.001s |
| Delete returns success                                          | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code> | Pass   | 0.003s |
| Deleted ref not in list                                         | <code>ref_id not in ids</code>                                                         | Pass   | 0.004s |
| Deleted doc not in list                                         | <code>doc_id not in ids</code>                                                         | Pass   | 0.004s |
| Not found returns 404                                           | <code>r.status_code == 404; r.status_code == 404</code>                                | Pass   | 0.001s |
| Valid delete returns success                                    | <code>r.status_code == 200; data["status"] == "success"</code>                         | Pass   | 0.004s |
| Delete save failure returns 500                                 | <code>r.status_code == 500; data["status"] == "error"</code>                           | Pass   | 0.030s |
| Gaugehd references gauges                                       | <code>gid in gauge_ids</code>                                                          | Pass   | 0.004s |
| Gaugets reads gauge                                             | <code>gid in gauge_ids</code>                                                          | Pass   | 0.002s |
| Hazard reads gauge and storms                                   | <code>len(hc_curves) &gt; 0; gid in gauge_ids</code>                                   | Pass   | 0.001s |
| Mortgage reads property                                         | <code>mid in prop_ids</code>                                                           | Pass   | 0.001s |
| Propertyhc references properties                                | <code>pid in prop_ids</code>                                                           | Pass   | 0.001s |
| Propertyts uses nearest gauges                                  | <code>ng["gauge_id"] in gauge_ids</code>                                               | Pass   | 0.001s |
| Commercial higher damage than residential                       | <code>com_damage &gt; res_damage</code>                                                | Pass   | 0.000s |
| Floor level reduces damage                                      | <code>damages[0] &gt;= damages[3]</code>                                               | Pass   | 0.000s |
| Large depth all near one                                        | <code>np.all(damages &gt; 0.5)</code>                                                  | Pass   | 0.000s |
| Output clipped to 0 1                                           | <code>np.all((damages &gt;= 0) &amp; (damages &lt;= 1))</code>                         | Pass   | 0.000s |
| Returns array same length                                       | <code>len(damages) == len(properties)</code>                                           | Pass   | 0.000s |
| Zero depths all zero damage                                     | —                                                                                      | Pass   | 0.030s |
| Line 150: commercial factor = 1.2.                              | <code>result[1] &gt; result[0]</code>                                                  | Pass   | 0.000s |
| Lines 159-160: adjusted_depths = max(depths - floor_levels, 0). | <code>result[3] == pytest.approx(0.0)</code>                                           | Pass   | 0.000s |
| Full function returns correct length                            | <code>len(result) == len(properties)</code>                                            | Pass   | 0.000s |
| Line 151: industrial factor = 0.9.                              | <code>result[2] &lt; result[0]</code>                                                  | Pass   | 0.000s |
| Line 165: np.clip(final_damage, 0, 1).                          | <code>np.all(result &lt;= 1.0); np.all(result &gt;= 0.0)</code>                        | Pass   | 0.000s |
| Line 155: .get(str(pt).lower(), 1.0) — 'other' defaults to 1.0. | <code>result[4] == pytest.approx(result[0])</code>                                     | Pass   | 0.000s |
| Zero depths all zero                                            | —                                                                                      | Pass   | 0.000s |
| Deep flood -i damage ratio approaches 1.                        | <code>result &gt; 0.5</code>                                                           | Pass   | 0.001s |
| Positive depth -i positive damage ratio.                        | <code>0.0 &lt; result &lt;= 1.0</code>                                                 | Pass   | 0.001s |
| 0m depth -i 0 damage.                                           | <code>gen..depth_to_damage(0.0) == 0.0</code>                                          | Pass   | 0.001s |

| Test Description                                                | Acceptance Criteria                                                        | Result | Time   |
|-----------------------------------------------------------------|----------------------------------------------------------------------------|--------|--------|
| Line 80: after loop exhausted, gauges_warning i 3 → 'baseline'. | <code>_derive_intensity_category(-1, 2) == "baseline"</code>               | Pass   | 0.000s |
| Negative severe with five warnings returns moderate             | <code>_derive_intensity_category(-1, 5) == "moderate"</code>               | Pass   | 0.000s |
| Lines 78-79: gauges_severe i 0 (doesn't match any threshold),   | <code>_derive_intensity_category(-1, 3) == "moderate"</code>               | Pass   | 0.000s |
| Negative severe zero warnings returns baseline                  | <code>_derive_intensity_category(-1, 0) == "baseline"</code>               | Pass   | 0.000s |
| When severe i= 1, warning count is irrelevant.                  | <code>_derive_intensity_category(1, 10) == "moderate"</code>               | Pass   | 0.000s |
| gauges_severe=0 matches (0, 'baseline') in the severity table.  | <code>_derive_intensity_category(0, 0) == "baseline"</code>                | Pass   | 0.000s |
| Both missing returns na                                         | <code>page._derive_tenor("", "") == "N/A"</code>                           | Pass   | 0.000s |
| Five year tenor                                                 | <code>page._derive_tenor("2024-01-01", "2029-01-01") == "5Y"</code>        | Pass   | 0.000s |
| Invalid date format returns na                                  | <code>page._derive_tenor("not-a-date", "2026-01-01") == "N/A"</code>       | Pass   | 0.000s |
| Missing end returns na                                          | <code>page._derive_tenor("2024-01-01", "") == "N/A"</code>                 | Pass   | 0.000s |
| Missing start returns na                                        | <code>page._derive_tenor("", "2026-01-01") == "N/A"</code>                 | Pass   | 0.000s |
| Valid dates returns years                                       | <code>page._derive_tenor("2024-01-01", "2026-01-01") == "2Y"</code>        | Pass   | 0.000s |
| Empty property info returns valid                               | <code>result in ["Residential", "Buy-to-Let", "Second Home", "H...]</code> | Pass   | 0.000s |
| Multi family returns btl or residential                         | <code>result in ["Buy-to-Let", "Residential"]</code>                       | Pass   | 0.000s |
| Rental history returns btl or residential                       | <code>result in ["Buy-to-Let", "Residential"]</code>                       | Pass   | 0.000s |
| Rental income returns btl or residential                        | <code>result in ["Buy-to-Let", "Residential"]</code>                       | Pass   | 0.000s |
| Vacant property                                                 | <code>result in ["Buy-to-Let", "Second Home", "Residential"]</code>        | Pass   | 0.000s |
| Buy to let returns investment or primary                        | <code>result in ["Investment", "PrimaryResidence"]</code>                  | Pass   | 0.000s |
| Holiday home returns second residence                           | <code>_determine_occupancy_type("Holiday Home") == "SecondResid..."</code> | Pass   | 0.000s |
| Residential returns primary or second                           | <code>result in ["PrimaryResidence", "SecondResidence"]</code>             | Pass   | 0.000s |
| Second home returns second residence                            | <code>_determine_occupancy_type("Second Home") == "SecondReside..."</code> | Pass   | 0.000s |

| Test Description                                                 | Acceptance Criteria                                                          | Result | Time   |
|------------------------------------------------------------------|------------------------------------------------------------------------------|--------|--------|
| Gauge IDs should be derived from location, not random UUIDs.     | —                                                                            | Skip   | 0.000s |
| Haversine known distance                                         | <code>6.0 &lt; distance &lt; 8.0</code>                                      | Pass   | 0.000s |
| Haversine same point                                             | <code>calculate_distance_km(lat, lon, lat, lon) == pytest.appro...</code>    | Pass   | 0.000s |
| Haversine teddington to richmond                                 | <code>2.5 &lt; distance &lt; 4.0</code>                                      | Pass   | 0.000s |
| Diagonal zero                                                    | <code>np.allclose(np.diag(D), 0.0)</code>                                    | Pass   | 0.000s |
| 51.5N 0.0 -j 51.5N 1.0E is approximately 69km.                   | <code>65.0 &lt; d &lt; 75.0</code>                                           | Pass   | 0.000s |
| Positive off diagonal                                            | <code>D[i, j] &gt; 0.0</code>                                                | Pass   | 0.000s |
| Shape                                                            | <code>D.shape == (n, n)</code>                                               | Pass   | 0.000s |
| Symmetric                                                        | <code>np.allclose(D, D.T)</code>                                             | Pass   | 0.000s |
| Thames gauges span ~80km corridor (Reading to Purfleet).         | <code>10.0 &lt; max_d &lt; 120.0</code>                                      | Pass   | 0.000s |
| Thames n gauges                                                  | <code>thames_model.n_gauges == 52</code>                                     | Pass   | 0.000s |
| At zero is max                                                   | <code>RiskAssessor.calculate_distance_risk_factor == 1.0</code>              | Pass   | 0.000s |
| Close band                                                       | <code>RiskAssessor.calculate_distance_risk_factor == 0.8</code>              | Pass   | 0.000s |
| Far band                                                         | <code>RiskAssessor.calculate_distance_risk_factor == 0.2</code>              | Pass   | 0.000s |
| Medium band                                                      | <code>RiskAssessor.calculate_distance_risk_factor == 0.4</code>              | Pass   | 0.000s |
| Monotone decrease                                                | <code>all(factors[i] &gt;= factors[i + 1] for i in range(len(facto...</code> | Pass   | 0.000s |
| Near band                                                        | <code>RiskAssessor.calculate_distance_risk_factor == 0.6</code>              | Pass   | 0.000s |
| Negative returns default                                         | <code>RiskAssessor.calculate_distance_risk_factor == 0.5</code>              | Pass   | 0.000s |
| None returns default                                             | <code>RiskAssessor.calculate_distance_risk_factor == 0.5</code>              | Pass   | 0.000s |
| Very close                                                       | <code>RiskAssessor.calculate_distance_risk_factor == 1.0</code>              | Pass   | 0.000s |
| Very far                                                         | <code>RiskAssessor.calculate_distance_risk_factor == 0.1</code>              | Pass   | 0.000s |
| Close distance label                                             | <code>"Very close to Thames" in str(elements)</code>                         | Pass   | 0.000s |
| Zero distance highest risk                                       | <code>"On Thames River" in str(elements)</code>                              | Pass   | 0.000s |
| STORM-d1a damage for PROP-A uses only that storm's damage_ratio. | <code>prop_a["damage_amount"] == pytest.approx(expected.damage)</code>       | Pass   | 0.004s |

| Test Description                                                         | Acceptance Criteria                                                                             | Result | Time   |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|--------|
| Query STORM-d1a: only PROP-A (depth=0) appears; PROP-B depth=0 excluded. | <code>r.status_code == 200;</code> <code>"PROP-A" in prop_ids (+1 more)</code>                  | Pass   | 0.004s |
| Total damage is different for each storm in the doublet.                 | <code>total2 &gt; total1</code>                                                                 | Pass   | 0.005s |
| STORM-d1b causes more damage to PROP-A (deeper flood).                   | <code>dmg2 &gt; dm1</code>                                                                      | Pass   | 0.005s |
| Query STORM-d1b: both PROP-A and PROP-B have positive depth.             | <code>r.status_code == 200;</code> <code>"PROP-A" in prop_ids (+1 more)</code>                  | Pass   | 0.004s |
| Response storm_id matches the queried storm.                             | <code>data["storm_id"] == "STORM-d1b"</code>                                                    | Pass   | 0.004s |
| Each storm computes its own damage_ratio from its own flood depth.       | <code>"damage_ratio" in event;</code> <code>0.0 &lt;= event["damage_ratio"] &lt;= 1.0</code>    | Pass   | 0.003s |
| Doublet events have distinct storm ids                                   | <code>"STORM-d001a" in event_storm_ids;</code><br><code>"STORM-d001b" in event_storm_ids</code> | Pass   | 0.003s |
| Doublet produces two flood events                                        | <code>result is not None; prop_file.exists()</code><br><code>(+1 more)</code>                   | Pass   | 0.003s |
| <code>floods_at_nearest_gauge == 2</code> for doublet sequence.          | <code>result["summary"]["floods_at_nearest_gauge"] == 2</code>                                  | Pass   | 0.001s |
| File not found returns 404                                               | <code>r.status_code == 404</code>                                                               | Pass   | 0.001s |
| Non pdf served as attachment                                             | <code>r.status_code == 200; "attachment" in r.headers.get("Content-Disposition", "")</code>     | Pass   | 0.001s |
| Path traversal returns 403                                               | <code>r.status_code in (403, 404)</code>                                                        | Pass   | 0.000s |
| Valid pdf served inline                                                  | <code>r.status_code == 200</code>                                                               | Pass   | 0.012s |
| Not found returns 404                                                    | <code>r.status_code == 404; r.status_code == 404</code>                                         | Pass   | 0.001s |
| Valid download returns 200                                               | <code>r.status_code == 200</code>                                                               | Pass   | 0.004s |
| File missing on disk returns 404                                         | <code>r.status_code == 404; "not found" in data["message"].lower()</code>                       | Pass   | 0.028s |
| Insufficient data message                                                | <code>any("Insufficient" in t for t in texts)</code>                                            | Pass   | 0.000s |
| Returns list                                                             | <code>isinstance(result, list); len(result) &gt; 0 (+6 more)</code>                             | Pass   | 0.001s |
| Single entry insufficient                                                | <code>any("Insufficient" in t for t in texts)</code>                                            | Pass   | 0.000s |
| Sufficient data creates content                                          | <code>len(result) &gt; 2</code>                                                                 | Pass   | 0.001s |
| More than 20 positions → only top 20 movers shown.                       | <code>isinstance(result, list)</code>                                                           | Pass   | 0.001s |
| Market move pnl creates table                                            | <code>len(tables) &gt;= 2</code>                                                                | Pass   | 0.000s |
| New trade pnl creates table                                              | <code>len(tables) &gt;= 1</code>                                                                | Pass   | 0.000s |

| Test Description                                                                                          | Acceptance Criteria                                                                                   | Result | Time   |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|--------|
| No positions shows messages                                                                               | <code>any("No new trades" in t for t in texts); any("No market move" in t for t in texts)</code>      | Pass   | 0.001s |
| Returns list                                                                                              | <code>isinstance(result, list); len(result) &gt; 0 (+6 more)</code>                                   | Pass   | 0.000s |
| Position with notional=0 triggers <code>fmt_gbp(0) → em-dash</code> path.                                 | <code>isinstance(result, list); len(result) &gt; 0</code>                                             | Pass   | 0.001s |
| Gauge IDs longer than 12 chars should not crash.                                                          | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Positions from 2 gauges should produce group subheaders.                                                  | <code>len(tables) &gt;= 1</code>                                                                      | Pass   | 0.000s |
| No positions shows message                                                                                | <code>any("No open positions" in t for t in texts)</code>                                             | Pass   | 0.000s |
| Returns list                                                                                              | <code>isinstance(result, list); len(result) &gt; 0 (+6 more)</code>                                   | Pass   | 0.000s |
| Single position                                                                                           | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| With positions creates table                                                                              | <code>any(isinstance(e, Table) for e in result)</code>                                                | Pass   | 0.000s |
| Line 48: <code>daily_pnl=0</code> and <code>running_pnl=0 → fmt_gbp</code> returns <code>em-dash</code> . | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Eod id in filename                                                                                        | <code>"EOD-2026-03-08" in path.name</code>                                                            | Pass   | 0.005s |
| Generate empty snapshot                                                                                   | <code>path.exists()</code>                                                                            | Pass   | 0.004s |
| Generate report returns path                                                                              | <code>path.exists(); path.suffix == ".pdf"</code>                                                     | Pass   | 0.010s |
| Generate report without pnl series                                                                        | <code>path.exists()</code>                                                                            | Pass   | 0.005s |
| Lines 49-50: <code>output_dir=None → uses config.get_eod_dir()</code> .                                   | <code>gen.output_dir == eod_dir</code>                                                                | Pass   | 0.000s |
| Output dir created                                                                                        | <code>new_dir.exists()</code>                                                                         | Pass   | 0.001s |
| Pdf not empty                                                                                             | <code>path.stat().st_size &gt; 1000</code>                                                            | Pass   | 0.009s |
| Date appears in paragraph                                                                                 | <code>any("2026-03-08" in t for t in texts)</code>                                                    | Pass   | 0.000s |
| Empty summary                                                                                             | <code>isinstance(result, list); len(result) &gt; 0</code>                                             | Pass   | 0.000s |
| Has paragraph and table                                                                                   | <code>any(isinstance(e, Paragraph) for e in result); any(isinstance(e, Table) for e in result)</code> | Pass   | 0.000s |
| Negative pnl snapshot                                                                                     | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Returns list                                                                                              | <code>isinstance(result, list); len(result) &gt; 0 (+6 more)</code>                                   | Pass   | 0.068s |
| All unknown pages still creates pdf                                                                       | <code>path.exists()</code>                                                                            | Pass   | 0.001s |
| Audit log single event                                                                                    | <code>len(story) &gt;= 5</code>                                                                       | Pass   | 0.000s |

| Test Description                                                      | Acceptance Criteria                                                                | Result | Time   |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------|--------|--------|
| Bcbs zero max score                                                   | <code>len(story) &gt;= 4</code>                                                    | Pass   | 0.000s |
| Completely empty flood data                                           | <code>path.exists()</code>                                                         | Pass   | 0.010s |
| Completely empty gauge data                                           | <code>path.exists()</code>                                                         | Pass   | 0.012s |
| Completely empty property data                                        | <code>path.exists()</code>                                                         | Pass   | 0.013s |
| Cover no coverage                                                     | <code>len(story) &gt;= 3</code>                                                    | Pass   | 0.000s |
| Gauge data with hazard curve and timeseries                           | <code>path.exists()</code>                                                         | Pass   | 0.015s |
| gauge.json with dict-format flood_gauges is handled.                  | <code>resp.status_code == 200;</code><br><code>data["status"] == "training"</code> | Pass   | 0.002s |
| Gauge with '-' for all numeric fields.                                | <code>result.exists()</code>                                                       | Pass   | 0.013s |
| Verify no crash or timeout with 500 property entries.                 | <code>path.exists()</code>                                                         | Pass   | 0.012s |
| Missing summary key                                                   | <code>path.exists()</code>                                                         | Pass   | 0.010s |
| Model with long name                                                  | <code>len(story) &gt;= 3</code>                                                    | Pass   | 0.000s |
| Model with no name                                                    | <code>len(story) &gt;= 3</code>                                                    | Pass   | 0.000s |
| Mortgage with None for numeric fields.                                | <code>result.exists()</code>                                                       | Pass   | 0.013s |
| Mrc actions with missing fields                                       | <code>len(story) &gt;= 3</code>                                                    | Pass   | 0.000s |
| Multiple models risk ratings                                          | <code>len(story) &gt;= 5</code>                                                    | Pass   | 0.000s |
| Negative value property                                               | <code>path.exists()</code>                                                         | Pass   | 0.017s |
| Negative/nonsensical values should not crash the generator.           | <code>path.exists()</code>                                                         | Pass   | 0.010s |
| None flood gauge header                                               | <code>path.exists()</code>                                                         | Pass   | 0.011s |
| None gauge data key                                                   | <code>path.exists()</code>                                                         | Pass   | 0.005s |
| Output is non empty bytes                                             | <code>isinstance(content, bytes);</code><br><code>len(content) &gt; 0</code>       | Pass   | 0.005s |
| Pdf content is bytes                                                  | <code>isinstance(content, bytes);</code><br><code>len(content) &gt; 0</code>       | Pass   | 0.019s |
| Property data with none header                                        | <code>path.exists()</code>                                                         | Pass   | 0.007s |
| Property with None for value field.                                   | <code>result.exists()</code>                                                       | Pass   | 0.013s |
| Raci no activities no triggers                                        | <code>len(story) &gt;= 3</code>                                                    | Pass   | 0.000s |
| Remediation null due date                                             | <code>len(story) &gt;= 4</code>                                                    | Pass   | 0.000s |
| Report with all pages                                                 | <code>path.exists(); path.stat().st_size &gt; 0</code>                             | Pass   | 0.016s |
| Report with all pages                                                 | <code>path.exists(); path.stat().st_size &gt; 0</code>                             | Pass   | 0.020s |
| Report with all pages                                                 | <code>path.exists(); path.stat().st_size &gt; 0</code>                             | Pass   | 0.017s |
| Single page report                                                    | <code>path.exists()</code>                                                         | Pass   | 0.003s |
| Single page report                                                    | <code>path.exists()</code>                                                         | Pass   | 0.003s |
| If joblib appears between status check and train call, returns ready. | <code>data["status"] == "ready"</code>                                             | Pass   | 0.001s |
| Training status of one gauge does not affect another.                 | <code>data["status"] == "not_trained"</code>                                       | Pass   | 0.001s |



| Test Description                                                          | Acceptance Criteria                                                                                                                   | Result | Time   |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Validation question no short label                                        | <code>len(story) &gt;= 1</code>                                                                                                       | Pass   | 0.000s |
| Zero properties at risk                                                   | <code>path.exists()</code>                                                                                                            | Pass   | 0.010s |
| Zero value property                                                       | <code>path.exists()</code>                                                                                                            | Pass   | 0.015s |
| Spec requirement: stronger storms<br>-¿ broader spatial footprint.        | <code>r.moderate &lt; r.severe &lt; r.extreme</code>                                                                                  | Pass   | 0.000s |
| intensity_factor ¿ 1 still gives at least<br>base range.                  | <code>r ==</code><br><code>pytest.approx(small_model.params.base_range_km)</code>                                                     | Pass   | 0.000s |
| intensity_factor = 1.0 -¿ range =<br>base_range.                          | <code>r ==</code><br><code>pytest.approx(small_model.params.base_range_km)</code>                                                     | Pass   | 0.000s |
| range = base * (1 + rho * (factor -<br>1)).                               | <code>m.effective_range_km(2.0) ==</code><br><code>pytest.approx(56.0)</code>                                                         | Pass   | 0.000s |
| Good elevation over 20m                                                   | <code>isinstance(result, list)</code>                                                                                                 | Pass   | 0.000s |
| Low elevation 5 to 10m                                                    | <code>isinstance(result, list)</code>                                                                                                 | Pass   | 0.000s |
| Moderate elevation 10 to 20m                                              | <code>isinstance(result, list)</code>                                                                                                 | Pass   | 0.000s |
| No ground level                                                           | <code>isinstance(result, list)</code>                                                                                                 | Pass   | 0.000s |
| Very low elevation under 5m                                               | <code>isinstance(result, list)</code>                                                                                                 | Pass   | 0.001s |
| Concerning under 1 year                                                   | <code>isinstance(result, list)</code>                                                                                                 | Pass   | 0.000s |
| Excellent 5 plus years                                                    | <code>isinstance(result, list)</code>                                                                                                 | Pass   | 0.000s |
| Fair 1 to 2 years                                                         | <code>isinstance(result, list)</code>                                                                                                 | Pass   | 0.000s |
| Good 2 to 5 years                                                         | <code>isinstance(result, list)</code>                                                                                                 | Pass   | 0.000s |
| No curves raises value error                                              | —                                                                                                                                     | Pass   | 0.001s |
| Empty storm_id returns 400.                                               | <code>resp.status_code == 400</code>                                                                                                  | Pass   | 0.000s |
| Line 160-161: responses with empty<br>storm_id are skipped.               | <code>result["total_storms"] == 1;</code><br><code>data["storms"][0]["storm_id"] ==</code><br><code>"STORM-valid001"</code>           | Pass   | 0.003s |
| storm_id key missing entirely →<br>resp.get('storm_id', "") == "" → skip. | <code>result["total_storms"] == 1</code>                                                                                              | Pass   | 0.001s |
| Full pipeline generates all outputs                                       | <code>(pipeline_dir / "gaugets").is_dir();</code><br><code>(pipeline_dir / "gaugehd").is_dir()</code><br>(+3 more)                    | Pass   | 0.000s |
| Gauge count propagates                                                    | <code>n == N_GAUGES;</code><br><code>len(list((pipeline_dir /</code><br><code>"gaugets").glob("GAUGE-*.json"))...</code><br>(+2 more) | Pass   | 0.001s |
| Pipeline output file integrity                                            | <code>expected_key in data</code>                                                                                                     | Pass   | 0.002s |
| Property count propagates                                                 | <code>n == N_PROPERTIES;</code><br><code>json.load(f)["summary"]["total_properties"]</code><br><code>== n</code>                      | Pass   | 0.000s |
| Analyze epc rating all bands                                              | <code>expected_word in result</code>                                                                                                  | Pass   | 0.000s |
| Analyze epc rating case insensitive                                       | <code>"Excellent" in</code><br><code>page..analyze_epc_rating("a")</code>                                                             | Pass   | 0.000s |

| Test Description                                               | Acceptance Criteria                                          | Result | Time   |
|----------------------------------------------------------------|--------------------------------------------------------------|--------|--------|
| Analyze epc rating unknown                                     | "Unknown" in result                                          | Pass   | 0.000s |
| Exercises energy intensity branch when area is known.          | len(tables) >= 1                                             | Pass   | 0.000s |
| Full data set exercises all branches without exception.        | isinstance(result, list); len(result) > 2                    | Pass   | 0.000s |
| No energy data fallback message                                | any("No energy performance data" in t for t in texts)        | Pass   | 0.000s |
| Returns list                                                   | isinstance(result, list); len(result) > 0                    | Pass   | 0.003s |
| With building fabric                                           | isinstance(result, list)                                     | Pass   | 0.000s |
| With energy usage                                              | isinstance(result, list); len(result) > 0                    | Pass   | 0.000s |
| With ratings data                                              | isinstance(result, list); len(result) > 0                    | Pass   | 0.000s |
| With ratings includes table                                    | len(tables) >= 1                                             | Pass   | 0.000s |
| Close point pushed away                                        | dist >= PropertyPortfolioGenerator.MIN_RIVER_DISTANCE * ...  | Pass   | 0.000s |
| Far point unchanged                                            | new_lat == pytest.approx(lat); new_lon == pytest.approx(lon) | Pass   | 0.000s |
| Returns tuple of two floats                                    | len(result) == 2; all(isinstance(v, float) for v in result)  | Pass   | 0.000s |
| Lines 353-355: FileNotFoundError → logger.error + sys.exit(1). | 1 in exited_with                                             | Pass   | 0.001s |
| Lines 451-453: FileNotFoundError → logger.error + sys.exit(1). | 1 in exited                                                  | Pass   | 0.001s |
| Lines 510-512: FileNotFoundError → logger.error + sys.exit(1). | 1 in exited                                                  | Pass   | 0.001s |
| Test handling of invalid JSON.                                 | response.status_code == 400                                  | Pass   | 0.000s |
| Lines 451-453: bad JSON → json.JSONDecodeError → sys.exit(1).  | 1 in exited                                                  | Pass   | 0.003s |
| Lines 510-512: bad JSON → json.JSONDecodeError → sys.exit(1).  | 1 in exited                                                  | Pass   | 0.002s |
| Lines 353-355: bad JSON → json.JSONDecodeError → sys.exit(1).  | 1 in exited_with                                             | Pass   | 0.002s |
| Test handling of missing content type.                         | response.status_code in [400, 415, 500]                      | Pass   | 0.000s |
| Kent multiplier                                                | result > 0                                                   | Pass   | 0.000s |

| Test Description                                | Acceptance Criteria                                                                                                           | Result | Time   |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| London multiplier                               | <code>london &gt; 0</code>                                                                                                    | Pass   | 0.000s |
| New property                                    | <code>result &gt; 0</code>                                                                                                    | Pass   | 0.000s |
| Old property                                    | <code>result &gt; 0</code>                                                                                                    | Pass   | 0.000s |
| Returns float                                   | <code>isinstance(result, float)</code>                                                                                        | Pass   | 0.000s |
| Surrey multiplier                               | <code>result &gt; 0</code>                                                                                                    | Pass   | 0.000s |
| With area                                       | <code>result &gt; 0</code>                                                                                                    | Pass   | 0.000s |
| With bedrooms                                   | <code>result &gt; 0</code>                                                                                                    | Pass   | 0.000s |
| Zero property value falls through               | <code>result &gt; 0</code>                                                                                                    | Pass   | 0.000s |
| Severe floods should be rare.                   | <code>prob &lt;= 0.05 + 1e-9</code>                                                                                           | Pass   | 0.000s |
| Normal levels should be common.                 | <code>prob &gt; 0.5</code>                                                                                                    | Pass   | 0.000s |
| Return period should be inverse of probability. | <code>return_period &gt; 1</code>                                                                                             | Pass   | 0.000s |
| Get gauge id missing key                        | <code>page._get_gauge_id({}) == "Unknown Gauge";</code><br><code>page._get_gauge_id(None) == "Unknown Gauge"</code>           | Pass   | 0.000s |
| Malformed gauge data returns error element      | <code>isinstance(elements, list);</code><br><code>len(elements) &gt; 0</code>                                                 | Pass   | 0.000s |
| Compliant verdict                               | <code>verdict_badge._cellvalues[1][0] == 'COMPLIANT'</code>                                                                   | Pass   | 0.000s |
| Non compliant verdict                           | <code>verdict_badge._cellvalues[1][0] in ('PARTIAL...</code>                                                                  | Pass   | 0.000s |
| Run history shown                               | <code>'Run History' in texts</code>                                                                                           | Pass   | 0.001s |
| Extracts ids                                    | <code>extract_property_ids(data) == {"P1", "P2"};</code><br><code>extract_mortgage_ids(data) == {"M1", "M2"} (+1 more)</code> | Pass   | 0.000s |
| Missing gauge id skipped                        | <code>extract_gauge_ids(data) == set()</code>                                                                                 | Pass   | 0.000s |
| None returns empty set                          | <code>extract_property_ids(None) == set();</code><br><code>extract_mortgage_ids(None) == set()</code><br>(+2 more)            | Pass   | 0.000s |
| Dict format                                     | <code>len(result) == 2</code>                                                                                                 | Pass   | 0.000s |
| Empty list                                      | <code>_extract_gauges({"flood_gauges": []}) == []</code>                                                                      | Pass   | 0.000s |
| List format                                     | <code>result == [_G1, _G2]</code>                                                                                             | Pass   | 0.000s |
| Missing key returns empty                       | <code>_extract_gauges({}) == []</code>                                                                                        | Pass   | 0.000s |
| Extracts ids                                    | <code>extract_property_ids(data) == {"P1", "P2"};</code><br><code>extract_mortgage_ids(data) == {"M1", "M2"} (+1 more)</code> | Pass   | 0.000s |
| None returns empty set                          | <code>extract_property_ids(None) == set();</code><br><code>extract_mortgage_ids(None) == set()</code><br>(+2 more)            | Pass   | 0.000s |

| Test Description                              | Acceptance Criteria                                                                                                        | Result | Time   |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Extracts property ids from mortgages          | <code>extract_mortgage_property_ids(data)</code><br><code>== {"P1", "P2"}</code>                                           | Pass   | 0.000s |
| None returns empty set                        | <code>extract_property_ids(None) == set();</code><br><code>extract_mortgage_ids(None) == set()</code><br>(+2 more)         | Pass   | 0.000s |
| Empty items returns empty set                 | <code>extract_property_ids({"items": []})</code><br><code>== set()</code>                                                  | Pass   | 0.000s |
| Extracts ids                                  | <code>extract_property_ids(data) == {"P1", "P2"};</code> <code>extract_mortgage_ids(data) == {"M1", "M2"} (+1 more)</code> | Pass   | 0.000s |
| Missing property id skipped                   | <code>extract_property_ids(data) == set()</code>                                                                           | Pass   | 0.000s |
| None returns empty set                        | <code>extract_property_ids(None) == set();</code><br><code>extract_mortgage_ids(None) == set()</code><br>(+2 more)         | Pass   | 0.000s |
| Construction year extracted                   | <code>gen._extract_property_info(self._full_record, "construction_year")</code>                                            | Pass   | 0.000s |
| Flood risk extracted                          | <code>gen._extract_property_info(self._full_record, "flood_risk")</code>                                                   | Pass   | 0.000s |
| Lat lon extracted                             | <code>abs(info["latitude"] - 51.45) &lt; 0.001;</code> <code>abs(info["longitude"] + 0.30) &lt; 0.001</code>               | Pass   | 0.000s |
| Missing attrs default construction year       | <code>gen._extract_property_info({"PropertyHeader": {"Header": ...}})</code>                                               | Pass   | 0.000s |
| Missing header defaults property id           | <code>gen._extract_property_info({"PropertyHeader": {"Header": ...}})</code>                                               | Pass   | 0.000s |
| Missing location defaults postcode and coords | <code>info["postcode"] == "SW1A 1AA";</code><br><code>abs(info["latitude"] - 51.5) &lt; 0.01</code><br>(+1 more)           | Pass   | 0.000s |
| Missing property header returns defaults      | <code>info["property_id"] == "";</code><br><code>info["property_value"] == 500000</code><br>(+1 more)                      | Pass   | 0.000s |
| Missing risk assessment defaults flood risk   | <code>gen._extract_property_info({"PropertyHeader": {"Header": ...}})</code>                                               | Pass   | 0.000s |
| Missing valuation defaults property value     | <code>gen._extract_property_info(record)["property_value"]</code><br><code>== 5...</code>                                  | Pass   | 0.000s |
| Postcode extracted                            | <code>gen._extract_property_info(self._full_record, "postcode")</code>                                                     | Pass   | 0.000s |
| Property id extracted from header             | <code>gen._extract_property_info(self._full_record, "property_id")</code>                                                  | Pass   | 0.000s |
| Property type extracted                       | <code>gen._extract_property_info(self._full_record, "property_type")</code>                                                | Pass   | 0.000s |
| Property value extracted                      | <code>gen._extract_property_info(self._full_record, "property_value")</code>                                               | Pass   | 0.000s |
| Age band factors within reasonable bounds     | <code>lo &gt; 0; hi &gt;= lo</code>                                                                                        | Pass   | 0.000s |
| Base area ranges all positive                 | <code>lo &gt; 0; hi &gt; lo</code>                                                                                         | Pass   | 0.000s |
| Base price per sqm all positive               | <code>lo &gt; 0; hi &gt; lo</code>                                                                                         | Pass   | 0.000s |

| Test Description                                                | Acceptance Criteria                                                                                                | Result | Time   |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------|--------|
| Condition factors ordered by severity                           | <code>excellent_lo &gt; poor_hi</code>                                                                             | Pass   | 0.000s |
| Epc factors a highest                                           | <code>a_lo &gt; g_hi</code>                                                                                        | Pass   | 0.000s |
| Flood risk factors decrease with risk                           | <code>very_low_lo &gt; very_high_hi</code>                                                                         | Pass   | 0.000s |
| Min max value bounds                                            | <code>MIN_PROPERTY_VALUE == 150.000;</code><br><code>MAX_PROPERTY_VALUE == 5_000.000</code>                        | Pass   | 0.000s |
| Proximity zones have range key                                  | <code>'range' in data</code>                                                                                       | Pass   | 0.000s |
| Rent per sqm all present                                        | <code>set(RENT_PER_SQM.keys()) ==</code><br><code>expected_types</code>                                            | Pass   | 0.000s |
| Rental yield rates all present                                  | <code>set(RENTAL_YIELD_RATES.keys()) ==</code><br><code>expected_types</code>                                      | Pass   | 0.000s |
| A failure matching FAILURE_METADATA renders a structured table. | <code>len(tables) &gt;= 3</code>                                                                                   | Pass   | 0.000s |
| A failure NOT in FAILURE_METADATA falls back to bullet point.   | <code>any('test_trade_gauges_exist_in_gauge_json'</code><br><code>in t ...</code>                                  | Pass   | 0.000s |
| Empty counterparty file uses fallback                           | <code>len(trades) &gt; 0;</code><br><code>all(cid.startswith("CTPY-") for</code><br><code>cid in ctpy_ids)</code>  | Pass   | 0.003s |
| Missing counterparty file uses fallback                         | <code>len(trades) &gt; 0</code>                                                                                    | Pass   | 0.002s |
| Exact lookup found                                              | <code>r.status_code == 200; data["status"]</code><br><code>== "success" (+2 more)</code>                           | Pass   | 0.002s |
| Exact lookup not found                                          | <code>r.status_code == 404; "Field not</code><br><code>found" in r.get_json()["message"]</code>                    | Pass   | 0.002s |
| Exact lookup wrong report                                       | <code>r.status_code == 404</code>                                                                                  | Pass   | 0.002s |
| Exact lookup wrong section                                      | <code>r.status_code == 404</code>                                                                                  | Pass   | 0.002s |
| No params 400                                                   | <code>r.status_code == 400; "Provide" in</code><br><code>r.get_json()["message"]</code>                            | Pass   | 0.002s |
| No registry 404                                                 | <code>r.status_code == 404; "not found" in</code><br><code>data["message"].lower() (+1 more)</code>                | Pass   | 0.000s |
| Search by cdm path                                              | <code>data["count"] &gt;= 1</code>                                                                                 | Pass   | 0.002s |
| Search by source field                                          | <code>data["count"] == 1;</code><br><code>data["results"][0]["field"] ==</code><br><code>"spread" (+1 more)</code> | Pass   | 0.002s |
| Search case insensitive                                         | <code>data["count"] &gt;= 1</code>                                                                                 | Pass   | 0.002s |
| Search finds matches                                            | <code>r.status_code == 200; data["status"]</code><br><code>== "success" (+2 more)</code>                           | Pass   | 0.003s |
| Search no results                                               | <code>r.status_code == 200; data["count"] ==</code><br><code>0 (+1 more)</code>                                    | Pass   | 0.002s |

| Test Description                                                               | Acceptance Criteria                                                                           | Result | Time   |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| Search result structure                                                        | <code>data["count"] &gt;= 1;</code> <code>"report" in</code><br><code>result (+4 more)</code> | Pass   | 0.003s |
| If only report and search are given<br>(not all 3 exact keys), search is used. | <code>data["count"] &gt;= 1</code>                                                            | Pass   | 0.002s |
| None field def returns empty string                                            | <code>result == ""</code>                                                                     | Pass   | 0.000s |
| Numeric field def returns empty<br>string                                      | <code>result == ""</code>                                                                     | Pass   | 0.000s |
| String field def returned as is                                                | <code>result == "literal.value"</code>                                                        | Pass   | 0.000s |
| No type defaults to text                                                       | <code>"SomeField" in result</code>                                                            | Pass   | 0.000s |
| Unknown type returns empty                                                     | <code>result == ""</code>                                                                     | Pass   | 0.000s |
| 5 year filter                                                                  | <code>len(result) &lt; len(flows); len(result)</code><br><code>&gt; 0</code>                  | Pass   | 0.002s |
| Empty flows returns empty                                                      | <code>GaugeHistoricalDaily().filter_by_years([],</code><br><code>years=10) == []</code>       | Pass   | 0.000s |
| Empty list returns empty                                                       | <code>result == []</code>                                                                     | Pass   | 0.000s |
| Filter 1 year keeps only recent                                                | <code>"2010-01-01" not in [f["date"] for f</code><br><code>in result]</code>                  | Pass   | 0.000s |
| Filter keeps recent                                                            | <code>"2000-06-01" not in dates;</code><br><code>"2020-06-01" in dates</code>                 | Pass   | 0.000s |
| Large years returns all                                                        | <code>len(result) == len(flows)</code>                                                        | Pass   | 0.001s |
| None years returns all                                                         | <code>len(gen.filter_by_years(flows,</code><br><code>years=None)) == 2</code>                 | Pass   | 0.000s |
| None years returns all                                                         | <code>len(result) == len(flows)</code>                                                        | Pass   | 0.000s |
| All dates within range kept                                                    | <code>len(gen.filter_by_years(flows,</code><br><code>years=5)) == 3</code>                    | Pass   | 0.000s |
| Zero years returns only most recent<br>year                                    | <code>"2020-01-01" in [f["date"] for f in</code><br><code>result]</code>                      | Pass   | 0.000s |
| Empty property does not crash                                                  | <code>isinstance(self._page().generate_elements()</code><br><code>list)</code>                | Pass   | 0.000s |
| Has financial header                                                           | <code>any("Financial" in t for t in texts)</code>                                             | Pass   | 0.000s |
| Has table                                                                      | <code>any(isinstance(e, Table) for e in</code><br><code>result)</code>                        | Pass   | 0.000s |
| No area skips value per sqm                                                    | <code>isinstance(result, list)</code>                                                         | Pass   | 0.000s |
| No property value                                                              | <code>isinstance(result, list)</code>                                                         | Pass   | 0.000s |
| No valuation data                                                              | <code>isinstance(result, list)</code>                                                         | Pass   | 0.000s |
| Other valuation field with value in<br>name                                    | <code>isinstance(result, list)</code>                                                         | Pass   | 0.000s |
| Other valuation field without value<br>in name                                 | <code>isinstance(result, list)</code>                                                         | Pass   | 0.000s |
| Returns list                                                                   | <code>isinstance(self._page().generate_elements()</code><br><code>list)</code>                | Pass   | 0.000s |



| Test Description                                | Acceptance Criteria                                                                                           | Result | Time   |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------|--------|
| Returns non empty list                          | <code>len(self._page().generate_elements(_make_property())) &gt; 0</code>                                     | Pass   | 0.000s |
| Value per sqm shown when area present           | <code>isinstance(result, list)</code>                                                                         | Pass   | 0.000s |
| Atl basin                                       | <code>"EVT-001" in ids; "EVT-002" in ids</code><br>(+1 more)                                                  | Pass   | 0.001s |
| Case insensitive                                | <code>result is not None; len(result) &gt;= 1</code>                                                          | Pass   | 0.001s |
| Unknown basin returns empty                     | <code>loader.find.by_basin("XYZ") == []</code>                                                                | Pass   | 0.001s |
| Wpac basin                                      | <code>len(result) == 1;</code><br><code>loader.get_entity_id(result[0]) == "EVT-003"</code>                   | Pass   | 0.001s |
| All category 4                                  | <code>all(s["TCEvent"]["Header"]["Category"] == 4 ...)</code>                                                 | Pass   | 0.001s |
| Category 3                                      | <code>"EVT-003" in ids</code>                                                                                 | Pass   | 0.001s |
| Category 5                                      | <code>len(result) == 1;</code><br><code>loader.get_entity_id(result[0]) == "EVT-002"</code>                   | Pass   | 0.001s |
| No matches returns empty                        | <code>loader.find.by_category(99) == []</code>                                                                | Pass   | 0.001s |
| Find by county case insensitive                 | <code>len(results) == 1</code>                                                                                | Pass   | 0.001s |
| Find by county empty portfolio                  | <code>results == []</code>                                                                                    | Pass   | 0.001s |
| Find by county exact match                      | <code>len(results) == 1;</code><br><code>results[0]["PropertyHeader"]["Header"]["PropertyID"] == "..."</code> | Pass   | 0.001s |
| Property with no County field should not crash. | <code>results == []</code>                                                                                    | Pass   | 0.001s |
| Find by county no match                         | <code>results == []</code>                                                                                    | Pass   | 0.001s |
| Find by county partial match                    | <code>len(results) == 2;</code> <code>ids == {"PROP-001", "PROP-003"}</code>                                  | Pass   | 0.001s |
| Case insensitive                                | <code>result is not None; len(result) &gt;= 1</code>                                                          | Pass   | 0.001s |
| Exact match                                     | <code>result is not None;</code><br><code>loader.get_entity_id(result) == "EVT-001"</code>                    | Pass   | 0.001s |
| No match returns none                           | <code>loader.find.by_name("Nonsense Storm")</code><br><code>is None</code>                                    | Pass   | 0.001s |
| Partial match                                   | <code>result is not None;</code><br><code>loader.get_entity_id(result) == "EVT-003"</code>                    | Pass   | 0.001s |
| Missing season returns empty                    | <code>loader.find.by_season(1900) == []</code>                                                                | Pass   | 0.001s |

| Test Description                                                           | Acceptance Criteria                                                                                         | Result | Time   |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|--------|
| Season 2017                                                                | <code>len(result) == 1;</code><br><code>loader.get_entity_id(result[0])</code><br><code>== "EVT-001"</code> | Pass   | 0.001s |
| Season with multiple                                                       | <code>len(result) == 1</code>                                                                               | Pass   | 0.001s |
| Dict with 'flood_gauges' list → finds first gauge correctly.               | <code>result["FloodGauge"]["Header"]["GaugeName"]</code><br><code>== "Gauge One"</code>                     | Pass   | 0.000s |
| Dict with 'flood_gauges' list → finds correct gauge by ID.                 | <code>result["FloodGauge"]["Header"]["GaugeID"]</code><br><code>== "GAUGE-002"</code>                       | Pass   | 0.000s |
| Plain list of gauge dicts → finds the correct one.                         | <code>result["FloodGauge"]["Header"]["GaugeID"]</code><br><code>== "GAUGE-B"</code>                         | Pass   | 0.000s |
| Dict with 'FloodGauge' key (single object) → returns the dict itself.      | <code>result is sample_gauge_data</code>                                                                    | Pass   | 0.000s |
| Flood gauges key found                                                     | <code>result == data["flood_gauges"][0]</code>                                                              | Pass   | 0.000s |
| Dict with neither 'flood_gauges' nor 'FloodGauge' key → ValueError raised. | —                                                                                                           | Pass   | 0.000s |
| Invalid dict structure raises                                              | —                                                                                                           | Pass   | 0.000s |
| Invalid type raises                                                        | —                                                                                                           | Pass   | 0.000s |
| Passing a string instead of dict/list → ValueError raised.                 | —                                                                                                           | Pass   | 0.000s |
| List input found                                                           | <code>result == gauges[0]</code>                                                                            | Pass   | 0.000s |
| Malformed gauge entry skipped                                              | <code>result == data["flood_gauges"][1]</code>                                                              | Pass   | 0.000s |
| Gauge ID not in plain list → ValueError raised.                            | —                                                                                                           | Pass   | 0.000s |
| Gauge ID not present in data → ValueError raised.                          | —                                                                                                           | Pass   | 0.000s |
| Not found raises value error                                               | —                                                                                                           | Pass   | 0.000s |
| Single gauge dict returns self                                             | <code>result is data</code>                                                                                 | Pass   | 0.000s |
| Line 361: dict with neither 'flood_gauges' nor 'FloodGauge' → ValueError.  | —                                                                                                           | Pass   | 0.000s |
| Flood gauges key first gauge                                               | <code>result["FloodGauge"]["Header"]["GaugeID"]</code><br><code>== "G-A"</code>                             | Pass   | 0.000s |
| Flood gauges key last gauge                                                | <code>result["FloodGauge"]["Header"]["GaugeID"]</code><br><code>== "G-C"</code>                             | Pass   | 0.000s |
| Line 375: gauge_id not present in any entry → ValueError.                  | —                                                                                                           | Pass   | 0.000s |
| Integer input raises                                                       | —                                                                                                           | Pass   | 0.000s |
| Line 363: list input → list used directly as gauge collection.             | <code>result["FloodGauge"]["Header"]["GaugeID"]</code><br><code>== "G-LIST"</code>                          | Pass   | 0.000s |

| Test Description                                        | Acceptance Criteria                                    | Result | Time   |
|---------------------------------------------------------|--------------------------------------------------------|--------|--------|
| Malformed entry missing flood gauge key skipped         | result["FloodGauge"]["Header"]["GaugeID"] == "G-REAL"  | Pass   | 0.000s |
| Malformed entry with none skipped                       | result["FloodGauge"]["Header"]["GaugeID"] == "G-OK"    | Pass   | 0.000s |
| Single flood gauge dict returns itself                  | result is data                                         | Pass   | 0.000s |
| String input raises                                     | —                                                      | Pass   | 0.000s |
| Finds existing meeting                                  | meeting is not None; meeting["id"] == "M1" (+1 more)   | Pass   | 0.028s |
| Returns error for missing meeting                       | meeting is None; err is not None                       | Pass   | 0.027s |
| Dict with mortgages found                               | result["amount"] == 100                                | Pass   | 0.000s |
| Dict with mortgages found                               | result["amount"] == 100                                | Pass   | 0.000s |
| Dict with mortgages not found returns none              | result is None                                         | Pass   | 0.000s |
| Dict with mortgages not found returns none              | result is None                                         | Pass   | 0.000s |
| Dict without mortgages key returns data itself          | result is not None                                     | Pass   | 0.000s |
| Empty mortgages list returns none                       | result is None                                         | Pass   | 0.000s |
| Lines 406-407: non-dict, non-list -¿ return data as-is. | result == "some_string"                                | Pass   | 0.000s |
| Invalid type returns data as is                         | result == "some_string"                                | Pass   | 0.000s |
| List input found                                        | result == data[0]; result["amount"] == 200             | Pass   | 0.000s |
| List input found                                        | result["amount"] == 200                                | Pass   | 0.000s |
| List input not found returns none                       | result is None                                         | Pass   | 0.000s |
| Dict without list key wraps itself                      | result is data                                         | Pass   | 0.000s |
| Dict without properties wraps in list                   | result == data                                         | Pass   | 0.000s |
| Empty properties list raises                            | —                                                      | Pass   | 0.000s |
| Invalid type raises                                     | —                                                      | Pass   | 0.000s |
| Invalid type raises value error                         | —                                                      | Pass   | 0.000s |
| List input                                              | result == data[0]                                      | Pass   | 0.000s |
| List input found                                        | result == data[0]; result["amount"] == 200             | Pass   | 0.000s |
| List not found raises                                   | —                                                      | Pass   | 0.000s |
| None raises value error                                 | —                                                      | Pass   | 0.000s |
| Not found raises                                        | —                                                      | Pass   | 0.000s |
| Not found raises value error                            | —                                                      | Pass   | 0.000s |
| Portfolio key                                           | result == data["portfolio"][0]                         | Pass   | 0.000s |
| Portfolio key found                                     | result["PropertyHeader"]["Header"]["Property"] == "PA" | Pass   | 0.000s |
| Properties key                                          | result == data["properties"][0]                        | Pass   | 0.000s |

| Test Description                                                                       | Acceptance Criteria                                                            | Result | Time   |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------|--------|
| Properties key found                                                                   | <code>result["PropertyHeader"]["Header"]["PropertyKey"] == "P-002"</code>      | Pass   | 0.000s |
| Train gauge creates joblib file                                                        | <code>(model_dir / "GAUGE-001.joblib").exists()</code>                         | Pass   | 0.081s |
| Train gauge has metrics                                                                | <code>"metrics" in result; "auc_roc" in result["metrics"] (+1 more)</code>     | Pass   | 0.081s |
| Train gauge returns dict                                                               | <code>isinstance(result, dict)</code>                                          | Pass   | 1.141s |
| Train gauge single class skipped                                                       | <code>result["status"] == "skipped"; result["reason"] == "single_class"</code> | Pass   | 0.002s |
| Train gauge status trained                                                             | <code>result["status"] == "trained"</code>                                     | Pass   | 0.080s |
| Flood above trigger                                                                    | <code>depth &gt; 0.0</code>                                                    | Pass   | 0.000s |
| Flood attenuates with distance                                                         | <code>depth_close &gt; depth_far</code>                                        | Pass   | 0.000s |
| High elevation no flood                                                                | <code>depth == 0.0</code>                                                      | Pass   | 0.000s |
| No flood below trigger                                                                 | <code>depth == 0.0</code>                                                      | Pass   | 0.000s |
| Calculate time above warning level.                                                    | <code>duration &gt; 0</code>                                                   | Pass   | 0.000s |
| Duration should be in reasonable range.                                                | <code>5 &lt; duration &lt; 30</code>                                           | Pass   | 0.000s |
| No time above threshold for normal readings.                                           | <code>duration == 0</code>                                                     | Pass   | 0.000s |
| Available gauges                                                                       | <code>"GAUGE-001" in gauges</code>                                             | Pass   | 0.000s |
| Available gauges empty dir                                                             | <code>predictor.available_gauges() == []</code>                                | Pass   | 0.000s |
| Lines 188-192: <code>_load_summary()</code> loads <code>training_summary.json</code> . | <code>'num_gauges' in summary; 'num_trained' in summary</code>                 | Pass   | 0.002s |
| <code>_load_summary</code> caches on second call.                                      | <code>s1 is s2</code>                                                          | Pass   | 0.001s |
| Predict caches model                                                                   | <code>"GAUGE-001" in predictor._models</code>                                  | Pass   | 0.008s |
| Predict high level higher probability                                                  | <code>p_high &gt; p_low</code>                                                 | Pass   | 0.010s |
| Predict in range                                                                       | <code>0.0 &lt;= prob &lt;= 1.0</code>                                          | Pass   | 0.011s |
| Predict missing model raises                                                           | —                                                                              | Pass   | 0.000s |
| Predict returns float                                                                  | <code>isinstance(prob, float)</code>                                           | Pass   | 0.009s |
| Predict series all in range                                                            | <code>all(0.0 &lt;= p &lt;= 1.0 for p in result)</code>                        | Pass   | 0.008s |
| Predict series default derivatives                                                     | <code>len(result) == 2</code>                                                  | Pass   | 0.008s |
| Predict series returns list                                                            | <code>isinstance(result, list); len(result) == 4</code>                        | Pass   | 0.010s |
| Predict series with derivatives                                                        | <code>len(result) == 3</code>                                                  | Pass   | 0.009s |
| <code>predict_surface</code> returns a 2D probability grid (hours × levels).           | <code>'water_levels' in result; 'hours' in result (+3 more)</code>             | Pass   | 0.009s |
| At high threshold                                                                      | <code>RiskAssessor.assess_flood_risk_level(1.0) == "High"</code>               | Pass   | 0.000s |
| At low threshold                                                                       | <code>RiskAssessor.assess_flood_risk_level(0.1) == "Low"</code>                | Pass   | 0.000s |

| Test Description                                                      | Acceptance Criteria                                                                                                                      | Result | Time   |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| At medium threshold                                                   | <code>RiskAssessor.assess_flood_risk_level(0.5)</code>                                                                                   | Pass   | 0.000s |
|                                                                       | <code>== "Medium"</code>                                                                                                                 |        |        |
| Below low threshold                                                   | <code>RiskAssessor.assess_flood_risk_level(0.05)</code>                                                                                  | Pass   | 0.000s |
|                                                                       | <code>== "Low"</code>                                                                                                                    |        |        |
| High range                                                            | <code>RiskAssessor.assess_flood_risk_level(0.75)</code>                                                                                  | Pass   | 0.000s |
|                                                                       | <code>== "High"</code>                                                                                                                   |        |        |
| Medium range                                                          | <code>RiskAssessor.assess_flood_risk_level(0.3)</code>                                                                                   | Pass   | 0.000s |
|                                                                       | <code>== "Medium"</code>                                                                                                                 |        |        |
| Monotone with depth                                                   | <code>levels == order</code>                                                                                                             | Pass   | 0.000s |
| Negative returns unknown                                              | <code>RiskAssessor.assess_flood_risk_level(-1.0)</code>                                                                                  | Pass   | 0.000s |
|                                                                       | <code>== "Unknown"</code>                                                                                                                |        |        |
| None returns unknown                                                  | <code>RiskAssessor.assess_flood_risk_level(None)</code>                                                                                  | Pass   | 0.000s |
|                                                                       | <code>== "Unknown";</code>                                                                                                               |        |        |
|                                                                       | <code>RiskAssessor.assess_ltv_risk_level(None)</code>                                                                                    |        |        |
|                                                                       | <code>== "Unknown"</code>                                                                                                                |        |        |
| Very high                                                             | <code>RiskAssessor.assess_flood_risk_level(2.5)</code>                                                                                   | Pass   | 0.000s |
|                                                                       | <code>== "Very High"</code>                                                                                                              |        |        |
| Zero is very low                                                      | <code>RiskAssessor.assess_flood_risk_level(0.0)</code>                                                                                   | Pass   | 0.000s |
|                                                                       | <code>== "Very Low"</code>                                                                                                               |        |        |
| Lines 59-74: model builds without error given valid inputs.           | <code>model.properties is not None;</code><br><code>model.flood_event == {"GAUGE-000": 1.5}</code>                                       | Pass   | 8.156s |
| Line 72-74: correlation_matrix is set after <code>__init__</code> .   | <code>model.correlation_matrix is not None</code>                                                                                        | Pass   | 0.001s |
| Lines 62-63: custom correlation_distance and base_correlation stored. | <code>model.correlation_distance == 500;</code><br><code>model.base_correlation == 0.6</code>                                            | Pass   | 0.001s |
| Lines 86-88: missing elevation → warning + default 20.0.              | <code>any("elevation" in str(warning.message).lower() for warni...;</code><br><code>(model.properties["elevation"] == 20.0).all()</code> | Pass   | 0.001s |
| Lines 94-95: missing floor_level_metres → default 0.0.                | <code>"floor_level_metres" in model.properties.columns;</code><br><code>(model.properties["floor_level_metres"] == 0.0).all()</code>     | Pass   | 0.001s |
| Line 81: missing 'geometry' column raises ValueError.                 | —                                                                                                                                        | Pass   | 0.000s |
| Lines 83-84: property_id auto-assigned when absent.                   | <code>"property_id" in model.properties.columns;</code><br><code>model.properties["property_id"].iloc[0].startswith("PROP-")</code>      | Pass   | 0.001s |

| Test Description                                                                      | Acceptance Criteria                                                                                                                  | Result | Time   |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Lines 90-92: missing property_type<br>→ warning + default 'residential'.              | any("property_type" in<br>str(warning.message).lower()<br>for w...;<br>(model.properties["property_type"] ==<br>"residential").all() | Pass   | 0.001s |
| Line 81: missing 'value' column<br>raises ValueError.                                 | —                                                                                                                                    | Pass   | 0.000s |
| Line 59: properties is a copy, origi-<br>nal not mutated.                             | set(props.columns) == original.cols                                                                                                  | Pass   | 0.001s |
| Line 71: kdtree built when<br>gauge_data provided.                                    | model.gauge_data is not None                                                                                                         | Pass   | 0.001s |
| Lines 131-139: ana-<br>lyze_risk_clusters() returns<br>DataFrame, sets risk_clusters. | isinstance(result, pd.DataFrame);<br>model.risk_clusters is not None                                                                 | Pass   | 0.053s |
| Lines 114-120: ana-<br>lyze_spatial_concentration() returns<br>DataFrame.             | isinstance(result, pd.DataFrame)                                                                                                     | Pass   | 0.002s |
| Lines 122-129: calcu-<br>late_advanced_metrics() returns<br>dict.                     | isinstance(result, dict)                                                                                                             | Pass   | 0.002s |
| Line 103: calculate_direct_impacts()<br>returns ndarray.                              | isinstance(impacts,<br>np.ndarray); len(impacts) ==<br>len(model.properties)                                                         | Pass   | 0.000s |
| Line 99: calculate_flood_depths() re-<br>turns ndarray.                               | isinstance(depths, np.ndarray);<br>len(depths) == len(model.properties)                                                              | Pass   | 0.000s |
| Line 151: _haversine_distance() del-<br>egates to haversine_distance.                 | isinstance(d, float); d > 0                                                                                                          | Pass   | 0.000s |
| Lines 107-112: simu-<br>late_portfolio_impact() returns<br>dict.                      | isinstance(result, dict)                                                                                                             | Pass   | 0.004s |
| Default params produces data                                                          | len(result) > 0; "readings" in<br>result[0]                                                                                          | Pass   | 0.001s |
| String start time is parsed                                                           | len(result) == 50                                                                                                                    | Pass   | 0.000s |
| Unknown zone                                                                          | "Unknown" in r or "Zone X" in r                                                                                                      | Pass   | 0.000s |
| Zone 1 low risk                                                                       | "Low" in r                                                                                                                           | Pass   | 0.000s |
| Zone 2 medium risk                                                                    | "Medium" in r                                                                                                                        | Pass   | 0.000s |
| Zone 3a high risk                                                                     | "High" in r                                                                                                                          | Pass   | 0.000s |
| Zone 3b floodplain                                                                    | "floodplain" in r.lower() or<br>"Functional" in r                                                                                    | Pass   | 0.000s |
| Zone in page output                                                                   | isinstance(result, list)                                                                                                             | Pass   | 0.002s |



| Test Description                                            | Acceptance Criteria                                                                   | Result | Time   |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|--------|
| Large value                                                 | '1,000,000' in result;<br>result.startswith('£')                                      | Pass   | 0.000s |
| Negative                                                    | '5,000' in result                                                                     | Pass   | 0.000s |
| Non numeric fallback                                        | fmt_gbp('N/A') == 'N/A'                                                               | Pass   | 0.000s |
| None fallback                                               | result == 'None'                                                                      | Pass   | 0.000s |
| Positive float                                              | fmt_gbp(1234.56) == '£1,235'                                                          | Pass   | 0.021s |
| Zero                                                        | fmt_gbp(0) == '£0'                                                                    | Pass   | 0.000s |
| line 286: f'{value:.6f}'.                                   | result == '51.500000'                                                                 | Pass   | 0.000s |
| Negative coordinate                                         | result == '-0.100000'                                                                 | Pass   | 0.000s |
| None delegates to format value                              | page._format.coordinate(None) == 'Not specified'                                      | Pass   | 0.000s |
| None currency returns not specified                         | result == 'Not specified'                                                             | Pass   | 0.000s |
| String currency delegates                                   | result == "N/A"                                                                       | Pass   | 0.000s |
| Abbreviation replacements                                   | 'GPS' in result                                                                       | Pass   | 0.000s |
| Camel case split                                            | 'Gauge' in result                                                                     | Pass   | 0.000s |
| Non numeric delegates                                       | page._format.frequency(None) == 'Not specified'                                       | Pass   | 0.000s |
| Positive returns times                                      | page._format.frequency(3) == '3 times'                                                | Pass   | 0.000s |
| line 294: 'Never'.                                          | page._format.frequency(0) == 'Never'                                                  | Pass   | 0.000s |
| Integer value                                               | result == '5.000 m'                                                                   | Pass   | 0.000s |
| Non numeric delegates to format value                       | result == 'Not specified'                                                             | Pass   | 0.000s |
| Numeric value with custom unit                              | result == '1.234 km'                                                                  | Pass   | 0.000s |
| line 280: f'{value:.3f} {unit}'.                            | result == '3.500 m'                                                                   | Pass   | 0.000s |
| None percentage returns not specified                       | result == 'Not specified'                                                             | Pass   | 0.000s |
| String percentage delegates                                 | result == "N/A"                                                                       | Pass   | 0.000s |
| Blank/whitespace-only string (line 272) -i 'Not specified'. | page._format.value('') == 'Not specified'; page._format.value(' ') == 'Not specified' | Pass   | 0.000s |
| False returns no                                            | page._format.value(False) == 'No'                                                     | Pass   | 0.000s |
| float.is_integer() -i str(int(value)).                      | result == '3'                                                                         | Pass   | 0.000s |
| float with fractional part -i 2 decimal places.             | result == '3.14'                                                                      | Pass   | 0.000s |
| int branch (line 270): str(value).                          | result == '42'                                                                        | Pass   | 0.000s |
| None returns not specified                                  | page._format.value(None) == 'Not specified'                                           | Pass   | 0.000s |
| Normal string returned                                      | page._format.value('hello') == 'hello'                                                | Pass   | 0.000s |
| True returns yes                                            | page._format.value(True) == 'Yes'                                                     | Pass   | 0.000s |

| Test Description                                                     | Acceptance Criteria                                                                                                                              | Result | Time   |
|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Empty string returns not specified                                   | <code>page._format_value('') == 'Not specified'</code>                                                                                           | Pass   | 0.000s |
| Whitespace string returns not specified                              | <code>page._format_value(' ') == 'Not specified'</code>                                                                                          | Pass   | 0.000s |
| None returns not specified                                           | <code>page._format_value(None) == 'Not specified'</code>                                                                                         | Pass   | 0.000s |
| Pad foundation                                                       | <code>isinstance(result, list)</code>                                                                                                            | Pass   | 0.000s |
| Pile foundation                                                      | <code>isinstance(result, list)</code>                                                                                                            | Pass   | 0.000s |
| Raft foundation                                                      | <code>isinstance(result, list)</code>                                                                                                            | Pass   | 0.000s |
| Strip foundation                                                     | <code>isinstance(result, list)</code>                                                                                                            | Pass   | 0.000s |
| Unknown foundation                                                   | <code>isinstance(result, list)</code>                                                                                                            | Pass   | 0.000s |
| Four exceedances high                                                | <code>"High frequency" in str(elements)</code>                                                                                                   | Pass   | 0.000s |
| One exceedance                                                       | <code>"One severe level exceedance" in str(elements)</code>                                                                                      | Pass   | 0.000s |
| Two exceedances moderate                                             | <code>"Moderate frequency" in str(elements)</code>                                                                                               | Pass   | 0.000s |
| Zero exceedances                                                     | <code>"No severe level exceedances recorded" in str(elements)</code>                                                                             | Pass   | 0.000s |
| Distance matrix built                                                | <code>thames_model.dist_matrix.shape == (52, 52)</code>                                                                                          | Pass   | 0.000s |
| From dict                                                            | <code>p.base_range_km == 50.0; p.nugget == 0.10 (+1 more)</code>                                                                                 | Pass   | 0.002s |
| Lines 323-325: construct model from a gauge.json file.               | <code>model.n_gauges == 3; model.dist_matrix.shape == (3, 3)</code>                                                                              | Pass   | 0.001s |
| Loads 52 gauges                                                      | <code>thames_model.n_gauges == 52</code>                                                                                                         | Pass   | 0.000s |
| Full js contains iife                                                | <code>'(function()'</code> in <code>full_governance_js</code> or <code>'(function ()'</code> in ...                                              | Pass   | 0.000s |
| The Leaflet control that adds the Regulatory shield must be present. | <code>'GovernanceControl'</code> in <code>full_governance_js;</code> <code>'Regulatory Compliance'</code> in <code>full_governance_js</code>     | Pass   | 0.000s |
| Full js contains render audit reports                                | <code>'renderAuditReports'</code> in <code>full_governance_js</code>                                                                             | Pass   | 0.000s |
| Full js contains script tags                                         | <code>'&lt;script&gt;'</code> in <code>full_governance_js;</code> <code>'&lt;/script&gt;'</code> in <code>full_governance_js</code>              | Pass   | 0.000s |
| Full js contains show mg panel                                       | <code>'showMgPanel'</code> in <code>full_governance_js</code>                                                                                    | Pass   | 0.000s |
| Terminal popup added in last session must be present.                | <code>'_openTerminalPopup'</code> in <code>full_governance_js;</code> <code>'_auditTerminalRefreshNow'</code> in <code>full_governance_js</code> | Pass   | 0.000s |
| Full js is str                                                       | <code>isinstance(full_governance_js, str)</code>                                                                                                 | Pass   | 0.000s |

| Test Description                                          | Acceptance Criteria                                                                 | Result | Time   |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------|--------|--------|
| Full js non empty                                         | <code>len(full_governance_js) &gt; 10.000</code>                                    | Pass   | 0.000s |
| Run node -check on the compiled governance script.        | <code>result.returncode == 0</code>                                                 | Pass   | 1.571s |
| Base level calculated                                     | <code>abs(gc.base_level - 1.4) &lt; 0.01</code>                                     | Pass   | 0.000s |
| Creates gauge config                                      | <code>isinstance(gc, GaugeConfig)</code>                                            | Pass   | 0.000s |
| Defaults when no flood stages                             | <code>gc.flood_alert == 4.0</code>                                                  | Pass   | 0.000s |
| Handles case where FloodGauge key is missing (flat data). | <code>gc.flood_alert == 3.5</code>                                                  | Pass   | 0.000s |
| Flood alert set                                           | <code>gc.flood_alert == 4.0</code>                                                  | Pass   | 0.000s |
| Gauge id set                                              | <code>gc.gauge_id == "GAUGE-099"</code>                                             | Pass   | 0.000s |
| Historical high set                                       | <code>gc.historical_high == 6.0</code>                                              | Pass   | 0.000s |
| Lines 32-34: loader.list_all() raises → 500 response.     | <code>response.status_code == 500;</code><br><code>data['status'] == 'error'</code> | Pass   | 0.000s |
| Line 19: OPTIONS on /api/v1/gauges returns 200.           | <code>response.status_code == 200;</code><br><code>data['status'] == 'ok'</code>    | Pass   | 0.000s |
| Empty historical daily shows fallback                     | <code>any(t for t in texts)</code>                                                  | Pass   | 0.000s |
| No timeseries data shows fallback                         | <code>any("No historical" in t or "No" in t for t in texts)</code>                  | Pass   | 0.000s |
| Returns list                                              | <code>isinstance(result, list); len(result) &gt; 0</code>                           | Pass   | 0.000s |
| With observations includes content                        | <code>len(result) &gt;= 2</code>                                                    | Skip   | 0.096s |
| With observations returns elements                        | <code>len(result) &gt; 2</code>                                                     | Skip   | 0.109s |
| Certified conclusion                                      | <code>"certification standards" in str(result)</code>                               | Pass   | 0.001s |
| Empty data does not crash                                 | <code>isinstance(result, list)</code>                                               | Pass   | 0.000s |
| Freq high significance                                    | <code>"Regular severe events" in str(result)</code>                                 | Pass   | 0.001s |
| Freq low significance                                     | <code>"Low frequency events" in str(result)</code>                                  | Pass   | 0.001s |
| Freq zero no exceedances conclusion                       | <code>"No historical severe flood events" in str(result)</code>                     | Pass   | 0.001s |
| Freq zero significance                                    | <code>"No severe exceedances" in str(result)</code>                                 | Pass   | 0.001s |
| Gauge overview summary section                            | <code>any("Gauge Overview Summary" in t for t in texts)</code>                      | Pass   | 0.001s |
| Get gauge id missing                                      | <code>page._get_gauge_id({}) == "Unknown Gauge"</code>                              | Pass   | 0.000s |
| Get gauge id present                                      | <code>page._get_gauge_id(_make_gauge()) == "GAUGE-001"</code>                       | Pass   | 0.000s |

| Test Description                                                                   | Acceptance Criteria                                                      | Result | Time   |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------|--------|
| Get gauge name missing                                                             | <code>page._get_gauge_name({}) == "Unknown Gauge Name"</code>            | Pass   | 0.000s |
| Get gauge name present                                                             | <code>page._get_gauge_name(_make_gauge()) == "Chelsea Bridge"</code>     | Pass   | 0.000s |
| Has tables                                                                         | <code>any(isinstance(e, Table) for e in result)</code>                   | Pass   | 0.001s |
| High risk conclusion                                                               | <code>"high flood risk" in str(result)</code>                            | Pass   | 0.001s |
| hist_high $\geq$ severe_level $\rightarrow$ 'Exceeds severe warning' significance. | <code>"Exceeds severe warning" in str(result)</code>                     | Pass   | 0.001s |
| hist_high $\leq$ severe_level $\rightarrow$ 'Within expected range' significance.  | <code>"Within expected range" in str(result)</code>                      | Pass   | 0.001s |
| Key metrics summary section                                                        | <code>any("Key Metrics Summary" in t for t in texts)</code>              | Pass   | 0.001s |
| Low risk conclusion                                                                | <code>"Low flood risk" in str(result)</code>                             | Pass   | 0.001s |
| Moderate risk conclusion                                                           | <code>"Moderate flood risk" in str(result)</code>                        | Pass   | 0.001s |
| Operational status conclusion                                                      | <code>"fully operational" in str(result)</code>                          | Pass   | 0.001s |
| Operational status summary section                                                 | <code>any("Operational Status Summary" in t for t in texts)</code>       | Pass   | 0.001s |
| Report conclusions section                                                         | <code>any("Report Conclusions" in t for t in texts)</code>               | Pass   | 0.001s |
| Returns list                                                                       | <code>isinstance(result, list); len(result) &gt; 0</code>                | Pass   | 0.001s |
| Timeseries alert conclusion                                                        | <code>"ALERT" in str(result)</code>                                      | Pass   | 0.001s |
| Timeseries normal conclusion                                                       | <code>"normal water levels" in str(result)</code>                        | Pass   | 0.001s |
| Timeseries severe conclusion                                                       | <code>"CRITICAL" in str(result)</code>                                   | Pass   | 0.001s |
| Timeseries warning conclusion                                                      | <code>"WARNING" in str(result)</code>                                    | Pass   | 0.001s |
| Title present                                                                      | <code>any("Comprehensive Data Summary" in t for t in texts)</code>       | Pass   | 0.001s |
| Historical graph no dates fallback                                                 | <code>isinstance(result, list)</code>                                    | Pass   | 0.001s |
| Historical graph with thresholds                                                   | <code>isinstance(result, list); len(result) &gt; 0</code>                | Pass   | 0.056s |
| No storm responses shows message                                                   | <code>any("No storm" in t for t in texts)</code>                         | Pass   | 0.001s |
| No timeseries data shows no data message                                           | <code>any("No historical" in t or "No storm" in t for t in texts)</code> | Pass   | 0.000s |
| Realised floods no exceedances                                                     | <code>any("No flood events" in t for t in texts)</code>                  | Pass   | 0.001s |
| Realised floods with exceedances                                                   | <code>len(tables) &gt;= 1</code>                                         | Pass   | 0.001s |
| Returns list                                                                       | <code>isinstance(result, list); len(result) &gt; 0</code>                | Pass   | 0.000s |
| With historical daily returns more elements                                        | <code>len(result) &gt; 2</code>                                          | Pass   | 0.417s |

| Test Description                                                           | Acceptance Criteria                                                     | Result | Time   |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------|--------|--------|
| With storm responses includes table                                        | <code>len(tables) &gt;= 1</code>                                        | Pass   | 0.001s |
| Worst storms severity classification                                       | <code>isinstance(result, list)</code>                                   | Pass   | 0.001s |
| Worst storms table top 10                                                  | <code>len(tables) == 1</code>                                           | Pass   | 0.001s |
| Current level at alert                                                     | <code>"FLOOD ALERT" in str(result)</code>                               | Pass   | 0.001s |
| Current level at severe                                                    | <code>"SEVERE FLOOD WARNING" in str(result)</code>                      | Pass   | 0.001s |
| Current level at warning                                                   | <code>"FLOOD WARNING" in str(result)</code>                             | Pass   | 0.001s |
| Current level below alert is normal                                        | <code>"Normal" in str(result)</code>                                    | Pass   | 0.001s |
| Empty data does not crash                                                  | <code>isinstance(result, list)</code>                                   | Pass   | 0.000s |
| Get gauge id missing                                                       | <code>page._get_gauge_id({}) == "Unknown Gauge"</code>                  | Pass   | 0.000s |
| Get gauge id present                                                       | <code>page._get_gauge_id(_make_gauge()) == "GAUGE-001"</code>           | Pass   | 0.000s |
| Has tables                                                                 | <code>any(isinstance(e, Table) for e in result)</code>                  | Pass   | 0.000s |
| Historical exceedance section                                              | <code>any("Historical Threshold Exceedance" in t for t in texts)</code> | Pass   | 0.000s |
| No timeseries shows message                                                | <code>any("No current timeseries data" in t for t in texts)</code>      | Pass   | 0.000s |
| Returns list                                                               | <code>isinstance(result, list); len(result) &gt; 0</code>               | Pass   | 0.000s |
| Threshold analysis section                                                 | <code>any("Threshold Analysis" in t for t in texts)</code>              | Pass   | 0.000s |
| Alert-to-warning and warning-to-severe ranges are added to analysis table. | <code>isinstance(result, list)</code>                                   | Pass   | 0.000s |
| Title present                                                              | <code>any("Flood Stages" in t for t in texts)</code>                    | Pass   | 0.000s |
| Uk thresholds section                                                      | <code>any("UK Flood Stage Thresholds" in t for t in texts)</code>       | Pass   | 0.000s |
| With timeseries shows current status section                               | <code>any("Current Status vs Thresholds" in t for t in texts)</code>    | Pass   | 0.000s |
| Lines 251-254: generate_basic_report() body on real generator.             | <code>path.exists(); path.suffix == ".pdf"</code>                       | Pass   | 0.013s |
| Line 315: analysis branch in gauge_generator.generate_gauge_report.        | <code>path.exists()</code>                                              | Pass   | 0.003s |
| Lines 321-328: auto_open=True — open attempt caught, does not raise.       | <code>path.exists()</code>                                              | Pass   | 0.013s |

| Test Description                                                           | Acceptance Criteria                                                            | Result | Time   |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------|--------|
| Lines 307-311: generate_gauge_report() in gauge_generator.py, basic type.  | path.exists()                                                                  | Pass   | 0.014s |
| Lines 316-317: unknown type falls back to generate_report.                 | path.exists()                                                                  | Pass   | 0.009s |
| Line 313: monitoring branch in gauge_generator.generate_gauge_report.      | path.exists()                                                                  | Pass   | 0.003s |
| Line 119: generate_report(pages_to_include=None) calls _auto_select_pages. | path.exists()                                                                  | Pass   | 0.009s |
| Lines 102-103 in __init__.py: auto_open=True, open returns False.          | result == fake_pdf                                                             | Pass   | 0.013s |
| Lines 285-286: configure(panel_height=...) updates panel_height.           | obj.panel_height == "600px";<br>obj.panel_height == "700px" (+1 more)          | Pass   | 0.000s |
| Lines 283-284: configure(panel_width=...) updates panel_width.             | obj.panel_width == "800px";<br>obj.panel_width == "900px" (+1 more)            | Pass   | 0.000s |
| Line 290: get_statistics() returns dict with panel_width and panel_height. | stats["panel_width"] == "700px";<br>stats["panel_height"] == "500px" (+4 more) | Pass   | 0.000s |
| Lines 480-481: configure(panel_height=...) updates panel_height.           | obj.panel_height == "600px";<br>obj.panel_height == "700px" (+1 more)          | Pass   | 0.000s |
| Lines 478-479: configure(panel_width=...) updates panel_width.             | obj.panel_width == "800px";<br>obj.panel_width == "900px" (+1 more)            | Pass   | 0.000s |
| Line 485: get_statistics() returns dict.                                   | stats["panel_width"] == "700px";<br>stats["panel_height"] == "500px" (+4 more) | Pass   | 0.000s |
| Empty return period levels no extra table                                  | len(tables) >= 1                                                               | Pass   | 0.000s |
| No hazard curve shows fallback                                             | any("No hazard" in t or "hazard" in t.lower() for t in te...                   | Pass   | 0.000s |
| Returns list                                                               | isinstance(result, list)                                                       | Pass   | 0.000s |
| With hazard curve basic                                                    | len(result) > 2                                                                | Pass   | 0.000s |
| With return period levels renders table                                    | len(tables) >= 2                                                               | Pass   | 0.000s |



| Test Description                                                          | Acceptance Criteria                                                                        | Result | Time   |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------|--------|
| GET /api/v1/gauges/{id}/hazard returns hazard data for known gauge.       | response.status_code == 200;<br>data['status'] == 'success' (+6 more)                      | Pass   | 0.003s |
| GEV parameters should be present and numeric.                             | gev['location'] is not None;<br>gev['scale'] is not None (+3 more)                         | Pass   | 0.003s |
| GET /api/v1/gauges/{id}/hazard returns 404 for nonexistent gauge.         | response.status_code == 404                                                                | Pass   | 0.002s |
| GET /api/v1/gauges/{id}/hazard returns 404 for gauge without hazard data. | response.status_code == 404                                                                | Pass   | 0.003s |
| GET /api/v1/gauges/{id}/storms returns storm data for known gauge.        | response.status_code == 200;<br>data['status'] == 'success' (+2 more)                      | Pass   | 0.006s |
| calculate_statistics_from_levels via class.                               | isinstance(stats, dict)                                                                    | Pass   | 0.002s |
| Calculate statistics via class                                            | isinstance(stats, dict)                                                                    | Pass   | 0.000s |
| Custom years                                                              | ghd.years_of_history == 30                                                                 | Pass   | 0.000s |
| Default years                                                             | ghd.years_of_history == 50                                                                 | Pass   | 0.000s |
| Filter by years explicit                                                  | isinstance(result, list)                                                                   | Pass   | 0.000s |
| Filter by years none                                                      | result == flows                                                                            | Pass   | 0.000s |
| generate_from_gauge_portfolio with explicit years overrides default.      | result["years_included"] == 3                                                              | Pass   | 0.208s |
| generate_from_gauge_portfolio via class method.                           | isinstance(result, dict);<br>result["years_included"] == 5                                 | Pass   | 0.205s |
| When years=None, uses years_of_history.                                   | result["years_included"] == 10                                                             | Pass   | 0.001s |
| Generate from nrfa via class                                              | output.exists(); isinstance(result, dict)                                                  | Pass   | 0.006s |
| generate_synthetic_timeseries via class method.                           | isinstance(result, list); len(result) > 0                                                  | Pass   | 0.003s |
| Generates per gauge files                                                 | gaugets_dir.exists();<br>len(list(gaugets_dir.glob("GAUGE-*.json"))) >= N_GAUGES (+2 more) | Pass   | 0.000s |
| Output has daily observations                                             | "daily_observations" in data;<br>len(data["daily_observations"]) > 0                       | Pass   | 0.014s |
| Parse nrfa csv via class                                                  | len(flows) == 10                                                                           | Pass   | 0.006s |
| Statistics computed                                                       | "statistics" in data; "mean_level" in data["statistics"] (+1 more)                         | Pass   | 0.002s |
| Cdm available flag is bool                                                | hasattr(mod, "CDM_AVAILABLE");<br>isinstance(mod.CDM_AVAILABLE, bool)                      | Pass   | 0.000s |

| Test Description                                                      | Acceptance Criteria                                                                      | Result | Time   |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------|--------|
| Response has count and gauges keys                                    | "count" in data; "gauges" in data (+1 more)                                              | Pass   | 0.001s |
| Returns 200 with no gaugehd files                                     | r.status_code == 200; data["status"] == "success" (+1 more)                              | Pass   | 0.001s |
| Returns correct count when files present                              | r.status_code == 200; data["status"] == "success" (+2 more)                              | Pass   | 0.001s |
| gauge.json must contain at least one gauge.                           | len(ids) > 0                                                                             | Pass   | 0.001s |
| Gauge IDs in gaugehc.json must exist in gauge.json.                   | not orphaned                                                                             | Pass   | 0.002s |
| Gauge IDs in gaugets/ files must exist in gauge.json.                 | not orphaned                                                                             | Pass   | 0.002s |
| Each gaugets file's internal gauge_id must match its filename.        | internal_id == gf.stem                                                                   | Pass   | 0.003s |
| gaugets/ files must reference gauge IDs from gauge.json.              | len(missing) == 0                                                                        | Pass   | 2.508s |
| Every gauge in gauge.json must have a hazard curve.                   | len(missing) == 0                                                                        | Pass   | 0.003s |
| gaugehc.json must not contain IDs absent from gauge.json.             | len(orphan) == 0                                                                         | Pass   | 0.003s |
| Every gauge referenced by a trade must exist in gauge.json.           | len(missing) == 0                                                                        | Pass   | 0.006s |
| Every gauge with trades must have a hazard curve (for PRS pricing).   | len(missing) == 0                                                                        | Pass   | 0.008s |
| gauge_random.py must NOT use GAUGE-{i:03d} style sequential IDs.      | 'GAUGE-{i:' not in src;<br>'GAUGE-{index' not in src (+3 more)                           | Pass   | 0.000s |
| gauge_random.py contains the GAUGE- prefix.                           | 'GAUGE-' in src                                                                          | Pass   | 0.001s |
| gauge_random.py uses uuid for ID generation (not sequential counter). | 'uuid' in src; 'uuid' in src (+1 more)                                                   | Pass   | 0.000s |
| uuid4()[8] always produces 8 lower-case hex chars — 100 samples.      | GAUGE_ID.RE.match(gauge_id);<br>PROP_ID.RE.match(prop_id) (+2 more)                      | Pass   | 0.000s |
| Accepts valid uuid derived ids                                        | GAUGE_ID.RE.match('GAUGE-9042bd95');<br>GAUGE_ID.RE.match('GAUGE-a1b2c3d4')<br>(+2 more) | Pass   | 0.000s |
| Rejects non hex chars                                                 | not GAUGE_ID.RE.match('GAUGE-9042xd95');<br>not GAUGE_ID.RE.match('GAUGE-9042gd95')      | Pass   | 0.000s |
| GAUGE-001, GAUGE-002 are NOT protocol-compliant.                      | not GAUGE_ID.RE.match('GAUGE-001');<br>not GAUGE_ID.RE.match('GAUGE-002')<br>(+5 more)   | Pass   | 0.000s |

| Test Description                                                    | Acceptance Criteria                                                                                                                         | Result | Time   |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Gauge IDs must be lowercase; GAUGE-9042BD95 is invalid.             | not GAUGE.ID.RE.match('GAUGE-9042BD95');                                                                                                    | Pass   | 0.000s |
| Rejects wrong length                                                | not GAUGE.ID.RE.match('GAUGE-A1B2C3D4')<br>not GAUGE.ID.RE.match('GAUGE-9042bd9');<br>not GAUGE.ID.RE.match('GAUGE-9042bd951')<br>(+1 more) | Pass   | 0.000s |
| Gauge id present in all responses                                   | "gauge_id" not in resp;<br>gr.get("gauge_id") ==<br>"GAUGE-deadbeef"                                                                        | Pass   | 0.001s |
| Multiple gauges have correct ids                                    | found_ids == {"GAUGE-00000001",<br>"GAUGE-00000002", "GAUGE-...                                                                             | Pass   | 0.004s |
| Corrupted gauge json does not crash                                 | result["processing_stats"]["successful_properties"]<br>== 2                                                                                 | Pass   | 0.595s |
| Gauge id map empty when no gauge json                               | isinstance(gen._gauge_id_map, dict)                                                                                                         | Pass   | 0.598s |
| When gauge.json exists with proper structure, gauge IDs are loaded. | 0 in gen._gauge_id_map;<br>gen._gauge_id_map[0] == "GAUGE-111"                                                                              | Pass   | 0.591s |
| Also supports 'gauges' as the top-level key.                        | gen._gauge_id_map.get(0) == "G-AAA"                                                                                                         | Pass   | 0.592s |
| Get most recent reading.                                            | current ==<br>flood_event_readings[-1]['level']                                                                                             | Pass   | 0.000s |
| Detect falling water levels.                                        | trend == "FALLING"                                                                                                                          | Pass   | 0.000s |
| Detect rising water levels.                                         | trend == "RISING"                                                                                                                           | Pass   | 0.000s |
| Detect stable water levels.                                         | trend == "STABLE"                                                                                                                           | Pass   | 0.000s |
| Detect peak level in time series.                                   | peak['level'] > 6.0; 'timestamp' in<br>peak                                                                                                 | Pass   | 0.000s |
| Peak should occur around hour 24.                                   | 20 < hours_to_peak < 28                                                                                                                     | Pass   | 0.000s |
| Count                                                               | GaugeLoader(tmp_path).count() == 4                                                                                                          | Pass   | 0.001s |
| Floodgauges key also works                                          | GaugeLoader(tmp_path).count() == 2                                                                                                          | Pass   | 0.002s |
| Load all returns list                                               | len(GaugeLoader(tmp_path).load_all())<br>== 3                                                                                               | Pass   | 0.001s |
| Load missing file returns empty                                     | GaugeLoader(tmp_path).load_all() ==<br>[]                                                                                                   | Pass   | 0.000s |
| Test GaugeID extraction from nested structure.                      | gauge_id is not None;<br>gauge_id.startswith("GAUGE-")                                                                                      | Pass   | 0.001s |
| Test gauge summary generation.                                      | len(summaries) == 3; 'gaugeId' in<br>summary (+5 more)                                                                                      | Pass   | 0.001s |
| Test loading gauge portfolio.                                       | len(gauges) == 3                                                                                                                            | Pass   | 0.001s |
| Caching works                                                       | loader.load_all() is<br>loader.load_all()                                                                                                   | Pass   | 0.001s |
| Clear cache                                                         | not loader._cache_valid                                                                                                                     | Pass   | 0.001s |

| Test Description                          | Acceptance Criteria                                                                           | Result | Time   |
|-------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| Force reload                              | <code>len(loader.load_all(force_reload=True)) == 5</code>                                     | Pass   | 0.002s |
| Test getting active gauges.               | <code>len(active) == 2; status == "Active"</code>                                             | Pass   | 0.001s |
| Test getting gauge coordinates.           | <code>coords is not None; lat == pytest.approx(51.4309, rel=1e-4)</code><br>(+1 more)         | Pass   | 0.001s |
| Test getting flood stage thresholds.      | <code>stages is not None; stages['ActionStage'] == 4.5</code> (+3 more)                       | Pass   | 0.001s |
| Test getting gauge name by ID.            | <code>name == "Thames at Teddington"</code>                                                   | Pass   | 0.001s |
| Test getting name for non-existent gauge. | <code>name is None</code>                                                                     | Pass   | 0.001s |
| Exists false                              | <code>GaugeLoader(tmp_path).exists("GAUGE-999") is False</code>                               | Pass   | 0.001s |
| Exists true                               | <code>GaugeLoader(tmp_path).exists("GAUGE-000") is True</code>                                | Pass   | 0.001s |
| Find by id missing returns none           | <code>GaugeLoader(tmp_path).find_by_id("GAUGE-999") is None</code>                            | Pass   | 0.001s |
| Find by id returns entity                 | <code>result is not None; result["FloodGauge"]["Header"]["GaugeID"] == "GAUGE-001"</code>     | Pass   | 0.001s |
| List all returns summaries                | <code>len(summaries) == 2; "gaugeId" in summaries[0]</code>                                   | Pass   | 0.001s |
| Test finding gauge by ID.                 | <code>gauge is not None; gauge['FloodGauge']['Header']['GaugeName'] == "Thames at ..."</code> | Pass   | 0.001s |
| Test finding gauge by name.               | <code>gauge is not None; gauge_loader.get_entity_id(gauge) == "GAUGE-001"</code>              | Pass   | 0.001s |
| Test name search is case insensitive.     | <code>gauge is not None</code>                                                                | Pass   | 0.002s |
| Test finding gauges by owner.             | <code>len(ea_gauges) == 2; len(usgs_gauges) == 1</code>                                       | Pass   | 0.001s |
| Test finding active gauges.               | <code>len(active) == 2</code>                                                                 | Pass   | 0.001s |
| Test finding inactive gauges.             | <code>len(inactive) == 1; gauge_loader.get_entity_id(inactive[0]) == "GAUGE-003"</code>       | Pass   | 0.001s |
| Test finding river gauges.                | <code>len(river) == 2</code>                                                                  | Pass   | 0.001s |
| Test finding tidal gauges.                | <code>len(tidal) == 1</code>                                                                  | Pass   | 0.001s |
| Test finding gauges within radius.        | <code>len(nearby) &gt;= 1; len(nearby) &gt;= 2</code>                                         | Pass   | 0.001s |
| Test radius search with no matches.       | <code>len(nearby) == 0</code>                                                                 | Pass   | 0.001s |

| Test Description                                                        | Acceptance Criteria                                   | Result | Time   |
|-------------------------------------------------------------------------|-------------------------------------------------------|--------|--------|
| Get status returns dict                                                 | "entity_name" in status; "path" in status (+1 more)   | Pass   | 0.000s |
| Distance $\leq 100 \rightarrow$ 'Very close to Thames'.                 | "Very close to Thames" in str(result)                 | Pass   | 0.001s |
| Distance $== 0 \rightarrow$ 'On Thames River' and 'Directly on Thames'. | "On Thames River" in str(result)                      | Pass   | 0.001s |
| Empty data does not crash                                               | isinstance(result, list)                              | Pass   | 0.000s |
| Flood risk section                                                      | any("Flood Risk" in t for t in texts)                 | Pass   | 0.000s |
| Geographic coordinates section                                          | any("Geographic Coordinates" in t for t in texts)     | Pass   | 0.000s |
| Has tables                                                              | any(isinstance(e, Table) for e in result)             | Pass   | 0.000s |
| Latitude 51.4–51.5 $\rightarrow$ 'Central London area'.                 | "Central London area" in str(result)                  | Pass   | 0.001s |
| Latitude $\geq 51.5 \rightarrow$ 'North London area'.                   | "North London area" in str(result)                    | Pass   | 0.001s |
| Latitude $\leq 51.4 \rightarrow$ 'South London area'.                   | "South London area" in str(result)                    | Pass   | 0.001s |
| No thames info does not crash                                           | isinstance(result, list)                              | Pass   | 0.000s |
| Positioning summary section                                             | any("Positioning Summary" in t for t in texts)        | Pass   | 0.000s |
| Returns list                                                            | isinstance(result, list); len(result) > 0             | Pass   | 0.000s |
| Risk category in location profile                                       | "High flood risk location" in str(result)             | Pass   | 0.001s |
| Risk score shown as 'X/10'.                                             | "7.5/10" in str(result)                               | Pass   | 0.001s |
| Thames context section                                                  | any("Thames River Context" in t for t in texts)       | Pass   | 0.000s |
| Title present                                                           | any("Geographic Location" in t for t in texts)        | Pass   | 0.000s |
| Lines 89-90: exception in create_gauge_mapping raises ValueError.       | —                                                     | Pass   | 0.000s |
| Line 100: get_nrfa_fields returns list of NRFA field names.             | isinstance(fields, list); len(fields) == 15 (+3 more) | Pass   | 0.000s |
| Empty data does not crash                                               | isinstance(result, list)                              | Pass   | 0.000s |
| Empty readings list does not crash                                      | isinstance(result, list)                              | Pass   | 0.000s |
| Get gauge id missing                                                    | page._get_gauge_id({}) == "Unknown Gauge"             | Pass   | 0.000s |

| Test Description                                          | Acceptance Criteria                                                        | Result | Time   |
|-----------------------------------------------------------|----------------------------------------------------------------------------|--------|--------|
| Get gauge id present                                      | <code>page._get_gauge_id(_make_gauge()) == "GAUGE-001"</code>              | Pass   | 0.000s |
| Has tables                                                | <code>any(isinstance(e, Table) for e in result)</code>                     | Pass   | 0.000s |
| Historical statistics section                             | <code>any("Historical Measurement Statistics" in t for t in texts)</code>  | Pass   | 0.000s |
| Measurement configuration section                         | <code>any("Measurement Configuration" in t for t in texts)</code>          | Pass   | 0.000s |
| No timeseries shows message                               | <code>any("No current timeseries data" in t for t in texts)</code>         | Pass   | 0.000s |
| Operational status section                                | <code>any("Operational Status" in t for t in texts)</code>                 | Pass   | 0.000s |
| Returns list                                              | <code>isinstance(result, list); len(result) &gt; 0</code>                  | Pass   | 0.000s |
| Title present                                             | <code>any("Current Measurements" in t for t in texts)</code>               | Pass   | 0.000s |
| Current readings table is added when timeseries provided. | <code>len(tables_with) &gt;= len(tables_without)</code>                    | Pass   | 0.001s |
| With timeseries shows current readings                    | <code>any("Current Readings" in t for t in texts)</code>                   | Pass   | 0.000s |
| Calculate level range.                                    | <code>level_range &gt; 2.0</code>                                          | Pass   | 0.000s |
| Calculate maximum level.                                  | <code>max_level &gt; 6.0</code>                                            | Pass   | 0.000s |
| Calculate mean water level.                               | <code>2.5 &lt; mean &lt; 3.5</code>                                        | Pass   | 0.000s |
| Calculate minimum level.                                  | <code>min_level &gt; 0</code>                                              | Pass   | 0.000s |
| Calculate maximum rate of rise.                           | <code>rate &gt; 0</code>                                                   | Pass   | 0.000s |
| When name is empty, uses area_name.                       | <code>"Westminster" in result; "Thames" in result</code>                   | Pass   | 0.000s |
| No name key falls back to area                            | <code>"Putney" in result</code>                                            | Pass   | 0.000s |
| No hazard curve shows fallback                            | <code>any("No hazard" in t or "hazard" in t.lower() for t in te...)</code> | Pass   | 0.000s |
| Returns list                                              | <code>isinstance(result, list)</code>                                      | Pass   | 0.000s |
| With nonzero rates renders spreads                        | <code>len(tables) &gt;= 1</code>                                           | Pass   | 0.000s |
| With zero rates renders na                                | <code>len(tables) &gt;= 1</code>                                           | Pass   | 0.000s |
| Basic gauge report produces a valid PDF file.             | <code>report_path.exists(); report_path.suffix == '.pdf' (+1 more)</code>  | Pass   | 0.016s |
| Full gauge report produces a valid PDF.                   | <code>report_path.exists(); report_path.suffix == '.pdf'</code>            | Pass   | 0.013s |
| Output directory is created automatically.                | <code>new_dir.exists(); report_path.exists()</code>                        | Pass   | 0.016s |



| Test Description                                                          | Acceptance Criteria                                                        | Result | Time   |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------|--------|--------|
| Generated filename includes the gauge ID.                                 | 'GAUGE-test0001' in report_path.name                                       | Pass   | 0.016s |
| Gauge report PDF should be non-trivial in size.                           | report_path.stat().st_size > 5000                                          | Pass   | 0.016s |
| Generated file is a valid PDF.                                            | header == b'%PDF-'                                                         | Pass   | 0.023s |
| Report handles minimal gauge data without crashing.                       | report_path.exists();<br>report_path.stat().st_size > 0                    | Pass   | 0.012s |
| Daily minus 15                                                            | r['reliability_score'] == 80                                               | Pass   | 0.000s |
| Decommissioned                                                            | r['reliability_score'] == 0;<br>r['category'] == "Unreliable"              | Pass   | 0.000s |
| Fully operational                                                         | r['reliability_score'] == 95;<br>r['category'] == "Highly Reliable"        | Pass   | 0.000s |
| Hourly minus 5                                                            | r['reliability_score'] == 90                                               | Pass   | 0.000s |
| Maintenance required                                                      | r['reliability_score'] == 70;<br>r['category'] == "Reliable"               | Pass   | 0.000s |
| Monthly minus 50                                                          | r['reliability_score'] == 20;<br>r['category'] == "Unreliable"             | Pass   | 0.000s |
| Realtime no adjustment                                                    | rt['reliability_score'] ==<br>base['reliability_score']                    | Pass   | 0.000s |
| Result keys                                                               | 'reliability_score' in r; 'category'<br>in r (+1 more)                     | Pass   | 0.000s |
| Score clamped at zero                                                     | r['reliability_score'] == 0                                                | Pass   | 0.000s |
| Temporarily offline                                                       | r['reliability_score'] == 30;<br>r['category'] == "Low Reliability"        | Pass   | 0.000s |
| Unknown status                                                            | r['reliability_score'] == 50;<br>r['category'] == "Moderately<br>Reliable" | Pass   | 0.000s |
| Weekly minus 30                                                           | r['reliability_score'] == 65                                               | Pass   | 0.000s |
| Generator initialises with output directory.                              | gen.output_dir == tmp_path                                                 | Pass   | 0.001s |
| Generator lists all registered gauge page modules.                        | 'title_overview' in pages;<br>'sensor_details' in pages (+7 more)          | Pass   | 0.001s |
| Validates known gauge page names.                                         | 'title_overview' in valid; 'bogus' in<br>invalid                           | Pass   | 0.001s |
| Line 163 (else branch): no time-series or hazard -i no analysis pages.    | not any(p in pages for p in<br>analysis_pages)                             | Pass   | 0.001s |
| Line 163 (if branch): gauge_data.get('hazard_curve') -i include analysis. | "risk_assessment" in pages                                                 | Pass   | 0.001s |
| Lines 269-270: generate_analysis_report creates PDF.                      | path.exists()                                                              | Pass   | 0.003s |

| Test Description                                                     | Acceptance Criteria                                                                                    | Result | Time   |
|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------|--------|
| Lines 208-210: exception in page -j logged, continue to next page.   | <code>isinstance(elements, list)</code>                                                                | Pass   | 0.002s |
| Lines 190-191: unknown page -j warning logged, skipped.              | <code>isinstance(elements, list)</code>                                                                | Pass   | 0.001s |
| Lines 176-177: missing GaugeID -j uses 'unknown' in filename.        | <code>"unknown" in name;</code><br><code>name.endswith(".pdf")</code>                                  | Pass   | 0.001s |
| Line 175: gauge_id extracted for file-name.                          | <code>"GAUGE-001" in name</code>                                                                       | Pass   | 0.001s |
| Lines 261-263: generate_monitoring_report creates PDF.               | <code>path.exists()</code>                                                                             | Pass   | 0.003s |
| Line 313: get_page_categories returns dict.                          | <code>isinstance(cats, dict); "gauge_info" in cats</code>                                              | Pass   | 0.001s |
| Line 279: list_available_pages returns list of strings.              | <code>isinstance(pages, list); "title_overview" in pages</code>                                        | Pass   | 0.001s |
| Lines 59-60: output_dir=None -j uses config.get_gauge_reports_dir(). | <code>gen.output_dir == gauge_dir</code>                                                               | Pass   | 0.001s |
| Line 315: validate_pages with all valid pages.                       | <code>"title_overview" in valid; invalid == []</code>                                                  | Pass   | 0.001s |
| validate_pages with some invalid page names.                         | <code>"title_overview" in valid;</code><br><code>"nonexistent_page" in invalid</code>                  | Pass   | 0.001s |
| Categories dict initialized                                          | <code>"gauge_info" in gen.categories;</code><br><code>"operational" in gen.categories</code> (+2 more) | Pass   | 0.001s |
| Explicit output dir path                                             | <code>isinstance(gen.output_dir, Path)</code>                                                          | Pass   | 0.001s |
| Explicit output dir string                                           | <code>gen.output_dir == tmp_path</code>                                                                | Pass   | 0.001s |
| No output dir uses config                                            | <code>gen.output_dir == gauge_dir</code>                                                               | Pass   | 0.001s |
| Output dir created                                                   | <code>new_dir.exists()</code>                                                                          | Pass   | 0.001s |
| Pages dict initialized                                               | <code>expected &lt;= set(gen.pages.keys())</code>                                                      | Pass   | 0.001s |
| Excludes timeseries when requested                                   | <code>"intensity_timeseries" not in d;</code><br><code>"level_timeseries" not in d</code>              | Pass   | 0.000s |
| Flooded is bool                                                      | <code>isinstance(d["flooded"], bool);</code><br><code>d["flooded"] is True</code>                      | Pass   | 0.000s |
| Has required keys                                                    | <code>"gauge_id" in d; "storm_id" in d</code> (+2 more)                                                | Pass   | 0.000s |
| Includes timeseries by default                                       | <code>"intensity_timeseries" in d;</code><br><code>"level_timeseries" in d</code> (+1 more)            | Pass   | 0.000s |
| Returns dict                                                         | <code>isinstance(d, dict);</code> <code>isinstance(d, dict)</code>                                     | Pass   | 0.000s |
| Values rounded                                                       | <code>d["peak_level"] == 5.2;</code><br><code>d["accumulation"] == 4.5</code>                          | Pass   | 0.000s |

| Test Description                                 | Acceptance Criteria                                                                             | Result | Time   |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|--------|
| Generates elements list                          | <code>isinstance(elements, list);</code><br><code>len(elements) &gt; 0</code>                   | Pass   | 0.000s |
| No timeseries shows placeholder                  | <code>any("No current timeseries" in t for t in texts)</code>                                   | Pass   | 0.000s |
| With timeseries data                             | <code>isinstance(elements, list);</code><br><code>len(elements) &gt; 0</code>                   | Pass   | 0.000s |
| Test POST /api/v1/gauges/report without gaugeId. | <code>response.status_code == 400; 'Gauge ID is required' in data['message']</code>             | Pass   | 0.000s |
| Test GET /api/v1/gauges/{id}.                    | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more) | Pass   | 0.002s |
| Test GET /api/v1/gauges/{id} with invalid ID.    | <code>response.status_code == 404</code>                                                        | Pass   | 0.002s |
| Test GET /api/v1/gauges/{id}/statistics.         | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more) | Pass   | 0.002s |
| Test GET /api/v1/gauges/{id}/timeseries.         | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more) | Pass   | 0.002s |
| Test legacy GET /api/v1/list_gauges endpoint.    | <code>response.status_code == 200;</code><br><code>data['count'] == 3</code>                    | Pass   | 0.002s |
| Test GET /api/v1/gauges.                         | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+2 more) | Pass   | 0.002s |
| Empty data does not crash                        | <code>isinstance(result, list)</code>                                                           | Pass   | 0.000s |
| Gauge id in subtitle                             | <code>any("GAUGE-001" in t for t in texts)</code>                                               | Pass   | 0.001s |
| Gauge information section                        | <code>any("Gauge Information" in t for t in texts)</code>                                       | Pass   | 0.001s |
| Get gauge id missing                             | <code>page._get_gauge_id({}) == "Unknown Gauge"</code>                                          | Pass   | 0.000s |
| Get gauge id present                             | <code>page._get_gauge_id(_make_gauge()) == "GAUGE-001"</code>                                   | Pass   | 0.000s |
| Has tables                                       | <code>any(isinstance(e, Table) for e in result)</code>                                          | Pass   | 0.001s |
| Historical sensor statistics section             | <code>any("Historical Sensor Statistics" in t for t in texts)</code>                            | Pass   | 0.001s |
| Location information section                     | <code>any("Location Information" in t for t in texts)</code>                                    | Pass   | 0.001s |
| Measurement configuration section                | <code>any("Measurement Configuration" in t for t in texts)</code>                               | Pass   | 0.001s |

| Test Description                                                                            | Acceptance Criteria                                                                                                                     | Result | Time   |
|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Missing SensorDetails → empty tables, no exception.                                         | <code>isinstance(result, list)</code>                                                                                                   | Pass   | 0.000s |
| Returns list                                                                                | <code>isinstance(result, list); len(result) &gt; 0</code>                                                                               | Pass   | 0.001s |
| Title present                                                                               | <code>any("Sensor Details" in t for t in texts)</code>                                                                                  | Pass   | 0.001s |
| With timeseries does not crash                                                              | <code>isinstance(result, list)</code>                                                                                                   | Pass   | 0.001s |
| Level at alert threshold.                                                                   | <code>status == "ALERT"</code>                                                                                                          | Pass   | 0.000s |
| Level between alert and warning.                                                            | <code>status == "ALERT"</code>                                                                                                          | Pass   | 0.000s |
| Level below alert = normal.                                                                 | <code>status == "NORMAL"</code>                                                                                                         | Pass   | 0.000s |
| Level at severe threshold.                                                                  | <code>status == "SEVERE"</code>                                                                                                         | Pass   | 0.000s |
| Level at warning threshold.                                                                 | <code>status == "WARNING"</code>                                                                                                        | Pass   | 0.000s |
| Lines 270-271: <code>configure(panel_height=...)</code> updates <code>panel_height</code> . | <code>obj.panel_height == "600px";</code><br><code>obj.panel_height == "700px"</code> (+1 more)                                         | Pass   | 0.000s |
| Lines 268-269: <code>configure(panel_width=...)</code> updates <code>panel_width</code> .   | <code>obj.panel_width == "800px";</code><br><code>obj.panel_width == "900px"</code> (+1 more)                                           | Pass   | 0.000s |
| Line 275: <code>get_statistics()</code> returns dict.                                       | <code>stats["panel_width"] == "700px";</code><br><code>stats["panel_height"] == "500px"</code> (+4 more)                                | Pass   | 0.000s |
| Empty timeseries returns short list                                                         | <code>isinstance(result, list); len(result) &gt; 0</code>                                                                               | Pass   | 0.000s |
| Gauge name in title                                                                         | <code>any("Chelsea Gauge" in t for t in texts)</code>                                                                                   | Pass   | 0.000s |
| Generates per gauge files                                                                   | <code>gaugets_dir.exists();</code><br><code>len(list(gaugets_dir.glob("GAUGE-*.json")))</code><br><code>&gt;= N_GAUGES</code> (+2 more) | Pass   | 0.000s |
| Missing gauge name does not crash                                                           | <code>isinstance(result, list)</code>                                                                                                   | Pass   | 0.000s |
| No timeseries message included                                                              | <code>any("No time series data" in t for t in texts)</code>                                                                             | Pass   | 0.000s |
| No timeseries returns short list                                                            | <code>len(result) &gt; 0</code>                                                                                                         | Pass   | 0.000s |
| Output file has readings                                                                    | <code>len(gauge_files) &gt; 0;</code><br><code>"flood_simulation" in data</code> (+1 more)                                              | Pass   | 0.001s |
| Requires gauge portfolio                                                                    | —                                                                                                                                       | Pass   | 0.000s |
| Returns list                                                                                | <code>isinstance(result, list)</code>                                                                                                   | Pass   | 0.004s |
| Timeseries mixed statuses                                                                   | <code>isinstance(result, list)</code>                                                                                                   | Pass   | 0.084s |
| Readings with non-zero thresholds exercising <code>fill_between</code> paths.               | <code>isinstance(result, list)</code>                                                                                                   | Pass   | 0.085s |

| Test Description                                                           | Acceptance Criteria                                                          | Result | Time   |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------|--------|--------|
| With timeseries returns more elements                                      | <code>len(result) &gt;= len(empty_result)</code>                             | Pass   | 0.089s |
| With timeseries table included                                             | <code>len(tables) &gt;= 1</code>                                             | Pass   | 0.085s |
| DEFAULT_PARAMS and generate() default must both be 168 hours (7 days).     | <code>GaugeTimeSeriesGenerator.DEFAULT_PARAMS["simulation_hours"]</code>     | Pass   | 0.000s |
| generate(simulation_hours) default parameter must be 168.                  | <code>default == 168</code>                                                  | Pass   | 0.000s |
| Convenience function generate_gaugets() default must be 168.               | <code>default == 168</code>                                                  | Pass   | 0.000s |
| peak_hour_max must be strictly less than simulation_hours.                 | <code>p["peak_hour_max"] &lt; p["simulation_hours"]</code>                   | Pass   | 0.000s |
| peak_hour_min must be less than peak_hour_max.                             | <code>p["peak_hour_min"] &lt; p["peak_hour_max"]</code>                      | Pass   | 0.000s |
| 168-hour simulation produces significantly more readings than 60-hour.     | <code>files60 and files168; count168 &gt; count60</code>                     | Pass   | 0.115s |
| Lines 127-130: configure() updates a known sim_param key.                  | <code>gen.sim_params['simulation_hours'] == 48</code>                        | Pass   | 0.000s |
| Lines 127-129: unknown key is silently ignored (if branch is False).       | <code>'nonexistent_param' not in gen.sim_params</code>                       | Pass   | 0.000s |
| generate(168) should produce ~168 readings per gauge.                      | <code>len(files) &gt; 0; sim.get("simulation_hours") == 168 (+1 more)</code> | Pass   | 0.184s |
| Explicit override stores the passed value in simulation_hours metadata.    | <code>sim.get("simulation_hours") == 60</code>                               | Pass   | 0.136s |
| Lines 157-159: gauge.json absent → FileNotFoundError re-raised.            | —                                                                            | Pass   | 0.000s |
| simulation_hours is written into each gaugets JSON.                        | <code>"simulation_hours" in data["flood_simulation"]</code>                  | Pass   | 0.177s |
| Line 175: stale GAUGE-*.json files in gaugets/ are deleted before writing. | <code>stale.exists(); not stale.exists()</code>                              | Pass   | 0.045s |
| Unparseable install date is silently skipped.                              | <code>isinstance(result, list)</code>                                        | Pass   | 0.000s |
| Empty data does not crash                                                  | <code>isinstance(result, list)</code>                                        | Pass   | 0.000s |
| Gauge id in output                                                         | <code>any("GAUGE-001" in t for t in texts)</code>                            | Pass   | 0.000s |
| Gauge overview section header                                              | <code>any("Gauge Overview" in t for t in texts)</code>                       | Pass   | 0.000s |
| Gauge summary section header                                               | <code>any("Gauge Summary" in t for t in texts)</code>                        | Pass   | 0.000s |

| Test Description                                                 | Acceptance Criteria                                                                                 | Result | Time   |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| Get gauge id missing                                             | <code>page._get_gauge_id({}) == "Unknown Gauge"</code>                                              | Pass   | 0.000s |
| Get gauge id present                                             | <code>page._get_gauge_id(_make_gauge()) == "GAUGE-001"</code>                                       | Pass   | 0.000s |
| Has tables                                                       | <code>any(isinstance(e, Table) for e in result)</code>                                              | Pass   | 0.000s |
| Valid install date triggers gauge age row in summary.            | <code>isinstance(result, list); len(result) &gt; 0</code>                                           | Pass   | 0.000s |
| No install date does not crash                                   | <code>isinstance(result, list)</code>                                                               | Pass   | 0.000s |
| Returns list                                                     | <code>isinstance(result, list); len(result) &gt; 0</code>                                           | Pass   | 0.000s |
| Title paragraph present                                          | <code>any("Comprehensive Flood Gauge Analysis Report" in t for ...</code>                           | Pass   | 0.000s |
| Unknown gauge fallback                                           | <code>any("Unknown Gauge" in t for t in texts)</code>                                               | Pass   | 0.000s |
| With timeseries does not crash                                   | <code>isinstance(result, list)</code>                                                               | Pass   | 0.000s |
| Lines 68-69: exception returns validation_error dict.            | <code>"validation_error" in result; isinstance(result["validation_error"], list) (+2 more)</code>   | Pass   | 0.000s |
| Lines 61,64: invalid menu field value appended to sensor_errors. | <code>"SensorDetails" in errors; any("DataSourceType" in e for e in errors["SensorDetails"])</code> | Pass   | 0.000s |
| Multiple invalid menu fields all collected.                      | <code>"SensorDetails" in errors; len(errors["SensorDetails"]) == 3</code>                           | Pass   | 0.000s |
| Valid menu field values produce no SensorDetails errors.         | <code>"SensorDetails" not in errors</code>                                                          | Pass   | 0.000s |
| Curve points present                                             | <code>not missing</code>                                                                            | Pass   | 0.000s |
| Each gauge has required fields                                   | <code>not missing</code>                                                                            | Pass   | 0.000s |
| File exists                                                      | <code>GAUGEHC_PATH.exists(); MARKET_STATE.exists()</code>                                           | Pass   | 0.000s |
| alert   warning   severe for every gauge.                        | <code>not violations</code>                                                                         | Pass   | 0.000s |
| Gauge count at least 40                                          | <code>count &gt;= 40</code>                                                                         | Pass   | 0.000s |
| Gauge ids are well formed                                        | <code>not bad</code>                                                                                | Pass   | 0.000s |
| Has hazard curves key                                            | <code>"hazard_curves" in gaugehc</code>                                                             | Pass   | 0.000s |
| alert   warning   severe (higher trigger = lower probability).   | <code>not violations</code>                                                                         | Pass   | 0.000s |
| Hazard rates positive                                            | <code>not violations</code>                                                                         | Pass   | 0.000s |
| Term structure present                                           | <code>not missing</code>                                                                            | Pass   | 0.000s |



| Test Description                                                                | Acceptance Criteria                                                            | Result | Time   |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------|--------|
| Lines 73-77: main() with no GDF files logs info.                                | —                                                                              | Pass   | 0.000s |
| Lines 70-71: main() processes files, handles errors.                            | —                                                                              | Pass   | 0.001s |
| Line 37: output_dir defaults to input_dir when None.                            | result == []                                                                   | Pass   | 0.000s |
| Lines 46-47: error processing a CSV file is logged and skipped.                 | result == []                                                                   | Pass   | 0.001s |
| directory.main() is callable.                                                   | callable(main)                                                                 | Pass   | 0.000s |
| Each file has required keys                                                     | not bad                                                                        | Pass   | 2.248s |
| Flood simulation has readings                                                   | not bad                                                                        | Pass   | 2.243s |
| Gauge id field matches filename                                                 | not bad                                                                        | Pass   | 2.221s |
| Gaugets dir exists                                                              | GAUGETS_DIR.exists()                                                           | Pass   | 0.000s |
| Gaugets gauge id format                                                         | not bad                                                                        | Pass   | 0.000s |
| Gaugets not empty                                                               | gaugets_files                                                                  | Pass   | 0.000s |
| simulation_hours should match number of reading entries.                        | not mismatches                                                                 | Pass   | 2.222s |
| Line 72: datetime → isoformat string.                                           | "2026-01-01" in result                                                         | Pass   | 0.000s |
| Lines 75-76: np.float32 is np.floating but not Python float → default() called. | "2.5" in result                                                                | Pass   | 0.000s |
| Lines 73-74: np.integer → int.                                                  | "42" in result                                                                 | Pass   | 0.000s |
| Lines 77-78: np.ndarray → list via tolist().                                    | "1.0" in result                                                                | Pass   | 0.000s |
| Line 79: unknown type → super().default() → TypeError.                          | —                                                                              | Pass   | 0.000s |
| When no gaugets file exists, hydrograph falls back to synthesis.                | resp.status_code == 200;<br>max(water_levels) > min(water_levels)              | Pass   | 0.008s |
| gaugets readings using 'level' key (alternate field name) are extracted.        | resp.status_code == 200;<br>max(water_levels) > min(water_levels)              | Pass   | 0.011s |
| gaugets readings using 'waterLevel' key are correctly extracted.                | resp.status_code == 200;<br>max(water_levels) > min(water_levels)<br>(+1 more) | Pass   | 0.009s |
| Lines 241-244: __main__ block runs generate_gaugets(168) and logs results.      | (tmp_path / 'gaugets').exists()                                                | Pass   | 0.004s |
| Random module peak_hour_max must be ≤ simulation_hours.                         | DEFAULT_PARAMS["peak_hour_max"] < DEFAULT_PARAMS["simulat...                   | Pass   | 0.000s |
| gaugets_random.DEFAULT_PARAMS simulation_hours must be 168.                     | DEFAULT_PARAMS["simulation_hours"] == 168                                      | Pass   | 0.000s |

| Test Description                                                        | Acceptance Criteria                                                          | Result | Time   |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------|--------|--------|
| The generator uses input_dir.name as the catchment name.                | gen.input_dir.name == 'thames'                                               | Pass   | 0.000s |
| Daily flows filtered by years                                           | len(result_3yr["daily_flows"]) ><br>len(result_1yr["daily_fl...])            | Pass   | 0.027s |
| Explicit gauge id overrides meta-data                                   | result["station_metadata"]["station_id"] == "OVERRIDE-ID"                    | Pass   | 0.015s |
| Even with completely empty input, PDF should still generate.            | result.exists(); result.stat().st_size > 0                                   | Pass   | 0.007s |
| Minimal data (empty arrays) should produce valid PDF.                   | result.exists()                                                              | Pass   | 0.013s |
| Output file written                                                     | output_path.exists();<br>json.load(f)["data.type"] == "GaugeHistoricalDaily" | Pass   | 0.017s |
| Output path creates parent dirs                                         | result.exists()                                                              | Pass   | 0.022s |
| With populated data, PDF should have multiple pages (from Page-Breaks). | out.stat().st_size > 5000                                                    | Pass   | 0.022s |
| Pdf starts with magic bytes                                             | f.read(5) == b'%PDF-'                                                        | Pass   | 0.028s |
| Produces pdf file                                                       | result == out; out.exists() (+1 more)                                        | Pass   | 0.034s |
| Returns dict with required keys                                         | key in result                                                                | Pass   | 0.015s |
| Returns path                                                            | isinstance(result, Path)                                                     | Pass   | 0.023s |
| Avg basis is float                                                      | isinstance(stats["avg_basis_bps"], float)                                    | Pass   | 0.356s |
| Avg transmission rate between 0 and 1                                   | 0.0 <= stats["avg_transmission_rate"] <= 1.0                                 | Pass   | 0.238s |
| Min spread bps in stats                                                 | stats["min_spread_bps"] == MIN_PRS_SPREAD_BPS                                | Pass   | 0.100s |
| Total flood events counted                                              | stats["total_flood_events"] == 7                                             | Pass   | 0.477s |
| No gauges in portfolio returns empty list.                              | result == []                                                                 | Pass   | 0.002s |
| Line 50: error path extracts GaugeID from FloodGauge.Header.            | len(result) == 0; "GAUGE-BADONE" in caplog.text                              | Pass   | 0.002s |
| Line 50: completely empty entry uses UNKNOWN as gauge_id.               | len(result) == 0; "UNKNOWN" in caplog.text                                   | Pass   | 0.002s |
| Good gauges succeed even when bad ones fail.                            | len(result) == 1                                                             | Pass   | 0.217s |
| Lines 83-110: generate_all_gauge_histories processes all gauges.        | isinstance(result, list); len(result) == 1                                   | Pass   | 0.194s |
| Lines 105-107: error on one gauge -i logged, continues.                 | isinstance(result, list)                                                     | Pass   | 0.433s |

| Test Description                                             | Acceptance Criteria                                                                                                                                 | Result | Time   |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Line 60-61: house with 3 bedrooms returns 1 or 2 bathrooms.  | <code>results.issubset({1, 2});<br/>len(results) == 2</code>                                                                                        | Pass   | 0.000s |
| Line 62-63: house with 4+ bedrooms returns 2 or 3 bathrooms. | <code>results.issubset({2, 3});<br/>len(results) == 2</code>                                                                                        | Pass   | 0.000s |
| 5 bedrooms also takes the 4+ path (line 62-63).              | <code>val in (2, 3)</code>                                                                                                                          | Pass   | 0.000s |
| Advised flag                                                 | <code>isinstance(generate_boolean_value("AdvisedFlag",<br/>self._fd...</code>                                                                       | Pass   | 0.000s |
| Btl commercial purpose                                       | <code>generate_boolean_value("BusinessOrCommercialPurpose",<br/>sel...;<br/>generate_boolean_value("BusinessOrCommercialPurpose",<br/>sel...</code> | Pass   | 0.000s |
| Default flag from data                                       | <code>generate_boolean_value("DefaultFlag",<br/>self._fd(is_default...;<br/>generate_boolean_value("DefaultFlag",<br/>self._fd(is_default...</code> | Pass   | 0.000s |
| Execution only flag                                          | <code>isinstance(generate_boolean_value("ExecutionOnlyFlag",<br/>se...</code>                                                                       | Pass   | 0.000s |
| First time buyer btl always false                            | <code>generate_boolean_value("FirstTimeBuyerFlag",<br/>self._fd(mor...</code>                                                                       | Pass   | 0.000s |
| First time buyer expensive property                          | <code>isinstance(result, bool)</code>                                                                                                               | Pass   | 0.000s |
| First time buyer normal property                             | <code>isinstance(result, bool)</code>                                                                                                               | Pass   | 0.000s |
| In arrears flag                                              | <code>generate_boolean_value("InArrearsFlag",<br/>self._fd(is_in_ar...</code>                                                                       | Pass   | 0.000s |
| Mmr compliant                                                | <code>generate_boolean_value("MMRCompliantFlag",<br/>self._fd()) is...</code>                                                                       | Pass   | 0.000s |
| Stress compliant                                             | <code>generate_boolean_value("StressTestCompliantFlag",<br/>self.f...</code>                                                                        | Pass   | 0.000s |
| Unknown field returns bool                                   | <code>isinstance(generate_boolean_value("Unknown",<br/>self._fd()),...</code>                                                                       | Pass   | 0.000s |
| Count respected                                              | <code>len(result["data"]) == 3</code>                                                                                                               | Pass   | 0.000s |
| Returns dict                                                 | <code>isinstance(result, dict); "data" in<br/>result</code>                                                                                         | Pass   | 0.000s |
| Fair section 50 to 75 gets enhancement rec                   | <code>any("OtherSection" in rec for rec in<br/>r)</code>                                                                                            | Pass   | 0.000s |
| Good sections maintain recommendation                        | <code>any("Maintain" in rec or "maintain"<br/>in rec for rec in r)</code>                                                                           | Pass   | 0.000s |
| Max 5 recommendations                                        | <code>len(r) &lt;= 5</code>                                                                                                                         | Pass   | 0.000s |
| Poor section under 50 gets recommendation                    | <code>any("SomeSection" in rec for rec in<br/>r)</code>                                                                                             | Pass   | 0.000s |

| Test Description               | Acceptance Criteria                                                                      | Result | Time   |
|--------------------------------|------------------------------------------------------------------------------------------|--------|--------|
| Returns list                   | <code>isinstance(self._page().generate_elements(<del>None</del>, property), list)</code> | Pass   | 0.000s |
| Application date is string     | <code>isinstance(result, str); len(result) == 10</code>                                  | Pass   | 0.000s |
| Default date not defaulted     | <code>result is None</code>                                                              | Pass   | 0.000s |
| Default date when defaulted    | <code>isinstance(result, str)</code>                                                     | Pass   | 0.000s |
| Disbursal date                 | <code>isinstance(result, str)</code>                                                     | Pass   | 0.000s |
| Maturity date                  | <code>isinstance(result, str)</code>                                                     | Pass   | 0.000s |
| Unknown field                  | <code>1 &lt;= result &lt;= 10; isinstance(result, str) (+1 more)</code>                  | Pass   | 0.000s |
| Application property valuation | <code>generate_decimal_value("ApplicationPropertyValuation", se...)</code>               | Pass   | 0.000s |
| Aprc initial rate              | <code>isinstance(result, float)</code>                                                   | Pass   | 0.000s |
| Aprc secondary rate            | <code>isinstance(result, float)</code>                                                   | Pass   | 0.000s |
| Borrower income                | <code>generate_decimal_value("BorrowerIncome", self._fd()) == 7...</code>                | Pass   | 0.000s |
| Current balance                | <code>generate_decimal_value("CurrentBalance", self._fd()) == 2...</code>                | Pass   | 0.000s |
| Current lending rate           | <code>abs(result - 4.0) &lt; 0.01</code>                                                 | Pass   | 0.000s |
| Current ltv                    | <code>abs(result - 75.0) &lt; 0.01</code>                                                | Pass   | 0.000s |
| Current payment                | <code>generate_decimal_value("CurrentPayment", self._fd()) == 1500</code>                | Pass   | 0.000s |
| Debt to income ratio           | <code>isinstance(result, float)</code>                                                   | Pass   | 0.000s |
| Debt to income zero income     | <code>result == 0.3</code>                                                               | Pass   | 0.000s |
| Default field                  | <code>isinstance(result, float)</code>                                                   | Pass   | 0.000s |
| Interest rate                  | <code>isinstance(result, float)</code>                                                   | Pass   | 0.000s |
| Loan to value ratio            | <code>abs(result - 80.0) &lt; 0.01</code>                                                | Pass   | 0.000s |
| Ltv field                      | <code>abs(result - 75.0) &lt; 0.01</code>                                                | Pass   | 0.000s |
| Original lending rate          | <code>abs(result - 4.0) &lt; 0.01</code>                                                 | Pass   | 0.000s |
| Original loan                  | <code>generate_decimal_value("OriginalLoan", self._fd()) == 320000</code>                | Pass   | 0.000s |
| Original loan amount           | <code>generate_decimal_value("OriginalLoanAmount", self._fd()) ...</code>                | Pass   | 0.000s |
| Original ltv                   | <code>abs(result - 80.0) &lt; 0.01</code>                                                | Pass   | 0.000s |
| Outstanding balance            | <code>generate_decimal_value("OutstandingBalance", self._fd()) ...</code>                | Pass   | 0.000s |
| Purchase value                 | <code>generate_decimal_value("PurchaseValue", self._fd()) == 40...</code>                | Pass   | 0.000s |
| Empty page list                | <code>elements == []</code>                                                              | Pass   | 0.002s |
| Empty page list                | <code>elements == []</code>                                                              | Pass   | 0.001s |
| Empty page list returns empty  | <code>elements == []</code>                                                              | Pass   | 0.001s |

| Test Description                                       | Acceptance Criteria                                                                                                          | Result | Time   |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Exception in page continues                            | <code>isinstance(elements, list);</code><br><code>len(elements) &gt; 0</code>                                                | Pass   | 0.004s |
| If one page raises, the remainder are still processed. | <code>isinstance(elements, list);</code><br><code>len(elements) &gt; 0</code>                                                | Pass   | 0.001s |
| Exception in page module continues                     | <code>isinstance(elements, list);</code><br><code>len(elements) &gt; 0</code>                                                | Pass   | 0.002s |
| First page no page break                               | <code>not any(isinstance(e, PageBreak) for e in elements)</code>                                                             | Pass   | 0.001s |
| First page no page break                               | <code>not any(isinstance(e, PageBreak) for e in elements)</code>                                                             | Pass   | 0.002s |
| First page no page break                               | <code>not any(isinstance(e, PageBreak) for e in elements)</code>                                                             | Pass   | 0.001s |
| Mixed valid and unknown pages                          | <code>len(elements) &gt; 0</code>                                                                                            | Pass   | 0.001s |
| Mixed valid and unknown pages                          | <code>len(elements) &gt; 0</code>                                                                                            | Pass   | 0.002s |
| Mixed valid and unknown pages                          | <code>len(elements) &gt; 0</code>                                                                                            | Pass   | 0.001s |
| Mortgage data passed to page                           | —                                                                                                                            | Pass   | 0.002s |
| Multiple pages have page break                         | <code>any(isinstance(e, PageBreak) for e in elements)</code>                                                                 | Pass   | 0.002s |
| Multiple pages have page breaks                        | <code>any(isinstance(e, PageBreak) for e in elements)</code>                                                                 | Pass   | 0.002s |
| Multiple pages inserts page breaks                     | <code>has_break</code>                                                                                                       | Pass   | 0.001s |
| No gauge id key returns message                        | <code>any("No gauge ID" in t for t in texts)</code>                                                                          | Pass   | 0.000s |
| None timeseries passed to page                         | —                                                                                                                            | Pass   | 0.001s |
| Returns list                                           | <code>isinstance(elements, list)</code>                                                                                      | Pass   | 0.002s |
| Returns list                                           | <code>isinstance(result, list); len(result) &gt; 0</code>                                                                    | Pass   | 0.001s |
| Returns list                                           | <code>isinstance(elements, list)</code>                                                                                      | Pass   | 0.002s |
| Returns list                                           | <code>isinstance(elements, list)</code>                                                                                      | Pass   | 0.001s |
| Timeseries passed to page                              | —                                                                                                                            | Pass   | 0.002s |
| Unknown page skipped                                   | <code>len(elements) == 0</code>                                                                                              | Pass   | 0.001s |
| Unknown page skipped                                   | <code>len(elements) == 0</code>                                                                                              | Pass   | 0.002s |
| Unknown page skipped                                   | <code>isinstance(elements, list);</code><br><code>len(elements) == 0</code>                                                  | Pass   | 0.001s |
| With gauge id returns trading header                   | <code>any("Trading Activity" in t for t in texts)</code>                                                                     | Pass   | 0.001s |
| When eod_date is None, uses to-day's date.             | <code>snapshot['date'] ==</code><br><code>date.today().isoformat();</code><br><code>snapshot['eod_id'] == expected_id</code> | Pass   | 0.000s |

| Test Description                            | Acceptance Criteria                                                                         | Result | Time   |
|---------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| Empty property list returns zero mortgages  | result["data"]["mortgages"] == [];<br>result["processing_stats"]["total_mortgages"]<br>== 0 | Pass   | 0.003s |
| Missing property file raises file not found | —                                                                                           | Pass   | 0.000s |
| Boolean type                                | isinstance(result, bool)                                                                    | Pass   | 0.000s |
| Date type                                   | isinstance(result, str)                                                                     | Pass   | 0.000s |
| Decimal type                                | isinstance(result, (int, float))                                                            | Pass   | 0.000s |
| Enum type                                   | result == "Monthly"                                                                         | Pass   | 0.000s |
| Integer type                                | result == 300                                                                               | Pass   | 0.000s |
| Menu type                                   | result == "Monthly"                                                                         | Pass   | 0.000s |
| Non dict field def returns string           | result == "literal"                                                                         | Pass   | 0.000s |
| Non dict non string returns empty           | result == ""                                                                                | Pass   | 0.000s |
| Text type                                   | result == "thames"                                                                          | Pass   | 0.000s |
| Unknown type returns empty                  | result == ""                                                                                | Pass   | 0.000s |
| Contains flood risk report                  | "flood_risk_report" in<br>gen._generate_filename()                                          | Pass   | 0.001s |
| Contains gauge id                           | "GAUGE-001" in name                                                                         | Pass   | 0.001s |
| Contains property id                        | "PROP-001" in name                                                                          | Pass   | 0.002s |
| Contains timestamp                          | digits >= 8                                                                                 | Pass   | 0.001s |
| Ends with pdf                               | gen._generate_filename(gauge_data).endswith(".pdf")                                         | Pass   | 0.001s |
| Ends with pdf                               | gen._generate_filename(prop_data).endswith(".pdf")                                          | Pass   | 0.002s |
| Ends with pdf                               | gen._generate_filename().endswith(".pdf")                                                   | Pass   | 0.001s |
| Missing gauge id uses unknown               | "unknown" in name;<br>name.endswith(".pdf")                                                 | Pass   | 0.001s |
| Missing property id uses unknown            | "unknown" in name;<br>name.endswith(".pdf")                                                 | Pass   | 0.002s |
| Partial path missing gauge id               | "unknown" in name                                                                           | Pass   | 0.001s |
| Partial path missing header                 | "unknown" in name                                                                           | Pass   | 0.001s |
| Partial path missing header                 | "unknown" in name                                                                           | Pass   | 0.002s |
| Partial path missing property id            | "unknown" in name                                                                           | Pass   | 0.002s |
| Returns string                              | isinstance(name, str)                                                                       | Pass   | 0.001s |
| Returns string                              | isinstance(name, str)                                                                       | Pass   | 0.002s |
| Returns string                              | isinstance(name, str)                                                                       | Pass   | 0.001s |
| Timestamp in filename                       | digits >= 8                                                                                 | Pass   | 0.001s |
| Timestamp in filename                       | digits >= 8                                                                                 | Pass   | 0.002s |
| Two calls produce string names              | isinstance(n1, str) and<br>isinstance(n2, str)                                              | Pass   | 1.112s |
| Estimates value when zero                   | result["loan.amount"] > 0                                                                   | Pass   | 0.000s |
| Flood risk passed through                   | result["flood_risk"] == "High"                                                              | Pass   | 0.000s |



| Test Description                                      | Acceptance Criteria                                                                       | Result | Time   |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| Returns dict with mortgage id                         | "mortgage_id" in result;                                                                  | Pass   | 0.000s |
|                                                       | result["mortgage_id"].startswith("MORT-")                                                 |        |        |
| Uses provided property value                          | result["property_value"] == 600000                                                        | Pass   | 0.000s |
| When output_dir=None, uses config.get_gaugehd_dir().  | isinstance(result, dict)                                                                  | Pass   | 0.214s |
| Daily observations generated                          | len(result["daily_observations"]) > 0                                                     | Pass   | 0.212s |
| Gauge metadata extracted                              | result["gauge_metadata"]["gauge_id"]<br>== "GAUGE-TEST01"                                 | Pass   | 0.208s |
| Gauge data without FloodGauge wrapper.                | result["gauge_metadata"]["gauge_id"]<br>== "GAUGE-FLAT"                                   | Pass   | 0.352s |
| Output file created                                   | (tmp_path /<br>"gauge-GAUGE-TEST01_hd.json").exists()                                     | Pass   | 0.213s |
| Returns dict                                          | isinstance(result, dict)                                                                  | Pass   | 0.232s |
| Schema version                                        | result["schema_version"] == "1.0"                                                         | Pass   | 0.212s |
| When output_path=None, uses config.get_gaugehd_dir(). | isinstance(result, dict);<br>(tmp_path / "gaugehd" /<br>"gauge-AUTO-01_hd.json").exists() | Pass   | 0.001s |
| Creates json file                                     | output.exists()                                                                           | Pass   | 0.002s |
| Gauge id override                                     | result["station_metadata"]["station_id"]<br>== "GAUGE-CUSTOM"                             | Pass   | 0.003s |
| Returns dict                                          | isinstance(result, dict)                                                                  | Pass   | 0.002s |
| Schema version in output                              | result["schema_version"] ==<br>"1.0"; result["data_type"] ==<br>"GaugeHistoricalDaily"    | Pass   | 0.001s |
| Station metadata extracted                            | result["station_metadata"]["station_name"]<br>== "Thames at ..."                          | Pass   | 0.001s |
| Current balance does not exceed original loan         | loan["CurrentBalance"] <=<br>loan["OriginalLoanAmount"]                                   | Pass   | 0.003s |
| Loan details values are positive                      | loan["OriginalLoanAmount"] > 0;<br>loan["CurrentBalance"] > 0 (+3 more)                   | Pass   | 0.002s |
| Ltv within valid range                                | 10 <=<br>m["Mortgage"]["LoanDetails"]["LTV"]<br>≤ 100                                     | Pass   | 0.002s |
| Mortgage data key is list                             | isinstance(result["data"]["mortgages"],<br>list)                                          | Pass   | 0.002s |
| Mortgage data structure has mortgage key              | "Mortgage" in m                                                                           | Pass   | 0.002s |
| Mortgage header has required fields                   | "MortgageID" in header; "PropertyID"<br>in header (+1 more)                               | Pass   | 0.002s |
| Mortgage ids key is list                              | isinstance(result["data"]["mortgage_ids"],<br>list)                                       | Pass   | 0.002s |

| Test Description                                                                                                   | Acceptance Criteria                                                                         | Result | Time   |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| Output file is valid json                                                                                          | <code>isinstance(json.load(f), dict)</code>                                                 | Pass   | 0.003s |
| Output json contains generation metadata                                                                           | <code>"generated_at" in meta; "catchment" in meta (+1 more)</code>                          | Pass   | 0.003s |
| Output json contains mortgages key                                                                                 | <code>"mortgages" in json.load(f)</code>                                                    | Pass   | 0.004s |
| Processing stats contain timing                                                                                    | <code>stats["start_time"] is not None;</code><br><code>stats["end_time"] is not None</code> | Pass   | 0.002s |
| Gauge reading returns all fields                                                                                   | <code>"gaugeId" in reading; "waterLevel" in reading (+1 more)</code>                        | Pass   | 0.000s |
| Gauge reading with string timestamp                                                                                | <code>reading["timestamp"] == "2024-01-01T00:00:00"</code>                                  | Pass   | 0.000s |
| <code>report_type='analysis'</code> calls <code>generate_analysis_report</code> .                                  | —                                                                                           | Pass   | 0.001s |
| <code>auto_open=True</code> with <code>open_pdf_file</code> raising <code>-i</code> logs warning, does not raise.  | <code>result is not None</code>                                                             | Pass   | 0.001s |
| <code>auto_open=True</code> tries to open the PDF (lines 95-106).                                                  | <code>mock_open.call_count &gt;= 0</code>                                                   | Pass   | 0.001s |
| <code>report_type='basic'</code> calls <code>generate_basic_report</code> .                                        | —                                                                                           | Pass   | 0.001s |
| Gauge not found returns 404                                                                                        | <code>r.status_code == 404; data["status"] == "error"</code>                                | Pass   | 0.002s |
| The <code>/generate_gauge_report</code> alias also works.                                                          | <code>r.status_code == 200</code>                                                           | Pass   | 0.021s |
| Missing gauge id returns 400                                                                                       | <code>r.status_code == 400; data["status"] == "error"</code>                                | Pass   | 0.000s |
| <code>report_type='monitoring'</code> with <code>timeseries</code> calls <code>generate_monitoring_report</code> . | —                                                                                           | Pass   | 0.001s |
| No json returns 400                                                                                                | <code>r.status_code in (400, 415, 500)</code>                                               | Pass   | 0.000s |
| Sending JSON null body makes <code>get_json()</code> return None → line 29.                                        | <code>r.status_code == 400</code>                                                           | Pass   | 0.000s |
| Options returns ok                                                                                                 | <code>r.status_code == 200</code>                                                           | Pass   | 0.000s |
| Response has file path                                                                                             | <code>"file_path" in data;</code><br><code>data["file_path"].endswith(".pdf")</code>        | Pass   | 0.021s |
| Response has pdf base64                                                                                            | <code>"pdf_base64" in data;</code><br><code>len(data["pdf_base64"]) &gt; 0</code>           | Pass   | 0.019s |
| Return value is the PDF path.                                                                                      | <code>result == tmp_path / "gauge_report.pdf"</code>                                        | Pass   | 0.001s |
| Unknown <code>report_type</code> calls <code>generate_report</code> .                                              | —                                                                                           | Pass   | 0.001s |

| Test Description                                                                                          | Acceptance Criteria                                                         | Result | Time   |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------|--------|
| Valid gauge generates report                                                                              | <code>r.status_code == 200; data["status"] == "success"</code>              | Pass   | 0.021s |
| gaugehd file exists → exercises lines 62-69.                                                              | <code>r.status_code == 200</code>                                           | Pass   | 0.106s |
| gaugehc.json exists → exercises lines 76-82.                                                              | <code>r.status_code == 200</code>                                           | Pass   | 0.050s |
| Analysis type calls generate analysis                                                                     | —                                                                           | Pass   | 0.001s |
| Auto open false returns path                                                                              | <code>result == fake_pdf</code>                                             | Pass   | 0.001s |
| Auto open true handles exception                                                                          | <code>result is not None</code>                                             | Pass   | 0.001s |
| Lines 343-344: <code>auto_open=True</code> + <code>open_pdf_file</code> succeeds → both lines covered.    | <code>result == fake_pdf</code>                                             | Pass   | 0.001s |
| Basic type calls generate basic                                                                           | —                                                                           | Pass   | 0.001s |
| Monitoring type with timeseries calls monitoring                                                          | —                                                                           | Pass   | 0.001s |
| Monitoring without timeseries falls back                                                                  | —                                                                           | Pass   | 0.001s |
| Returns path                                                                                              | <code>result == fake_pdf</code>                                             | Pass   | 0.001s |
| Unknown type calls generate report                                                                        | —                                                                           | Pass   | 0.001s |
| Lines 236-237: <code>generate_gaugets()</code> instantiates generator and calls <code>generate()</code> . | <code>result['data']['num_gauges'] == 5</code>                              | Pass   | 0.000s |
| Creates output file                                                                                       | <code>output.exists()</code>                                                | Pass   | 0.002s |
| History entries have date                                                                                 | <code>'date' in entry</code>                                                | Pass   | 0.002s |
| Metadata lists gauges                                                                                     | <code>'GAUGE-005' in data['metadata']['gauges']</code>                      | Pass   | 0.002s |
| Multiple gauges                                                                                           | <code>'GAUGE-001' in data['history']; 'GAUGE-002' in data['history']</code> | Pass   | 0.002s |
| No eod files returns early                                                                                | <code>not output.exists()</code>                                            | Pass   | 0.000s |
| Output has history                                                                                        | <code>'history' in data; 'GAUGE-001' in data['history']</code>              | Pass   | 0.002s |
| Output has metadata                                                                                       | <code>'metadata' in data; data['metadata']['num_snapshots'] == 2</code>     | Pass   | 0.002s |
| Tenors sorted numerically                                                                                 | <code>len(entries) == 1; tenor_keys == sorted(tenor_keys, key=int)</code>   | Pass   | 0.002s |
| Two eod files two history entries                                                                         | <code>len(entries) == 2</code>                                              | Pass   | 0.003s |
| Empty dir produces no file                                                                                | <code>not output.exists(); output.exists()</code>                           | Pass   | 0.000s |
| History entries have date                                                                                 | <code>entry["date"] == "2026-01-02"</code>                                  | Pass   | 0.003s |

| Test Description                                                          | Acceptance Criteria                                                                                              | Result | Time   |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------|--------|
| History entries have tenors                                               | "1" in entry and "2" in entry and<br>"3" in entry                                                                | Pass   | 0.003s |
| History keyed by gauge                                                    | "GAUGE-042" in data["history"]                                                                                   | Pass   | 0.003s |
| History keyed by trigger                                                  | "alert" in gauge_hist; "warning" in<br>gauge_hist                                                                | Pass   | 0.002s |
| Metadata has generated at                                                 | "generated_at" in data["metadata"];<br>"generated_at" in data["metadata"]                                        | Pass   | 0.002s |
| Metadata num snapshots                                                    | data["metadata"]["num_snapshots"] ==<br>3; data["metadata"]["num_snapshots"]<br>== 3                             | Pass   | 0.004s |
| Multiple gauges in metadata                                               | set(data["metadata"]["gauges"]) ==<br>{ "GAUGE-001", "GAUGE-0...                                                 | Pass   | 0.002s |
| Multiple snapshots produce multi-<br>ple entries                          | len(alert_entries) == 3                                                                                          | Pass   | 0.004s |
| Output has metadata and history                                           | "metadata" in data; "history" in data                                                                            | Pass   | 0.003s |
| Single snapshot creates file                                              | output.exists(); output.exists()                                                                                 | Pass   | 0.002s |
| Standalone check: _business_days<br>always returns Mon-Fri dates.         | len(result) == 63; d.weekday() < 5                                                                               | Pass   | 0.000s |
| Pre-existing EOD-*.json files are<br>deleted before simulation.           | not stale.exists()                                                                                               | Pass   | 0.004s |
| Creates eod dir                                                           | (trading_dir / 'eod').exists()                                                                                   | Pass   | 0.002s |
| Empty trades returns zero                                                 | count == 0                                                                                                       | Pass   | 0.002s |
| A trade that fails enrich_trade is<br>skipped gracefully (except branch). | count == 0                                                                                                       | Pass   | 0.017s |
| More trades should generally yield<br>more (or equal) snapshots.          | count > 0                                                                                                        | Pass   | 0.166s |
| Returns integer                                                           | isinstance(result, int)                                                                                          | Pass   | 0.002s |
| Same seed should produce same ran-<br>dom walk (deterministic).           | count1 == count2                                                                                                 | Pass   | 0.004s |
| With a real trade + real engine, we<br>expect 0 snapshots.                | count > 0                                                                                                        | Pass   | 0.124s |
| EOD snapshot JSON files are cre-<br>ated in eod/ directory.               | len(eod_files) == count                                                                                          | Pass   | 0.135s |
| hazard_curve_history.json and<br>trade_pnl_history.json are created.      | (trading_dir /<br>'hazard_curve_history.json').exists();<br>(trading_dir /<br>'trade_pnl_history.json').exists() | Pass   | 0.163s |
| Arrears months current                                                    | generate_integer_value("ArrearsMonths",<br>self._fd()) == 0                                                      | Pass   | 0.000s |
| Arrears months in arrears                                                 | 1 <= result <= 6                                                                                                 | Pass   | 0.000s |
| Behavioral score                                                          | 30 <= result <= 100                                                                                              | Pass   | 0.000s |

| Test Description                                                     | Acceptance Criteria                                       | Result | Time   |
|----------------------------------------------------------------------|-----------------------------------------------------------|--------|--------|
| Borrower age btl                                                     | 35 <= result <= 65                                        | Pass   | 0.000s |
| Borrower age expensive                                               | 35 <= result <= 65                                        | Pass   | 0.000s |
| Borrower age normal                                                  | 25 <= result <= 60                                        | Pass   | 0.000s |
| Credit score expensive property                                      | 700 <= result <= 800                                      | Pass   | 0.000s |
| Credit score low rate                                                | 750 <= result <= 850                                      | Pass   | 0.000s |
| Credit score normal                                                  | 650 <= result <= 750                                      | Pass   | 0.000s |
| Days in arrears current                                              | generate_integer_value("DaysInArrears", self._fd()) == 0  | Pass   | 0.000s |
| Days in arrears defaulted                                            | 90 <= result <= 365                                       | Pass   | 0.000s |
| Days in arrears in arrears                                           | 1 <= result <= 89                                         | Pass   | 0.000s |
| Family members                                                       | result in [1, 2, 3, 4]                                    | Pass   | 0.000s |
| Loan term                                                            | generate_integer_value("LoanTerm", self._fd()) == 300     | Pass   | 0.000s |
| Number of borrowers                                                  | result in [1, 2]                                          | Pass   | 0.000s |
| Original term                                                        | generate_integer_value("OriginalTerm", self._fd()) == 300 | Pass   | 0.000s |
| Prepayment risk high rate                                            | 7 <= result <= 10                                         | Pass   | 0.000s |
| Prepayment risk low rate                                             | 1 <= result <= 5                                          | Pass   | 0.000s |
| Remaining term                                                       | result == 240                                             | Pass   | 0.000s |
| Total payments                                                       | result == 60                                              | Pass   | 0.000s |
| Unknown field                                                        | 1 <= result <= 10; isinstance(result, str) (+1 more)      | Pass   | 0.000s |
| Each location has required keys                                      | "lat" in loc; "lon" in loc (+4 more)                      | Pass   | 0.000s |
| Elevation is non negative                                            | loc["elevation"] >= 0                                     | Pass   | 0.000s |
| With j 3 gauge points, fallback generator is called.                 | len(locs) == 3                                            | Pass   | 0.000s |
| Fallback used when no gauge points                                   | len(locs) == 3                                            | Pass   | 0.000s |
| When primary_idx == 0, triangle is gauges 0, 1, 2 -i sorted [0,1,2]. | set(ref) == {0, 1, 2}                                     | Pass   | 0.000s |
| Reference gauge indices sorted                                       | idxs == sorted(idxs)                                      | Pass   | 0.001s |
| Reference gauge indices three elements                               | len(loc["reference_gauge_indices"]) == 3                  | Pass   | 0.000s |
| Returns correct count                                                | len(locs) == 10                                           | Pass   | 0.000s |
| When params has no get_elevation, elevation is sampled from          | len(locs) == 5; 0 <= loc["elevation"] <= 40               | Pass   | 0.000s |
| Application channel                                                  | result in ["Retail", "Broker", "Correspondent"]           | Pass   | 0.000s |
| Loan purpose btl                                                     | result in ["Purchase", "Refinancing"]                     | Pass   | 0.000s |
| Loan purpose residential                                             | result in ["Purchase", "Refinancing", "Home Improvement"] | Pass   | 0.000s |

| Test Description                                                              | Acceptance Criteria                                                                   | Result | Time   |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|--------|
| Mortgage type                                                                 | <code>result == "Buy-to-Let"</code>                                                   | Pass   | 0.000s |
| Occupancy type                                                                | <code>result == "Investment"</code>                                                   | Pass   | 0.000s |
| Original rate type                                                            | <code>isinstance(result, str) and len(result) &gt; 0</code>                           | Pass   | 0.000s |
| Payment frequency                                                             | <code>generate_menu_value("PaymentFrequency", {}, 0, self._fd()...</code>             | Pass   | 0.000s |
| Repayment type btl                                                            | <code>result in ["Interest only", "Repayment", "Part and part"]</code>                | Pass   | 0.000s |
| Repayment type residential                                                    | <code>result in ["Repayment", "Interest only", "Part and part"]</code>                | Pass   | 0.000s |
| Unknown field no options                                                      | <code>result == ""</code>                                                             | Pass   | 0.000s |
| Unknown field with options                                                    | <code>result in ["X", "Y"]</code>                                                     | Pass   | 0.000s |
| Convenience function creates pdf                                              | <code>output.exists()</code>                                                          | Pass   | 0.010s |
| Convenience function returns path                                             | <code>isinstance(result, Path)</code>                                                 | Pass   | 0.012s |
| Lines 203-208: exception → 500.                                               | <code>r.status_code in (200, 500)</code>                                              | Pass   | 0.001s |
| Lines 174-193: full success path with mocked mortgage report generator.       | <code>r.status_code in (200, 500)</code>                                              | Pass   | 0.002s |
| Missing property id returns 400                                               | <code>r.status_code == 400; "Property ID" in r.get_json()["message"] (+2 more)</code> | Pass   | 0.000s |
| Mortgage not found returns 404                                                | <code>r.status_code == 404; "mortgage" in r.get_json()["message"].lower()</code>      | Pass   | 0.001s |
| Line 149: <code>get_json()</code> returns None → 400.                         | <code>r.status_code in (400, 415, 500); r.status_code in (400, 415, 500)</code>       | Pass   | 0.000s |
| No json returns error                                                         | <code>r.status_code in (400, 415); r.status_code in (400, 415)</code>                 | Pass   | 0.000s |
| Options returns ok                                                            | <code>r.status_code == 200; r.get_json()["status"] == "ok" (+2 more)</code>           | Pass   | 0.000s |
| Property not found returns 404                                                | <code>r.status_code == 404; r.status_code == 404</code>                               | Pass   | 0.001s |
| Mortgage import error returns 500                                             | <code>r.status_code == 500; data["status"] == "error"</code>                          | Pass   | 0.001s |
| Raises without property file                                                  | —                                                                                     | Pass   | 0.001s |
| Returns dict when property file exists                                        | <code>isinstance(result, dict); "data" in result (+1 more)</code>                     | Pass   | 0.022s |
| Lines 332-334: <code>open_pdf_file</code> raises -i warning logged, no raise. | <code>result is not None</code>                                                       | Pass   | 0.001s |
| Lines 335-336: <code>auto_open=False</code> -i skips opening.                 | —                                                                                     | Pass   | 0.001s |



| Test Description                                                | Acceptance Criteria                                                         | Result | Time   |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------|--------|--------|
| Lines 327-334: auto_open=True -i calls open_pdf_file.           | mock_open.called                                                            | Pass   | 0.001s |
| Lines 128-133: exception in report generation → 500.            | r.status_code in (500, 404, 400)                                            | Pass   | 0.002s |
| Lines 94-119: full success path with mocked report generator.   | r.status_code in (200, 500);<br>data["status"] == "success" (+1 more)       | Pass   | 0.002s |
| Missing property id returns 400                                 | r.status_code == 400; "Property ID" in<br>r.get_json()["message"] (+2 more) | Pass   | 0.000s |
| Lines 319-320: 'mortgage-focused' with mortgage_data.           | —                                                                           | Pass   | 0.001s |
| Line 74: get_json() returns None → 400.                         | r.status_code in (400, 415, 500);<br>r.status_code in (400, 415, 500)       | Pass   | 0.000s |
| No json returns error                                           | r.status_code in (400, 415);<br>r.status_code in (400, 415)                 | Pass   | 0.000s |
| Options returns ok                                              | r.status_code == 200;<br>r.get_json()["status"] == "ok" (+2 more)           | Pass   | 0.000s |
| Property not found returns 404                                  | r.status_code == 404; r.status_code == 404                                  | Pass   | 0.001s |
| Line 317-318: 'property-only' -i generate_property_only_report. | —                                                                           | Pass   | 0.001s |
| Lines 321-322: 'risk-focused'.                                  | —                                                                           | Pass   | 0.001s |
| Lines 323-324: unknown type -i generate_report.                 | —                                                                           | Pass   | 0.001s |
| Auto open exception does not raise                              | result is not None                                                          | Pass   | 0.001s |
| Auto open false skips open                                      | —                                                                           | Pass   | 0.001s |
| Auto open true calls open                                       | —                                                                           | Pass   | 0.001s |
| Full type falls back to generate report                         | —                                                                           | Pass   | 0.001s |
| Mortgage focused type with mortgage                             | —                                                                           | Pass   | 0.001s |
| Mortgage focused without mortgage falls back                    | —                                                                           | Pass   | 0.001s |
| Property only type                                              | —                                                                           | Pass   | 0.001s |
| Returns path                                                    | result == fake_pdf                                                          | Pass   | 0.001s |
| Risk focused type                                               | —                                                                           | Pass   | 0.001s |
| Unknown type falls back to generate report                      | —                                                                           | Pass   | 0.001s |
| Line 167: log milestone every 50 properties.                    | any("50" in m for m in log_messages)                                        | Pass   | 0.013s |

| Test Description                                            | Acceptance Criteria                                                                  | Result | Time   |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|--------|
| Line 171: ratio branch when total_storms_at_gauge $\neq$ 0. | "gauge_to_property_ratio" in result;<br>result["gauge_to_property_ratio"] $\geq$ 0.0 | Pass   | 0.202s |
| Line 176: ratio=0 branch when total_storms_at_gauge == 0.   | result["gauge_to_property_ratio"] == 0.0                                             | Pass   | 0.006s |
| All items are strings                                       | all(isinstance(r, str) for r in recs)                                                | Pass   | 0.000s |
| Deep flood extra items                                      | len(deep) > len(shallow)                                                             | Pass   | 0.000s |
| High risk four or more                                      | len(recs) $\geq$ 4                                                                   | Pass   | 0.000s |
| High risk mentions flood insurance                          | any("flood insurance" in r.lower() for r in recs)                                    | Pass   | 0.000s |
| Low risk                                                    | len(recs) $\geq$ 2                                                                   | Pass   | 0.000s |
| Medium risk                                                 | len(recs) $\geq$ 3                                                                   | Pass   | 0.000s |
| Very low risk                                               | len(recs) $\geq$ 1                                                                   | Pass   | 0.000s |
| All unknown pages still creates pdf                         | path.exists()                                                                        | Pass   | 0.002s |
| All unknown pages still creates pdf                         | path.exists()                                                                        | Pass   | 0.002s |
| Auto select called when pages none                          | —                                                                                    | Pass   | 0.011s |
| Auto select called when pages none                          | —                                                                                    | Pass   | 0.015s |
| Auto select not called when pages provided                  | —                                                                                    | Pass   | 0.003s |
| Auto select not called when pages provided                  | —                                                                                    | Pass   | 0.003s |
| Auto select pages when none                                 | —                                                                                    | Pass   | 0.012s |
| Auto select skipped when provided                           | —                                                                                    | Pass   | 0.003s |
| Custom output filename                                      | path.name == "my_gauge.pdf";<br>path.exists()                                        | Pass   | 0.011s |
| Custom output filename                                      | path.name == "my_prop.pdf";<br>path.exists()                                         | Pass   | 0.015s |
| Custom output filename                                      | path.name == "my_report.pdf";<br>path.exists()                                       | Pass   | 0.012s |
| Custom pages to include                                     | path.exists()                                                                        | Pass   | 0.005s |
| Custom pages to include                                     | path.exists()                                                                        | Pass   | 0.005s |
| Custom pages to include                                     | path.exists()                                                                        | Pass   | 0.007s |
| Empty flood data                                            | path.exists()                                                                        | Pass   | 0.012s |
| Empty gauge data                                            | path.exists()                                                                        | Pass   | 0.009s |
| Empty property data                                         | path.exists()                                                                        | Pass   | 0.014s |
| Full data report                                            | path.exists(); path.stat().st_size > 0                                               | Pass   | 0.021s |
| Full data report                                            | path.exists(); path.stat().st_size > 0                                               | Pass   | 0.018s |
| Pdf exists                                                  | gen.generate_report(gauge_data).exists()                                             | Pass   | 0.011s |
| Pdf exists                                                  | gen.generate_report(prop_data).exists()                                              | Pass   | 0.015s |
| Pdf exists                                                  | path.exists()                                                                        | Pass   | 0.012s |

| Test Description                                               | Acceptance Criteria                                                           | Result | Time   |
|----------------------------------------------------------------|-------------------------------------------------------------------------------|--------|--------|
| Pdf is bytes                                                   | <code>isinstance(content, bytes);</code><br><code>len(content) &gt; 0</code>  | Pass   | 0.006s |
| Pdf non empty                                                  | <code>path.stat().st_size &gt; 0</code>                                       | Pass   | 0.012s |
| Pdf non empty                                                  | <code>path.stat().st_size &gt; 0</code>                                       | Pass   | 0.016s |
| Pdf non empty                                                  | <code>path.stat().st_size &gt; 0</code>                                       | Pass   | 0.012s |
| Pdf suffix                                                     | <code>gen.generate_report(gauge_data).suffix</code><br><code>== ".pdf"</code> | Pass   | 0.011s |
| Pdf suffix                                                     | <code>gen.generate_report(prop_data).suffix</code><br><code>== ".pdf"</code>  | Pass   | 0.018s |
| Pdf suffix                                                     | <code>path.suffix == ".pdf"</code>                                            | Pass   | 0.012s |
| Returns path                                                   | <code>isinstance(result, Path)</code>                                         | Pass   | 0.011s |
| Returns path                                                   | <code>isinstance(result, Path)</code>                                         | Pass   | 0.015s |
| Returns path                                                   | <code>isinstance(result, Path)</code>                                         | Pass   | 0.012s |
| With mortgage data                                             | <code>path.exists()</code>                                                    | Pass   | 0.021s |
| With timeseries data                                           | <code>path.exists()</code>                                                    | Pass   | 0.016s |
| gauge_integrator should handle flood_gauges key too.           | <code>result is not None</code>                                               | Pass   | 0.001s |
| Found gauge returns path                                       | <code>result is not None;</code><br><code>isinstance(result, Path)</code>     | Pass   | 0.001s |
| Gauge not found returns none                                   | <code>result is None</code>                                                   | Pass   | 0.002s |
| Missing file returns none                                      | <code>result is None</code>                                                   | Pass   | 0.000s |
| Output contains gauge id                                       | <code>"GAUGE-001" in content</code>                                           | Pass   | 0.003s |
| Output dir created                                             | <code>output_dir.exists()</code>                                              | Pass   | 0.001s |
| Output file created                                            | <code>result.exists()</code>                                                  | Pass   | 0.003s |
| Corrupt json returns none                                      | <code>result is None</code>                                                   | Pass   | 0.005s |
| Non json file returns none                                     | <code>result is None</code>                                                   | Pass   | 0.001s |
| Found property returns path                                    | <code>result is not None;</code><br><code>isinstance(result, Path)</code>     | Pass   | 0.002s |
| Missing property file returns none                             | <code>result is None</code>                                                   | Pass   | 0.002s |
| No mortgage file still generates                               | <code>result is not None</code>                                               | Pass   | 0.001s |
| Output contains property id                                    | <code>"PROP-001" in content</code>                                            | Pass   | 0.004s |
| Output dir created                                             | <code>output_dir.exists()</code>                                              | Pass   | 0.001s |
| Output file created                                            | <code>result.exists()</code>                                                  | Pass   | 0.002s |
| Property not found returns none                                | <code>result is None</code>                                                   | Pass   | 0.001s |
| A file with invalid JSON triggers an exception -j return None. | <code>result is None</code>                                                   | Pass   | 0.001s |
| Simulated write permission error returns None.                 | <code>result is None</code>                                                   | Pass   | 0.002s |
| Import error returns 500                                       | <code>r.status_code == 500; data["status"]</code><br><code>== "error"</code>  | Pass   | 0.008s |
| Has report generated timestamp                                 | <code>"Report Generated" in r</code>                                          | Pass   | 0.000s |

| Test Description                                                          | Acceptance Criteria                                                                   | Result | Time   |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|--------|
| Last updated extracted                                                    | <code>r["Data Last Updated"] == "2025-01-01"</code>                                   | Pass   | 0.000s |
| Mortgage id extracted                                                     | <code>r.get("Mortgage ID") == "MORT-001"</code>                                       | Pass   | 0.000s |
| Property id extracted                                                     | <code>r["Property ID"] == "PROP-999"</code>                                           | Pass   | 0.000s |
| Returns dict                                                              | <code>isinstance(r, dict); isinstance(r, dict)</code>                                 | Pass   | 0.000s |
| Scope combined with mortgage                                              | <code>"Mortgage" in r["Analysis Scope"]</code>                                        | Pass   | 0.000s |
| Scope property only                                                       | <code>"Property-Only" in r["Analysis Scope"]</code>                                   | Pass   | 0.000s |
| Unknown property id when missing                                          | <code>r["Property ID"] == "Unknown"</code>                                            | Pass   | 0.000s |
| Upnrn extracted                                                           | <code>r["UPRN"] == "123456789"</code>                                                 | Pass   | 0.000s |
| Lines 263-264: 'analysis' type.                                           | <code>path.exists()</code>                                                            | Pass   | 0.017s |
| Line 257-258: 'basic' type.                                               | <code>path.exists()</code>                                                            | Pass   | 0.017s |
| Lines 259-260: 'detailed' type.                                           | <code>path.exists()</code>                                                            | Pass   | 0.022s |
| Lines 261-262: 'summary' type.                                            | <code>path.exists()</code>                                                            | Pass   | 0.009s |
| Lines 265-266: unknown type → generate_report.                            | <code>path.exists()</code>                                                            | Pass   | 0.022s |
| Analysis type returns path                                                | <code>path.exists()</code>                                                            | Pass   | 0.010s |
| Basic type returns path                                                   | <code>isinstance(path, Path); path.exists()</code>                                    | Pass   | 0.010s |
| Omitting report_type should default to 'basic'.                           | —                                                                                     | Pass   | 0.011s |
| Detailed type returns path                                                | <code>path.exists()</code>                                                            | Pass   | 0.012s |
| Output dir as string                                                      | <code>path.exists()</code>                                                            | Pass   | 0.010s |
| Pdf is non empty bytes on disk                                            | <code>isinstance(content, bytes); len(content) &gt; 0</code>                          | Pass   | 0.015s |
| Summary type returns path                                                 | <code>path.exists()</code>                                                            | Pass   | 0.008s |
| Unknown type falls back to generate report                                | <code>path.exists()</code>                                                            | Pass   | 0.012s |
| Empty gaugets directory must raise FileNotFoundError.                     | —                                                                                     | Pass   | 0.003s |
| Each index entry must have gauge_ids_alert for directory-based filtering. | <code>"gauge_ids_alert" in entry; "GAUGE-00000001" in entry["gauge_ids_alert"]</code> | Pass   | 0.002s |
| Index has required top level keys                                         | <code>key in data</code>                                                              | Pass   | 0.002s |
| Individual storm files must contain the full gauge_responses array.       | <code>"gauge_responses" in storm; len(storm["gauge_responses"]) == 1</code>           | Pass   | 0.001s |
| Storm id prefix uses config constant                                      | <code>data["storm_id_prefix"] == STORM_ID_PREFIX</code>                               | Pass   | 0.002s |
| Storms where no gauge exceeded alert must NOT appear in output.           | <code>result["total_storms"] == 0</code>                                              | Pass   | 0.001s |

| Test Description                                                                | Acceptance Criteria                                                                 | Result | Time   |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------|--------|
| Total storms matches list length                                                | <code>data["total_storms"] == len(data["storms"])</code>                            | Pass   | 0.003s |
| Output directory must contain index.json and per-storm files.                   | <code>(out_dir / "_index.json").exists(); (out_dir / f"{sid}.json").exists()</code> | Pass   | 0.002s |
| Gauge summary gauge ids                                                         | <code>len(d["gauge_ids"]) == 3</code>                                               | Pass   | 0.000s |
| Gauge summary json written                                                      | <code>(gauge_dir / GAUGE_SUMMARY_FILENAME).exists()</code>                          | Pass   | 0.000s |
| Gauge summary num gauges                                                        | <code>d["num_gauges"] == 3</code>                                                   | Pass   | 0.000s |
| Gauge summary schema version                                                    | <code>d["schema_version"] == SCHEMA_VERSION_SPATIAL</code>                          | Pass   | 0.002s |
| Gauge summary sequence count                                                    | <code>len(d["sequences"]) == 40</code>                                              | Pass   | 0.000s |
| Sequences file loadable                                                         | <code>len(seqs) == 40</code>                                                        | Pass   | 0.001s |
| Sequences json written                                                          | <code>(gauge_dir / SEQUENCES_FILENAME).exists()</code>                              | Pass   | 0.000s |
| Summary json written                                                            | <code>(gauge_dir / SUMMARY_FILENAME).exists()</code>                                | Pass   | 0.000s |
| Alert count non negative                                                        | <code>full_run["alert_sequences"] &gt;= 0</code>                                    | Pass   | 0.000s |
| Elapsed positive                                                                | <code>full_run["elapsed_seconds"] &gt; 0</code>                                     | Pass   | 0.000s |
| Num gauges                                                                      | <code>full_run["num_gauges"] == 3</code>                                            | Pass   | 0.000s |
| Num sequences                                                                   | <code>full_run["num_sequences"] == 40</code>                                        | Pass   | 0.000s |
| Returns dict                                                                    | <code>isinstance(full_run, dict)</code>                                             | Pass   | 0.000s |
| Severe lte warning                                                              | <code>full_run["severe_sequences"] &lt;= full_run["warning_sequenc...</code>        | Pass   | 0.000s |
| Type counts present                                                             | <code>"type_counts" in full_run; sum(full_run["type_counts"].values()) == 40</code> | Pass   | 0.000s |
| Warning lte alert                                                               | <code>full_run["warning_sequences"] &lt;= full_run["alert_sequences"]</code>        | Pass   | 0.000s |
| All levels positive                                                             | <code>all(obs["level_meters"] &gt; 0 for obs in result)</code>                      | Pass   | 0.006s |
| Correct length approximately                                                    | <code>728 &lt;= len(result) &lt;= 732</code>                                        | Pass   | 0.003s |
| Dates are sequential                                                            | <code>(d_curr - d_prev).days == 1</code>                                            | Pass   | 0.003s |
| Different seeds different output                                                | <code>levels1 != levels2</code>                                                     | Pass   | 0.012s |
| Each obs has date and level                                                     | <code>"date" in obs; "level_meters" in obs (+1 more)</code>                         | Pass   | 0.001s |
| Gauge data without FloodGauge wrapper uses defaults.                            | <code>isinstance(result, list)</code>                                               | Pass   | 0.001s |
| When historical high date is within range, that day gets historical_high level. | <code>match[0]["level_meters"] == 7.5</code>                                        | Pass   | 0.001s |
| Invalid date format doesn't crash.                                              | <code>isinstance(result, list)</code>                                               | Pass   | 0.001s |

| Test Description                                                                    | Acceptance Criteria                                                               | Result | Time   |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------|--------|
| Historical high date before start_date is ignored.                                  | <code>isinstance(result, list)</code>                                             | Pass   | 0.001s |
| Lines 176-179: Historical high injected via fallback when date not matched in loop. | <code>len(match) == 1;</code><br><code>match[0]["level_meters"] == 8.0</code>     | Pass   | 0.001s |
| Gauge data with no FloodStages uses sensible defaults.                              | <code>isinstance(result, list); len(result) &gt; 0</code>                         | Pass   | 0.001s |
| Lines 97-100: Partially tidal gauge uses dampened tidal signal.                     | <code>len(result) &gt; 700;</code> <code>all(lev &gt; 0 for lev in levels)</code> | Pass   | 0.003s |
| Returns list                                                                        | <code>isinstance(result, list)</code>                                             | Pass   | 0.001s |
| Seed reproducible                                                                   | <code>r1 == r2</code>                                                             | Pass   | 0.003s |
| Lines 92-95: Tidal gauge uses wider daily_std, tidal_amplitude $\neq$ 0.            | <code>len(result) &gt; 700;</code> <code>max(levels) &gt; min(levels)</code>      | Pass   | 0.003s |
| Lines 147-149: Tidal component with spring-neap cycle.                              | <code>l1 != l2</code>                                                             | Pass   | 0.003s |
| Account status defaulted                                                            | <code>result == "Default"</code>                                                  | Pass   | 0.000s |
| Account status normal                                                               | <code>result in ["Current", "Arrears", "Closed"]</code>                           | Pass   | 0.000s |
| Borrower employment btl                                                             | <code>isinstance(result, str)</code>                                              | Pass   | 0.000s |
| Catchment id                                                                        | <code>generate_text_value("CatchmentID", 0, self._fd()) == "tha..."</code>        | Pass   | 0.000s |
| Currency                                                                            | <code>generate_text_value("currency", 0, self._fd()) == "GBP"</code>              | Pass   | 0.000s |
| Employment type residential                                                         | <code>isinstance(result, str)</code>                                              | Pass   | 0.000s |
| Flood risk category                                                                 | <code>result == "Medium"</code>                                                   | Pass   | 0.000s |
| Lastatus defaulted                                                                  | <code>result == "Defaulted"</code>                                                | Pass   | 0.000s |
| Lastatus normal                                                                     | <code>result in ["Current", "Completed", "Redeemed"]</code>                       | Pass   | 0.000s |
| Marital status older                                                                | <code>isinstance(result, str)</code>                                              | Pass   | 0.000s |
| Marital status young                                                                | <code>result in ["Single", "Married", "Civil Partnership"]</code>                 | Pass   | 0.000s |
| Member id                                                                           | <code>result.startswith("MEMBER-")</code>                                         | Pass   | 0.000s |
| Mortgage id                                                                         | <code>generate_text_value("MortgageID", 0, self._fd()) == "MORT..."</code>        | Pass   | 0.000s |
| Mortgage provider                                                                   | <code>isinstance(result, str)</code>                                              | Pass   | 0.000s |
| Property id                                                                         | <code>generate_text_value("PropertyID", 0, self._fd()) == "PROP..."</code>        | Pass   | 0.000s |
| Unknown field                                                                       | <code>1 &lt;= result &lt;= 10; isinstance(result, str) (+1 more)</code>           | Pass   | 0.000s |
| Uprn no existing                                                                    | <code>result.startswith("UPRN-") or isinstance(result, str)</code>                | Pass   | 0.000s |



| Test Description                                                                     | Acceptance Criteria                                                                                   | Result | Time   |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|--------|
| Upnrn with existing                                                                  | <code>result == "UPRN-12345"</code>                                                                   | Pass   | 0.000s |
| Alias returns same structure                                                         | <code>len(result) == 50;</code> <code>"readings" in result[0]</code>                                  | Pass   | 0.000s |
| CloseOutDate with Payable SettlementDirection.                                       | <code>pdf_path.exists()</code>                                                                        | Pass   | 0.003s |
| <code>_generate_trade_pdf</code> creates a PDF without cashflows.                    | <code>pdf_path.exists();</code><br><code>pdf_path.stat().st_size &gt; 1000</code>                     | Pass   | 0.002s |
| <code>_generate_trade_pdf</code> with cashflows exercises CF schedule section.       | <code>pdf_path.exists()</code>                                                                        | Pass   | 0.002s |
| <code>_generate_trade_pdf</code> with CloseOutOf exercises close-out section.        | <code>pdf_path.exists()</code>                                                                        | Pass   | 0.003s |
| <code>_generate_trade_pdf</code> with CloseOutDate (original trade that was closed). | <code>pdf_path.exists()</code>                                                                        | Pass   | 0.003s |
| Pdf failure returns empty list                                                       | <code>pdfs == []</code>                                                                               | Pass   | 0.000s |
| Pdf partial failure returns successful only                                          | <code>len(pdfs) == 1</code>                                                                           | Pass   | 0.000s |
| Creates output file                                                                  | <code>output.exists()</code>                                                                          | Pass   | 0.002s |
| History entry has required keys                                                      | <code>'date' in entry; 'mark' in entry (+2 more)</code>                                               | Pass   | 0.002s |
| Multiple eods accumulate                                                             | <code>len(trade['history']) == 3</code>                                                               | Pass   | 0.003s |
| Multiple trades in same eod                                                          | <code>'PRS-A' in data['trades']; 'PRS-B' in data['trades'] (+1 more)</code>                           | Pass   | 0.002s |
| No eod files returns early                                                           | <code>not output.exists()</code>                                                                      | Pass   | 0.000s |
| Num snapshots in metadata                                                            | <code>data['metadata']['num_snapshots'] == 2</code>                                                   | Pass   | 0.002s |
| Output has metadata                                                                  | <code>'metadata' in data; 'num_trades' in data['metadata']</code>                                     | Pass   | 0.002s |
| Output has trades                                                                    | <code>'trades' in data; 'PRS-001' in data['trades']</code>                                            | Pass   | 0.002s |
| Position without swap id skipped                                                     | <code>data['trades'] == {};</code><br><code>data['metadata']['num_trades'] == 0</code>                | Pass   | 0.002s |
| Trade has history                                                                    | <code>'history' in trade;</code><br><code>len(trade['history']) == 1</code>                           | Pass   | 0.001s |
| Trade metadata preserved                                                             | <code>trade['gauge_id'] == 'GAUGE-001';</code><br><code>trade['trigger'] == 'severe' (+3 more)</code> | Pass   | 0.002s |
| Empty dir produces no file                                                           | <code>not output.exists();</code> <code>not output.exists()</code>                                    | Pass   | 0.000s |

| Test Description                                                  | Acceptance Criteria                                                                                                                     | Result | Time   |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| History entry structure                                           | <code>entry["date"] == "2026-01-02";</code>                                                                                             | Pass   | 0.003s |
| Metadata has generated at                                         | <code>entry["mark"] == 215.0 (+2 more)</code>                                                                                           | Pass   | 0.002s |
| Metadata num snapshots                                            | <code>"generated_at" in data["metadata"];</code><br><code>"generated_at" in data["metadata"]</code>                                     | Pass   | 0.005s |
| Multiple days builds history vector                               | <code>data["metadata"]["num_snapshots"] == 3;</code><br><code>data["metadata"]["num_snapshots"] == 3</code>                             | Pass   | 0.005s |
| Multiple trades in snapshot                                       | <code>len(data["trades"]["PRS-001"]["history"]) == 3</code>                                                                             | Pass   | 0.003s |
| Output has metadata and trades                                    | <code>data["metadata"]["num_trades"] == 3</code>                                                                                        | Pass   | 0.002s |
| Position missing swap id is skipped                               | <code>"metadata" in data; "trades" in data</code><br><code>"PRS-001" in data["trades"];</code><br><code>len(data["trades"]) == 1</code> | Pass   | 0.002s |
| Single snapshot creates file                                      | <code>output.exists(); output.exists()</code>                                                                                           | Pass   | 0.002s |
| Trade keyed by swap id                                            | <code>"PRS-001" in data["trades"]</code>                                                                                                | Pass   | 0.003s |
| Trade metadata fields                                             | <code>"gauge_id" in t; "trigger" in t (+3 more)</code>                                                                                  | Pass   | 0.003s |
| Explicit years overrides instance default                         | <code>result["years_included"] == 1</code>                                                                                              | Pass   | 0.013s |
| Instance years used when no years arg                             | <code>result["years_included"] == 2</code>                                                                                              | Pass   | 0.015s |
| If all sequence storms are sub-alert, portfolio flood count is 0. | <code>result["total_storms_at_gauge"] == 0;</code><br><code>result["properties_with_floods"] == 0</code>                                | Pass   | 0.006s |
| Portfolio summary total_storms_at_gauge counts all alert storms.  | <code>result["total_storms_at_gauge"] == 2</code>                                                                                       | Pass   | 0.006s |
| portfolio_flood_summary.json is written with per-property data.   | <code>summary_path.exists();</code><br><code>summary["summary"]["total_gauges"] == 1 (+1 more)</code>                                   | Pass   | 0.006s |
| Generate output path none uses config input dir                   | <code>isinstance(result, dict); (tmp_path / "gaugehd.39001.json").exists()</code>                                                       | Pass   | 0.014s |
| Banks always have CSAAgreement = True.                            | <code>platform["CSAAgreement"] is True</code>                                                                                           | Pass   | 0.001s |
| Counterparties list has two parties                               | <code>len(counterparties) == 2</code>                                                                                                   | Pass   | 0.000s |
| Counterparty has counterparty set                                 | <code>"CounterpartySet" in ctpy</code>                                                                                                  | Pass   | 0.000s |
| Isda master agreement is true                                     | <code>platform["ISDAMasterAgreement"] is True</code>                                                                                    | Pass   | 0.000s |
| Party has accounts                                                | <code>"Accounts" in party;</code><br><code>len(party["Accounts"]) &gt;= 1</code>                                                        | Pass   | 0.000s |

| Test Description                                                         | Acceptance Criteria                                                             | Result    | Time   |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------|--------|
| Party has contact info                                                   | "ContactInformation" in party;<br>"PrimaryEmail" in party["ContactInformation"] | Pass      | 0.000s |
| Party has id                                                             | "PartyID" in party;<br>party["PartyID"].startswith("CTPY-")                     | Pass      | 0.000s |
| Party has name                                                           | "PartyName" in party;<br>isinstance(party["PartyName"], str)                    | Pass      | 0.000s |
| Party has natural person                                                 | "NaturalPersons" in party;<br>len(party["NaturalPersons"]) == 1                 | Pass      | 0.000s |
| Platform has credit rating                                               | "CreditRating" in platform;<br>platform["CreditRating"] in [ "AAA", "AA+..."    | Pass      | 0.000s |
| Unique party ids                                                         | len(set(ids)) == 5                                                              | Pass      | 0.001s |
| Generator exception falls through to type                                | isinstance(result, float);<br>result <= 1000                                    | 0 <= Pass | 0.000s |
| Gauge importable without quantlib                                        | hasattr(mod, "GaugePortfolioGenerator")                                         | Pass      | 0.015s |
| models.hazard.builder must not drag in QuantLib pricing at import.       | mod is not None                                                                 | Pass      | 0.005s |
| Mortgage importable without quantlib                                     | hasattr(mod, "MortgagePortfolioGenerator")                                      | Pass      | 0.002s |
| Importing port.src (package __init__) must not require QuantLib.         | hasattr(mod, "GaugePortfolioGenerator")                                         | Pass      | 0.002s |
| Storm multi importable without quantlib                                  | mod is not None                                                                 | Pass      | 0.022s |
| Depth damage importable without geopandas                                | hasattr(mod, 'scalar_depth_damage')                                             | Pass      | 0.001s |
| __init__.py must not have an executable FloodRiskModel import statement. | 'FloodRiskModel' not in executable                                              | Pass      | 0.001s |
| propertyts imports scalar_depth_damage — must work without geopandas.    | hasattr(mod, 'PropertyTimeSeriesGenerator')                                     | Pass      | 0.001s |
| Scalar depth damage works without geopandas                              | 0.0 < result <= 1.0                                                             | Pass      | 0.001s |
| Spatial importable without geopandas                                     | hasattr(mod, 'haversine_distance')                                              | Pass      | 0.004s |
| 100 years is heritage                                                    | get_age.band(1925, current_year=2025) == 'heritage'                             | Pass      | 0.000s |

| Test Description                | Acceptance Criteria                                                                       | Result | Time   |
|---------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| 10 years is modern              | <code>get_age.band(2015, current_year=2025)</code><br><code>== 'modern'</code>            | Pass   | 0.000s |
| 25 years is established         | <code>get_age.band(2000, current_year=2025)</code><br><code>== 'established'</code>       | Pass   | 0.000s |
| 50 years is period              | <code>get_age.band(1975, current_year=2025)</code><br><code>== 'period'</code>            | Pass   | 0.000s |
| Default current year            | <code>result in AGE_BAND_FACTORS</code>                                                   | Pass   | 0.000s |
| Established                     | <code>get_age.band(1985, current_year=2025)</code><br><code>== 'established'</code>       | Pass   | 0.000s |
| Exactly 24 years is modern      | <code>get_age.band(2001, current_year=2025)</code><br><code>== 'modern'</code>            | Pass   | 0.000s |
| Exactly 49 years is established | <code>get_age.band(1976, current_year=2025)</code><br><code>== 'established'</code>       | Pass   | 0.000s |
| Exactly 99 years is period      | <code>get_age.band(1926, current_year=2025)</code><br><code>== 'period'</code>            | Pass   | 0.000s |
| Exactly 9 years is new build    | <code>get_age.band(2016, current_year=2025)</code><br><code>== 'new_build'</code>         | Pass   | 0.000s |
| Heritage old                    | <code>get_age.band(1800, current_year=2025)</code><br><code>== 'heritage'</code>          | Pass   | 0.000s |
| Modern                          | <code>get_age.band(2005, current_year=2025)</code><br><code>== 'modern'</code>            | Pass   | 0.000s |
| New build                       | <code>get_age.band(2020, current_year=2025)</code><br><code>== 'new_build'</code>         | Pass   | 0.000s |
| Period                          | <code>get_age.band(1950, current_year=2025)</code><br><code>== 'period'</code>            | Pass   | 0.000s |
| 100 years is heritage           | <code>get_age.premium.band(1925,</code><br><code>current_year=2025) == 'heritage'</code>  | Pass   | 0.000s |
| 10 years is modern              | <code>get_age.premium.band(2015,</code><br><code>current_year=2025) == 'modern'</code>    | Pass   | 0.000s |
| 30 years is established         | <code>get_age.premium.band(1995,</code><br><code>current_year=2025) == 'establi...</code> | Pass   | 0.000s |
| Default current year            | <code>result in AGE_PREMIUM_FACTORS</code>                                                | Pass   | 0.000s |
| Established                     | <code>get_age.premium.band(1960,</code><br><code>current_year=2025) == 'establi...</code> | Pass   | 0.000s |
| Exactly 29 years modern         | <code>get_age.premium.band(1996,</code><br><code>current_year=2025) == 'modern'</code>    | Pass   | 0.000s |
| Exactly 99 years established    | <code>get_age.premium.band(1926,</code><br><code>current_year=2025) == 'establi...</code> | Pass   | 0.000s |
| Exactly 9 years new build       | <code>get_age.premium.band(2016,</code><br><code>current_year=2025) == 'new_build'</code> | Pass   | 0.000s |

| Test Description                             | Acceptance Criteria                                                                       | Result | Time   |
|----------------------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| Heritage old                                 | <code>get_age_premium_band(1850,</code><br><code>current_year=2025) == 'heritage'</code>  | Pass   | 0.000s |
| Modern                                       | <code>get_age_premium_band(2000,</code><br><code>current_year=2025) == 'modern'</code>    | Pass   | 0.000s |
| New build                                    | <code>get_age_premium_band(2020,</code><br><code>current_year=2025) == 'new_build'</code> | Pass   | 0.000s |
| Exactly 100 third band                       | <code>(lo, hi) == AREA_PREMIUM_BANDS[2][1]</code>                                         | Pass   | 0.000s |
| Exactly 50 second band                       | <code>(lo, hi) == AREA_PREMIUM_BANDS[1][1]</code>                                         | Pass   | 0.000s |
| Just below 100 second band                   | <code>(lo, hi) == AREA_PREMIUM_BANDS[1][1]</code>                                         | Pass   | 0.000s |
| Just below 50 first band                     | <code>(lo, hi) == AREA_PREMIUM_BANDS[0][1]</code>                                         | Pass   | 0.000s |
| Large area fourth band                       | <code>(lo, hi) == AREA_PREMIUM_BANDS[3][1]</code>                                         | Pass   | 0.000s |
| Medium area second band                      | <code>(lo, hi) == AREA_PREMIUM_BANDS[1][1]</code>                                         | Pass   | 0.000s |
| Returns tuple                                | <code>isinstance(result, tuple);</code><br><code>len(result) == 2</code>                  | Pass   | 0.000s |
| Small area first band                        | <code>(lo, hi) == AREA_PREMIUM_BANDS[0][1]</code>                                         | Pass   | 0.000s |
| After log entry appears                      | <code>data["total_entries"] == 1;</code><br><code>len(data["entries"]) == 1</code>        | Pass   | 0.004s |
| Empty trail returns 200                      | <code>r.status_code == 200; data["status"]</code><br><code>== "success" (+1 more)</code>  | Pass   | 0.001s |
| Filter by event type                         | <code>all(e["event_type"] == "calibration"</code><br><code>for e in entries)</code>       | Pass   | 0.005s |
| Filter by model id                           | <code>r.get_json()["total_entries"] == 1</code>                                           | Pass   | 0.004s |
| Filter by model id no match                  | <code>r.get_json()["total_entries"] == 0</code>                                           | Pass   | 0.003s |
| Has returned field                           | <code>"returned" in r.get_json()</code>                                                   | Pass   | 0.001s |
| Limit parameter                              | <code>r.get_json()["returned"] == 2</code>                                                | Pass   | 0.009s |
| Missing file returns empty list              | <code>result == []</code>                                                                 | Pass   | 0.000s |
| File with two gauges should return both IDs. | <code>set(result) == {"GAUGE-A",</code><br><code>"GAUGE-B"}</code>                        | Pass   | 0.002s |
| Returns correct id                           | <code>"GAUGE-001" in result</code>                                                        | Pass   | 0.001s |
| Returns list                                 | <code>isinstance(result, list)</code>                                                     | Pass   | 0.001s |
| Binary file returns empty                    | <code>result == []</code>                                                                 | Pass   | 0.001s |
| Corrupt json returns empty                   | <code>result == []</code>                                                                 | Pass   | 0.001s |
| No data returns 404                          | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code>      | Pass   | 0.000s |
| Principles count                             | <code>len(r.get_json()["assessment"]["principles"])</code><br><code>== 2</code>           | Pass   | 0.003s |
| Response has assessment                      | <code>"assessment" in data; "principles" in</code><br><code>data["assessment"]</code>     | Pass   | 0.002s |
| With data returns 200                        | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>    | Pass   | 0.002s |

| Test Description                                                       | Acceptance Criteria                                                                                      | Result | Time   |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------|--------|
| Returns 200 if PDF exists, 404 if not yet generated.                   | <code>r.status_code in (200, 404)</code>                                                                 | Pass   | 0.022s |
| Empty references                                                       | <code>r.get_json()["references"] == []</code>                                                            | Pass   | 0.002s |
| Has categories key                                                     | <code>"categories" in r.get_json()</code>                                                                | Pass   | 0.002s |
| Returns 200                                                            | <code>r.status_code == 200</code>                                                                        | Pass   | 0.002s |
| Returns success status                                                 | <code>r.get_json()["status"] == "success"</code>                                                         | Pass   | 0.002s |
| Found returns 200                                                      | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                                 | Pass   | 0.002s |
| Found second counterparty                                              | <code>r.status_code == 200</code>                                                                        | Pass   | 0.002s |
| Not found returns 404                                                  | <code>r.status_code == 404; data["status"] == "error"</code>                                             | Pass   | 0.002s |
| Format                                                                 | <code>rid.startswith("run-"); len(rid) == 19</code>                                                      | Pass   | 0.000s |
| All pipeline outputs exist and are fresh.                              | <code>data["status"] == "success"; data["summary"]["fresh"] == 10 (+4 more)</code>                       | Pass   | 0.004s |
| All present but some stale = degraded.                                 | <code>data["summary"]["missing"] == 0; data["summary"]["stale"] == 1 (+1 more)</code>                    | Pass   | 0.003s |
| Response includes as_of timestamp.                                     | <code>"as_of" in data</code>                                                                             | Pass   | 0.000s |
| Mix of fresh, stale, and missing.                                      | <code>data["summary"]["fresh"] == 1; data["summary"]["stale"] == 1 (+2 more)</code>                      | Pass   | 0.002s |
| When lineage file does not exist, manifest is None but route succeeds. | <code>r.status_code == 200; data["status"] == "success" (+3 more)</code>                                 | Pass   | 0.001s |
| Empty list returns success                                             | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                                 | Pass   | 0.001s |
| Has fields                                                             | <code>"owner" in fields; "rag_rating" in fields</code>                                                   | Pass   | 0.000s |
| Returns 200                                                            | <code>r.status_code == 200; r.status_code == 200</code>                                                  | Pass   | 0.000s |
| Returns success status                                                 | <code>r.get_json()["status"] == "success"; r.get_json()["status"] == "success"</code>                    | Pass   | 0.000s |
| Empty entity returns none                                              | <code>loader.get_entity_id({}) is None</code>                                                            | Pass   | 0.000s |
| Extracts event id                                                      | <code>loader.get_entity_id(storms[0]) == "EVT-001"</code>                                                | Pass   | 0.001s |
| Missing id returns none                                                | <code>loader.get_entity_id(entity) is None; loader.get_track("EVT-NONEXISTENT") is None (+1 more)</code> | Pass   | 0.000s |
| Missing fields return none                                             | <code>summary["eventId"] is None; summary["name"] is None</code>                                         | Pass   | 0.000s |



| Test Description                                                    | Acceptance Criteria                                                                    | Result | Time   |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------|--------|
| Returns summary dict                                                | summary["eventId"] == "EVT-001";<br>summary["name"] == "Harvey" (+3 more)              | Pass   | 0.001s |
| get_eod_snapshot returns snapshot data for existing date.           | result is not None; result['date'] == '2026-03-15'                                     | Pass   | 0.002s |
| get_eod_snapshot returns None when no snapshot exists for the date. | result is None                                                                         | Pass   | 0.000s |
| Filter by report                                                    | r.status_code == 200;<br>data["total_reports"] == 1 (+3 more)                          | Pass   | 0.002s |
| Filter unknown report 404                                           | r.status_code == 404;<br>"available_reports" in data                                   | Pass   | 0.002s |
| Full registry                                                       | r.status_code == 200; data["status"] == "success" (+4 more)                            | Pass   | 0.002s |
| No field lineage file returns 404.                                  | r.status_code == 404; "not found" in data["message"].lower() (+1 more)                 | Pass   | 0.000s |
| Summary contents                                                    | fr_summary["label"] == "Flood Risk Report"; fr_summary["section_count"] == 2 (+1 more) | Pass   | 0.002s |
| Producer step exists in STEP_IO but not in manifest.                | result["produced.by"] == "gauges";<br>result["producer_run_id"] is None                | Pass   | 0.000s |
| Known file                                                          | result["produced.by"] == "gauges";<br>result["producer_run_id"] == "r1" (+1 more)      | Pass   | 0.000s |
| Unknown file                                                        | result["produced.by"] is None;<br>result["producer_run_id"] is None (+1 more)          | Pass   | 0.000s |
| Gauge not found returns 404                                         | r.status_code == 404;<br>r.get_json()["status"] == "error"                             | Pass   | 0.002s |
| No curve for gauge returns 404                                      | r.status_code == 404                                                                   | Pass   | 0.003s |
| No hazard file returns 404                                          | r.status_code == 404; "not available" in r.get_json()["message"].lower() (+1 more)     | Pass   | 0.002s |
| Response has annual flood probs                                     | "alert" in probs; "warning" in probs (+1 more)                                         | Pass   | 0.003s |
| Response has curve points                                           | len(r.get_json()["curve_points"]) == 3                                                 | Pass   | 0.003s |
| Response has flood stages                                           | "FloodAlert" in stages;<br>"FloodWarning" in stages (+1 more)                          | Pass   | 0.002s |
| Response has gauge id                                               | r.get_json()["gauge_id"] == GAUGE_ID                                                   | Pass   | 0.003s |

| Test Description                       | Acceptance Criteria                                                        | Result | Time   |
|----------------------------------------|----------------------------------------------------------------------------|--------|--------|
| Response has gev parameters            | "location" in gev; "scale" in gev (+1 more)                                | Pass   | 0.003s |
| Response has term structures           | "alert" in ts; "warning" in ts (+1 more)                                   | Pass   | 0.003s |
| Valid hazard returns 200               | r.status_code == 200;<br>r.get_json()["status"] == "success"               | Pass   | 0.003s |
| Days larger than available returns all | len(data["daily_observations"]) == 12                                      | Pass   | 0.001s |
| Days param limits observations         | len(data["daily_observations"]) == 5                                       | Pass   | 0.002s |
| Days param zero returns all            | len(data["daily_observations"]) == 12                                      | Pass   | 0.002s |
| No history file returns 404            | r.status_code == 404;<br>r.get_json()["status"] == "error"                 | Pass   | 0.000s |
| Response has daily observations        | "daily_observations" in data;<br>len(data["daily_observations"]) == 12     | Pass   | 0.002s |
| Response has flood stages              | "flood_stages" in data                                                     | Pass   | 0.002s |
| Response has gauge metadata            | "gauge_metadata" in data                                                   | Pass   | 0.002s |
| Response has statistics                | "statistics" in data;<br>data["statistics"]["mean_level"] == 2.5           | Pass   | 0.002s |
| Valid history returns 200              | r.status_code == 200;<br>r.get_json()["status"] == "success"               | Pass   | 0.002s |
| Missing key defaults to zero rate      | result == [0.0] * len(TENORS)                                              | Pass   | 0.000s |
| Negative rate returns all zeros        | result == [0.0] * len(TENORS)                                              | Pass   | 0.000s |
| Positive rate returns positive spreads | len(result) == len(TENORS); all(s > 0 for s in result)                     | Pass   | 0.000s |
| Spread list has one entry per tenor    | len(result) == len(TENORS)                                                 | Pass   | 0.000s |
| Zero rate returns all zeros            | result == [0.0] * len(TENORS)                                              | Pass   | 0.000s |
| Gauge not found returns 404            | r.status_code == 404;<br>r.get_json()["status"] == "error"<br>(+2 more)    | Pass   | 0.002s |
| Valid gauge returns success            | r.status_code == 200; data["status"] == "success" (+4 more)                | Pass   | 0.003s |
| Gauge not found returns 404            | r.status_code == 404;<br>r.get_json()["status"] == "error"                 | Pass   | 0.001s |
| Response has flood simulation          | "flood_simulation" in data;<br>"num_timesteps" in data["flood_simulation"] | Pass   | 0.004s |
| Response has flood stages              | "flood_stages" in r.get_json()                                             | Pass   | 0.003s |
| Response has gauge id                  | r.get_json()["gauge_id"] == GAUGE_ID                                       | Pass   | 0.003s |

| Test Description                                                           | Acceptance Criteria                                                                         | Result | Time   |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| Response has storm responses                                               | "storm_responses" in data; "num_storms" in data["storm_responses"]                          | Pass   | 0.003s |
| Storm responses enriched with metadata from storm_sequences.json.          | "name" in responses[0] or responses[0].get("name") == ""                                    | Pass   | 0.002s |
| Valid gauge returns 200                                                    | r.status_code == 200; r.get_json()["status"] == "success"                                   | Pass   | 0.003s |
| When exceeded_severe=True, severe count computed from gauges dir.          | r.status_code == 200                                                                        | Pass   | 0.002s |
| Gauge not found returns 404                                                | r.status_code == 404; r.get_json()["status"] == "error" (+2 more)                           | Pass   | 0.001s |
| Response has count and readings                                            | "count" in data; "readings" in data (+1 more)                                               | Pass   | 0.003s |
| Valid gauge returns success                                                | r.status_code == 200; data["status"] == "success" (+4 more)                                 | Pass   | 0.005s |
| When term structure doesn't have the requested tenor, falls back           | rate == base_rate                                                                           | Pass   | 0.001s |
| Unknown gauge with no term structure returns 0.0 from flat rate fall-back. | rate == 0.0                                                                                 | Pass   | 0.006s |
| Missing id returns none                                                    | loader.get_entity_id(entity) is None; loader.get_track("EVT-NONEXISTENT") is None (+1 more) | Pass   | 0.001s |
| Returns impact dict                                                        | isinstance(impact, dict); "TotalDamage" in impact                                           | Pass   | 0.001s |
| Contains timeout                                                           | '30000' in js                                                                               | Pass   | 0.000s |
| Returns script tag                                                         | '<script>' in js; '</script>' in js (+2 more)                                               | Pass   | 0.003s |
| All modules non empty                                                      | len(js) > 50                                                                                | Pass   | 0.000s |
| All modules return str                                                     | isinstance(js, str)                                                                         | Pass   | 0.000s |
| Excludes category 1 and 2                                                  | "EVT-004" not in ids                                                                        | Pass   | 0.001s |
| Includes cat 3 4 5                                                         | "EVT-001" in ids; "EVT-002" in ids (+1 more)                                                | Pass   | 0.001s |
| Returns category 3 plus                                                    | all(c >= 3 for c in categories)                                                             | Pass   | 0.001s |
| Get created meeting                                                        | r.status_code == 200; data["status"] == "success" (+1 more)                                 | Pass   | 0.003s |
| Unknown id returns 404                                                     | r.status_code == 404; r.get_json()["status"] == "error"                                     | Pass   | 0.002s |

| Test Description               | Acceptance Criteria                                                                                                   | Result | Time   |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------|--------|
| Meeting not found returns 404  | <code>r.status_code == 404;</code>                                                                                    | Pass   | 0.001s |
|                                | <code>r.get_json()["status"] == "error"</code>                                                                        |        |        |
| Valid meeting content type pdf | <code>"application/pdf" in r.content_type</code>                                                                      | Pass   | 0.005s |
| Valid meeting returns bytes    | <code>len(r.data) &gt; 100</code>                                                                                     | Pass   | 0.004s |
| Valid meeting returns pdf      | <code>r.status_code == 200</code>                                                                                     | Pass   | 0.004s |
| Model has validation questions | <code>"validation_questions" in model;</code><br><code>len(model["validation_questions"]) == 9</code>                 | Pass   | 0.003s |
| Model not found returns 404    | <code>r.status_code == 404; r.status_code == 404 (+1 more)</code>                                                     | Pass   | 0.001s |
| Response has audit entries     | <code>"audit_entries" in r.get_json()</code>                                                                          | Pass   | 0.003s |
| Valid model returns success    | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                                              | Pass   | 0.003s |
| No inventory returns 404       | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br>(+3 more)                     | Pass   | 0.001s |
| Category distribution          | <code>"category_distribution" in data</code>                                                                          | Pass   | 0.002s |
| Model has review status        | <code>"review_status" in model;</code><br><code>model["review_status"] in ("On Track", "Upcoming", "Due S...</code>   | Pass   | 0.002s |
| Model has risk rating          | <code>"risk_rating" in model</code>                                                                                   | Pass   | 0.002s |
| Model has validation coverage  | <code>"validation_coverage" in model</code>                                                                           | Pass   | 0.002s |
| Response has metadata          | <code>"metadata" in data</code>                                                                                       | Pass   | 0.002s |
| Returns 200                    | <code>r.status_code == 200; r.status_code == 200</code>                                                               | Pass   | 0.002s |
| Returns models list            | <code>"models" in data; data["total_models"] &gt;= 1</code>                                                           | Pass   | 0.002s |
| Returns success status         | <code>r.get_json()["status"] == "success";</code><br><code>r.get_json()["status"] == "success"</code>                 | Pass   | 0.002s |
| Tier distribution              | <code>"tier_distribution" in data</code>                                                                              | Pass   | 0.002s |
| No inventory returns 404       | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br>(+3 more)                     | Pass   | 0.000s |
| Not scheduled review status    | <code>r.status_code == 200;</code><br><code>model["review_status"] == "Not Scheduled"</code>                          | Pass   | 0.002s |
| Overdue review status          | <code>r.status_code == 200;</code><br><code>model["review_status"] == "Overdue"</code>                                | Pass   | 0.002s |
| Returns expected keys          | <code>result["NRFASStationID"] == "39001";</code><br><code>result["MeanFlow"] == pytest.approx(55.0) (+2 more)</code> | Pass   | 0.000s |

| Test Description                                                         | Acceptance Criteria                                                                    | Result | Time   |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------|--------|
| Truncates data quality notes                                             | <code>len(result["DataQualityNotes"]) &lt;= 500</code>                                 | Pass   | 0.000s |
| All expected keys present                                                | <code>key in result</code>                                                             | Pass   | 0.000s |
| Catchment area always none                                               | <code>result["CatchmentArea"] is None</code>                                           | Pass   | 0.000s |
| Custom flow units                                                        | <code>gen.get_nrfa_metadata_for_cdm({}, {"units": "ft3/s"}, "f....</code>              | Pass   | 0.000s |
| Database source from station meta-data                                   | <code>result["DatabaseSource"] == "TestDB"</code>                                      | Pass   | 0.000s |
| Default flow units m3 per s                                              | <code>gen.get_nrfa_metadata_for_cdm({}, {}, "f.json")["FlowUnit...</code>              | Pass   | 0.000s |
| Empty data quality notes when no description                             | <code>gen.get_nrfa_metadata_for_cdm({}, {}, "f.json")["DataQual...</code>              | Pass   | 0.000s |
| Missing database source returns none                                     | <code>gen.get_nrfa_metadata_for_cdm({}, {}, "f.json")["Database...</code>              | Pass   | 0.000s |
| Missing stats keys return none                                           | <code>result[key] is None</code>                                                       | Pass   | 0.000s |
| Response has 'storms' list with count $\geq 0$ .                         | <code>'storms' in data; data['count'] &gt; 0</code>                                    | Pass   | 0.002s |
| Returns 200 with status=success.                                         | <code>resp.status_code == 200; data['status'] == 'success'</code>                      | Pass   | 0.001s |
| Each storm has storm_id, name, intensity_category, gauges_severe,        | <code>'storm_id' in s; 'name' in s (+5 more)</code>                                    | Pass   | 0.004s |
| Returns 404 when stress_storms not found.                                | <code>resp.status_code == 404</code>                                                   | Pass   | 0.000s |
| When gauges_severe is equal, storms sorted by gauges_warning desc.       | <code>s1['gauges_warning'] &gt;= s2['gauges_warning']</code>                           | Pass   | 0.003s |
| Storms sorted by gauges_severe desc — STORM_PORT_SEVERE must come first. | <code>len(storms) &gt;= 2; storms[0]['storm_id'] == STORM_PORT_SEVERE (+1 more)</code> | Pass   | 0.003s |
| Lines 91-96: empty stressm dir falls back to legacy stress/ dir.         | <code>result is mock_predictor</code>                                                  | Pass   | 0.001s |
| Lines 86-90: stressm dir with joblib files loads predictor.              | <code>result is mock_predictor</code>                                                  | Pass   | 0.001s |
| Line 97: no models in either directory logs warning, returns None.       | <code>result is None</code>                                                            | Pass   | 0.001s |
| Get preloader js matches get js                                          | <code>_get_preloader_js() == get_js()</code>                                           | Pass   | 0.000s |
| Get preloader js returns string                                          | <code>isinstance(result, str); 'DOMContentLoaded' in result</code>                     | Pass   | 0.000s |
| Found returns property                                                   | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>               | Pass   | 0.001s |

| Test Description                            | Acceptance Criteria                                                                                         | Result | Time   |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|--------|
| Not found returns 404                       | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br>(+2 more)           | Pass   | 0.001s |
| Close high risk high                        | <code>get_proximity_zone(50, 'High') ==</code><br><code>'close_high_risk'</code>                            | Pass   | 0.000s |
| Close high risk medium                      | <code>get_proximity_zone(100, 'Medium') ==</code><br><code>'close_high_risk'</code>                         | Pass   | 0.000s |
| Close high risk very high                   | <code>get_proximity_zone(10, 'Very High')</code><br><code>== 'close_high_risk'</code>                       | Pass   | 0.000s |
| Close low risk low                          | <code>get_proximity_zone(150, 'Low') ==</code><br><code>'close_low_risk'</code>                             | Pass   | 0.000s |
| Close low risk very low                     | <code>get_proximity_zone(100, 'Very Low')</code><br><code>== 'close_low_risk'</code>                        | Pass   | 0.000s |
| Exactly 200m is medium zone                 | <code>get_proximity_zone(200, 'Low') ==</code><br><code>'medium'</code>                                     | Pass   | 0.000s |
| Exactly 500m is far                         | <code>get_proximity_zone(500, 'Low') ==</code><br><code>'far'</code>                                        | Pass   | 0.000s |
| Far zone                                    | <code>get_proximity_zone(600, 'Medium') ==</code><br><code>'far'</code>                                     | Pass   | 0.000s |
| Medium zone                                 | <code>get_proximity_zone(300, 'Very Low')</code><br><code>== 'medium'</code>                                | Pass   | 0.000s |
| Medium zone upper bound                     | <code>get_proximity_zone(499, 'Very High')</code><br><code>== 'medium'</code>                               | Pass   | 0.000s |
| Returns none when quantlib un-<br>available | <code>result is None</code>                                                                                 | Pass   | 0.001s |
| Model not found returns 404                 | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br>(+3 more)           | Pass   | 0.002s |
| Response has risk rating                    | <code>"risk_rating" in data;</code><br><code>"calculated_rating" in rr</code> (+1 more)                     | Pass   | 0.002s |
| Risk rating has component scores            | <code>"component_scores" in rr</code>                                                                       | Pass   | 0.002s |
| Valid model returns success                 | <code>r.status_code == 200; data["status"]</code><br><code>== "success"</code> (+1 more)                    | Pass   | 0.002s |
| Gauges affects many                         | <code>"properties" in downstream; "gaugehd"</code><br><code>in downstream</code> (+1 more)                  | Pass   | 0.000s |
| Leaf has no downstream                      | <code>downstream == []</code>                                                                               | Pass   | 0.000s |
| Mid chain                                   | <code>"hazard" in downstream; "propertyts"</code><br><code>in downstream</code> (+2 more)                   | Pass   | 0.000s |
| Custom statistics                           | <code>stats['property_menu_items'] == 2;</code><br><code>stats['gauge_menu_items'] == 1</code> (+1<br>more) | Pass   | 0.000s |



| Test Description                | Acceptance Criteria                                                                                                                 | Result | Time   |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Default statistics              | stats['property_menu_items']<br>== len(DEFAULT_PROPERTY_MENU);<br>stats['gauge_menu_items'] ==<br>len(DEFAULT_GAUGE_MENU) (+1 more) | Pass   | 0.000s |
| Get statistics keys             | stats['server_url']<br>== 'http://test:1234';<br>stats['total_endpoints'] == 2 (+1<br>more)                                         | Pass   | 0.000s |
| Get statistics single endpoint  | stats['total_endpoints'] == 1;<br>stats['endpoints'] == ['only']                                                                    | Pass   | 0.000s |
| Returns dict with expected keys | 'notifications' in stats;<br>'backend_handler' in stats (+1<br>more)                                                                | Pass   | 0.000s |
| Returns expected keys           | 'default_position' in stats;<br>'auto_dismiss_timeout' in stats<br>(+3 more)                                                        | Pass   | 0.000s |
| Statistics are dicts            | isinstance(stats['notifications'],<br>dict);<br>isinstance(stats['backend_handler'],<br>dict) (+1 more)                             | Pass   | 0.000s |
| Statistics reflect config       | stats['default_position']<br>== 'bottom-center';<br>stats['auto_dismiss_timeout'] ==<br>9000 (+1 more)                              | Pass   | 0.000s |
| Mid step                        | "gauges" in result["upstream"];<br>"mortgages" in result["downstream"]<br>(+1 more)                                                 | Pass   | 0.000s |
| Root step                       | result["upstream"] == [];<br>"properties" in result["downstream"]<br>(+3 more)                                                      | Pass   | 0.000s |
| Step not in manifest            | result["last_run"] is None;<br>result["last_timestamp"] is None                                                                     | Pass   | 0.000s |
| Cluster returns 3 or 4          | count in (3, 4)                                                                                                                     | Pass   | 0.000s |
| Doublet returns 2               | get_storm_count(SequenceType.DOUBLET,<br>rng) == 2                                                                                  | Pass   | 0.000s |
| Isolated returns 1              | get_storm_count(SequenceType.ISOLATED,<br>rng) == 1                                                                                 | Pass   | 0.000s |
| Persistent returns 4 or 5       | count in (4, 5)                                                                                                                     | Pass   | 0.000s |
| Each storm has peak_level.m.    | 'peak_level.m' in s; s['peak_level.m']<br>> 0                                                                                       | Pass   | 0.208s |

| Test Description                                                  | Acceptance Criteria                                                                            | Result | Time   |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------|--------|
| Storms have required identification fields.                       | 'storm_id' in s; 'name' in s (+3 more)                                                         | Pass   | 0.003s |
| Returns 404 when stress_storms not found.                         | resp.status_code == 404                                                                        | Pass   | 0.001s |
| Missing gauge_id returns total count (used by startup preloader). | resp.status_code == 200;<br>data['status'] == 'success' (+2 more)                              | Pass   | 0.034s |
| Returns empty list for gauge with no storms.                      | resp.status_code == 200;<br>data['storms'] == []                                               | Pass   | 0.002s |
| Returns storm list for specified gauge.                           | resp.status_code == 200;<br>data['status'] == 'success' (+2 more)                              | Pass   | 0.011s |
| Storms sorted by peak level descending (worst first).             | peaks == sorted(peaks, reverse=True)                                                           | Pass   | 0.003s |
| Each storm returned at most once                                  | len(result) == 1                                                                               | Pass   | 0.001s |
| Empty for remote location                                         | result == []                                                                                   | Pass   | 0.001s |
| Finds nearby storm                                                | "EVT-001" in ids                                                                               | Pass   | 0.001s |
| Large radius finds more                                           | len(large) >= len(small)                                                                       | Pass   | 0.001s |
| Each gauge has gauge_id, gauge_name, lon, lat, trade_count.       | 'gauge_id' in g; 'gauge_name' in g (+3 more)                                                   | Pass   | 0.175s |
| Gauges with trades show non-zero trade_count.                     | sum(counts) > 0                                                                                | Pass   | 0.127s |
| Returns gauge list with trade counts.                             | resp.status_code == 200;<br>data['status'] == 'success' (+2 more)                              | Pass   | 0.040s |
| Gauges sorted west to east (ascending longitude).                 | lons == sorted(lons)                                                                           | Pass   | 0.026s |
| Missing id returns none                                           | loader.get_entity_id(entity) is None;<br>loader.get_track("EVT-NONEXISTENT") is None (+1 more) | Pass   | 0.001s |
| Returns track for existing storm                                  | track is not None; isinstance(track, list) (+1 more)                                           | Pass   | 0.001s |
| Track has lat lon                                                 | "Latitude" in track[0]; "Longitude" in track[0]                                                | Pass   | 0.001s |
| get_trade_mark returns mark data for an existing trade.           | mark['fair_spread_bps'] == 250.0                                                               | Pass   | 0.001s |
| get_trade_mark returns empty dict for unknown trade.              | mark == {}                                                                                     | Pass   | 0.000s |
| get_trade_mark returns merged data after multiple updates.        | mark['fair_spread_bps'] == 200.0;<br>mark['mtm'] == 5000.0                                     | Pass   | 0.002s |

| Test Description                                                                | Acceptance Criteria                                                                  | Result | Time   |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|--------|
| If the PDF exists, it should be served with 200.                                | <code>r.status_code == 200; r.content_type == "application/pdf"</code>               | Pass   | 0.028s |
| Missing pdf returns 404                                                         | <code>r.status_code == 404; r.get_json()["status"] == "error"</code>                 | Pass   | 0.000s |
| When yield_curve is empty dict, falls back to risk_free_rate.                   | <code>rate == 0.042</code>                                                           | Pass   | 0.000s |
| When yield_curve key is missing, falls back to risk_free_rate.                  | <code>rate == 0.05</code>                                                            | Pass   | 0.000s |
| When tenor falls exactly between integer keys but loop exits, returns fallback. | <code>0.035 &lt;= rate &lt;= 0.045</code>                                            | Pass   | 0.000s |
| Apply documents function exists                                                 | <code>'_applyDocuments' in src</code>                                                | Pass   | 0.000s |
| Checks gov docs cache                                                           | <code>'window._tdPreGovDocs' in src</code>                                           | Pass   | 0.000s |
| Clears cache after use                                                          | <code>'window._preStorms = null' in src; 'window._tdPreGovDocs = null' in src</code> | Pass   | 0.000s |
| Hash directory                                                                  | <code>result["type"] == "directory"; result["hash"] is not None (+1 more)</code>     | Pass   | 0.001s |
| Hash file                                                                       | <code>result["type"] == "file"; result["hash"] is not None</code>                    | Pass   | 0.001s |
| Hash missing                                                                    | <code>result == {"hash": None, "type": "missing"}</code>                             | Pass   | 0.000s |
| Hash empty directory                                                            | <code>count == 0; len(digest) == 64</code>                                           | Pass   | 0.000s |
| Hash multiple files deterministic                                               | <code>d1 == d2; c1 == c2 == 2</code>                                                 | Pass   | 0.001s |
| Hash respects pattern                                                           | <code>count_json == 1; count_csv == 1</code>                                         | Pass   | 0.002s |
| Hash single file                                                                | <code>count == 1; len(digest) == 64</code>                                           | Pass   | 0.001s |
| Hash binary file                                                                | <code>result == expected</code>                                                      | Pass   | 0.002s |
| Hash empty file                                                                 | <code>result == expected</code>                                                      | Pass   | 0.000s |
| Hash known content                                                              | <code>result == expected</code>                                                      | Pass   | 0.001s |
| Approximate one degree lat                                                      | <code>110.000 &lt; d &lt; 112.000</code>                                             | Pass   | 0.000s |
| Approximate one degree lon at equator                                           | <code>110.000 &lt; d &lt; 112.000</code>                                             | Pass   | 0.000s |
| Same point returns zero                                                         | <code>d == pytest.approx(0.0, abs=1e-6)</code>                                       | Pass   | 0.000s |
| Symmetry                                                                        | <code>d1 == pytest.approx(d2, rel=1e-9)</code>                                       | Pass   | 0.000s |
| Equatorial degree approx 111km                                                  | <code>111.000 &lt; d &lt; 112.000</code>                                             | Pass   | 0.000s |
| London paris approx 340km                                                       | <code>330.000 &lt; d &lt; 350.000</code>                                             | Pass   | 0.000s |
| North south degree approx 111km                                                 | <code>110.000 &lt; d &lt; 112.000</code>                                             | Pass   | 0.000s |
| Returns float                                                                   | <code>isinstance(haversine_distance(51.0, -0.1, 52.0, 0.0), float)</code>            | Pass   | 0.000s |
| Same point is zero                                                              | <code>haversine_distance(51.5, -0.1, 51.5, -0.1) == pytest.appr...</code>            | Pass   | 0.000s |
| Symmetry                                                                        | <code>d1 == pytest.approx(d2, rel=1e-9)</code>                                       | Pass   | 0.000s |

| Test Description                                               | Acceptance Criteria                                                                                                                                    | Result | Time   |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| hazard.py lines 100-102: exception reading hazard file -i 500. | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>                                                                   | Pass   | 0.029s |
| Custom day count                                               | <code>curve is not None</code>                                                                                                                         | Pass   | 0.000s |
| High hazard gives low survival                                 | <code>sp &lt; 0.1</code>                                                                                                                               | Pass   | 0.000s |
| Returns credit curve                                           | <code>curve is not None</code>                                                                                                                         | Pass   | 0.002s |
| Survival prob decreases                                        | <code>sp2 &lt; sp1</code>                                                                                                                              | Pass   | 0.000s |
| Zero hazard gives survival near one                            | <code>sp &gt;= 0.999</code>                                                                                                                            | Pass   | 0.000s |
| Header footer callable                                         | —                                                                                                                                                      | Pass   | 0.004s |
| Header footer callable                                         | —                                                                                                                                                      | Pass   | 0.003s |
| Header footer callable                                         | —                                                                                                                                                      | Pass   | 0.002s |
| Header footer draws confidential                               | <code>mock_canvas.drawCentredString.called</code>                                                                                                      | Pass   | 0.003s |
| Header footer draws confidential footer                        | <code>mock_canvas.drawCentredString.called</code>                                                                                                      | Pass   | 0.002s |
| Header footer draws header strings                             | <code>mock_canvas.drawString.call_count</code><br><code>&gt;= 1;</code><br><code>mock_canvas.drawRightString.call_count</code><br><code>&gt;= 2</code> | Pass   | 0.003s |
| Header footer draws strings                                    | <code>mock_canvas.drawString.call_count</code><br><code>&gt;= 1;</code><br><code>mock_canvas.drawRightString.call_count</code><br><code>&gt;= 2</code> | Pass   | 0.002s |
| Health 200 at api v1                                           | <code>r.status_code == 200</code>                                                                                                                      | Pass   | 0.001s |
| Health 200 at root                                             | <code>r.status_code == 200</code>                                                                                                                      | Pass   | 0.000s |
| A single function must handle both route decorators.           | <code>api_idx != -1 and plain_idx != -1 and</code><br><code>fn_idx != -1; api_idx &lt; fn_idx (+1</code><br><code>more)</code>                         | Pass   | 0.000s |
| health.py must declare /api/v1/health as a route alias.        | <code>'/api/v1/health' in src</code>                                                                                                                   | Pass   | 0.003s |
| Backward-compat: /health must still exist.                     | <code>"/health" in src</code>                                                                                                                          | Pass   | 0.000s |
| Health returns ok status                                       | <code>data['status'] == 'ok'</code>                                                                                                                    | Pass   | 0.000s |
| Test GET /health/detailed returns detailed status.             | <code>response.status_code in [200, 503];</code><br><code>data['status'] in ['healthy',</code><br><code>'degraded'] (+2 more)</code>                   | Pass   | 0.000s |
| Test GET /health returns health status.                        | <code>response.status_code == 200;</code><br><code>data['status'] == 'ok' (+1 more)</code>                                                             | Pass   | 0.000s |
| Test GET / redirects to catchment selector.                    | <code>response.status_code == 302</code>                                                                                                               | Pass   | 0.000s |
| Medium height 10 to 15m                                        | <code>isinstance(result, list)</code>                                                                                                                  | Pass   | 0.000s |
| No height skips classification                                 | <code>isinstance(result, list)</code>                                                                                                                  | Pass   | 0.000s |
| Standard height under 10m                                      | <code>isinstance(result, list)</code>                                                                                                                  | Pass   | 0.000s |

| Test Description                                         | Acceptance Criteria                                                                                       | Result | Time   |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------|--------|
| Tall structure over 15m                                  | <code>isinstance(result, list)</code>                                                                     | Pass   | 0.000s |
| Generate monitoring high score                           | <code>"Weekly" in r.values()</code>                                                                       | Pass   | 0.000s |
| Generate monitoring low score                            | <code>"Annually" in r.values()</code>                                                                     | Pass   | 0.000s |
| Generate monitoring medium score                         | <code>"Monthly" in r.values()</code>                                                                      | Pass   | 0.000s |
| Generate recommendations high score                      | <code>any("URGENT" in rec for rec in r)</code>                                                            | Pass   | 0.000s |
| Generate recommendations low score                       | <code>any("LOW" in rec for rec in r)</code>                                                               | Pass   | 0.000s |
| Generate recommendations medium score                    | <code>any("HIGH PRIORITY" in rec or "MODERATE" in rec for rec i...</code>                                 | Pass   | 0.000s |
| Identify key factors high risk                           | <code>any("High Risk Factor" in k for k in r.keys())</code>                                               | Pass   | 0.000s |
| Identify key factors medium risk                         | <code>any("Medium Risk Factor" in k for k in r.keys())</code>                                             | Pass   | 0.000s |
| Status colour fresh                                      | <code>col == dl.mod.GREEN</code>                                                                          | Pass   | 0.000s |
| Status colour missing                                    | <code>col == dl.mod.RED</code>                                                                            | Pass   | 0.000s |
| Status colour stale                                      | <code>col == dl.mod.AMBER</code>                                                                          | Pass   | 0.000s |
| Status colour unknown                                    | <code>col == dl.mod.STEEL</code>                                                                          | Pass   | 0.000s |
| Status label unknown uppcased                            | <code>dl.mod._status_label('degraded') == 'DEGRADED'</code>                                               | Pass   | 0.000s |
| Status label values                                      | <code>dl.mod._status_label('fresh') == 'FRESH'; dl.mod._status_label('stale') == 'STALE' (+1 more)</code> | Pass   | 0.000s |
| Alert line uses amber (#FFC107).                         | <code>'#FFC107' in hist_js</code>                                                                         | Pass   | 0.000s |
| Chart uses daily_observations from history API response. | <code>'daily_observations' in hist_js</code>                                                              | Pass   | 0.000s |
| Historical chart canvas element defined.                 | <code>'hist-timeseries-chart' in hist_js</code>                                                           | Pass   | 0.000s |
| Severe line uses red (#F44336).                          | <code>'#F44336' in hist_js</code>                                                                         | Pass   | 0.000s |
| Alert threshold line on the 50yr chart.                  | <code>'FloodAlert' in hist_js</code>                                                                      | Pass   | 0.000s |
| Warning threshold line on the 50yr chart.                | <code>'FloodWarning' in hist_js</code>                                                                    | Pass   | 0.000s |
| Severe threshold line on the 50yr chart.                 | <code>'SevereFloodWarning' in hist_js</code>                                                              | Pass   | 0.000s |
| Subsamples every 7th day for performance.                | <code>'7' in hist_js; 'sampleObs' in hist_js</code>                                                       | Pass   | 0.000s |
| Warning line uses orange (#FF9800).                      | <code>'#FF9800' in hist_js</code>                                                                         | Pass   | 0.000s |
| Cache is keyed by gauge ID (_gaugeId).                   | <code>'_gaugeId' in hist_js</code>                                                                        | Pass   | 0.000s |

| Test Description                                                         | Acceptance Criteria                                              | Result | Time   |
|--------------------------------------------------------------------------|------------------------------------------------------------------|--------|--------|
| Historical data is cached to avoid repeated fetches.                     | '_histData' in hist_js                                           | Pass   | 0.000s |
| If cache exists for same gauge, skip API fetch.                          | '_histData.gaugeId' in hist_js or<br>\{\} '_his...               | Pass   | 0.000s |
| Exceeded severe label                                                    | "Exceeded severe warning level" in<br>str(elements)              | Pass   | 0.000s |
| Within severe level                                                      | "4.000m" in str(elements)                                        | Pass   | 0.000s |
| Lines 138-141: historical high injected after main loop.                 | len(match) == 1;<br>match[0]["level_meters"] == 9.99             | Pass   | 0.010s |
| When historical_high_dt is None (out of range), post-loop block skipped. | all(o["level_meters"] != 8.0 for o in<br>result)                 | Pass   | 0.001s |
| When historical_high_date is None, both branches skipped.                | isinstance(result, list); len(result)<br>> 0                     | Pass   | 0.001s |
| Chart cleanup function defined.                                          | '_cleanupHistCharts' in hist_js                                  | Pass   | 0.000s |
| Stats bar shows flood exceedances.                                       | 'flood_exceedances' in hist_js                                   | Pass   | 0.000s |
| Stats bar shows mean level.                                              | 'mean_level' in hist_js or 'Mean' in<br>hist_js                  | Pass   | 0.000s |
| Stats bar shows record years.                                            | 'record_years' in hist_js or 'Record'<br>in hist_js              | Pass   | 0.000s |
| Should fetch from<br>/api/v1/gauges/{gauge_id}/history.                  | '/api/v1/gauges/' in js; '/history'<br>in js                     | Pass   | 0.000s |
| Should fetch from<br>/api/v1/trading/stress/storms.                      | '/api/v1/trading/stress/storms' in<br>js                         | Pass   | 0.000s |
| Right panel should show storm scenarios list.                            | 'hist-storms-panel' in js;<br>'hist-storms-list' in js (+1 more) | Pass   | 0.000s |
| Should render a time series chart.                                       | 'hist-timeseries-chart' in js;<br>'renderHistorical' in js       | Pass   | 0.000s |
| Historical tab JS should render without errors.                          | len(js) > 0                                                      | Pass   | 0.000s |
| Should have function to render storm list.                               | '_renderStormList' in js;<br>'hist-storm-row' in js              | Pass   | 0.000s |
| Chart should show flood stage threshold lines.                           | 'FloodAlert' in js; 'FloodWarning' in<br>js (+1 more)            | Pass   | 0.000s |
| Storm rows should show trigger-colored dots.                             | '#FFC107' in js; '#FF9800' in js (+1<br>more)                    | Pass   | 0.000s |
| Storm row click should navigate to Stress Test tab.                      | '_navigateToStress' in js;<br>'_stressStormHint' in js (+1 more) | Pass   | 0.000s |
| Should use addEventListener (not inline onclick) for storms.             | 'addEventListener' in js                                         | Pass   | 0.000s |



| Test Description                                                      | Acceptance Criteria                                                                                         | Result | Time   |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|--------|
| ghc_historical.get_js() returns substantial JS output.                | <code>len(hist_js) &gt; 1000</code>                                                                         | Pass   | 0.000s |
| Output long enough to contain real function definitions.              | <code>len(hist_js) &gt; 3000</code>                                                                         | Pass   | 0.000s |
| Historical tab fetches from <code>/gauges/{gauge_id}/history</code> . | <code>'/api/v1/gauges/'</code> in <code>hist_js</code> ;<br><code>'/history'</code> in <code>hist_js</code> | Pass   | 0.000s |
| Fetch URL is built using <code>gaugeId</code> variable.               | <code>'gaugeId'</code> in <code>hist_js</code>                                                              | Pass   | 0.000s |
| history.py lines 88-90: corrupted gaugehd file -i 500.                | <code>r.status_code == 500</code> ;<br><code>r.get_json()["status"] == "error"</code>                       | Pass   | 0.028s |
| history.py lines 144-146: statistics loader exception -i 500.         | <code>r.status_code == 500</code>                                                                           | Pass   | 0.027s |
| history.py lines 119-121: timeseries loader exception -i 500.         | <code>r.status_code == 500</code>                                                                           | Pass   | 0.028s |
| Assess historical risks active subsidence                             | <code>"Ground Stability Risk" in result</code> ;<br><code>"High" in result["Ground Stability Risk"]</code>  | Pass   | 0.000s |
| Assess historical risks high env                                      | <code>"Environmental Risk" in result</code> ;<br><code>"High" in result["Environmental Risk"]</code>        | Pass   | 0.000s |
| Assess historical risks low env                                       | <code>"Low" in result["Environmental Risk"]</code>                                                          | Pass   | 0.000s |
| Assess historical risks minor subsidence                              | <code>"Medium" in result["Ground Stability Risk"]</code>                                                    | Pass   | 0.000s |
| Assess historical risks moderate env                                  | <code>"Medium" in result["Environmental Risk"]</code>                                                       | Pass   | 0.000s |
| Assess historical risks moderate flood                                | <code>"Medium" in result["Historical Flood Risk"]</code>                                                    | Pass   | 0.000s |
| Assess historical risks no flood damage                               | <code>"Low" in result["Historical Flood Risk"]</code>                                                       | Pass   | 0.000s |
| Assess historical risks severe flood                                  | <code>"Historical Flood Risk" in result</code> ;<br><code>"High" in result["Historical Flood Risk"]</code>  | Pass   | 0.000s |
| Assess historical risks stable ground                                 | <code>"Low" in result["Ground Stability Risk"]</code>                                                       | Pass   | 0.000s |
| Build flood history bad json                                          | <code>any("Error" in t or "error" in t.lower() for t in texts)</code>                                       | Pass   | 0.001s |
| Build flood history no flood days                                     | <code>any("No flood events" in t for t in texts)</code>                                                     | Pass   | 0.001s |
| Build flood history no hd file                                        | <code>any("No historical" in t or "historical" in t.lower() for...</code>                                   | Pass   | 0.000s |

| Test Description                                                                                   | Acceptance Criteria                                                         | Result | Time   |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------|--------|
| Build flood history no reference gauges                                                            | <code>any("No reference gauges" in t for t in texts)</code>                 | Pass   | 0.000s |
| Build flood history with flood data                                                                | <code>isinstance(result, list); len(tables) &gt;= 1</code>                  | Pass   | 0.002s |
| Empty flood events                                                                                 | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| Empty property does not crash                                                                      | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| Has history header                                                                                 | <code>any("History" in t or "Flood" in t for t in texts)</code>             | Pass   | 0.000s |
| No flood history                                                                                   | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| Returns list                                                                                       | <code>isinstance(result, list); len(result) &gt; 0</code>                   | Pass   | 0.000s |
| With environmental issues                                                                          | <code>isinstance(result, list); len(tables) &gt;= 1</code>                  | Pass   | 0.000s |
| With fire incidents                                                                                | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| With ground conditions                                                                             | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| With mortgage data                                                                                 | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| Hmda minimal fields                                                                                | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| Hmda rate spread formatted                                                                         | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| Hmda section creates table                                                                         | <code>len(tables) &gt;= 1</code>                                            | Pass   | 0.000s |
| When gauges file has invalid JSON, exception is caught (lines 249-250)                             | <code>resp.status_code == 200; data['status'] == 'success'</code>           | Pass   | 0.072s |
| When gauges file has <code>STORM_HOURS</code> readings, hydrograph is truncated (line 245).        | <code>resp.status_code == 200</code>                                        | Pass   | 0.063s |
| When gauges JSON is corrupt, exception caught (lines 109-110),                                     | <code>resp.status_code == 200; data['status'] == 'success' (+1 more)</code> | Pass   | 0.030s |
| When gauges has fewer than <code>STORM_HOURS</code> readings, pads with last value (line 106-108). | <code>resp.status_code == 200; len(data['hourly']) == 168</code>            | Pass   | 0.034s |
| When <code>peak_hour == num_hours</code> (remaining=0), <code>frac=1.0</code> for post-peak hours. | <code>len(levels) == 5; abs(levels[0] - 3.0) &lt; 0.01</code>               | Pass   | 0.000s |
| <code>run_series</code> result matches manual step-by-step.                                        | <code>qf == pytest.approx(qf_series[h], rel=1e-6)</code>                    | Pass   | 0.000s |
| Wet initial state produces higher early quickflow.                                                 | <code>qf_wet[:6].mean() &gt; qf_dry[:6].mean()</code>                       | Pass   | 0.000s |
| No rain no quickflow                                                                               | <code>np.all(qf == pytest.approx(0.0))</code>                               | Pass   | 0.000s |
| Quickflow positive                                                                                 | <code>np.all(qf &gt;= 0.0)</code>                                           | Pass   | 0.002s |
| Shape                                                                                              | <code>qf.shape == (168,); len(states) == 168</code>                         | Pass   | 0.000s |

| Test Description                                                 | Acceptance Criteria                                                                                                            | Result | Time   |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Soil moisture drains to near zero after long dry period.         | <code>states[-1].soil_moisture &lt; 1.0</code>                                                                                 | Pass   | 0.000s |
| Dry start no precip no quickflow                                 | <code>qf == pytest.approx(0.0);</code><br><code>new_state.soil_moisture ==</code><br><code>pytest.approx(0.0) (+1 more)</code> | Pass   | 0.000s |
| Groundwater accumulates from soil                                | <code>new_state.groundwater &gt; 0.0</code>                                                                                    | Pass   | 0.000s |
| Negative precip treated as zero                                  | <code>qf &gt;= 0.0; new_state.soil_moisture &gt;=</code><br><code>0.0</code>                                                   | Pass   | 0.000s |
| Precip increases soil moisture                                   | <code>new_state.soil_moisture &gt; 0.0</code>                                                                                  | Pass   | 0.000s |
| Precip produces quickflow                                        | <code>qf &gt; 0.0</code>                                                                                                       | Pass   | 0.000s |
| Soil moisture drains without rain                                | <code>new_state.soil_moisture &lt; 50.0</code>                                                                                 | Pass   | 0.000s |
| Elevated antecedent state produces more quickflow per mm.        | <code>qf_wet &gt; qf_dry</code>                                                                                                | Pass   | 0.000s |
| All zero weights returns zero                                    | <code>result == pytest.approx(7.5)</code>                                                                                      | Pass   | 0.000s |
| Closer gauge has higher weight                                   | <code>result &lt; 5.0</code>                                                                                                   | Pass   | 0.000s |
| Empty list returns zero                                          | <code>idw_interpolate.from_gauges([]) ==</code><br><code>0.0</code>                                                            | Pass   | 0.000s |
| Equal distance gauges averages wse                               | <code>result == pytest.approx(3.0)</code>                                                                                      | Pass   | 0.000s |
| Near gauge returns wse directly                                  | <code>result == pytest.approx(5.0)</code>                                                                                      | Pass   | 0.000s |
| Power one changes result                                         | <code>r2 != pytest.approx(r1, rel=0.01)</code>                                                                                 | Pass   | 0.000s |
| Single far gauge                                                 | <code>result == pytest.approx(3.0)</code>                                                                                      | Pass   | 0.000s |
| Line 158: power parameter affects weights.                       | <code>r3 &lt; r1</code>                                                                                                        | Pass   | 0.000s |
| Line 149-150: empty input → 0.0.                                 | <code>idw_interpolate.from_gauges([]) ==</code><br><code>0.0</code>                                                            | Pass   | 0.000s |
| Gauge at exactly min_distance=1.0 is NOT less than, so IDW used. | <code>result == pytest.approx(5.0)</code>                                                                                      | Pass   | 0.000s |
| Multiple gauges produce weighted result.                         | <code>2.0 &lt; result &lt; 6.0; result &lt; 4.0</code>                                                                         | Pass   | 0.000s |
| Line 156-157: dist ; min_distance → return wse.                  | <code>result == pytest.approx(42.0)</code>                                                                                     | Pass   | 0.000s |
| Idw gauge spreads computed with two gauges                       | <code>"any_flood" in idw;</code><br><code>len(idw["any_flood"]) == len(TENORS)</code>                                          | Pass   | 0.235s |
| With empty nearest_gauges, idw_gauge_spreads is still a dict.    | <code>isinstance(result.get("idw_gauge_spreads"),</code><br><code>dict)</code>                                                 | Pass   | 0.240s |
| Fair income 30k to 50k                                           | <code>isinstance(result, list)</code>                                                                                          | Pass   | 0.000s |
| Good income 50k to 100k                                          | <code>isinstance(result, list)</code>                                                                                          | Pass   | 0.000s |
| High income over 100k                                            | <code>isinstance(result, list)</code>                                                                                          | Pass   | 0.000s |
| Limited income under 30k                                         | <code>isinstance(result, list)</code>                                                                                          | Pass   | 0.000s |
| Total income with secondary                                      | <code>isinstance(result, list)</code>                                                                                          | Pass   | 0.000s |
| Input dir accepts string                                         | <code>isinstance(gen.input_dir, Path)</code>                                                                                   | Pass   | 0.000s |

| Test Description                                               | Acceptance Criteria                                                                                                            | Result | Time   |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Input dir set                                                  | <code>gen.input_dir == tmp_path</code>                                                                                         | Pass   | 0.000s |
| Output path explicit                                           | <code>gen.output_path == out</code>                                                                                            | Pass   | 0.000s |
| Styles initialised                                             | <code>gen.title_style is not None;</code><br><code>gen.body_style is not None</code> (+1 more)                                 | Pass   | 0.000s |
| Known inputs                                                   | <code>"properties" in m["gauge.json"];</code><br><code>"gaugehd" in m["gauge.json"]</code>                                     | Pass   | 0.000s |
| Area = float('inf') skips all threshold checks, hits fallback. | <code>isinstance(result, tuple);</code><br><code>len(result) == 2</code>                                                       | Pass   | 0.000s |
| Age premium factors all positive                               | <code>lo &gt; 0; hi &gt;= lo</code>                                                                                            | Pass   | 0.000s |
| Area bands are sorted                                          | <code>thresholds == sorted(thresholds)</code>                                                                                  | Pass   | 0.000s |
| Base rate range                                                | <code>lo &gt; 0; hi &gt; lo</code>                                                                                             | Pass   | 0.000s |
| Construction type factors present                              | <code>'Brick' in CONSTRUCTION_TYPE_FACTORS;</code><br><code>'Timber Frame' in</code><br><code>CONSTRUCTION_TYPE_FACTORS</code> | Pass   | 0.000s |
| Flood risk factors increase with risk                          | <code>very_high_hi &gt; very_low_lo</code>                                                                                     | Pass   | 0.000s |
| Flood risk factors present                                     | <code>set(FLOOD_RISK_PREMIUM_FACTORS.keys())</code><br><code>== expected</code>                                                | Pass   | 0.000s |
| Min max premium bounds                                         | <code>MIN_PREMIUM == 200;</code> <code>MAX_PREMIUM ==</code><br><code>20.000</code>                                            | Pass   | 0.000s |
| Property type factors present                                  | <code>set(PROPERTY_TYPE_PREMIUM_FACTORS.keys())</code><br><code>== expected</code>                                             | Pass   | 0.000s |
| Base rates by level                                            | <code>result == pytest.approx(rate)</code>                                                                                     | Pass   | 0.000s |
| Capped at 10 pct                                               | <code>result == pytest.approx(0.10)</code>                                                                                     | Pass   | 0.000s |
| Depth increases premium                                        | <code>with_depth &gt; no_depth</code>                                                                                          | Pass   | 0.000s |
| Depth multiplier formula                                       | <code>rate == pytest.approx(expected)</code>                                                                                   | Pass   | 0.000s |
| Unknown level uses default                                     | <code>result == pytest.approx(0.005)</code>                                                                                    | Pass   | 0.000s |
| Zero depth no effect                                           | <code>base == zero</code>                                                                                                      | Pass   | 0.000s |
| Unknown integer field returns int                              | <code>isinstance(result, int); 1 &lt;= result</code><br><code>&lt;= 10</code>                                                  | Pass   | 0.000s |
| Integer type returns int                                       | <code>isinstance(result, int); 0 &lt;= result</code><br><code>&lt;= 100</code>                                                 | Pass   | 0.000s |
| Baseline alert only                                            | <code>self.run(tmp_path, n_severe=0,</code><br><code>n_warning=0) == "baseline"</code>                                         | Pass   | 0.003s |
| Catastrophic 10 severe                                         | <code>self.run(tmp_path, n_severe=10) ==</code><br><code>"catastrophic"</code>                                                 | Pass   | 0.012s |
| Extreme 6 severe                                               | <code>self.run(tmp_path, n_severe=6) ==</code><br><code>"extreme"</code>                                                       | Pass   | 0.008s |
| Moderate 1 severe                                              | <code>self.run(tmp_path, n_severe=1) ==</code><br><code>"moderate"</code>                                                      | Pass   | 0.002s |

| Test Description                                          | Acceptance Criteria                                                                                  | Result | Time   |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------|--------|
| Severe 3 severe                                           | <code>self._run(tmp_path, n_severe=3) == "severe"</code>                                             | Pass   | 0.005s |
| Batch generator populates category                        | <code>all(s.intensity_category in valid for s in seqs)</code>                                        | Pass   | 0.001s |
| From dict missing category defaults empty                 | <code>reconstructed.intensity_category == ""</code>                                                  | Pass   | 0.000s |
| From dict round trip                                      | <code>reconstructed.intensity_category == s.intensity_category</code>                                | Pass   | 0.000s |
| Serialized sequences have category                        | <code>all(s.intensity_category in valid for s in seqs)</code>                                        | Pass   | 0.001s |
| To dict includes intensity category                       | <code>"intensity_category" in d; d["intensity_category"] == s.intensity_category</code>              | Pass   | 0.001s |
| Running ‘python -m models.intensity’ executes cli.main(). | <code>result.returncode == 0; "Storm Intensity" in result.stdout or "usage" in result.s...</code>    | Pass   | 0.523s |
| Beta value                                                | <code>IntensityProfile.BETA.value == "beta"</code>                                                   | Pass   | 0.000s |
| Gaussian value                                            | <code>IntensityProfile.GAUSSIAN.value == "gaussian"; DecayKernel.GAUSSIAN.value == "gaussian"</code> | Pass   | 0.000s |
| Triangular value                                          | <code>IntensityProfile.TRIANGULAR.value == "triangular"</code>                                       | Pass   | 0.000s |
| All profiles produce valid storm                          | <code>isinstance(storm, Storm); len(storm.track) &gt; 0</code>                                       | Pass   | 0.000s |
| Beta profile creates storm                                | <code>storm is not None; len(storm.track) &gt; 0</code>                                              | Pass   | 0.000s |
| Gaussian profile peak near midpoint                       | <code>8 &lt;= peak_idx &lt;= 16</code>                                                               | Pass   | 0.000s |
| Triangular max equals peak intensity                      | <code>max(tp.intensity for tp in storm.track) == pytest.approx(...</code>                            | Pass   | 0.000s |
| Triangular profile peak near midpoint                     | <code>8 &lt;= peak_idx &lt;= 16</code>                                                               | Pass   | 0.000s |
| Unknown sequence type falls back to 1 storm.              | <code>count == 1</code>                                                                              | Pass   | 0.008s |
| ContextMenuHandler JS renders without f-string errors.    | <code>len(js) &gt; 0; '&lt;script&gt;' in js</code>                                                  | Pass   | 0.002s |
| GaugeHazardCurve JS renders without f-string errors.      | <code>len(js) &gt; 0; '&lt;script&gt;' in js</code>                                                  | Pass   | 0.000s |
| GaugePDFPanel JS renders without f-string errors.         | <code>len(js) &gt; 0; 'GaugePDFPanel' in js</code>                                                   | Pass   | 0.000s |

| Test Description                                                      | Acceptance Criteria                                                                     | Result | Time   |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------|--------|
| GaugeStormAnalysis JS renders without f-string errors.                | <code>len(js) &gt; 0; '&lt;script&gt;' in js</code>                                     | Pass   | 0.000s |
| NotificationSystem JS renders without f-string errors.                | <code>len(js) &gt; 0; '&lt;script&gt;' in js</code>                                     | Pass   | 0.001s |
| PropertyHazardCurvePanel JS renders without f-string errors.          | <code>len(js) &gt; 0; '&lt;script&gt;' in js</code>                                     | Pass   | 0.000s |
| PropertyPDFPanel JS renders without f-string errors.                  | <code>len(js) &gt; 0; 'PropertyPDFPanel' in js</code>                                   | Pass   | 0.000s |
| PropertyStormAnalysis JS renders without f-string errors.             | <code>len(js) &gt; 0; '&lt;script&gt;' in js (+1 more)</code>                           | Pass   | 0.392s |
| PropertyStormAnalysis JS must guard against empty flood_events array. | <code>'flood_events.length === 0' in js</code>                                          | Pass   | 0.000s |
| TradingDeskPanel should be in __all__.                                | <code>'TradingDeskPanel' in __all__</code>                                              | Pass   | 0.000s |
| InteractivityManager should have trading_desk attribute.              | <code>hasattr(mgr, 'trading_desk')</code>                                               | Pass   | 0.000s |
| Setup map adds preloader before notifications                         | <code>preloader_idx != -1; notifications_idx != -1 (+1 more)</code>                     | Pass   | 0.000s |
| Startup preloader imported in init                                    | <code>'from .startup import StartupPreloader' in src</code>                             | Pass   | 0.002s |
| Startup preloader instantiated in init                                | <code>'self.startup_preloader = StartupPreloader()' in src</code>                       | Pass   | 0.000s |
| Elevated rate 5 to 7                                                  | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| High rate over 7                                                      | <code>isinstance(result, list)</code>                                                   | Pass   | 0.000s |
| Low rate under 3                                                      | <code>isinstance(result, list)</code>                                                   | Pass   | 0.000s |
| Moderate rate 3 to 5                                                  | <code>isinstance(result, list)</code>                                                   | Pass   | 0.000s |
| No rate unknown                                                       | <code>isinstance(result, list)</code>                                                   | Pass   | 0.000s |
| Line 64: empty yield_curve returns default fallback 0.04.             | <code>result == 0.04</code>                                                             | Pass   | 0.000s |
| Line 64: empty yield_curve returns fallback.                          | <code>result == 0.05</code>                                                             | Pass   | 0.000s |
| Line 77: fallthrough at end of for loop returns last key rate.        | <code>interpolate_yield_rate(curve, 4.0) == 0.04; result == pytest.approx(0.035)</code> | Pass   | 0.000s |
| Line 71: <code>t != keys[-1]</code> returns the last key's rate.      | <code>result == 0.045</code>                                                            | Pass   | 0.000s |
| Line 71: <code>t == keys[-1]</code> returns the last key's rate.      | <code>result == 0.045</code>                                                            | Pass   | 0.000s |



| Test Description                                                                                               | Acceptance Criteria                                                                                                                               | Result | Time   |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Invalidate clears cache                                                                                        | <code>stress_helpers._predictor_cache</code><br><code>is not None;</code><br><code>stress_helpers._predictor_cache</code><br><code>is None</code> | Pass   | 0.000s |
| File exists                                                                                                    | <code>INVENTORY_PATH.exists()</code>                                                                                                              | Pass   | 0.000s |
| File is valid json                                                                                             | <code>isinstance(data, dict)</code>                                                                                                               | Pass   | 0.001s |
| Has at least one model                                                                                         | <code>len(models) &gt; 0</code>                                                                                                                   | Pass   | 0.000s |
| Has metadata                                                                                                   | <code>"metadata" in inventory;</code> <code>"framework"</code><br><code>in inventory["metadata"]</code>                                           | Pass   | 0.000s |
| Has model chain                                                                                                | <code>"model_chain" in inventory</code>                                                                                                           | Pass   | 0.000s |
| Has models key                                                                                                 | <code>"models" in inventory</code>                                                                                                                | Pass   | 0.000s |
| Get models includes risk fields                                                                                | <code>response.status_code == 200;</code><br><code>data["status"] == "success"</code> (+4<br>more)                                                | Pass   | 0.002s |
| Line 125: <code>_inverse_normal(1.5)</code> re-<br>turns <code>+inf</code> .                                   | <code>result == float('inf')</code>                                                                                                               | Pass   | 0.000s |
| Line 123: <code>_inverse_normal(-0.5)</code> re-<br>turns <code>-inf</code> .                                  | <code>result == float('-inf')</code>                                                                                                              | Pass   | 0.000s |
| Line 125: <code>_inverse_normal(1.0)</code> re-<br>turns <code>+inf</code> .                                   | <code>result == float('inf')</code>                                                                                                               | Pass   | 0.000s |
| Line 123: <code>_inverse_normal(0)</code> re-<br>turns <code>-inf</code> .                                     | <code>result == float('-inf')</code>                                                                                                              | Pass   | 0.000s |
| Lines 170-171: <code>p &lt; p_high (0.97575)</code><br>uses the upper tail formula.                            | <code>result == pytest.approx(2.326,</code><br><code>abs=0.02)</code>                                                                             | Pass   | 0.000s |
| Lines 170-171: <code>p=0.98</code> also triggers<br>high-p branch.                                             | <code>result == pytest.approx(2.054,</code><br><code>abs=0.02)</code>                                                                             | Pass   | 0.000s |
| Lines 170-171: high-p branch re-<br>turns positive value.                                                      | <code>result &gt; 0</code>                                                                                                                        | Pass   | 0.000s |
| Lines 170-171: verify roundtrip<br>CDF( <code>inverse.CDF(p)</code> ) <code>p</code> for high <code>p</code> . | <code>recovered == pytest.approx(p,</code><br><code>abs=0.001)</code>                                                                             | Pass   | 0.000s |
| Below average 2 to 4pct                                                                                        | <code>isinstance(result, list)</code>                                                                                                             | Pass   | 0.000s |
| Excellent over 8pct                                                                                            | <code>isinstance(result, list)</code>                                                                                                             | Pass   | 0.000s |
| Fair 4 to 6pct                                                                                                 | <code>isinstance(result, list)</code>                                                                                                             | Pass   | 0.000s |
| Good 6 to 8pct                                                                                                 | <code>isinstance(result, list)</code>                                                                                                             | Pass   | 0.000s |
| Poor under 2pct                                                                                                | <code>isinstance(result, list)</code>                                                                                                             | Pass   | 0.000s |
| Get raises on unknown key                                                                                      | —                                                                                                                                                 | Pass   | 0.000s |
| <code>JSONFileConfig.get('hazard_curves')</code><br>returns the hazard curves filename.                        | <code>result ==</code><br><code>JSONFileConfig.HAZARD_CURVES</code>                                                                               | Pass   | 0.000s |
| Datetime encoded as isoformat                                                                                  | <code>"2024-01-15" in result</code>                                                                                                               | Pass   | 0.000s |
| Numpy float32 encoded                                                                                          | <code>abs(result - 1.5) &lt; 0.01</code>                                                                                                          | Pass   | 0.000s |
| Numpy float encoded                                                                                            | <code>abs(result - 3.14) &lt; 1e-6</code>                                                                                                         | Pass   | 0.000s |

| Test Description                                              | Acceptance Criteria                                                                                           | Result | Time   |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------|--------|
| Numpy int32 encoded                                           | <code>result == 7</code>                                                                                      | Pass   | 0.000s |
| Numpy int encoded                                             | <code>result == 42; isinstance(result, int)</code>                                                            | Pass   | 0.000s |
| Numpy ndarray encoded as list                                 | <code>result == [1, 2, 3]</code>                                                                              | Pass   | 0.000s |
| Unknown type raises type error                                | —                                                                                                             | Pass   | 0.000s |
| Charts reference <code>first_trigger_hour</code> for KO line. | <code>'first_trigger_hour' in stress_js</code>                                                                | Pass   | 0.000s |
| Charts have KO annotation line.                               | <code>'koLine' in stress_js</code>                                                                            | Pass   | 0.000s |
| Shows KO $H\{n\}$ label for triggered trades.                 | <code>'KO H' in stress_js</code>                                                                              | Pass   | 0.000s |
| References <code>triggered_hour</code> for KO status.         | <code>'triggered_hour' in stress_js</code>                                                                    | Pass   | 0.000s |
| Custom label width                                            | <code>t._colWidths[0] ==<br/>pytest.approx(1w); t._colWidths[1] ==<br/>pytest.approx(AVAIL.WIDTH - 1w)</code> | Pass   | 0.000s |
| Default label width                                           | <code>t._colWidths[0] == pytest.approx(2.8<br/>* inch)</code>                                                 | Pass   | 0.000s |
| Multiple pairs                                                | <code>len(t._cellvalues) == 3</code>                                                                          | Pass   | 0.000s |
| Returns table                                                 | <code>isinstance(t, Table); isinstance(t,<br/>Table)</code>                                                   | Pass   | 0.000s |
| Numeric values are <code>str()</code> -ified in Paragraph.    | <code>isinstance(t, Table)</code>                                                                             | Pass   | 0.000s |
| LTV $\geq 1$ (e.g. 75 instead of 0.75) should still work.     | <code>isinstance(result, list)</code>                                                                         | Pass   | 0.000s |
| Ltv over 60 low medium                                        | <code>isinstance(result, list)</code>                                                                         | Pass   | 0.000s |
| Ltv over 70 medium                                            | <code>isinstance(result, list)</code>                                                                         | Pass   | 0.000s |
| Ltv over 80 medium high                                       | <code>isinstance(result, list)</code>                                                                         | Pass   | 0.000s |
| Ltv over 90 high risk                                         | <code>isinstance(result, list)</code>                                                                         | Pass   | 0.000s |
| Ltv under 60 low risk                                         | <code>isinstance(result, list)</code>                                                                         | Pass   | 0.000s |
| High ltv 90 to 95                                             | <code>isinstance(result, list)</code>                                                                         | Pass   | 0.001s |
| Low ltv under 70                                              | <code>isinstance(result, list)</code>                                                                         | Pass   | 0.000s |
| Low medium ltv 70 to 80                                       | <code>isinstance(result, list)</code>                                                                         | Pass   | 0.000s |
| Medium ltv 80 to 90                                           | <code>isinstance(result, list)</code>                                                                         | Pass   | 0.001s |
| Very high ltv over 95                                         | <code>isinstance(result, list)</code>                                                                         | Pass   | 0.001s |
| At high boundary                                              | <code>RiskAssessor.assess_ltv_risk_level(0.95)<br/>== "High"</code>                                           | Pass   | 0.000s |
| At low boundary                                               | <code>RiskAssessor.assess_ltv_risk_level(0.6)<br/>== "Low"</code>                                             | Pass   | 0.000s |
| At moderate boundary                                          | <code>RiskAssessor.assess_ltv_risk_level(0.8)<br/>== "Moderate"</code>                                        | Pass   | 0.000s |
| Critical ltv                                                  | <code>RiskAssessor.assess_ltv_risk_level(0.98)<br/>== "Critical"</code>                                       | Pass   | 0.000s |

| Test Description                                                                                 | Acceptance Criteria                                                                                                                               | Result | Time   |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| High ltv                                                                                         | <code>RiskAssessor.assess_ltv_risk_level(0.88)</code><br>== "High"                                                                                | Pass   | 0.000s |
| Low ltv                                                                                          | <code>RiskAssessor.assess_ltv_risk_level(0.5)</code><br>== "Low"                                                                                  | Pass   | 0.000s |
| Moderate ltv                                                                                     | <code>RiskAssessor.assess_ltv_risk_level(0.7)</code><br>== "Moderate"                                                                             | Pass   | 0.000s |
| None returns unknown                                                                             | <code>RiskAssessor.assess_flood_risk_level(None)</code><br>== "Unknown";<br><code>RiskAssessor.assess_ltv_risk_level(None)</code><br>== "Unknown" | Pass   | 0.000s |
| Normalises 95 pct                                                                                | <code>RiskAssessor.assess_ltv_risk_level(95)</code><br>== "High"                                                                                  | Pass   | 0.000s |
| Normalises percentage input                                                                      | <code>RiskAssessor.assess_ltv_risk_level(70)</code><br>== "Moderate"                                                                              | Pass   | 0.000s |
| Bad date string gracefully skipped                                                               | <code>isinstance(elements, list)</code>                                                                                                           | Pass   | 0.000s |
| Moderate recent activity                                                                         | "Moderate recent activity" in<br><code>str(elements)</code>                                                                                       | Pass   | 0.000s |
| No recent activity                                                                               | "No recent severe activity" in<br><code>str(elements)</code>                                                                                      | Pass   | 0.000s |
| Recent activity within 1 year                                                                    | "Recent severe level activity" in<br><code>str(elements)</code>                                                                                   | Pass   | 0.000s |
| Body style alignment                                                                             | <code>s.alignment == TA_CENTER</code>                                                                                                             | Pass   | 0.000s |
| Body style bold                                                                                  | <code>s.fontName == 'Helvetica-Bold'</code>                                                                                                       | Pass   | 0.000s |
| Body style default                                                                               | <code>isinstance(s, ParagraphStyle);</code><br><code>s.fontName == 'Helvetica'</code>                                                             | Pass   | 0.000s |
| Build header footer callable                                                                     | <code>callable(cb)</code>                                                                                                                         | Pass   | 0.000s |
| White hdr style returns paragraph style                                                          | <code>isinstance(s, ParagraphStyle);</code><br><code>s.fontName == 'Helvetica-Bold'</code>                                                        | Pass   | 0.000s |
| Lines 67-68: legacy file uses mtime cache — second call returns cached.                          | <code>r1["storms"][0]["storm_id"] ==</code><br>"LEGACY-002"; <code>r2</code> is <code>r1</code>                                                   | Pass   | 0.005s |
| Lines 61-73: falls back to <code>stress_storms.json</code> when <code>_index.json</code> absent. | <code>result</code> is not <code>None</code> ;<br><code>result["storms"][0]["storm_id"]</code><br>== "LEGACY-001"                                 | Pass   | 0.005s |
| Lines 64-66: <code>OSError</code> on legacy file stat returns existing cache.                    | <code>result == {"storms": [{"storm_id":</code><br>"cached-legacy"}]}                                                                             | Pass   | 0.022s |
| Coverage available false when no index                                                           | <code>r.get_json()["coverage_available"]</code> is<br><code>False</code>                                                                          | Pass   | 0.001s |
| Coverage available true when index present                                                       | <code>r.get_json()["coverage_available"]</code> is<br><code>True</code>                                                                           | Pass   | 0.001s |
| Empty dir returns empty list                                                                     | <code>data["reports"] == []</code>                                                                                                                | Pass   | 0.001s |

| Test Description                                                                                                       | Acceptance Criteria                                                                             | Result | Time   |
|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|--------|
| File has size and modified                                                                                             | <code>reports[0]["size"] &gt; 0; "modified" in reports[0]</code>                                | Pass   | 0.001s |
| File type label json                                                                                                   | <code>r.get_json()["reports"][0]["type"] == "JSON"</code>                                       | Pass   | 0.001s |
| File type label pdf                                                                                                    | <code>r.get_json()["reports"][0]["type"] == "PDF"</code>                                        | Pass   | 0.001s |
| File type label unknown                                                                                                | <code>r.get_json()["reports"][0]["type"] == "XYZ"</code>                                        | Pass   | 0.001s |
| No dir returns empty list                                                                                              | <code>r.status_code == 200; data["status"] == "success" (+2 more)</code>                        | Pass   | 0.000s |
| Pdf file listed                                                                                                        | <code>len(data["reports"]) == 1; data["reports"][0]["filename"] == "report.pdf"</code>          | Pass   | 0.001s |
| Subdirectories not listed                                                                                              | <code>r.get_json()["reports"] == []</code>                                                      | Pass   | 0.001s |
| <code>sys.exit(0)</code> fires before the gauge file is opened.                                                        | —                                                                                               | Pass   | 0.002s |
| <code>sys.exit(0)</code> fires before the property file is opened.                                                     | —                                                                                               | Pass   | 0.002s |
| Lines 324-327: all category/page pairs are logged.                                                                     | —                                                                                               | Pass   | 0.002s |
| Lines 412-419: <code>-list-categories</code> iterates category dict, <code>sys.exit(0)</code> .                        | —                                                                                               | Pass   | 0.002s |
| Lines 469-476: <code>-list-categories</code> iterates category dict, <code>sys.exit(0)</code> .                        | —                                                                                               | Pass   | 0.002s |
| Lines 321-328: <code>-list-categories</code> → <code>get_page_categories()</code> iterated, <code>sys.exit(0)</code> . | —                                                                                               | Pass   | 0.001s |
| No file returns empty list                                                                                             | <code>r.status_code == 200; data["status"] == "success" (+2 more)</code>                        | Pass   | 0.000s |
| Summary fields                                                                                                         | <code>first["counterparty_id"] == "CTPY-001"; first["name"] == "Test Bank PLC" (+2 more)</code> | Pass   | 0.002s |
| With data returns summary                                                                                              | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                        | Pass   | 0.002s |
| No storms file returns 404                                                                                             | <code>r.status_code == 404</code>                                                               | Pass   | 0.000s |
| Options returns ok                                                                                                     | <code>r.status_code == 200</code>                                                               | Pass   | 0.000s |
| Storm has expected fields                                                                                              | <code>len(storms) &gt;= 1; "storm_id" in s (+3 more)</code>                                     | Pass   | 0.005s |
| Storm name is capitalised intensity_category, not empty.                                                               | <code>s["name"] == s["intensity_category"].capitalize()</code>                                  | Pass   | 0.005s |
| Storms list count                                                                                                      | <code>data["count"] &gt;= 1</code>                                                              | Pass   | 0.005s |

| Test Description                                                              | Acceptance Criteria                                                                                                       | Result | Time   |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Storms sorted by gauges_severe descending (most severe first).                | <code>counts == sorted(counts, reverse=True)</code>                                                                       | Pass   | 0.005s |
| With data returns success                                                     | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                                                  | Pass   | 0.004s |
| Lines 218-219: invalid JSON in gaugets file triggers except continue.         | <code>data["status"] == "success"; data["count"] &gt;= 1</code>                                                           | Pass   | 0.031s |
| Lines 235-239: invalid metadata JSON triggers except continue.                | <code>r.status_code == 200</code>                                                                                         | Pass   | 0.030s |
| Lines 280-292: property flooding enriches storm with depth/damage.            | <code>storms["STORM-D"]["properties_flooded"] == 1; storms["STORM-D"]["max_depth_m"] == 1.2 (+1 more)</code>              | Pass   | 0.032s |
| Lines 280-281: property.json read error triggers except pass.                 | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                                                  | Pass   | 0.031s |
| Line 288: property not in valued_ids is skipped; storm filtered out.          | <code>"STORM-E" not in storm_ids</code>                                                                                   | Pass   | 0.030s |
| Line 204: storm responses with empty storm_id are skipped.                    | <code>data["status"] == "success"; "" not in storm_ids (+1 more)</code>                                                   | Pass   | 0.029s |
| Lines 235-237: storms.json enrichment path (not sequences).                   | <code>s["intensity_category"] == "extreme"; s["effective_precipitation_mm"] == 80.0</code>                                | Pass   | 0.030s |
| Lines 254-266: storms without metadata get classified by gauges_severe count. | <code>storms["S-CAT"]["intensity_category"] == "catastrophic"; storms["S-CAT"]["name"] == "Catastrophic" (+4 more)</code> | Pass   | 0.051s |
| Line 292: flood events with depth != 0 are skipped; storm filtered out.       | <code>"STORM-F" not in storm_ids</code>                                                                                   | Pass   | 0.030s |
| Created meeting appears in list                                               | <code>len(data["meetings"]) == 1; data["meetings"][0]["title"] == "Board Meeting"</code>                                  | Pass   | 0.003s |
| Empty list returns success                                                    | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                                                  | Pass   | 0.001s |
| List summary has required keys                                                | <code>key in m</code>                                                                                                     | Pass   | 0.002s |
| Sorted by date desc                                                           | <code>dates == sorted(dates, reverse=True)</code>                                                                         | Pass   | 0.003s |
| List mortgages empty                                                          | <code>r.status_code == 200; r.get_json()["count"] == 0</code>                                                             | Pass   | 0.001s |
| List mortgages exception returns 500                                          | <code>r.status_code == 500; r.get_json()["status"] == "error"</code>                                                      | Pass   | 0.001s |
| List mortgages success                                                        | <code>r.status_code == 200; data["status"] == "success" (+2 more)</code>                                                  | Pass   | 0.001s |
| sys.exit(0) fires before the flood file is opened.                            | —                                                                                                                         | Pass   | 0.001s |

| Test Description                                                           | Acceptance Criteria                                                     | Result | Time   |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------|--------|--------|
| sys.exit(0) fires before the gauge file is opened.                         | —                                                                       | Pass   | 0.002s |
| sys.exit(0) fires before the property file is opened.                      | —                                                                       | Pass   | 0.002s |
| Lines 405-410: -list-pages iterates available pages, sys.exit(0).          | —                                                                       | Pass   | 0.003s |
| Lines 462-467: -list-pages iterates available pages, sys.exit(0).          | —                                                                       | Pass   | 0.002s |
| Lines 314-319: -list-pages → list_available_pages() iterated, sys.exit(0). | —                                                                       | Pass   | 0.002s |
| Loader exception returns 500                                               | r.status_code == 500;<br>r.get_json()["status"] == "error"<br>(+1 more) | Pass   | 0.001s |
| Success empty list                                                         | r.status_code == 200; data["status"] == "success" (+2 more)             | Pass   | 0.001s |
| Success with properties                                                    | r.status_code == 200;<br>r.get_json()["count"] == 2                     | Pass   | 0.001s |
| Empty dir returns zero count                                               | r.status_code == 200; data["status"] == "success" (+2 more)             | Pass   | 0.001s |
| Returns trades list key                                                    | "trades" in r.get_json()                                                | Pass   | 0.001s |
| Corrupt trade file returns 500                                             | r.status_code == 500;<br>r.get_json()["status"] == "error"              | Pass   | 0.028s |
| list_prs_trades reads PRS-*.json files and returns them.                   | r.status_code == 200; data["count"] == 1 (+1 more)                      | Pass   | 0.028s |
| Loads list json                                                            | result == [1, 2, 3]                                                     | Pass   | 0.001s |
| Loads nested json                                                          | _load(p) == data                                                        | Pass   | 0.001s |
| Loads valid json                                                           | _load(p) == {'key': 'value'}                                            | Pass   | 0.001s |
| Returns empty dict for directory                                           | _load(tmp_path) == {}                                                   | Pass   | 0.000s |
| Returns empty dict for invalid json                                        | _load(p) == {}                                                          | Pass   | 0.002s |
| Returns empty dict for missing file                                        | _load(tmp_path / 'nonexistent.json') == {}                              | Pass   | 0.000s |
| Classifier count                                                           | data['classifier.count'] == 2                                           | Pass   | 0.010s |
| Eod count defaults zero on import error                                    | isinstance(data['eod.count'], int)                                      | Pass   | 0.007s |
| Gaugehd baselines loaded                                                   | len(data['gaugehd.baselines']) == 2                                     | Pass   | 0.007s |
| Gaugehd count                                                              | data['gaugehd.count'] == 2                                              | Pass   | 0.007s |
| Gaugets count                                                              | data['gaugets.count'] == 2                                              | Pass   | 0.008s |
| Loads counterparties                                                       | len(data['counterparties']) == 1                                        | Pass   | 0.010s |
| Loads gaugehc                                                              | 'GAUGE-0001' in<br>data['gaugehc']['hazard.curves']                     | Pass   | 0.009s |



| Test Description                                                 | Acceptance Criteria                                                                    | Result | Time   |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------|--------|
| Loads gauges                                                     | <code>len(data['gauges']) == 2</code>                                                  | Pass   | 0.010s |
| Loads mortgages                                                  | <code>len(data['mortgages']) == 1</code>                                               | Pass   | 0.011s |
| Loads properties                                                 | <code>len(data['properties']) == 1</code>                                              | Pass   | 0.009s |
| Loads propertyhc                                                 | <code>'PROP-0001' in data['propertyhc']['property_hazard_curves']</code>               | Pass   | 0.016s |
| Loads seq summary                                                | <code>data['seq_summary']['num_sequences'] == 10000</code>                             | Pass   | 0.009s |
| Loads stress storms index                                        | <code>len(data['stress_storms']['storms']) == 1</code>                                 | Pass   | 0.009s |
| Falls back to stress_storms.json when index.json is absent.      | <code>len(data['stress_storms']['storms']) == 2</code>                                 | Pass   | 0.009s |
| When input dir is empty, everything defaults safely.             | <code>data['gauges'] == [];</code><br><code>data['properties'] == [] (+10 more)</code> | Pass   | 0.001s |
| Propertyts count                                                 | <code>data['propertyts_count'] == 1</code>                                             | Pass   | 0.007s |
| Stress storms empty when neither exists                          | <code>data['stress_storms'] == {}</code>                                               | Pass   | 0.005s |
| config import may fail in test env — defaults to 0.              | <code>isinstance(data['trade_count'], int)</code>                                      | Pass   | 0.013s |
| Training summary                                                 | <code>data['training_summary']['avg_auc_roc'] == pytest.approx(...)</code>             | Pass   | 0.011s |
| Loads entries                                                    | <code>result[0]["event"] == "test"</code>                                              | Pass   | 0.001s |
| Returns empty list on bad json                                   | <code>_load_audit_log() == []</code>                                                   | Pass   | 0.001s |
| Returns empty list when missing                                  | <code>_load_audit_log() == [];</code><br><code>_load_meetings() == []</code>           | Pass   | 0.000s |
| Loads valid data                                                 | <code>"principles" in result;</code> <code>"roles" in result</code>                    | Pass   | 0.002s |
| Returns none when missing                                        | <code>_load_bcbs239() is None;</code> <code>_load_raci() is None</code>                | Pass   | 0.000s |
| Loads bibliography                                               | <code>result["references"][0]["id"] == "R1"</code>                                     | Pass   | 0.001s |
| Returns default when missing                                     | <code>result == {"references": [], "categories": []}</code>                            | Pass   | 0.000s |
| Line 208: file exists with empty counterparties → fallback list. | <code>len(result) == 21;</code> <code>result[0]["id"] == "CTPY-001" (+1 more)</code>   | Pass   | 0.001s |
| Line 207-209: file doesn't exist → fallback list.                | <code>len(result) == 21</code>                                                         | Pass   | 0.000s |
| When file has entries, those are returned (not fallback).        | <code>len(result) == 1;</code> <code>result[0]["id"] == "CTPY-REAL" (+1 more)</code>   | Pass   | 0.001s |
| Corrupt counterparty file leaves data none                       | <code>d1.loaded_data.counterparty_data is None</code>                                  | Pass   | 0.006s |

| Test Description                                                   | Acceptance Criteria                                                                                             | Result | Time   |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------|--------|
| Missing counterparty file leaves data none                         | <code>dl.loaded_data.counterparty_data is None</code>                                                           | Pass   | 0.000s |
| Valid counterparty file loads                                      | <code>dl.loaded_data.counterparty_data is not None; "counterparties" in dl.loaded_data.counterparty_data</code> | Pass   | 0.001s |
| Missing gaugehc returns empty and default storms                   | <code>curves == {}; n_storms == 1000</code>                                                                     | Pass   | 0.000s |
| Present gaugehc returns curves                                     | <code>"GAUGE-001" in curves; n_storms == 500</code>                                                             | Pass   | 0.001s |
| Present gaugehc returns num storms                                 | <code>n_storms == 250</code>                                                                                    | Pass   | 0.001s |
| Lines 74-75: missing file raises FileNotFoundError.                | —                                                                                                               | Pass   | 0.000s |
| Lines 72-80: load_gauge_portfolio reads gauge.json.                | <code>isinstance(result, list); len(result) == 1</code>                                                         | Pass   | 0.012s |
| Lines 59-64: get_reports_dir raises → fallback to get_output_path. | <code>result == []</code>                                                                                       | Pass   | 0.000s |
| Line 67: prs_dir doesn't exist → return empty list.                | <code>result == []</code>                                                                                       | Pass   | 0.000s |
| Lines 75-76: trade with matching gauge_id → appended to list.      | <code>len(result) == 1</code>                                                                                   | Pass   | 0.002s |
| Trade for different gauge not appended.                            | <code>result == []</code>                                                                                       | Pass   | 0.002s |
| Computes summer mean                                               | <code>bl['summer'] == pytest.approx(1.0667, abs=0.001)</code>                                                   | Pass   | 0.002s |
| Computes winter mean                                               | <code>bl['winter'] == pytest.approx(1.8)</code>                                                                 | Pass   | 0.002s |
| Corrupt file skipped                                               | <code>len(baselines) == 1; "GAUGE-OK" in baselines</code>                                                       | Pass   | 0.002s |
| Empty directory                                                    | <code>_load_gaugehd.baselines(d) == {}</code>                                                                   | Pass   | 0.000s |
| Handles missing monthly means                                      | <code>len(baselines) == 1; baselines[0]['winter'] is None</code><br>(+1 more)                                   | Pass   | 0.001s |
| Loads baselines                                                    | <code>len(baselines) == 2; bl['gauge_id'] in ('GAUGE-0001', 'GAUGE-0002')</code> (+1 more)                      | Pass   | 0.003s |
| Missing directory                                                  | <code>_load_gaugehd.baselines(tmp_path / "nonexistent") == {}</code>                                            | Pass   | 0.000s |
| Missing mean level skipped                                         | <code>len(baselines) == 0</code>                                                                                | Pass   | 0.001s |
| Multiple gauges                                                    | <code>len(baselines) == 3</code>                                                                                | Pass   | 0.003s |
| Returns empty for nonexistent dir                                  | <code>loader._load_gaugehd.baselines(tmp_path / 'nope') == []</code>                                            | Pass   | 0.000s |

| Test Description                                  | Acceptance Criteria                                                                                                                | Result | Time   |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Skips files without mean level                    | <code>loader._load_gaugehd.baselines(ghd)</code><br><code>== []</code>                                                             | Pass   | 0.001s |
| Skips invalid files                               | <code>loader._load_gaugehd.baselines(ghd)</code><br><code>== []</code>                                                             | Pass   | 0.001s |
| Valid gaugehd files                               | <code>"GAUGE-001" in baselines;</code><br><code>baselines["GAUGE-001"]["mean_level"]</code><br><code>== 1.25 (+1 more)</code>      | Pass   | 0.002s |
| Empty gauge list raises                           | —                                                                                                                                  | Pass   | 0.001s |
| File not found raises                             | —                                                                                                                                  | Pass   | 0.000s |
| Gauge has id                                      | <code>result[0]["gauge_id"] == "GAUGE-999"</code>                                                                                  | Pass   | 0.001s |
| Gauge that maps to no gauge_id raises ValueError. | —                                                                                                                                  | Pass   | 0.002s |
| Loads gauges                                      | <code>isinstance(result, list); len(result)</code><br><code>== 1</code>                                                            | Pass   | 0.001s |
| Empty portfolio returns empty list                | <code>result == []</code>                                                                                                          | Pass   | 0.001s |
| Loads gauge from portfolio                        | <code>len(result) == 1;</code><br><code>isinstance(result[0], GaugeConfig)</code><br>(+1 more)                                     | Pass   | 0.001s |
| Loads documents                                   | <code>_load_gov_documents()[0]["doc_id"] ==</code><br><code>"D1"</code>                                                            | Pass   | 0.001s |
| Returns empty list when missing                   | <code>_load_gov_documents() == []</code>                                                                                           | Pass   | 0.000s |
| Corrupt hazard file leaves hazard data none       | <code>dl.loaded_data.hazard_data is None</code>                                                                                    | Pass   | 0.005s |
| Corrupt property hazard file leaves phc none      | <code>dl.loaded_data.property_hazard_data</code><br><code>is None</code>                                                           | Pass   | 0.002s |
| Loaded data has gauge                             | <code>"TEST-G001" in</code><br><code>result["hazard_curves"]</code>                                                                | Pass   | 0.001s |
| Loads from file                                   | <code>"hazard_curves" in result</code>                                                                                             | Pass   | 0.009s |
| Missing hazard file leaves hazard data none       | <code>dl.loaded_data.hazard_data is None</code>                                                                                    | Pass   | 0.000s |
| Missing property hazard file leaves phc none      | <code>dl.loaded_data.property_hazard_data</code><br><code>is None</code>                                                           | Pass   | 0.000s |
| Valid hazard file populates hazard data           | <code>dl.loaded_data.hazard_data is</code><br><code>not None; "hazard_curves" in</code><br><code>dl.loaded_data.hazard_data</code> | Pass   | 0.002s |
| Empty json object returns empty dict              | <code>result == {}</code>                                                                                                          | Pass   | 0.001s |
| Invalid json raises json decode error             | —                                                                                                                                  | Pass   | 0.001s |
| Missing file raises file not found                | —                                                                                                                                  | Pass   | 0.000s |
| Returns full structure                            | <code>result["meta"]["version"] == 2;</code><br><code>"G-001" in result["hazard_curves"]</code>                                    | Pass   | 0.001s |

| Test Description                                             | Acceptance Criteria                                                          | Result | Time   |
|--------------------------------------------------------------|------------------------------------------------------------------------------|--------|--------|
| Loads valid file                                             | <code>result["models"][0]["model_id"] == "M1"</code>                         | Pass   | 0.001s |
| Returns none on bad json                                     | <code>_load_inventory() is None</code>                                       | Pass   | 0.001s |
| Returns none when missing                                    | <code>_load_inventory() is None</code>                                       | Pass   | 0.000s |
| Loads existing results                                       | <code>result['total'] == 15;</code><br><code>result['failed'] == 1</code>    | Pass   | 0.001s |
| Returns dict when missing                                    | <code>isinstance(result, dict); len(result) == 0</code>                      | Pass   | 0.000s |
| Returns empty skeleton for corrupt JSON.                     | <code>result == {"runs": {}, "steps": {}}</code>                             | Pass   | 0.001s |
| Load corrupt json returns skeleton                           | <code>result == {"runs": {}, "steps": {}}</code>                             | Pass   | 0.001s |
| Load missing returns skeleton                                | <code>result == {"runs": {}, "steps": {}}</code>                             | Pass   | 0.000s |
| Load valid manifest                                          | <code>result == data</code>                                                  | Pass   | 0.001s |
| Reads and parses an existing manifest.                       | <code>'gauges' in result['steps']</code>                                     | Pass   | 0.002s |
| Returns empty skeleton when file doesn't exist.              | <code>isinstance(result, dict); 'runs' in result (+1 more)</code>            | Pass   | 0.000s |
| Lines 89-91: exception → returns {}.                         | <code>result == {}</code>                                                    | Pass   | 0.000s |
| Loads list                                                   | <code>_load_meetings()[0]["id"] == "M1"</code>                               | Pass   | 0.001s |
| Returns empty list when missing                              | <code>_load_audit_log() == [];</code><br><code>_load_meetings() == []</code> | Pass   | 0.000s |
| When mortgage.json has invalid JSON, the exception is caught | <code>r.status_code == 200; len(props) == 1 (+1 more)</code>                 | Pass   | 0.030s |
| Missing property json returns 404                            | <code>r.status_code == 404</code>                                            | Pass   | 0.002s |
| Missing property json sequence returns 404                   | <code>r.status_code == 404</code>                                            | Pass   | 0.002s |
| Loads valid data                                             | <code>"principles" in result; "roles" in result</code>                       | Pass   | 0.001s |
| Returns none when missing                                    | <code>_load_bcbs239() is None; _load_raci() is None</code>                   | Pass   | 0.000s |
| Corrupt storm file leaves storm data none                    | <code>dl.loaded_data.storm_data is None</code>                               | Pass   | 0.002s |
| Missing storm file leaves storm data none                    | <code>dl.loaded_data.storm_data is None</code>                               | Pass   | 0.000s |
| Storms dict with storms key                                  | <code>dl.loaded_data.storm_data is not None</code>                           | Pass   | 0.001s |
| Storms list format                                           | <code>dl.loaded_data.storm_data is not None</code>                           | Pass   | 0.001s |
| Storms other type                                            | <code>dl.loaded_data.storm_data is not None</code>                           | Pass   | 0.001s |
| File not found raises                                        | —                                                                            | Pass   | 0.000s |

| Test Description                                                                                    | Acceptance Criteria                                                                                                               | Result | Time   |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Loads list                                                                                          | <code>isinstance(result, list); len(result) == 5</code>                                                                           | Pass   | 0.002s |
| Storm has id                                                                                        | <code>result[0]["storm_id"] == "STORM-0000"</code>                                                                                | Pass   | 0.001s |
| Lines 90-96: storm loaded by searching legacy storms list.                                          | <code>result is not None; result["storm_id"] == "LEGACY-STORM-001" (+1 more)</code>                                               | Pass   | 0.000s |
| When <code>path.stat()</code> raises <code>OSError</code> , return existing cache.                  | <code>result == {"storms": [{"storm_id": "cached"}]}</code>                                                                       | Pass   | 0.000s |
| When <code>path.stat()</code> raises <code>OSError</code> and no cache, returns <code>None</code> . | <code>result is None</code>                                                                                                       | Pass   | 0.000s |
| Empty data list records invalid result                                                              | <code>dl._validation_results["empty_type"].is_valid is False; "No data loaded" in dl._validation_results["empty_type"]...</code>  | Pass   | 0.000s |
| Empty data list returns none                                                                        | <code>dl._load_with_validation(mock_loader, "empty_type") is None</code>                                                          | Pass   | 0.000s |
| Exception in loader returns none                                                                    | <code>dl._load_with_validation(mock_loader, "error_type") is None</code>                                                          | Pass   | 0.021s |
| Exception records error in validation                                                               | <code>dl._validation_results["error_type"].is_valid is False; "disk error" in dl._validation_results["error_type"].erro...</code> | Pass   | 0.001s |
| None return from loader returns none                                                                | <code>dl._load_with_validation(mock_loader, "none_type") is None</code>                                                           | Pass   | 0.000s |
| Successful load records valid result                                                                | <code>"gauge_test" in dl._validation_results; dl._validation_results["gauge_test"].is_valid is True</code>                        | Pass   | 0.001s |
| Successful load returns data dict                                                                   | <code>result is not None; result["count"] == 2 (+1 more)</code>                                                                   | Pass   | 0.001s |
| Validation summary count matches items                                                              | <code>dl._validation_results["three_items"].summary["count"] == 3</code>                                                          | Pass   | 0.000s |
| Validation summary has type field                                                                   | <code>dl._validation_results["xtype"].summary["type"] == "xtype"</code>                                                           | Pass   | 0.000s |
| Can set gauge data                                                                                  | <code>d.gauge_data is not None</code>                                                                                             | Pass   | 0.000s |
| Can set multiple fields                                                                             | <code>d.gaugets_count == 3; d.gaugehd_count == 5</code>                                                                           | Pass   | 0.000s |
| Counts default to zero                                                                              | <code>d.gaugets_count == 0; d.gaugehd_count == 0 (+1 more)</code>                                                                 | Pass   | 0.000s |
| Defaults are none                                                                                   | <code>getattr(d, field) is None</code>                                                                                            | Pass   | 0.000s |
| Is dataclass                                                                                        | <code>dataclasses.is_dataclass(LoadedData)</code>                                                                                 | Pass   | 0.000s |
| Lookup defaults are none                                                                            | <code>getattr(d, field) is None</code>                                                                                            | Pass   | 0.000s |

| Test Description                               | Acceptance Criteria                                                                       | Result | Time   |
|------------------------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| Test creating a registry.                      | registry.data_dir ==<br>populated.data_dir; registry is<br>not None (+1 more)             | Pass   | 0.000s |
| Test getting list of available loader types.   | 'property' in loaders; 'mortgage' in<br>loaders (+3 more)                                 | Pass   | 0.000s |
| Clear all caches                               | not loader._cache_valid                                                                   | Pass   | 0.001s |
| Get all status                                 | "gauge" in status; "entity_name" in<br>status["gauge"]                                    | Pass   | 0.001s |
| Invalidate all caches                          | —                                                                                         | Pass   | 0.000s |
| Test clearing all loader caches.               | prop_loader._cache is None;<br>gauge_loader._cache is None                                | Pass   | 0.001s |
| Test that different loader types are separate. | prop_loader is not gauge_loader                                                           | Pass   | 0.000s |
| Test that loader instances are cached.         | loader1 is loader2                                                                        | Pass   | 0.000s |
| Check data files all missing                   | all(v is False for v in<br>result.values())                                               | Pass   | 0.000s |
| Check data files present                       | result["property"] is True;<br>result["gauge"] is False                                   | Pass   | 0.000s |
| File overrides                                 | registry.get_gauge_loader().count()<br>== 3                                               | Pass   | 0.001s |
| Get data dir                                   | LoaderRegistry(data_dir=tmp_path).get_data_dir()<br>== tmp_path                           | Pass   | 0.000s |
| Test providing custom filenames.               | len(props) == 1;<br>loader.get_entity_id(props[0]) ==<br>"CUSTOM-001"                     | Pass   | 0.001s |
| Test getting loader by name string.            | loader is not None; loader.count() ==<br>3                                                | Pass   | 0.001s |
| Test getting unknown loader type.              | loader is None                                                                            | Pass   | 0.000s |
| Get available loaders                          | "property" in available; "gauge" in<br>available (+1 more)                                | Pass   | 0.000s |
| Get gauge loader                               | isinstance(LoaderRegistry(data_dir=tmp_path).get_loader("gauge"),<br>LoaderRegistry)      | Pass   | 0.000s |
| Get loader by name                             | LoaderRegistry(data_dir=tmp_path).get_loader("gauge")<br>is ...                           | Pass   | 0.000s |
| Get loader unknown name returns none           | LoaderRegistry(data_dir=tmp_path).get_loader("nonexistent")                               | Pass   | 0.000s |
| Get mortgage loader                            | isinstance(LoaderRegistry(data_dir=tmp_path).get_loader("mortgage"),<br>LoaderRegistry)   | Pass   | 0.000s |
| Get property loader                            | isinstance(LoaderRegistry(data_dir=tmp_path).get_loader("property"),<br>LoaderRegistry)   | Pass   | 0.000s |
| Get storm loader                               | isinstance(LoaderRegistry(data_dir=tmp_path).get_loader("storm"),<br>LoaderRegistry)      | Pass   | 0.000s |
| Get timeseries loader                          | isinstance(LoaderRegistry(data_dir=tmp_path).get_loader("timeseries"),<br>LoaderRegistry) | Pass   | 0.000s |



| Test Description                                               | Acceptance Criteria                                                                                                                      | Result | Time   |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Loader cached on repeated access                               | <code>registry.get_property_loader()</code> is                                                                                           | Pass   | 0.000s |
| Test checking data file existence.                             | <code>registry.get_property.l...</code><br><code>status['property']</code> is True;<br><code>status['mortgage']</code> is True (+3 more) | Pass   | 0.000s |
| Test getting status for all loaders.                           | <code>'property'</code> in <code>status</code> ; <code>'gauge'</code> in <code>status</code> (+1 more)                                   | Skip   | 0.000s |
| Test getting GaugeLoader.                                      | <code>isinstance(loader, GaugeLoader);</code><br><code>loader.count() == 3</code>                                                        | Pass   | 0.001s |
| Test getting MortgageLoader.                                   | <code>isinstance(loader, MortgageLoader);</code><br><code>loader.count() == 3</code>                                                     | Pass   | 0.001s |
| Test getting PropertyLoader.                                   | <code>isinstance(loader, PropertyLoader);</code><br><code>loader.count() == 3</code>                                                     | Pass   | 0.001s |
| Test getting StormLoader.                                      | <code>isinstance(loader, StormLoader)</code>                                                                                             | Pass   | 0.000s |
| Test getting TimeseriesLoader.                                 | <code>isinstance(loader, TimeseriesLoader)</code>                                                                                        | Pass   | 0.000s |
| Address subsection header                                      | <code>any("Primary Address" in t for t in texts)</code>                                                                                  | Pass   | 0.000s |
| Administrative subsection header                               | <code>any("Administrative Areas" in t for t in texts)</code>                                                                             | Pass   | 0.000s |
| Area classification subsection header                          | <code>any("Area Classification" in t for t in texts)</code>                                                                              | Pass   | 0.000s |
| Density high                                                   | <code>isinstance(result, list)</code>                                                                                                    | Pass   | 0.000s |
| Density low                                                    | <code>isinstance(result, list)</code>                                                                                                    | Pass   | 0.000s |
| Density low medium                                             | <code>isinstance(result, list)</code>                                                                                                    | Pass   | 0.000s |
| Density medium                                                 | <code>isinstance(result, list)</code>                                                                                                    | Pass   | 0.000s |
| Density very high                                              | <code>isinstance(result, list)</code>                                                                                                    | Pass   | 0.000s |
| Empty property does not crash                                  | <code>isinstance(result, list)</code>                                                                                                    | Pass   | 0.000s |
| Geographic subsection header                                   | <code>any("Geographic Information" in t for t in texts)</code>                                                                           | Pass   | 0.000s |
| Has section header                                             | <code>any("Location Details" in t for t in texts)</code>                                                                                 | Pass   | 0.000s |
| Has tables                                                     | <code>any(isinstance(e, Table) for e in result)</code>                                                                                   | Pass   | 0.000s |
| Missing lat/lon means coordinates rows are omitted — no crash. | <code>isinstance(result, list)</code>                                                                                                    | Pass   | 0.000s |
| Returns list                                                   | <code>isinstance(result, list); len(result) &gt; 0</code>                                                                                | Pass   | 0.000s |
| With mortgage data does not crash                              | <code>isinstance(result, list)</code>                                                                                                    | Pass   | 0.000s |
| Debug does not raise                                           | —                                                                                                                                        | Pass   | 0.000s |
| Debug level does not raise                                     | —                                                                                                                                        | Pass   | 0.000s |
| Error does not raise                                           | —                                                                                                                                        | Pass   | 0.000s |

| Test Description                                                 | Acceptance Criteria                                                                       | Result | Time   |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| Error level does not raise                                       | —                                                                                         | Pass   | 0.000s |
| Info does not raise                                              | —                                                                                         | Pass   | 0.000s |
| Info level does not raise                                        | —                                                                                         | Pass   | 0.000s |
| Log does not raise                                               | —                                                                                         | Pass   | 0.000s |
| Log with empty string                                            | —                                                                                         | Pass   | 0.000s |
| Success falls back gracefully                                    | —                                                                                         | Pass   | 0.000s |
| Success level falls back gracefully                              | —                                                                                         | Pass   | 0.000s |
| Unknown level falls back to info                                 | —                                                                                         | Pass   | 0.000s |
| Warning does not raise                                           | —                                                                                         | Pass   | 0.000s |
| Warning level does not raise                                     | —                                                                                         | Pass   | 0.000s |
| Appends entry                                                    | len(entries) == 1;<br>entries[0]["action"] == "test_action"<br>(+1 more)                  | Pass   | 0.002s |
| Failures in audit logging must not propagate.                    | —                                                                                         | Pass   | 0.000s |
| Corrupted audit log → start fresh, don't raise.                  | len(entries) == 1                                                                         | Pass   | 0.002s |
| Creates audit file                                               | os.path.exists(audit_path)                                                                | Pass   | 0.000s |
| Defaults applied when no json                                    | r.status_code == 200;<br>entry["event_type"] == "usage"<br>(+1 more)                      | Pass   | 0.003s |
| Entry has expected fields                                        | entry["model_id"] == "MKM-TEST-001";<br>entry["event_type"] == "calibration"<br>(+2 more) | Pass   | 0.002s |
| Entry has required keys                                          | "timestamp" in entry; "model_id" in entry (+3 more)                                       | Pass   | 0.001s |
| Existing file appended                                           | len(entries) == 2; "existing" in entries[0]                                               | Pass   | 0.002s |
| Model not found returns 404                                      | r.status_code == 404;<br>r.get_json()["status"] == "error"<br>(+3 more)                   | Pass   | 0.002s |
| Multiple calls accumulate                                        | len(entries) == 2                                                                         | Pass   | 0.002s |
| Parameters stored                                                | r.get_json()["entry"]["parameters"] == {"key": "value"}                                   | Pass   | 0.002s |
| Valid model returns 200                                          | r.status_code == 200;<br>r.get_json()["status"] == "success"                              | Pass   | 0.002s |
| ltv_ratio > 1 is interpreted as a percentage and divided by 100. | score > 0                                                                                 | Pass   | 0.000s |
| Compare mode                                                     | "Scenario Comparison" in captured.out                                                     | Pass   | 0.365s |
| Custom notional and spread                                       | rc == 0                                                                                   | Pass   | 0.002s |

| Test Description                                                                                        | Acceptance Criteria                                                                                                                                                            | Result | Time   |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Default describe mode                                                                                   | <code>len(captured.out) &gt; 0</code>                                                                                                                                          | Pass   | 0.073s |
| Default gauge selection returns zero                                                                    | <code>rc == 0</code>                                                                                                                                                           | Pass   | 0.002s |
| Exceedance json mode                                                                                    | <code>"exceedance_probability" in data</code>                                                                                                                                  | Pass   | 0.000s |
| Exceedance mode                                                                                         | <code>"70" in captured.out</code>                                                                                                                                              | Pass   | 0.000s |
| Explicit gauge returns zero                                                                             | <code>rc == 0</code>                                                                                                                                                           | Pass   | 0.002s |
| Flat flag dispatches flat mode                                                                          | <code>rc == 0</code>                                                                                                                                                           | Pass   | 0.002s |
| Json default mode                                                                                       | <code>"summary_statistics" in data</code>                                                                                                                                      | Pass   | 0.279s |
| List flag logs gauge names                                                                              | <code>"THAMES-TEST-99" in combined</code>                                                                                                                                      | Pass   | 0.002s |
| List flag returns zero                                                                                  | <code>rc == 0</code>                                                                                                                                                           | Pass   | 0.002s |
| <code>main()</code> creates audit directory if it doesn't exist.                                        | <code>audit.exists(); out.exists()</code>                                                                                                                                      | Pass   | 0.023s |
| Lines 120-122: default (no <code>-nrfa-dir</code> ) calls <code>generate_all_gauge_histories</code> .   | —                                                                                                                                                                              | Pass   | 0.328s |
| <code>main()</code> exits cleanly if <code>jscpd</code> fails.                                          | —                                                                                                                                                                              | Pass   | 0.000s |
| Lines 107-108: without <code>-catchment</code> , <code>config.CATCHMENT</code> is untouched.            | <code>config.CATCHMENT == original</code>                                                                                                                                      | Pass   | 0.322s |
| <code>main()</code> prints file/clone/pct stats to <code>stdout</code> .                                | <code>'Clones' in out; 'PDF saved' in out</code>                                                                                                                               | Pass   | 0.231s |
| <code>main()</code> prints step count and consistency status.                                           | <code>'pipeline steps' in output;</code><br><code>'Consistency' in output (+1 more)</code>                                                                                     | Pass   | 0.023s |
| <code>main()</code> writes <code>data_lineage_report.pdf</code> to audit dir.                           | <code>out.exists(); out.stat().st_size &gt; 5000</code>                                                                                                                        | Pass   | 0.024s |
| <code>main()</code> writes <code>code_duplication_report.pdf</code> to audit dir.                       | <code>(tmp_path /</code><br><code>'code_duplication_report.pdf').exists();</code><br><code>(tmp_path /</code><br><code>'code_duplication_report.pdf').stat().st_size...</code> | Pass   | 0.230s |
| Line 102: <code>-c</code> short flag for catchment.                                                     | —                                                                                                                                                                              | Pass   | 0.325s |
| Line 101: <code>-y</code> short flag for years.                                                         | —                                                                                                                                                                              | Pass   | 0.204s |
| <code>main()</code> logs errors for unreadable files and continues.                                     | —                                                                                                                                                                              | Pass   | 0.001s |
| Lines 107-108: <code>-catchment</code> sets <code>config.CATCHMENT</code> .                             | <code>config.CATCHMENT == "severn"</code>                                                                                                                                      | Pass   | 0.327s |
| <code>main()</code> processes any <code>*_gdf.csv</code> files found in <code>config.input_dir</code> . | —                                                                                                                                                                              | Pass   | 0.016s |
| <code>main()</code> handles an empty directory gracefully.                                              | —                                                                                                                                                                              | Pass   | 0.000s |
| Lines 117-119: <code>-nrfa-dir</code> calls <code>process_nrfa_directory</code> .                       | <code>list(gaugehd_dir.glob("gauge.*_hd.json"))</code>                                                                                                                         | Pass   | 0.002s |

| Test Description                                              | Acceptance Criteria                                                        | Result | Time   |
|---------------------------------------------------------------|----------------------------------------------------------------------------|--------|--------|
| Line 101: -years flag is passed through.                      | —                                                                          | Pass   | 0.226s |
| Missing gauge logs available gauges                           | "THAMES-G042" in combined                                                  | Pass   | 0.001s |
| Missing gauge logs error                                      | any("GHOST-GAUGE" in r.message for r in caplog.records)                    | Pass   | 0.001s |
| Missing gauge returns one                                     | rc == 1                                                                    | Pass   | 0.002s |
| Return period json mode                                       | "return_period_years" in data;<br>"intensity" in data                      | Pass   | 0.000s |
| Return period mode                                            | "100" in captured.out                                                      | Pass   | 0.000s |
| Samples json mode                                             | "samples" in data;<br>len(data["samples"]) == 3                            | Pass   | 0.000s |
| Samples mode                                                  | "5 samples" in captured.out                                                | Pass   | 0.000s |
| Scenario mode                                                 | len(captured.out) > 0                                                      | Pass   | 0.074s |
| Seed reproducible                                             | out1 == out2                                                               | Pass   | 0.001s |
| Trigger alert explicit                                        | rc == 0                                                                    | Pass   | 0.002s |
| Trigger severe explicit                                       | rc == 0                                                                    | Pass   | 0.002s |
| Trigger warning explicit                                      | rc == 0                                                                    | Pass   | 0.001s |
| Add to map calls add child                                    | —                                                                          | Pass   | 0.001s |
| Get js returns script                                         | "<script>" in js;<br>"refreshMainMapFS01" in js                            | Pass   | 0.000s |
| Get statistics returns dict                                   | isinstance(result, dict);<br>result["component"] ==<br>"main_map_fs01"     | Pass   | 0.000s |
| Report with 2% duplication should render without error.       | analysis['total']['percentage'] < 3;<br>len(pdf) > 0                       | Pass   | 0.240s |
| Report with 8% duplication renders correctly.                 | pdf[:4] == b'%PDF'                                                         | Pass   | 0.228s |
| Pdf header                                                    | pdf[:4] == b'%PDF'                                                         | Pass   | 0.247s |
| Pdf nonempty                                                  | len(pdf) > 2000                                                            | Pass   | 0.229s |
| Returns bytes                                                 | isinstance(result, bytes)                                                  | Pass   | 1.923s |
| Peak precipitation is in the storm's active window.           | peak.in.storm >= peak.outside                                              | Pass   | 0.000s |
| Drainage window (hours 156-167) must be precipitation-free.   | np.all(p[156:] ==<br>pytest.approx(0.0))                                   | Pass   | 0.001s |
| Non negative                                                  | np.all(p >= 0.0)                                                           | Pass   | 0.000s |
| Output shape                                                  | p.shape == (EVENT_WINDOW_HOURS,);<br>levels.shape == (EVENT_WINDOW_HOURS,) | Pass   | 0.000s |
| Sum of hourly series == sum of storm precipitation_mm values. | p.sum() == pytest.approx(expected,<br>rel=1e-4)                            | Pass   | 0.001s |
| Col widths propagated                                         | t._colWidths == cw                                                         | Pass   | 0.000s |
| Custom alt color                                              | isinstance(t, Table)                                                       | Pass   | 0.000s |

| Test Description                                                           | Acceptance Criteria                                            | Result | Time   |
|----------------------------------------------------------------------------|----------------------------------------------------------------|--------|--------|
| When no alt_color passed, ALT_ROWS.BLUE is applied.                        | isinstance(t, Table)                                           | Pass   | 0.000s |
| Empty rows                                                                 | isinstance(t, Table);<br>len(t._cellvalues) == 1               | Pass   | 0.000s |
| Repeat rows set                                                            | t.repeatRows == 1                                              | Pass   | 0.000s |
| Returns table                                                              | isinstance(t, Table);<br>isinstance(t, Table)                  | Pass   | 0.000s |
| Table has correct row count                                                | len(t._cellvalues) == 4                                        | Pass   | 0.000s |
| Add action with malformed existing id                                      | r.status_code == 200; "A-002" in ids                           | Pass   | 0.002s |
| Malformed decision ID is skipped; new decision still gets a valid auto-ID. | r.status_code == 200; "D-002" in ids                           | Pass   | 0.002s |
| Add participant with malformed existing id                                 | r.status_code == 200; "P-002" in ids                           | Pass   | 0.002s |
| Click-through should hide trading desk before opening gauge.               | 'TradingDesk.hide' in js                                       | Pass   | 0.000s |
| Map should have New Trade link.                                            | 'window.tdNewTrade' in js; 'New Trade' in js                   | Pass   | 0.000s |
| Map should have tdViewHazardCurve navigating to gauge panel.               | 'window.tdViewHazardCurve' in js; 'GaugeHazardCurve' in js     | Pass   | 0.000s |
| Any flood maps to alert                                                    | gen._map.threshold.to_gauge.trigger("any_flood") == "alert"    | Pass   | 0.000s |
| Moderate maps to warning                                                   | gen._map.threshold.to_gauge.trigger("moderate") == "warning"   | Pass   | 0.000s |
| Severe maps to severe                                                      | gen._map.threshold.to_gauge.trigger("severe") == "severe"      | Pass   | 0.000s |
| Unknown key returns warning                                                | gen._map.threshold.to_gauge.trigger("nonexistent") == "wa..."  | Pass   | 0.000s |
| P&L bar must show 'From Market' metric.                                    | 'From Market' in js; 'fromMarket' in js                        | Pass   | 0.000s |
| Blotter must have a Mkt P&L column.                                        | "'market_pnl'" in js or<br>'"market_pnl"' in js; 'Mkt P' in js | Pass   | 0.000s |
| Market tab commit notification must show P&L impact.                       | 'total_pnl_impact' in js; 'gross_pnl_impact' in js             | Pass   | 0.000s |
| Hazard column must show up/down arrows when fair != prev_fair.             | 'prev_fair_spread.bps' in js; '\\{\\}\\{u25b2' in js (+1 more) | Pass   | 0.000s |
| Entry level under 250k                                                     | isinstance(result, list)                                       | Pass   | 0.000s |
| Middle 250k to 500k                                                        | isinstance(result, list)                                       | Pass   | 0.000s |
| Premium over 1m                                                            | isinstance(result, list)                                       | Pass   | 0.000s |

| Test Description                                                     | Acceptance Criteria                                                                                                                                                                    | Result | Time   |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Upper middle 500k to 1m                                              | <code>isinstance(result, list)</code>                                                                                                                                                  | Pass   | 0.000s |
| Base rates not empty                                                 | <code>br</code>                                                                                                                                                                        | Pass   | 0.000s |
| File exists                                                          | <code>GAUGEHC_PATH.exists();</code><br><code>MARKET_STATE.exists()</code>                                                                                                              | Pass   | 0.000s |
| Has required keys                                                    | <code>not missing</code>                                                                                                                                                               | Pass   | 0.000s |
| Hazard term structure has triggers                                   | <code>"alert" in triggers</code>                                                                                                                                                       | Pass   | 0.000s |
| Hazard term structure not empty                                      | <code>hts</code>                                                                                                                                                                       | Pass   | 0.000s |
| Last updated present                                                 | <code>"last.updated" in ms</code>                                                                                                                                                      | Pass   | 0.000s |
| Yield curve has all tenors                                           | <code>not missing</code>                                                                                                                                                               | Pass   | 0.000s |
| Yield curve rates in range                                           | <code>not bad</code>                                                                                                                                                                   | Pass   | 0.000s |
| <code>computeMaturityDate</code> function exists.                    | <code>'computeMaturityDate' in prs_js</code>                                                                                                                                           | Pass   | 0.000s |
| <code>currentRollDate</code> function exists.                        | <code>'currentRollDate' in prs_js</code>                                                                                                                                               | Pass   | 0.000s |
| <code>lastFridayOfMonth</code> function exists.                      | <code>'lastFridayOfMonth' in prs_js</code>                                                                                                                                             | Pass   | 0.000s |
| <code>showMaturityPopup</code> function exists.                      | <code>'showMaturityPopup' in prs_js</code>                                                                                                                                             | Pass   | 0.000s |
| Mcob advice rejection reason                                         | <code>isinstance(result, list)</code>                                                                                                                                                  | Pass   | 0.000s |
| Mcob all fields returns list                                         | <code>isinstance(result, list); len(result)</code><br><code>&gt; 5</code>                                                                                                              | Pass   | 0.000s |
| Mcob section creates tables                                          | <code>len(tables) &gt;= 2</code>                                                                                                                                                       | Pass   | 0.000s |
| Mcob with minimal fields                                             | <code>isinstance(result, list)</code>                                                                                                                                                  | Pass   | 0.000s |
| Serve missing document returns 404                                   | <code>r.status_code == 404</code>                                                                                                                                                      | Pass   | 0.002s |
| Upload empty filename returns 400                                    | <code>r.status_code == 400</code>                                                                                                                                                      | Pass   | 0.003s |
| Upload no file returns 400                                           | <code>r.status_code == 400;</code><br><code>r.get_json()["status"] == "error"</code>                                                                                                   | Pass   | 0.003s |
| Upload not found meeting returns 404                                 | <code>r.status_code == 404</code>                                                                                                                                                      | Pass   | 0.001s |
| Unknown menu no options returns empty                                | <code>result == ""</code>                                                                                                                                                              | Pass   | 0.000s |
| Unknown menu with options cycles                                     | <code>result == "OptionB"</code>                                                                                                                                                       | Pass   | 0.000s |
| Unknown menu with options wraps                                      | <code>result == "X"</code>                                                                                                                                                             | Pass   | 0.000s |
| Duration from storms json                                            | <code>storm["duration_hours"] ==</code><br><code>72;</code> <code>storm["peak_position"] ==</code><br><code>pytest.approx(0.3) (+1 more)</code>                                        | Pass   | 0.004s |
| Fallback to config defaults                                          | <code>storm["duration_hours"] ==</code><br><code>STRESS_STORM_DEFAULT_DURATION...;</code><br><code>storm["peak_position"] ==</code><br><code>pytest.approx(STRESS_STORM_DEFA...</code> | Pass   | 0.003s |
| Passing a non-existent <code>storms_json_path</code> must not crash. | <code>out.exists() and (out /</code><br><code>"_index.json").exists()</code>                                                                                                           | Pass   | 0.002s |



| Test Description                                                 | Acceptance Criteria                                                                                                                    | Result | Time   |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Distant point returns larger distance                            | <code>d.far &gt; d.near</code>                                                                                                         | Pass   | 0.000s |
| Point on gauge returns small distance                            | <code>d &lt; 10.000</code>                                                                                                             | Pass   | 0.000s |
| Returns float                                                    | <code>isinstance(d, float); d &gt;= 0</code>                                                                                           | Pass   | 0.000s |
| Add minute item                                                  | <code>r.status_code == 200;</code><br><code>isinstance(minutes, list) (+1 more)</code>                                                 | Pass   | 0.002s |
| Delete minute item legacy format returns 400                     | <code>r.status_code == 400</code>                                                                                                      | Pass   | 0.003s |
| Update minute item legacy format returns 400                     | <code>r.status_code == 400</code>                                                                                                      | Pass   | 0.003s |
| Update minute item not found returns 404                         | <code>r.status_code == 404</code>                                                                                                      | Pass   | 0.003s |
| Update minutes text                                              | <code>r.status_code == 200;</code><br><code>r.get_json()["meeting"]["minutes"] ==</code><br><code>"# Minutes\{\}n\{\}nAll a..."</code> | Pass   | 0.002s |
| Update minutes unknown meeting returns 404                       | <code>r.status_code == 404</code>                                                                                                      | Pass   | 0.001s |
| Add minute item missing meeting returns 404                      | <code>r.status_code == 404</code>                                                                                                      | Pass   | 0.002s |
| Add minute item returns 200                                      | <code>r.status_code == 200;</code><br><code>isinstance(meeting["minutes"],</code><br><code>list) (+1 more)</code>                      | Pass   | 0.002s |
| When minutes is a string, deleting returns 400.                  | <code>r.status_code == 400</code>                                                                                                      | Pass   | 0.004s |
| Delete minute item missing meeting returns 404                   | <code>r.status_code == 404</code>                                                                                                      | Pass   | 0.002s |
| Delete minute item returns 200                                   | <code>r.status_code == 200;</code><br><code>len(meeting["minutes"]) == 1</code> (+1 more)                                              | Pass   | 0.003s |
| When minutes is a string (legacy), updating an item returns 400. | <code>r.status_code == 400</code>                                                                                                      | Pass   | 0.004s |
| Update minute item not found returns 404                         | <code>r.status_code == 404</code>                                                                                                      | Pass   | 0.002s |
| Update minute item returns 200                                   | <code>r.status_code == 200; item["text"] ==</code><br><code>"Updated text."</code>                                                     | Pass   | 0.002s |
| Set minutes to a string via update, then add a minutes-item.     | <code>r.status_code == 200;</code><br><code>isinstance(meeting["minutes"],</code><br><code>list) (+2 more)</code>                      | Pass   | 0.003s |
| Update minutes missing meeting returns 404                       | <code>r.status_code == 404</code>                                                                                                      | Pass   | 0.002s |

| Test Description                                                                                       | Acceptance Criteria                                                                                                               | Result | Time   |
|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Update minutes returns 200                                                                             | <code>r.status_code == 200;</code><br><code>r.get_json()["meeting"]["minutes"] ==</code><br><code>"New minutes text"</code>       | Pass   | 0.002s |
| Property hazard returns 404 when<br>propertyhc.json missing.                                           | <code>response.status_code == 404;</code><br><code>data['status'] == 'error'</code>                                               | Pass   | 0.000s |
| PropertyHC summary returns 404<br>with helpful message when file miss-<br>ing.                         | <code>response.status_code == 404;</code><br><code>data['status'] == 'error' (+1 more)</code>                                     | Pass   | 0.000s |
| PropertyTS summary returns 404<br>when directory missing.                                              | <code>response.status_code == 404;</code><br><code>data['status'] == 'error'</code>                                               | Pass   | 0.000s |
| No crash when gauge json absent                                                                        | <code>isinstance(result, dict)</code>                                                                                             | Pass   | 0.002s |
| Num gauges zero without gauge json                                                                     | <code>result["num_gauges"] == 0</code>                                                                                            | Pass   | 0.002s |
| Sequences still written without<br>gauge json                                                          | <code>(tmp_path /</code><br><code>SEQUENCES.FILENAME).exists()</code>                                                             | Pass   | 0.002s |
| Only alert storms create flood<br>events                                                               | <code>result["summary"]["floods_at_nearest_gauge"]</code><br><code>== 2</code>                                                    | Pass   | 0.001s |
| Sub alert storms not in flood events                                                                   | <code>"STORM-m001b" not in event_storm_ids;</code><br><code>"STORM-m001d" not in event_storm_ids</code><br><code>(+2 more)</code> | Pass   | 0.003s |
| Has kv table                                                                                           | <code>callable(gen._kv_table)</code>                                                                                              | Pass   | 0.000s |
| Has load all                                                                                           | <code>callable(gen._load_all)</code>                                                                                              | Pass   | 0.000s |
| Has make table                                                                                         | <code>callable(gen._make_table)</code>                                                                                            | Pass   | 0.000s |
| Has section methods                                                                                    | <code>hasattr(gen, name)</code>                                                                                                   | Pass   | 0.000s |
| <code>_load_all()</code> output can be consumed<br>by all section methods.                             | —                                                                                                                                 | Pass   | 0.014s |
| When <code>predictor.load_summary</code> has<br>no matching gauge, <code>model_auc</code> is<br>None.  | <code>resp.status_code == 200;</code><br><code>data['model_auc'] is None</code>                                                   | Pass   | 0.039s |
| When <code>predictor.load_summary</code><br>raises, <code>model_auc</code> is None (lines<br>136-137). | <code>resp.status_code == 200;</code><br><code>data['model_auc'] is None</code>                                                   | Pass   | 0.039s |
| When <code>predictor.load_summary</code> has<br>matching gauge, <code>model_auc</code> is set.         | <code>resp.status_code == 200;</code><br><code>data['model_auc'] == 0.95</code>                                                   | Pass   | 0.047s |
| When gauge status is not 'trained',<br><code>model_auc</code> remains None.                            | <code>resp.status_code == 200;</code><br><code>data['model_auc'] is None</code>                                                   | Pass   | 0.043s |
| All models have model id                                                                               | <code>"model_id" in m;</code><br><code>isinstance(m["model_id"], str)</code><br><code>(+1 more)</code>                            | Pass   | 0.000s |
| Model ids are unique                                                                                   | <code>len(ids) == len(set(ids))</code>                                                                                            | Pass   | 0.000s |
| Model ids follow naming convention                                                                     | <code>mid.startswith("MKM-")</code>                                                                                               | Pass   | 0.000s |

| Test Description                                                               | Acceptance Criteria                                                                          | Result | Time   |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------|--------|
| Lines 362-363: exception in create_mapping raises ValueError.                  | —                                                                                            | Pass   | 0.000s |
| Line 280: valid data returns empty dict.                                       | mortgage_cdm.validate(data) == {}                                                            | Pass   | 0.000s |
| Lines 282-283: exception in validate returns validation_error.                 | "validation_error" in result;<br>isinstance(result["validation_error"],<br>list) (+1 more)   | Pass   | 0.000s |
| Line 268: header_errors list appended to errors dict when non-empty.           | "Header" in errors;<br>len(errors["Header"]) >= 2                                            | Pass   | 0.000s |
| Line 262: Missing CatchmentID produces header error.                           | "Header" in errors; any("CatchmentID"<br>in e for e in errors["Header"]) (+2<br>more)        | Pass   | 0.000s |
| Line 259: Missing MortgageID produces header error.                            | "Header" in errors; any("MortgageID"<br>in e for e in errors["Header"])                      | Pass   | 0.000s |
| Line 275: Missing OriginalLoan produces FinancialTerms error.                  | "FinancialTerms" in errors;<br>any("OriginalLoan" in e for e in<br>errors["FinancialTerms"]) | Pass   | 0.000s |
| Line 265: Missing PropertyID produces header error.                            | "Header" in errors; any("PropertyID"<br>in e for e in errors["Header"])                      | Pass   | 0.000s |
| Line 278: terms_errors appended when non-empty.                                | "FinancialTerms" in errors                                                                   | Pass   | 0.000s |
| Cost analysis subsection                                                       | any("Cost Analysis" in t for t in<br>texts)                                                  | Pass   | 0.000s |
| Net upfront ; 5% of loan → High assessment.                                    | isinstance(result, list)                                                                     | Pass   | 0.000s |
| Net upfront ; 0.5% → Low assessment.                                           | isinstance(result, list)                                                                     | Pass   | 0.000s |
| Net upfront 0.5–1.5% → Low-Medium assessment.                                  | isinstance(result, list)                                                                     | Pass   | 0.000s |
| Net upfront 1.5–3% → Medium assessment.                                        | isinstance(result, list)                                                                     | Pass   | 0.000s |
| Net upfront 3–5% → Medium-High assessment.                                     | isinstance(result, list)                                                                     | Pass   | 0.000s |
| Discount points + original rate triggers estimated points/rate reduction rows. | isinstance(result, list)                                                                     | Pass   | 0.000s |
| Empty mortgage dict does not crash                                             | isinstance(result, list)                                                                     | Pass   | 0.000s |
| Has tables                                                                     | any(isinstance(e, Table) for e in<br>result)                                                 | Pass   | 0.000s |
| Lender credits reduces net cost                                                | isinstance(result, list)                                                                     | Pass   | 0.000s |

| Test Description                                                      | Acceptance Criteria                                                       | Result | Time   |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------|--------|--------|
| No financial terms returns message                                    | <code>any("No cost information available" in t for t in texts)</code>     | Pass   | 0.000s |
| No mortgage data returns message                                      | <code>any("No mortgage data available" in t for t in texts)</code>        | Pass   | 0.000s |
| Origination setup subsection                                          | <code>any("Origination" in t for t in texts)</code>                       | Pass   | 0.000s |
| Penalties subsection                                                  | <code>any("Penalties" in t for t in texts)</code>                         | Pass   | 0.000s |
| Returns list with data                                                | <code>isinstance(result, list); len(result) &gt; 0</code>                 | Pass   | 0.000s |
| Section header present                                                | <code>any("Mortgage Costs" in t for t in texts)</code>                    | Pass   | 0.000s |
| TotalLoanCosts + OriginalLoan triggers percentage row.                | <code>isinstance(result, list)</code>                                     | Pass   | 0.000s |
| Total costs overview subsection                                       | <code>any("Total Loan Costs" in t for t in texts)</code>                  | Pass   | 0.000s |
| LoanToValueRatio shown when different from OriginalLTV.               | <code>isinstance(result, list)</code>                                     | Pass   | 0.000s |
| Core loan terms subsection                                            | <code>any("Core Loan Terms" in t for t in texts)</code>                   | Pass   | 0.000s |
| Dti high risk                                                         | <code>isinstance(result, list)</code>                                     | Pass   | 0.000s |
| Dti low risk                                                          | <code>isinstance(result, list)</code>                                     | Pass   | 0.000s |
| Dti medium high risk                                                  | <code>isinstance(result, list)</code>                                     | Pass   | 0.000s |
| Dti medium risk                                                       | <code>isinstance(result, list)</code>                                     | Pass   | 0.000s |
| Dti subsection                                                        | <code>any("Debt-to-Income" in t for t in texts)</code>                    | Pass   | 0.001s |
| Has tables                                                            | <code>any(isinstance(e, Table) for e in result)</code>                    | Pass   | 0.000s |
| Interest rates subsection                                             | <code>any("Interest Rates" in t for t in texts)</code>                    | Pass   | 0.000s |
| Loan as % of purchase value row is computed when both fields present. | <code>isinstance(result, list)</code>                                     | Pass   | 0.000s |
| Ltv high risk                                                         | <code>isinstance(result, list)</code>                                     | Pass   | 0.000s |
| Ltv low medium risk                                                   | <code>isinstance(result, list)</code>                                     | Pass   | 0.000s |
| Ltv low risk                                                          | <code>isinstance(result, list)</code>                                     | Pass   | 0.000s |
| Ltv medium risk                                                       | <code>isinstance(result, list)</code>                                     | Pass   | 0.000s |
| Ltv subsection                                                        | <code>any("Loan-to-Value" in t for t in texts)</code>                     | Pass   | 0.000s |
| Ltv very high risk                                                    | <code>isinstance(result, list)</code>                                     | Pass   | 0.000s |
| No financial terms returns message                                    | <code>any("No detailed financial terms available" in t for t in...</code> | Pass   | 0.000s |

| Test Description                                  | Acceptance Criteria                                                                           | Result | Time   |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| No mortgage data returns message                  | <code>any("No mortgage data available" in t for t in texts)</code>                            | Pass   | 0.000s |
| Returns list with data                            | <code>isinstance(result, list); len(result) &gt; 0</code>                                     | Pass   | 0.000s |
| Section header present                            | <code>any("Detailed Mortgage Financial Terms" in t for t in texts)</code>                     | Pass   | 0.000s |
| Init no output dir uses config                    | <code>gen.output_dir == reports_dir; reports_dir.exists()</code>                              | Pass   | 0.001s |
| Generate filename non dict raises attribute error | <code>"unknown" in fname</code>                                                               | Pass   | 0.001s |
| Generate filename none mortgage data              | <code>"unknown" in fname; fname.endswith(".pdf")</code>                                       | Pass   | 0.001s |
| Page exception is caught                          | <code>isinstance(elements, list); len(elements) &gt; 0</code>                                 | Pass   | 0.002s |
| Unknown page in order is skipped                  | <code>isinstance(elements, list); len(elements) &gt; 0</code>                                 | Pass   | 0.002s |
| Generator does not use sequential format          | <code>'GAUGE-{i:}' not in src; 'GAUGE-{index}' not in src (+3 more)</code>                    | Pass   | 0.000s |
| Generator source uses mort prefix                 | <code>'MORT-' in src</code>                                                                   | Pass   | 0.004s |
| Generator source uses uuid                        | <code>'uuid' in src; 'uuid' in src (+1 more)</code>                                           | Pass   | 0.000s |
| Uuid4 produces protocol matching ids              | <code>GAUGE_ID.RE.match(gauge_id); PROP_ID.RE.match(prop_id) (+2 more)</code>                 | Pass   | 0.000s |
| Accepts valid                                     | <code>PROP_ID.RE.match('PROP-a3f7bc12'); PROP_ID.RE.match('PROP-00000000') (+6 more)</code>   | Pass   | 0.000s |
| Rejects sequential                                | <code>not PROP_ID.RE.match('PROP-001'); not PROP_ID.RE.match('PROP-1234') (+3 more)</code>    | Pass   | 0.000s |
| Test getting delinquent mortgages.                | <code>len(delinquent) == 1; delinquent[0]['MortgageID'] == "MORT-003"</code>                  | Pass   | 0.001s |
| Test exposure aggregation by lender.              | <code>"First National Bank" in exposure; "Regional Credit Union" in exposure (+2 more)</code> | Pass   | 0.001s |
| Test getting property IDs that have mortgages.    | <code>len(prop_ids) == 3; "PROP-001" in prop_ids (+2 more)</code>                             | Pass   | 0.001s |
| Test total loan exposure calculation.             | <code>total == 1640000</code>                                                                 | Pass   | 0.001s |
| List all returns summaries                        | <code>len(MortgageLoader(tmp_path).list_all()) == 2</code>                                    | Pass   | 0.001s |
| Load all returns list                             | <code>len(MortgageLoader(tmp_path).load_all()) == 3</code>                                    | Pass   | 0.001s |

| Test Description                                                                       | Acceptance Criteria                                                                      | Result | Time   |
|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------|--------|
| Test MortgageID extraction.                                                            | <code>mort.id is not None;</code><br><code>mort.id.startswith("MORT-")</code>            | Pass   | 0.001s |
| Test mortgage summary generation.                                                      | <code>len(summaries) == 3;</code> 'mortgageId' in<br>summary (+3 more)                   | Pass   | 0.001s |
| Test loading mortgage portfolio.                                                       | <code>len(mortgages) == 3</code>                                                         | Pass   | 0.001s |
| Find by id                                                                             | <code>MortgageLoader(tmp_path).find_by_id("MORT-001")</code><br><code>is not None</code> | Pass   | 0.001s |
| Mortgage without top-level PropertyID falls back to nested Mortgage.Header.PropertyID. | <code>MortgageLoader(tmp_path).find_by_property_id("PROP-NESTED")</code>                 | Pass   | 0.002s |
| Missing returns none                                                                   | <code>MortgageLoader(tmp_path).find_by_id("MORT-999")</code><br><code>is None</code>     | Pass   | 0.001s |
| Test finding mortgage by ID.                                                           | <code>mort is not None;</code> mort['PropertyID']<br>== "PROP-001" (+1 more)             | Pass   | 0.001s |
| Test finding mortgages by lender.                                                      | <code>len(morts) == 2</code>                                                             | Pass   | 0.001s |
| Test lender search is case insensitive.                                                | <code>len(morts) == 2</code>                                                             | Pass   | 0.001s |
| Test finding mortgage by property ID.                                                  | <code>mort is not None;</code> mort['MortgageID']<br>== "MORT-001"                       | Pass   | 0.001s |
| Test finding mortgage for property without mortgage.                                   | <code>mort is None</code>                                                                | Pass   | 0.001s |
| Test finding active mortgages.                                                         | <code>len(morts) == 2</code>                                                             | Pass   | 0.001s |
| Test finding delinquent mortgages.                                                     | <code>len(morts) == 1;</code><br><code>morts[0]['MortgageID'] == "MORT-003"</code>       | Pass   | 0.001s |
| Application subsection                                                                 | <code>any("Application Information" in t</code><br><code>for t in texts)</code>          | Pass   | 0.000s |
| Denial reason hidden when default                                                      | <code>isinstance(result, list)</code>                                                    | Pass   | 0.000s |
| Denial reason shown when not default                                                   | <code>isinstance(result, list)</code>                                                    | Pass   | 0.000s |
| Empty mortgage dict does not crash                                                     | <code>isinstance(result, list)</code>                                                    | Pass   | 0.000s |
| mortgage_data without outer 'Mortgage' key is handled via fallback.                    | <code>isinstance(result, list)</code>                                                    | Pass   | 0.000s |
| Has tables                                                                             | <code>any(isinstance(e, Table) for e in</code><br><code>result)</code>                   | Pass   | 0.000s |
| Key financial terms subsection                                                         | <code>any("Key Financial Terms" in t for t</code><br><code>in texts)</code>              | Pass   | 0.000s |
| LTV row is computed when both PurchaseValue and OriginalLoan are present.              | <code>isinstance(result, list)</code>                                                    | Pass   | 0.000s |
| Mortgage features subsection                                                           | <code>any("Mortgage Features" in t for t</code><br><code>in texts)</code>                | Pass   | 0.000s |



| Test Description                                          | Acceptance Criteria                                        | Result | Time   |
|-----------------------------------------------------------|------------------------------------------------------------|--------|--------|
| Mortgage identification subsection                        | any("Mortgage Identification" in t<br>for t in texts)      | Pass   | 0.000s |
| No mortgage data returns message                          | any("No mortgage data available" in<br>t for t in texts)   | Pass   | 0.000s |
| None mortgage data returns message                        | any("No mortgage data available" in<br>t for t in texts)   | Pass   | 0.000s |
| Returns list with data                                    | isinstance(result, list); len(result)<br>> 0               | Pass   | 0.000s |
| Section header present                                    | any("Mortgage Overview" in t for t<br>in texts)            | Pass   | 0.000s |
| Test that mortgage PropertyID<br>links to valid property. | prop is not None                                           | Pass   | 0.001s |
| Test finding mortgage for each prop-<br>erty.             | prop is not None; mort is not None                         | Pass   | 0.001s |
| Generate elements returns list                            | isinstance(elements, list);<br>len(elements) > 0           | Pass   | 0.002s |
| Generate filename fallback                                | fname.endswith(".pdf"); "unknown" in<br>fname              | Pass   | 0.001s |
| Generate filename with id                                 | "MORT-test0001" in fname;<br>fname.endswith(".pdf")        | Pass   | 0.001s |
| Generate report creates pdf                               | output.exists(); output.suffix ==<br>".pdf"                | Pass   | 0.011s |
| Generate report returns path                              | isinstance(result, Path)                                   | Pass   | 0.011s |
| Init creates output dir                                   | output_dir.exists()                                        | Pass   | 0.001s |
| Page order set                                            | "title" in gen.page_order;<br>gen.page_order[0] == "title" | Pass   | 0.001s |
| Pages initialised                                         | "title" in gen.pages;<br>"mortgage_overview" in gen.pages  | Pass   | 0.001s |
| Elevated ltv medium flood                                 | "Moderate Risk" in r                                       | Pass   | 0.000s |
| High flood overrides                                      | "High Risk" in r and "flood" in<br>r.lower()               | Pass   | 0.000s |
| High ltv medium flood                                     | "High Risk" in r                                           | Pass   | 0.000s |
| Low flood risk                                            | "Low Risk" in r                                            | Pass   | 0.000s |
| Medium flood default                                      | "Moderate Risk" in r                                       | Pass   | 0.000s |
| Minimal risk                                              | "Minimal" in r                                             | Pass   | 0.000s |
| Negative value critical                                   | "Critical Risk" in r                                       | Pass   | 0.000s |
| Negative value high                                       | "High Risk" in r                                           | Pass   | 0.000s |
| Negative value moderate                                   | "Moderate Risk" in r                                       | Pass   | 0.000s |
| None ltv no crash                                         | isinstance(r, str)                                         | Pass   | 0.000s |
| Very high flood overrides                                 | "High Risk" in r                                           | Pass   | 0.000s |
| Elements non empty                                        | len(elements) > 0                                          | Pass   | 0.001s |

| Test Description                                                      | Acceptance Criteria                                                                                   | Result | Time   |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|--------|
| Empty property data does not crash                                    | <code>isinstance(elements, list)</code>                                                               | Pass   | 0.000s |
| Extracts mortgage id                                                  | <code>"MORT-test0001" in full_text</code>                                                             | Pass   | 0.000s |
| No mortgage data returns elements                                     | <code>isinstance(elements, list);</code><br><code>len(elements) &gt; 0</code>                         | Pass   | 0.000s |
| Returns list                                                          | <code>isinstance(elements, list)</code>                                                               | Pass   | 0.059s |
| Table elements included                                               | <code>len(tables) &gt;= 1</code>                                                                      | Pass   | 0.000s |
| Line 108: <code>_save_meetings</code> fails → 500.                    | <code>r.status_code == 500;</code> <code>"Failed" in</code><br><code>r.get_json()["message"]</code>   | Pass   | 0.001s |
| Line 195: document file not found → 404.                              | <code>r.status_code == 404</code>                                                                     | Pass   | 0.002s |
| Line 196: serve an uploaded document.                                 | <code>r.status_code == 200</code>                                                                     | Pass   | 0.004s |
| Line 162: file with empty filename → 400.                             | <code>r.status_code == 400</code>                                                                     | Pass   | 0.003s |
| Line 155: meeting not found → 404.                                    | <code>r.status_code == 404</code>                                                                     | Pass   | 0.001s |
| Line 158: no file in request → 400.                                   | <code>r.status_code == 400</code>                                                                     | Pass   | 0.003s |
| Line 184: save failure after upload → 500.                            | <code>r.status_code == 500</code>                                                                     | Pass   | 0.004s |
| Lines 164-186: upload a file to a meeting.                            | <code>r.status_code == 200; doc["filename"]</code><br><code>== "report.pdf"</code>                    | Pass   | 0.003s |
| Line 179: meeting without 'documents' key gets it created.            | <code>r.status_code == 200</code>                                                                     | Pass   | 0.004s |
| Line 144: <code>_save_meetings</code> fails on update → 500.          | <code>r.status_code == 500</code>                                                                     | Pass   | 0.003s |
| Count gauges exceeding warning level.                                 | <code>count &gt;= 0;</code> <code>count &lt;=</code><br><code>len(sample_gauges_df)</code>            | Pass   | 0.004s |
| Find gauge with highest current level.                                | <code>highest is not None;</code> <code>'gauge_id' in</code><br><code>highest (+1 more)</code>        | Pass   | 0.005s |
| First storm gets greek name                                           | <code>name; name[0].isupper()</code>                                                                  | Pass   | 0.002s |
| Name from storms json takes priority                                  | <code>data["storms"][0]["name"] ==</code><br><code>"CustomStormName"</code>                           | Pass   | 0.005s |
| When segment is a point (seg2 j 1e-18), returns start point distance. | <code>abs(nx - 51.5) &lt; 1e-9;</code> <code>abs(ny + 0.1)</code><br><code>&lt; 1e-9 (+1 more)</code> | Pass   | 0.000s |
| Distance is non negative                                              | <code>d &gt;= 0</code>                                                                                | Pass   | 0.000s |
| Midpoint projects to midpoint                                         | <code>abs(nx - 51.5) &lt; 0.01;</code> <code>abs(ny - 0.0)</code><br><code>&lt; 0.01 (+1 more)</code> | Pass   | 0.000s |
| Point before start clamps to start                                    | <code>abs(nx - 51.0) &lt; 0.01;</code> <code>abs(ny - 0.0)</code><br><code>&lt; 0.01</code>           | Pass   | 0.000s |
| Point past end clamps to end                                          | <code>abs(nx - 52.0) &lt; 0.01;</code> <code>abs(ny - 0.0)</code><br><code>&lt; 0.01</code>           | Pass   | 0.000s |

| Test Description                                                                                        | Acceptance Criteria                                                                               | Result | Time   |
|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------|--------|
| Damage amount computed correctly                                                                        | <code>prop["damage.amount"] ==<br/>pytest.approx(expected)</code>                                 | Pass   | 0.004s |
| Negative equity flag set on property                                                                    | <code>data["properties"][0]["negative.equity"]<br/>is True</code>                                 | Pass   | 0.004s |
| LTV $\geq$ 100% when outstanding $\geq$ post-damage value.                                              | <code>prop["post_damage_ltv"] &gt; 100.0</code>                                                   | Pass   | 0.004s |
| Severe storm triggers negative equity                                                                   | <code>r.status_code == 200;<br/>data["portfolio"]["mortgages_in_negative.equity"]<br/>== 1</code> | Pass   | 0.004s |
| Same for mortgage-report: null JSON body $\rightarrow$ line 168 $\rightarrow$ 400.                      | <code>r.status_code in (400, 500)</code>                                                          | Pass   | 0.000s |
| Sending content-type application/json with null body $\rightarrow$ get_json() = None $\rightarrow$ 400. | <code>r.status_code in (400, 500)</code>                                                          | Pass   | 0.000s |
| All modules no newline in js strings                                                                    | <code>failures == {}</code>                                                                       | Pass   | 0.005s |
| Audit reports no newline in js strings                                                                  | <code>offenders == []</code>                                                                      | Pass   | 0.001s |
| Full compiled js no newline in js strings                                                               | <code>offenders == []</code>                                                                      | Pass   | 0.030s |
| No mortgage message                                                                                     | <code>any("No mortgage data" in t for t in<br/>texts)</code>                                      | Pass   | 0.000s |
| No mortgage returns list                                                                                | <code>isinstance(result, list); len(result)<br/>&gt; 0</code>                                     | Pass   | 0.000s |
| Gauge that exists in storm data but has no trades returns 404.                                          | <code>resp.status_code == 404; 'No open<br/>trades' in data.get('message', '')</code>             | Pass   | 0.020s |
| Empty regulatory returns message                                                                        | <code>any("No regulatory information" in t<br/>for t in texts)</code>                             | Pass   | 0.000s |
| Returns list                                                                                            | <code>isinstance(result, list)</code>                                                             | Pass   | 0.000s |
| gauge.json present but every record missing gauge_id $\rightarrow$ n_all == 0.                          | <code>isinstance(result, dict);<br/>result["num_gauges"] == 0 (+1 more)</code>                    | Pass   | 0.002s |
| Sequences still written when all gauges invalid                                                         | <code>(tmp_path /<br/>SEQUENCES.FILENAME).exists()</code>                                         | Pass   | 0.002s |
| Line 95: <code>_normal_pdf(0)</code> should return $1/\sqrt{2\pi}$ .                                    | <code>result == pytest.approx(expected,<br/>rel=1e-10)</code>                                     | Pass   | 0.000s |
| Line 95: PDF decreases as $ x $ increases.                                                              | <code>d._normal_pdf(0) &gt; d._normal_pdf(1) &gt;<br/>d._normal_pdf(2)</code>                     | Pass   | 0.000s |
| Line 95: <code>_normal_pdf</code> at positive value.                                                    | <code>result == pytest.approx(expected,<br/>rel=1e-10)</code>                                     | Pass   | 0.000s |
| Line 95: <code>_normal_pdf</code> is symmetric.                                                         | <code>d._normal_pdf(2.0) ==<br/>pytest.approx(d._normal_pdf(-2.0))</code>                         | Pass   | 0.000s |
| Normalize float returns empty                                                                           | <code>loader._normalize_to_list(3.14) == []</code>                                                | Pass   | 0.000s |

| Test Description                                                                           | Acceptance Criteria                                                                    | Result | Time   |
|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------|--------|
| Normalize integer returns empty                                                            | <code>result == []</code>                                                              | Pass   | 0.001s |
| Normalize string returns empty                                                             | <code>result == []</code>                                                              | Pass   | 0.000s |
| Normalize true returns empty                                                               | <code>loader._normalize_to_list(True) == []</code>                                     | Pass   | 0.000s |
| <code>-stressm</code> alone must always run stressm regardless of <code>-nostress</code> . | —                                                                                      | Pass   | 0.006s |
| <code>-nostress</code> must be a registered argument on the port subparser.                | <code>args.nostress is True</code>                                                     | Pass   | 0.018s |
| With <code>-all -nostress</code> , <code>generate_stressm</code> must not be called.       | —                                                                                      | Pass   | 0.083s |
| Number type returns float                                                                  | <code>isinstance(result, float);</code><br><code>result &lt;= 1000</code>              | Pass   | 0.000s |
| CalledProcessError → returns False.                                                        | <code>result is False</code>                                                           | Pass   | 0.000s |
| macOS: <code>subprocess.run(['open', ...])</code> is called.                               | <code>result is True;</code><br><code>mock_run.call_args[0][0][0] == "open"</code>     | Pass   | 0.001s |
| Darwin success                                                                             | <code>result is True;</code><br><code>mock_run.call_args[0][0][0] == "open"</code>     | Pass   | 0.000s |
| Darwin success                                                                             | <code>result is True</code>                                                            | Pass   | 0.000s |
| Generic exception returns false                                                            | <code>result is False</code>                                                           | Pass   | 0.001s |
| Generic exception returns false                                                            | <code>result is False</code>                                                           | Pass   | 0.000s |
| Any other exception → returns False.                                                       | <code>result is False</code>                                                           | Pass   | 0.000s |
| Linux: <code>subprocess.run(['xdg-open', ...])</code> is called.                           | <code>result is True;</code><br><code>mock_run.call_args[0][0][0] == "xdg-open"</code> | Pass   | 0.000s |
| Linux success                                                                              | <code>result is True; "xdg-open" in args</code>                                        | Pass   | 0.000s |
| Linux success                                                                              | <code>result is True</code>                                                            | Pass   | 0.000s |
| Subprocess called process error returns false                                              | <code>result is False</code>                                                           | Pass   | 0.000s |
| Subprocess failure returns false                                                           | <code>result is False</code>                                                           | Pass   | 0.000s |
| Unknown platform uses webbrowser                                                           | <code>result is True</code>                                                            | Pass   | 0.000s |
| Unknown platform uses webbrowser                                                           | <code>result is True</code>                                                            | Pass   | 0.000s |
| Unknown platform: falls back to <code>webbrowser.open()</code> .                           | <code>result is True</code>                                                            | Pass   | 0.000s |
| Windows: <code>subprocess.run(['start', ...])</code> is called.                            | <code>result is True;</code><br><code>mock_run.call_args[0][0][0] == "start"</code>    | Pass   | 0.001s |
| Windows case insensitive                                                                   | <code>isinstance(result, bool)</code>                                                  | Pass   | 0.000s |
| Windows success                                                                            | <code>result is True</code>                                                            | Pass   | 0.000s |

| Test Description                                                                                           | Acceptance Criteria                                                                                       | Result | Time   |
|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------|--------|
| Windows success                                                                                            | <code>result is True</code>                                                                               | Pass   | 0.000s |
| Lines 432-433: <code>-gauge-id</code> → <code>_find_gauge_by_id</code> called.                             | —                                                                                                         | Pass   | 0.005s |
| Lines 484-486: <code>-mortgage-file</code> → mortgage JSON is opened and parsed.                           | —                                                                                                         | Pass   | 0.006s |
| Lines 489-490: <code>-property-id</code> with no mortgage file → <code>_find_property_by_id</code> called. | —                                                                                                         | Pass   | 0.005s |
| Lines 491-492: both <code>-property-id</code> and <code>-mortgage-file</code> → both finders called.       | —                                                                                                         | Pass   | 0.006s |
| Lines 427-429: <code>-timeseries-file</code> → timeseries JSON opened and parsed.                          | —                                                                                                         | Pass   | 0.005s |
| Known outputs                                                                                              | <code>m["gauge.json"] == "gauges";</code><br><code>m["property.json"] == "properties"</code><br>(+1 more) | Pass   | 0.000s |
| Dir output empty                                                                                           | <code>not _outputs_exist("stressm", tmp_path)</code>                                                      | Pass   | 0.001s |
| Dir output missing                                                                                         | <code>not _outputs_exist("stressm", tmp_path)</code>                                                      | Pass   | 0.000s |
| Dir output present non empty                                                                               | <code>_outputs_exist("stressm", tmp_path)</code>                                                          | Pass   | 0.001s |
| File output missing                                                                                        | <code>not _outputs_exist("gauges", tmp_path)</code>                                                       | Pass   | 0.000s |
| File output present                                                                                        | <code>_outputs_exist("gauges", tmp_path)</code>                                                           | Pass   | 0.000s |
| Partial outputs missing                                                                                    | <code>not _outputs_exist("stressm", tmp_path)</code>                                                      | Pass   | 0.000s |
| Unknown step                                                                                               | <code>not _outputs_exist("nonexistent", tmp_path)</code>                                                  | Pass   | 0.000s |
| Very high flood risk + arrears + very high LTV + poor credit → CRITICAL.                                   | <code>isinstance(result, list)</code>                                                                     | Pass   | 0.001s |
| Very low flood risk + good payment + low LTV + excellent credit → LOW.                                     | <code>isinstance(result, list)</code>                                                                     | Pass   | 0.000s |
| Property only high flood risk                                                                              | <code>isinstance(result, list)</code>                                                                     | Pass   | 0.000s |
| Property only very low flood risk                                                                          | <code>isinstance(result, list)</code>                                                                     | Pass   | 0.000s |
| Effective rating is valid                                                                                  | <code>rating in valid_ratings</code>                                                                      | Pass   | 0.000s |
| Overall risk rating has component scores                                                                   | <code>isinstance(cs, dict)</code>                                                                         | Pass   | 0.000s |

| Test Description                                                   | Acceptance Criteria                                                                        | Result | Time   |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------|--------|
| Overall risk rating has required keys                              | key in orr                                                                                 | Pass   | 0.000s |
| Clear override with null                                           | r.status_code == 200;<br>r.get_json()["status"] == "success"                               | Pass   | 0.005s |
| Invalid rating returns 400                                         | r.status_code == 400;<br>r.get_json()["status"] == "error"                                 | Pass   | 0.003s |
| Model not found returns 404                                        | r.status_code == 404;<br>r.get_json()["status"] == "error"<br>(+3 more)                    | Pass   | 0.002s |
| No reason field returns 400                                        | r.status_code == 400                                                                       | Pass   | 0.002s |
| No reason returns 400                                              | r.status_code == 400                                                                       | Pass   | 0.002s |
| Override stored in model                                           | rr["mrc_override"] == "Conditional";<br>rr["effective_rating"] ==<br>"Conditional"         | Pass   | 0.004s |
| Response has audit entry                                           | "audit_entry" in data;<br>data["audit_entry"]["event_type"]<br>== "risk_rating_override"   | Pass   | 0.003s |
| Valid override returns success                                     | r.status_code == 200;<br>r.get_json()["status"] == "success"                               | Pass   | 0.003s |
| Line 301: audit overflow during<br>override.                       | r.status_code == 200                                                                       | Pass   | 0.032s |
| Line 248: inventory not found →<br>404.                            | r.status_code == 404                                                                       | Pass   | 0.001s |
| Line 282: save failure → 500.                                      | r.status_code == 500                                                                       | Pass   | 0.002s |
| Custom gauge basket size                                           | cdm.gauge.basket_size == 10                                                                | Pass   | 0.000s |
| Default gauge basket size                                          | cdm.gauge.basket_size == 20                                                                | Pass   | 0.000s |
| Lines 333-334: exception in cre-<br>ate_mapping raises ValueError. | —                                                                                          | Pass   | 0.000s |
| Verify all sections of create_mapping<br>produce correct keys.     | mapping["swap_id"] == "PRS-1";<br>mapping["notional"] == 1000000<br>(+3<br>more)           | Pass   | 0.000s |
| validate({}) returns errors.                                       | isinstance(errors, dict)                                                                   | Pass   | 0.000s |
| Lines 255-256: Empty GaugeBas-<br>ket.                             | "GaugeSet" in errors;<br>any("GaugeBasket" in e for e in<br>errors["GaugeSet"])            | Pass   | 0.000s |
| Lines 263-264: exception returns<br>validation_error.              | "validation_error" in result;<br>isinstance(result["validation_error"],<br>list) (+1 more) | Pass   | 0.000s |
| Lines 258-259: gauge errors col-<br>lected.                        | "GaugeSet" in errors                                                                       | Pass   | 0.000s |
| Lines 244-245: header errors col-<br>lected.                       | "Header" in errors;<br>len(errors["Header"]) >= 2                                          | Pass   | 0.000s |



| Test Description                                                        | Acceptance Criteria                                                             | Result | Time   |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------|--------|
| Line 241-242: Missing CatchmentID.                                      | "Header" in errors; any("CatchmentID" in e for e in errors["Header"]) (+2 more) | Pass   | 0.000s |
| Line 251-252: Missing GaugeSetID.                                       | "GaugeSet" in errors; any("GaugeSetID" in e for e in errors["GaugeSet"])        | Pass   | 0.000s |
| Line 238-239: Missing SwapID.                                           | "Header" in errors; any("SwapID" in e for e in errors["Header"])                | Pass   | 0.000s |
| Line 261: valid data returns empty dict.                                | prs_cdm.validate(data) == {}                                                    | Pass   | 0.000s |
| buildPRSControls function exists.                                       | 'buildPRSControls' in prs_js                                                    | Pass   | 0.000s |
| Counterparty selector present.                                          | 'prs-counterparty' in prs_js                                                    | Pass   | 0.000s |
| Payer/Receiver direction selector present.                              | 'payer' in prs_js.lower(); 'receiver' in prs_js.lower() (+1 more)               | Pass   | 0.001s |
| Maturity/tenor selector present.                                        | 'prs-maturity' in prs_js                                                        | Pass   | 0.000s |
| Notional input field present.                                           | 'prs-notional' in prs_js                                                        | Pass   | 0.000s |
| Trigger level selector present.                                         | 'prs-trigger' in prs_js                                                         | Pass   | 0.000s |
| ghc_prs.get_js() returns valid JS.                                      | len(prs_js) > 1000                                                              | Pass   | 0.000s |
| Fetches yield curve from /api/v1/trading/yield-curve.                   | '/api/v1/trading/yield-curve' in prs_js                                         | Pass   | 0.000s |
| Each trade has physical swap root                                       | not bad                                                                         | Pass   | 0.009s |
| Each trade has swap id                                                  | not bad                                                                         | Pass   | 0.004s |
| Prs directory exists                                                    | PRS_DIR.exists()                                                                | Pass   | 0.000s |
| Swap id matches filename                                                | not bad                                                                         | Pass   | 0.004s |
| Thames Central book has 16 trades.                                      | len(prs_files) >= 16                                                            | Pass   | 0.000s |
| Trade statuses are valid                                                | not bad                                                                         | Pass   | 0.004s |
| Trades are present                                                      | len(prs_files) >= 1                                                             | Pass   | 0.000s |
| Trades have gauge set                                                   | not bad                                                                         | Pass   | 0.004s |
| Book thames uses prs prefix                                             | 'PRS-' in src                                                                   | Pass   | 0.001s |
| Book thames uses uuid                                                   | 'uuid' in src                                                                   | Pass   | 0.000s |
| uuid4().hex[:8].upper() — the book generator pattern — always matches.  | PRS_ID_RE.match(swap_id)                                                        | Pass   | 0.000s |
| Some paths may produce lowercase — also accepted.                       | PRS_ID_RE.match('PRS-a8b3c4d5'); PRS_ID_RE.match('PRS-ffffffff')                | Pass   | 0.000s |
| book.py uses                                                            | PRS_ID_RE.match('PRS-A8B3C4D5');                                                | Pass   | 0.000s |
| uuid4().hex[:8].upper() → uppercase hex.                                | PRS_ID_RE.match('PRS-FFFFFFFF') (+1 more)                                       | Pass   | 0.000s |
| PRS-TEST-001 style IDs used in test fixtures are NOT production format. | not PRS_ID_RE.match('PRS-TEST-001'); not PRS_ID_RE.match('PRS-VAUXHALL')        | Pass   | 0.000s |

| Test Description                                                                             | Acceptance Criteria                                 | Result | Time   |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------|--------|--------|
| Port src all does not include book symbols                                                   | not overlap                                         | Pass   | 0.000s |
| The package <code>__init__.py</code> must not contain 'from .book import'.                   | "from .book import" not in src                      | Pass   | 0.001s |
| Package <code>__init__.py</code> must not expose <code>generate_thames_central_book</code> . | not hasattr(pkg, "generate_thames_central_book")    | Pass   | 0.000s |
| Contains claim ref paragraph                                                                 | len(tables) > 0                                     | Pass   | 0.001s |
| No flood events still renders                                                                | len(result) > 0                                     | Pass   | 0.001s |
| Coordinates that are not floats should not raise.                                            | isinstance(result, list)                            | Pass   | 0.001s |
| Returns list                                                                                 | isinstance(result, list); len(result) > 0 (+4 more) | Pass   | 0.001s |
| Stats reflect events                                                                         | any(isinstance(e, Paragraph) for e in result)       | Pass   | 0.001s |
| Without mortgage                                                                             | isinstance(result, list); len(result) > 0           | Pass   | 0.001s |
| Events with no sequence_id (None) should render without error.                               | isinstance(result, list)                            | Pass   | 0.001s |
| No flood events                                                                              | isinstance(result, list); len(result) > 0 (+2 more) | Pass   | 0.000s |
| Returns list                                                                                 | isinstance(result, list); len(result) > 0 (+4 more) | Pass   | 0.001s |
| Stats bar present                                                                            | len(paragraphs) > 0                                 | Pass   | 0.001s |
| Table included                                                                               | any(isinstance(e, Table) for e in result)           | Pass   | 0.001s |
| Events with sequence_id not in lookup default to 'isolated'.                                 | isinstance(result, list)                            | Pass   | 0.001s |
| Contains tables                                                                              | len(tables) >= 2                                    | Pass   | 0.001s |
| Events with damage ratio $\geq 5\%$ trigger red background rows.                             | isinstance(result, list)                            | Pass   | 0.001s |
| Multiple sequences grouped                                                                   | len(tables) >= 3                                    | Pass   | 0.001s |
| No flood events                                                                              | isinstance(result, list); len(result) > 0 (+2 more) | Pass   | 0.000s |
| Returns list                                                                                 | isinstance(result, list); len(result) > 0 (+4 more) | Pass   | 0.001s |
| Zero property value should not raise a ZeroDivisionError.                                    | isinstance(result, list)                            | Pass   | 0.001s |
| Contains tables                                                                              | any(isinstance(e, Table) for e in result)           | Pass   | 0.001s |

| Test Description                                                    | Acceptance Criteria                                                                                                                               | Result | Time   |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| High damage + large mortgage triggers negative equity highlighting. | <code>isinstance(result, list);</code><br><code>any(isinstance(e, Table) for e</code><br><code>in result)</code>                                  | Pass   | 0.001s |
| No flood events with mortgage                                       | <code>isinstance(result, list)</code>                                                                                                             | Pass   | 0.001s |
| With mortgage                                                       | <code>isinstance(result, list); len(result)</code><br><code>&gt; 0</code>                                                                         | Pass   | 0.001s |
| Without mortgage                                                    | <code>isinstance(result, list);</code> <code>any('No</code><br><code>mortgage' in t for t in texts)</code> <code>(+2</code><br><code>more)</code> | Pass   | 0.000s |
| Contains reference table                                            | <code>len(tables) &gt;= 2</code>                                                                                                                  | Pass   | 0.001s |
| Assessed damage = prop_value * max damage_ratio per sequence.       | <code>isinstance(result, list)</code>                                                                                                             | Pass   | 0.001s |
| When damage exceeds property value, LTV is very high.               | <code>isinstance(result, list)</code>                                                                                                             | Pass   | 0.001s |
| Returns list                                                        | <code>isinstance(result, list); len(result)</code><br><code>&gt; 0</code>                                                                         | Pass   | 0.001s |
| Signature block present                                             | <code>len(paragraphs) &gt; 3</code>                                                                                                               | Pass   | 0.001s |
| Without mortgage                                                    | <code>isinstance(result, list);</code> <code>any('No</code><br><code>mortgage' in t for t in texts)</code> <code>(+2</code><br><code>more)</code> | Pass   | 0.001s |
| sp-panel-title element ID must be referenced in the JS.             | <code>'sp-panel-title' in table_js</code>                                                                                                         | Pass   | 0.000s |
| Title must start with 'Portfolio Storm Impact'.                     | <code>'Portfolio Storm Impact' in table_js</code>                                                                                                 | Pass   | 0.000s |
| Panel title must use options[selectedIndex].textContent.            | <code>'selectedIndex' in table_js;</code><br><code>'textContent' in table_js</code>                                                               | Pass   | 0.000s |
| Title assignment must NOT directly concatenate bare stormId.        | <code>"Portfolio Storm Impact \{\}\u2014</code><br><code>' + stormId" not in table_js</code>                                                      | Pass   | 0.000s |
| Base level derived from alert                                       | <code>r["base_level"] == pytest.approx(3.5</code><br><code>* 0.35)</code>                                                                         | Pass   | 0.000s |
| Flood alert                                                         | <code>_parse_gauge(_G1)["flood.alert"] ==</code><br><code>pytest.approx(3.5)</code>                                                               | Pass   | 0.000s |
| Flood warning                                                       | <code>_parse_gauge(_G1)["flood.warning"] ==</code><br><code>pytest.approx(4.6)</code>                                                             | Pass   | 0.000s |
| Gauge id                                                            | <code>_parse_gauge(_G1)["gauge.id"] ==</code><br><code>"GAUGE-test001"</code>                                                                     | Pass   | 0.000s |
| Lat lon                                                             | <code>r["lat"] == pytest.approx(51.46);</code><br><code>r["lon"] == pytest.approx(-0.30)</code>                                                   | Pass   | 0.000s |
| Missing alert returns none                                          | <code>_parse_gauge(bad) is None</code>                                                                                                            | Pass   | 0.000s |
| Missing gauge id returns none                                       | <code>_parse_gauge(bad) is None</code>                                                                                                            | Pass   | 0.000s |
| Missing lat returns none                                            | <code>_parse_gauge(bad) is None</code>                                                                                                            | Pass   | 0.000s |

| Test Description                             | Acceptance Criteria                                                                                       | Result | Time   |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------|--------|
| Severe warning                               | <code>.parse_gauge(_G1)["severe_warning"] == pytest.approx(5.5)</code>                                    | Pass   | 0.000s |
| Valid record returns dict                    | <code>result is not None</code>                                                                           | Pass   | 0.000s |
| Warning defaults when absent                 | <code>r is not None; r["flood_warning"] == pytest.approx(4.0 * 1.33) (+1 more)</code>                     | Pass   | 0.000s |
| Baseline without monthly means               | <code>result["base_level"] == 1.5; result["monthly_means"] is None</code>                                 | Pass   | 0.000s |
| With baseline                                | <code>result["base_level"] == 1.25; result["monthly_means"] == {"01": 1.4}</code>                         | Pass   | 0.000s |
| Without baseline uses heuristic              | <code>result["base_level"] == pytest.approx(3.5 * 0.35); result["monthly_means"] is None</code>           | Pass   | 0.000s |
| Empty file                                   | <code>metadata == {}; flows == []</code>                                                                  | Pass   | 0.000s |
| Flow values are floats                       | <code>isinstance(f["flow_cumeecs"], float)</code>                                                         | Pass   | 0.006s |
| Flows have date and flow                     | <code>len(flows) == 5; "date" in f (+1 more)</code>                                                       | Pass   | 0.001s |
| Flows have date and flow cumeecs             | <code>len(flows) &gt; 0; "date" in flows[0] (+1 more)</code>                                              | Pass   | 0.006s |
| Invalid rows skipped                         | <code>len(flows) == 2</code>                                                                              | Pass   | 0.002s |
| Metadata contains station id                 | <code>"station_id" in metadata; metadata["station_id"] == "39001"</code>                                  | Pass   | 0.001s |
| Metadata contains station name               | <code>metadata.get("station_name") == "My Test Gauge"</code>                                              | Pass   | 0.007s |
| Metadata contains station name               | <code>metadata.get("station_name") == "Test River"</code>                                                 | Pass   | 0.001s |
| Metadata station id                          | <code>metadata.get("station_id") == "40007"</code>                                                        | Pass   | 0.006s |
| Returns metadata and flows                   | <code>isinstance(metadata, dict); isinstance(flows, list)</code>                                          | Pass   | 0.006s |
| Returns tuple                                | <code>isinstance(metadata, dict); isinstance(flows, list)</code>                                          | Pass   | 0.002s |
| Short row skipped                            | <code>len(flows) == 2</code>                                                                              | Pass   | 0.001s |
| Lines with fewer than 2 columns are skipped. | <code>len(flows) == 1</code>                                                                              | Pass   | 0.001s |
| All metadata categories parsed               | <code>metadata.get("file_timestamp") == "2024-01-01"; metadata.get("database_id") == "7" (+4 more)</code> | Pass   | 0.001s |
| Flow row with non numeric value skipped      | <code>len(flows) == 2</code>                                                                              | Pass   | 0.001s |

| Test Description                                   | Acceptance Criteria                                                                                                 | Result | Time   |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------|--------|
| Metadata row with no value uses empty string       | <code>metadata.get("station_name") == "";</code><br><code>metadata.get("station_id") == "39001"</code><br>(+1 more) | Pass   | 0.001s |
| Metadata row with three columns stores value       | <code>metadata.get("dataType_units") == "m3/s"</code>                                                               | Pass   | 0.001s |
| Add participant auto generates id                  | <code>r.status_code == 200;</code><br><code>new_p["id"].startswith("P-")</code>                                     | Pass   | 0.001s |
| Add participant unknown meeting returns 404        | <code>r.status_code == 404</code>                                                                                   | Pass   | 0.001s |
| Delete participant                                 | <code>r.status_code == 200;</code> <code>pid not in remaining</code>                                                | Pass   | 0.003s |
| Delete participant unknown meeting returns 404     | <code>r.status_code == 404</code>                                                                                   | Pass   | 0.001s |
| Update missing participant returns 404             | <code>r.status_code == 404</code>                                                                                   | Pass   | 0.002s |
| Update participant                                 | <code>r.status_code == 200;</code><br><code>updated["status"] == "Attended"</code>                                  | Pass   | 0.003s |
| Add participant auto id                            | <code>"P-001" in ids; "P-002" in ids</code>                                                                         | Pass   | 0.002s |
| Add participant missing meeting returns 404        | <code>r.status_code == 404</code>                                                                                   | Pass   | 0.001s |
| Add participant returns 200                        | <code>r.status_code == 200;</code><br><code>len(meeting["participants"]) == 2</code>                                | Pass   | 0.002s |
| Delete participant missing meeting returns 404     | <code>r.status_code == 404</code>                                                                                   | Pass   | 0.002s |
| Delete participant returns 200                     | <code>r.status_code == 200;</code><br><code>len(r.get_json()["meeting"]["participants"]) == 0</code>                | Pass   | 0.002s |
| Update participant not found returns 404           | <code>r.status_code == 404</code>                                                                                   | Pass   | 0.002s |
| Update participant returns 200                     | <code>r.status_code == 200;</code> <code>p["role"] == "CF0"</code>                                                  | Pass   | 0.002s |
| All holiday fields shown                           | <code>isinstance(result, list)</code>                                                                               | Pass   | 0.000s |
| Holidays taken shown                               | <code>isinstance(result, list)</code>                                                                               | Pass   | 0.000s |
| Last holiday date shown                            | <code>isinstance(result, list)</code>                                                                               | Pass   | 0.000s |
| Without holiday fields, section should not appear. | <code>not any("Holiday" in t for t in texts)</code>                                                                 | Pass   | 0.000s |
| Total holidays shown                               | <code>isinstance(result, list)</code>                                                                               | Pass   | 0.000s |
| Arrears balance shown                              | <code>isinstance(result, list)</code>                                                                               | Pass   | 0.000s |
| Five missed payments poor                          | <code>isinstance(result, list)</code>                                                                               | Pass   | 0.000s |
| Four missed payments concerning                    | <code>isinstance(result, list)</code>                                                                               | Pass   | 0.000s |

| Test Description                                                           | Acceptance Criteria                                                               | Result | Time   |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------|--------|
| Highest arrears shown                                                      | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| In arrears flag true                                                       | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| In arrears score 5                                                         | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Many missed payments over 3                                                | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| No missed payments good                                                    | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| One missed payment good                                                    | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Some missed payments 1 to 3                                                | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Three missed payments concerning                                           | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Two missed payments good                                                   | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Zero missed payments excellent                                             | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Documentation pdf known model no file 404                                  | <code>r.status_code == 404</code>                                                 | Pass   | 0.000s |
| Documentation pdf unknown model 404                                        | <code>r.status_code == 404</code>                                                 | Pass   | 0.001s |
| Mrc tor pdf responds                                                       | <code>r.status_code in (200, 404)</code>                                          | Pass   | 0.010s |
| Parameter inventory pdf responds                                           | <code>r.status_code in (200, 404)</code>                                          | Pass   | 0.012s |
| Results pdf known model no file 404                                        | <code>r.status_code == 404</code>                                                 | Pass   | 0.000s |
| Results pdf unknown model 404                                              | <code>r.status_code == 404</code>                                                 | Pass   | 0.000s |
| Line 323: valid model dir but PDF file missing → 404.                      | <code>r.status_code == 404; "not found" in r.get_json()["message"].lower()</code> | Pass   | 0.000s |
| Line 325: model documentation PDF served when file exists.                 | <code>r.status_code == 200</code>                                                 | Pass   | 0.001s |
| Line 357: MRC ToR PDF missing → 404.                                       | <code>r.status_code == 404</code>                                                 | Pass   | 0.000s |
| Line 359: MRC ToR PDF served.                                              | <code>r.status_code == 200</code>                                                 | Pass   | 0.001s |
| Line 347: parameter inventory PDF missing → 404.                           | <code>r.status_code == 404</code>                                                 | Pass   | 0.000s |
| Line 349: parameter inventory PDF served.                                  | <code>r.status_code == 200</code>                                                 | Pass   | 0.002s |
| Line 337: valid model dir but test_results.pdf missing → 404.              | <code>r.status_code == 404</code>                                                 | Pass   | 0.000s |
| Line 339: test results PDF served when file exists.                        | <code>r.status_code == 200</code>                                                 | Pass   | 0.002s |
| STORM-SEVERE-001 (peak 6.5m) should come before STORM-MODERATE-002 (4.8m). | <code>severe_idx &lt; moderate_idx</code>                                         | Pass   | 0.004s |
| GET /trading/stress/storms returns storms sorted by peak_level_m DESC.     | <code>len(storms) &gt;= 2; sorted(peaks, reverse=True)</code>                     | Pass   | 0.003s |
| storms[0] must have the highest peak_level_m.                              | <code>storms[0]['peak_level_m'] &gt;= storms[-1]['peak_level_m']</code>           | Pass   | 0.004s |



| Test Description                                                               | Acceptance Criteria                                                                                   | Result | Time   |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|--------|
| When <code>seg.start == seg_end</code> , norm ~ 0 -i degenerate fallback path. | <code>abs(lon - (-0.30)) &lt; 1e-9</code>                                                             | Pass   | 0.000s |
| Offset moves point                                                             | <code>dist &gt;=</code><br><code>PropertyPortfolioGenerator.MIN_OFFSET_M</code><br><code>* 0.5</code> | Pass   | 0.000s |
| Returns two floats                                                             | <code>isinstance(lat, float);</code><br><code>isinstance(lon, float)</code>                           | Pass   | 0.000s |
| Persistent all events have readings                                            | <code>"flood_depth_m" in event;</code><br><code>"damage_ratio" in event (+2 more)</code>              | Pass   | 0.003s |
| Persistent four events recorded                                                | <code>result["summary"]["floods_at_nearest_gauge"]</code><br><code>== 4</code>                        | Pass   | 0.002s |
| FAIL if any output from STEP_IO is missing or empty on disk.                   | —                                                                                                     | Pass   | 0.036s |
| At least one trained classifier must be documented in training_summary.        | <code>stress_storms_dir.exists();</code><br><code>summary_path.exists() (+1 more)</code>              | Pass   | 0.002s |
| gauges/ must have one file per gauge in gauge.json.                            | <code>n_files == n_gauges</code>                                                                      | Pass   | 0.001s |
| gaugehc.json must have one curve per gauge in gauge.json.                      | <code>n_curves == n_gauges</code>                                                                     | Pass   | 0.003s |
| Directories that exist but are empty are just as broken as missing.            | <code>not empty</code>                                                                                | Pass   | 0.004s |
| propertyhc.json must have one curve per property in property.json.             | <code>n_curves == n_props</code>                                                                      | Pass   | 0.010s |
| Every gauge referenced by PRS trades must exist in gauge.json.                 | <code>not missing</code>                                                                              | Pass   | 0.009s |
| Line 48: <code>fmt_gbp(0)</code> returns em-dash.                              | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Missing positions key returns drawing                                          | <code>isinstance(result, Drawing)</code>                                                              | Pass   | 0.000s |
| Negative gauge P&L -i bar drawn to the left.                                   | <code>isinstance(result, Drawing)</code>                                                              | Pass   | 0.000s |
| No positions returns drawing                                                   | <code>isinstance(result, Drawing)</code>                                                              | Pass   | 0.000s |
| Positions exist but all running_pnl = 0 -i gauge_pnl has values.               | <code>isinstance(result, Drawing)</code>                                                              | Pass   | 0.000s |
| In -all mode, <code>resolve_prerequisites</code> should not be called.         | <code>run_all is True</code>                                                                          | Pass   | 0.000s |
| Running -propertyts with stale pre-reqs should enable their flags.             | <code>args.gauges is True; args.properties</code><br><code>is True (+3 more)</code>                   | Pass   | 0.000s |

| Test Description                                                                                                                                                                  | Acceptance Criteria                                                                                                                     | Result           | Time                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------|
| In <code>–strict</code> mode, <code>resolve_prerequisites</code> should not be called.                                                                                            | <code>not run_all; args.strict is True</code>                                                                                           | Pass             | 0.000s               |
| JS must POST to <code>/api/v1/trading/stress/portfolio-run</code> .                                                                                                               | <code>'/api/v1/trading/stress/portfolio-run' in ps_js</code>                                                                            | Pass             | 0.000s               |
| JS must fetch from <code>/api/v1/trading/stress/portfolio-storms</code> .                                                                                                         | <code>'/api/v1/trading/stress/portfolio-storms' in ps_js</code>                                                                         | Pass             | 0.000s               |
| <code>portfolio-run</code> endpoint must use POST method.                                                                                                                         | <code>'POST' in ps_js</code>                                                                                                            | Pass             | 0.000s               |
| Full Detail button must navigate to Tab 7 via <code>switchTab('stress')</code> .                                                                                                  | <code>"switchTab('stress')"</code> in <code>ps_js</code>                                                                                | Pass             | 0.000s               |
| 'Full Detail' link button must be present — navigates to Tab 7 gauge view.                                                                                                        | <code>'Full Detail' in ps_js</code>                                                                                                     | Pass             | 0.000s               |
| <code>_psRenderGaugeTable</code> must be defined — renders per-gauge trade rows.                                                                                                  | <code>'_psRenderGaugeTable' in ps_js</code>                                                                                             | Pass             | 0.000s               |
| <code>_psRenderGaugePnlTab</code> must be defined.                                                                                                                                | <code>'_psRenderGaugePnlTab' in ps_js</code>                                                                                            | Pass             | 0.000s               |
| <code>tdStressGaugeHint</code> must be set — passes selected gauge to Tab 7 pre-selection.                                                                                        | <code>'tdStressGaugeHint' in ps_js</code>                                                                                               | Pass             | 0.000s               |
| <code>_stressStormHint</code> must be set — passes selected storm to Tab 7 auto-selection.                                                                                        | <code>'_stressStormHint' in ps_js</code>                                                                                                | Pass             | 0.000s               |
| ps- IDs must not collide with Tab 7's <code>td-stress-gauge</code> ID.                                                                                                            | <code>'td-stress-gauge' not in ps_js</code>                                                                                             | Pass             | 0.000s               |
| Content area uses ps-content ID.                                                                                                                                                  | <code>'ps-content' in ps_js</code>                                                                                                      | Pass             | 0.000s               |
| Storm selector uses ps- prefix (not <code>td-stress-*</code> ).                                                                                                                   | <code>'ps-storm-sel' in ps_js</code>                                                                                                    | Pass             | 0.000s               |
| Sub-tab button IDs are built using ps-tab- prefix.                                                                                                                                | <code>"ps-tab-" in ps_js or "'ps-tab-' +"</code><br><code>in ps_js or 'ps-tab-...; "'pflood'"</code><br><code>in ps_js (+3 more)</code> | Pass             | 0.000s               |
| Rendered full JS must be of substantial length — sanity check that <code>loadPortStressData()</code> must be called in the <code>portstress</code> <code>switchTab</code> branch. | <code>len(full_js) &gt; 200000</code><br><br><code>'loadPortStressData' in full_js</code>                                               | Pass<br><br>Pass | 0.000s<br><br>0.000s |

| Test Description                                                                | Acceptance Criteria                                 | Result | Time   |
|---------------------------------------------------------------------------------|-----------------------------------------------------|--------|--------|
| psResult must survive f-string rendering — absent if brace escaping broke.      | 'psResult' in full_js                               | Pass   | 0.000s |
| 'Port Stress' tab label must appear in the rendered trading desk.               | 'Port Stress' in full_js                            | Pass   | 0.000s |
| portStressView variable must appear — panel appends it to DOM.                  | 'portStressView' in full_js                         | Pass   | 0.000s |
| createPortStressView() call must appear in panel construction.                  | 'createPortStressView' in full_js                   | Pass   | 0.000s |
| switchTab must handle tab === 'portstress' branch.                              | "tab === 'portstress'" in full_js                   | Pass   | 0.000s |
| tabs array must include the portstress entry.                                   | "id: 'portstress'" in full_js                       | Pass   | 0.000s |
| switchTab views array must include 'portstress'.                                | "'portstress'" in full_js                           | Pass   | 0.000s |
| hidePanel must call psCleanupCharts() to destroy chart instances.               | 'psCleanupCharts' in full_js                        | Pass   | 0.000s |
| psResult state variable must appear inside the IIFE, not after it.              | idx > 0; idx < iife_close                           | Pass   | 0.000s |
| Canvas element uses ps-pflood-canvas ID.                                        | 'ps-pflood-canvas' in ps_js                         | Pass   | 0.000s |
| _psRenderPFloodTab must be defined.                                             | '_psRenderPFloodTab' in ps_js                       | Pass   | 0.000s |
| Threshold bar colours must be present: severe red, warning orange, alert amber. | '#c62828' in ps_js; '#e65100' in ps_js<br>(+1 more) | Pass   | 0.000s |
| P(flood) tab creates a Chart.js instance.                                       | 'Chart(' in ps_js                                   | Pass   | 0.000s |
| Canvas element uses ps-portpnl-canvas ID.                                       | 'ps-portpnl-canvas' in ps_js                        | Pass   | 0.000s |
| Portfolio P&L tab uses the shared fmtGBP formatter.                             | 'fmtGBP' in ps_js                                   | Pass   | 0.000s |
| _psRenderPortPnlTab must be defined.                                            | '_psRenderPortPnlTab' in ps_js                      | Pass   | 0.000s |
| Horizontal bar chart needs indexAxis: 'y' in Chart.js options.                  | 'indexAxis' in ps_js                                | Pass   | 0.000s |
| Severity rows must navigate to Gauge P&L sub-tab via psSwitchSubTab.            | 'psSwitchSubTab' in ps_js                           | Pass   | 0.000s |

| Test Description                                                        | Acceptance Criteria                                                           | Result | Time    |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|---------|
| All four severity group labels must be present in the JS output.        | 'SEVERE' in ps_js; 'WARNING' in ps_js<br>(+2 more)                            | Pass   | 0.000s  |
| _psRenderSeverityTab must be defined.                                   | '_psRenderSeverityTab' in ps_js                                               | Pass   | 0.000s  |
| Severity row click must set psSelectedGaugeId before switching sub-tab. | 'psSelectedGaugeId' in ps_js                                                  | Pass   | 0.000s  |
| Severe group must use the #ffebee background colour.                    | ps_js.lower().find('#ffebee') >= 0                                            | Pass   | 0.000s  |
| All four sub-tab render functions must be present.                      | '_psRenderPFloodTab' in ps_js;<br>'_psRenderPortPnlTab' in ps_js<br>(+2 more) | Pass   | 0.000s  |
| 'psCleanupCharts' must be defined — destroys Chart.js instances.        | 'psCleanupCharts' in ps_js                                                    | Pass   | 0.000s  |
| 'createPortStressView' must be defined — builds the Tab 8 DOM.          | 'createPortStressView' in ps_js                                               | Pass   | 0.000s  |
| 'loadPortStressData' must be defined — fetches storm list on tab open.  | 'loadPortStressData' in ps_js                                                 | Pass   | 0.000s  |
| 'psStormChanged' must be defined — handles storm dropdown selection.    | 'psStormChanged' in ps_js                                                     | Pass   | 0.000s  |
| 'psSwitchSubTab' must be defined — controls the 4 sub-tabs.             | 'psSwitchSubTab' in ps_js                                                     | Pass   | 0.000s  |
| port_stress.get_js() returns non-empty string without errors.           | len(ps_js) > 0                                                                | Pass   | 0.000s  |
| Module-level state variables must be declared.                          | 'psResult' in ps_js; 'psActiveSubTab' in ps_js (+1 more)                      | Pass   | 0.000s  |
| Return value still has the correct structure after classifier training. | result["num_sequences"] == 20;<br>"type_counts" in result                     | Pass   | 17.938s |
| train_classifier=True without gauge_id covers lines 364-384.            | result["num_gauges"] == 3;<br>stressm_dir.exists() (+1 more)                  | Pass   | 18.338s |
| Known storm returns success                                             | r.status_code == 200;<br>r.get_json()["status"] == "success"                  | Pass   | 0.005s  |
| Mortgaged property has ltv fields                                       | len(mortgaged) >= 1;<br>"outstanding_balance" in p<br>(+3 more)               | Pass   | 0.004s  |
| No pts dir returns 404                                                  | r.status_code == 404;<br>r.get_json()["status"] == "error"<br>(+2 more)       | Pass   | 0.000s  |

| Test Description                                                             | Acceptance Criteria                                                                                          | Result | Time   |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------|--------|
| Options returns ok                                                           | <code>r.status_code == 200; r.status_code == 200</code>                                                      | Pass   | 0.000s |
| Portfolio damage pct non negative                                            | <code>pct &gt;= 0</code>                                                                                     | Pass   | 0.005s |
| Portfolio totals count                                                       | <code>port["properties_affected"] == len(r.get_json()["propertyi...</code>                                   | Pass   | 0.005s |
| Properties sorted by damage descending                                       | <code>amounts == sorted(amounts, reverse=True); amounts == sorted(amounts, reverse=True)</code>              | Pass   | 0.005s |
| Property entry has financial fields                                          | <code>"property_id" in prop; "property_value" in prop (+9 more)</code>                                       | Pass   | 0.005s |
| Response has portfolio                                                       | <code>"portfolio" in data; "total_properties" in port (+4 more)</code>                                       | Pass   | 0.005s |
| Response has properties list                                                 | <code>"properties" in data; len(data["properties"]) &gt;= 1 (+2 more)</code>                                 | Pass   | 0.004s |
| Response has storm id                                                        | <code>r.get_json()["storm_id"] == sid</code>                                                                 | Pass   | 0.005s |
| Unknown storm returns 404                                                    | <code>r.status_code == 404; r.get_json()["status"] == "error"</code>                                         | Pass   | 0.005s |
| No pts dir returns 404                                                       | <code>r.status_code == 404; data["status"] == "error"</code>                                                 | Pass   | 0.000s |
| Options returns 200                                                          | <code>r.status_code == 200</code>                                                                            | Pass   | 0.000s |
| Unknown storm returns 404                                                    | <code>r.status_code == 404; data["status"] == "error"</code>                                                 | Pass   | 0.003s |
| Mortgage enrichment present                                                  | <code>entry["has_mortgage"] is True; entry["outstanding_balance"] == pytest.approx(280.000) (+2 more)</code> | Pass   | 0.004s |
| Property entry has damage fields                                             | <code>entry["property_id"] == "PROP-001"; entry["flood_depth_m"] == pytest.approx(0.5) (+2 more)</code>      | Pass   | 0.005s |
| Response has portfolio key                                                   | <code>"portfolio" in data; key in portfolio</code>                                                           | Pass   | 0.004s |
| Response has properties list                                                 | <code>"properties" in data; len(props) &gt;= 1</code>                                                        | Pass   | 0.004s |
| Storm id in response                                                         | <code>data["storm_id"] == "STORM-001"</code>                                                                 | Pass   | 0.003s |
| Valid storm returns 200                                                      | <code>r.status_code == 200; data["status"] == "success"</code>                                               | Pass   | 0.004s |
| API must not return name == storm_id (produces 'ID (ID)' in UI).             | <code>s['name']</code>                                                                                       | Pass   | 0.002s |
| At least one storm must have a human-readable name (not a raw STORM-xxx id). | <code>named</code>                                                                                           | Pass   | 0.003s |

| Test Description                                                          | Acceptance Criteria                                                                        | Result | Time   |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------|--------|
| In gauge_results, severe gauges appear before warning gauges.             | levels == sorted(levels)                                                                   | Pass   | 0.034s |
| Warning gauges appear before alert gauges in results.                     | levels[i] <= levels[i + 1]                                                                 | Pass   | 0.031s |
| Gauges with num_trades != 0 have non-empty trades list.                   | len(g['trades']) > 0                                                                       | Pass   | 0.036s |
| Each trade has swap_id, trigger, notional, mtm, cash_price, stress_pnl,   | found_trade; 'swap_id' in t (+8 more)                                                      | Pass   | 0.033s |
| Exception in _load_stress_storms causes 500.                              | resp.status_code == 500                                                                    | Pass   | 0.008s |
| When two storms have equal gauges_severe, the one with                    | storms[0]['intensity_category'] == 'catastrophic'                                          | Pass   | 0.000s |
| patch _get_engines to raise RuntimeError → 500.                           | resp.status_code == 500                                                                    | Pass   | 0.004s |
| No stress_storms/ + cache cleared → portfolio-run returns 404.            | resp.status_code == 404                                                                    | Pass   | 0.001s |
| POST with no body returns 400 or 500.                                     | resp.status_code in (400, 500)                                                             | Pass   | 0.001s |
| Payer (is_payer=True) has positive notional in trade details.             | t['notional'] > 0                                                                          | Pass   | 0.029s |
| portfolio_stress_pnl == sum(g['stress_pnl'] for g in gauges).             | abs(data['portfolio_stress_pnl'] - expected_total) < 0.01                                  | Pass   | 0.026s |
| Receiver (is_payer=False) has negative notional in trade details.         | t['notional'] < 0                                                                          | Pass   | 0.042s |
| When peak_level_m != severe_level, p_flood should be 1.0.                 | gauge_001 is not None;<br>gauge_001['p_flood'] == 1.0                                      | Pass   | 0.031s |
| stress_pnl = cash_price - mtm;<br>cash_price = signed_notional * p_flood. | data['status'] == 'success';<br>abs(t['stress_pnl'] - (t['cash_price'] - t['mtm'])) < 0.01 | Pass   | 0.031s |
| Gauge not in storm's gauge_responses has p_flood=0.0 and impacted=False.  | lambeth['p_flood'] == 0.0;<br>lambeth['impacted'] is False                                 | Pass   | 0.041s |
| With 3 impacted gauges in STORM_PORT_SEVERE, predictor.predict is called  | pred.predict.call_count >= 1                                                               | Pass   | 0.033s |
| _stressm_predictor_cache is reused on second call.                        | result is sentinel                                                                         | Pass   | 0.000s |
| If stressm/ dir missing, falls back gracefully — no crash.                | result is None or<br>callable(getattr(result, 'predict', None))                            | Pass   | 0.000s |



| Test Description                               | Acceptance Criteria                                                                                                         | Result | Time   |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Missing sequences message mentions doublet     | "storm_sequences" in msg or "not found" in msg                                                                              | Pass   | 0.028s |
| Missing sequences returns 404                  | r.status_code == 404                                                                                                        | Pass   | 0.026s |
| No pts dir returns 404                         | r.status_code == 404                                                                                                        | Pass   | 0.000s |
| No pts dir status error                        | pts_client_no_data.get(URL).get_json()["status"] == "error"                                                                 | Pass   | 0.000s |
| Options returns ok                             | r.status_code == 200                                                                                                        | Pass   | 0.000s |
| Missing mortgage json returns 200              | client.get(URL).status_code == 200                                                                                          | Pass   | 0.129s |
| Missing mortgage json zero portfolio mortgages | client.get(URL).get_json()["total_portfolio_mortgages"] == ...                                                              | Pass   | 0.128s |
| Missing property json returns 200              | client.get(URL).status_code == 200                                                                                          | Pass   | 0.129s |
| Missing property json status success           | client.get(URL).get_json()["status"] == "success"                                                                           | Pass   | 0.129s |
| Missing property json zero portfolio value     | client.get(URL).get_json()["total_portfolio_value"] == 0.0                                                                  | Pass   | 0.129s |
| Mortgage impairment max is positive            | mi["max"] > 0                                                                                                               | Pass   | 0.109s |
| Mortgage impairment value                      | mi["max"] == pytest.approx(140000.0, rel=1e-3)                                                                              | Pass   | 0.107s |
| Tail storm has mortgage impairment             | len(tail) >= 1;<br>tail[0]["mortgage_impairment"] > 0                                                                       | Pass   | 0.102s |
| Prob loss pct is 50                            | client_two_storms.get(URL).get_json()["prob_loss_pct"] == ...                                                               | Pass   | 0.101s |
| Storm count is two                             | client_two_storms.get(URL).get_json()["storm_count"] == 2                                                                   | Pass   | 0.105s |
| Storms with damage is one                      | pts_env["client"].get(URL).get_json()["storms_with_damage"];<br>client_two_storms.get(URL).get_json()["storms_with_damage"] | Pass   | 0.104s |
| Storms zero damage is one                      | client_two_storms.get(URL).get_json()["storms_zero_damage"]                                                                 | Pass   | 0.102s |
| Tail storms contains the damaged storm         | "STORM-0001" in storm_ids                                                                                                   | Pass   | 0.108s |
| Tail storms limited to 50                      | len(client_two_storms.get(URL).get_json()["tail_storms"])                                                                   | Pass   | 0.102s |
| No mortgage impairment when above water        | mi["max"] == 0.0                                                                                                            | Pass   | 0.103s |
| Prob loss pct is 100                           | pts_env["client"].get(URL).get_json()["prob_loss_pct"] == ...                                                               | Pass   | 0.104s |
| Property damage cond mean positive             | pts_env["client"].get(URL).get_json()["property_damage"]                                                                    | Pass   | 0.102s |
| Property damage max is positive                | pts_env["client"].get(URL).get_json()["property_damage"]                                                                    | Pass   | 0.104s |
| Storm count is one                             | pts_env["client"].get(URL).get_json()["storm_count"] == 1                                                                   | Pass   | 0.104s |

| Test Description                                                           | Acceptance Criteria                                                                          | Result | Time   |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------|--------|
| Storms with damage is one                                                  | <code>pts_env["client"].get(URL).get_json()["storms_with_damage"] == 1</code>                | Pass   | 0.104s |
| Storms zero plus with damage equals total                                  | <code>data["storms_with_damage"] + data["storms_zero_damage"] == ...</code>                  | Pass   | 0.105s |
| Total portfolio mortgages                                                  | <code>pts_env["client"].get(URL).get_json()["total_portfolio_mortgages"]</code>              | Pass   | 0.105s |
| Total portfolio value                                                      | <code>pts_env["client"].get(URL).get_json()["total_portfolio_value"]</code>                  | Pass   | 0.105s |
| Histogram bin has lo hi count                                              | <code>"lo" in bin0 and "hi" in bin0 and "count" in bin0</code>                               | Pass   | 0.106s |
| Mort histogram has 50 bins                                                 | <code>len(pts_env["client"].get(URL).get_json()["mort_histogram"]) == 50</code>              | Pass   | 0.109s |
| Mortgage impairment sub keys                                               | <code>key in mi</code>                                                                       | Pass   | 0.104s |
| Prop histogram has 50 bins                                                 | <code>len(pts_env["client"].get(URL).get_json()["prop_histogram"]) == 50</code>              | Pass   | 0.105s |
| Property damage sub keys                                                   | <code>key in pd</code>                                                                       | Pass   | 0.104s |
| Returns 200                                                                | <code>pts_env["client"].get(URL).status_code == 200</code>                                   | Pass   | 0.091s |
| Status success                                                             | <code>pts_env["client"].get(URL).get_json()["status"] == "success"</code>                    | Pass   | 0.104s |
| Tail storm has required fields                                             | <code>len(storms) &gt;= 1; f in s</code>                                                     | Pass   | 0.105s |
| Tail storms is list                                                        | <code>isinstance(pts_env["client"].get(URL).get_json()["tail_storms"], list)</code>          | Pass   | 0.105s |
| Tail storms sorted descending                                              | <code>damages == sorted(damages, reverse=True)</code>                                        | Pass   | 0.106s |
| Top level keys present                                                     | <code>key in data</code>                                                                     | Pass   | 0.105s |
| Portfolio value from known property                                        | <code>client_unknown_prop.get(URL).get_json()["total_portfolio_value"]</code>                | Pass   | 0.102s |
| Unknown prop id is skipped                                                 | <code>data["storms_with_damage"] == 0</code>                                                 | Pass   | 0.102s |
| Zero depth cond mean is zero                                               | <code>client_zero_depth.get(URL).get_json()["property_damage"]</code>                        | Pass   | 0.105s |
| Zero depth prob loss is zero                                               | <code>client_zero_depth.get(URL).get_json()["probability_loss_pct"] == ...</code>            | Pass   | 0.104s |
| Zero depth property damage max is zero                                     | <code>client_zero_depth.get(URL).get_json()["property_damage"]</code>                        | Pass   | 0.101s |
| Zero depth storms with damage is zero                                      | <code>client_zero_depth.get(URL).get_json()["storms_with_damage"]</code>                     | Pass   | 0.099s |
| If predictor.predict raises, p_flood defaults to 0.0.                      | <code>resp.status_code == 200; g001['p_flood'] == 0.0</code>                                 | Pass   | 0.038s |
| When _get_predictor returns None, returns 404.                             | <code>resp.status_code == 404; 'models not found' in data.get('message', '').lower()</code>  | Pass   | 0.033s |
| trading/preloader.py must set window._tdPreloadDone (not bare assignment). | <code>'window._tdPreloadDone = true' in src</code>                                           | Pass   | 0.000s |
| startup.py must initialise via window._tdPreloadDone (not bare var).       | <code>'window._tdPreloadDone = false' in src; 'var _tdPreloadDone = false' not in src</code> | Pass   | 0.001s |

| Test Description                                                                    | Acceptance Criteria                                                                                         | Result | Time   |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|--------|
| startup.py must set window.tdPreloadDone = true after all fetches settle.           | 'window.tdPreloadDone = true' in src                                                                        | Pass   | 0.000s |
| tradingdesk.py showPanel() must read window.tdPreloadDone, not bare .tdPreloadDone. | 'window.tdPreloadDone' in js                                                                                | Pass   | 0.000s |
| All three tabs must fetch from /trading/stress/storms.                              | '/trading/stress/storms' in js                                                                              | Pass   | 0.000s |
| Gauge Stress tab auto-selects storms[0] (worst case).                               | 'storms[0]' in ghc_stress.js or<br>'storms.length > 0' in gh...                                             | Pass   | 0.000s |
| Gauge Stress tab storm dropdown uses all required storm fields.                     | field in ghc_stress.js                                                                                      | Pass   | 0.000s |
| Historical tab storm list uses all required storm fields.                           | field in hist.js                                                                                            | Pass   | 0.000s |
| Trading Desk storm dropdown includes peak_level_m in the label.                     | 'peak_level_m' in td_stress.js                                                                              | Pass   | 0.000s |
| Trading Desk Stress tab also auto-selects worst case storm.                         | 'withTrades' in td_stress.js or \{\}                                                                        | Pass   | 0.000s |
| Trading Desk Stress tab storm dropdown uses all required storm fields.              | 'storms....                                                                                                 | Pass   | 0.000s |
| Alert trigger                                                                       | field in td_stress.js                                                                                       | Pass   | 0.000s |
| Custom notional                                                                     | result["trigger_level"] == "alert";<br>result["trigger_threshold_m"] == 4.0                                 | Pass   | 0.000s |
| Flat hazard curve                                                                   | result["notional"] == 5.000.000<br>"fair_spread_bps" in result;<br>result["use_term_structure"] is<br>False | Pass   | 0.000s |
| Gauge not in hazard returns 404                                                     | r.status_code == 404                                                                                        | Pass   | 0.002s |
| Invalid trigger level returns 400                                                   | r.status_code == 400;<br>r.get_json()["status"] == "error"                                                  | Pass   | 0.000s |
| Missing gauge id returns 400                                                        | r.status_code == 400                                                                                        | Pass   | 0.000s |
| No hazard file returns 404                                                          | r.status_code == 404; "not available"<br>in r.get_json()["message"].lower()<br>(+1 more)                    | Pass   | 0.001s |
| No json returns 400                                                                 | r.status_code in (400, 500)                                                                                 | Pass   | 0.000s |
| Options returns ok                                                                  | r.status_code == 200                                                                                        | Pass   | 0.000s |
| Result has fair spread bps                                                          | "fair_spread_bps" in result;<br>result["fair_spread_bps"] > 0                                               | Pass   | 0.000s |
| Result has npv                                                                      | "npv" in result;<br>isinstance(result["npv"], float)                                                        | Pass   | 0.000s |
| Result has survival probabilities                                                   | len(sp) == 3; "1yr" in sp                                                                                   | Pass   | 0.000s |
| Returns dict                                                                        | isinstance(result, dict)                                                                                    | Pass   | 0.001s |

| Test Description                                          | Acceptance Criteria                                                                                 | Result | Time   |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| Severe trigger                                            | <code>result["trigger_level"] == "severe";</code>                                                   | Pass   | 0.000s |
|                                                           | <code>result["trigger_threshold_m"] == 5.0</code>                                                   |        |        |
| Term structure true                                       | <code>result["use_term_structure"] is True</code>                                                   | Pass   | 0.000s |
| Annual hazard rate matches input                          | <code>r["annual_hazard_rate"] ==</code><br><code>gauge["annual_hazard_rate_warn...</code>           | Pass   | 0.000s |
| Gauge id in result                                        | <code>r["gauge_id"] == gauge["gauge_id"]</code>                                                     | Pass   | 0.000s |
| Gauge name in result                                      | <code>r["gauge_name"] ==</code><br><code>gauge["gauge_name"]</code>                                 | Pass   | 0.000s |
| Severe annual hazard rate matches input                   | <code>r["annual_hazard_rate"] ==</code><br><code>gauge["annual_hazard_rate_seve...</code>           | Pass   | 0.000s |
| hazard.py lines 174-176: exception during pricing -i 500. | <code>r.status_code == 500</code>                                                                   | Pass   | 0.029s |
| hazard.py lines 146-147: gauge not in hazard data -i 404. | <code>r.status_code == 404</code>                                                                   | Pass   | 0.031s |
| hazard.py lines 139-140: no gaugehc.json -i 404.          | <code>r.status_code == 404</code>                                                                   | Pass   | 0.027s |
| hazard.py line 123: no JSON body -i 400.                  | <code>r.status_code in (400, 500)</code>                                                            | Pass   | 0.027s |
| hazard.py lines 121-123: JSON null body -i 400.           | <code>r.status_code == 400;</code><br><code>r.get_json()["status"] == "error"</code>                | Pass   | 0.029s |
| All trigger levels work: alert, warning, severe.          | <code>r.status_code == 200</code>                                                                   | Pass   | 0.128s |
| hazard.py lines 133-172: successful PRS pricing.          | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                            | Pass   | 0.116s |
| hazard.py: PRS pricing with default params.               | <code>r.status_code == 200</code>                                                                   | Pass   | 0.140s |
| Npv is sum of legs                                        | <code>r["npv"] == pytest.approx(</code><br><code>r["protection_leg...</code>                        | Pass   | 0.000s |
| Premium leg negative                                      | <code>r["premium_leg_npv"] &lt; 0</code>                                                            | Pass   | 0.000s |
| Protection leg positive                                   | <code>r["protection_leg_npv"] &gt; 0</code>                                                         | Pass   | 0.000s |
| Nonzero recovery rate stored                              | <code>r["recovery_rate"] ==</code><br><code>pytest.approx(0.4)</code>                               | Pass   | 0.000s |
| Recovery rate affects fair spread                         | <code>r0["fair_spread_bps"] &gt;</code><br><code>r40["fair_spread_bps"]</code>                      | Pass   | 0.000s |
| Higher risk free rate changes npv                         | <code>r_low["npv"] !=</code><br><code>pytest.approx(r_high["npv"],</code><br><code>rel=1e-4)</code> | Pass   | 0.000s |
| Risk free rate not in output                              | <code>"risk_free_rate" not in r</code>                                                              | Pass   | 0.000s |
| Fair spread is float                                      | <code>isinstance(r["fair_spread"], float)</code>                                                    | Pass   | 0.000s |
| Fair upfront is float                                     | <code>isinstance(r["fair_upfront"], float)</code>                                                   | Pass   | 0.000s |
| Npv is float                                              | <code>isinstance(r["npv"], float)</code>                                                            | Pass   | 0.000s |

| Test Description                                                            | Acceptance Criteria                                                                                                      | Result | Time   |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Running spread bps identity                                                 | <code>r["running_spread_bps"] ==</code><br><code>pytest.approx(r["running_sprea...</code>                                | Pass   | 0.000s |
| Survival probs all in zero one                                              | <code>0.0 &lt; val &lt;= 1.0</code>                                                                                      | Pass   | 0.000s |
| Tenor 1yr gives one survival entry                                          | <code>len(r["survival_probabilities"])</code><br><code>== 1;</code> "1yr" in<br><code>r["survival_probabilities"]</code> | Pass   | 0.000s |
| Use term structure stored false                                             | <code>r["use_term_structure"] is False</code>                                                                            | Pass   | 0.000s |
| Use term structure stored true                                              | <code>r["use_term_structure"] is True</code>                                                                             | Pass   | 0.000s |
| Not trained no metrics printed                                              | "AUC" not in out                                                                                                         | Pass   | 0.000s |
| Not trained status                                                          | "insufficient_data" in out;<br>"GAUGE-bad" in out (+1 more)                                                              | Pass   | 0.000s |
| Trained shows feature importance                                            | "log_h_s" in out; "log_t_end" in out                                                                                     | Pass   | 0.000s |
| Trained shows label info                                                    | "severe_warning" in out; "5.50" in out                                                                                   | Pass   | 0.000s |
| Trained status prints metrics                                               | "trained" in out; "0.9200" in out (+1<br>more)                                                                           | Pass   | 0.000s |
| Percentile between alert and warn-<br>ing → '← ALERT' marker (line<br>480). | "ALERT" in out                                                                                                           | Pass   | 0.001s |
| All sequence types reported                                                 | "isolated" in out; "doublet" in out<br>(+1 more)                                                                         | Pass   | 0.000s |
| Categories with zero peaks are<br>skipped via 'continue' (line 493).        | "moderate" in out; "catastrophic" not<br>in out                                                                          | Pass   | 0.000s |
| Output contains gauge id                                                    | "GAUGE-test-pgp" in out                                                                                                  | Pass   | 0.000s |
| Percentile $i$ = severe → '← SE-<br>VERE' marker.                           | "SEVERE" in out                                                                                                          | Pass   | 0.000s |
| Percentile between warning and se-<br>vere → '← WARNING' marker.            | "WARNING" in out                                                                                                         | Pass   | 0.000s |
| Summary with full data does not<br>crash                                    | —                                                                                                                        | Pass   | 0.004s |
| Summary with no data does not<br>crash                                      | —                                                                                                                        | Pass   | 0.000s |
| Summary with partial data does not<br>crash                                 | —                                                                                                                        | Pass   | 0.000s |
| Logs gauge name                                                             | <code>isinstance(result["gauge_name"], str)</code>                                                                       | Pass   | 0.000s |
| Logs without error                                                          | <code>len(caplog.records) &gt;= 0</code>                                                                                 | Pass   | 0.000s |
| Alert trigger in log                                                        | "ALERT" in combined                                                                                                      | Pass   | 0.000s |
| All survival years in log                                                   | key in combined                                                                                                          | Pass   | 0.000s |
| Contract terms header in log                                                | "Contract Terms" in combined                                                                                             | Pass   | 0.000s |
| Fair spread in log                                                          | "Fair Spread" in combined                                                                                                | Pass   | 0.000s |
| Gauge id in log                                                             | <code>result["gauge_id"]</code> in combined                                                                              | Pass   | 0.000s |
| Gauge name in log                                                           | <code>result["gauge_name"]</code> in combined                                                                            | Pass   | 0.000s |

| Test Description                                                 | Acceptance Criteria                                                       | Result | Time   |
|------------------------------------------------------------------|---------------------------------------------------------------------------|--------|--------|
| Npv value in log                                                 | "NPV" in combined                                                         | Pass   | 0.000s |
| Premium leg in log                                               | "Premium Leg" in combined                                                 | Pass   | 0.000s |
| Protection leg in log                                            | "Protection Leg" in combined                                              | Pass   | 0.000s |
| Recovery rate in log                                             | "Recovery Rate" in combined                                               | Pass   | 0.000s |
| Severe trigger in log                                            | "SEVERE" in combined                                                      | Pass   | 0.000s |
| Survival curve label in log                                      | "Survival Curve" in combined                                              | Pass   | 0.000s |
| Trigger level in log                                             | "WARNING" in combined                                                     | Pass   | 0.000s |
| Empty directory returns empty list                               | result == []                                                              | Pass   | 0.000s |
| Non matching files skipped                                       | result == []                                                              | Pass   | 0.001s |
| Output dir none uses input dir                                   | isinstance(result, list);<br>any("gaugehd.39999" in r for r in<br>result) | Pass   | 0.014s |
| Output files created                                             | all(Path(p).exists() for p in<br>result)                                  | Pass   | 0.015s |
| Process directory error continues                                | isinstance(result, list)                                                  | Pass   | 0.015s |
| Processes matching files                                         | len(result) == 2                                                          | Pass   | 0.029s |
| Lines 143-144: bad CSV -i error<br>logged, continues.            | isinstance(result, list)                                                  | Pass   | 0.001s |
| Lines 113-147: process_nrfa_directory scans CSV<br>files.        | isinstance(result, list); len(result)<br>== 1                             | Pass   | 0.002s |
| Line 134: output_dir=None uses<br>config.get_gaugehd_dir().      | isinstance(result, list)                                                  | Pass   | 0.002s |
| Lines 131-147: no matching files -i<br>empty list.               | result == []                                                              | Pass   | 0.000s |
| Line 61/80: custom glob pattern se-<br>lects different files.    | len(result) == 1                                                          | Pass   | 0.002s |
| Line 87-88: error on one file doesn't<br>stop processing others. | len(result) >= 1                                                          | Pass   | 0.003s |
| Multiple CSV files in directory are<br>all processed.            | len(result) == 2                                                          | Pass   | 0.003s |
| Line 81: station_id extracted by<br>stripping _gdf from stem.    | len(result) == 1;<br>result[0]                                            | Pass   | 0.001s |
| Line 288: empty gauge_lookup -i no<br>nearest -i None.           | result is None                                                            | Pass   | 0.001s |
| Line 283: empty prop_id -i None.                                 | result is None                                                            | Pass   | 0.001s |
| Line 283: prop_lat == 0 -i None.                                 | result is None                                                            | Pass   | 0.001s |
| Non flooded events excluded from<br>depth count                  | result["flood_count"] == 4                                                | Pass   | 0.231s |



| Test Description                                                     | Acceptance Criteria                                                                                               | Result | Time   |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------|--------|
| Zero flooded events uses floor pricing                               | result["has_gev"] is False;<br>result["pricing_method"] == "floor"<br>(+1 more)                                   | Pass   | 0.100s |
| Exactly min events triggers gev                                      | result["flood_count"] == MIN_EVENTS_FOR_GEV                                                                       | Pass   | 0.237s |
| Many events enables gev                                              | result["flood_count"] == 10;<br>result["gev_params"] is not None<br>or result["has_gev"] is ...                   | Pass   | 0.217s |
| One below min events uses floor                                      | result["has_gev"] is False;<br>result["pricing_method"] == "floor"                                                | Pass   | 0.099s |
| Max depth positive for flooded property                              | result["summary"]["max_depth_m"] > 0;<br>result["summary"]["mean_depth_m"] > 0                                    | Pass   | 0.238s |
| Transmission rate zero when no gauge floods                          | result["nearest_gauges"][0]["flood_transmission_rate"] == ...                                                     | Pass   | 0.100s |
| Zero depths for non flooded                                          | result["summary"]["max_depth_m"] == 0.0; result["summary"]["mean_depth_m"] == 0.0                                 | Pass   | 0.103s |
| Property with elevation $i$ gauge peak $-i$ depth 0.                 | result["summary"]["floods_at_property"] == 0                                                                      | Pass   | 0.001s |
| Gauge response with exceeded_alert=False $-i$ 0 gauge storms.        | result["summary"]["floods_at_nearest_gauge"] == 0                                                                 | Pass   | 0.001s |
| Summary dict has floods_at_nearest_gauge etc.                        | "property_id" in s;<br>"floods_at_nearest_gauge" in s<br>(+2 more)                                                | Pass   | 0.001s |
| Full processing writes a JSON file for the property.                 | result is not None; (pts_dir / "PROP-001.json").exists()                                                          | Pass   | 0.001s |
| If a single property raises during generation, failed_properties     | result["processing_stats"]["failed_properties"] >= 1;<br>result["processing_stats"]["successful_properties"] >= 2 | Pass   | 0.595s |
| getSampleStyleSheet() includes 'Title' by default, so line 65 branch | 'Title' in page.styles                                                                                            | Pass   | 0.000s |
| Trade with property_id computes basis_delta_bps and basis_dv01.      | enriched['property_id'] == 'PROP-001';<br>'basis_delta_bps' in enriched (+2 more)                                 | Pass   | 0.002s |
| Existing property returns success                                    | r.status_code == 200;<br>r.get_json()["status"] == "success"<br>(+3 more)                                         | Pass   | 0.003s |
| Include readings false strips readings                               | "readings" not in event                                                                                           | Pass   | 0.003s |

| Test Description                                                                       | Acceptance Criteria                                                                             | Result | Time   |
|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|--------|
| Include readings true keeps readings                                                   | "readings" in<br>data["flood_events"][0];<br>len(data["flood_events"][0]["readings"])<br>== 168 | Pass   | 0.003s |
| Missing property returns 404                                                           | r.status_code == 404;<br>r.get_json()["status"] == "error"<br>(+1 more)                         | Pass   | 0.000s |
| Options returns ok                                                                     | r.status_code == 200; r.status_code ==<br>200 (+1 more)                                         | Pass   | 0.000s |
| GET /api/v1/properties/{id}/hazard<br>returns hazard data for property<br>with curves. | response.status_code == 200;<br>data['status'] == 'success' (+2<br>more)                        | Pass   | 0.002s |
| GET /api/v1/properties/{id}/hazard<br>returns 404 for unknown property.                | response.status_code == 404                                                                     | Pass   | 0.002s |
| GET /api/v1/properties/{id}/hazard<br>returns 404 for property without<br>curves.      | response.status_code == 404;<br>data['status'] == 'error' (+1 more)                             | Pass   | 0.002s |
| GET /api/v1/propertyhc/basis re-<br>turns basis table.                                 | response.status_code == 200;<br>data['status'] == 'success' (+5<br>more)                        | Pass   | 0.004s |
| GET /api/v1/propertyhc/summary<br>returns portfolio hazard summary.                    | response.status_code == 200;<br>data['status'] == 'success' (+3<br>more)                        | Pass   | 0.002s |
| Data contains hazard curve                                                             | "hazard_curve" in<br>r.get_json()["data"]                                                       | Pass   | 0.002s |
| Data contains nearest gauges                                                           | "nearest_gauges" in<br>r.get_json()["data"]                                                     | Pass   | 0.001s |
| Data contains property id                                                              | r.get_json()["data"]["property_id"]<br>== "PROP-001"                                            | Pass   | 0.002s |
| Data contains summary                                                                  | "summary" in r.get_json()["data"]                                                               | Pass   | 0.001s |
| Known property returns 200                                                             | r.status_code == 200                                                                            | Pass   | 0.002s |
| Known property status success                                                          | r.get_json()["status"] == "success"                                                             | Pass   | 0.002s |
| No data returns 404                                                                    | r.status_code == 404; r.status_code ==<br>404 (+1 more)                                         | Pass   | 0.000s |
| No data status error                                                                   | r.get_json()["status"] == "error";<br>r.get_json()["status"] == "error"                         | Pass   | 0.000s |
| Options returns 200                                                                    | r.status_code == 200; r.status_code ==<br>200 (+1 more)                                         | Pass   | 0.000s |
| Options status ok                                                                      | r.get_json()["status"] == "ok";<br>r.get_json()["status"] == "ok" (+1<br>more)                  | Pass   | 0.000s |

| Test Description                                                           | Acceptance Criteria                                                           | Result | Time   |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|--------|
| Response has data key                                                      | "data" in r.get_json();<br>"data" in r.get_json()                             | Pass   | 0.002s |
| Second property also works                                                 | r.status_code == 200;<br>r.get_json()["data"]["property_id"]<br>== "PROP-002" | Pass   | 0.002s |
| Unknown property message contains id                                       | "PROP-GHOST" in r.get_json()["message"]                                       | Pass   | 0.002s |
| Unknown property returns 404                                               | r.status_code == 404                                                          | Pass   | 0.002s |
| Unknown property status error                                              | r.get_json()["status"] == "error"                                             | Pass   | 0.002s |
| configure(panel_width, panel_height) updates both.                         | obj.panel_width == "1000px";<br>obj.panel_height == "800px"                   | Pass   | 0.000s |
| configure() with no args leaves values unchanged.                          | obj.panel_width == original_width;<br>obj.panel_height == original_height     | Pass   | 0.000s |
| Lines 303-304: configure(panel_height=...) updates panel_height.           | obj.panel_height == "700px"                                                   | Pass   | 0.000s |
| Lines 301-302: configure(panel_width=...) updates panel_width.             | obj.panel_width == "900px"                                                    | Pass   | 0.000s |
| Line 308: get_statistics() returns dict with panel_width and panel_height. | stats["panel_width"] == "750px";<br>stats["panel_height"] == "550px"          | Pass   | 0.000s |
| get_statistics() always returns a dict.                                    | isinstance(stats, dict); "panel_width" in stats (+1 more)                     | Pass   | 0.000s |
| PropertyID must be populated in Header (not just PropertyAttributes).      | len(empty_header_ids) == 0                                                    | Pass   | 0.003s |
| property.json must contain at least one property.                          | len(ids) > 0                                                                  | Pass   | 0.003s |
| Property IDs in propertyhc.json must exist in property.json.               | not orphaned                                                                  | Pass   | 0.002s |
| Properties with hazard curves should also have timeseries data.            | not missing                                                                   | Pass   | 0.001s |
| propertyts/ files must reference property IDs from property.json.          | len(missing) == 0                                                             | Pass   | 0.897s |
| Property IDs in propertyts/ files must exist in property.json.             | not orphaned                                                                  | Pass   | 0.004s |
| Generator does not use sequential format                                   | 'GAUGE-{i:}' not in src;<br>'GAUGE-{index}' not in src (+3 more)              | Pass   | 0.000s |
| Generator source uses prop prefix                                          | 'PROP-' in src                                                                | Pass   | 0.001s |
| Generator source uses uuid                                                 | 'uuid' in src; 'uuid' in src (+1 more)                                        | Pass   | 0.000s |

| Test Description                                    | Acceptance Criteria                                                                                                             | Result | Time   |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Uuid4 produces protocol matching ids                | <code>GAUGE_ID.RE.match(gauge_id);</code><br><code>PROP_ID.RE.match(prop_id) (+2 more)</code>                                   | Pass   | 0.000s |
| Accepts valid                                       | <code>PROP_ID.RE.match('PROP-a3f7bc12');</code><br><code>PROP_ID.RE.match('PROP-00000000') (+6 more)</code>                     | Pass   | 0.000s |
| Rejects sequential                                  | <code>not PROP_ID.RE.match('PROP-001');</code> <code>not</code><br><code>PROP_ID.RE.match('PROP-1234') (+3 more)</code>         | Pass   | 0.000s |
| Rejects uppercase                                   | <code>not PROP_ID.RE.match('PROP-A3F7BC12');</code><br><code>not STORM_ID.RE.match('STORM-C7F4B8E2')</code>                     | Pass   | 0.000s |
| Test total portfolio value calculation.             | <code>total == 2050000</code>                                                                                                   | Pass   | 0.001s |
| Test value aggregation by state.                    | <code>'FL' in values;</code> <code>'TX' in values (+2 more)</code>                                                              | Pass   | 0.001s |
| Custom filename                                     | <code>PropertyLoader(tmp_path,</code><br><code>filename="my_props.json").count(...</code>                                       | Pass   | 0.001s |
| Load all returns list                               | <code>len(PropertyLoader(tmp_path).load_all())</code><br><code>== 3</code>                                                      | Pass   | 0.001s |
| Missing file empty                                  | <code>PropertyLoader(tmp_path).load_all()</code><br><code>== []</code>                                                          | Pass   | 0.000s |
| Test PropertyID extraction from nested structure.   | <code>prop_id is not None;</code><br><code>prop_id.startswith("PROP-")</code>                                                   | Pass   | 0.001s |
| Test property summary generation.                   | <code>len(summaries) == 3;</code> <code>'propertyId' in</code><br><code>summary (+4 more)</code>                                | Pass   | 0.001s |
| Test loading property portfolio.                    | <code>len(properties) == 3</code>                                                                                               | Pass   | 0.001s |
| Test getting coordinates.                           | <code>coords is not None;</code> <code>lat ==</code><br><code>pytest.approx(25.7617, rel=1e-4)</code><br><code>(+1 more)</code> | Pass   | 0.001s |
| Test getting coordinates for non-existent property. | <code>coords is None</code>                                                                                                     | Pass   | 0.001s |
| Test getting location data.                         | <code>location is not None;</code><br><code>location['City'] == "Miami" (+1 more)</code>                                        | Pass   | 0.001s |
| Test getting location for non-existent property.    | <code>location is None</code>                                                                                                   | Pass   | 0.001s |
| Test finding properties within radius.              | <code>len(nearby) &gt;= 1;</code> <code>len(nearby) == 2</code><br><code>(+1 more)</code>                                       | Pass   | 0.001s |
| Find by id                                          | <code>PropertyLoader(tmp_path).find_by_id("PROP-00000000")</code><br><code>is not None</code>                                   | Pass   | 0.001s |
| List all has property id                            | <code>"propertyId" in summaries[0]</code>                                                                                       | Pass   | 0.001s |

| Test Description                                                                                     | Acceptance Criteria                                                                                                  | Result | Time   |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------|--------|
| Test finding property by address.                                                                    | <code>prop is not None;</code> <code>"PROP-001" ==</code><br><code>property_loader.get_entity_id(prop)</code>        | Pass   | 0.001s |
| Test address search is case insensitive.                                                             | <code>prop is not None</code>                                                                                        | Pass   | 0.001s |
| Test address search with no match.                                                                   | <code>prop is None</code>                                                                                            | Pass   | 0.001s |
| Test finding property by ID.                                                                         | <code>prop is not None;</code><br><code>prop['PropertyHeader']['Location']['City']</code><br><code>== "Miami"</code> | Pass   | 0.001s |
| Test finding properties by state.                                                                    | <code>len(fl_props) == 2; len(tx_props) == 1</code>                                                                  | Pass   | 0.001s |
| Test state search is case insensitive.                                                               | <code>len(props) == 2</code>                                                                                         | Pass   | 0.001s |
| Test finding properties by zip code.                                                                 | <code>len(props) == 1;</code><br><code>property_loader.get_entity_id(props[0])</code><br><code>== "PROP-001"</code>  | Pass   | 0.001s |
| Test finding multiple properties in same zip.                                                        | <code>len(props) == 1</code>                                                                                         | Pass   | 0.001s |
| Passing an unhandled type triggers <code>super().default() → TypeError</code> .                      | —                                                                                                                    | Pass   | 0.000s |
| Lines 62-64: <code>ground_level</code> falls back to <code>default_elevation</code> .                | <code>mapping["ground_level_meters"] ==</code><br><code>10.0; mapping["elevation"] == 10.0</code>                    | Pass   | 0.000s |
| Lines 136-137: exception in <code>create_mapping</code> raises <code>ValueError</code> .             | —                                                                                                                    | Pass   | 0.000s |
| Found returns mortgage                                                                               | <code>r.status_code == 200; data["status"]</code><br><code>== "success" (+2 more)</code>                             | Pass   | 0.001s |
| Loader exception returns 500                                                                         | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code><br>(+1 more)                    | Pass   | 0.001s |
| Not found returns 404                                                                                | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br>(+2 more)                    | Pass   | 0.001s |
| Options returns ok                                                                                   | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "ok"</code><br>(+2 more)                       | Pass   | 0.000s |
| Line 258: continue when <code>prop_id</code> not in <code>prop_values → no properties → 404</code> . | <code>r.status_code == 404</code>                                                                                    | Pass   | 0.004s |
| Line 193: break when <code>prop_id</code> not in <code>prop_values → no properties → 404</code> .    | <code>r.status_code == 404</code>                                                                                    | Pass   | 0.004s |
| Output json contains generation metadata                                                             | <code>"generated_at" in meta; "catchment" in</code><br><code>meta (+1 more)</code>                                   | Pass   | 0.590s |
| Output json contains properties key                                                                  | <code>"properties" in data</code>                                                                                    | Pass   | 0.931s |

| Test Description                                                | Acceptance Criteria                                                            | Result | Time   |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------|--------|--------|
| Property header has property id                                 | "PropertyHeader" in prop;<br>"PropertyAttributes" in<br>prop["PropertyHeader"] | Pass   | 0.895s |
| Result data contains locations                                  | len(result["data"]["locations"]) ==<br>3                                       | Pass   | 0.895s |
| Full property+mortgage report produces a valid PDF file.        | report_path.exists();<br>report_path.suffix == '.pdf' (+1<br>more)             | Pass   | 0.028s |
| Mortgage-focused report produces a valid PDF.                   | report_path.exists();<br>report_path.suffix == '.pdf' (+1<br>more)             | Pass   | 0.022s |
| Output directory is created automatically if it doesn't exist.  | new_dir.exists(); report_path.exists()                                         | Pass   | 0.020s |
| Generated filename includes the property ID.                    | 'PROP-test0001' in report_path.name                                            | Pass   | 0.019s |
| Full report PDF should be non-trivial in size (multi-page).     | report_path.stat().st_size > 10000                                             | Pass   | 0.028s |
| Generated file is a valid PDF (starts with %PDF).               | header == b'%PDF-'                                                             | Pass   | 0.024s |
| Property-only report (no mortgage) produces a valid PDF.        | report_path.exists();<br>report_path.suffix == '.pdf' (+1<br>more)             | Pass   | 0.020s |
| Report handles minimal/sparse property data without crashing.   | report_path.exists();<br>report_path.stat().st_size > 0                        | Pass   | 0.014s |
| Risk-focused report produces a valid PDF.                       | report_path.exists();<br>report_path.suffix == '.pdf' (+1<br>more)             | Pass   | 0.015s |
| Auto-select includes mortgage pages when mortgage data present. | 'mortgage_overview' in pages;<br>'mortgage_details' in pages                   | Pass   | 0.002s |
| Auto-select excludes mortgage pages when no mortgage data.      | 'title_overview' in pages;<br>'mortgage_overview' not in pages<br>(+1 more)    | Pass   | 0.002s |
| Generator initialises with output directory.                    | gen.output_dir == tmp_path                                                     | Pass   | 0.002s |
| Generator lists all registered page modules.                    | 'title_overview' in pages; 'location'<br>in pages (+2 more)                    | Pass   | 0.002s |
| Validates known page names.                                     | 'title_overview' in valid; 'location'<br>in valid (+1 more)                    | Pass   | 0.002s |
| Line 174-175: mortgage_data -i includes mortgage pages.         | "mortgage_overview" in pages                                                   | Pass   | 0.002s |



| Test Description                                                        | Acceptance Criteria                                                       | Result | Time   |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------|--------|--------|
| Line 174 (else): no mortgage_data -i no mortgage pages.                 | "mortgage.overview" not in pages                                          | Pass   | 0.002s |
| Lines 214-216: exception in page -i logged, continue.                   | isinstance(elements, list)                                                | Pass   | 0.002s |
| Lines 198-199: unknown page -i skip.                                    | isinstance(elements, list)                                                | Pass   | 0.002s |
| Lines 184-185: missing PropertyID -i 'unknown'.                         | "unknown" in name                                                         | Pass   | 0.002s |
| Lines 257-264: mortgage-focused report.                                 | path.exists()                                                             | Pass   | 0.016s |
| Lines 251-255: property-only report.                                    | path.exists()                                                             | Pass   | 0.014s |
| Lines 266-276: risk-focused report without mortgage data.               | path.exists()                                                             | Pass   | 0.009s |
| Lines 266-276: risk-focused report with mortgage data.                  | path.exists()                                                             | Pass   | 0.011s |
| Line 313: get_page_categories.                                          | "property" in cats; "mortgage" in cats                                    | Pass   | 0.002s |
| Line 285: list_available_pages.                                         | isinstance(pages, list); "title.overview" in pages                        | Pass   | 0.002s |
| Lines 69-70: output_dir=None -i uses config.get_property_reports_dir(). | gen.output_dir == prop_dir                                                | Pass   | 0.002s |
| Line 315: validate_pages.                                               | "title.overview" in valid; "bogus_page" in invalid                        | Pass   | 0.002s |
| Categories dict initialized                                             | "property" in gen.categories; "mortgage" in gen.categories (+1 more)      | Pass   | 0.002s |
| Explicit output dir path                                                | isinstance(gen.output_dir, Path)                                          | Pass   | 0.002s |
| Explicit output dir string                                              | gen.output_dir == tmp_path                                                | Pass   | 0.002s |
| No output dir uses config                                               | gen.output_dir == prop_dir                                                | Pass   | 0.002s |
| Output dir created                                                      | new_dir.exists()                                                          | Pass   | 0.002s |
| Property pages dict initialized                                         | expected <= set(gen.property_pages.keys())                                | Pass   | 0.002s |
| Test POST /api/v1/properties/report without body.                       | response.status_code == 400                                               | Pass   | 0.000s |
| Test POST /api/v1/properties/report without propertyId.                 | response.status_code == 400; 'Property ID is required' in data['message'] | Pass   | 0.000s |

| Test Description                                                     | Acceptance Criteria                                                                                 | Result | Time   |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| Test POST /api/v1/properties/report with invalid property.           | <code>response.status_code == 404</code>                                                            | Pass   | 0.002s |
| Test GET /api/v1/properties/{id}.                                    | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)     | Pass   | 0.002s |
| Test GET /api/v1/properties/{id} with invalid ID.                    | <code>response.status_code == 404;</code><br><code>data['status'] == 'error'</code>                 | Pass   | 0.002s |
| Test legacy POST /api/v1/generate_property_report endpoint.          | <code>response.status_code == 404</code>                                                            | Pass   | 0.002s |
| Test GET /api/v1/properties.                                         | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+2 more)     | Pass   | 0.002s |
| Test OPTIONS preflight request.                                      | <code>response.status_code == 200</code>                                                            | Pass   | 0.000s |
| Existing property returns success                                    | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code><br>(+3 more) | Pass   | 0.003s |
| Flood events tagged with sequence type                               | <code>len(events) &gt;= 1; "sequence_type" in events[0]</code>                                      | Pass   | 0.004s |
| Missing property returns 404                                         | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br>(+1 more)   | Pass   | 0.000s |
| Nearest gauges enriched with flood stages                            | <code>len(gauges) &gt;= 1; "flood_stages" in gauges[0]</code>                                       | Pass   | 0.004s |
| Options returns ok                                                   | <code>r.status_code == 200; r.status_code == 200</code> (+1 more)                                   | Pass   | 0.000s |
| Response has flood events                                            | <code>"flood_events" in data;</code><br><code>len(data["flood_events"]) &gt;= 1</code>              | Pass   | 0.004s |
| Response has nearest gauges                                          | <code>"nearest_gauges" in data</code>                                                               | Pass   | 0.004s |
| Response has property info                                           | <code>"property_info" in data; "elevation_m" in data["property_info"]</code>                        | Pass   | 0.005s |
| Lines 157-158: gauge.json missing triggers except pass.              | <code>r.status_code == 200;</code><br><code>ng["flood_stages"] == {}</code>                         | Pass   | 0.028s |
| Lines 134-135: storm_sequences.json missing triggers except pass.    | <code>r.status_code == 200; data["status"] == "success"</code> (+1 more)                            | Pass   | 0.031s |
| GET /api/v1/propertyts/storms returns storms with correct structure. | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+8 more)     | Pass   | 0.009s |

| Test Description                                                            | Acceptance Criteria                                                                                  | Result | Time   |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------|--------|
| Storm list must only include storms with property damage.                   | <code>len(non_flooding) == 0</code>                                                                  | Pass   | 0.009s |
| GET /api/v1/properties/{id}/floods returns flood events for known property. | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+2 more)      | Pass   | 0.001s |
| Property with few flood events should still return data from propertyts.    | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)      | Pass   | 0.002s |
| GET /api/v1/properties/{id}/floods returns 404 for unknown property.        | <code>response.status_code == 404;</code><br><code>data['status'] == 'error'</code>                  | Pass   | 0.000s |
| GET /api/v1/properties/{id}/storms returns storm analysis data.             | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+4 more)      | Pass   | 0.004s |
| GET /api/v1/properties/{id}/storms returns 404 for unknown property.        | <code>response.status_code == 404</code>                                                             | Pass   | 0.000s |
| Property with zero flood events should return empty array, not error.       | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+4 more)      | Pass   | 0.003s |
| GET /api/v1/propertyts/summary returns portfolio flood summary.             | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)      | Pass   | 0.002s |
| Generates per property files                                                | <code>pts_dir.exists();</code> (pts_dir / "portfolio_flood_summary.json").exists()                   | Pass   | 0.000s |
| Per property files have flood data                                          | "property_id" in data;<br>"nearest_gauges" in data (+1 more)                                         | Pass   | 0.001s |
| Portfolio summary has statistics                                            | "total_properties" in s;<br>s["total_properties"] == N_PROPERTIES (+1 more)                          | Pass   | 0.002s |
| np.float64 is converted to Python float by encoder.                         | <code>isinstance(result, float); abs(result - 3.14) &lt; 1e-10</code>                                | Pass   | 0.000s |
| Lines 69-70: exception returns validation_error dict.                       | "validation_error" in result;<br><code>isinstance(result["validation_error"], list)</code> (+2 more) | Pass   | 0.000s |
| get_required_fields returns expected paths.                                 | <code>len(fields) == 4;</code><br>"PropertyHeader.Header.PropertyID" in fields (+1 more)             | Pass   | 0.000s |
| Line 65: location_errors appended when non-empty.                           | "Location" in errors;<br><code>len(errors["Location"]) == 2</code>                                   | Pass   | 0.000s |
| Line 60: Missing LatitudeDegrees.                                           | "Location" in errors;<br><code>any("LatitudeDegrees" in e for e in errors["Location"])</code>        | Pass   | 0.000s |

| Test Description                                       | Acceptance Criteria                                                                  | Result | Time   |
|--------------------------------------------------------|--------------------------------------------------------------------------------------|--------|--------|
| Line 62: Missing LongitudeDegrees.                     | "Location" in errors;<br>any("LongitudeDegrees" in e for<br>e in errors["Location"]) | Pass   | 0.000s |
| Valid data returns no errors.                          | property.validate(data) == {}                                                        | Pass   | 0.000s |
| Close to water increases score                         | near['vulnerability_score'] ><br>far['vulnerability_score']                          | Pass   | 0.000s |
| Flooded depth computed                                 | r['flood_depth'] ==<br>pytest.approx(1.0);<br>r['vulnerability_score'] > 0           | Pass   | 0.000s |
| High risk recommendations present                      | any("flood" in rec.lower() for rec<br>in r['recommendations'])                       | Pass   | 0.000s |
| No flood                                               | r['flood_depth'] == 0.0;<br>r['vulnerability_score'] == 0                            | Pass   | 0.000s |
| None elevation returns unknown                         | r['risk_level'] == 'Unknown';<br>r['flood_depth'] is None (+1 more)                  | Pass   | 0.000s |
| None flood level returns unknown                       | r['risk_level'] == 'Unknown'                                                         | Pass   | 0.000s |
| Result dict keys                                       | set(r.keys()) == {'flood_depth',<br>'risk_level', 'vulnerabi...                      | Pass   | 0.000s |
| Score capped at 100                                    | r['vulnerability_score'] <= 100                                                      | Pass   | 0.000s |
| No mortgage returns zero fill                          | r.status_code == 200;<br>prop["has_mortgage"] is False (+4<br>more)                  | Pass   | 0.004s |
| No mortgage sequence                                   | r.status_code == 200;<br>prop["has_mortgage"] is False                               | Pass   | 0.004s |
| Basis entry has avg basis bps                          | "avg_basis_bps" in entry                                                             | Pass   | 0.002s |
| Basis entry has flood count                            | "flood_count" in entry                                                               | Pass   | 0.002s |
| Basis entry has nearest gauges                         | "nearest_gauges" in entry                                                            | Pass   | 0.001s |
| Basis entry has property id                            | "property_id" in entry                                                               | Pass   | 0.002s |
| Basis table count matches                              | data["count"] ==<br>len(data["basis_table"])                                         | Pass   | 0.002s |
| Basis table length                                     | r.get_json()["count"] == 2                                                           | Pass   | 0.002s |
| Results must be sorted highest<br>avg_basis_bps first. | bases == sorted(bases, reverse=True)                                                 | Pass   | 0.002s |
| Nearest gauge has distance km                          | "distance_km" in<br>entry["nearest_gauges"][0]                                       | Pass   | 0.002s |
| Nearest gauge has gauge id                             | "gauge_id" in<br>entry["nearest_gauges"][0]                                          | Pass   | 0.002s |
| No data returns 404                                    | r.status_code == 404; r.status_code ==<br>404 (+1 more)                              | Pass   | 0.000s |
| No data status error                                   | r.get_json()["status"] == "error";<br>r.get_json()["status"] == "error"              | Pass   | 0.000s |

| Test Description                 | Acceptance Criteria                                                                   | Result | Time   |
|----------------------------------|---------------------------------------------------------------------------------------|--------|--------|
| Options returns 200              | <code>r.status_code == 200; r.status_code == 200 (+1 more)</code>                     | Pass   | 0.000s |
| Options status ok                | <code>r.get_json()["status"] == "ok"; r.get_json()["status"] == "ok" (+1 more)</code> | Pass   | 0.000s |
| Response has basis table         | <code>"basis_table" in r.get_json()</code>                                            | Pass   | 0.002s |
| Response has count               | <code>"count" in r.get_json()</code>                                                  | Pass   | 0.002s |
| Returns 200 with data            | <code>r.status_code == 200; r.status_code == 200</code>                               | Pass   | 0.002s |
| Status is success                | <code>r.get_json()["status"] == "success"; r.get_json()["status"] == "success"</code> | Pass   | 0.002s |
| Empty basis table is list        | <code>isinstance(r.get_json()["basis_table"], list)</code>                            | Pass   | 0.002s |
| Empty count is zero              | <code>r.get_json()["count"] == 0</code>                                               | Pass   | 0.002s |
| Empty returns 200                | <code>r.status_code == 200</code>                                                     | Pass   | 0.002s |
| Distribution avg basis bps       | <code>"avg_basis_bps" in dist; dist["avg_basis_bps"] &gt;= 0</code>                   | Pass   | 0.002s |
| Distribution avg flood count     | <code>"avg_flood_count" in dist; dist["avg_flood_count"] &gt; 0</code>                | Pass   | 0.002s |
| Distribution max depth m         | <code>"max_depth_m" in dist; dist["max_depth_m"] &gt; 0</code>                        | Pass   | 0.002s |
| Distribution num properties      | <code>dist["num_properties"] == 2</code>                                              | Pass   | 0.002s |
| No data message mentions command | <code>"port" in r.get_json()["message"].lower() or "propertyhc"...</code>             | Pass   | 0.001s |
| No data returns 404              | <code>r.status_code == 404; r.status_code == 404 (+1 more)</code>                     | Pass   | 0.000s |
| No data status is error          | <code>r.get_json()["status"] == "error"</code>                                        | Pass   | 0.000s |
| Options returns 200              | <code>r.status_code == 200; r.status_code == 200 (+1 more)</code>                     | Pass   | 0.000s |
| Options status ok                | <code>r.get_json()["status"] == "ok"; r.get_json()["status"] == "ok" (+1 more)</code> | Pass   | 0.000s |
| Response has data key            | <code>"data" in r.get_json(); "data" in r.get_json()</code>                           | Pass   | 0.002s |
| Response has distribution        | <code>"distribution" in r.get_json()["data"]</code>                                   | Pass   | 0.002s |
| Response has metadata            | <code>"metadata" in r.get_json()["data"]</code>                                       | Pass   | 0.002s |
| Response has summary             | <code>"summary" in r.get_json()["data"]</code>                                        | Pass   | 0.002s |
| Returns 200 with data            | <code>r.status_code == 200; r.status_code == 200</code>                               | Pass   | 0.002s |

| Test Description                                        | Acceptance Criteria                                                                                   | Result | Time   |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|--------|
| Status is success                                       | <code>r.get_json()["status"] == "success";</code><br><code>r.get_json()["status"] == "success"</code> | Pass   | 0.002s |
| Empty curves avg flood count is zero                    | <code>r.get_json()["data"]["distribution"]["avg_flood_count"] == 0</code>                             | Pass   | 0.002s |
| Empty curves num properties is zero                     | <code>r.get_json()["data"]["distribution"]["num_properties"] == 0</code>                              | Pass   | 0.002s |
| Empty curves returns 200                                | <code>r.status_code == 200</code>                                                                     | Pass   | 0.002s |
| When no natural measures, recommendation should appear. | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Natural measures section rendered                       | <code>any("Natural" in t for t in texts)</code>                                                       | Pass   | 0.000s |
| Empty property returns list                             | <code>isinstance(result, list); len(result) &gt; 0</code>                                             | Pass   | 0.000s |
| Empty protection measures dict shows fallback           | <code>any("No protection" in t for t in texts)</code>                                                 | Pass   | 0.000s |
| No protection measures shows fallback                   | <code>any("No protection" in t or "Protection" in t for t in te...)</code>                            | Pass   | 0.000s |
| Full protection fewer recommendations                   | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.001s |
| Generate protection recommendations returns dict        | <code>isinstance(result, dict)</code>                                                                 | Pass   | 0.000s |
| Missing critical measures in recommendations            | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Recommendations all natural false                       | <code>"Natural Solutions" in result</code>                                                            | Pass   | 0.000s |
| Recommendations flood re eligible                       | <code>"Insurance" in result</code>                                                                    | Pass   | 0.000s |
| Recommendations missing critical                        | <code>"Priority Installations" in result</code>                                                       | Pass   | 0.000s |
| Recommendations section always rendered                 | <code>any("Recommendation" in t for t in texts)</code>                                                | Pass   | 0.000s |
| All installed excellent rating                          | <code>any("Resilience" in t or "Protection" in t for t in texts)</code>                               | Pass   | 0.000s |
| Building measures listed                                | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| 8 of 10 = 80% → Excellent.                              | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| 4 of 10 = 40% → Fair.                                   | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| 6 of 10 = 60% → Good.                                   | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| 2 of 10 = 20% → Limited.                                | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| 0 of 10 = 0% → Poor.                                    | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Critical measures listed                                | <code>any(isinstance(e, Table) for e in result)</code>                                                | Pass   | 0.000s |
| Emergency measures listed                               | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Uncategorised measure still shown                       | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Excess amount shown                                     | <code>any(isinstance(e, Table) for e in result)</code>                                                | Pass   | 0.000s |

| Test Description                                                                       | Acceptance Criteria                                                                                                                 | Result | Time   |
|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Insurance premium shown                                                                | <code>any(isinstance(e, Table) for e in result)</code>                                                                              | Pass   | 0.000s |
| Other risk fields shown                                                                | <code>isinstance(result, list); len(result) &gt; 1</code>                                                                           | Pass   | 0.000s |
| Recommendations include flood re                                                       | <code>isinstance(result, list)</code>                                                                                               | Pass   | 0.000s |
| Risk assessment section rendered                                                       | <code>any("Insurance" in t for t in texts)</code>                                                                                   | Pass   | 0.000s |
| All entity ID suffixes are exactly 8 characters.                                       | <code>all(v == 8 for v in suffix_lengths.values())</code>                                                                           | Pass   | 0.000s |
| <code>str(uuid.uuid4())[8]</code> always yields lowercase hex.                         | <code>suffix == suffix.lower(); re.match(r'^[0-9a-f]{8}\$', suffix)</code>                                                          | Pass   | 0.001s |
| All four entity types use <code>str(uuid4())[8]</code> — same derivation, same format. | <code>pattern.match(entity_id)</code>                                                                                               | Pass   | 0.000s |
| All exports                                                                            | <code>'ClaimReportGenerator' in pkg.__all__</code>                                                                                  | Pass   | 0.000s |
| Import from package                                                                    | <code>ClaimReportGenerator is not None</code>                                                                                       | Pass   | 0.000s |
| Current exceeds original is capped                                                     | <code>result["Mortgage"]["LoanDetails"]["CurrentBalance"] &lt;= 40...</code>                                                        | Pass   | 0.000s |
| Interest over 15 capped                                                                | <code>result["Mortgage"]["LoanDetails"]["InterestRate"] == 12.0</code>                                                              | Pass   | 0.000s |
| Interest under 1 raised                                                                | <code>result["Mortgage"]["LoanDetails"]["InterestRate"] == 2.0</code>                                                               | Pass   | 0.000s |
| Ltv over 100 capped at 95                                                              | <code>result["Mortgage"]["LoanDetails"]["LTV"] == 95</code>                                                                         | Pass   | 0.000s |
| Ltv under 10 set to 60                                                                 | <code>result["Mortgage"]["LoanDetails"]["LTV"] == 60</code>                                                                         | Pass   | 0.000s |
| Valid data unchanged                                                                   | <code>result["Mortgage"]["LoanDetails"]["CurrentBalance"] == 35...; result["Mortgage"]["LoanDetails"]["LTV"] == 80 (+1 more)</code> | Pass   | 0.000s |
| Current exceeds original is capped to 95pct                                            | <code>result["Mortgage"]["LoanDetails"]["CurrentBalance"] &lt;= 30...</code>                                                        | Pass   | 0.000s |
| Current within original unchanged                                                      | <code>result["Mortgage"]["LoanDetails"]["CurrentBalance"] == 35...</code>                                                           | Pass   | 0.000s |
| Delegates to random module when method present                                         | <code>result is sentinel</code>                                                                                                     | Pass   | 0.000s |
| Ltv over 100 set to 95                                                                 | <code>result["Mortgage"]["LoanDetails"]["LTV"] == 95</code>                                                                         | Pass   | 0.000s |
| Ltv under 10 set to 60                                                                 | <code>result["Mortgage"]["LoanDetails"]["LTV"] == 60</code>                                                                         | Pass   | 0.000s |
| Valid ltv unchanged                                                                    | <code>result["Mortgage"]["LoanDetails"]["LTV"] == 75</code>                                                                         | Pass   | 0.000s |



| Test Description                          | Acceptance Criteria                                                                                                             | Result | Time   |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Get raci matrix                           | <code>response.status_code == 200;</code><br><code>data["status"] == "success"</code> (+3 more)                                 | Pass   | 0.002s |
| Get raci matrix file missing              | <code>response.status_code == 404</code>                                                                                        | Pass   | 0.000s |
| Update raci activity                      | <code>response.status_code == 200;</code><br><code>data["status"] == "success"</code> (+3 more)                                 | Pass   | 0.003s |
| Update raci activity invalid role         | <code>response.status_code == 400</code>                                                                                        | Pass   | 0.002s |
| Update raci escalation                    | <code>response.status_code == 200;</code><br><code>data["status"] == "success"</code> (+3 more)                                 | Pass   | 0.003s |
| Update raci role                          | <code>response.status_code == 200;</code><br><code>data["status"] == "success"</code> (+3 more)                                 | Pass   | 0.004s |
| Update raci role not found                | <code>response.status_code == 404</code>                                                                                        | Pass   | 0.002s |
| No raci data returns 404                  | <code>r.status_code == 404; r.status_code == 404</code>                                                                         | Pass   | 0.000s |
| Save raci failure returns 500             | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code> (+4 more)                                  | Pass   | 0.003s |
| Tier emphasis updated                     | <code>r.status_code == 200;</code><br><code>activities[0]["tier_emphasis"] == "Tier 1 focus"</code>                             | Pass   | 0.003s |
| No raci data returns 404                  | <code>r.status_code == 404; r.status_code == 404</code>                                                                         | Pass   | 0.000s |
| Save raci failure returns 500             | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code> (+4 more)                                  | Pass   | 0.002s |
| Save raci failure returns 500             | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code> (+4 more)                                  | Pass   | 0.003s |
| Remaining zero uses base amplitude        | <code>isinstance(level, float)</code>                                                                                           | Pass   | 0.000s |
| Failure metadata covers key tests         | <code>name in dl_mod.FAILURE.METADATA;</code><br><code>'principle' in meta</code> (+2 more)                                     | Pass   | 0.000s |
| Recon table has principle column          | <code>'BCBS Principle(s)' in header_row</code>                                                                                  | Pass   | 0.000s |
| Record and reload                         | <code>"gaugehd" in manifest["steps"];</code><br><code>manifest["steps"]["gaugehd"]["run_id"] == "run-test-001"</code> (+4 more) | Pass   | 0.003s |
| Record with auto run id                   | <code>entry["run_id"].startswith("run-")</code>                                                                                 | Pass   | 0.001s |
| Empty EndDate returns default tenor of 5. | <code>result == 5</code>                                                                                                        | Pass   | 0.000s |

| Test Description                                                         | Acceptance Criteria                                                                                                  | Result | Time   |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------|--------|
| Invalid date format returns default tenor of 5 (ValueError path).        | <code>result == 5</code>                                                                                             | Pass   | 0.000s |
| Missing EndDate key returns default tenor of 5.                          | <code>result == 5</code>                                                                                             | Pass   | 0.000s |
| None EndDate returns default tenor of 5 (TypeError path).                | <code>result == 5</code>                                                                                             | Pass   | 0.000s |
| Valid far-future EndDate returns positive tenor.                         | <code>result &gt;= 1</code>                                                                                          | Pass   | 0.000s |
| Compliant run is suitable                                                | <code>'suitable for regulatory risk reporting' in texts</code>                                                       | Pass   | 0.000s |
| Issues run not suitable                                                  | <code>'not suitable for regulatory risk reporting' in texts; 'COMPLIANT' not in texts.replace( 'PARTIALLY ...</code> | Pass   | 0.001s |
| Remediation lists stale steps                                            | <code>'Stale data' in texts; 'hazard' in texts</code>                                                                | Pass   | 0.001s |
| When lineage_results has a known failure, remediation cites it.          | <code>'PARTIALLY COMPLIANT' in texts or 'NON-COMPLIANT' in texts</code>                                              | Pass   | 0.001s |
| Shows strengths and weaknesses                                           | <code>'Strengths' in texts; 'Weaknesses' in texts</code>                                                             | Pass   | 0.001s |
| Steps with empty output directories are skipped.                         | <code>"propertyts" in result["skipped"]</code>                                                                       | Pass   | 0.013s |
| Repair with no existing manifest creates entries for all steps.          | <code>len(result["repaired"]) == len(STEP_IO); result["skipped"] == [] (+2 more)</code>                              | Pass   | 0.028s |
| Repaired hashes must match current file content.                         | <code>recorded_hash == hash_file(gauge_file)</code>                                                                  | Pass   | 0.005s |
| Running repair twice produces identical hashes.                          | <code>out_info["hash"] == data2["steps"][step]["outputs"][out.n...</code>                                            | Pass   | 0.015s |
| Input hashes recorded by repair should match the producer's output hash. | <code>gauges_out == props_in</code>                                                                                  | Pass   | 0.010s |
| Repair keeps run_id, timestamp, generator from existing entries.         | <code>"gauges" in result["repaired"]; gauges_entry["run_id"] == "run-original" (+3 more)</code>                      | Pass   | 0.021s |
| Repair uses repair-YYYYMMDD-HHMMSS as run_id for new entries.            | <code>gauges_entry["run_id"].startswith("repair-Pass</code>                                                          | Pass   | 0.002s |
| Steps whose outputs don't exist on disk are skipped.                     | <code>"gauges" in result["repaired"]; "properties" in result["skipped"] (+1 more)</code>                             | Pass   | 0.001s |

| Test Description                                                                    | Acceptance Criteria                                                                                                              | Result | Time   |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Repair adds a runs entry listing all repaired steps.                                | <code>len(repair_runs) == 1;</code><br><code>set(data["runs"][repair_runs[0]])</code><br><code>== set(result["repaired"])</code> | Pass   | 0.010s |
| Lines 503-504: default 'full' report_type → args.property_pages AttributeError      | <code>1 in exited</code>                                                                                                         | Pass   | 0.003s |
| Lines 349-351: success-logging block executes without error.                        | —                                                                                                                                | Pass   | 0.004s |
| Lines 444-445: monitoring without timeseries_data → condition False → else.         | —                                                                                                                                | Pass   | 0.004s |
| Lines 442-443: -report-type analysis → generate_analysis_report path.               | —                                                                                                                                | Pass   | 0.003s |
| Lines 344-345: -report-type analysis → generate_analysis_report path executed.      | —                                                                                                                                | Pass   | 0.004s |
| Lines 438-439: -report-type basic → generate_basic_report path.                     | —                                                                                                                                | Pass   | 0.004s |
| Lines 338-339: -report-type basic → generate_basic_report path executed.            | —                                                                                                                                | Pass   | 0.004s |
| Lines 340-341: -report-type detailed → generate_detailed_report path executed.      | —                                                                                                                                | Pass   | 0.004s |
| Lines 346-347: unmapped report_type → generate_report(pages.to_include=args.pages). | —                                                                                                                                | Pass   | 0.005s |
| Lines 444-445: -report-type full (not basic/monitoring/analysis) → else branch.     | —                                                                                                                                | Pass   | 0.006s |
| Lines 440-441: -report-type monitoring + timeseries → monitoring path.              | —                                                                                                                                | Pass   | 0.005s |
| Lines 499-500: -report-type mortgage-focused + mortgage file → focused report.      | —                                                                                                                                | Pass   | 0.006s |
| Lines 499 condition False (no mortgage_data) → falls to elif risk-focused / else.   | <code>1 in exited</code>                                                                                                         | Pass   | 0.007s |

| Test Description                                                                          | Acceptance Criteria                                                                   | Result | Time   |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|--------|
| Lines 497-498: <code>-report-type property-only → generate_property_only_report.</code>   | —                                                                                     | Pass   | 0.005s |
| Lines 501-502: <code>-report-type risk-focused → generate_risk_focused_report.</code>     | —                                                                                     | Pass   | 0.004s |
| Lines 342-343: <code>-report-type summary → generate_summary_report</code> path executed. | —                                                                                     | Pass   | 0.003s |
| Lines 447-449: success-logging block runs after generation.                               | —                                                                                     | Pass   | 0.004s |
| Lines 506-508: success-logging block runs after a successful generation.                  | —                                                                                     | Pass   | 0.004s |
| reports.py lines 87-88: gaugehd file read error -i warning, continues.                    | <code>r.status_code == 200</code>                                                     | Pass   | 0.029s |
| reports.py lines 135-140: general Exception -i 500.                                       | <code>r.status_code == 500; "disk full" in r.get_json()["message"]</code>             | Pass   | 0.027s |
| reports.py lines 100-101: hazard file read error -i warning, continues.                   | <code>r.status_code == 200</code>                                                     | Pass   | 0.030s |
| reports.py lines 128-133: ImportError -i 500.                                             | <code>r.status_code == 500; "not available" in r.get_json()["message"].lower()</code> | Pass   | 0.032s |
| reports.py lines 70-72: storm_responses loaded -i time-series_data set.                   | <code>r.status_code == 200</code>                                                     | Pass   | 0.032s |
| reports.py lines 73-74: timeseries load failure -i warning, continues.                    | <code>r.status_code == 200</code>                                                     | Pass   | 0.028s |
| Field is dict[overall risk rating] [field='overall_risk_rating']                          | <code>field in m; isinstance(m[field], dict)</code>                                   | Pass   | 0.000s |
| Field is dict[coverage] [field='test_coverage']                                           | <code>field in m; isinstance(m[field], dict)</code>                                   | Pass   | 0.000s |
| Field is list[assumptions] [field='assumptions']                                          | <code>field in m; isinstance(m[field], list)</code>                                   | Pass   | 0.000s |
| Field is list[downstream models] [field='downstream_models']                              | <code>field in m; isinstance(m[field], list)</code>                                   | Pass   | 0.000s |
| Field is list[limitations] [field='limitations']                                          | <code>field in m; isinstance(m[field], list)</code>                                   | Pass   | 0.000s |
| Field is list[upstream models] [field='upstream_models']                                  | <code>field in m; isinstance(m[field], list)</code>                                   | Pass   | 0.000s |
| Field is string[category] [field='category']                                              | <code>field in m; isinstance(m[field], str)</code><br>(+1 more)                       | Pass   | 0.000s |

| Test Description                                                  | Acceptance Criteria                                     | Result | Time   |
|-------------------------------------------------------------------|---------------------------------------------------------|--------|--------|
| Field is string[description] [field='description']                | field in m; isinstance(m[field], str)<br>(+1 more)      | Pass   | 0.000s |
| Field is string[name] [field='name']                              | field in m; isinstance(m[field], str)<br>(+1 more)      | Pass   | 0.000s |
| Field is string[owner] [field='owner']                            | field in m; isinstance(m[field], str)<br>(+1 more)      | Pass   | 0.000s |
| Field is string[status] [field='status']                          | field in m; isinstance(m[field], str)<br>(+1 more)      | Pass   | 0.000s |
| Field is string[version] [field='version']                        | field in m; isinstance(m[field], str)<br>(+1 more)      | Pass   | 0.000s |
| Case insensitive                                                  | resolve_model_id("GEV") ==<br>"MKM-GH-001"              | Pass   | 0.000s |
| Depth damage maps to dd                                           | resolve_model_id("depth_damage") ==<br>"MKM-DD-001"     | Pass   | 0.000s |
| Distribution maps to si                                           | resolve_model_id("distribution") ==<br>"MKM-SI-001"     | Pass   | 0.000s |
| Mkm prefix returned as is                                         | resolve_model_id("MKM-GH-001") ==<br>"MKM-GH-001"       | Pass   | 0.003s |
| Shorthand gev resolved                                            | resolve_model_id("gev") ==<br>"MKM-GH-001"              | Pass   | 0.000s |
| Shorthand intensity resolved                                      | resolve_model_id("intensity") ==<br>"MKM-SI-001"        | Pass   | 0.000s |
| Shorthand mortgage resolved                                       | resolve_model_id("mortgage") ==<br>"MKM-MP-001"         | Pass   | 0.000s |
| Shorthand prs resolved                                            | resolve_model_id("prs") ==<br>"MKM-PR-001"              | Pass   | 0.000s |
| Spatial maps to sp                                                | resolve_model_id("spatial") ==<br>"MKM-SP-001"          | Pass   | 0.000s |
| Unknown shorthand returned as is                                  | resolve_model_id("unknown_model") ==<br>"unknown_model" | Pass   | 0.000s |
| If all prerequisites are fresh, returns empty list.               | result == []                                            | Pass   | 0.000s |
| Upstream never recorded but outputs exist → skip it.              | "gauges" not in result                                  | Pass   | 0.000s |
| Upstream never recorded and outputs missing → needs to run.       | "gauges" in result                                      | Pass   | 0.000s |
| Multiple targets: prerequisites are union of both.                | "gauges" in result; "counterparties" in result          | Pass   | 0.000s |
| Without data_dir, missing manifest entries → prerequisite needed. | "gauges" in result                                      | Pass   | 0.000s |

| Test Description                                              | Acceptance Criteria                                                                                                                                | Result | Time   |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Root steps (gauges) have no prerequisites.                    | <code>result == []</code>                                                                                                                          | Pass   | 0.000s |
| Upstream recorded but inputs stale → needs to run.            | <code>"properties" in result</code>                                                                                                                | Pass   | 0.000s |
| Target steps themselves should never appear in prerequisites. | <code>"propertyts" not in result</code>                                                                                                            | Pass   | 0.000s |
| Prerequisites are returned in valid topological order.        | <code>result.index("gauges") &lt; result.index("properties");</code><br><code>result.index("gauges") &lt; result.index("gaugehd") (+1 more)</code> | Pass   | 0.000s |
| Deep chain: if gauges missing, all downstream prereqs needed. | <code>"gauges" in result;</code> <code>"properties" in result (+3 more)</code>                                                                     | Pass   | 0.000s |
| Unknown step returns empty list.                              | <code>result == []</code>                                                                                                                          | Pass   | 0.000s |
| Retention mentions reproducibility                            | <code>'Principle 5/6' in texts;</code><br><code>'Reproducibility' in texts</code>                                                                  | Pass   | 0.000s |
| Returns dict with total storms                                | <code>isinstance(result, dict);</code><br><code>"total_storms" in result (+1 more)</code>                                                          | Pass   | 0.002s |
| Returns elapsed s                                             | <code>"elapsed_s" in result;</code><br><code>result["elapsed_s"] &gt;= 0</code>                                                                    | Pass   | 0.001s |
| revalue_all(trades, None) loads market state internally.      | <code>len(enriched) == 1;</code><br><code>enriched[0]['swap_id'] == 'PRS-TEST001' (+1 more)</code>                                                 | Pass   | 0.002s |
| Combined mode with mortgage                                   | <code>isinstance(result, list) and len(result) &gt; 0</code>                                                                                       | Pass   | 0.001s |
| Empty property does not crash                                 | <code>isinstance(self._page().generate_elements(list), list)</code>                                                                                | Pass   | 0.000s |
| Has risk header                                               | <code>any("Risk" in t for t in texts)</code>                                                                                                       | Pass   | 0.000s |
| Has table                                                     | <code>any(isinstance(e, Table) for e in result)</code>                                                                                             | Pass   | 0.000s |
| Property only mode                                            | <code>isinstance(result, list) and len(result) &gt; 0</code>                                                                                       | Pass   | 0.000s |
| Returns list                                                  | <code>isinstance(self._page().generate_elements(list), list)</code>                                                                                | Pass   | 0.000s |
| Canal distance shown                                          | <code>isinstance(result, list)</code>                                                                                                              | Pass   | 0.000s |
| Empty property does not crash                                 | <code>isinstance(self._page().generate_elements(list), list)</code>                                                                                | Pass   | 0.000s |
| Flood risk type shown                                         | <code>isinstance(result, list)</code>                                                                                                              | Pass   | 0.000s |
| Has risk header                                               | <code>any("Risk" in t for t in texts)</code>                                                                                                       | Pass   | 0.000s |
| Has table                                                     | <code>any(isinstance(e, Table) for e in result)</code>                                                                                             | Pass   | 0.000s |
| Lake distance shown                                           | <code>isinstance(result, list)</code>                                                                                                              | Pass   | 0.000s |
| Last flood date shown                                         | <code>isinstance(result, list)</code>                                                                                                              | Pass   | 0.000s |

| Test Description                                                  | Acceptance Criteria                                                            | Result | Time   |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------|--------|--------|
| No defence scheme                                                 | <code>isinstance(result, list)</code>                                          | Pass   | 0.000s |
| No ea zone                                                        | <code>isinstance(result, list)</code>                                          | Pass   | 0.000s |
| No last flood date                                                | <code>isinstance(result, list)</code>                                          | Pass   | 0.000s |
| Returns list                                                      | <code>isinstance(self._page().generate_elements(Lines, property), list)</code> | Pass   | 0.001s |
| Line 257: non-numeric → delegates to <code>_format_value</code> . | <code>page._format.count(None) == "Not specified"</code>                       | Pass   | 0.000s |
| Line 256: numeric → comma-formatted.                              | <code>"1,234,567" in result</code>                                             | Pass   | 0.000s |
| Line 239: non-numeric → delegates to <code>_format_value</code> . | <code>result == "Not specified"</code>                                         | Pass   | 0.000s |
| Line 238: numeric → £ formatted.                                  | <code>"£" in result; "1,234,567" in result</code>                              | Pass   | 0.000s |
| Line 251: non-numeric → delegates to <code>_format_value</code> . | <code>page._format.depth("N/A") == "N/A"</code>                                | Pass   | 0.000s |
| Line 250: numeric → formatted with 'm'.                           | <code>"1.50m" in result</code>                                                 | Pass   | 0.000s |
| Lines 204-215: camelCase → 'Camel Case'.                          | <code>"Risk" in result</code>                                                  | Pass   | 0.000s |
| Lines 207-213: abbreviation replacements applied.                 | <code>"ID" in result</code>                                                    | Pass   | 0.000s |
| Line 245: non-numeric → delegates to <code>_format_value</code> . | <code>page._format.percentage(None) == "Not specified"</code>                  | Pass   | 0.000s |
| Line 244: numeric → formatted with %.                             | <code>"%" in result; "12.3" in result</code>                                   | Pass   | 0.000s |
| Line 222: False → 'No'.                                           | <code>page._format.value(False) == "No"</code>                                 | Pass   | 0.000s |
| Line 222: True → 'Yes'.                                           | <code>page._format.value(True) == "Yes"</code>                                 | Pass   | 0.000s |
| Lines 230-231: blank string → 'Not specified'.                    | <code>page._format.value(" ") == "Not specified"</code>                        | Pass   | 0.000s |
| Lines 226-227: float with fractional part → 2dp string.           | <code>result == "3.14"</code>                                                  | Pass   | 0.000s |
| Lines 224-225: float with no fractional part → int string.        | <code>page._format.value(5.0) == "5"</code>                                    | Pass   | 0.000s |
| Lines 228-229: integer → plain string.                            | <code>page._format.value(42) == "42"</code>                                    | Pass   | 0.000s |
| Lines 232-233: non-empty string → passed through.                 | <code>page._format.value("hello") == "hello"</code>                            | Pass   | 0.000s |
| Line 220: None → 'Not specified'.                                 | <code>page._format.value(None) == "Not specified"</code>                       | Pass   | 0.000s |
| High risk                                                         | <code>classify_risk_level(3.0) == "HIGH"</code>                                | Pass   | 0.000s |
| Low risk                                                          | <code>classify_risk_level(0.3) == "LOW"</code>                                 | Pass   | 0.000s |
| Medium risk                                                       | <code>classify_risk_level(1.0) == "MEDIUM"</code>                              | Pass   | 0.000s |
| Minimal risk                                                      | <code>classify_risk_level(0.0) == "MINIMAL"</code>                             | Pass   | 0.000s |



| Test Description                                                                              | Acceptance Criteria                                                                                         | Result | Time   |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|--------|
| Get risk rating calculation                                                                   | <code>response.status_code == 200;</code><br><code>data["status"] == "success"</code> (+5 more)             | Pass   | 0.002s |
| Risk rating clear override                                                                    | <code>response.status_code == 200;</code><br><code>rr["mrc_override"] is None</code> (+1 more)              | Pass   | 0.007s |
| Risk rating override                                                                          | <code>response.status_code == 200;</code><br><code>rr["mrc_override"] == "Unacceptable"</code><br>(+3 more) | Pass   | 0.003s |
| Risk rating override invalid rating                                                           | <code>response.status_code == 400</code>                                                                    | Pass   | 0.002s |
| Risk rating override requires reason                                                          | <code>response.status_code == 400</code>                                                                    | Pass   | 0.002s |
| Line 210: inventory not found → 404.                                                          | <code>r.status_code == 404</code>                                                                           | Pass   | 0.001s |
| Lines 126-134: <code>_auto_select_pages</code> returns known pages.                           | <code>isinstance(pages, list); len(pages) &gt; 0</code>                                                     | Pass   | 0.001s |
| Empty gauge data                                                                              | <code>RiskReportGenerator(output_dir).generate_basic_report()</code>                                        | Pass   | 0.015s |
| Lines 162-164: page raises → continue to next.                                                | <code>isinstance(elements, list)</code>                                                                     | Pass   | 0.001s |
| Lines 148-149: unknown page → warning, skipped.                                               | <code>isinstance(elements, list)</code>                                                                     | Pass   | 0.001s |
| Line 265: <code>get_page_categories</code> .                                                  | <code>isinstance(cats, dict)</code>                                                                         | Pass   | 0.001s |
| High risk portfolio report                                                                    | <code>RiskReportGenerator(output_dir).generate_detailed_report()</code>                                     | Pass   | 0.022s |
| Line 261: <code>list_available_pages</code> .                                                 | <code>isinstance(pages, list); len(pages) &gt; 0</code>                                                     | Pass   | 0.001s |
| Minimal data report                                                                           | <code>RiskReportGenerator(output_dir).generate_basic_report()</code>                                        | Pass   | 0.014s |
| Missing mortgage data                                                                         | <code>RiskReportGenerator(output_dir).generate_detailed_report()</code>                                     | Pass   | 0.021s |
| Lines 60-62: <code>output_dir=None</code> → uses <code>config.get_risk_reports_dir()</code> . | <code>gen.output_dir == risk_dir</code>                                                                     | Pass   | 0.000s |
| Lines 270-336: <code>validate_pages</code> .                                                  | <code>"title" in valid;</code> <code>"nonexistent" in invalid</code>                                        | Pass   | 0.001s |
| Analysis report generation                                                                    | <code>report_path.exists();</code><br><code>report_path.stat().st_size &gt; 0</code>                        | Pass   | 0.017s |
| Basic report generation                                                                       | <code>report_path.exists();</code><br><code>report_path.suffix == ".pdf"</code> (+1 more)                   | Pass   | 0.042s |
| Custom filename                                                                               | <code>report_path.name == custom_name;</code><br><code>report_path.exists()</code>                          | Pass   | 0.016s |
| Custom page selection                                                                         | <code>report_path.exists()</code>                                                                           | Pass   | 0.012s |
| Detailed report generation                                                                    | <code>report_path.exists();</code><br><code>report_path.stat().st_size &gt; 0</code>                        | Pass   | 0.023s |

| Test Description                                                              | Acceptance Criteria                                                                                                                                 | Result | Time   |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Summary report generation                                                     | <code>report_path.exists();</code>                                                                                                                  | Pass   | 0.010s |
| Categories dict initialized                                                   | <code>report_path.stat().st_size &gt; 0</code><br>"overview" in <code>gen.categories;</code><br>"analysis" in <code>gen.categories</code> (+1 more) | Pass   | 0.001s |
| Explicit output dir path                                                      | <code>isinstance(gen.output_dir, Path)</code>                                                                                                       | Pass   | 0.001s |
| Explicit output dir string                                                    | <code>gen.output_dir == tmp_path</code>                                                                                                             | Pass   | 0.001s |
| output_dir=None → falls back to <code>config.get_reports_dir('risk')</code> . | <code>gen.output_dir == risk_dir</code>                                                                                                             | Pass   | 0.001s |
| Output dir created                                                            | <code>new_dir.exists()</code>                                                                                                                       | Pass   | 0.001s |
| Pages dict initialized                                                        | <code>expected &lt;= set(gen.pages.keys())</code>                                                                                                   | Pass   | 0.001s |
| Get page categories                                                           | <code>isinstance(categories, dict); key in categories</code>                                                                                        | Pass   | 0.001s |
| List available pages                                                          | <code>isinstance(pages, list); key in pages</code>                                                                                                  | Pass   | 0.001s |
| Validate pages invalid                                                        | "title" in <code>valid;</code> "nonexistent-page" in <code>invalid</code>                                                                           | Pass   | 0.001s |
| Validate pages valid                                                          | <code>len(valid) == 2; len(invalid) == 0</code>                                                                                                     | Pass   | 0.001s |
| Complete workflow all types                                                   | <code>p.exists(); p.stat().st_size &gt; 0</code>                                                                                                    | Pass   | 0.062s |
| Complete workflow basic                                                       | <code>report_path.exists();</code><br><code>report_path.suffix == ".pdf"</code> (+1 more)                                                           | Pass   | 0.016s |
| High risk label                                                               | "High Risk" in <code>str(elements)</code>                                                                                                           | Pass   | 0.000s |
| Low risk label                                                                | "Low Risk" in <code>str(elements)</code>                                                                                                            | Pass   | 0.000s |
| Moderate risk label                                                           | "Moderate Risk" in <code>str(elements)</code>                                                                                                       | Pass   | 0.000s |
| Non numeric score formatted                                                   | <code>isinstance(elements, list)</code>                                                                                                             | Pass   | 0.000s |
| Very high risk label                                                          | "Very High Risk" in <code>str(elements)</code>                                                                                                      | Pass   | 0.000s |
| Clay soil adds 1                                                              | <code>result["score"] &gt;= 1</code>                                                                                                                | Pass   | 0.000s |
| Close river adds 1                                                            | <code>result["score"] &gt;= 1</code>                                                                                                                | Pass   | 0.000s |
| High flood risk adds 3                                                        | <code>result["score"] &gt;= 3</code>                                                                                                                | Pass   | 0.000s |
| High score level                                                              | <code>result["level"] == "High Risk"</code>                                                                                                         | Pass   | 0.000s |
| Low elevation adds 2                                                          | <code>result["score"] &gt;= 2</code>                                                                                                                | Pass   | 0.000s |
| Low flood risk adds 1                                                         | <code>result["score"] &gt;= 1</code>                                                                                                                | Pass   | 0.000s |
| Low medium score level                                                        | <code>result["level"] in ("Medium Risk", "Low-Medium Risk")</code>                                                                                  | Pass   | 0.000s |
| Low risk level                                                                | <code>result["level"] == "Low Risk"</code>                                                                                                          | Pass   | 0.000s |
| Medium flood risk adds 2                                                      | <code>result["score"] &gt;= 2</code>                                                                                                                | Pass   | 0.000s |
| Medium score level                                                            | "Medium" in <code>result["level"]</code> or <code>result["score"] &gt;= 4</code>                                                                    | Pass   | 0.000s |
| Moderate elevation adds 1                                                     | <code>result["score"] &gt;= 1</code>                                                                                                                | Pass   | 0.000s |
| No risk factors message                                                       | "No major" in <code>result["primary_factors"]</code>                                                                                                | Pass   | 0.000s |

| Test Description                                                                                                                            | Acceptance Criteria                                                                                                   | Result | Time   |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------|--------|
| Peat soil adds 2                                                                                                                            | <code>result["score"] &gt;= 2</code>                                                                                  | Pass   | 0.000s |
| Returns all keys                                                                                                                            | <code>k in result</code>                                                                                              | Pass   | 0.000s |
| Very close river adds 2                                                                                                                     | <code>result["score"] &gt;= 2</code>                                                                                  | Pass   | 0.000s |
| Very high flood risk adds 4                                                                                                                 | <code>result["score"] &gt;= 4</code>                                                                                  | Pass   | 0.000s |
| Very high score level                                                                                                                       | <code>result["level"] == "Very High Risk"</code>                                                                      | Pass   | 0.000s |
| Very low elevation adds 3                                                                                                                   | <code>result["score"] &gt;= 3</code>                                                                                  | Pass   | 0.000s |
| Close 100 to 500m                                                                                                                           | <code>isinstance(result, list)</code>                                                                                 | Pass   | 0.000s |
| Good distance over 1000m                                                                                                                    | <code>isinstance(result, list)</code>                                                                                 | Pass   | 0.000s |
| Moderate 500 to 1000m                                                                                                                       | <code>isinstance(result, list)</code>                                                                                 | Pass   | 0.000s |
| Very close under 100m                                                                                                                       | <code>isinstance(result, list)</code>                                                                                 | Pass   | 0.000s |
| <code>_run_and_save</code> should append startup header lines to the output buffer.                                                         | <code>any("=" * 10 in ln for ln in lines);</code><br><code>any("Audit run complete" in ln for ln in lines)</code>     | Pass   | 0.002s |
| When an exception is raised, an ERROR line should appear in output.                                                                         | <code>any("ERROR" in ln for ln in lines)</code>                                                                       | Pass   | 0.002s |
| <code>_run_and_save</code> should catch exceptions and still set <code>_running=False</code> .                                              | <code>tr_mod._running is False</code>                                                                                 | Pass   | 0.002s |
| <code>_run_and_save</code> should set <code>_running=False</code> in the finally block.                                                     | <code>tr_mod._running is False</code>                                                                                 | Pass   | 0.002s |
| Setting <code>_batch_job</code> to None mid-run stops processing.                                                                           | <code>call_count == 1; cl_mod._batch_job is None</code>                                                               | Pass   | 0.000s |
| Failed gauge training is recorded as 'failed' in results.                                                                                   | <code>cl_mod._batch_job["status"] == "complete"; results[0]["status"] == "trained" (+2 more)</code>                   | Pass   | 0.000s |
| Batch trains gauges and records timings.                                                                                                    | <code>cl_mod._batch_job["status"] == "complete";</code><br><code>cl_mod._batch_job["completed"] == 2 (+5 more)</code> | Pass   | 0.001s |
| Each gauge has <code>gauge_id</code> , <code>gauge_name</code> , <code>p_flood</code> , <code>p_flood_pct</code> , <code>threshold</code> , | <code>'gauge_id' in g; 'gauge_name' in g (+8 more)</code>                                                             | Pass   | 0.039s |
| Response has <code>gauges_severe</code> as list of <code>gauge_ids</code> ; for STORM_PORT_SEVERE                                           | <code>'gauges_severe' in data;</code><br><code>isinstance(data['gauges_severe'], list) (+2 more)</code>               | Pass   | 0.027s |
| <code>gauges_warning</code> list includes GAUGE-002.                                                                                        | <code>'gauges_warning' in data; 'GAUGE-002' in data['gauges_warning']</code>                                          | Pass   | 0.032s |
| Response has 'gauges' list.                                                                                                                 | <code>'gauges' in data;</code><br><code>isinstance(data['gauges'], list)</code>                                       | Pass   | 0.033s |
| Response has <code>portfolio_stress_pnl</code> , <code>portfolio_mtm</code> , <code>num_gauges</code> .                                     | <code>'portfolio_stress_pnl' in data;</code><br><code>'portfolio_mtm' in data (+1 more)</code>                        | Pass   | 0.039s |

| Test Description                                               | Acceptance Criteria                                                                             | Result | Time   |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|--------|
| POST with non-existent storm_id returns 404.                   | <code>resp.status_code == 404</code>                                                            | Pass   | 0.008s |
| POST with no body or empty storm_id returns 400.               | <code>resp.status_code == 400</code>                                                            | Pass   | 0.000s |
| POST STORM-PORT-001 returns 200 with status=success.           | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>               | Pass   | 0.038s |
| Each hour has water_level, p_flood, portfolio fields.          | <code>'hour' in h; 'water_level' in h (+4 more)</code>                                          | Pass   | 0.032s |
| Non-existent storm returns 404.                                | <code>resp.status_code == 404</code>                                                            | Pass   | 0.002s |
| Severe storm triggers knock-out (P(flood)=100%).               | <code>data['hourly'][first_breach]['p_flood'] == 1.0; h['p_flood'] is None</code>               | Pass   | 0.032s |
| Missing gauge_id or storm_id returns 400.                      | <code>resp.status_code == 400</code>                                                            | Pass   | 0.001s |
| No JSON body returns error.                                    | <code>resp.status_code in (400, 500)</code>                                                     | Pass   | 0.001s |
| Response includes portfolio-level stress summary.              | <code>'num_trades' in summary;</code><br><code>'total_notional' in summary (+5 more)</code>     | Pass   | 0.026s |
| Response includes probability surface grid.                    | <code>'probability_surface' in data;</code><br><code>'water_levels' in surface (+7 more)</code> | Pass   | 0.025s |
| POST returns 60-hour forecast array.                           | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success' (+1 more)</code>     | Pass   | 0.031s |
| Response includes per-trade summary.                           | <code>'trades' in data; len(data['trades']) == 2 (+4 more)</code>                               | Pass   | 0.029s |
| Surface water levels cap at severe – no rows above severe.     | <code>max(surface['water_levels']) &lt;= severe</code>                                          | Pass   | 0.027s |
| Surface must cover full 168h storm horizon regardless of KO.   | <code>max(surface['hours']) &gt;= 164</code>                                                    | Pass   | 0.024s |
| Empty gauge_id returns 400.                                    | <code>resp.status_code == 400</code>                                                            | Pass   | 0.000s |
| No open trades at gauge returns 404.                           | <code>resp.status_code == 404</code>                                                            | Pass   | 0.005s |
| Null JSON body returns 400.                                    | <code>resp.status_code == 400</code>                                                            | Pass   | 0.000s |
| Predictor None (no model dir) returns 404.                     | <code>resp.status_code == 404</code>                                                            | Pass   | 0.032s |
| Gaugets file present uses hydrograph built from real readings. | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>               | Pass   | 0.040s |
| Already running message                                        | <code>data["status"] == "running";</code><br><code>"message" in data (+1 more)</code>           | Pass   | 0.000s |
| Already running returns 202                                    | <code>resp.status_code == 202</code>                                                            | Pass   | 0.000s |
| Response has started status                                    | <code>data["status"] == "started"</code>                                                        | Pass   | 0.001s |
| Starts run returns 202                                         | <code>resp.status_code == 202</code>                                                            | Pass   | 0.001s |

| Test Description                                                                          | Acceptance Criteria                                                                                         | Result | Time   |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|--------|
| Empty running pnl returns drawing                                                         | <code>isinstance(result, Drawing)</code>                                                                    | Pass   | 0.000s |
| All positive                                                                              | <code>np.all(M &gt; 0.0)</code>                                                                             | Pass   | 0.000s |
| Empirical pairwise correlation approximately matches model correlation for large samples. | <code>abs(emp_corr[i, j] - model_corr[i, j]) &lt; 0.15</code>                                               | Pass   | 0.006s |
| Higher intensity -i longer range -i more spatially coherent (lower variance).             | <code>np.mean(vars_extreme) &lt; np.mean(vars_moderate)</code>                                              | Pass   | 0.008s |
| Normalisation ensures catchment mean is preserved.                                        | <code>M.mean() == pytest.approx(1.0, abs=1e-10)</code>                                                      | Pass   | 0.000s |
| Pre-normalisation, $E[M] = \exp(\sigma^2/2) / \exp(\sigma^2/2) = 1$ theoretically.        | <code>np.mean(pre_norms) == pytest.approx(1.0, abs=0.05)</code>                                             | Pass   | 0.002s |
| Reproducible with seed                                                                    | <code>np.allclose(M1, M2)</code>                                                                            | Pass   | 0.000s |
| Shape                                                                                     | <code>M.shape == (small_model.n_gauges,)</code>                                                             | Pass   | 0.000s |
| When rng is None, falls back to <code>np.random.randn</code> (line 251).                  | <code>M.shape == (model.n_gauges,); M.mean() == pytest.approx(1.0, abs=1e-10) (+1 more)</code>              | Pass   | 0.000s |
| Catastrophic weights: 15% doublet, 45% cluster, 40% persistent.                           | <code>doublets &lt; 150</code>                                                                              | Pass   | 0.003s |
| Defined categories[catastrophic] [cat='catastrophic']                                     | <code>result in [SequenceType.DOUBLET, SequenceType.CLUSTER, Se...</code>                                   | Pass   | 0.000s |
| Defined categories[extreme] [cat='extreme']                                               | <code>result in [SequenceType.DOUBLET, SequenceType.CLUSTER, Se...</code>                                   | Pass   | 0.000s |
| Defined categories[moderate] [cat='moderate']                                             | <code>result in [SequenceType.DOUBLET, SequenceType.CLUSTER, Se...</code>                                   | Pass   | 0.000s |
| Defined categories[severe] [cat='severe']                                                 | <code>result in [SequenceType.DOUBLET, SequenceType.CLUSTER, Se...</code>                                   | Pass   | 0.000s |
| Returns sequence type                                                                     | <code>isinstance(result, SequenceType); result in [SequenceType.DOUBLET, SequenceType.CLUSTER, Se...</code> | Pass   | 0.000s |
| Unknown category uses DEFAULT_TYPE_WEIGHTS.                                               | <code>isinstance(result, SequenceType)</code>                                                               | Pass   | 0.000s |
| Lines 122-123: 30% chance subsequent storm goes down.                                     | <code>below_base &gt; 10</code>                                                                             | Pass   | 0.000s |
| Storm index 0 has downward variation (up to -20%).                                        | <code>all(r &lt;= base + 0.001 for r in results)</code>                                                     | Pass   | 0.001s |
| First storm returns positive                                                              | <code>result &gt; 0</code>                                                                                  | Pass   | 0.000s |
| Result should be within +/-20% of base for first storm.                                   | <code>all(r &gt;= lower - 0.001 for r in results); all(r &lt;= upper + 0.001 for r in results)</code>       | Pass   | 0.001s |

| Test Description                                                 | Acceptance Criteria                                                                               | Result | Time   |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------|--------|
| 70% chance of maintaining/increasing subsequent storm intensity. | <code>above_base &gt; 100</code>                                                                  | Pass   | 0.000s |
| Subsequent storm positive                                        | <code>result &gt; 0</code>                                                                        | Pass   | 0.000s |
| <code>max(0.1, ...)</code> prevents zero or negative intensity.  | <code>result &gt;= 0.1</code>                                                                     | Pass   | 0.000s |
| Bool passed through                                              | <code>result["flag"] is True</code>                                                               | Pass   | 0.000s |
| Empty params                                                     | <code>result == {}</code>                                                                         | Pass   | 0.000s |
| Float passed through                                             | <code>result["rate"] == 0.05</code>                                                               | Pass   | 0.000s |
| Int passed through                                               | <code>result["n"] == 42</code>                                                                    | Pass   | 0.000s |
| Long list summarised                                             | <code>isinstance(result["items"], str); "15 items" in result["items"]</code>                      | Pass   | 0.000s |
| None passed through                                              | <code>result["x"] is None</code>                                                                  | Pass   | 0.000s |
| Object converted to string                                       | <code>result["obj"] == "my-object"</code>                                                         | Pass   | 0.000s |
| Short list converted to str                                      | <code>isinstance(result["items"], list); len(result["items"]) == 3</code>                         | Pass   | 0.000s |
| String passed through                                            | <code>result["name"] == "test"</code>                                                             | Pass   | 0.000s |
| Tuple treated like list                                          | <code>isinstance(result["t"], list)</code>                                                        | Pass   | 0.000s |
| Saves and returns response                                       | <code>resp is not None</code>                                                                     | Pass   | 0.028s |
| Returns false on oerror                                          | <code>result is False; _save_audit_log([]) is False (+1 more)</code>                              | Pass   | 0.000s |
| Saves entries                                                    | <code>result is True; json.loads(p.read_text())[0]["event"] == "created"</code>                   | Pass   | 0.001s |
| Returns false on oerror                                          | <code>_save_bcbs239({}) is False; _save_raci({}) is False (+2 more)</code>                        | Pass   | 0.000s |
| Saves data                                                       | <code>_save_bcbs239({"principles": []}) is True; _save_raci({"roles": []}) is True</code>         | Pass   | 0.000s |
| Returns false on oerror                                          | <code>_save_bcbs239({}) is False; _save_raci({}) is False (+2 more)</code>                        | Pass   | 0.000s |
| Saves bibliography                                               | <code>_save_bibliography({"references": [], "categories": []}) ...</code>                         | Pass   | 0.000s |
| Save error is reraised                                           | —                                                                                                 | Pass   | 0.001s |
| Creates gauge file                                               | <code>(tmp_path / "gaugets" / "GAUGE-001.json").exists()</code>                                   | Pass   | 0.000s |
| If gauge file already exists, merge storm_responses into it.     | <code>"flood_simulation" in data; "storm_responses" in data</code>                                | Pass   | 0.002s |
| Multiple gauges                                                  | <code>(gaugets_dir / "GAUGE-001.json").exists(); (gaugets_dir / "GAUGE-002.json").exists()</code> | Pass   | 0.000s |

| Test Description                    | Acceptance Criteria                                                                       | Result | Time   |
|-------------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| Saved file has responses            | "storm_responses" in data;<br><br>len(data["storm_responses"]["responses"])<br><br>== 3   | Pass   | 0.001s |
| Returns false on oserror            | _save_bcbs239({}) is False;<br><br>_save_raci({}) is False (+2 more)                      | Pass   | 0.000s |
| Saves documents                     | _save_gov_documents([{"doc_id":<br>"D1"}]) is True                                        | Pass   | 0.000s |
| Creates file                        | output_path.exists();      result ==<br><br>output_path                                   | Pass   | 0.000s |
| Creates parent dirs                 | output_path.exists()                                                                      | Pass   | 0.000s |
| File has hazard curves              | "GAUGE-001" in data["hazard_curves"]                                                      | Pass   | 0.002s |
| File has metadata                   | "metadata" in data;<br><br>data["metadata"]["catchment_id"]<br><br>== "thames"            | Pass   | 0.002s |
| Metadata merged                     | data["metadata"]["num_storms"] == 100                                                     | Pass   | 0.002s |
| Returns false on oserror            | result is False;    _save_audit_log()<br><br>is False (+1 more)                           | Pass   | 0.000s |
| Saves data                          | result is True;<br><br>json.loads(p.read_text())["models"]<br><br>== []                   | Pass   | 0.001s |
| Save cleans up on failure           | len(list(tmp_path.glob("*.tmp"))) ==<br><br>0                                             | Pass   | 0.001s |
| Save creates file                   | result == data                                                                            | Pass   | 0.001s |
| Save overwrites existing            | result == new_data                                                                        | Pass   | 0.001s |
| Returns false on oserror            | result is False;    _save_audit_log()<br><br>is False (+1 more)                           | Pass   | 0.002s |
| Saves meetings                      | result is True;<br><br>json.loads(p.read_text())[0]["id"] ==<br>"M1"                      | Pass   | 0.003s |
| Returns false on oserror            | _save_bcbs239({}) is False;<br><br>_save_raci({}) is False (+2 more)                      | Pass   | 0.000s |
| Saves data                          | _save_bcbs239({"principles": []}) is<br>True;    _save_raci({"roles": []}) is<br><br>True | Pass   | 0.000s |
| Above max depth returns one         | scalar_depth_damage(100.0) ==<br><br>pytest.approx(1.0)                                   | Pass   | 0.000s |
| Between control points interpolated | 0.25 < d < 0.40                                                                           | Pass   | 0.000s |
| Damage between 0 and 1              | 0 <= scalar_depth_damage(depth) <=<br><br>1.0                                             | Pass   | 0.000s |
| Half meter damage                   | d == pytest.approx(DAMAGE_POINTS[2],<br><br>abs=1e-6)                                     | Pass   | 0.000s |



| Test Description                                                         | Acceptance Criteria                                                                               | Result | Time   |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------|--------|
| Max depth returns one                                                    | <code>scalar_depth_damage(DEPTH_POINTS[-1]) == pytest.approx(1.0)</code>                          | Pass   | 0.000s |
| Monotonically increasing                                                 | <code>all(damages[i] &lt;= damages[i + 1] for i in range(len(damag...</code>                      | Pass   | 0.000s |
| Negative depth returns zero                                              | <code>scalar_depth_damage(-1.0) == 0.0;</code><br><code>scalar_depth_damage(-100.0) == 0.0</code> | Pass   | 0.000s |
| One meter damage                                                         | <code>d == pytest.approx(DAMAGE_POINTS[3], abs=1e-6)</code>                                       | Pass   | 0.000s |
| Returns float                                                            | <code>isinstance(scalar_depth_damage(0.75), float)</code>                                         | Pass   | 0.000s |
| Small depth returns small damage                                         | <code>d == pytest.approx(DAMAGE_POINTS[1], abs=1e-9)</code>                                       | Pass   | 0.000s |
| Zero depth returns zero                                                  | <code>scalar_depth_damage(0.0) == 0.0</code>                                                      | Pass   | 0.000s |
| Every control point must return a value in [0, 1].                       | <code>0.0 &lt;= val &lt;= 1.0</code>                                                              | Pass   | 0.000s |
| Very small positive depth (between 0 and first control point).           | <code>0.0 &lt; d &lt; 0.05</code>                                                                 | Pass   | 0.000s |
| 'base_level_m' field — used in Stress tab storm info bar.                | <code>'base_level_m' in s</code>                                                                  | Pass   | 0.004s |
| 'duration_hours' field — used in Stress tab storm info bar.              | <code>'duration_hours' in s</code>                                                                | Pass   | 0.004s |
| 'effective_precipitation_mm' field — shown in storm row.                 | <code>'effective_precipitation_mm' in s</code>                                                    | Pass   | 0.003s |
| 'gauges_severe' field — severity indicator in storm row.                 | <code>'gauges_severe' in s</code>                                                                 | Pass   | 0.004s |
| 'intensity_category' field — colour-coded in storm row.                  | <code>'intensity_category' in s</code>                                                            | Pass   | 0.004s |
| 'level_change_m' field — used in Stress tab storm info bar.              | <code>'level_change_m' in s</code>                                                                | Pass   | 0.004s |
| 'max_trigger' field — used for trigger colour coding in Historical list. | <code>'max_trigger' in s</code>                                                                   | Pass   | 0.004s |
| 'name' field — shown in storm row title.                                 | <code>'name' in s</code>                                                                          | Pass   | 0.003s |
| 'peak_level_m' field — primary sort key and shown in Trading Desk label. | <code>'peak_level_m' in s</code>                                                                  | Pass   | 0.004s |
| 'peak_position' field — used in hydrograph synthesis.                    | <code>'peak_position' in s</code>                                                                 | Pass   | 0.005s |
| Hour label element must be present to display current hour.              | <code>'sp-sim-hour-label' in sim.js</code>                                                        | Pass   | 0.000s |

| Test Description                                                | Acceptance Criteria                                                                                                                  | Result | Time   |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Scrubber element must have the expected DOM ID.                 | 'sp-sim-scrubber' in sim_js                                                                                                          | Pass   | 0.000s |
| scrubber.max must equal STORM_SIMULATION_HOURS - 1 (0-indexed). | f"scrubber.max = '{expected}'" in sim_js                                                                                             | Pass   | 0.000s |
| Regression guard: the old wrong value '59' must not be present. | "scrubber.max = '59'" not in sim_js                                                                                                  | Pass   | 0.000s |
| scrubber.min must be '0' so the timeline starts at hour 0.      | "scrubber.min = '0'" in sim_js                                                                                                       | Pass   | 0.000s |
| Scrubber element must be an input[type=range].                  | "scrubber.type = 'range'" in sim_js                                                                                                  | Pass   | 0.000s |
| Empty monthly means                                             | _seasonal_base_level({},<br>mean_level=1.0, month=6) == 1.0                                                                          | Pass   | 0.000s |
| Missing month falls back                                        | _seasonal_base_level(monthly,<br>mean_level=1.0, month=3) ==...                                                                      | Pass   | 0.000s |
| No monthly means                                                | _seasonal_base_level(None,<br>mean_level=1.0, month=6) == 1.0                                                                        | Pass   | 0.000s |
| With monthly means                                              | _seasonal_base_level(monthly,<br>mean_level=1.0, month=1) ==...;<br>_seasonal_base_level(monthly,<br>mean_level=1.0, month=7) ==...  | Pass   | 0.000s |
| Heading                                                         | _has_paragraph_containing(els,<br>'7. Property Flood Time Se...;<br>_has_paragraph_containing(els, '9.<br>Counterparties') (+2 more) | Pass   | 0.000s |
| Kv table                                                        | _has_table(els)                                                                                                                      | Pass   | 0.000s |
| Returns list                                                    | isinstance(els, list);<br>isinstance(els, list) (+2 more)                                                                            | Pass   | 0.000s |
| Cover healthy                                                   | len(story) > 0                                                                                                                       | Pass   | 0.000s |
| Cover issues                                                    | len(story) > 0                                                                                                                       | Pass   | 0.000s |
| Exec summary healthy                                            | len(story) > 0                                                                                                                       | Pass   | 0.000s |
| Exec summary issues                                             | len(story) > 0                                                                                                                       | Pass   | 0.001s |
| Quality metrics healthy                                         | len(story) > 0                                                                                                                       | Pass   | 0.000s |
| Quality metrics with stale details                              | len(story) > 0                                                                                                                       | Pass   | 0.001s |
| Reconciliation no results                                       | len(story) > 0                                                                                                                       | Pass   | 0.000s |
| Reconciliation with failures                                    | len(story) > 0                                                                                                                       | Pass   | 0.000s |
| Reconciliation with results                                     | len(story) > 0                                                                                                                       | Pass   | 0.000s |
| Remediation compliant                                           | len(story) > 0                                                                                                                       | Pass   | 0.000s |
| Remediation with issues                                         | len(story) > 2                                                                                                                       | Pass   | 0.001s |
| Retention policy                                                | len(story) > 0                                                                                                                       | Pass   | 0.000s |
| Source documentation                                            | len(story) > 0                                                                                                                       | Pass   | 0.001s |

| Test Description                                     | Acceptance Criteria                                                                                                                                      | Result | Time   |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Step with no parameters should still render.         | <code>len(story) &gt; 0</code>                                                                                                                           | Pass   | 0.000s |
| Topology                                             | <code>len(story) &gt; 0</code>                                                                                                                           | Pass   | 0.000s |
| Handles Counterparty key instead of CounterpartySet. | <code>.has_table(els)</code>                                                                                                                             | Pass   | 0.000s |
| Empty message                                        | <code>.has_paragraph_containing(els, 'No counterparties generat...</code>                                                                                | Pass   | 0.000s |
| Heading                                              | <code>.has_paragraph_containing(els, '7. Property Flood Time Se...;</code><br><code>.has_paragraph_containing(els, '9. Counterparties') (+2 more)</code> | Pass   | 0.000s |
| Returns list                                         | <code>isinstance(els, list);</code><br><code>isinstance(els, list) (+2 more)</code>                                                                      | Pass   | 0.000s |
| Table present                                        | <code>.has_table(els)</code>                                                                                                                             | Pass   | 0.000s |
| Heading                                              | <code>.has_paragraph_containing(els, '3. Mortgages');</code><br><code>.has_paragraph_containing(els, '4. Historical Gauge Data') (+2 more)</code>        | Pass   | 0.000s |
| No baselines message                                 | <code>.has_paragraph_containing(els, '5 gaugehd files')</code>                                                                                           | Pass   | 0.000s |
| Range computed                                       | <code>.has_paragraph_containing(els, 'Seasonal range')</code>                                                                                            | Pass   | 0.000s |
| Returns list                                         | <code>isinstance(els, list);</code><br><code>isinstance(els, list) (+4 more)</code>                                                                      | Pass   | 0.000s |
| Seasonal averages                                    | <code>.has_paragraph_containing(els, 'Avg winter baseline')</code>                                                                                       | Pass   | 0.000s |
| Table when baselines                                 | <code>.has_table(els)</code>                                                                                                                             | Pass   | 0.000s |
| Empty gauges message                                 | <code>.has_paragraph_containing(els, 'No gauges generated');</code> not<br><code>.has_table(els)</code>                                                  | Pass   | 0.000s |
| Multiple gauges                                      | <code>.has_paragraph_containing(els, '2 gauges')</code>                                                                                                  | Pass   | 0.000s |
| Returns list                                         | <code>isinstance(els, list);</code><br><code>isinstance(els, list) (+4 more)</code>                                                                      | Pass   | 0.000s |
| Section heading                                      | <code>.has_paragraph_containing(els, '1. Gauge Network')</code>                                                                                          | Pass   | 0.000s |
| Summary line counts                                  | <code>.has_paragraph_containing(els, '5 time series');</code><br><code>.has_paragraph_containing(els, '3 historical daily')</code>                       | Pass   | 0.000s |
| Table present                                        | <code>.has_table(els); .has_table(els)</code>                                                                                                            | Pass   | 0.000s |

| Test Description                                             | Acceptance Criteria                                                                                                                                    | Result | Time   |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Handles exceedance_probabilities + return_levels format.     | <code>has_table(els)</code>                                                                                                                            | Pass   | 0.000s |
| Non-dict gauge curve entry should produce dashes, not crash. | <code>has_table(els)</code>                                                                                                                            | Pass   | 0.000s |
| Heading                                                      | <code>has_paragraph_containing(els, '3. Mortgages');</code><br><code>has_paragraph_containing(els, '4. Historical Gauge Data') (+2 more)</code>        | Pass   | 0.000s |
| No curves message                                            | <code>has_paragraph_containing(els, 'No hazard curves available')</code>                                                                               | Pass   | 0.000s |
| Returns list                                                 | <code>isinstance(els, list);</code><br><code>isinstance(els, list) (+4 more)</code>                                                                    | Pass   | 0.000s |
| Table with curves                                            | <code>has_table(els);</code><br><code>has_paragraph_containing(els, '1 property hazard curves')</code>                                                 | Pass   | 0.000s |
| Count in summary                                             | <code>has_paragraph_containing(els, '2 mortgages linked')</code>                                                                                       | Pass   | 0.000s |
| Empty mortgages message                                      | <code>has_paragraph_containing(els, 'No mortgages generated')</code>                                                                                   | Pass   | 0.000s |
| Heading                                                      | <code>has_paragraph_containing(els, '3. Mortgages');</code><br><code>has_paragraph_containing(els, '4. Historical Gauge Data') (+2 more)</code>        | Pass   | 0.000s |
| Returns list                                                 | <code>isinstance(els, list);</code><br><code>isinstance(els, list) (+4 more)</code>                                                                    | Pass   | 0.000s |
| Table present                                                | <code>has_table(els); has_table(els)</code>                                                                                                            | Pass   | 0.000s |
| Empty properties message                                     | <code>has_paragraph_containing(els, 'No properties generated')</code>                                                                                  | Pass   | 0.000s |
| Heading present                                              | <code>has_paragraph_containing(els, '2. Properties')</code>                                                                                            | Pass   | 0.000s |
| Property count in summary                                    | <code>has_paragraph_containing(els, '2 properties')</code>                                                                                             | Pass   | 0.000s |
| Returns list                                                 | <code>isinstance(els, list);</code><br><code>isinstance(els, list) (+4 more)</code>                                                                    | Pass   | 0.000s |
| Table with properties                                        | <code>has_table(els)</code>                                                                                                                            | Pass   | 0.000s |
| Heading                                                      | <code>has_paragraph_containing(els, '7. Property Flood Time Se...;</code><br><code>has_paragraph_containing(els, '9. Counterparties') (+2 more)</code> | Pass   | 0.000s |
| No phc section when empty                                    | <code>not has_paragraph_containing(els, '8. Property Hazard')</code>                                                                                   | Pass   | 0.000s |

| Test Description                                    | Acceptance Criteria                                                                                                                                      | Result | Time   |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Phc section added                                   | <code>.has_paragraph_containing(els, '8. Property Hazard Curves');</code><br><code>.has_paragraph_containing(els, '2 property hazard curves')</code>     | Pass   | 0.000s |
| Returns list                                        | <code>isinstance(els, list);</code><br><code>isinstance(els, list) (+2 more)</code>                                                                      | Pass   | 0.000s |
| Heading                                             | <code>.has_paragraph_containing(els, '3. Mortgages');</code><br><code>.has_paragraph_containing(els, '4. Historical Gauge Data') (+2 more)</code>        | Pass   | 0.000s |
| Kv table present                                    | <code>.has_table(els)</code>                                                                                                                             | Pass   | 0.000s |
| Returns list                                        | <code>isinstance(els, list);</code><br><code>isinstance(els, list) (+4 more)</code>                                                                      | Pass   | 0.000s |
| With full summary                                   | <code>.has_table(els)</code>                                                                                                                             | Pass   | 0.000s |
| If input_dir is a real path, data size is appended. | <code>.has_table(els)</code>                                                                                                                             | Pass   | 0.001s |
| Heading                                             | <code>.has_paragraph_containing(els, '7. Property Flood Time Se...;</code><br><code>.has_paragraph_containing(els, '9. Counterparties') (+2 more)</code> | Pass   | 0.001s |
| Kv table present                                    | <code>.has_table(els)</code>                                                                                                                             | Pass   | 0.001s |
| Returns list                                        | <code>isinstance(els, list);</code><br><code>isinstance(els, list) (+2 more)</code>                                                                      | Pass   | 0.001s |
| All builders exported                               | <code>hasattr(sections_mod, name);</code><br><code>callable(getattr(sections_mod, name))</code>                                                          | Pass   | 0.000s |
| All list                                            | <code>len(sections_mod._all_) == 12</code>                                                                                                               | Pass   | 0.000s |
| Case insensitive                                    | <code>seq_type_color('DOUBLET') == colors.lightyellow</code>                                                                                             | Pass   | 0.000s |
| Cluster                                             | <code>seq_type_color('cluster') == colors.lightsalmon</code>                                                                                             | Pass   | 0.000s |
| Doublet                                             | <code>seq_type_color('doublet') == colors.lightyellow</code>                                                                                             | Pass   | 0.000s |
| Isolated                                            | <code>seq_type_color('isolated') == colors.lightblue</code>                                                                                              | Pass   | 0.000s |
| None returns white                                  | <code>seq_type_color(None) == colors.white</code>                                                                                                        | Pass   | 0.000s |
| Persistent                                          | <code>seq_type_color('persistent') == colors.mistyrose</code>                                                                                            | Pass   | 0.000s |
| Unknown returns white                               | <code>seq_type_color('unknown') == colors.white</code>                                                                                                   | Pass   | 0.000s |

| Test Description                                                                | Acceptance Criteria                                                                                                                                                            | Result | Time   |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Response declares the max_depth damage model.                                   | <code>data["damage_model"] == "max_depth"</code>                                                                                                                               | Pass   | 0.006s |
| Each property entry includes num_sequence_floods.                               | <code>prop_a["num_sequence_floods"] &gt;= 1</code>                                                                                                                             | Pass   | 0.006s |
| Response includes num_storms_in_sequence.                                       | <code>"num_storms_in_sequence" in data;</code><br><code>data["num_storms_in_sequence"] &gt;= 1</code>                                                                          | Pass   | 0.007s |
| Damage is max depth, not $0.10 + 0.18 = 0.28$ .                                 | <code>prop_a["damage_ratio"] &lt; 0.20</code>                                                                                                                                  | Pass   | 0.006s |
| PROP-A: second storm (0.65m) is worst — sequence damage uses 0.18 ratio.        | <code>prop_a["damage_ratio"]</code><br><code>== pytest.approx(0.18);</code><br><code>prop_a["damage_amount"] ==</code><br><code>pytest.approx(350_000 * 0.18)</code> (+1 more) | Pass   | 0.006s |
| PROP-B: first storm depth=0 so excluded, second storm 0.55m -i included.        | <code>"PROP-B" in prop_ids</code>                                                                                                                                              | Pass   | 0.006s |
| PROP-C is flooded by STORM-other, not STORM-test1 — must not appear.            | <code>"PROP-C" not in prop_ids</code>                                                                                                                                          | Pass   | 0.005s |
| Sequence total damage i sum of individual storm damages (not additive).         | <code>seq_damage &lt; dmg.d1a + dmg.d1b</code>                                                                                                                                 | Pass   | 0.007s |
| Sequence endpoint options                                                       | <code>r.status_code == 200</code>                                                                                                                                              | Pass   | 0.000s |
| Sequence endpoint returns 200                                                   | <code>r.status_code == 200;</code> <code>data["status"]</code><br><code>== "success"</code>                                                                                    | Pass   | 0.005s |
| Sequence id in response                                                         | <code>data["sequence_id"] == "STORM-test1"</code>                                                                                                                              | Pass   | 0.006s |
| Unknown sequence returns 404                                                    | <code>r.status_code == 404</code>                                                                                                                                              | Pass   | 0.006s |
| Worst storm (17% damage on 300k) -i 249k i 280k outstanding -i negative equity. | <code>data["portfolio"]["mortgages_in_negative_equality"]</code><br><code>== 1; prop["negative_equity"] is True</code>                                                         | Pass   | 0.004s |
| Sequence num floods is four                                                     | <code>data["properties"][0]["num_sequence_floods"]</code><br><code>== 4</code>                                                                                                 | Pass   | 0.005s |
| Sequence damage uses storm p3 (1.8m, ratio=0.17), not p1 or p2.                 | <code>prop["worst_storm_id"] ==</code><br><code>"STORM-p3"; prop["damage_ratio"]</code><br><code>== pytest.approx(0.17)</code>                                                 | Pass   | 0.004s |
| Querying seqA1 does not return seqB damage.                                     | <code>dmg_a != dmg_b; dmg_a != dmg_a + dmg_b</code>                                                                                                                            | Pass   | 0.004s |
| Seqa first storm returns only its damage                                        | <code>prop["damage_amount"] ==</code><br><code>pytest.approx(400_000 * 0.12)</code>                                                                                            | Pass   | 0.008s |
| Seqb storm returns only seqb damage                                             | <code>prop["damage_amount"] ==</code><br><code>pytest.approx(400_000 * 0.06)</code>                                                                                            | Pass   | 0.003s |

| Test Description                       | Acceptance Criteria                                                                                                                   | Result | Time   |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Known sequence returns success         | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>                                                | Pass   | 0.004s |
| No pts dir returns 404                 | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br>(+2 more)                                     | Pass   | 0.001s |
| Options returns ok                     | <code>r.status_code == 200; r.status_code == 200</code>                                                                               | Pass   | 0.000s |
| Properties sorted by damage descending | <code>amounts == sorted(amounts, reverse=True);</code><br><code>amounts == sorted(amounts, reverse=True)</code>                       | Pass   | 0.005s |
| Property entry has financial fields    | <code>"property_id" in prop;</code><br><code>"property_value" in prop</code> (+9 more)                                                | Pass   | 0.005s |
| Property entry has num sequence floods | <code>"num_sequence_floods" in prop;</code><br><code>prop["num_sequence_floods"] &gt;= 1</code>                                       | Pass   | 0.005s |
| Property entry has worst storm id      | <code>"worst_storm_id" in prop</code>                                                                                                 | Pass   | 0.004s |
| Response has damage model              | <code>r.get_json()["damage_model"] == "max_depth"</code>                                                                              | Pass   | 0.004s |
| Response has num storms                | <code>"num_storms_in_sequence"</code><br><code>in r.get_json();</code><br><code>r.get_json()["num_storms_in_sequence"] &gt;= 1</code> | Pass   | 0.004s |
| Response has portfolio                 | <code>"portfolio" in data;</code><br><code>"total_properties" in port</code> (+4 more)                                                | Pass   | 0.005s |
| Response has properties list           | <code>"properties" in data;</code><br><code>len(data["properties"]) &gt;= 1</code> (+2 more)                                          | Pass   | 0.005s |
| Response has sequence id               | <code>r.get_json()["sequence_id"] == seq_id</code>                                                                                    | Pass   | 0.004s |
| Unknown sequence returns 404           | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code>                                                  | Pass   | 0.005s |
| Alert flags length matches num gauges  | <code>all(len(r["alert"]) == 3 for r in records)</code>                                                                               | Pass   | 0.000s |
| All have intensity category            | <code>all(r["intensity_category"] in valid for r in records)</code>                                                                   | Pass   | 0.000s |
| All have sequence id                   | <code>all("sequence_id" in r for r in records)</code>                                                                                 | Pass   | 0.000s |
| All have sequence type                 | <code>all(r["sequence_type"] in valid for r in records)</code>                                                                        | Pass   | 0.000s |



| Test Description                                                    | Acceptance Criteria                                                                                          | Result | Time   |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------|--------|
| All have total precip                                               | <code>all("total_precip_mm" in r for r in records); all(r["total_precip_mm"] &gt; 0 for r in records)</code> | Pass   | 0.000s |
| Peak hours in valid range                                           | <code>all(0 &lt;= h &lt;= 167 for h in r["peak_hours"])</code>                                               | Pass   | 0.000s |
| Peak hours length matches num gauges                                | <code>all(len(r["peak_hours"]) == 3 for r in records)</code>                                                 | Pass   | 0.000s |
| Peaks length matches num gauges                                     | <code>all(len(r["peaks_m"]) == 3 for r in records)</code>                                                    | Pass   | 0.000s |
| Peaks positive                                                      | <code>all(p &gt; 0 for p in r["peaks_m"])</code>                                                             | Pass   | 0.000s |
| Severe flags length matches num gauges                              | <code>all(len(r["severe"]) == 3 for r in records)</code>                                                     | Pass   | 0.000s |
| Severe implies warning                                              | <code>r["warning"][i]</code>                                                                                 | Pass   | 0.000s |
| Warning flags length matches num gauges                             | <code>all(len(r["warning"]) == 3 for r in records)</code>                                                    | Pass   | 0.000s |
| Every sequence that breached warning must also have breached alert. | <code>r["alert"][i]</code>                                                                                   | Pass   | 0.000s |
| Returns 404 when file missing                                       | <code>r.status_code in (200, 404, 500)</code>                                                                | Pass   | 0.001s |
| Lines 33-35: send_file when HTML file actually exists.              | <code>r.status_code in (200, 404, 500)</code>                                                                | Pass   | 0.031s |
| Returns json error or html                                          | <code>r.status_code in (200, 404, 500)</code>                                                                | Pass   | 0.001s |
| No coverage dir returns 404                                         | <code>r.status_code == 404</code>                                                                            | Pass   | 0.000s |
| With coverage index returns 200                                     | <code>r.status_code == 200</code>                                                                            | Pass   | 0.002s |
| The response body is valid JSON with an error status.               | <code>r.status_code in (500, 503); data is not None (+1 more)</code>                                         | Pass   | 0.001s |
| Without gauge.json, visualization fails with 500.                   | <code>r.status_code in (500, 503)</code>                                                                     | Pass   | 0.002s |
| No gauge data returns error status                                  | <code>data["status"] == "error"</code>                                                                       | Pass   | 0.001s |
| Lines 97-99: exception in serve_visualization_for_catchment → 500.  | <code>r.status_code in (302, 500)</code>                                                                     | Pass   | 0.029s |
| Matching is case-insensitive.                                       | <code>r.status_code == 302</code>                                                                            | Pass   | 0.001s |
| 'thames' catchment redirects to main visualization route.           | <code>r.status_code == 302</code>                                                                            | Pass   | 0.000s |
| Unknown catchment returns 400                                       | <code>r.status_code == 400; data["status"] == "error" (+1 more)</code>                                       | Pass   | 0.001s |
| Unsupported catchment message                                       | <code>r.status_code == 400; "thames" in r.get_json()["message"].lower()</code>                               | Pass   | 0.001s |
| Lines 59-65: ImportError → 503.                                     | <code>r.status_code in (500, 503)</code>                                                                     | Pass   | 0.028s |
| Lines 50-57: mock TCEventVisualization to return existing file.     | <code>r.status_code in (200, 500)</code>                                                                     | Pass   | 0.030s |

| Test Description                                            | Acceptance Criteria                                                                                                                                     | Result | Time   |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| main() calls app.run() with configured host/port/debug.     | <code>len(run_calls) == 1;</code><br><code>run_calls[0]["host"] ==</code><br><code>config.SERVER_HOST (+1 more)</code>                                  | Pass   | 0.026s |
| server.main() is callable (test by importing, not running). | <code>callable(main)</code>                                                                                                                             | Pass   | 0.000s |
| Simulate what routes/gauges.py:generate_report() does.      | <code>report_path.exists();</code><br><code>report_path.suffix == '.pdf' (+1 more)</code>                                                               | Pass   | 0.016s |
| Test with actual Thames gauge data if available.            | <code>report_path.exists();</code><br><code>report_path.stat().st_size &gt; 5000</code>                                                                 | Pass   | 0.017s |
| Simulate what routes/properties.py:generate_report() does.  | <code>report_path.exists();</code><br><code>report_path.suffix == '.pdf' (+1 more)</code>                                                               | Pass   | 0.028s |
| Test with actual Thames portfolio data if available.        | <code>report_path.exists();</code><br><code>report_path.stat().st_size &gt; 10000</code>                                                                | Pass   | 0.037s |
| Case insensitive thames                                     | <code>r.status_code == 200</code>                                                                                                                       | Pass   | 0.001s |
| Missing catchment id returns 400                            | <code>r.status_code == 400</code>                                                                                                                       | Pass   | 0.000s |
| No json returns 400                                         | <code>r.status_code in (400, 415, 500)</code>                                                                                                           | Pass   | 0.000s |
| Line 64: non-empty body but null/empty catchment → 400.     | <code>r.status_code == 400; "Catchment ID is required" in data["message"]</code>                                                                        | Pass   | 0.000s |
| Options returns 200                                         | <code>r.status_code == 200</code>                                                                                                                       | Pass   | 0.000s |
| Unknown catchment returns 400                               | <code>r.status_code == 400; data["status"] == "error"</code>                                                                                            | Pass   | 0.000s |
| Valid thames returns success                                | <code>r.status_code == 200; data["status"] == "success" (+2 more)</code>                                                                                | Pass   | 0.001s |
| Existing mortgage dict updated                              | <code>data["Mortgage"]["Header"]["ExistingField"] == "keep";</code><br><code>data["Mortgage"]["Header"]["MortgageID"] == "MORT-new"</code>              | Pass   | 0.000s |
| Initialises header key when absent                          | <code>"Header" in data["Mortgage"]</code>                                                                                                               | Pass   | 0.000s |
| Initialises mortgage key when absent                        | <code>"Mortgage" in data</code>                                                                                                                         | Pass   | 0.000s |
| Missing property id key defaults empty string               | <code>data["Mortgage"]["Header"]["PropertyID"] == ""</code>                                                                                             | Pass   | 0.000s |
| Sets catchment id                                           | <code>data["Mortgage"]["Header"]["CatchmentID"] == "thames"</code>                                                                                      | Pass   | 0.000s |
| Sets loan details fields                                    | <code>loan["OriginalLoanAmount"] == fd["loan.amount"];</code><br><code>loan["CurrentBalance"] ==</code><br><code>fd["current.balance"] (+3 more)</code> | Pass   | 0.000s |

| Test Description                                   | Acceptance Criteria                                                                                          | Result | Time   |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------|--------|
| Sets mortgage id                                   | <code>data["Mortgage"]["Header"]["MortgageID"] == "MORT-abc12"</code>                                        | Pass   | 0.000s |
| Sets property id in header                         | <code>data["Mortgage"]["Header"]["PropertyID"] == "PROP-xyz"</code>                                          | Pass   | 0.000s |
| Catchment id set in attrs                          | <code>data["PropertyHeader"]["PropertyAttributes"]["CatchmentID"]</code>                                     | Pass   | 0.000s |
| Elevation sets ground level                        | <code>loc["RiskAssessment"]["GroundLevelMeters"] == pytest.appr...</code>                                    | Pass   | 0.000s |
| Empty reference gauge indices no ref key           | <code>"ReferenceGauges" not in data.get("PropertyHeader", {})</code>                                         | Pass   | 0.000s |
| Lat lon set in location                            | <code>loc["LatitudeDegrees"] == pytest.approx(51.45); loc["LongitudeDegrees"] == pytest.approx(-0.31)</code> | Pass   | 0.000s |
| No elevation in location does not set ground level | <code>"RiskAssessment" not in loc</code>                                                                     | Pass   | 0.000s |
| No reference gauge indices no ref key              | <code>"ReferenceGauges" not in data.get("PropertyHeader", {})</code>                                         | Pass   | 0.000s |
| Property id set in attrs                           | <code>data["PropertyHeader"]["PropertyAttributes"]["PropertyID"]</code>                                      | Pass   | 0.000s |
| Reference gauges fallback when not in map          | <code>ref == ["GAUGE-001", "GAUGE-002", "GAUGE-003"]</code>                                                  | Pass   | 0.000s |
| Reference gauges set from gauge id map             | <code>ref == ["GAUGE-abc", "GAUGE-def", "GAUGE-ghi"]</code>                                                  | Pass   | 0.000s |
| Adds all components                                | <code>mock_html.add_child.call_count &gt; 0</code>                                                           | Pass   | 0.072s |
| Copyright element added                            | <code>len(calls) &gt; 0</code>                                                                               | Pass   | 0.081s |
| Returns self                                       | <code>result is mgr</code>                                                                                   | Pass   | 0.073s |
| Body style font size                               | <code>styled.body_style.fontSize == 9</code>                                                                 | Pass   | 0.000s |
| Kv style font size                                 | <code>styled.kv_style.fontSize == 9</code>                                                                   | Pass   | 0.000s |
| Kv style left indent                               | <code>styled.kv_style.leftIndent == 12</code>                                                                | Pass   | 0.000s |
| Required keys                                      | <code>key in styles</code>                                                                                   | Pass   | 0.000s |
| Returns stylesheet                                 | <code>isinstance(styles, StyleSheet1)</code>                                                                 | Pass   | 0.000s |
| Section style font size                            | <code>styled.section_style.fontSize == 13</code>                                                             | Pass   | 0.000s |
| Section style text color                           | <code>styled.section_style.textColor == DARK.BLUE</code>                                                     | Pass   | 0.000s |
| Styles are paragraph styles                        | <code>isinstance(styles[key], ParagraphStyle)</code>                                                         | Pass   | 0.000s |
| Subsection style font size                         | <code>styled.subsection_style.fontSize == 10</code>                                                          | Pass   | 0.000s |
| Subsection style text color                        | <code>styled.subsection_style.textColor == BLUE</code>                                                       | Pass   | 0.000s |
| Subtitle style font size                           | <code>styled.subtitle_style.fontSize == 10</code>                                                            | Pass   | 0.000s |

| Test Description                                                          | Acceptance Criteria                                                                                               | Result | Time   |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------|--------|
| Subtitle style text color                                                 | <code>styled.subtitle.style.textColor == GREY</code>                                                              | Pass   | 0.000s |
| Title style font name                                                     | <code>styled.title.style.fontName == 'Helvetica-Bold'</code>                                                      | Pass   | 0.000s |
| Title style font size                                                     | <code>styled.title.style.fontSize == 18</code>                                                                    | Pass   | 0.000s |
| Title style text color                                                    | <code>styled.title.style.textColor == DARK.BLUE</code>                                                            | Pass   | 0.000s |
| All defined categories                                                    | <code>isinstance(result, bool)</code>                                                                             | Pass   | 0.000s |
| Catastrophic has 85% chance — over many trials, majority should be True.  | <code>sum(results) &gt; 140</code>                                                                                | Pass   | 0.000s |
| Minimal has 5% chance — most should be False.                             | <code>sum(results) &lt; 30</code>                                                                                 | Pass   | 0.000s |
| Returns bool                                                              | <code>isinstance(result, bool)</code>                                                                             | Pass   | 0.000s |
| Unknown category uses default 0.15 probability.                           | <code>50 &lt; sum(results) &lt; 350</code>                                                                        | Pass   | 0.001s |
| <code>sp_sim.get_js()</code> must return a non-empty string.              | <code>len(sim_js) &gt; 100</code>                                                                                 | Pass   | 0.000s |
| <code>spSeekTo()</code> must be defined — called by scrubber oninput.     | <code>'spSeekTo' in sim_js</code>                                                                                 | Pass   | 0.000s |
| <code>spSetSpeed()</code> must be defined — called by speed buttons.      | <code>'spSetSpeed' in sim_js</code>                                                                               | Pass   | 0.000s |
| <code>spStopAnim()</code> must be defined — cleans up animation interval. | <code>'spStopAnim' in sim_js</code>                                                                               | Pass   | 0.000s |
| Business days is three months                                             | <code>60 &lt;= NUM.BUSINESS.DAYS &lt;= 66</code>                                                                  | Pass   | 0.000s |
| Daily vol is small                                                        | <code>0.0001 &lt;= DAILY_HAZARD.VOL &lt;= 0.001</code>                                                            | Pass   | 0.000s |
| Random walk floors at 0.0001 — vol must not exceed that floor.            | <code>DAILY_HAZARD.VOL &gt; 0; MAX.TOTAL.MOVE &gt; 0</code>                                                       | Pass   | 0.000s |
| Cumulative cap should allow meaningful drift.                             | <code>MAX.TOTAL.MOVE &gt;= 5 * DAILY_HAZARD.VOL</code>                                                            | Pass   | 0.000s |
| Max move bounds vol                                                       | <code>MAX.TOTAL.MOVE &gt; DAILY_HAZARD.VOL; MAX.TOTAL.MOVE &lt;= 0.01</code>                                      | Pass   | 0.000s |
| Test complete gauge analysis workflow.                                    | <code>results['success'] is True; results['gauge']['gauge_id'] == "THAMES.TEDDINGTON" (+2 more)</code>            | Pass   | 0.004s |
| Handle non-existent gauge gracefully.                                     | <code>results['success'] is False; len(results['errors']) &gt; 0</code>                                           | Pass   | 0.002s |
| Analysis should include flood status.                                     | <code>'status' in results['flood_analysis']; results['flood_analysis']['status'] in ["NORMAL", "ALERT"]...</code> | Pass   | 0.004s |

| Test Description                                                                | Acceptance Criteria                                                                                                                                                                 | Result | Time   |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Test loading gauge metadata from JSON.                                          | <code>gauges is not None; len(gauges) == 4</code><br>(+2 more)                                                                                                                      | Pass   | 0.002s |
| Test loading gauge time series.                                                 | <code>readings is not None;</code><br><code>"THAMES_TEDDINGTON" in readings</code><br>(+1 more)                                                                                     | Pass   | 0.002s |
| Error message lists available ids                                               | <code>"GAUGE-test001" in</code><br><code>str(exc_info.value)</code>                                                                                                                 | Pass   | 0.001s |
| Invalid gauge id raises                                                         | —                                                                                                                                                                                   | Pass   | 0.001s |
| Named file gauge ids                                                            | <code>d["gauge_ids"] == ["GAUGE-test002"]</code>                                                                                                                                    | Pass   | 0.001s |
| Named file peaks length one                                                     | <code>all(len(r["peaks_m"]) == 1 for r in</code><br><code>d["sequences"])</code>                                                                                                    | Pass   | 0.000s |
| Named file schema version                                                       | <code>d["schema_version"] ==</code><br><code>SCHEMA.VERSION_SPATIAL</code>                                                                                                          | Pass   | 0.002s |
| Named file sequence count                                                       | <code>len(d["sequences"]) == 40</code>                                                                                                                                              | Pass   | 0.000s |
| Named output file written                                                       | <code>(gauge_dir /</code><br><code>"sequence_gauge_GAUGE-test002.json").exists()</code>                                                                                             | Pass   | 0.000s |
| Summary num gauges is one                                                       | <code>single_run["num_gauges"] == 1</code>                                                                                                                                          | Pass   | 0.000s |
| Single gauge classifier result dict                                             | <code>isinstance(result, dict)</code>                                                                                                                                               | Pass   | 6.183s |
| Single gauge + train_classifier=True covers lines 353-361.                      | <code>result["num_gauges"] == 1;</code><br><code>stressm_dir.exists()</code>                                                                                                        | Pass   | 6.142s |
| All mortgages fail returns empty list                                           | <code>result["processing_stats"]["failed_mortgages"]</code><br><code>== 2; result["data"]["mortgages"] ==</code><br><code>[]</code>                                                 | Pass   | 0.002s |
| When <code>_generate_single_mortgage</code> raises, the generator continues and | <code>result["processing_stats"]["failed_mortgages"]</code><br><code>== 1;</code><br><code>result["processing_stats"]["successful_mortgages"]</code><br><code>== 2 (+1 more)</code> | Pass   | 0.003s |
| Find nearest gauge                                                              | <code>gauge_id == "THAMES_TEDDINGTON";</code><br><code>distance &lt; 1.0</code>                                                                                                     | Pass   | 0.000s |
| Full analysis high risk property                                                | <code>results["success"] is True;</code><br><code>results["property"]["property_id"] ==</code><br><code>"THAMES_TEST_002" (+1 more)</code>                                          | Pass   | 0.003s |
| Full analysis low risk property                                                 | <code>results["success"] is True;</code><br><code>results["property"]["property_id"] ==</code><br><code>"THAMES_TEST_003" (+2 more)</code>                                          | Pass   | 0.003s |
| Load gauge data                                                                 | <code>gauge_meta is not None;</code><br><code>len(gauge_meta) == 3 (+2 more)</code>                                                                                                 | Pass   | 0.003s |
| Load property data                                                              | <code>properties is not None;</code><br><code>len(properties) == 3 (+3 more)</code>                                                                                                 | Pass   | 0.002s |
| Clay medium high                                                                | <code>"Medium" in r or "high" in r.lower()</code>                                                                                                                                   | Pass   | 0.000s |
| Peat high risk                                                                  | <code>"High" in r</code>                                                                                                                                                            | Pass   | 0.000s |

| Test Description                                                           | Acceptance Criteria                                       | Result | Time   |
|----------------------------------------------------------------------------|-----------------------------------------------------------|--------|--------|
| Rock keyword                                                               | "Low" in r                                                | Pass   | 0.000s |
| Rock low risk                                                              | "Low" in r                                                | Pass   | 0.000s |
| Sand medium risk                                                           | "Medium" in r                                             | Pass   | 0.000s |
| Unknown soil                                                               | "Assessment" in r or "chalk" in r                         | Pass   | 0.000s |
| STORM-cccccccc (alert-only, no warning/severe) must be last.               | sorted_storms[-1]["storm_id"] == "STORM-cccccccc"         | Pass   | 0.000s |
| STORM-bbbbbbbb (2 severe) must rank before all others.                     | sorted_storms[0]["storm_id"] == "STORM-bbbbbbbb"          | Pass   | 0.000s |
| gauges_severe must be non-increasing.                                      | severe_counts == sorted(severe_counts, reverse=True)      | Pass   | 0.000s |
| Defaults                                                                   | p.base_range_km == 40.0; p.nugget == 0.05 (+2 more)       | Pass   | 0.000s |
| From dict                                                                  | p.base_range_km == 50.0; p.nugget == 0.10 (+1 more)       | Pass   | 0.000s |
| Load from file                                                             | p.base_range_km == 40.0; p.nugget == 0.05                 | Pass   | 0.002s |
| At gauge location returns wse directly                                     | result == pytest.approx(3.0)                              | Pass   | 0.000s |
| Empty gauge data returns zero                                              | result == 0.0                                             | Pass   | 0.000s |
| Idw returns weighted value                                                 | 2.5 <= result <= 4.5                                      | Pass   | 0.000s |
| Single gauge                                                               | result > 0                                                | Pass   | 0.000s |
| Lines 119-120: distance ; 1m → direct return.                              | result == pytest.approx(7.0)                              | Pass   | 0.000s |
| Line 97-98: no gauge data → 0.0.                                           | spatial_interpolate_wse(51.5, -0.1, {}, {}) == 0.0        | Pass   | 0.000s |
| Lines 113-124: IDW with two equidistant gauges.                            | result == pytest.approx(2.0, rel=0.01)                    | Pass   | 0.000s |
| Detailed report uses more pages than basic, so should produce larger file. | detailed.stat().st_size >= basic.stat().st_size           | Pass   | 0.030s |
| Generate analysis report custom filename                                   | path.name == "analysis.pdf"                               | Pass   | 0.003s |
| Generate analysis report custom filename                                   | path.name == "analysis.pdf"                               | Pass   | 0.010s |
| Generate analysis report exists                                            | gen.generate_analysis_report(gauge_data).exists()         | Pass   | 0.003s |
| Generate analysis report exists                                            | gen.generate_analysis_report(minimal_flood_data).exists() | Pass   | 0.010s |
| Generate analysis report with time-series                                  | path.exists()                                             | Pass   | 0.003s |
| Generate basic report custom filename                                      | path.name == "basic.pdf"                                  | Pass   | 0.015s |
| Generate basic report exists                                               | gen.generate_basic_report(gauge_data).exists()            | Pass   | 0.014s |

| Test Description                                                            | Acceptance Criteria                                                        | Result | Time   |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------|--------|--------|
| Generate basic report exists                                                | <code>gen.generate_basic_report(minimal_flood_data).exists()</code>        | Pass   | 0.044s |
| Generate basic report non empty                                             | <code>path.stat().st_size &gt; 0</code>                                    | Pass   | 0.014s |
| Generate basic report non empty                                             | <code>path.stat().st_size &gt; 0</code>                                    | Pass   | 0.023s |
| Generate basic with timeseries                                              | <code>path.exists()</code>                                                 | Pass   | 0.015s |
| Generate detailed report exists                                             | <code>gen.generate_detailed_report(minimal_flood_data).exists()</code>     | Pass   | 0.016s |
| Detailed report includes mortgage_analysis — verify call reaches that page. | —                                                                          | Pass   | 0.019s |
| Generate monitoring report custom filename                                  | <code>path.name == "mon.pdf"</code>                                        | Pass   | 0.003s |
| Generate monitoring report exists                                           | <code>gen.generate_monitoring_report(gauge_data, ts_data).exists()</code>  | Pass   | 0.003s |
| Generate monitoring report no timeseries                                    | <code>path.exists()</code>                                                 | Pass   | 0.003s |
| Generate mortgage focused report custom filename                            | <code>path.name == "mort_focused.pdf"</code>                               | Pass   | 0.017s |
| Generate mortgage focused report exists                                     | <code>gen.generate_mortgage_focused_report(prop_data, mort_data...)</code> | Pass   | 0.016s |
| Generate property only report custom filename                               | <code>path.name == "prop_only.pdf"</code>                                  | Pass   | 0.015s |
| Generate property only report exists                                        | <code>gen.generate_property_only_report(prop_data).exists()</code>         | Pass   | 0.015s |
| Generate risk focused report custom filename                                | <code>path.name == "risk_only.pdf"</code>                                  | Pass   | 0.009s |
| Generate risk focused report with mortgage                                  | <code>path.exists()</code>                                                 | Pass   | 0.012s |
| Generate risk focused report without mortgage                               | <code>path.exists()</code>                                                 | Pass   | 0.010s |
| Generate summary report custom filename                                     | <code>path.name == "summary.pdf"</code>                                    | Pass   | 0.008s |
| Generate summary report exists                                              | <code>gen.generate_summary_report(minimal_flood_data).exists()</code>      | Pass   | 0.008s |
| Mortgage focused includes mortgage context pages                            | —                                                                          | Pass   | 0.016s |
| Risk focused with mortgage includes current status                          | —                                                                          | Pass   | 0.011s |
| Risk focused without mortgage excludes current status                       | —                                                                          | Pass   | 0.012s |
| Only gauge_GAUGE-*_hd.json files are removed, not other files.              | <code>other.exists()</code>                                                | Pass   | 0.212s |
| Multiple stale files are all removed.                                       | <code>not s.exists()</code>                                                | Pass   | 0.232s |
| Lines 37-39: old gauge_GAUGE-*_hd.json files are removed.                   | <code>stale.exists(); not stale.exists()</code>                            | Pass   | 0.213s |



| Test Description                                                    | Acceptance Criteria                                                                                              | Result | Time   |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------|--------|
| All fresh full health                                               | <code>data["health_pct"] == 100.0</code>                                                                         | Pass   | 0.003s |
| All missing zero health                                             | <code>r.status_code == 200; data["status"] == "success" (+2 more)</code>                                         | Pass   | 0.000s |
| Has as of                                                           | <code>"as_of" in data</code>                                                                                     | Pass   | 0.000s |
| Partial health                                                      | <code>data["health_pct"] == 10.0</code>                                                                          | Pass   | 0.001s |
| Stale does not count as fresh                                       | <code>data["health_pct"] == 0.0</code>                                                                           | Pass   | 0.001s |
| Every dataset key must appear in _startupDetail — no silent blanks. | <code>f"key === '{k}'" in src</code>                                                                             | Pass   | 0.000s |
| Blotter stat uses trades length                                     | <code>"key === '_tdPreBlotter'" in src;</code><br><code>"data.trades.length + ' trades'" in src</code>           | Pass   | 0.000s |
| Eod history stat uses history length                                | <code>"key === '_tdPreEodHistory'" in src;</code><br><code>"data.history.length + ' snapshots'" in src</code>    | Pass   | 0.000s |
| Gauge hist stat uses count                                          | <code>"key === '_preGaugeHist'" in src;</code><br><code>'data.count' in src</code>                               | Pass   | 0.000s |
| Gauges stat uses gauges length                                      | <code>"key === '_tdPreGauges'" in src;</code><br><code>"data.gauges.length + ' gauges'" in src</code>            | Pass   | 0.000s |
| Gov audit stat uses events length                                   | <code>"key === '_preGovAudit'" in src;</code><br><code>"data.total_entries + ' events'" in src</code>            | Pass   | 0.000s |
| Gov bib stat uses references length                                 | <code>"key === '_preGovBib'" in src;</code><br><code>"data.references.length + ' refs'" in src</code>            | Pass   | 0.001s |
| Gov docs stat uses documents length                                 | <code>"key === '_tdPreGovDocs'" in src;</code><br><code>"data.documents.length + ' docs'" in src</code>          | Pass   | 0.000s |
| Market stat uses object keys gauges                                 | <code>"key === '_tdPreMarket'" in src;</code><br><code>"Object.keys(data.gauges).length" in src (+1 more)</code> | Pass   | 0.000s |
| Mortgages stat uses mortgages length                                | <code>"key === '_preMortgages'" in src;</code><br><code>"data.mortgages" in src</code>                           | Pass   | 0.000s |
| Port storms stat uses storms length                                 | <code>"key === '_tdPrePortStorms'" in src;</code><br><code>"data.storms.length + ' storms'" in src</code>        | Pass   | 0.000s |
| Pre storms stat uses storms length                                  | <code>"key === '_preStorms'" in src;</code><br><code>"data.storms.length + ' storms'" in src</code>              | Pass   | 0.000s |

| Test Description                                                     | Acceptance Criteria                                                                                  | Result | Time   |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------|--------|
| Property ts stat uses total properties                               | "key === '_prePropertyTS'" in src;<br>'data.data.summary.properties_with_floods'<br>in src (+1 more) | Pass   | 0.000s |
| Stress storms stat uses storms length                                | "key === '_tdPreStressStorms'" in src;<br>"data.storms.length + '<br>scenarios'" in src              | Pass   | 0.000s |
| Yield curve stat uses object keys                                    | "key === '_tdPreYieldCurve'" in src;<br>"Object.keys(data.yield.curve).length"<br>in src (+1 more)   | Pass   | 0.000s |
| Add to map calls add child                                           | —                                                                                                    | Pass   | 0.002s |
| Add to map output contains script                                    | '<script>' in rendered;<br>'window._tdPreloadDone' in rendered                                       | Pass   | 0.002s |
| Add to map method defined                                            | 'def add_to_map' in src                                                                              | Pass   | 0.000s |
| All fourteen endpoints present                                       | ep in js                                                                                             | Pass   | 0.000s |
| Must NOT defer to a button click — runs on DOMContentLoaded.         | 'DOMContentLoaded' in js                                                                             | Pass   | 0.000s |
| 'Market state' was renamed to 'Hazard curves'.                       | 'Hazard curves' in js;        "'Market state'" not in js and "'Market state'" not in js              | Pass   | 0.000s |
| Sets window._tdPreloadDone so any IIFE can read it via global scope. | 'window._tdPreloadDone = true' in _src('src/visual/intera...                                         | Pass   | 0.000s |
| Shows progress popup                                                 | 'startup-preloader-popup' in js;<br>'startup-pre-bar' in js                                          | Pass   | 0.000s |
| Startup preloader class defined                                      | 'class StartupPreloader' in src                                                                      | Pass   | 0.000s |
| Declares pre gauge hist                                              | 'window._preGaugeHist' in _src('src/visual/interactivity/...                                         | Pass   | 0.000s |
| Declares pre gov audit                                               | 'window._preGovAudit' in _src('src/visual/interactivity/s...                                         | Pass   | 0.000s |
| Declares pre gov bib                                                 | 'window._preGovBib' in _src('src/visual/interactivity/sta...                                         | Pass   | 0.000s |
| Declares pre property ts                                             | 'window._prePropertyTS' in _src('src/visual/interactivity...                                         | Pass   | 0.000s |
| Declares pre storms                                                  | 'window._preStorms' in _src('src/visual/interactivity/sta...                                         | Pass   | 0.000s |
| Declares td pre blotter                                              | 'window._tdPreBlotter' in _src('src/visual/interactivity/...                                         | Pass   | 0.001s |
| Declares td pre eod history                                          | 'window._tdPreEodHistory' in _src('src/visual/interactivi...                                         | Pass   | 0.000s |
| Declares td pre gauges                                               | 'window._tdPreGauges' in _src('src/visual/interactivity/s...                                         | Pass   | 0.000s |

| Test Description                                                          | Acceptance Criteria                                                                      | Result | Time   |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------|--------|
| Declares td pre gov docs                                                  | 'window._tdPreGovDocs' in<br>_src('src/visual/interactivity/...                          | Pass   | 0.000s |
| Declares td pre market                                                    | 'window._tdPreMarket' in<br>_src('src/visual/interactivity/s...                          | Pass   | 0.000s |
| Declares td pre port storms                                               | 'window._tdPrePortStorms' in<br>_src('src/visual/interactivi...                          | Pass   | 0.000s |
| Declares td pre stress storms                                             | 'window._tdPreStressStorms' in<br>_src('src/visual/interacti...                          | Pass   | 0.000s |
| Declares td pre yield curve                                               | 'window._tdPreYieldCurve' in<br>_src('src/visual/interactivi...                          | Pass   | 0.000s |
| Line 46: GET /apple-touch-icon.png returns the PNG.                       | r.status_code == 200; "image/png" in<br>r.content_type                                   | Pass   | 0.002s |
| Line 51: GET /apple-touch-icon-precomposed.png returns the PNG.           | r.status_code == 200; "image/png" in<br>r.content_type                                   | Pass   | 0.001s |
| Line 41: GET /favicon.ico returns the icon file.                          | r.status_code == 200;<br>r.content_type in ("image/x-icon",<br>"image/vnd.microsoft.i... | Pass   | 0.003s |
| _navigateToStress function must be defined.                               | '_navigateToStress' in hist_js                                                           | Pass   | 0.000s |
| Click handlers use addEventListener, not inline onclick.                  | 'addEventListener' in hist_js;<br>'hist-storm-row' in hist_js                            | Pass   | 0.000s |
| Must call switchTab(5), NOT switchTab(4) or any other index.              | 'switchTab(5)' in hist_js;<br>'switchTab(4)' not in<br>hist_js.replace('switc...         | Pass   | 0.000s |
| _navigateToStress must set window._stressStormHint before switching tab.  | 'window._stressStormHint' in hist_js                                                     | Pass   | 0.000s |
| Hint must be set BEFORE switchTab is called — otherwise Stress tab        | idx_hint > 0; idx_switch > 0 (+1 more)                                                   | Pass   | 0.000s |
| data-storm-id attribute used to pass storm_id to _navigateToStress.       | 'data-storm-id' in hist_js;<br>'getAttribute' in hist_js                                 | Pass   | 0.000s |
| Every gauge in gaugehc.json should have a trained classifier.             | not missing                                                                              | Skip   | 0.000s |
| Every gauge referenced in stress_storms should have a trained classifier. | not missing                                                                              | Skip   | 0.000s |
| Every gauge in stress_storms should have a hazard curve in gaugehc.json.  | not missing                                                                              | Skip   | 0.000s |

| Test Description                                                         | Acceptance Criteria                                                     | Result | Time   |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------|--------|--------|
| Storm IDs in stress_storms must overlap with storm IDs in proper-tyts/.  | <code>len(overlap) &gt; 0</code>                                        | Skip   | 0.000s |
| Every storm_id in stress_storms.json must exist in gaugets/GAUGE-*.json. | not missing                                                             | Skip   | 0.000s |
| Every storm dropdown must show the storm_id in the option text.          | not violations                                                          | Pass   | 0.001s |
| All dropdown option builders must use sentinel or pipe separators.       | not violations                                                          | Pass   | 0.001s |
| The storm display format is centralised in config/format.py.             | '   ' in js; 'storm_id' in js (+2 more)                                 | Pass   | 0.000s |
| 'precipitation' as a label suffix must not appear.                       | not violations                                                          | Pass   | 0.001s |
| 'severe levels' (old space-separated format) must not appear.            | not violations                                                          | Pass   | 0.011s |
| port_stress/setup.py uses sentinel with show_warning=True.               | '_storm_opt' in src;<br>'show_warning=True' in src                      | Pass   | 0.000s |
| sp_table.py delegates storm label to config.format via sentinel.         | '_storm_opt' in src; '__STORM_OPT__' in src (+1 more)                   | Pass   | 0.000s |
| Custom variable name is used throughout — gsa_timeline uses 'r'.         | js.startswith('(r.name'); 's.storm_id' not in js                        | Pass   | 0.000s |
| show_peak=True adds peak water level after mm.                           | 'peak' in js_peak                                                       | Pass   | 0.000s |
| show_warning=True adds gauges_warning count between severe and mm.       | 'warn' in js_warn; 'warn' not in js_base                                | Pass   | 0.000s |
| trading/stress/setup.py uses sentinel with show_peak=True.               | '_storm_opt' in src; 'show_peak=True' in src                            | Pass   | 0.000s |
| Visual files must not contain hard-coded storm display format strings.   | pat not in src                                                          | Pass   | 0.001s |
| Unknown intensity profile falls to else: intensity = peak * t_frac.      | storm.storm_id.startswith("STORM-");<br>len(storm.track) == 5 (+1 more) | Pass   | 0.000s |
| Both tabs use 'effective-precipitation_mm' field.                        | 'effective-precipitation_mm' in hist_js                                 | Pass   | 0.000s |
| Both tabs use 'gauges_severe' field.                                     | 'gauges_severe' in hist_js                                              | Pass   | 0.000s |
| Both tabs use 'intensity_category' field.                                | 'intensity_category' in hist_js                                         | Pass   | 0.000s |
| Both tabs use 'peak_level_m' field.                                      | 'peak_level_m' in hist_js                                               | Pass   | 0.000s |
| Both tabs use 'storm_id' (not 'id' or 'stormId').                        | 'storm_id' in hist_js                                                   | Pass   | 0.000s |

| Test Description                                                                          | Acceptance Criteria                                                                              | Result | Time   |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------|--------|
| Storm IDs in gaugets/ must come from storm_sequences.json.                                | <code>len(overlap) &gt; 0</code>                                                                 | Pass   | 0.423s |
| Storm IDs in propertyts must come from storm_sequences.json.                              | <code>len(overlap) &gt; 0</code>                                                                 | Pass   | 1.058s |
| <code>storm_multi.make_storm_id()</code> produces STORM- <code>{8hex}</code> IDs.         | <code>STORM_ID.RE.match(sid)</code>                                                              | Pass   | 0.000s |
| storm generator must NOT use STORM- <code>{i:03d}</code> or STORM- <code>{i:04d}</code> . | <code>'STORM-{i:03d}' not in src; 'STORM-{i:04d}' not in src</code>                              | Pass   | 0.000s |
| Single storm generator uses storm prefix                                                  | <code>'STORM-' in src or 'STORM_ID_PREFIX' in src</code>                                         | Pass   | 0.001s |
| Single storm generator uses uuid                                                          | <code>'uuid' in src</code>                                                                       | Pass   | 0.000s |
| Uuid4 produces protocol matching ids                                                      | <code>GAUGE_ID.RE.match(gauge_id); PROP_ID.RE.match(prop_id) (+2 more)</code>                    | Pass   | 0.000s |
| Accepts valid                                                                             | <code>PROP_ID.RE.match('PROP-a3f7bc12'); PROP_ID.RE.match('PROP-00000000') (+6 more)</code>      | Pass   | 0.000s |
| Human-readable names like STORM-FLOOD are not protocol-compliant.                         | <code>not STORM_ID.RE.match('STORM-FLOOD'); not STORM_ID.RE.match('STORM-SAFE') (+1 more)</code> | Pass   | 0.000s |
| STORM-001, STORM-0001 violate the agreed protocol.                                        | <code>not GAUGE_ID.RE.match('GAUGE-001'); not GAUGE_ID.RE.match('GAUGE-002') (+5 more)</code>    | Pass   | 0.000s |
| Rejects uppercase                                                                         | <code>not PROP_ID.RE.match('PROP-A3F7BC12'); not STORM_ID.RE.match('STORM-C7F4B8E2')</code>      | Pass   | 0.000s |
| STORM-PORT-001 uses a compound prefix — not protocol-compliant.                           | <code>not STORM_ID.RE.match('STORM-PORT-001'); not STORM_ID.RE.match('STORM-PORT-002')</code>    | Pass   | 0.000s |
| Storm from list is runnable                                                               | <code>resp.status_code == 200; len(data['storms']) &gt; 0 (+2 more)</code>                       | Pass   | 0.017s |
| Worst storm auto selection is runnable                                                    | <code>run_resp.status_code == 200</code>                                                         | Pass   | 0.012s |
| Storm list endpoint must be <code>/api/v1/propertyts/storms</code> .                      | <code>'/api/v1/propertyts/storms' in table_js</code>                                             | Pass   | 0.000s |
| Empty storm list must display a user-readable message.                                    | <code>'No flooding storms found' in table_js</code>                                              | Pass   | 0.000s |
| JS must check <code>window._preStorms</code> cache before fetching.                       | <code>'_preStorms' in table_js</code>                                                            | Pass   | 0.000s |
| When more than 30 storms exist, shows <code>'+ N more'</code> message.                    | <code>'more storms' in hist_js or '+ ' in hist_js</code>                                         | Pass   | 0.000s |

| Test Description                                                   | Acceptance Criteria                                                        | Result | Time   |
|--------------------------------------------------------------------|----------------------------------------------------------------------------|--------|--------|
| Historical tab limits display to 30 storms.                        | '30' in hist_js                                                            | Pass   | 0.000s |
| Comment or logic references peak_level sort for display.           | 'sorted by peak level' in hist_js or<br>\{\} 'pe...                        | Pass   | 0.000s |
| Storm with empty name falls back to storm_id.                      | resp.status_code == 200; s['name'] ==<br>s['storm_id']                     | Pass   | 0.037s |
| Apply storm list function exists                                   | '_applyStormList' in src                                                   | Pass   | 0.000s |
| Checks pre storms cache                                            | 'window._preStorms' in src                                                 | Pass   | 0.001s |
| Cache var should be nulled after consuming to allow refresh.       | 'window._preStorms = null' in src;<br>'window._tdPreGovDocs = null' in src | Pass   | 0.000s |
| Storm display shows effective_precipitation_mm.                    | 'effective_precipitation_mm' in<br>hist_js                                 | Pass   | 0.000s |
| Storm display shows gauges_severe count.                           | 'gauges_severe' in hist_js                                                 | Pass   | 0.000s |
| Storm display shows intensity_category.                            | 'intensity_category' in hist_js                                            | Pass   | 0.000s |
| Storm rows use max_trigger field for colour coding.                | 'max_trigger' in hist_js                                                   | Pass   | 0.000s |
| Storm display shows peak_level_m — the primary sort key.           | 'peak_level_m' in hist_js                                                  | Pass   | 0.000s |
| Storm display uses storm_id.                                       | 'storm_id' in hist_js                                                      | Pass   | 0.000s |
| Storm display uses 'name' field from the stress/storms response.   | 's.name' in hist_js or 'storm.name'<br>in hist_js                          | Pass   | 0.000s |
| Historical tab must NOT use /gauges/id_/storms for the storm list. | '_loadStormScenarios' in hist_js;<br>fn_idx >= 0 (+1 more)                 | Pass   | 0.000s |
| _loadStormScenarios function must be defined.                      | '_loadStormScenarios' in hist_js                                           | Pass   | 0.000s |
| Storm list renders in hist-storms-list element.                    | 'hist-storms-list' in hist_js;<br>'hist-storms-panel' in hist_js           | Pass   | 0.000s |
| Storm list must be loaded from the same endpoint as Stress tab.    | '/api/v1/trading/stress/storms' in<br>hist_js                              | Pass   | 0.000s |
| Sorts by peak level for auto-selection.                            | 'peak_level_m' in stress_js                                                | Pass   | 0.000s |
| Storm select element present.                                      | 'stress-storm-select' in stress_js                                         | Pass   | 0.000s |
| Custom kernel                                                      | d["decay_kernel"] == "exponential"                                         | Pass   | 0.000s |
| Decay kernel serialised as string                                  | isinstance(d["decay_kernel"], str);<br>d["decay_kernel"] == "gaussian"     | Pass   | 0.000s |
| Has storm id                                                       | storm.to_dict()["storm_id"] ==<br>"STORM-0001"                             | Pass   | 0.000s |
| Has track                                                          | "track" in d; len(d["track"]) == 2                                         | Pass   | 0.000s |

| Test Description                                                                | Acceptance Criteria                                                                          | Result | Time   |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------|--------|
| Intensity profile serialised as string                                          | <code>isinstance(d["intensity_profile"], str); d["intensity_profile"] == "triangular"</code> | Pass   | 0.000s |
| Returns dict                                                                    | <code>isinstance(d, dict); isinstance(d, dict)</code>                                        | Pass   | 0.000s |
| Track has coords                                                                | <code>"longitude" in point; "latitude" in point (+2 more)</code>                             | Pass   | 0.000s |
| storms.py lines 91-92: corrupted gauges file -i skipped in severe count loop.   | <code>r.status_code == 200</code>                                                            | Pass   | 0.030s |
| storms.py line 80: gauges dir doesn't exist -i skip severe counting.            | <code>r.status_code == 200</code>                                                            | Pass   | 0.029s |
| storms.py line 61: storm_sequences.json doesn't exist -i skip enrichment.       | <code>r.status_code == 200; data["storm_responses"]["num_storms"] == 1</code>                | Pass   | 0.031s |
| storms.py lines 75-76: corrupted storm_sequences.json -i silently skipped.      | <code>r.status_code == 200</code>                                                            | Pass   | 0.031s |
| storms.py lines 113-115: exception in storm loading -i 500.                     | <code>r.status_code == 500; r.get_json()["status"] == "error"</code>                         | Pass   | 0.027s |
| _stream_proc should call proc.wait() after exhausting stdout.                   | —                                                                                            | Pass   | 0.000s |
| _stream_proc should push each line from proc.stdout into _output_lines.         | <code>"line one" in lines; "line two" in lines (+1 more)</code>                              | Pass   | 0.000s |
| _stream_proc should strip trailing newlines before appending.                   | <code>"stripped" in lines; "stripped\\{\\n" not in lines</code>                              | Pass   | 0.000s |
| _get_predictor returns cached value without re-importing.                       | <code>result is sentinel</code>                                                              | Pass   | 0.000s |
| _get_predictor returns None when both stressm and legacy stress dirs are empty. | <code>pred is None</code>                                                                    | Pass   | 0.000s |
| All gauges trained                                                              | <code>n_skipped == 0; n_trained == n_gauges</code>                                           | Pass   | 0.000s |
| Avg auc above threshold                                                         | <code>auc &gt;= MIN_AUC</code>                                                               | Pass   | 0.000s |
| Classifier count                                                                | <code>len(classifier_files) &gt;= MIN_CLASSIFIERS</code>                                     | Pass   | 0.000s |
| Classifier count matches summary                                                | <code>n_files &gt;= n_summary</code>                                                         | Pass   | 0.000s |
| Classifier filenames well formed                                                | <code>not bad</code>                                                                         | Pass   | 0.000s |
| Classifiers not empty                                                           | <code>not empty</code>                                                                       | Pass   | 0.000s |
| Stress model dir exists                                                         | <code>STRESS_MODEL_DIR.exists()</code>                                                       | Pass   | 0.000s |
| Training summary exists                                                         | <code>TRAINING_SUMMARY.exists()</code>                                                       | Pass   | 0.000s |
| Training summary has required keys                                              | <code>not missing</code>                                                                     | Pass   | 0.000s |



| Test Description                                                            | Acceptance Criteria                                      | Result | Time   |
|-----------------------------------------------------------------------------|----------------------------------------------------------|--------|--------|
| Hits /trading/stress/run.                                                   | '/trading/stress/run' in stress_js                       | Pass   | 0.000s |
| Hits /trading/stress/storms.                                                | '/trading/stress/storms' in stress_js                    | Pass   | 0.000s |
| stress/run returns 500 on engine error.                                     | resp.status_code == 500                                  | Pass   | 0.003s |
| stress/gauges returns 500 when gauge loader raises.                         | resp.status_code == 500                                  | Pass   | 0.001s |
| stress/storms returns 500 when storm loader raises.                         | resp.status_code == 500                                  | Pass   | 0.016s |
| _tdPopulateGaugeDropdown must survive f-string rendering.                   | '_tdPopulateGaugeDropdown' in full_js                    | Pass   | 0.000s |
| tdStressGaugeChanged must survive f-string rendering in tradingdesk.py.     | 'tdStressGaugeChanged' in full_js                        | Pass   | 0.000s |
| Rendered JS must not have unescaped f-string brace artifacts.               | len(full_js) > 10000                                     | Pass   | 0.000s |
| td-stress-gauge element must be present for JS to find it.                  | "getElementById('td-stress-gauge')" in stress_js         | Pass   | 0.000s |
| _tdPopulateGaugeDropdown must exist — builds gauge selector.                | '_tdPopulateGaugeDropdown' in stress_js                  | Pass   | 0.000s |
| _tdPopulateStormDropdown must exist — builds storm list for selected gauge. | '_tdPopulateStormDropdown' in stress_js                  | Pass   | 0.000s |
| td-stress-storm element must be present for JS to find it.                  | "getElementById('td-stress-storm')" in stress_js         | Pass   | 0.000s |
| Stress functions must be INSIDE the IIFE, not after it.                     | idx_gauge_changed > 0;<br>idx_gauge_changed < iife_close | Pass   | 0.000s |
| gauge onchange must call tdStressGaugeChanged.                              | 'tdStressGaugeChanged()' in stress_js                    | Pass   | 0.000s |
| tdStressGaugeChanged must exist — called when gauge is selected.            | 'function tdStressGaugeChanged' in stress_js             | Pass   | 0.000s |
| storm onchange must call tdStressStormChanged.                              | 'tdStressStormChanged()' in stress_js                    | Pass   | 0.000s |
| tdStressStormChanged must exist — called when storm is selected.            | 'function tdStressStormChanged' in stress_js             | Pass   | 0.000s |
| 'First with trades' fallback must only run when sel.value is still empty.   | "if (!sel.value && tdStressGauges.length > 0)" in js     | Pass   | 0.000s |
| Hint selection must happen before the 'first with trades' fallback.         | idx_hint > 0; idx_fallback > idx_hint                    | Pass   | 0.000s |
| tdStressGaugeHint must be set to null after use to prevent stale hint.      | 'tdStressGaugeHint = null' in js                         | Pass   | 0.000s |

| Test Description                                                                    | Acceptance Criteria                                                                | Result | Time   |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------|--------|
| When no market hint, tdBlotterFilters.gauge_id must be the fallback.                | 'tdBlotterFilters' in js;<br>'tdBlotterFilters.gauge_id' in js                     | Pass   | 0.000s |
| loadStressData must accept a gaugeHint parameter.                                   | 'function loadStressData(gaugeHint)' in js                                         | Pass   | 0.000s |
| loadStressData must store gaugeHint in tdStressGaugeHint.                           | 'if (gaugeHint) tdStressGaugeHint = gaugeHint' in js                               | Pass   | 0.000s |
| _tdPopulateGaugeDropdown must use tdStressGaugeHint for pre-selection.              | 'tdStressGaugeHint' in js;<br>idx_hint.use > idx_populate                          | Pass   | 0.000s |
| tdStressGaugeHint state variable must exist in stress JS.                           | 'tdStressGaugeHint' in js                                                          | Pass   | 0.000s |
| switchTab('stress') must use blotter filter as primary hint.                        | idx_blotter > 0; idx_market > 0 (+4 more)                                          | Pass   | 0.000s |
| switchTab('stress') in tradingdesk must pass gauge hint to loadStressData.          | "loadStressData(stressHint)" in js;<br>"tdSelectedGauge" in js (+1 more)           | Pass   | 0.000s |
| switchTab('stress') must fall back to first blotter trade gauge when no filter set. | "tdBlotterData" in js;<br>"tdBlotterData[0].gauge_id" in js                        | Pass   | 0.000s |
| peak_position=0 uses decay-only path.                                               | len(levels) == 10; all(isinstance(v, float) for v in levels)                       | Pass   | 0.000s |
| remaining=0 uses frac=1.0 path.                                                     | len(levels) == 5                                                                   | Pass   | 0.000s |
| Both stress charts must show KO vertical annotation line.                           | 'koLine' in js; 'first_trigger_hour' in js                                         | Pass   | 0.000s |
| Stress table must show KO status per trade.                                         | 'triggered_hour' in js; 'KO H' in js                                               | Pass   | 0.000s |
| Stats bar must show knock-out count.                                                | 'num.triggered' in js; 'Knocked Out' in js                                         | Pass   | 0.000s |
| Storm id exact match required                                                       | resp.status_code == 200;<br>resp.status_code == 404                                | Pass   | 0.010s |
| Storm list has all run inputs                                                       | data['gauge_id'] == 'GAUGE-001';<br>'storm_id' in s                                | Pass   | 0.003s |
| Response includes trigger levels for threshold annotation on charts.                | 'alert_level' in data; 'warning_level' in data (+1 more)                           | Pass   | 0.009s |
| Response includes hydrograph_source string describing the data origin.              | 'hydrograph_source' in data                                                        | Pass   | 0.009s |
| Response includes storm_name and intensity_category for header display.             | data['storm_name'] == 'Severe Storm Alpha'; data['intensity_category'] == 'severe' | Pass   | 0.010s |

| Test Description                                                          | Acceptance Criteria                                                               | Result | Time   |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------|--------|
| Hourly array must be 168 elements (7-day storm window).                   | <code>len(data['hourly']) == 168</code>                                           | Pass   | 0.007s |
| Surface table must cover the full 168h window regardless of knock-out.    | <code>max(surface['hours']) &gt;= 164</code>                                      | Pass   | 0.013s |
| If the index has not changed, the cached object is returned (no re-read). | <code>id(data1) == id(data2)</code>                                               | Pass   | 0.004s |
| Replacing stress_storms/_index.json on disk is picked up on next          | <code>len(data1['storms']) == 1;</code><br><code>len(data2['storms']) == 2</code> | Pass   | 0.021s |
| Thames catchment has 52 gauges; all must be represented.                  | <code>len(gauge_ids) == 52</code>                                                 | Pass   | 0.012s |
| All gauge_id values in gauge_responses must match GAUGE-xxxx format.      | <code>not bad</code>                                                              | Pass   | 0.029s |
| All storm_id values must use STORM- prefix.                               | <code>not bad</code>                                                              | Pass   | 0.002s |
| At least one storm exceeds alert                                          | <code>any_alert</code>                                                            | Pass   | 0.000s |
| Each storm has gauge responses                                            | <code>not empty</code>                                                            | Pass   | 0.001s |
| Each storm has required fields                                            | <code>not missing</code>                                                          | Pass   | 0.003s |
| File exists                                                               | —                                                                                 | Pass   | 0.000s |
| Gauge ids well formed                                                     | <code>not bad</code>                                                              | Pass   | 0.018s |
| Gauge responses have required fields                                      | <code>not missing</code>                                                          | Pass   | 0.089s |
| Has storms                                                                | <code>len(storms) &gt; 0</code>                                                   | Pass   | 8.931s |
| Intensity categories present                                              | <code>cats</code>                                                                 | Pass   | 0.002s |
| Minimum storm count                                                       | <code>len(storms) &gt;= MIN_STORM_COUNT</code>                                    | Pass   | 0.000s |
| Peak level exceeds base                                                   | <code>not violations</code>                                                       | Pass   | 0.021s |
| Peak levels positive                                                      | <code>not violations</code>                                                       | Pass   | 0.015s |
| Storm ids unique                                                          | <code>not dupes</code>                                                            | Pass   | 0.517s |
| Storm ids well formed                                                     | <code>not bad</code>                                                              | Pass   | 0.001s |
| Storms cover multiple gauges                                              | <code>len(gauges) &gt;= 10</code>                                                 | Pass   | 0.015s |
| Storms must be sorted: gauges_severe DESC, gauges_warning DESC.           | <code>severe_counts ==</code><br><code>sorted(severe_counts, reverse=True)</code> | Pass   | 0.001s |
| Total storms key matches list length                                      | <code>data["total_storms"] ==</code><br><code>len(data["storms"])</code>          | Pass   | 0.032s |
| Trigger summary max trigger valid                                         | <code>not bad</code>                                                              | Pass   | 0.001s |
| Every storm must have a fully-populated trigger_summary dict.             | <code>not missing</code>                                                          | Pass   | 0.004s |
| No gauge id response has count                                            | <code>isinstance(data['count'], int);</code><br><code>data['count'] &gt; 0</code> | Pass   | 0.002s |

| Test Description                                                          | Acceptance Criteria                                                                               | Result | Time   |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------|--------|
| Startup preloader fetches without gauge_id — must not return 400.         | <code>r.status_code == 200</code>                                                                 | Pass   | 0.002s |
| All storms in file returned — not filtered by gauge.                      | <code>'storms' in data;</code> <code>data['count'] == len(data['storms'])</code> (+1 more)        | Pass   | 0.002s |
| No gauge id returns success                                               | <code>data['status'] == 'success'</code>                                                          | Pass   | 0.002s |
| gauge_id param still filters to matching storms.                          | <code>data['status'] == 'success'; all(True for _ in data['storms'])</code>                       | Pass   | 0.004s |
| When stress_storms.json is absent, returns 404.                           | <code>resp.status_code == 404;</code> <code>'not found' in data.get('message', '').lower()</code> | Pass   | 0.001s |
| ghc_stress.get_js() returns valid JS.                                     | <code>len(stress_js) &gt; 1000</code>                                                             | Pass   | 0.000s |
| Chart canvas element must use td-prefixed ID.                             | <code>'td-stress-chart-canvas' in js</code>                                                       | Pass   | 0.000s |
| Chart sub-tabs must exist for Flood Probability, Stress P&L, and Surface. | <code>'td-stress-ctab-0' in js;</code> <code>'td-stress-ctab-1' in js</code> (+3 more)            | Pass   | 0.000s |
| Stress tab must have both chart renderers.                                | <code>'_tdRenderProbabilityChart' in js;</code> <code>'_tdRenderStressPnlChart' in js</code>      | Pass   | 0.000s |
| Stress tab must have cleanup function.                                    | <code>'tdCleanupStressCharts' in js</code>                                                        | Pass   | 0.000s |
| Stress tab must have gauge selector dropdown.                             | <code>'td-stress-gauge' in js;</code> <code>'tdStressGaugeChanged' in js</code>                   | Pass   | 0.000s |
| Stress tab must fetch from stress/-gauges endpoint.                       | <code>'/api/v1/trading/stress/gauges' in js</code>                                                | Pass   | 0.000s |
| Stress tab must have createStressView and loadStressData.                 | <code>'createStressView' in js;</code> <code>'loadStressData' in js</code>                        | Pass   | 0.000s |
| Charts must have KO annotation line.                                      | <code>'koLine' in js;</code> <code>'first_trigger_hour' in js</code>                              | Pass   | 0.000s |
| Stress tab must have Property Damage link.                                | <code>'tdStressOpenDamage' in js;</code> <code>'Property Damage' in js</code>                     | Pass   | 0.000s |
| Stress tab must have Storm Simulation link.                               | <code>'tdStressOpenGauge' in js;</code> <code>'Storm Simulation' in js</code>                     | Pass   | 0.000s |
| Stress sub-module should render without errors.                           | <code>len(js) &gt; 0</code>                                                                       | Pass   | 0.000s |
| Stress tab must have stats bar with KO info.                              | <code>'_tdRenderStressStats' in js;</code> <code>'td-stress-stats-bar' in js</code> (+1 more)     | Pass   | 0.000s |
| Stress tab must have storm selector dropdown.                             | <code>'td-stress-storm' in js;</code> <code>'tdStressStormChanged' in js</code>                   | Pass   | 0.000s |
| Stress tab must render probability surface table.                         | <code>'probability_surface' in js;</code> <code>'td-stress-surface-wrap' in js</code> (+2 more)   | Pass   | 0.000s |

| Test Description                                                   | Acceptance Criteria                                           | Result | Time   |
|--------------------------------------------------------------------|---------------------------------------------------------------|--------|--------|
| Stress tab must have trade table renderer.                         | 'tdRenderStressTable' in js;                                  | Pass   | 0.000s |
| Surface table must be horizontally scrollable for 168h of columns. | 'td-stress-table-wrap' in js                                  | Pass   | 0.000s |
| Should fetch from /api/v1/trading/stress/storms.                   | 'overflow:auto' in js; 'width:100%' not in snippet            | Pass   | 0.000s |
| Should use cash pricing formula.                                   | '/api/v1/trading/stress/storms' in js                         | Pass   | 0.000s |
| Should have two chart sub-tabs: Flood Probability and Stress P&L.  | 'cash_price' in js or 'cashPrice' in js or 'portfolio.cas...  | Pass   | 0.000s |
| Should have storm selector drop-down.                              | 'Flood Probability' in js; 'Stress P' in js (+2 more)         | Pass   | 0.000s |
| Stress test JS should render without errors.                       | 'stress-storm-select' in js; 'populateStormSelect' in js      | Pass   | 0.000s |
| Should POST to /api/v1/trading/stress/run.                         | len(js) > 0                                                   | Pass   | 0.000s |
| Should show P(flood) in chart.                                     | '/api/v1/trading/stress/run' in js; 'runStressScenario' in js | Pass   | 0.000s |
| Should pick up _stressStormHint from Historical tab.               | 'p_flood' in js                                               | Pass   | 0.000s |
| Should render trade table at peak hour.                            | '_stressStormHint' in js; '_stressStormHint = null' in js     | Pass   | 0.000s |
| Chart tab clicks should use addEventListener (IIFE-safe).          | 'stress-table-wrap' in js; 'renderStressTable' in js          | Pass   | 0.000s |
| When stressm dir has .joblib files, predictor is loaded from it.   | 'addEventListener' in js                                      | Pass   | 0.000s |
| String with options picks from list                                | result is mock_predictor                                      | Pass   | 0.001s |
| String without options returns empty                               | result in ["Brick", "Stone", "Timber"]                        | Pass   | 0.000s |
| Colour constants                                                   | result == ""                                                  | Pass   | 0.000s |
| Get styles keys                                                    | hasattr(styles_mod, name)                                     | Pass   | 0.000s |
| Rag colour maps                                                    | expected == set(S.keys())                                     | Pass   | 0.000s |
| Section rule appends flowables                                     | key in styles_mod.RAG.COLOURS; key in styles_mod.RAG.BG       | Pass   | 0.000s |
| Tbl style returns tablestyle                                       | len(story) == 3                                               | Pass   | 0.000s |
| No data returns 404                                                | isinstance(ts, TableStyle)                                    | Pass   | 0.000s |
| Options returns ok                                                 | r.status_code == 404; r.get_json()["status"] == "error"       | Pass   | 0.000s |
| With data returns success                                          | r.status_code == 200; r.status_code == 200 (+1 more)          | Pass   | 0.000s |
|                                                                    | r.status_code == 200; r.get_json()["status"] == "success"     | Pass   | 0.001s |

| Test Description                                                       | Acceptance Criteria                                                     | Result | Time   |
|------------------------------------------------------------------------|-------------------------------------------------------------------------|--------|--------|
| avg_auc_roc is None when no gauges are trained.                        | data["avg_auc_roc"] is None                                             | Pass   | 0.002s |
| Verify all expected gauge fields are in the response.                  | field in g                                                              | Pass   | 0.004s |
| avg_auc_roc averages correctly over multiple trained gauges.           | data["num_trained"] == 2;<br>data["avg_auc_roc"] == 0.85                | Pass   | 0.005s |
| Internal error → 500.                                                  | resp.status_code == 500;<br>data["status"] == "error"                   | Pass   | 0.001s |
| Gauge with .joblib but no auc_roc in summary still counted as trained. | data["num_trained"] == 1;<br>data["avg_auc_roc"] == 0.0                 | Pass   | 0.005s |
| Surface footer references KO trimming.                                 | 'trimmed at KO' in stress_js                                            | Pass   | 0.000s |
| Surface table reads probability_surface from data.                     | 'probability_surface' in stress_js                                      | Pass   | 0.000s |
| Renders an HTML table with monospace font.                             | 'border-collapse' in stress_js;<br>'font-family:monospace' in stress_js | Pass   | 0.000s |
| Surface table applies trigger-band background colours.                 | '#FFEBEE' in stress_js; '#FFF3E0' in stress_js (+1 more)                | Pass   | 0.000s |
| Surface tab creates a wrapping div.                                    | 'stress-surface-wrap' in stress_js                                      | Pass   | 0.000s |
| Lines 72-76: water_level != severe_level → no flooding.                | —                                                                       | Pass   | 0.000s |
| Line 96: flood_depths[i] = min(depth, 5.0).                            | np.all(depths <= 5.0)                                                   | Pass   | 0.000s |
| Lines 89-92: distance_factor reduces depth proportionally.             | depths[0] > 0; depths[1] > 0                                            | Pass   | 0.000s |
| Lines 72: len(gauge_data) == 0.                                        | —                                                                       | Pass   | 0.000s |
| Lines 93-94: distance != max_distance → depth = 0.0.                   | depths[0] == 0.0                                                        | Pass   | 0.000s |
| Lines 72-96: water_level != severe_level, close to Thames.             | np.any(depths > 0)                                                      | Pass   | 0.000s |
| Lines 72: gauge_data is None.                                          | —                                                                       | Pass   | 0.000s |
| loadPortfolioImpact() must be defined.                                 | 'loadPortfolioImpact' in table_js                                       | Pass   | 0.000s |
| Portfolio impact fetch must use the correct endpoint path.             | 'portfolio-impact' in table_js                                          | Pass   | 0.000s |
| Sort state variables must be initialised.                              | 'spSortCol' in table_js; 'spSortAsc' in table_js                        | Pass   | 0.000s |
| Summary card labels must be present.                                   | 'Properties Affected' in table_js;<br>'Total Damage' in table_js        | Pass   | 0.000s |

| Test Description                                                       | Acceptance Criteria                                                                                             | Result | Time   |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------|--------|
| Alt rows amber                                                         | <code>len(ALT_ROWS_AMBER) == 1;</code><br><code>ALT_ROWS_AMBER[0][0] ==</code><br><code>'ROWBACKGROUNDS'</code> | Pass   | 0.000s |
| Alt rows blue                                                          | <code>len(ALT_ROWS_BLUE) == 1;</code><br><code>ALT_ROWS_BLUE[0][0] ==</code><br><code>'ROWBACKGROUNDS'</code>   | Pass   | 0.000s |
| Alt rows green                                                         | <code>len(ALT_ROWS_GREEN) == 1;</code><br><code>ALT_ROWS_GREEN[0][0] ==</code><br><code>'ROWBACKGROUNDS'</code> | Pass   | 0.000s |
| Hdr style first command is back-ground                                 | <code>HDR_STYLE[0][0] == 'BACKGROUND'</code>                                                                    | Pass   | 0.000s |
| Hdr style has grid                                                     | <code>'GRID' in cmds; 'BOX' in cmds</code>                                                                      | Pass   | 0.000s |
| Hdr style is list                                                      | <code>isinstance(HDR_STYLE, list);</code><br><code>len(HDR_STYLE) &gt; 0</code>                                 | Pass   | 0.000s |
| any_flood has a lower threshold than severe -i higher or equal spread. | <code>any_flood_5yr &gt;= severe_5yr - 1e-6</code>                                                              | Pass   | 0.216s |
| PRS spread should be non-negative for all tenors.                      | <code>all(s &gt;= 0 for s in spreads)</code>                                                                    | Pass   | 0.213s |
| Survival between zero and one                                          | <code>0.0 &lt;= s &lt;= 1.0</code>                                                                              | Pass   | 0.218s |
| Coverage booleans are bool                                             | <code>isinstance(val, bool)</code>                                                                              | Pass   | 0.000s |
| Coverage has required keys                                             | <code>key in tc</code>                                                                                          | Pass   | 0.000s |
| Coverage has file                                                      | <code>"test_file" in tc;</code><br><code>isinstance(tc["test_file"], str)</code>                                | Pass   | 0.000s |
| GAUGE-001, GAUGE-002, GAUGE-9042bd95 are backward-compat exceptions.   | <code>compat_ids == {'GAUGE-001', 'GAUGE-002', 'GAUGE-9042bd95'}</code>                                         | Pass   | 0.000s |
| gauge.json and gaugehc.json gauge IDs must be identical sets.          | <code>gauge_json_ids == gaugehc_ids</code>                                                                      | Pass   | 0.000s |
| The new gauge entries in SAMPLE_GAUGEHC follow GAUGE-{8hex}.           | <code>not violations</code>                                                                                     | Pass   | 0.000s |
| New gauge IDs added in conftest (GAUGE-{8hex}) follow the protocol.    | <code>GAUGE_ID_RE.match(gid)</code>                                                                             | Pass   | 0.000s |
| Gauge IDs in SAMPLE_STRESS_STORMS gauge_responses are from _data.      | <code>not violations</code>                                                                                     | Pass   | 0.000s |
| _data.py STORM_PORT_* constants follow the protocol.                   | <code>STORM_ID_RE.match(STORM_PORT_SEVERE);</code><br><code>STORM_ID_RE.match(STORM_PORT_ALERT)</code>          | Pass   | 0.000s |



| Test Description                                                      | Acceptance Criteria                                                  | Result | Time   |
|-----------------------------------------------------------------------|----------------------------------------------------------------------|--------|--------|
| All storm.id values in SAMPLE_PORT_STRESS_STORMS follow STORM-{8hex}. | STORM.ID.RE.match(sid)                                               | Pass   | 0.000s |
| All storm.id values in SAMPLE_STRESS_STORMS follow STORM-{8hex}.      | STORM.ID.RE.match(sid)                                               | Pass   | 0.000s |
| _data.py STORM_SEVERE and STORM_WARNING follow the protocol.          | STORM.ID.RE.match(STORM_SEVERE);<br>STORM.ID.RE.match(STORM_WARNING) | Pass   | 0.000s |
| Has lines field                                                       | "lines" in data;<br>isinstance(data["lines"], list)                  | Pass   | 0.000s |
| Has running field                                                     | "running" in data;<br>isinstance(data["running"], bool)              | Pass   | 0.000s |
| Lines includes appended                                               | "test line" in data["lines"]                                         | Pass   | 0.000s |
| Not running initially                                                 | data["running"] is False                                             | Pass   | 0.000s |
| Returns 200                                                           | resp.status_code == 200                                              | Pass   | 0.000s |
| Returns not found when no file                                        | data["status"] == "not_found"                                        | Pass   | 0.000s |
| Returns ok when file exists                                           | data["status"] == "ok"                                               | Pass   | 0.003s |
| Returns summary fields                                                | "summary" in data; "failures" in data<br>(+3 more)                   | Pass   | 0.002s |
| Running field is bool                                                 | "running" in data;<br>isinstance(data["running"], bool)              | Pass   | 0.000s |
| Historical high exceeds severe by less than 1m                        | "Elevated risk" in str(elements)                                     | Pass   | 0.000s |
| Historical high exceeds severe by more than 1m                        | "High risk" in str(elements)                                         | Pass   | 0.001s |
| Historical high within severe                                         | "within severe warning levels" in<br>str(elements)                   | Pass   | 0.000s |
| Moderate alert range                                                  | "Moderate alert range" in<br>str(elements)                           | Pass   | 0.000s |
| Narrow alert range                                                    | "Narrow alert range" in<br>str(elements)                             | Pass   | 0.000s |
| Wide alert range                                                      | "Wide alert range" in str(elements)                                  | Pass   | 0.000s |
| Alert status                                                          | "ELEVATED RISK" in<br>str(self._elements("Alert"))                   | Pass   | 0.000s |
| Normal status                                                         | "NORMAL" in<br>str(self._elements("Normal"))                         | Pass   | 0.001s |
| Severe status                                                         | "CRITICAL RISK" in<br>str(self._elements("Severe"))                  | Pass   | 0.000s |
| Timeseries with no level                                              | isinstance(elements, list)                                           | Pass   | 0.000s |
| Timeseries with no readings                                           | isinstance(elements, list)                                           | Pass   | 0.000s |

| Test Description                                                    | Acceptance Criteria                                         | Result | Time   |
|---------------------------------------------------------------------|-------------------------------------------------------------|--------|--------|
| Warning status                                                      | "HIGH RISK" in<br>str(self._elements("Warning"))            | Pass   | 0.000s |
| Test getting timeseries for a gauge from portfolio.                 | len(active_gauges) > 0; len(readings) > 0                   | Pass   | 0.001s |
| Test that timeseries gauge IDs exist in portfolio.                  | gauge is not None                                           | Pass   | 0.001s |
| Count returns number of gauge files                                 | TimeseriesLoader(tmp_path).count() == 2                     | Pass   | 0.001s |
| Get available gauge ids                                             | "GAUGE-001" in ids; "GAUGE-002" in ids                      | Pass   | 0.001s |
| Get available gauge ids missing dir                                 | TimeseriesLoader(tmp_path / "nonexistent").get_available... | Pass   | 0.000s |
| List all returns summaries                                          | isinstance(TimeseriesLoader(tmp_path).list_all(), list)     | Pass   | 0.002s |
| Load all from directory                                             | len(TimeseriesLoader(tmp_path).load_all()) >= 1             | Pass   | 0.002s |
| Missing dir returns empty                                           | TimeseriesLoader(tmp_path / "nonexistent").load_all() == [] | Pass   | 0.000s |
| Test that entity ID is the timestamp.                               | entity_id is not None; "2025-01-01" in entity_id            | Pass   | 0.001s |
| Test timeseries summary generation.                                 | len(summaries) == 3; 'timestamp' in summary (+3 more)       | Pass   | 0.001s |
| Test loading timeseries data.                                       | len(records) == 3                                           | Pass   | 0.001s |
| Clear cache                                                         | not loader._cache_valid                                     | Pass   | 0.001s |
| Force reload                                                        | "GAUGE-002" in loader.get_available_gauge_ids()             | Pass   | 0.003s |
| Invalidate cache                                                    | not loader._cache_valid                                     | Pass   | 0.000s |
| Test getting most recent reading.                                   | latest is not None; latest['hour'] == 2 (+1 more)           | Pass   | 0.001s |
| Test getting peak (highest) reading.                                | peak is not None; peak['level'] == 4.8                      | Pass   | 0.001s |
| Test getting readings for a specific gauge.                         | len(readings) == 3; 'timestamp' in reading (+1 more)        | Pass   | 0.001s |
| Test getting readings using gauge name.                             | len(readings) == 3                                          | Pass   | 0.001s |
| Test getting readings for non-existent gauge.                       | readings == []                                              | Pass   | 0.000s |
| Readings with unparseable timestamps are skipped (all-gauges path). | len(result) == 1                                            | Pass   | 0.001s |
| Readings with unparseable timestamps are skipped (gauge_id path).   | len(result) == 1                                            | Pass   | 0.001s |

| Test Description                                            | Acceptance Criteria                                                                                             | Result | Time   |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------|--------|
| Get readings in range all gauges                            | <code>isinstance(result, list)</code>                                                                           | Pass   | 0.001s |
| Get readings in range for gauge                             | <code>isinstance(result, list)</code>                                                                           | Pass   | 0.001s |
| Get lareading                                               | <code>TimeseriesLoader(tmp_path).get_latest_reading(gauge="GAUGE-001")</code>                                   | Pass   | 0.001s |
| Get lareading missing returns none                          | <code>TimeseriesLoader(tmp_path).get_latest_reading(gauge="GAUGE-999")</code>                                   | Pass   | 0.000s |
| Get peak reading                                            | <code>TimeseriesLoader(tmp_path).get_peak_reading(gauge="GAUGE-001")</code>                                     | Pass   | 0.001s |
| ...                                                         | ...                                                                                                             |        |        |
| Get peak reading missing returns none                       | <code>TimeseriesLoader(tmp_path).get_peak_reading(gauge="GAUGE-999")</code>                                     | Pass   | 0.000s |
| ...                                                         | ...                                                                                                             |        |        |
| Get readings for gauge                                      | <code>isinstance(result, list)</code> and <code>len(result) &gt;= 1</code>                                      | Pass   | 0.001s |
| Gauge ID not found — fallback to gauge_name search.         | <code>isinstance(result, list)</code>                                                                           | Pass   | 0.001s |
| Get readings missing gauge returns empty                    | <code>TimeseriesLoader(tmp_path).get_readings_for_gauge("GAUGE-999")</code>                                     | Pass   | 0.000s |
| Get storm responses                                         | <code>len(responses) == 1;</code><br><code>responses[0]["storm_id"] == "STORM-0001"</code>                      | Pass   | 0.001s |
| Get storm responses missing gauge                           | <code>TimeseriesLoader(tmp_path).get_storm_responses(gauge="GAUGE-999")</code>                                  | Pass   | 0.000s |
| Test getting list of gauge IDs in timeseries.               | <code>len(gauge_ids) == 2;</code> "GAUGE-001" in <code>gauge_ids</code> (+1 more)                               | Pass   | 0.000s |
| Test calculating gauge statistics.                          | <code>stats['gauge.id'] == "GAUGE-001";</code><br><code>stats['reading_count'] == 3</code> (+4 more)            | Pass   | 0.001s |
| Get gauge statistics missing gauge                          | <code>stats["reading_count"] == 0;</code> "error" in <code>stats</code>                                         | Pass   | 0.000s |
| Readings with no numeric level field return error in stats. | "error" in <code>stats</code>                                                                                   | Pass   | 0.001s |
| Test statistics for non-existent gauge.                     | <code>stats['reading_count'] == 0;</code> 'error' in <code>stats</code>                                         | Pass   | 0.000s |
| Get gauge statistics with data                              | "reading_count" in <code>stats</code> ;<br><code>stats["reading_count"] &gt;= 1</code>                          | Pass   | 0.001s |
| Exists by timestamp                                         | <code>loader.exists(result[0]["timestamp"])</code> is True                                                      | Pass   | 0.001s |
| Find by id not found returns none                           | <code>TimeseriesLoader(tmp_path).find_by_id("1999-01-01T00:00:00")</code>                                       | Pass   | 0.001s |
| Get file status alias                                       | "entity_name" in <code>TimeseriesLoader(tmp_path).get_file_stat...</code>                                       | Pass   | 0.000s |
| Get status returns dict                                     | "entity_name" in <code>status</code> ; "exists" in <code>status</code>                                          | Pass   | 0.000s |
| Test getting readings in time range.                        | <code>len(readings) == 2</code>                                                                                 | Skip   | 0.000s |
| Test getting readings in range for specific gauge.          | <code>len(readings) &gt;= 2;</code><br><code>reading.get('gaugeId') == "GAUGE-001" or 'level' in reading</code> | Skip   | 0.000s |

| Test Description                                             | Acceptance Criteria                                                                         | Result | Time   |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| Bad gauge data is skipped                                    | <code>len(result["readings"]) &gt;= 1</code>                                                | Pass   | 0.000s |
| Avg per gauge seconds                                        | <code>abs(result - 160 / 15) &lt; 0.001</code>                                              | Pass   | 0.000s |
| Avg per gauge seconds empty                                  | <code>_avg_per_gauge_seconds({"runs": []}) == 0.0</code>                                    | Pass   | 0.000s |
| Load timings corrupt json                                    | <code>result == {"runs": []}</code>                                                         | Pass   | 0.001s |
| Load timings missing file                                    | <code>result == {"runs": []}</code>                                                         | Pass   | 0.000s |
| Load timings valid                                           | <code>len(result["runs"]) == 1;</code><br><code>result["runs"][0]["num-gauges"] == 5</code> | Pass   | 0.002s |
| Save timings truncates to 20                                 | <code>len(saved["runs"]) == 20;</code><br><code>saved["runs"][0]["i"] == 10</code>          | Pass   | 0.001s |
| ConstructionYear triggers building age row in summary table. | <code>isinstance(result, list); len(result) &gt; 0</code>                                   | Pass   | 0.000s |
| Empty property does not crash                                | <code>isinstance(result, list)</code>                                                       | Pass   | 0.000s |
| Missing nested key falls back to 'Unknown Property'.         | <code>result == "Unknown Property"</code>                                                   | Pass   | 0.000s |
| Get property id present                                      | <code>result == "PROP-001"</code>                                                           | Pass   | 0.000s |
| Has tables                                                   | <code>any(isinstance(e, Table) for e in result)</code>                                      | Pass   | 0.000s |
| Has title paragraph                                          | <code>any("Comprehensive Property Analysis Report" in t for t i...</code>                   | Pass   | 0.000s |
| Property id in output                                        | <code>any("PROP-001" in t for t in texts)</code>                                            | Pass   | 0.000s |
| Property overview section header                             | <code>any("Property Overview" in t for t in texts)</code>                                   | Pass   | 0.000s |
| Property summary section header                              | <code>any("Property Summary" in t for t in texts)</code>                                    | Pass   | 0.000s |
| Returns list                                                 | <code>isinstance(result, list); len(result) &gt; 0</code>                                   | Pass   | 0.000s |
| Unknown property id fallback                                 | <code>any("Unknown Property" in t for t in texts)</code>                                    | Pass   | 0.000s |
| With mortgage data does not crash                            | <code>isinstance(result, list)</code>                                                       | Pass   | 0.000s |
| Step owners covers all graph steps                           | <code>step in dl_mod.STEP.OWNERS</code>                                                     | Pass   | 0.000s |
| Topology renders owner                                       | <code>len(tables) &gt;= 1;</code> <code>'Owner' in header_row</code>                        | Pass   | 0.001s |
| Direct trace found                                           | <code>len(result) == 2;</code> <code>result[0]["step"] == "gauges" (+1 more)</code>         | Pass   | 0.000s |
| When no direct trace exists, scan step outputs for the ID.   | <code>len(result) == 2;</code> <code>"output" in roles (+1 more)</code>                     | Pass   | 0.000s |
| Fallback scan no match                                       | <code>result == []</code>                                                                   | Pass   | 0.000s |
| Non-string outputs are converted via str() for matching.     | <code>len(result) == 1;</code> <code>result[0]["role"] == "output" (+1 more)</code>         | Pass   | 0.000s |

| Test Description                                              | Acceptance Criteria                                                       | Result | Time   |
|---------------------------------------------------------------|---------------------------------------------------------------------------|--------|--------|
| None lineage returns empty                                    | <code>_trace.data(None, "gauge", "GAUGE-001") == []</code>                | Pass   | 0.000s |
| Trace id not found                                            | <code>result == []</code>                                                 | Pass   | 0.000s |
| Trace type not found                                          | <code>result == []</code>                                                 | Pass   | 0.000s |
| Missing data id                                               | <code>r.status_code == 400</code>                                         | Pass   | 0.000s |
| Missing data type                                             | <code>r.status_code == 400</code>                                         | Pass   | 0.000s |
| Missing data_type or data_id returns 400.                     | <code>r.status_code == 400; "required" in r.get_json()["message"]</code>  | Pass   | 0.000s |
| Fallback scan when no direct traces key matches.              | <code>r.status_code == 200; data["found"] is True (+1 more)</code>        | Pass   | 0.002s |
| No lineage file <code>=i</code> found=False, empty trace.     | <code>r.status_code == 200; data["found"] is False</code>                 | Pass   | 0.000s |
| Trace not found                                               | <code>r.status_code == 200; data["found"] is False (+1 more)</code>       | Pass   | 0.002s |
| Valid trace                                                   | <code>r.status_code == 200; data["status"] == "success" (+4 more)</code>  | Pass   | 0.002s |
| Whitespace only params                                        | <code>r.status_code == 400</code>                                         | Pass   | 0.000s |
| Gauge id chain                                                | <code>len(chain) == 4; chain[0]["step"] == "gauges" (+4 more)</code>      | Pass   | 0.000s |
| Property id chain                                             | <code>len(chain) == 3; all(c["recorded"] is False for c in chain)</code>  | Pass   | 0.000s |
| Unknown data type                                             | <code>chain == []</code>                                                  | Pass   | 0.000s |
| Posts to <code>/api/v1/prs/commit</code> .                    | <code>'/api/v1/prs/commit' in prs_js</code>                               | Pass   | 0.000s |
| Commit payload includes payer flag.                           | <code>'isPayer' in prs_js</code>                                          | Pass   | 0.000s |
| Panel should add to a Folium map without error.               | <code>len(html._children) &gt; 0</code>                                   | Pass   | 0.024s |
| Blotter sub-module should render with FS01 column.            | <code>'createBlotterView' in js; 'loadBlotterData' in js (+2 more)</code> | Pass   | 0.000s |
| EOD sub-module should render.                                 | <code>'createEodView' in js; 'loadEodData' in js</code>                   | Pass   | 0.000s |
| Aggregate View sub-module should render.                      | <code>'createMapView' in js; 'loadMapData' in js (+2 more)</code>         | Pass   | 0.000s |
| Market sub-module should render.                              | <code>'createMarketView' in js; 'loadMarketData' in js</code>             | Pass   | 0.000s |
| Ensure no unescaped single braces that could cause NameError. | <code>len(js) &gt; 1000</code>                                            | Pass   | 0.000s |
| Panel should contain all 7 tab IDs and labels.                | <code>'blotter' in js; 'market' in js (+7 more)</code>                    | Pass   | 0.000s |
| Panel should export TradingDesk to window.                    | <code>'window.TradingDesk' in js; 'window.showTradingDesk' in js</code>   | Pass   | 0.000s |
| Panel should contain the Pi button.                           | <code>'&amp;Pi;' in js</code>                                             | Pass   | 0.000s |

| Test Description                                                     | Acceptance Criteria                                                                          | Result | Time   |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------|--------|
| TradingDeskPanel JS should render without errors.                    | len(js) > 0;    '<script>' in js    (+1 more)                                                | Pass   | 0.000s |
| FS01 Matrix sub-module should render.                                | 'createRiskView' in js;<br>'loadRiskData' in js (+1 more)                                    | Pass   | 0.000s |
| get_statistics() returns panel_width and panel_height.               | stats["panel_width"] == "420px";<br>stats["panel_height"] == "600px"                         | Pass   | 0.000s |
| Td preloader does not redeclare td pre blotter                       | 'window._tdPreBlotter = null' not in src                                                     | Pass   | 0.000s |
| Td preloader does not redeclare td pre market                        | 'window._tdPreMarket = null' not in src                                                      | Pass   | 0.000s |
| Td preloader does not redeclare td preload done                      | 'var _tdPreloadDone = false' not in src                                                      | Pass   | 0.000s |
| If training is already in progress, return training status.          | resp.status_code == 200;<br>data["status"] == "training"    (+1 more)                        | Pass   | 0.001s |
| If joblib already exists, POST returns ready immediately.            | resp.status_code == 200;<br>data["status"] == "ready" (+1 more)                              | Pass   | 0.001s |
| Corrupt gauge.json = 500.                                            | resp.status_code == 500;<br>data["status"] == "error" (+1 more)                              | Pass   | 0.002s |
| Gauge not in gauge.json = 404.                                       | resp.status_code == 404;<br>data["status"] == "error" (+1 more)                              | Pass   | 0.001s |
| Missing gauge.json = 404.                                            | resp.status_code == 404;    "gauge.json not found" in data["message"]                        | Pass   | 0.001s |
| Missing storm_sequences.json = 404.                                  | resp.status_code == 404;<br>"storm_sequences.json not found" in data["message"]              | Pass   | 0.001s |
| A previously failed job can be restarted.                            | resp.status_code == 200;<br>data["status"] == "training"                                     | Pass   | 0.002s |
| Successful POST starts training and returns training status.         | resp.status_code == 200;<br>data["status"] == "training"    (+4 more)                        | Pass   | 0.002s |
| severe_warning=100m ensures no events breach it → fallback to alert. | result["label_threshold"] == "flood_alert"                                                   | Pass   | 6.067s |
| Model file written                                                   | stressm_dir.exists()                                                                         | Pass   | 6.191s |
| Result has gauge id                                                  | result.get("gauge_id") == "GAUGE-clf001"                                                     | Pass   | 6.120s |
| Result has label threshold                                           | "label_threshold" in result;<br>result["label_threshold"] in ("severe_warning", "flood_al... | Pass   | 6.109s |
| Returns result dict                                                  | isinstance(result, dict)                                                                     | Pass   | 6.137s |

| Test Description                                                  | Acceptance Criteria                                                           | Result  | Time   |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------|---------|--------|
| severe_warning=0 exercises the else branch in log_hs computation. | isinstance(result, dict)                                                      | Pass    | 6.170s |
| Exception during training sets job status to 'failed'.            | job["status"] == "failed";<br>job["error"] is not None (+1 more)              | Pass    | 0.003s |
| Successful training sets job status to 'ready'.                   | job["status"] == "ready";<br>job["error"] is None                             | Pass    | 0.011s |
| Monthly rent and annual rent                                      | isinstance(result, list)                                                      | Pass    | 0.000s |
| No transaction data                                               | isinstance(self.page().generate_elements(prop, list)                          | Pass    | 0.000s |
| Rental yield computed                                             | isinstance(result, list)                                                      | Pass    | 0.000s |
| Sale date shown                                                   | isinstance(result, list)                                                      | Pass    | 0.000s |
| Sale price and appreciation                                       | any(isinstance(e, Table) for e in result)                                     | Pass    | 0.000s |
| Sales section rendered                                            | any("Transaction" in t for t in texts)                                        | Pass    | 0.000s |
| Assess yield average                                              | page.assess_yield(5.0) == "Average yield"                                     | Pass    | 0.000s |
| Assess yield below average                                        | page.assess_yield(3.0) == "Below average yield"                               | Pass    | 0.000s |
| Assess yield excellent                                            | page.assess_yield(9.0) == "Excellent yield"                                   | Pass    | 0.000s |
| Assess yield good                                                 | page.assess_yield(7.0) == "Good yield"                                        | Pass    | 0.000s |
| Assess yield poor                                                 | page.assess_yield(1.0) == "Poor yield"                                        | Pass    | 0.000s |
| Empty property does not crash                                     | isinstance(result, list)                                                      | Pass    | 0.000s |
| Empty transactions                                                | isinstance(result, list)                                                      | Pass    | 0.000s |
| Has transactions header                                           | any("Transaction" in t or "History" in t for t in texts)                      | Pass    | 0.000s |
| Market analysis appreciation                                      | "Capital Appreciation" in result                                              | Pass    | 0.000s |
| Market analysis few vacancies                                     | "Good" in result["Rental Stability"]                                          | Pass    | 0.000s |
| Market analysis many vacancies                                    | "Concerning" in result["Rental Stability"]                                    | Pass    | 0.000s |
| Market analysis marketing above 30                                | "Marketing Efficiency" in result;<br>"Good" in result["Marketing Efficiency"] | Pass    | 0.000s |
| Market analysis marketing above 90                                | "Below average" in result["Marketing Efficiency"]                             | Pass    | 0.000s |
| Market analysis never rented                                      | "Rental History" in result;<br>rental history" in result["Rental History"]    | No Pass | 0.000s |



| Test Description                       | Acceptance Criteria                                                           | Result | Time   |
|----------------------------------------|-------------------------------------------------------------------------------|--------|--------|
| Market analysis no vacancy             | "Rental Stability" in result;<br>"Excellent" in result["Rental<br>Stability"] | Pass   | 0.000s |
| Market analysis yield                  | "Gross Rental Yield" in result                                                | Pass   | 0.000s |
| No transaction history                 | isinstance(result, list)                                                      | Pass   | 0.000s |
| Returns list                           | isinstance(result, list); len(result)<br>> 0                                  | Pass   | 0.000s |
| Sales extended marketing               | isinstance(result, list)                                                      | Pass   | 0.000s |
| Sales quick sale                       | isinstance(result, list)                                                      | Pass   | 0.000s |
| Sales very long marketing              | isinstance(result, list)                                                      | Pass   | 0.000s |
| With mortgage data                     | isinstance(result, list)                                                      | Pass   | 0.000s |
| With rental data                       | isinstance(result, list);<br>any(isinstance(e, Table) for e<br>in result)     | Pass   | 0.000s |
| With sales data                        | isinstance(result, list);<br>any(isinstance(e, Table) for e<br>in result)     | Pass   | 0.000s |
| Gauges alert count                     | single_storm_data["trigger_summary"]["gauges_alert"]<br>== 2                  | Pass   | 0.000s |
| Gauges impacted matches alert          | ts["gauges_impacted"] ==<br>ts["gauges_alert"]                                | Pass   | 0.000s |
| Gauges severe count                    | single_storm_data["trigger_summary"]["gauges_severe"]<br>== 1                 | Pass   | 0.000s |
| Gauges warning count                   | single_storm_data["trigger_summary"]["gauges_warning"]<br>== 1                | Pass   | 0.000s |
| Max trigger is alert when alert only   | data["storms"][0]["trigger_summary"]["max_trigger"]<br>== "a..."              | Pass   | 0.002s |
| Max trigger is severe                  | single_storm_data["trigger_summary"]["max_trigger"]<br>== "s..."              | Pass   | 0.000s |
| Max trigger is warning when no severe  | data["storms"][0]["trigger_summary"]["max_trigger"]<br>== "w..."              | Pass   | 0.004s |
| Trigger summary present                | "trigger_summary" in<br>single_storm_data                                     | Pass   | 0.000s |
| Assessment date updated                | r.get_json()["assessment"]["assessment_date"]<br>is not None                  | Pass   | 0.002s |
| No data file returns 404               | r.status_code == 404                                                          | Pass   | 0.000s |
| Principle not found returns 404        | r.status_code == 404                                                          | Pass   | 0.002s |
| Score 1 derives non compliant status   | r.status_code == 200; p["status"] ==<br>"Non-compliant"                       | Pass   | 0.002s |
| Score 4 derives fully compliant status | p["status"] == "Fully Compliant"                                              | Pass   | 0.002s |

| Test Description                                        | Acceptance Criteria                                                                             | Result | Time   |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|--------|
| Update evidence field                                   | <code>r.status_code == 200; p["evidence"] == "New evidence gathered"</code>                     | Pass   | 0.003s |
| Update gaps and remediation                             | <code>p["gaps"] == "Minor gap";<br/>p["remediation"] == "Action plan created"</code>            | Pass   | 0.002s |
| Valid update returns 200                                | <code>r.status_code == 200;<br/>r.get_json()["status"] == "success"</code>                      | Pass   | 0.002s |
| Not found returns 404                                   | <code>r.status_code == 404</code>                                                               | Pass   | 0.002s |
| Update changes field                                    | <code>ref["author"] == "New Author"</code>                                                      | Pass   | 0.004s |
| Update returns success                                  | <code>r.status_code == 200;<br/>r.get_json()["status"] == "success"</code>                      | Pass   | 0.003s |
| Update with new category adds to list                   | <code>"NewCat" in<br/>r.get_json()["categories"]</code>                                         | Pass   | 0.004s |
| Update sets updated at                                  | <code>r.get_json()["meeting"]["updated_at"]<br/>is not None</code>                              | Pass   | 0.003s |
| Update title                                            | <code>r.status_code == 200;<br/>r.get_json()["meeting"]["title"]<br/>== "Updated Title"</code>  | Pass   | 0.004s |
| Update unknown id returns 404                           | <code>r.status_code == 404</code>                                                               | Pass   | 0.001s |
| Line 211: update_action with non-existent meeting.      | <code>r.status_code == 404</code>                                                               | Pass   | 0.002s |
| Line 147: update_decision with non-existent meeting.    | <code>r.status_code == 404</code>                                                               | Pass   | 0.001s |
| Line 333: update_minute_item with non-existent meeting. | <code>r.status_code == 404</code>                                                               | Pass   | 0.002s |
| Line 275: update_participant with non-existent meeting. | <code>r.status_code == 404</code>                                                               | Pass   | 0.002s |
| Empty value returns 400                                 | <code>r.status_code == 400</code>                                                               | Pass   | 0.002s |
| Invalid choice value returns 400                        | <code>r.status_code == 400</code>                                                               | Pass   | 0.002s |
| Invalid field returns 400                               | <code>r.status_code == 400;<br/>r.get_json()["status"] == "error"</code>                        | Pass   | 0.002s |
| Model not found returns 404                             | <code>r.status_code == 404; r.status_code == 404 (+1 more)</code>                               | Pass   | 0.002s |
| No reason returns 400                                   | <code>r.status_code == 400</code>                                                               | Pass   | 0.002s |
| Response has audit entry                                | <code>"audit_entry" in data;<br/>data["audit_entry"]["event_type"]<br/>== "field_update"</code> | Pass   | 0.004s |
| Update lifecycle stage                                  | <code>r.status_code == 200</code>                                                               | Pass   | 0.004s |
| Update validation status                                | <code>r.status_code == 200</code>                                                               | Pass   | 0.004s |
| Valid choice field update returns success               | <code>r.status_code == 200;<br/>r.get_json()["new_value"] == "Amber"</code>                     | Pass   | 0.003s |

| Test Description                                      | Acceptance Criteria                                                                               | Result | Time   |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------|--------|
| Valid text field update returns success               | <code>r.status_code == 200; data["status"] == "success" (+2 more)</code>                          | Pass   | 0.003s |
| No inventory returns 404                              | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br>(+3 more) | Pass   | 0.000s |
| Save failure returns 500                              | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>              | Pass   | 0.002s |
| Activity not found returns 404                        | <code>r.status_code == 404</code>                                                                 | Pass   | 0.002s |
| Invalid role id returns 400                           | <code>r.status_code == 400</code>                                                                 | Pass   | 0.002s |
| Notes updated                                         | <code>activities[0]["notes"] == "New note"</code>                                                 | Pass   | 0.002s |
| Setting C or I to None (null) is valid.               | <code>r.status_code == 200</code>                                                                 | Pass   | 0.003s |
| Valid activity update returns success                 | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>            | Pass   | 0.004s |
| Escalation tier threshold updated                     | <code>triggers[0]["tier_threshold"]["1"] == "Immediate"</code>                                    | Pass   | 0.003s |
| Response has audit entry                              | <code>"audit_entry" in r.get_json();</code><br><code>"audit_entry" in r.get_json()</code>         | Pass   | 0.003s |
| Trigger not found returns 404                         | <code>r.status_code == 404</code>                                                                 | Pass   | 0.002s |
| Valid escalation update returns success               | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>            | Pass   | 0.003s |
| Backup field updated                                  | <code>owner["backup"] == "Owner B"</code>                                                         | Pass   | 0.003s |
| No raci file returns 404                              | <code>r.status_code == 404</code>                                                                 | Pass   | 0.000s |
| Response has audit entry                              | <code>"audit_entry" in r.get_json();</code><br><code>"audit_entry" in r.get_json()</code>         | Pass   | 0.003s |
| Role not found returns 404                            | <code>r.status_code == 404</code>                                                                 | Pass   | 0.001s |
| Valid role update returns success                     | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>            | Pass   | 0.002s |
| When all gauges are skipped, avg_auc_roc is 0.        | <code>summary["num_trained"] == 0;</code><br><code>summary["avg_auc_roc"] == 0.0</code>           | Pass   | 0.001s |
| Creates training_summary.json when it does not exist. | <code>summary_path.exists();</code><br><code>summary["num_gauges"] == 1</code> (+5 more)          | Pass   | 0.001s |
| Creates new summary                                   | <code>summary["num_gauges"] == 1;</code><br><code>summary["num_trained"] == 1</code> (+1 more)    | Pass   | 0.004s |
| Merges a new gauge result into an existing summary.   | <code>summary["num_gauges"] == 2;</code><br><code>summary["num_trained"] == 2</code> (+2 more)    | Pass   | 0.001s |
| Merges into existing                                  | <code>summary["num_gauges"] == 2;</code><br><code>summary["num_trained"] == 2</code> (+1 more)    | Pass   | 0.002s |

| Test Description                                                  | Acceptance Criteria                                                               | Result | Time   |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------|--------|
| Re-training a gauge replaces its entry, not duplicates it.        | summary["num_gauges"] == 1;<br>summary["num_trained"] == 1 (+2 more)              | Pass   | 0.001s |
| Replaces existing gauge                                           | summary["num_gauges"] == 1;<br>summary["gauges"][0]["metrics"]["auc_roc"] == 0.95 | Pass   | 0.002s |
| Skipped gauge not counted as trained                              | summary["num_trained"] == 0;<br>summary["num_skipped"] == 1                       | Pass   | 0.001s |
| A 'skipped' gauge does not contribute to avg_auc_roc.             | summary["num_gauges"] == 2;<br>summary["num_trained"] == 1 (+2 more)              | Pass   | 0.001s |
| All valid statuses                                                | r.status_code == 200                                                              | Pass   | 0.013s |
| Invalid status returns 400                                        | r.status_code == 400;<br>r.get_json()["status"] == "error"                        | Pass   | 0.002s |
| Model not found returns 404                                       | r.status_code == 404;<br>r.get_json()["status"] == "error" (+3 more)              | Pass   | 0.002s |
| Question not found returns 404                                    | r.status_code == 404                                                              | Pass   | 0.002s |
| Risk rating recalculated                                          | rr["last_calculated"] is not None                                                 | Pass   | 0.004s |
| Valid update has model and audit entry                            | "model" in data; "audit_entry" in data                                            | Pass   | 0.003s |
| Valid update returns success                                      | r.status_code == 200;<br>r.get_json()["status"] == "success"                      | Pass   | 0.004s |
| update_yield_curve creates yield_curve from default when missing. | '1' in updated['yield_curve'];<br>updated['yield_curve']['1'] == 0.055            | Pass   | 0.003s |
| Empty filename returns 400                                        | r.status_code == 400                                                              | Pass   | 0.001s |
| No file returns 400                                               | r.status_code == 400; data["status"] == "error"                                   | Pass   | 0.000s |
| Upload appears in list                                            | len(data["documents"]) == 1                                                       | Pass   | 0.003s |
| Upload response has id                                            | "id" in data["document"]                                                          | Pass   | 0.002s |
| Upload stores description                                         | data["document"]["description"] == "My description"                               | Pass   | 0.002s |
| Valid upload returns 200                                          | r.status_code == 200; data["status"] == "success" (+1 more)                       | Pass   | 0.002s |
| Upload save failure returns 500                                   | r.status_code == 500; data["status"] == "error" (+1 more)                         | Pass   | 0.028s |
| Get page categories contains expected keys                        | key in cats                                                                       | Pass   | 0.001s |
| Get page categories contains expected keys                        | key in cats                                                                       | Pass   | 0.001s |

| Test Description                                                           | Acceptance Criteria                                               | Result | Time   |
|----------------------------------------------------------------------------|-------------------------------------------------------------------|--------|--------|
| Get page categories returns copy                                           | "_new_key_" not in gen.categories                                 | Pass   | 0.001s |
| Get page categories returns copy                                           | "_new_key_" not in gen.categories                                 | Pass   | 0.002s |
| Mutating the returned dict must not affect the generator's internal state. | "_new_key_" not in gen.categories                                 | Pass   | 0.001s |
| Get page categories returns dict                                           | isinstance(cats, dict)                                            | Pass   | 0.001s |
| Get page categories returns dict                                           | isinstance(cats, dict)                                            | Pass   | 0.002s |
| Get page categories returns dict                                           | isinstance(cats, dict)                                            | Pass   | 0.001s |
| List available pages contains all expected                                 | expected <= set(pages)                                            | Pass   | 0.002s |
| List available pages contains all expected                                 | expected <= set(pages)                                            | Pass   | 0.001s |
| List available pages contains expected                                     | expected <= set(pages)                                            | Pass   | 0.001s |
| List available pages non empty                                             | len(gen.list.available_pages()) > 0                               | Pass   | 0.001s |
| List available pages non empty                                             | len(gen.list.available_pages()) > 0                               | Pass   | 0.001s |
| List available pages returns list                                          | isinstance(pages, list)                                           | Pass   | 0.001s |
| List available pages returns list                                          | isinstance(pages, list)                                           | Pass   | 0.002s |
| List available pages returns list                                          | isinstance(pages, list)                                           | Pass   | 0.001s |
| Validate pages all invalid                                                 | valid == []; "bogus_one" in invalid<br>(+1 more)                  | Pass   | 0.001s |
| Validate pages all invalid                                                 | valid == []; "no_such_page" in invalid<br>(+1 more)               | Pass   | 0.002s |
| Validate pages all invalid                                                 | valid == []; "bogus_one" in invalid<br>(+1 more)                  | Pass   | 0.001s |
| Validate pages all valid                                                   | "title_overview" in valid;<br>"sensor_details" in valid (+1 more) | Pass   | 0.001s |
| Validate pages all valid                                                   | "title_overview" in valid;<br>"data_summary" in valid (+1 more)   | Pass   | 0.002s |
| Validate pages all valid                                                   | "title" in valid; "appendix" in valid<br>(+1 more)                | Pass   | 0.001s |
| Validate pages empty list                                                  | valid == []; invalid == []                                        | Pass   | 0.001s |
| Validate pages empty list                                                  | valid == []; invalid == []                                        | Pass   | 0.002s |
| Validate pages empty list                                                  | valid == []; invalid == []                                        | Pass   | 0.001s |
| Validate pages mixed                                                       | "title_overview" in valid;<br>"nonexistent_page" in invalid       | Pass   | 0.001s |
| Validate pages mixed                                                       | "title_overview" in valid;<br>"nonexistent_page" in invalid       | Pass   | 0.002s |
| Validate pages mixed                                                       | "title" in valid; "nonexistent_page"<br>in invalid                | Pass   | 0.001s |

| Test Description                                                      | Acceptance Criteria                                                                                                 | Result | Time   |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------|--------|
| The error message should be a string representation of the exception. | "validation_error" in result;<br>isinstance(result["validation_error"][0],<br>str)                                  | Pass   | 0.000s |
| Another path to trigger the except branch.                            | "validation_error" in result                                                                                        | Pass   | 0.000s |
| Lines 251-252: passing data that causes an exception in the try       | "validation_error" in result;<br>isinstance(result["validation_error"],<br>list) (+1 more)                          | Pass   | 0.000s |
| Consistent chain                                                      | result["is_consistent"];<br>result["stale_steps"] == [] (+1<br>more)                                                | Pass   | 0.000s |
| Empty manifest                                                        | not result["is_consistent"];<br>len(result["missing_steps"]) ==<br>10                                               | Pass   | 0.000s |
| Stale step detected                                                   | not result["is_consistent"];<br>"properties" in result["stale_steps"]                                               | Pass   | 0.000s |
| Alt key structure                                                     | validate_gauge_exists("GAUGE-002",<br>gauge_file) is True                                                           | Pass   | 0.001s |
| Existing gauge returns true                                           | validate_gauge_exists("GAUGE-001",<br>gauge_file) is True                                                           | Pass   | 0.002s |
| Missing file returns false                                            | validate_gauge_exists("GAUGE-001",<br>tmp_path / "missing.js...)                                                    | Pass   | 0.000s |
| Nonexistent gauge returns false                                       | validate_gauge_exists("GAUGE-XXX",<br>gauge_file) is False                                                          | Pass   | 0.001s |
| Binary file returns false                                             | result is False                                                                                                     | Pass   | 0.001s |
| Corrupt json returns false                                            | result is False                                                                                                     | Pass   | 0.001s |
| Line 196: audit log overflow during VQ update.                        | r.status_code == 200                                                                                                | Pass   | 0.055s |
| Line 130: inventory not found → 404.                                  | r.status_code == 404                                                                                                | Pass   | 0.001s |
| Line 176: save failure → 500.                                         | r.status_code == 500                                                                                                | Pass   | 0.002s |
| Update triggers risk recalculation                                    | rr["calculated_score"] is not<br>None; rr["calculated_rating"] in<br>["Acceptable", "Conditional", ...<br>(+1 more) | Pass   | 0.005s |
| Update validation question                                            | response.status_code == 200;<br>data["status"] == "success" (+5<br>more)                                            | Pass   | 0.003s |
| Update validation question invalid status                             | response.status_code == 400                                                                                         | Pass   | 0.002s |

| Test Description                                                                          | Acceptance Criteria                                                                  | Result | Time   |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|--------|
| Update validation question not found                                                      | <code>response.status_code == 404</code>                                             | Pass   | 0.002s |
| Base pct wins when higher than depth factor                                               | <code>var == pytest.approx(350.000)</code>                                           | Pass   | 0.000s |
| Depth 1 to 2m                                                                             | <code>var == pytest.approx(500.000)</code>                                           | Pass   | 0.000s |
| Depth half to 1m                                                                          | <code>var == pytest.approx(250.000)</code>                                           | Pass   | 0.000s |
| Depth over 2m                                                                             | <code>var == pytest.approx(800.000)</code>                                           | Pass   | 0.000s |
| Depth under half m                                                                        | <code>var == pytest.approx(100.000)</code>                                           | Pass   | 0.000s |
| Negative property value                                                                   | <code>RiskAssessor.calculate_value_at_risk(-100.000, "High") == ...</code>           | Pass   | 0.000s |
| Risk level percentages                                                                    | <code>var == pytest.approx(1.000_000 * pct)</code>                                   | Pass   | 0.000s |
| Zero depth uses base pct                                                                  | <code>var == pytest.approx(150.000)</code>                                           | Pass   | 0.000s |
| Zero property value                                                                       | <code>RiskAssessor.calculate_value_at_risk(0, "High") == 0.0</code>                  | Pass   | 0.000s |
| When verbose=False, the per-counterparty log lines are absent.                            | <code>"[1/2]" not in caplog.text; "Wrote 2 counterparties" not in caplog.text</code> | Pass   | 0.000s |
| Lines 169-170: when verbose=True, each counterparty is logged.                            | <code>len(result["data"]) == 3; "[1/3]" in caplog.text (+2 more)</code>              | Pass   | 0.001s |
| Line 187: verbose=True logs final summary with count and path.                            | <code>"Wrote 2 counterparties" in caplog.text</code>                                 | Pass   | 0.001s |
| With count ; 2000 and verbose=False, no mid-loop progress line.                           | <code>"ETA" not in out or "[" not in out</code>                                      | Pass   | 0.072s |
| count=501 with verbose=True triggers the progress line at i=500.                          | <code>result["num_sequences"] == 501; "[500" in out or "500/" in out</code>          | Pass   | 0.607s |
| <code>create_visualization()</code> returns a <code>TCEventVisualization</code> instance. | <code>isinstance(viz, TCEventVisualization)</code>                                   | Pass   | 0.000s |
| All addressed                                                                             | <code>_vq_summary(model) == "3/3"</code>                                             | Pass   | 0.000s |
| Empty questions                                                                           | <code>_vq_summary(model) == "0/0"</code>                                             | Pass   | 0.000s |
| None addressed                                                                            | <code>_vq_summary(model) == "0/2"</code>                                             | Pass   | 0.000s |
| Some not applicable                                                                       | <code>_vq_summary(model) == "1/2"</code>                                             | Pass   | 0.000s |
| <code>_walk_upstream</code> should not infinite loop on revisited nodes.                  | <code>result.count("gauges") == 1</code>                                             | Pass   | 0.000s |
| Deep chain                                                                                | <code>"gauges" in result; "propertyhc" in result</code>                              | Pass   | 0.000s |
| Mid chain                                                                                 | <code>"hazard" in downstream; "propertyts" in downstream (+2 more)</code>            | Pass   | 0.000s |
| Root has no upstream                                                                      | <code>result == ["gauges"]</code>                                                    | Pass   | 0.000s |